

fong-spivak-invitation: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

| Identifier                    | Page | Conf.   | LaTeX source                           | Rendered                               | Image                                                                                                                                                   |
|-------------------------------|------|---------|----------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| fong-spivak-invitation_F00001 | 17   | ■ 0.998 | a                                      | $a$                                    | Connections are symmetric, so if $a$ is connected to $b$ , then $b$ is connected to $a$ . Connec-                                                       |
| fong-spivak-invitation_F00002 | 17   | ■ 0.998 | b                                      | $b$                                    | Connections are symmetric, so if $a$ is connected to $b$ , then $b$ is connected to $a$ . Connec-                                                       |
| fong-spivak-invitation_F00003 | 53   | ■ 1.000 | $\mathcal{V}$                          | $\mathcal{V}$                          | preorders $\mathcal{V}$ .                                                                                                                               |
| fong-spivak-invitation_F00004 | 15   | ■ 1.000 | $f: X \rightarrow Y$                   | $f: X \rightarrow Y$                   | maps. A map $f: X \rightarrow Y$ is a kind of observation of object $X$ via a specified relationship                                                    |
| fong-spivak-invitation_F00005 | 15   | ■ 1.000 | $X$                                    | $X$                                    | maps. A map $f: X \rightarrow Y$ is a kind of observation of object $X$ via a specified relationship                                                    |
| fong-spivak-invitation_F00006 | 15   | ■ 1.000 | $Y$                                    | $Y$                                    | it has with another object, $Y$ . For example, think of $X$ as the subject of an experiment                                                             |
| fong-spivak-invitation_F00007 | 15   | ■ 1.000 | $f$                                    | $f$                                    | Asking which aspects of $X$ one wants to preserve under the observation $f$ becomes                                                                     |
| fong-spivak-invitation_F00008 | 15   | ■ 1.000 | $\mathbb{R}$                           | $\mathbb{R}$                           | tions $f$ from $\mathbb{R}$ to $\mathbb{R}$ (where $\mathbb{R}$ is the set of real numbers), and we can think of them                                   |
| fong-spivak-invitation_F00009 | 15   | ■ 0.980 | $x$                                    | $x$                                    | as observations: rather than view $x$ “directly,” we only observe $f(x)$ . Out of all the                                                               |
| fong-spivak-invitation_F00010 | 15   | ■ 0.980 | $f(x)$                                 | $f(x)$                                 | as observations: rather than view $x$ “directly,” we only observe $f(x)$ . Out of all the                                                               |
| fong-spivak-invitation_F00011 | 16   | ■ 1.000 | $f: \mathbb{R} \rightarrow \mathbb{R}$ | $f: \mathbb{R} \rightarrow \mathbb{R}$ | functions $f: \mathbb{R} \rightarrow \mathbb{R}$ , only some of them preserve the order of numbers, only some of                                        |
| fong-spivak-invitation_F00012 | 16   | ■ 0.412 | $x \leq y$                             | $x \leq y$                             | (a) <i>order-preserving</i> if $x \leq y$ implies $f(x) \leq f(y)$ , for all $x, y \in \mathbb{R}$ ; <sup>1</sup>                                       |
| fong-spivak-invitation_F00013 | 16   | ■ 0.412 | $f(x) \leq f(y)$                       | $f(x) \leq f(y)$                       | (a) <i>order-preserving</i> if $x \leq y$ implies $f(x) \leq f(y)$ , for all $x, y \in \mathbb{R}$ ; <sup>1</sup>                                       |
| fong-spivak-invitation_F00014 | 16   | ■ 0.412 | $x, y \in \mathbb{R}$                  | $x, y \in \mathbb{R}$                  | (a) <i>order-preserving</i> if $x \leq y$ implies $f(x) \leq f(y)$ , for all $x, y \in \mathbb{R}$ ; <sup>1</sup>                                       |
| fong-spivak-invitation_F00015 | 16   | ■ 1.000 | $ x - y  =  f(x) - f(y) $              | $ x - y  =  f(x) - f(y) $              | (b) <i>metric-preserving</i> if $ x - y  =  f(x) - f(y) $ ;                                                                                             |
| fong-spivak-invitation_F00016 | 16   | ■ 1.000 | $f(x + y) = f(x) + f(y)$               | $f(x + y) = f(x) + f(y)$               | (c) <i>addition-preserving</i> if $f(x + y) = f(x) + f(y)$ .                                                                                            |
| fong-spivak-invitation_F00017 | 16   | —       | $\square$                              | $\square$                              | —                                                                                                                                                       |
| fong-spivak-invitation_F00018 | 16   | ■ 0.942 | $(x, y)$                               | $(x, y)$                               | <sup>1</sup> We are often taught to view functions $f: \mathbb{R} \rightarrow \mathbb{R}$ as plots in the $(x, y)$ -coordinate system, where $x$ is the |
| fong-spivak-invitation_F00019 | 16   | ■ 1.000 | $y$                                    | $y$                                    | domain (independent) variable and $y$ is the codomain (dependent) variable. In this book, we do not adhere                                              |
| fong-spivak-invitation_F00020 | 16   | ■ 1.000 | $f(x)=x^2$                             | $f(x) = x^2$                           | example consider the function $f(x) = x^2$ . Then $f$ being order-preserving would say that, for any                                                    |
| fong-spivak-invitation_F00021 | 16   | ■ 1.000 | $x, y \in \mathbb{R}$                  | $x, y \in \mathbb{R}$                  | $x, y \in \mathbb{R}$ , if $x \leq y$ then $x^2 \leq y^2$ ; is that true?                                                                               |
| fong-spivak-invitation_F00022 | 16   | ■ 1.000 | $x^2 \leq y^2$                         | $x^2 \leq y^2$                         | $x, y \in \mathbb{R}$ , if $x \leq y$ then $x^2 \leq y^2$ ; is that true?                                                                               |
| fong-spivak-invitation_F00023 | 17   | ■ 1.000 | c                                      | $c$                                    | tions are also transitive, meaning that if $a$ is connected to $b$ , and $b$ is connected to $c$ , then                                                 |

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| fong-spivak-invitation_F00024 | 17   | ■ 1.000 | a, b                                                                                    | $a, b$                                                | $a$ is connected to $c$ ; that is, all $a, b$ , and $c$ are connected. Friendship is not transitive – my                                                                |
| fong-spivak-invitation_F00025 | 17   | ■ 1.000 | $\Phi$                                                                                  | $\Phi$                                                | is observing this system. Her observation of interest, which we call $\Phi$ , extracts a single                                                                         |
| fong-spivak-invitation_F00026 | 17   | ■ 1.000 | $A$                                                                                     | $A$                                                   | Joining two systems $A$ and $B$ is performed simply by combining their connections. That                                                                                |
| fong-spivak-invitation_F00027 | 17   | ■ 1.000 | $B$                                                                                     | $B$                                                   | Joining two systems $A$ and $B$ is performed simply by combining their connections. That                                                                                |
| fong-spivak-invitation_F00028 | 17   | ■ 1.000 | $A \vee B$                                                                              | $A \vee B$                                            | is, we shall say the <i>join</i> of systems $A$ and $B$ , denoted $A \vee B$ , has a connection between                                                                 |
| fong-spivak-invitation_F00029 | 17   | ■ 1.000 | $z_{\{1\}}, \ldots, z_{\{n\}}$                                                          | $z_1, \dots, z_n$                                     | points $x$ and $y$ if there are some points <span style="border: 1px solid red; padding: 0 2px;"><math>z_1, \dots, z_n</math></span> such that each of the following is |
| fong-spivak-invitation_F00030 | 17   | ■ 0.999 | $B: x$                                                                                  | $B: x$                                                | true in at least one of $A$ or $B$ : $x$ is connected to $z_1$ , $z_i$ is connected to $z_{i+1}$ , and $z_n$ is                                                         |
| fong-spivak-invitation_F00031 | 17   | ■ 0.999 | $z_{\{1\}}, z_{\{i\}}$                                                                  | $z_1, z_i$                                            | true in at least one of $A$ or $B$ : $x$ is connected to $z_1$ , $z_i$ is connected to $z_{i+1}$ , and $z_n$ is                                                         |
| fong-spivak-invitation_F00032 | 17   | ■ 0.999 | $z_{\{i+1\}}$                                                                           | $z_{i+1}$                                             | true in at least one of $A$ or $B$ : $x$ is connected to $z_1$ , $z_i$ is connected to $z_{i+1}$ , and $z_n$ is                                                         |
| fong-spivak-invitation_F00033 | 17   | ■ 0.999 | $z_{\{n\}}$                                                                             | $z_n$                                                 | true in at least one of $A$ or $B$ : $x$ is connected to $z_1$ , $z_i$ is connected to $z_{i+1}$ , and $z_n$ is                                                         |
| fong-spivak-invitation_F00034 | 18   | ■ 0.493 | $\Phi(\left\{ \dot{\phantom{x}} \right\}) = \Phi(\left\{ \dot{\phantom{x}} \right\}^6)$ | $\Phi(\circlearrowleft) = \Phi(\circlearrowleft^6) =$ | We see above in Eq. (1.1) that $\Phi(\circlearrowleft) = \Phi(\circlearrowright) = \text{false}$ : in both cases $\bullet$ is not                                       |
| fong-spivak-invitation_F00035 | 18   | ■ 0.999 | $*$                                                                                     | $*$                                                   | connected to $*$ . On the other hand, when we join these two systems as in Eq. (1.2), we                                                                                |
| fong-spivak-invitation_F00036 | 18   | ■ 0.444 | $\Phi(\oplus \vee) = \Phi(\odot \oplus) =$                                              | $\Phi(\oplus \vee) = \Phi(\odot \oplus) =$            | see that $\Phi(\circlearrowleft \vee \circlearrowright) = \Phi(\circlearrowright) = \text{true}$ : in the joined system, $\bullet$ is connected to $*$ . The            |
| fong-spivak-invitation_F00037 | 18   | ■ 0.444 | $\bullet$                                                                               | $\bullet$                                             | see that $\Phi(\circlearrowleft \vee \circlearrowright) = \Phi(\circlearrowright) = \text{true}$ : in the joined system, $\bullet$ is connected to $*$ . The            |
| fong-spivak-invitation_F00038 | 19   | ■ 1.000 | $A \leq B$                                                                              | $A \leq B$                                            | tems $A$ and $B$ , we say that $A \leq B$ if, whenever $x$ is connected to $y$ in $A$ , then $x$ is                                                                     |
| fong-spivak-invitation_F00039 | 19   | ■ 1.000 | $\leq$                                                                                  | $\leq$                                                | This notion of $\leq$ leads to the following diagram:                                                                                                                   |
| fong-spivak-invitation_F00040 | 19   | ■ 1.000 | $A \leq (A \vee B)$                                                                     | $A \leq (A \vee B)$                                   | system that is bigger than both $A$ and $B$ . That is, $A \leq (A \vee B)$ and $B \leq (A \vee B)$ , and                                                                |
| fong-spivak-invitation_F00041 | 19   | ■ 1.000 | $B \leq (A \vee B)$                                                                     | $B \leq (A \vee B)$                                   | system that is bigger than both $A$ and $B$ . That is, $A \leq (A \vee B)$ and $B \leq (A \vee B)$ , and                                                                |
| fong-spivak-invitation_F00042 | 19   | ■ 0.996 | $C$                                                                                     | $C$                                                   | for any $C$ , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$ . Let’s walk through this with an                                                                   |
| fong-spivak-invitation_F00043 | 19   | ■ 0.996 | $A \leq C$                                                                              | $A \leq C$                                            | for any $C$ , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$ . Let’s walk through this with an                                                                   |
| fong-spivak-invitation_F00044 | 19   | ■ 0.996 | $B \leq C$                                                                              | $B \leq C$                                            | for any $C$ , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$ . Let’s walk through this with an                                                                   |
| fong-spivak-invitation_F00045 | 19   | ■ 0.996 | $(A \vee B) \leq C$                                                                     | $(A \vee B) \leq C$                                   | for any $C$ , if $A \leq C$ and $B \leq C$ then $(A \vee B) \leq C$ . Let’s walk through this with an                                                                   |
| fong-spivak-invitation_F00046 | 19   | ■ 0.999 | $\{\bullet, *\}$                                                                        | $\{\bullet, *\}$                                      | 1. Write down all the partitions of a two-element set $\{\bullet, *\}$ , order them as above, and                                                                       |
| fong-spivak-invitation_F00047 | 20   | ■ 0.883 | $\mathbb{B} = \{$                                                                       | $\mathbb{B} = \{$                                     | The set $\mathbb{B} = \{\text{true}, \text{false}\}$ of booleans also has an order, $\text{false} \leq \text{true}$ :                                                   |
| fong-spivak-invitation_F00048 | 20   | ■ 0.883 | $\}$                                                                                    | $\}$                                                  | The set $\mathbb{B} = \{\text{true}, \text{false}\}$ of booleans also has an order, $\text{false} \leq \text{true}$ :                                                   |
| fong-spivak-invitation_F00049 | 20   | ■ 0.997 | $\not \leq$                                                                             | $\not \leq$                                           | Thus $\text{false} \leq \text{false}$ , $\text{false} \leq \text{true}$ , and $\text{true} \leq \text{true}$ , but $\text{true} \not \leq \text{false}$ .               |
| fong-spivak-invitation_F00050 | 20   | ■ 0.999 | $B.^2$                                                                                  | $B.^2$                                                | In other words, $A \leq B$ if $A$ implies $B$ . <sup>2</sup>                                                                                                            |
| fong-spivak-invitation_F00051 | 20   | ■ 1.000 | $A, B$                                                                                  | $A, B$                                                | For any $A, B$ in $\mathbb{B}$ , we can again write $A \vee B$ to mean the least element that is greater                                                                |

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| fong-spivak-invitation_F00052 | 20   | ■ 1.000 | $\mathbb{B}$                                     | $\mathbb{B}$                                         | For any $A, B$ in $\mathbb{B}$ , we can again write $A \vee B$ to mean the least element that is greater                                                               |
| fong-spivak-invitation_F00053 | 20   | ■ 1.000 | $\Phi(A)=\Phi(B)=$                               | $\Phi(A) = \Phi(B) =$                                | with $\Phi(A) = \Phi(B) = \text{false}$ , so $\Phi(A) \vee \Phi(B) = \text{false}$ , but where $\Phi(A \vee B) =$                                                      |
| fong-spivak-invitation_F00054 | 20   | ■ 1.000 | $\Phi(A) \vee \Phi(B)=$                          | $\Phi(A) \vee \Phi(B) =$                             | with $\Phi(A) = \Phi(B) = \text{false}$ , so $\Phi(A) \vee \Phi(B) = \text{false}$ , but where $\Phi(A \vee B) =$                                                      |
| fong-spivak-invitation_F00055 | 20   | ■ 1.000 | $\Phi(A \vee B)=$                                | $\Phi(A \vee B) =$                                   | with $\Phi(A) = \Phi(B) = \text{false}$ , so $\Phi(A) \vee \Phi(B) = \text{false}$ , but where $\Phi(A \vee B) =$                                                      |
| fong-spivak-invitation_F00056 | 20   | ■ 1.000 | P                                                | $P$                                                  | <sup>2</sup> In mathematical logic, <code>false</code> implies <code>true</code> but <code>true</code> does not imply <code>false</code> . That is “ $P$ implies $Q$ ” |
| fong-spivak-invitation_F00057 | 20   | ■ 1.000 | Q                                                | $Q$                                                  | <sup>2</sup> In mathematical logic, <code>false</code> implies <code>true</code> but <code>true</code> does not imply <code>false</code> . That is “ $P$ implies $Q$ ” |
| fong-spivak-invitation_F00058 | 21   | ■ 1.000 | a, b, c                                          | $a, b, c$                                            | tree, or the names of your favorite fruits, or simply some symbols $a, b, c$ . For example,                                                                            |
| fong-spivak-invitation_F00059 | 21   | ■ 1.000 | $A=\{h, 1\}$                                     | $A = \{h, 1\}$                                       | we write $A = \{h, 1\}$ to denote the set, called $A$ , that contains exactly two elements, one                                                                        |
| fong-spivak-invitation_F00060 | 21   | ■ 0.935 | h                                                | $h$                                                  | called $h$ and one called 1. The set $\{h, h, 1, h, 1\}$ is exactly the same as $A$ because they                                                                       |
| fong-spivak-invitation_F00061 | 21   | ■ 0.935 | $\{h, h, 1, h, 1\}$                              | $\{h, h, 1, h, 1\}$                                  | called $h$ and one called 1. The set $\{h, h, 1, h, 1\}$ is exactly the same as $A$ because they                                                                       |
| fong-spivak-invitation_F00062 | 21   | ■ 1.000 | $x \in X$                                        | $x \in X$                                            | the notation doesn’t change the set. <sup>3</sup> For an arbitrary set $X$ , we write $x \in X$ if $x$ is an                                                           |
| fong-spivak-invitation_F00063 | 21   | ■ 1.000 | $h \in A$                                        | $h \in A$                                            | element of $X$ ; so we have $h \in A$ and $1 \in A$ , but $0 \notin A$ .                                                                                               |
| fong-spivak-invitation_F00064 | 21   | ■ 1.000 | $1 \in A$                                        | $1 \in A$                                            | element of $X$ ; so we have $h \in A$ and $1 \in A$ , but $0 \notin A$ .                                                                                               |
| fong-spivak-invitation_F00065 | 21   | ■ 1.000 | $0 \notin A$                                     | $0 \notin A$                                         | element of $X$ ; so we have $h \in A$ and $1 \in A$ , but $0 \notin A$ .                                                                                               |
| fong-spivak-invitation_F00066 | 21   | ■ 1.000 | $\varnothing$                                    | $\varnothing$                                        | • $\varnothing$ denotes the empty set; it has no elements.                                                                                                             |
| fong-spivak-invitation_F00067 | 21   | ■ 1.000 | $\mathbb{N}$                                     | $\mathbb{N}$                                         | • $\mathbb{N}$ denotes the set of <i>natural numbers</i> ; it has elements $0, 1, 2, 3, \dots, 90^{717}, \dots$                                                        |
| fong-spivak-invitation_F00068 | 21   | ■ 1.000 | $0, 1, 2, 3, \ldots, 90^{717}, \ldots$           | $0, 1, 2, 3, \dots, 90^{717}, \dots$                 | • $\mathbb{N}$ denotes the set of <i>natural numbers</i> ; it has elements $0, 1, 2, 3, \dots, 90^{717}, \dots$ .                                                      |
| fong-spivak-invitation_F00069 | 21   | ■ 0.996 | $\underline{n}$                                  | $\underline{n}$                                      | • $\underline{n}$ , for any $n \in \mathbb{N}$ , denotes the <i>nth ordinal</i> ; it has $n$ elements $1, 2, \dots, n$ . For example,                                  |
| fong-spivak-invitation_F00070 | 21   | ■ 0.996 | $n \in \mathbb{N}$                               | $n \in \mathbb{N}$                                   | • $\underline{n}$ , for any $n \in \mathbb{N}$ , denotes the <i>nth ordinal</i> ; it has $n$ elements $1, 2, \dots, n$ . For example,                                  |
| fong-spivak-invitation_F00071 | 21   | ■ 0.996 | $n$                                              | $n$                                                  | • $\underline{n}$ , for any $n \in \mathbb{N}$ , denotes the <i>nth ordinal</i> ; it has $n$ elements $1, 2, \dots, n$ . For example,                                  |
| fong-spivak-invitation_F00072 | 21   | ■ 0.996 | $1, 2, \ldots, n$                                | $1, 2, \dots, n$                                     | • $\underline{n}$ , for any $n \in \mathbb{N}$ , denotes the <i>nth ordinal</i> ; it has $n$ elements $1, 2, \dots, n$ . For example,                                  |
| fong-spivak-invitation_F00073 | 21   | ■ 0.999 | $\underline{0}=\varnothing, \underline{1}=\{1\}$ | $\underline{0} = \varnothing, \underline{1} = \{1\}$ | $\underline{0} = \varnothing, \underline{1} = \{1\}$ , and $\underline{5} = \{1, 2, 3, 4, 5\}$ .                                                                       |
| fong-spivak-invitation_F00074 | 21   | ■ 0.999 | $\underline{5}=\{1,2,3,4,5\}$                    | $\underline{5} = \{1, 2, 3, 4, 5\}$                  | $\underline{0} = \varnothing, \underline{1} = \{1\}$ , and $\underline{5} = \{1, 2, 3, 4, 5\}$ .                                                                       |
| fong-spivak-invitation_F00075 | 21   | ■ 0.398 | $\mathbb{Z}$                                     | $\mathbb{Z}$                                         | • $\mathbb{Z}$ , the set of <i>integers</i> ; it has elements $\dots, -2, -1, 0, 1, 2, \dots, 90^{717}, \dots$                                                         |
| fong-spivak-invitation_F00076 | 21   | ■ 0.398 | $90^{717}, \ldots$                               | $90^{717}, \dots$                                    | • $\mathbb{Z}$ , the set of <i>integers</i> ; it has elements $\dots, -2, -1, 0, 1, 2, \dots, 90^{717}, \dots$                                                         |
| fong-spivak-invitation_F00077 | 21   | ■ 1.000 | $\pi, 3.14, 5 * \sqrt{2}, e, e^2, -1457$         | $\pi, 3.14, 5 * \sqrt{2}, e, e^2, -1457$             | • $\mathbb{R}$ , the set of <i>real numbers</i> ; it has elements like $\pi, 3.14, 5 * \sqrt{2}, e, e^2, -1457$ ,                                                      |
| fong-spivak-invitation_F00078 | 21   | ■ 0.977 | $90^{717}$                                       | $90^{717}$                                           | $90^{717}$ , etc.                                                                                                                                                      |
| fong-spivak-invitation_F00079 | 21   | ■ 1.000 | $X \subseteq Y$                                  | $X \subseteq Y$                                      | Given sets $X$ and $Y$ , we say that $X$ is a <i>subset</i> of $Y$ , and write $X \subseteq Y$ , if every element                                                      |

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| fong-spivak-invitation_F00080 | 21   | ■ 1.000 | $\{h\} \subseteq A$                                                        | $\{h\} \subseteq A$                                        | in $X$ is also in $Y$ . For example $\{h\} \subseteq A$ . Note that the empty set $\emptyset := \{\}$ is a subset                                   |
| fong-spivak-invitation_F00081 | 21   | ■ 1.000 | $\varnothing := \{\}$                                                      | $\emptyset := \{\}$                                        | in $X$ is also in $Y$ . For example $\{h\} \subseteq A$ . Note that the empty set $\emptyset := \{\}$ is a subset                                   |
| fong-spivak-invitation_F00082 | 21   | ■ 1.000 | $\{y \in Y \mid P(y)\}$                                                    | $\{y \in Y \mid P(y)\}$                                    | element of $Y$ , we write $\{y \in Y \mid P(y)\}$ to mean the subset of those $y$ 's that satisfy $P$ .                                             |
| fong-spivak-invitation_F00083 | 21   | ■ 1.000 | $\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 0\}$                          | $\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 0\}$          | 1. Is it true that $\mathbb{N} = \{n \in \mathbb{Z} \mid n > 0\}$ ?                                                                                 |
| fong-spivak-invitation_F00084 | 21   | ■ 0.741 | $h, 1$                                                                     | $h, 1$                                                     | <sup>3</sup> If you want a notion where “ $h, 1$ ” is different from “ $h, h, 1, h, 1$ ,” you can use something called <i>bags</i> ,                |
| fong-spivak-invitation_F00085 | 21   | ■ 0.741 | $h, h, 1, h, 1$                                                            | $h, h, 1, h, 1$                                            | <sup>3</sup> If you want a notion where “ $h, 1$ ” is different from “ $h, h, 1, h, 1$ ,” you can use something called <i>bags</i> ,                |
| fong-spivak-invitation_F00086 | 21   | ■ 0.995 | $Z :=$                                                                     | $Z :=$                                                     | <sup>4</sup> When we write $Z := f \circ \circ$ , it means “assign the meaning $f \circ \circ$ to variable $Z$ ,” whereas $Z = f \circ \circ$ means |
| fong-spivak-invitation_F00087 | 21   | ■ 0.995 | $Z$                                                                        | $Z$                                                        | <sup>4</sup> When we write $Z := f \circ \circ$ , it means “assign the meaning $f \circ \circ$ to variable $Z$ ,” whereas $Z = f \circ \circ$ means |
| fong-spivak-invitation_F00088 | 21   | ■ 0.995 | $Z =$                                                                      | $Z =$                                                      | <sup>4</sup> When we write $Z := f \circ \circ$ , it means “assign the meaning $f \circ \circ$ to variable $Z$ ,” whereas $Z = f \circ \circ$ means |
| fong-spivak-invitation_F00089 | 21   | ■ 1.000 | $3+2:=5$                                                                   | $3+2:=5$                                                   | $Z = f \circ \circ$ but not vice versa; indeed it <i>would not</i> be proper to write $3+2:=5$ or $\{\} := \emptyset$ .                             |
| fong-spivak-invitation_F00090 | 21   | ■ 1.000 | $\} := \varnothing$                                                        | $\} := \emptyset$                                          | $Z = f \circ \circ$ but not vice versa; indeed it <i>would not</i> be proper to write $3+2:=5$ or $\{\} := \emptyset$ .                             |
| fong-spivak-invitation_F00091 | 22   | ■ 1.000 | $\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 1\}$                          | $\mathbb{N} = \{n \in \mathbb{Z} \mid n \geq 1\}$          | 2. Is it true that $\mathbb{N} = \{n \in \mathbb{Z} \mid n > 1\}$ ?                                                                                 |
| fong-spivak-invitation_F00092 | 22   | ■ 1.000 | $\varnothing = \{n \in \mathbb{Z} \mid 1 < n < 2\}$                        | $\emptyset = \{n \in \mathbb{Z} \mid 1 < n < 2\}$          | 3. Is it true that $\emptyset = \{n \in \mathbb{Z} \mid 1 < n < 2\}$ ?                                                                              |
| fong-spivak-invitation_F00093 | 22   | ■ 1.000 | $X_{\{1\}}$                                                                | $X_1$                                                      | If both $X_1$ and $X_2$ are subsets of $Y$ , their <i>union</i> , denoted $X_1 \cup X_2$ , is also a subset of $Y$ ,                                |
| fong-spivak-invitation_F00094 | 22   | ■ 1.000 | $X_{\{2\}}$                                                                | $X_2$                                                      | If both $X_1$ and $X_2$ are subsets of $Y$ , their <i>union</i> , denoted $X_1 \cup X_2$ , is also a subset of $Y$ ,                                |
| fong-spivak-invitation_F00095 | 22   | ■ 1.000 | $X_{\{1\}} \cup X_{\{2\}}$                                                 | $X_1 \cup X_2$                                             | If both $X_1$ and $X_2$ are subsets of $Y$ , their <i>union</i> , denoted $X_1 \cup X_2$ , is also a subset of $Y$ ,                                |
| fong-spivak-invitation_F00096 | 22   | ■ 1.000 | $Y = \{1, 2, 3, 4\}$                                                       | $Y = \{1, 2, 3, 4\}$                                       | example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$ , then $X_1 \cup X_2 = \{1, 2, 4\}$ .                                     |
| fong-spivak-invitation_F00097 | 22   | ■ 1.000 | $X_{\{1\}} = \{1, 2\}$                                                     | $X_1 = \{1, 2\}$                                           | example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$ , then $X_1 \cup X_2 = \{1, 2, 4\}$ .                                     |
| fong-spivak-invitation_F00098 | 22   | ■ 1.000 | $X_{\{2\}} = \{2, 4\}$                                                     | $X_2 = \{2, 4\}$                                           | example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$ , then $X_1 \cup X_2 = \{1, 2, 4\}$ .                                     |
| fong-spivak-invitation_F00099 | 22   | ■ 1.000 | $X_{\{1\}} \cup X_{\{2\}} = \{1, 2, 4\}$                                   | $X_1 \cup X_2 = \{1, 2, 4\}$                               | example if $Y = \{1, 2, 3, 4\}$ and $X_1 = \{1, 2\}$ and $X_2 = \{2, 4\}$ , then $X_1 \cup X_2 = \{1, 2, 4\}$ .                                     |
| fong-spivak-invitation_F00100 | 22   | ■ 1.000 | $\varnothing \cup X = X$                                                   | $\emptyset \cup X = X$                                     | Note that $\emptyset \cup X = X$ for any $X \subseteq Y$ .                                                                                          |
| fong-spivak-invitation_F00101 | 22   | ■ 1.000 | $X_{\{1\}} \cap X_{\{2\}}$                                                 | $X_1 \cap X_2$                                             | Similarly, if both $X_1$ and $X_2$ are subsets of $Y$ , then their <i>intersection</i> , denoted $X_1 \cap X_2$ ,                                   |
| fong-spivak-invitation_F00102 | 22   | ■ 1.000 | $\{1, 2, 3\} \cap \{2, 5\} = \{2\}$                                        | $\{1, 2, 3\} \cap \{2, 5\} = \{2\}$                        | $X_1$ and in $X_2$ , and no others. So $\{1, 2, 3\} \cap \{2, 5\} = \{2\}$ .                                                                        |
| fong-spivak-invitation_F00103 | 22   | ■ 1.000 | $X_{\{\emptyset\}} = \varnothing, X_{\{1\}} = \{1\}, X_{\{2\}} = \{1, 2\}$ | $X_0 = \emptyset, X_1 = \{1\}, X_2 = \{1, 2\}$             | the sets $X_0 = \emptyset, X_1 = \{1\}, X_2 = \{1, 2\}$ , etc. as subsets of $\mathbb{N}$ , and we want to know                                     |
| fong-spivak-invitation_F00104 | 22   | ■ 1.000 | $\bigcup_{n \in \mathbb{N}} X_n$                                           | $\bigcup_{n \in \mathbb{N}} X_n$                           | what the union of all of them is. This union is written $\bigcup_{n \in \mathbb{N}} X_n$ , and it is the subset of $\mathbb{N}$                     |
| fong-spivak-invitation_F00105 | 22   | ■ 0.994 | $X_{\{n\}}$                                                                | $X_n$                                                      | that contains every element of every $X_n$ , but no others. Namely, $\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid$                      |
| fong-spivak-invitation_F00106 | 22   | ■ 0.994 | $\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid$                 | $\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid$ | that contains every element of every $X_n$ , but no others. Namely, $\bigcup_{n \in \mathbb{N}} X_n = \{n \in \mathbb{N} \mid$                      |
| fong-spivak-invitation_F00107 | 22   | ■ 1.000 | $n \geq 1\}$                                                               | $n \geq 1\}$                                               | $n \geq 1\}$ . Similarly one can write $\bigcap_{n \in \mathbb{N}} X_n$ for the intersection of all of them, which will                             |

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| fong-spivak-invitation_F00108 | 22   | ■ 1.000 | $\bigcap_{n \in \mathbb{N}} X_n$ | $\bigcap_{n \in \mathbb{N}} X_n$ | $n \geq 1$ ). Similarly one can write $\bigcap_{n \in \mathbb{N}} X_n$ for the intersection of all of them, which will  |
| fong-spivak-invitation_F00109 | 22   | ■ 0.998 | $X \times Y$                     | $X \times Y$                     | Given two sets $X$ and $Y$ , the <i>product</i> $X \times Y$ of $X$ and $Y$ is the set of pairs $(x, y)$ ,              |
| fong-spivak-invitation_F00110 | 22   | ■ 1.000 | $y \in Y$                        | $y \in Y$                        | where $x \in X$ and $y \in Y$ .                                                                                         |
| fong-spivak-invitation_F00111 | 22   | ■ 1.000 | $X \sqcup Y$                     | $X \sqcup Y$                     | in common. Given two sets $X$ and $Y$ , their <i>disjoint union</i> $X \sqcup Y$ is the set of pairs of the             |
| fong-spivak-invitation_F00112 | 22   | ■ 1.000 | $(x, 1)$                         | $(x, 1)$                         | form $(x, 1)$ or $(y, 2)$ , where $x \in X$ and $y \in Y$ .                                                             |
| fong-spivak-invitation_F00113 | 22   | ■ 1.000 | $(y, 2)$                         | $(y, 2)$                         | form $(x, 1)$ or $(y, 2)$ , where $x \in X$ and $y \in Y$ .                                                             |
| fong-spivak-invitation_F00114 | 22   | ■ 1.000 | $A := \{h, 1\}$                  | $A := \{h, 1\}$                  | <b>Exercise 1.7.</b> Let $A := \{h, 1\}$ and $B := \{1, 2, 3\}$ .                                                       |
| fong-spivak-invitation_F00115 | 22   | ■ 1.000 | $B := \{1, 2, 3\}$               | $B := \{1, 2, 3\}$               | <b>Exercise 1.7.</b> Let $A := \{h, 1\}$ and $B := \{1, 2, 3\}$ .                                                       |
| fong-spivak-invitation_F00116 | 22   | ■ 1.000 | $A \times B$                     | $A \times B$                     | 3. There are six elements in $A \times B$ ; write them out.                                                             |
| fong-spivak-invitation_F00117 | 22   | ■ 1.000 | $A \sqcup B$                     | $A \sqcup B$                     | 4. There are five elements of $A \sqcup B$ ; write them out.                                                            |
| fong-spivak-invitation_F00118 | 22   | ■ 0.711 | $\{h, 1, 2, 3\}$                 | $\{h, 1, 2, 3\}$                 | 5. If we consider $A$ and $B$ as subsets of the set $\{h, 1, 2, 3\}$ , there are four elements of                       |
| fong-spivak-invitation_F00119 | 22   | ■ 1.000 | $A \cup B$                       | $A \cup B$                       | $A \cup B$ ; write them out.                                                                                            |
| fong-spivak-invitation_F00120 | 22   | ■ 0.998 | $R \subseteq X \times Y$         | $R \subseteq X \times Y$         | $R \subseteq X \times Y$ . A <i>binary relation on <math>X</math></i> is a relation between $X$ and $X$ , i.e. a subset |
| fong-spivak-invitation_F00121 | 22   | ■ 1.000 | $R \subseteq X \times X$         | $R \subseteq X \times X$         | $R \subseteq X \times X$ .                                                                                              |
| fong-spivak-invitation_F00122 | 22   | ■ 1.000 | $R \subseteq A \times A$         | $R \subseteq A \times A$         | It is convenient to use something called <i>infix notation</i> for binary relations $R \subseteq A \times A$ .          |
| fong-spivak-invitation_F00123 | 22   | ■ 0.992 | $a \star b$                      | $a \star b$                      | This means one picks a symbol, say $\star$ , and writes $a \star b$ to mean $(a, b) \in R$ .                            |
| fong-spivak-invitation_F00124 | 22   | ■ 0.992 | $(a, b) \in R$                   | $(a, b) \in R$                   | This means one picks a symbol, say $\star$ , and writes $a \star b$ to mean $(a, b) \in R$ .                            |
| fong-spivak-invitation_F00125 | 22   | ■ 1.000 | $(5, 6) \in R$                   | $(5, 6) \in R$                   | $(5, 6) \in R$ , we write $5 \leq 6$ .                                                                                  |
| fong-spivak-invitation_F00126 | 22   | ■ 1.000 | $5 \leq 6$                       | $5 \leq 6$                       | $(5, 6) \in R$ , we write $5 \leq 6$ .                                                                                  |
| fong-spivak-invitation_F00127 | 23   | ■ 0.997 | $=, \approx, <, >$               | $=, \approx, <, >$               | Other examples of infix notation for relations are $=, \approx, <, >$ . In number theory, we                            |
| fong-spivak-invitation_F00128 | 23   | ■ 1.000 | $p \in P$                        | $p \in P$                        | <b>Definition 1.10.</b> If $A$ is a set, a <i>partition</i> of $A$ consists of a set $P$ and, for each $p \in P$ ,      |
| fong-spivak-invitation_F00129 | 23   | ■ 1.000 | $A_p \subseteq A$                | $A_p \subseteq A$                | a nonempty subset $A_p \subseteq A$ , such that                                                                         |
| fong-spivak-invitation_F00130 | 23   | ■ 1.000 | $\{A_p\}_{p \in P}$              | $\{A_p\}_{p \in P}$              | We may denote the partition by $\{A_p\}_{p \in P}$ . We refer to $P$ as the set of <i>part labels</i> and if            |
| fong-spivak-invitation_F00131 | 23   | ■ 1.000 | $A_p$                            | $A_p$                            | $p \in P$ is a part label, we refer to $A_p$ as the <i>pth part</i> . The condition (1.5) says that each                |
| fong-spivak-invitation_F00132 | 23   | ■ 1.000 | $p$                              | $p$                              | $p \in P$ is a part label, we refer to $A_p$ as the <i>pth part</i> . The condition (1.5) says that each                |
| fong-spivak-invitation_F00133 | 23   | ■ 1.000 | $a \in A$                        | $a \in A$                        | element $a \in A$ is in exactly one part.                                                                               |
| fong-spivak-invitation_F00134 | 23   | ■ 1.000 | $\{A_{p'}^{'}\}_{p' \in P'}$     | $\{A_{p'}^{'}\}_{p' \in P'}$     | We consider two different partitions $\{A_p\}_{p \in P}$ and $\{A_{p'}^{'}\}_{p' \in P'}$ of $A$ to be the same if      |



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| fong-spivak-invitation_F00161 | 25   | ■ 1.000 | $S$                                                                               | $S$                                                                               | <b>Definition 1.17.</b> Let $S$ and $T$ be sets. A function from $S$ to $T$ is a subset $F \subseteq S \times T$                                         |
| fong-spivak-invitation_F00162 | 25   | ■ 1.000 | $T$                                                                               | $T$                                                                               | <b>Definition 1.17.</b> Let $S$ and $T$ be sets. A function from $S$ to $T$ is a subset $F \subseteq S \times T$                                         |
| fong-spivak-invitation_F00163 | 25   | ■ 1.000 | $F \subseteq S \times T$                                                          | $F \subseteq S \times T$                                                          | <b>Definition 1.17.</b> Let $S$ and $T$ be sets. A function from $S$ to $T$ is a subset $F \subseteq S \times T$                                         |
| fong-spivak-invitation_F00164 | 25   | ■ 1.000 | $s \in S$                                                                         | $s \in S$                                                                         | such that for all $s \in S$ , there exists a unique $t \in T$ with $(s, t) \in F$ .                                                                      |
| fong-spivak-invitation_F00165 | 25   | ■ 1.000 | $t \in T$                                                                         | $t \in T$                                                                         | such that for all $s \in S$ , there exists a unique $t \in T$ with $(s, t) \in F$ .                                                                      |
| fong-spivak-invitation_F00166 | 25   | ■ 1.000 | $(s, t) \in F$                                                                    | $(s, t) \in F$                                                                    | such that for all $s \in S$ , there exists a unique $t \in T$ with $(s, t) \in F$ .                                                                      |
| fong-spivak-invitation_F00167 | 25   | ■ 1.000 | $F$                                                                               | $F$                                                                               | The function $F$ is often denoted $F: S \rightarrow T$ . From now on, we write $F(s) = t$ , or                                                           |
| fong-spivak-invitation_F00168 | 25   | ■ 1.000 | $F: S \rightarrow T$                                                              | $F: S \rightarrow T$                                                              | The function $F$ is often denoted $F: S \rightarrow T$ . From now on, we write $F(s) = t$ , or                                                           |
| fong-spivak-invitation_F00169 | 25   | ■ 1.000 | $F(s)=t$                                                                          | $F(s) = t$                                                                        | The function $F$ is often denoted $F: S \rightarrow T$ . From now on, we write $F(s) = t$ , or                                                           |
| fong-spivak-invitation_F00170 | 25   | ■ 1.000 | $s \mapsto t$                                                                     | $s \mapsto t$                                                                     | sometimes $s \mapsto t$ , to mean $(s, t) \in F$ . For any $t \in T$ , the preimage of $t$ along $F$ is the                                              |
| fong-spivak-invitation_F00171 | 25   | ■ 1.000 | $t$                                                                               | $t$                                                                               | sometimes $s \mapsto t$ , to mean $(s, t) \in F$ . For any $t \in T$ , the preimage of $t$ along $F$ is the                                              |
| fong-spivak-invitation_F00172 | 25   | ■ 1.000 | $f^{-1}(t) := \{s \in S \mid F(s)=t\}$                                            | $f^{-1}(t) := \{s \in S \mid F(s) = t\}$                                          | subset $f^{-1}(t) := \{s \in S \mid F(s) = t\}$ .                                                                                                        |
| fong-spivak-invitation_F00173 | 25   | ■ 1.000 | $s_1, s_2 \in S$                                                                  | $s_1, s_2 \in S$                                                                  | $F(s) = t$ . A function is called <i>injective</i> , or an <i>injection</i> , if for all $t \in T$ and $s_1, s_2 \in S$                                  |
| fong-spivak-invitation_F00174 | 25   | ■ 1.000 | $F(s_1)=t$                                                                        | $F(s_1) = t$                                                                      | with $F(s_1) = t$ and $F(s_2) = t$ , we have $s_1 = s_2$ . A function is called <i>bijective</i> if it is                                                |
| fong-spivak-invitation_F00175 | 25   | ■ 1.000 | $F(s_2)=t$                                                                        | $F(s_2) = t$                                                                      | with $F(s_1) = t$ and $F(s_2) = t$ , we have $s_1 = s_2$ . A function is called <i>bijective</i> if it is                                                |
| fong-spivak-invitation_F00176 | 25   | ■ 1.000 | $s_1=s_2$                                                                         | $s_1 = s_2$                                                                       | with $F(s_1) = t$ and $F(s_2) = t$ , we have $s_1 = s_2$ . A function is called <i>bijective</i> if it is                                                |
| fong-spivak-invitation_F00177 | 25   | ■ 0.918 | $\rightarrow, \rightarrowtail, \twoheadrightarrow, \stackrel{\cong}{\rightarrow}$ | $\rightarrow, \rightarrowtail, \twoheadrightarrow, \stackrel{\cong}{\rightarrow}$ | We use various decorations on arrows, $\rightarrow, \rightarrowtail, \twoheadrightarrow, \stackrel{\cong}{\rightarrow}$ to denote these special sorts of |
| fong-spivak-invitation_F00178 | 25   | ■ 0.680 | $\mathrm{id}_X$                                                                   | $\mathrm{id}_X$                                                                   | a set $X$ , denoted $\mathrm{id}_X$ . It is the bijective function $\mathrm{id}_X(x) = x$ .                                                              |
| fong-spivak-invitation_F00179 | 25   | ■ 0.680 | $\mathrm{id}_X(x)=x$                                                              | $\mathrm{id}_X(x) = x$                                                            | a set $X$ , denoted $\mathrm{id}_X$ . It is the bijective function $\mathrm{id}_X(x) = x$ .                                                              |
| fong-spivak-invitation_F00180 | 25   | ■ 1.000 | $f: A \rightarrow B$                                                              | $f: A \rightarrow B$                                                              | 1. Find two sets $A$ and $B$ and a function $f: A \rightarrow B$ that is injective but not surjective.                                                   |
| fong-spivak-invitation_F00181 | 26   | ■ 1.000 | $f: A \rightarrow \varnothing$                                                    | $f: A \rightarrow \varnothing$                                                    | <b>Exercise 1.20.</b> Suppose that $A$ is a set and $f: A \rightarrow \varnothing$ is a function to the empty set.                                       |
| fong-spivak-invitation_F00182 | 26   | ■ 1.000 | $f: A \rightarrow P$                                                              | $f: A \rightarrow P$                                                              | functions out of $A$ . Given a surjective function $f: A \twoheadrightarrow P$ , where $P$ is any other set,                                             |
| fong-spivak-invitation_F00183 | 26   | ■ 1.000 | $f^{-1}(p) \subseteq A$                                                           | $f^{-1}(p) \subseteq A$                                                           | the preimages $f^{-1}(p) \subseteq A$ , one for each element $p \in P$ , form a partition of $A$ . Here is                                               |
| fong-spivak-invitation_F00184 | 26   | ■ 1.000 | $S := \{11, 12, 13, 21, 22, 23\}$                                                 | $S := \{11, 12, 13, 21, 22, 23\}$                                                 | Consider the partition of $S := \{11, 12, 13, 21, 22, 23\}$ shown below:                                                                                 |
| fong-spivak-invitation_F00185 | 26   | ■ 1.000 | $P = \{a, b, c, d\}$                                                              | $P = \{a, b, c, d\}$                                                              | It has been partitioned into four parts, so let $P = \{a, b, c, d\}$ and let $f: S \twoheadrightarrow P$ be                                              |
| fong-spivak-invitation_F00186 | 26   | ■ 1.000 | $f: S \rightarrow P$                                                              | $f: S \rightarrow P$                                                              | It has been partitioned into four parts, so let $P = \{a, b, c, d\}$ and let $f: S \twoheadrightarrow P$ be                                              |
| fong-spivak-invitation_F00187 | 26   | ■ 1.000 | $F: X \rightarrow Y$                                                              | $F: X \rightarrow Y$                                                              | <b>Definition 1.23.</b> If $F: X \rightarrow Y$ is a function and $G: Y \rightarrow Z$ is a function, their com-                                         |

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| fong-spivak-invitation_F00188 | 26   | ■ 1.000 | $G: Y \rightarrow Z$              | $G: Y \rightarrow Z$              | <b>Definition 1.23.</b> If $F: X \rightarrow Y$ is a function and $G: Y \rightarrow Z$ is a function, their <i>com-</i> |
| fong-spivak-invitation_F00189 | 26   | ■ 0.999 | $X \rightarrow Z$                 | $X \rightarrow Z$                 | <i>posite</i> is the function $X \rightarrow Z$ defined to be $G(F(x))$ for any $x \in X$ . It is often denoted         |
| fong-spivak-invitation_F00190 | 26   | ■ 0.999 | $G(F(x))$                         | $G(F(x))$                         | <i>posite</i> is the function $X \rightarrow Z$ defined to be $G(F(x))$ for any $x \in X$ . It is often denoted         |
| fong-spivak-invitation_F00191 | 26   | ■ 0.996 | $G \circ F$                       | $G \circ F$                       | $G \circ F$ , but we prefer to denote it $F \circ G$ . It takes any element $x \in X$ , evaluates $F$ to get            |
| fong-spivak-invitation_F00192 | 26   | ■ 0.996 | $F; G$                            | $F; G$                            | $G \circ F$ , but we prefer to denote it $F \circ G$ . It takes any element $x \in X$ , evaluates $F$ to get            |
| fong-spivak-invitation_F00193 | 26   | ■ 1.000 | $F(x) \in Y$                      | $F(x) \in Y$                      | an element $F(x) \in Y$ and then evaluates $G$ to get an element $G(F(x))$ .                                            |
| fong-spivak-invitation_F00194 | 26   | ■ 1.000 | $G$                               | $G$                               | an element $F(x) \in Y$ and then evaluates $G$ to get an element $G(F(x))$ .                                            |
| fong-spivak-invitation_F00195 | 26   | ■ 0.997 | $\{1\} \rightarrow X$             | $\{1\} \rightarrow X$             | $\{1\} \rightarrow X$ , namely the function sending 1 to $x$ . For example, the three functions $\{1\} \rightarrow$     |
| fong-spivak-invitation_F00196 | 26   | ■ 0.997 | $\{1\} \rightarrow$               | $\{1\} \rightarrow$               | $\{1\} \rightarrow X$ , namely the function sending 1 to $x$ . For example, the three functions $\{1\} \rightarrow$     |
| fong-spivak-invitation_F00197 | 27   | ■ 1.000 | $x: \{1\} \rightarrow X$          | $x: \{1\} \rightarrow X$          | function $x: \{1\} \rightarrow X$ . Then evaluating $F$ at $x$ is given by the composite $F(x) = x \circ F$ .           |
| fong-spivak-invitation_F00198 | 27   | ■ 1.000 | $F(x)=x; F$                       | $F(x) = x; F$                     | function $x: \{1\} \rightarrow X$ . Then evaluating $F$ at $x$ is given by the composite $F(x) = x \circ F$ .           |
| fong-spivak-invitation_F00199 | 27   | ■ 1.000 | $x \leq x$                        | $x \leq x$                        | (a) $x \leq x$ ; and                                                                                                    |
| fong-spivak-invitation_F00200 | 27   | ■ 1.000 | $y \leq z$                        | $y \leq z$                        | (b) if $x \leq y$ and $y \leq z$ , then $x \leq z$ .                                                                    |
| fong-spivak-invitation_F00201 | 27   | ■ 1.000 | $x \leq z$                        | $x \leq z$                        | (b) if $x \leq y$ and $y \leq z$ , then $x \leq z$ .                                                                    |
| fong-spivak-invitation_F00202 | 27   | ■ 0.837 | $y \leq x$                        | $y \leq x$                        | $y \leq x$ , we write $x \cong y$ and say $x$ and $y$ are <i>equivalent</i> . We call a pair $(X, \leq)$ consisting     |
| fong-spivak-invitation_F00203 | 27   | ■ 0.837 | $x \cong y$                       | $x \cong y$                       | $y \leq x$ , we write $x \cong y$ and say $x$ and $y$ are <i>equivalent</i> . We call a pair $(X, \leq)$ consisting     |
| fong-spivak-invitation_F00204 | 27   | ■ 0.837 | $X, \leq$                         | $X, \leq$                         | $y \leq x$ , we write $x \cong y$ and say $x$ and $y$ are <i>equivalent</i> . We call a pair $(X, \leq)$ consisting     |
| fong-spivak-invitation_F00205 | 27   | ■ 0.981 | $(X, =)$                          | $(X, =)$                          | $(X, =)$ . This means that the only order relationships on $X$ are of the form $x \leq x$ ; if                          |
| fong-spivak-invitation_F00206 | 27   | ■ 1.000 | $x \neq y$                        | $x \neq y$                        | $x \neq y$ then neither $x \leq y$ nor $y \leq x$ hold.                                                                 |
| fong-spivak-invitation_F00207 | 27   | ■ 0.931 | $X \times X \subseteq X \times X$ | $X \times X \subseteq X \times X$ | preorder $(X, \leq)$ by equipping it with the total binary relation $X \times X \subseteq X \times X$ . This is a       |
| fong-spivak-invitation_F00208 | 28   | ■ 0.999 | $x=y$                             | $x = y$                           | (c) $x \cong y$ implies $x = y$ .                                                                                       |
| fong-spivak-invitation_F00209 | 28   | ■ 1.000 | $x, y$                            | $x, y$                            | any two elements $x, y$ for which $x \cong y$ , i.e. for which $x \leq y$ and $y \leq x$ .                              |
| fong-spivak-invitation_F00210 | 28   | ■ 1.000 | $G=(V, A, s, t)$                  | $G = (V, A, s, t)$                | <b>Definition 1.31.</b> A graph $G = (V, A, s, t)$ consists of a set $V$ whose elements are called                      |
| fong-spivak-invitation_F00211 | 28   | ■ 1.000 | $V$                               | $V$                               | <b>Definition 1.31.</b> A graph $G = (V, A, s, t)$ consists of a set $V$ whose elements are called                      |
| fong-spivak-invitation_F00212 | 28   | ■ 1.000 | $s, t: A \rightarrow V$           | $s, t: A \rightarrow V$           | <i>vertices</i> , a set $A$ whose elements are called <i>arrows</i> , and two functions $s, t: A \rightarrow V$         |
| fong-spivak-invitation_F00213 | 28   | ■ 1.000 | $s(a)=v$                          | $s(a) = v$                        | known as the <i>source</i> and <i>target</i> functions respectively. Given $a \in A$ with $s(a) = v$ and                |
| fong-spivak-invitation_F00214 | 28   | ■ 1.000 | $t(a)=w$                          | $t(a) = w$                        | $t(a) = w$ , we say that $a$ is an arrow from $v$ to $w$ .                                                              |

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| fong-spivak-invitation_F00215 | 28   | ■ 1.000 | $v$                                 | $v$                                   | $t(a) = w$ , we say that $a$ is an arrow from $v$ to $w$ .                                                            |
| fong-spivak-invitation_F00216 | 28   | ■ 1.000 | $w$                                 | $w$                                   | $t(a) = w$ , we say that $a$ is an arrow from $v$ to $w$ .                                                            |
| fong-spivak-invitation_F00217 | 28   | ■ 1.000 | $V=\{1,2,3,4\}$                     | $V = \{1, 2, 3, 4\}$                  | It has $V = \{1, 2, 3, 4\}$ and $A = \{a, b, c, d, e\}$ . The source and target functions,                            |
| fong-spivak-invitation_F00218 | 28   | ■ 1.000 | $A=\{a, b, c, d, e\}$               | $A = \{a, b, c, d, e\}$               | It has $V = \{1, 2, 3, 4\}$ and $A = \{a, b, c, d, e\}$ . The source and target functions,                            |
| fong-spivak-invitation_F00221 | 120  | ■ 0.999 | $e$                                 | $e$                                   | The functor $\Pi_F$ puts into Airline Seat the set of those pairs $(e, f)$ where $e$ is an Econ-                      |
| fong-spivak-invitation_F00222 | 29   | ■ 0.928 | $1 \rightarrow 2$                   | $1 \rightarrow 2$                     | of length 0. There are infinitely many paths $1 \rightarrow 2$ because one can loop and loop and                      |
| fong-spivak-invitation_F00223 | 29   | ■ 1.000 | $d$                                 | $d$                                   | loop through $d$ as many times as one pleases.                                                                        |
| fong-spivak-invitation_F00224 | 29   | ■ 0.531 | $P, \leq$                           | $P, \leq$                             | graph $G = (V, A, s, t)$ that gives a <i>presentation</i> of a preorder $(P, \leq)$ . The elements of                 |
| fong-spivak-invitation_F00225 | 29   | ■ 1.000 | $v \leq w$                          | $v \leq w$                            | $P$ are the vertices $V$ in $G$ , and the order $\leq$ is given by $v \leq w$ iff there is a path $v \rightarrow w$ . |
| fong-spivak-invitation_F00226 | 29   | ■ 1.000 | $v \rightarrow w$                   | $v \rightarrow w$                     | $P$ are the vertices $V$ in $G$ , and the order $\leq$ is given by $v \leq w$ iff there is a path $v \rightarrow w$ . |
| fong-spivak-invitation_F00227 | 29   | ■ 1.000 | $v \rightarrow v$                   | $v \rightarrow v$                     | For any vertex $v$ , there is always a path $v \rightarrow v$ , and this translates into the reflexivity              |
| fong-spivak-invitation_F00228 | 29   | ■ 1.000 | $u \rightarrow v$                   | $u \rightarrow v$                     | law from Definition 1.25. The fact that paths $u \rightarrow v$ and $v \rightarrow w$ can be concatenated to          |
| fong-spivak-invitation_F00229 | 29   | ■ 1.000 | $u \rightarrow w$                   | $u \rightarrow w$                     | a path $u \rightarrow w$ translates into the transitivity law.                                                        |
| fong-spivak-invitation_F00230 | 29   | ■ 0.998 | $(P, \leq)$                         | $(P, \leq)$                           | <b>Exercise 1.35.</b> What preorder relation $(P, \leq)$ is depicted by the graph $G$ in Exam-                        |
| fong-spivak-invitation_F00231 | 29   | ■ 0.898 | $\left(p_{\{1\}}, p_{\{2\}}\right)$ | $(p_1, p_2)$                          | ple 1.32? That is, write down the elements of $P$ and write down every pair $(p_1, p_2)$ for                          |
| fong-spivak-invitation_F00232 | 29   | ■ 1.000 | $p_{\{1\}} \leq p_{\{2\}}$          | $p_1 \leq p_2$                        | which $p_1 \leq p_2$ .                                                                                                |
| fong-spivak-invitation_F00233 | 29   | ■ 1.000 | $\{\bullet, \circ, *\}$             | $\{\bullet, \circ, *\}$               | <b>Exercise 1.37.</b> Let $X$ be the set of partitions of $\{\bullet, \circ, *\}$ ; it has five elements and an       |
| fong-spivak-invitation_F00234 | 29   | ■ 1.000 | $\mathbb{N}:=\{0,1,2,3,\ldots\}$    | $\mathbb{N} := \{0, 1, 2, 3, \dots\}$ | <b>Example 1.40</b> (Natural numbers). The natural numbers $\mathbb{N} := \{0, 1, 2, 3, \dots\}$ are a pre-           |
| fong-spivak-invitation_F00235 | 29   | ■ 1.000 | $0 \leq 1$                          | $0 \leq 1$                            | order with the order given by the usual size ordering, e.g. $0 \leq 1$ and $5 \leq 100$ . This is                     |
| fong-spivak-invitation_F00236 | 29   | ■ 1.000 | $5 \leq 100$                        | $5 \leq 100$                          | order with the order given by the usual size ordering, e.g. $0 \leq 1$ and $5 \leq 100$ . This is                     |
| fong-spivak-invitation_F00237 | 29   | ■ 1.000 | $m \leq n$                          | $m \leq n$                            | a total order: either $m \leq n$ or $n \leq m$ for all $m, n$ . One can see that its Hasse diagram                    |
| fong-spivak-invitation_F00238 | 29   | ■ 1.000 | $n \leq m$                          | $n \leq m$                            | a total order: either $m \leq n$ or $n \leq m$ for all $m, n$ . One can see that its Hasse diagram                    |
| fong-spivak-invitation_F00239 | 29   | ■ 1.000 | $m, n$                              | $m, n$                                | a total order: either $m \leq n$ or $n \leq m$ for all $m, n$ . One can see that its Hasse diagram                    |
| fong-spivak-invitation_F00240 | 30   | ■ 0.989 | $a \leq b$                          | $a \leq b$                            | and $b$ – namely all those in the middle row – which satisfy neither $a \leq b$ nor $b \leq a$ .                      |
| fong-spivak-invitation_F00241 | 30   | ■ 0.989 | $b \leq a$                          | $b \leq a$                            | and $b$ – namely all those in the middle row – which satisfy neither $a \leq b$ nor $b \leq a$ .                      |
| fong-spivak-invitation_F00242 | 30   | ■ 1.000 | $n=m$                               | $n = m$                               | $n = m$ . Another ordering is the reverse ordering, such as $5 \leq 3$ and $3 \leq 2$ , like how                      |
| fong-spivak-invitation_F00243 | 30   | ■ 1.000 | $5 \leq 3$                          | $5 \leq 3$                            | $n = m$ . Another ordering is the reverse ordering, such as $5 \leq 3$ and $3 \leq 2$ , like how                      |
| fong-spivak-invitation_F00244 | 30   | ■ 1.000 | $3 \leq 2$                          | $3 \leq 2$                            | $n = m$ . Another ordering is the reverse ordering, such as $5 \leq 3$ and $3 \leq 2$ , like how                      |

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| fong-spivak-invitation_F00245 | 30   | ■ 1.000 | m                                                  | $m$                                              | $m$ without remainder. In this ordering $2 \leq 4$ , for example, but $2 \not\leq 3$ , since there is a                                                           |
| fong-spivak-invitation_F00246 | 30   | ■ 1.000 | $2 \leq 4$                                         | $2 \leq 4$                                       | $m$ without remainder. In this ordering $2 \leq 4$ , for example, but $2 \not\leq 3$ , since there is a                                                           |
| fong-spivak-invitation_F00247 | 30   | ■ 1.000 | $2 \not\leq 3$                                     | $2 \not\leq 3$                                   | $m$ without remainder. In this ordering $2 \leq 4$ , for example, but $2 \not\leq 3$ , since there is a                                                           |
| fong-spivak-invitation_F00248 | 30   | ■ 0.994 | $1, 2, \ldots, 10$                                 | $1, 2, \dots, 10$                                | <b>Exercise 1.41.</b> Write down the numbers <span style="border: 1px solid red; padding: 0 2px;">1, 2, ..., 10</span> and draw an arrow $a \rightarrow b$ if $a$ |
| fong-spivak-invitation_F00249 | 30   | ■ 0.994 | $a \rightarrowtail b$                              | $a \rightarrow b$                                | <b>Exercise 1.41.</b> Write down the numbers $1, 2, \dots, 10$ and draw an arrow $a \rightarrow b$ if $a$                                                         |
| fong-spivak-invitation_F00250 | 30   | ■ 0.999 | $-500 \leq -499 \leq 0 \leq \sqrt{2} \leq 100 / 3$ | $-500 \leq -499 \leq 0 \leq \sqrt{2} \leq 100/3$ | ordering,” e.g. <span style="border: 1px solid red; padding: 0 2px;"><math>-500 &lt; -499 &lt; 0 &lt; \sqrt{2} &lt; 100/3</math></span> .                         |
| fong-spivak-invitation_F00251 | 30   | ■ 1.000 | $\mathrm{P}(X)$                                    | $P(X)$                                           | set of $X$ ; we denote it $P(X)$ . The power set can naturally be given an order by inclusion                                                                     |
| fong-spivak-invitation_F00252 | 31   | ■ 1.000 | $X = \{0, 1, 2\}$                                  | $X = \{0, 1, 2\}$                                | For example, taking $X = \{0, 1, 2\}$ , we depict $P(X)$ as                                                                                                       |
| fong-spivak-invitation_F00253 | 31   | ■ 0.933 | $\mathrm{P}(\{1, 2, \ldots, n\})$                  | $P\{1, 2, \dots, n\}$ , <sup>7</sup>             | See the cube? The Hasse diagram for the power set of a finite set, say $P\{1, 2, \dots, n\}$ , <sup>7</sup>                                                       |
| fong-spivak-invitation_F00254 | 31   | ■ 0.998 | $\mathrm{P}(\varnothing), \mathrm{P}\{1\}$         | $P(\varnothing), P\{1\}$                         | <b>Exercise 1.46.</b> Draw the Hasse diagrams for $P(\varnothing)$ , $P\{1\}$ , and $P\{1, 2\}$ .                                                                 |
| fong-spivak-invitation_F00255 | 31   | ■ 0.998 | $\mathrm{P}\{1, 2\}$                               | $P\{1, 2\}$                                      | <b>Exercise 1.46.</b> Draw the Hasse diagrams for $P(\varnothing)$ , $P\{1\}$ , and $P\{1, 2\}$ .                                                                 |
| fong-spivak-invitation_F00256 | 31   | ■ 0.981 | $\operatorname{Prt}(A)$                            | $\operatorname{Prt}(A)$                          | think about how we might order the set $\operatorname{Prt}(A)$ of <i>all partitions</i> of $A$ , for some set $A$ . In                                            |
| fong-spivak-invitation_F00257 | 31   | ■ 0.999 | $q \in Q$                                          | $q \in Q$                                        | partition $P$ is <i>finer</i> than a partition $Q$ if, for every part $p \in P$ , there is a part $q \in Q$                                                       |
| fong-spivak-invitation_F00258 | 31   | ■ 1.000 | $A_p \subseteq A_q$                                | $A_p \subseteq A_q$                              | such that $A_p \subseteq A_q$ . We could also say that $Q$ is <i>coarser</i> than $P$ .                                                                           |
| fong-spivak-invitation_F00259 | 31   | ■ 1.000 | $g: A \rightarrowtail Q$                           | $g: A \rightarrow Q$                             | tions out of $A$ . Then $f: A \twoheadrightarrow P$ is finer than $g: A \twoheadrightarrow Q$ if there is a function $h: P \rightarrow Q$                         |
| fong-spivak-invitation_F00260 | 31   | ■ 1.000 | $h: P \rightarrowtail Q$                           | $h: P \rightarrow Q$                             | tions out of $A$ . Then $f: A \twoheadrightarrow P$ is finer than $g: A \twoheadrightarrow Q$ if there is a function $h: P \rightarrow Q$                         |
| fong-spivak-invitation_F00261 | 31   | ■ 1.000 | $f; h = g$                                         | $f; h = g$                                       | such that $f \circ h = g$ .                                                                                                                                       |
| fong-spivak-invitation_F00262 | 31   | ■ 0.519 | $U$                                                | $U$                                              | <b>Example 1.49</b> (Upper sets). Given a preorder $(P, \leq)$ , an <i>upper set</i> in $P$ is a subset $U$ of                                                    |
| fong-spivak-invitation_F00263 | 31   | ■ 1.000 | $p \in U$                                          | $p \in U$                                        | $P$ satisfying the condition that if $p \in U$ and $p \leq q$ , then $q \in U$ . “If $p$ is an element                                                            |
| fong-spivak-invitation_F00264 | 31   | ■ 1.000 | $p \leq q$                                         | $p \leq q$                                       | $P$ satisfying the condition that if $p \in U$ and $p \leq q$ , then $q \in U$ . “If $p$ is an element                                                            |
| fong-spivak-invitation_F00265 | 31   | ■ 1.000 | $q \in U$                                          | $q \in U$                                        | $P$ satisfying the condition that if $p \in U$ and $p \leq q$ , then $q \in U$ . “If $p$ is an element                                                            |
| fong-spivak-invitation_F00266 | 31   | ■ 0.913 | $U(P)$                                             | $U(P)$                                           | then so is anything bigger.” Write $U(P)$ for the set of upper sets in $P$ . We can give the                                                                      |
| fong-spivak-invitation_F00267 | 31   | ■ 0.906 | $U \leq V$                                         | $U \leq V$                                       | set $U$ an order by letting $U \leq V$ if $U$ is contained in $V$ .                                                                                               |
| fong-spivak-invitation_F00268 | 31   | ■ 0.973 | $(\mathbb{B}, \leq)$                               | $(\mathbb{B}, \leq)$                             | For example, if $(\mathbb{B}, \leq)$ is the booleans (Example 1.29), then its preorder of upper sets                                                              |
| fong-spivak-invitation_F00269 | 31   | ■ 0.997 | $U(\mathbb{B})$                                    | $U(\mathbb{B})$                                  | $U(\mathbb{B})$ is                                                                                                                                                |
| fong-spivak-invitation_F00270 | 31   | ■ 1.000 | $\mathrm{P} X$                                     | $PX$                                             | <sup>7</sup> Note that we omit the parentheses here, writing $PX$ instead of $P(X)$ ; throughout this book we will omit                                           |
| fong-spivak-invitation_F00271 | 32   | ■ 0.845 | $\subseteq \mathbb{B}$                             | $\subseteq \mathbb{B}$                           | The subset $\{\text{false}\} \subseteq \mathbb{B}$ is not an upper set, because $\text{false} \leq \text{true}$ and $\text{true} \notin$                          |
| fong-spivak-invitation_F00272 | 32   | ■ 0.845 | $\notin$                                           | $\notin$                                         | The subset $\{\text{false}\} \subseteq \mathbb{B}$ is not an upper set, because $\text{false} \leq \text{true}$ and $\text{true} \notin$                          |

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| fong-spivak-invitation_F00273 | 32   | ■ 0.996 | $(Q, \leq)$                                      | $(Q, \leq)$                                      | <b>Example 1.51</b> (Product preorder). Given preorders $(P, \leq)$ and $(Q, \leq)$ , we may define a                          |
| fong-spivak-invitation_F00274 | 32   | ■ 1.000 | $P \times Q$                                     | $P \times Q$                                     | preorder structure on the product set $P \times Q$ by setting $(p, q) \leq (p', q')$ if and only if                            |
| fong-spivak-invitation_F00275 | 32   | ■ 1.000 | $(p, q) \leq (p', q')$                           | $(p, q) \leq (p', q')$                           | preorder structure on the product set $P \times Q$ by setting $\boxed{p, q} < \boxed{p', q'}$ if and only if                   |
| fong-spivak-invitation_F00276 | 32   | ■ 1.000 | $p \leq p'$                                      | $p \leq p'$                                      | $p \leq p'$ and $q \leq q'$ . We call this the <i>product preorder</i> . This is a basic example of a more                     |
| fong-spivak-invitation_F00277 | 32   | ■ 1.000 | $q \leq q'$                                      | $q \leq q'$                                      | $p \leq p'$ and $q \leq q'$ . We call this the <i>product preorder</i> . This is a basic example of a more                     |
| fong-spivak-invitation_F00278 | 32   | ■ 0.763 | $\left(P, \leq^{\mathrm{op}}\right)$             | $(P, \leq^{\mathrm{op}})$                        | preorder $(P, \leq^{\mathrm{op}})$ to have the same set of elements, but with $p \leq^{\mathrm{op}} q$ if and only if          |
| fong-spivak-invitation_F00279 | 32   | ■ 0.763 | $p \leq^{\mathrm{op}} q$                         | $p \leq^{\mathrm{op}} q$                         | preorder $(P, \leq^{\mathrm{op}})$ to have the same set of elements, but with $p \leq^{\mathrm{op}} q$ if and only if          |
| fong-spivak-invitation_F00280 | 32   | ■ 1.000 | $q \leq p$                                       | $q \leq p$                                       | $q \leq p$ .                                                                                                                   |
| fong-spivak-invitation_F00281 | 32   | ■ 0.999 | $A, \leq_A$                                      | $A, \leq_A$                                      | <b>Definition 1.54.</b> A monotone map between preorders $(A, \leq_A)$ and $(B, \leq_B)$ is a function                         |
| fong-spivak-invitation_F00282 | 32   | ■ 0.999 | $B, \leq_B$                                      | $B, \leq_B$                                      | <b>Definition 1.54.</b> A monotone map between preorders $(A, \leq_A)$ and $(B, \leq_B)$ is a function                         |
| fong-spivak-invitation_F00283 | 32   | ■ 0.993 | $x, y \in A$                                     | $x, y \in A$                                     | $f: A \rightarrow B$ such that, for all elements $x, y \in A$ , if $x \leq_A y$ then $f(x) \leq_B f(y)$ .                      |
| fong-spivak-invitation_F00284 | 32   | ■ 0.993 | $x \leq_A y$                                     | $x \leq_A y$                                     | $f: A \rightarrow B$ such that, for all elements $x, y \in A$ , if $x \leq_A y$ then $f(x) \leq_B f(y)$ .                      |
| fong-spivak-invitation_F00285 | 32   | ■ 0.993 | $f(x) \leq_B f(y)$                               | $f(x) \leq_B f(y)$                               | $f: A \rightarrow B$ such that, for all elements $x, y \in A$ , if $x \leq_A y$ then $\boxed{f(x) \leq_B f(y)}$ .              |
| fong-spivak-invitation_F00286 | 32   | ■ 1.000 | $A \rightarrow B$                                | $A \rightarrow B$                                | A monotone map $A \rightarrow B$ between two preorders associates to each element of pre-                                      |
| fong-spivak-invitation_F00287 | 33   | ■ 1.000 | $x \in A$                                        | $x \in A$                                        | each element $x \in A$ to its image $f(x) \in B$ . Note that the order must be preserved in                                    |
| fong-spivak-invitation_F00288 | 33   | ■ 1.000 | $f(x) \in B$                                     | $f(x) \in B$                                     | each element $x \in A$ to its image $f(x) \in B$ . Note that the order must be preserved in                                    |
| fong-spivak-invitation_F00289 | 33   | ■ 1.000 | $f(y)$                                           | $f(y)$                                           | hand preorder $A$ , then the image $f(x)$ will be above the image $f(y)$ in the right-hand                                     |
| fong-spivak-invitation_F00290 | 33   | ■ 1.000 | $\mathbb{B} \rightarrow \mathbb{N}$              | $\mathbb{B} \rightarrow \mathbb{N}$              | the preorder of natural numbers from Example 1.40. The map $\mathbb{B} \rightarrow \mathbb{N}$ sending <i>false</i>            |
| fong-spivak-invitation_F00291 | 34   | ■ 1.000 | $x \rightarrow y$                                | $x \rightarrow y$                                | because it preserves order: whenever there is a path $x \rightarrow y$ upstairs, there is a path                               |
| fong-spivak-invitation_F00292 | 34   | ■ 1.000 | $F(x) \rightarrow F(y)$                          | $F(x) \rightarrow F(y)$                          | $\boxed{F(x) \rightarrow F(y)}$ downstairs.                                                                                    |
| fong-spivak-invitation_F00293 | 34   | ■ 0.998 | $ \cdot : \mathcal{P}(X) \rightarrow \mathbb{N}$ | $ \cdot : \mathcal{P}(X) \rightarrow \mathbb{N}$ | tion from Example 1.45. The map $ \cdot : \mathcal{P}(X) \rightarrow \mathbb{N}$ sending each subset $S$ to its number         |
| fong-spivak-invitation_F00294 | 34   | ■ 0.996 | $ S $                                            | $ S $                                            | of elements $ S $ , also called its <i>cardinality</i> , is a monotone map.                                                    |
| fong-spivak-invitation_F00295 | 34   | ■ 1.000 | $0 \leq 1 \leq 2 \leq 3$                         | $0 \leq 1 \leq 2 \leq 3$                         | 2. Draw the Hasse diagram for the preorder $0 \leq 1 \leq 2 \leq 3$ .                                                          |
| fong-spivak-invitation_F00296 | 34   | ■ 0.948 | $ \cdot $                                        | $ \cdot $                                        | 3. Draw the cardinality map $ \cdot $ from Example 1.57 as dashed lines between them.                                          |
| fong-spivak-invitation_F00297 | 34   | ■ 0.993 | $\mathcal{U}(P) \rightarrow \mathcal{P}(P)$      | $\mathcal{U}(P) \rightarrow \mathcal{P}(P)$      | $(P, \leq)$ , the map $\mathcal{U}(P) \rightarrow \mathcal{P}(P)$ sending each upper set of $(P, \leq)$ to itself – considered |
| fong-spivak-invitation_F00298 | 34   | ■ 0.998 | $\mathcal{P}(\mathbb{B})$                        | $\mathcal{P}(\mathbb{B})$                        | Example 1.49, and you drew the Hasse diagram for $\mathcal{P}(\mathbb{B})$ in Exercise 1.46. Now draw                          |
| fong-spivak-invitation_F00299 | 34   | ■ 1.000 | $\uparrow p := \{p' \in P \mid p \leq p'\}$      | $\uparrow p := \{p' \in P \mid p \leq p'\}$      | 1. Show that the set $\boxed{\uparrow p := \{p' \in P \mid p \leq p'\}}$ is an upper set, for any $p \in P$ .                  |

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| fong-spivak-invitation_F00300 | 34   | ■ 0.801 | $\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$ | $\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$ | 2. Show that this construction defines a monotone map $\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$ .                       |
| fong-spivak-invitation_F00301 | 34   | ■ 0.999 | $\uparrow(p') \subseteq \uparrow(p)$                  | $\uparrow(p') \subseteq \uparrow(p)$                  | 3. Show that $p \leq p'$ in $P$ if and only if $\uparrow(p') \subseteq \uparrow(p)$ .                                               |
| fong-spivak-invitation_F00302 | 34   | ■ 0.952 | $b \geq a \leq c$                                     | $b \geq a \leq c$                                     | 4. Draw a picture of the map $\uparrow$ in the case where $P$ is the preorder ( $b \geq a \leq c$ ) from                            |
| fong-spivak-invitation_F00303 | 34   | ■ 1.000 | $f: (P, \leq_P) \rightarrow (Q, \leq_Q)$              | $f: (P, \leq_P) \rightarrow (Q, \leq_Q)$              | <b>Exercise 1.62.</b> As you know, a monotone map $f: (P, \leq_P) \rightarrow (Q, \leq_Q)$ consists of a                            |
| fong-spivak-invitation_F00304 | 34   | ■ 0.999 | $f: P \rightarrow Q$                                  | $f: P \rightarrow Q$                                  | function $f: P \rightarrow Q$ that satisfies a “monotonicity” property. Show that when $(P, \leq_P)$                                |
| fong-spivak-invitation_F00305 | 34   | ■ 0.999 | $(P, \leq_P)$                                         | $(P, \leq_P)$                                         | function $f: P \rightarrow Q$ that satisfies a “monotonicity” property. Show that when $(P, \leq_P)$                                |
| fong-spivak-invitation_F00306 | 34   | ■ 1.000 | $P \rightarrow Q$                                     | $P \rightarrow Q$                                     | is a discrete preorder, then <i>every</i> function $P \rightarrow Q$ satisfies the monotonicity property,                           |
| fong-spivak-invitation_F00307 | 34   | ■ 1.000 | $\leq_Q$                                              | $\leq_Q$                                              | regardless of the order $\leq_Q$ .                                                                                                  |
| fong-spivak-invitation_F00308 | 35   | ■ 1.000 | $\mathrm{Prt}(X)$                                     | $\mathrm{Prt}(X)$                                     | <b>Example 1.63.</b> Recall from Example 1.47 that given a set $X$ we define $\mathrm{Prt}(X)$ to be the                            |
| fong-spivak-invitation_F00309 | 35   | ■ 1.000 | $s: X \rightarrow P$                                  | $s: X \rightarrow P$                                  | $s: X \twoheadrightarrow P$ for some set $P$ .                                                                                      |
| fong-spivak-invitation_F00310 | 35   | ■ 1.000 | $f^*: \mathrm{Prt}(Y) \rightarrow \mathrm{Prt}(X)$    | $f^*: \mathrm{Prt}(Y) \rightarrow \mathrm{Prt}(X)$    | Any surjective function $f: X \twoheadrightarrow Y$ induces a monotone map $f^*: \mathrm{Prt}(Y) \rightarrow \mathrm{Prt}(X)$ ,     |
| fong-spivak-invitation_F00311 | 35   | ■ 1.000 | $s: Y \rightarrow P$                                  | $s: Y \rightarrow P$                                  | going “backwards.” It is defined by sending a partition $s: Y \twoheadrightarrow P$ to the composite                                |
| fong-spivak-invitation_F00312 | 35   | ■ 0.727 | $s$                                                   | $s$                                                   | $f \circ s: X \twoheadrightarrow P$ . <sup>8</sup>                                                                                  |
| fong-spivak-invitation_F00313 | 35   | ■ 0.727 | $X \rightarrow P$                                     | $X \rightarrow P$                                     | $f \circ s: X \twoheadrightarrow P$ . <sup>8</sup>                                                                                  |
| fong-spivak-invitation_F00314 | 35   | ■ 1.000 | $f^*(P)$                                              | $f^*(P)$                                              | partitions $P$ and $Q$ of $Y$ , and then find $f^*(P)$ and $f^*(Q)$ .                                                               |
| fong-spivak-invitation_F00315 | 35   | ■ 1.000 | $f^*(Q)$                                              | $f^*(Q)$                                              | partitions $P$ and $Q$ of $Y$ , and then find $f^*(P)$ and $f^*(Q)$ .                                                               |
| fong-spivak-invitation_F00316 | 35   | ■ 0.865 | $P, \leq_P$                                           | $P, \leq_P$                                           | <b>Proposition 1.65.</b> For any preorder $(P, \leq_P)$ , the identity function is monotone.                                        |
| fong-spivak-invitation_F00317 | 35   | ■ 0.794 | $(Q, \leq_Q)$                                         | $(Q, \leq_Q)$                                         | If $(Q, \leq_Q)$ and $(R, \leq_R)$ are preorders and $f: P \rightarrow Q$ and $g: Q \rightarrow R$ are monotone,                    |
| fong-spivak-invitation_F00318 | 35   | ■ 0.794 | $(R, \leq_R)$                                         | $(R, \leq_R)$                                         | If $(Q, \leq_Q)$ and $(R, \leq_R)$ are preorders and $f: P \rightarrow Q$ and $g: Q \rightarrow R$ are monotone,                    |
| fong-spivak-invitation_F00319 | 35   | ■ 0.794 | $g: Q \rightarrow R$                                  | $g: Q \rightarrow R$                                  | If $(Q, \leq_Q)$ and $(R, \leq_R)$ are preorders and $f: P \rightarrow Q$ and $g: Q \rightarrow R$ are monotone,                    |
| fong-spivak-invitation_F00320 | 35   | ■ 0.991 | $(f; g): P \rightarrow R$                             | $(f; g): P \rightarrow R$                             | then $(f \circ g): P \rightarrow R$ is also monotone.                                                                               |
| fong-spivak-invitation_F00321 | 35   | ■ 0.514 | $\mathrm{id}_P: P \rightarrow P$                      | $\mathrm{id}_P: P \rightarrow P$                      | identity function $\mathrm{id}_P: P \rightarrow P$ is a monotone map $(P, \leq) \rightarrow (P, \leq^{\mathrm{op}})$ if and only if |
| fong-spivak-invitation_F00322 | 35   | ■ 0.514 | $(P, \leq) \rightarrow (P, \leq^{\mathrm{op}})$       | $(P, \leq) \rightarrow (P, \leq^{\mathrm{op}})$       | identity function $\mathrm{id}_P: P \rightarrow P$ is a monotone map $(P, \leq) \rightarrow (P, \leq^{\mathrm{op}})$ if and only if |
| fong-spivak-invitation_F00323 | 35   | ■ 1.000 | $p, q \in P$                                          | $p, q \in P$                                          | for all $p, q \in P$ we have $q \leq p$ whenever $p \leq q$ . For historical reasons connected to                                   |
| fong-spivak-invitation_F00324 | 35   | ■ 0.795 | $p_1 = p_2$                                           | $p_1 = p_2$                                           | be made into a discrete preorder ( $p_1 \leq p_2$ iff $p_1 = p_2$ ), and the round-trips return us                                  |
| fong-spivak-invitation_F00325 | 36   | ■ 1.000 | $g: Q \rightarrow P$                                  | $g: Q \rightarrow P$                                  | $f: P \rightarrow Q$ is called an <i>isomorphism</i> if there exists a monotone function $g: Q \rightarrow P$                       |
| fong-spivak-invitation_F00326 | 36   | ■ 0.957 | $f; g = \mathrm{id}_P$                                | $f; g = \mathrm{id}_P$                                | such that $f \circ g = \mathrm{id}_P$ and $g \circ f = \mathrm{id}_Q$ . This means that, for any $p \in P$ and $q \in Q$ , we       |

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| fong-spivak-invitation_F00327 | 36   | ■ 0.957 | $g ; f=\operatorname{id}_{\{Q\}}$                                      | $g; f = \operatorname{id}_Q$                                       | such that $f \circ g = \operatorname{id}_P$ and $g \circ f = \operatorname{id}_Q$ . This means that, for any $p \in P$ and $q \in Q$ , we          |
| fong-spivak-invitation_F00328 | 36   | ■ 1.000 | $g$                                                                    | $g$                                                                | We refer to $g$ as the <i>inverse</i> of $f$ , and vice versa: $f$ is the inverse of $g$ .                                                         |
| fong-spivak-invitation_F00329 | 36   | ■ 1.000 | $P, Q$                                                                 | $P, Q$                                                             | <b>Example 1.71.</b> Here are the Hasse diagrams for three preorders $P, Q$ , and $R$ , all of                                                     |
| fong-spivak-invitation_F00330 | 36   | ■ 1.000 | $R$                                                                    | $R$                                                                | <b>Example 1.71.</b> Here are the Hasse diagrams for three preorders $P, Q$ , and $R$ , all of                                                     |
| fong-spivak-invitation_F00331 | 36   | ■ 1.000 | $f(a)=v, f(b)=w, f(c)=x, f(d)=y$                                       | $f(a) = v, f(b) = w, f(c) = x, f(d) = y$                           | The map $f: P \rightarrow Q$ given by $f(a) = v, f(b) = w, f(c) = x, f(d) = y$ , and                                                               |
| fong-spivak-invitation_F00332 | 36   | ■ 1.000 | $f(e)=z$                                                               | $f(e) = z$                                                         | $f(e) = z$ has an inverse.                                                                                                                         |
| fong-spivak-invitation_F00333 | 36   | ■ 0.743 | $x \rightarrowtail z$                                                  | $x \rightarrow z$                                                  | is an arrow $x \rightarrow z$ in the Hasse diagram for $R$ and not one in $Q$ , but in fact this arrow                                             |
| fong-spivak-invitation_F00334 | 36   | ■ 1.000 | $v \rightarrowtail y$                                                  | $v \rightarrow y$                                                  | drawn an arrow $v \rightarrow y$ in either $Q$ or $R$ and it would not have changed the preorder.                                                  |
| fong-spivak-invitation_F00335 | 36   | ■ 0.657 | $\operatorname{Prt}(\{*, \bullet, \circ\}) \rightarrowtail \mathbb{B}$ | $\operatorname{Prt}(\{*, \bullet, \circ\}) \rightarrow \mathbb{B}$ | “Is $\bullet$ connected to $*$ ?” is a monotone map $\operatorname{Prt}(\{*, \bullet, \circ\}) \rightarrow \mathbb{B}$ ; see also Eq. (1.3).       |
| fong-spivak-invitation_F00336 | 36   | ■ 1.000 | $P \rightarrowtail \mathbb{B}$                                         | $P \rightarrow \mathbb{B}$                                         | <b>Proposition 1.73.</b> Let $P$ be a preorder. Monotone maps $P \rightarrow \mathbb{B}$ are in one-to-one                                         |
| fong-spivak-invitation_F00337 | 37   | ■ 1.000 | $f: P \rightarrowtail \mathbb{B}$                                      | $f: P \rightarrow \mathbb{B}$                                      | <i>Proof.</i> Let $f: P \rightarrow \mathbb{B}$ be a monotone map. We will show that the subset                                                    |
| fong-spivak-invitation_F00338 | 37   | ■ 1.000 | $f^{-1}$                                                               | $f^{-1}$                                                           | $f^{-1}(\operatorname{true}) \subseteq P$ is an upper set. Suppose $p \in f^{-1}(\operatorname{true})$ and $p \leq q$ ; then                       |
| fong-spivak-invitation_F00339 | 37   | ■ 1.000 | $\subseteq P$                                                          | $\subseteq P$                                                      | $f^{-1}(\operatorname{true}) \subseteq P$ is an upper set. Suppose $p \in f^{-1}(\operatorname{true})$ and $p \leq q$ ; then                       |
| fong-spivak-invitation_F00340 | 37   | ■ 1.000 | $p \in f^{-1}$                                                         | $p \in f^{-1}$                                                     | $f^{-1}(\operatorname{true}) \subseteq P$ is an upper set. Suppose $p \in f^{-1}(\operatorname{true})$ and $p \leq q$ ; then                       |
| fong-spivak-invitation_F00341 | 37   | ■ 0.617 | $=f(p) \leq f(q)$                                                      | $= f(p) \leq f(q)$                                                 | $\operatorname{true} = f(p) \leq f(q)$ . But in $\mathbb{B}$ , if $\operatorname{true} \leq f(q)$ then $\operatorname{true} = f(q)$ . This implies |
| fong-spivak-invitation_F00342 | 37   | ■ 0.617 | $\leq f(q)$                                                            | $\leq f(q)$                                                        | $\operatorname{true} = f(p) \leq f(q)$ . But in $\mathbb{B}$ , if $\operatorname{true} \leq f(q)$ then $\operatorname{true} = f(q)$ . This implies |
| fong-spivak-invitation_F00343 | 37   | ■ 0.617 | $=f(q)$                                                                | $= f(q)$                                                           | $\operatorname{true} = f(p) \leq f(q)$ . But in $\mathbb{B}$ , if $\operatorname{true} \leq f(q)$ then $\operatorname{true} = f(q)$ . This implies |
| fong-spivak-invitation_F00344 | 37   | ■ 1.000 | $q \in f^{-1}$                                                         | $q \in f^{-1}$                                                     | $q \in f^{-1}(\operatorname{true})$ and thus shows that $f^{-1}(\operatorname{true})$ is an upper set.                                             |
| fong-spivak-invitation_F00345 | 37   | ■ 0.522 | $f_{\{U\}}: P \rightarrowtail \mathbb{B}$                              | $f_U: P \rightarrow \mathbb{B}$                                    | Conversely, if $U$ is an upper set in $P$ , define $f_U: P \rightarrow \mathbb{B}$ such that $f_U(p) = \operatorname{true}$                        |
| fong-spivak-invitation_F00346 | 37   | ■ 0.522 | $f_{\{U\}}(p)=\operatorname{true}$                                     | $f_U(p) = \operatorname{true}$                                     | Conversely, if $U$ is an upper set in $P$ , define $f_U: P \rightarrow \mathbb{B}$ such that $f_U(p) = \operatorname{true}$                        |
| fong-spivak-invitation_F00347 | 37   | ■ 1.000 | $f_{\{U\}}(p)=$                                                        | $f_U(p) =$                                                         | when $p \in U$ , and $f_U(p) = \operatorname{false}$ when $p \notin U$ . This is a monotone map, because                                           |
| fong-spivak-invitation_F00348 | 37   | ■ 1.000 | $p \notin U$                                                           | $p \notin U$                                                       | when $p \in U$ , and $f_U(p) = \operatorname{false}$ when $p \notin U$ . This is a monotone map, because                                           |
| fong-spivak-invitation_F00349 | 37   | ■ 1.000 | $f_{\{U\}}(p)=\operatorname{true}=f(q)$                                | $f_U(p) = \operatorname{true} = f(q)$                              | if $p \leq q$ , then either $p \in U$ , so $q \in U$ and $f_U(p) = \operatorname{true} = f(q)$ , or $p \notin U$ , so                              |
| fong-spivak-invitation_F00350 | 37   | ■ 1.000 | $\leq f_{\{U\}}(q)$                                                    | $\leq f_U(q)$                                                      | $f_U(p) = \operatorname{false} \leq f_U(q)$ .                                                                                                      |
| fong-spivak-invitation_F00351 | 37   | ■ 0.890 | $f^*: \operatorname{U}(Q) \rightarrowtail \operatorname{U}(P)$         | $f^*: \operatorname{U}(Q) \rightarrow \operatorname{U}(P)$         | map. Then we can define a monotone map $f^*: \operatorname{U}(Q) \rightarrow \operatorname{U}(P)$ sending an upper set                             |
| fong-spivak-invitation_F00352 | 37   | ■ 1.000 | $U \subseteqtail Q$                                                    | $U \subseteq Q$                                                    | $U \subseteq Q$ to the upper set $f^{-1}(U) \subseteq P$ . We call this the <i>pullback along <math>f</math></i> .                                 |
| fong-spivak-invitation_F00353 | 37   | ■ 1.000 | $f^{-1}(U) \subseteqtail P$                                            | $f^{-1}(U) \subseteq P$                                            | $U \subseteq Q$ to the upper set $f^{-1}(U) \subseteq P$ . We call this the <i>pullback along <math>f</math></i> .                                 |

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| fong-spivak-invitation_F00354 | 37   | ■ 1.000 | $f^{\{*\}}$                                                                 | $f^*$                                                                       | understood in terms of composition. Indeed, show that $f^*$ is defined by taking $u: Q \rightarrow$                       |
| fong-spivak-invitation_F00355 | 37   | ■ 1.000 | $u: Q \rightarrow$                                                          | $u: Q \rightarrow$                                                          | understood in terms of composition. Indeed, show that $f^*$ is defined by taking $u: Q \rightarrow$                       |
| fong-spivak-invitation_F00356 | 37   | ■ 1.000 | $(f; u): P \rightarrow \mathbb{B}$                                          | $(f; u): P \rightarrow \mathbb{B}$                                          | $\mathbb{B}$ to $(f \circ u): P \rightarrow \mathbb{B}$ .                                                                 |
| fong-spivak-invitation_F00357 | 37   | ■ 0.798 | $(\mathbb{R}, \leq)$                                                        | $(\mathbb{R}, \leq)$                                                        | Consider the preorder $(\mathbb{R}, \leq)$ of real numbers ordered in the usual way. The subset                           |
| fong-spivak-invitation_F00358 | 37   | ■ 1.000 | $\mathbb{N} \subseteq \mathbb{R}$                                           | $\mathbb{N} \subseteq \mathbb{R}$                                           | $\mathbb{N} \subseteq \mathbb{R}$ has many lower bounds, namely $-1.5$ is a lower bound: every element of $\mathbb{N}$ is |
| fong-spivak-invitation_F00359 | 37   | ■ 1.000 | $\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} \subseteq \mathbb{R}$ | $\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} \subseteq \mathbb{R}$ | 1. Why is 0 a lower bound for $\{\frac{1}{n+1} \mid n \in \mathbb{N}\} \subseteq \mathbb{R}$ ?                            |
| fong-spivak-invitation_F00360 | 38   | ■ 0.752 | $A \subseteq P$                                                             | $A \subseteq P$                                                             | <b>Definition 1.76.</b> Let $(P, \leq)$ be a preorder, and let $A \subseteq P$ be a subset. We say that an                |
| fong-spivak-invitation_F00361 | 38   | ■ 1.000 | $p \leq a$                                                                  | $p \leq a$                                                                  | (a) for all $a \in A$ , we have $p \leq a$ ,                                                                              |
| fong-spivak-invitation_F00362 | 38   | ■ 1.000 | $q$                                                                         | $q$                                                                         | (b) for all $q$ such that $q \leq a$ for all $a \in A$ , we have that $q \leq p$ .                                        |
| fong-spivak-invitation_F00363 | 38   | ■ 1.000 | $q \leq a$                                                                  | $q \leq a$                                                                  | (b) for all $q$ such that $q \leq a$ for all $a \in A$ , we have that $q \leq p$ .                                        |
| fong-spivak-invitation_F00364 | 38   | ■ 0.998 | $p = \bigwedge A, p = \bigwedge_{a \in A} a$                                | $p = \bigwedge A, p = \bigwedge_{a \in A} a$                                | We write $p = \bigwedge A$ , $p = \bigwedge_{a \in A} a$ , or, if the dummy variable $a$ is clear from context,           |
| fong-spivak-invitation_F00365 | 38   | ■ 0.999 | $p = \bigwedge_{A} a$                                                       | $p = \bigwedge_A a$                                                         | just $p = \bigwedge_A a$ . If $A$ just consists of two elements, say $A = \{a, b\}$ , we can denote $\bigwedge A$         |
| fong-spivak-invitation_F00366 | 38   | ■ 0.999 | $A = \{a, b\}$                                                              | $A = \{a, b\}$                                                              | just $p = \bigwedge_A a$ . If $A$ just consists of two elements, say $A = \{a, b\}$ , we can denote $\bigwedge A$         |
| fong-spivak-invitation_F00367 | 38   | ■ 0.999 | $\bigwedge A$                                                               | $\bigwedge A$                                                               | just $p = \bigwedge_A a$ . If $A$ just consists of two elements, say $A = \{a, b\}$ , we can denote $\bigwedge A$         |
| fong-spivak-invitation_F00368 | 38   | ■ 1.000 | $a \wedge b$                                                                | $a \wedge b$                                                                | simply by $a \wedge b$ .                                                                                                  |
| fong-spivak-invitation_F00369 | 38   | ■ 1.000 | $a \leq p$                                                                  | $a \leq p$                                                                  | (a) for all $a \in A$ we have $a \leq p$ ,                                                                                |
| fong-spivak-invitation_F00370 | 38   | ■ 1.000 | $a \leq q$                                                                  | $a \leq q$                                                                  | (b) for all $q$ such that $a \leq q$ for all $a \in A$ , we have that $p \leq q$ .                                        |
| fong-spivak-invitation_F00371 | 38   | ■ 0.777 | $p = \bigvee A$                                                             | $p = \bigvee A$                                                             | We write $p = \bigvee A$ or $p = \bigvee_{a \in A} a$ , or when $A = \{a, b\}$ we may simply write                        |
| fong-spivak-invitation_F00372 | 38   | ■ 0.777 | $p = \bigvee_{a \in A} a$                                                   | $p = \bigvee_{a \in A} a$                                                   | We write $p = \bigvee A$ or $p = \bigvee_{a \in A} a$ , or when $A = \{a, b\}$ we may simply write                        |
| fong-spivak-invitation_F00373 | 38   | ■ 1.000 | $p = a \vee b$                                                              | $p = a \vee b$                                                              | $p = a \vee b$ .                                                                                                          |
| fong-spivak-invitation_F00374 | 38   | ■ 1.000 | $p = \bigwedge A$                                                           | $p = \bigwedge A$                                                           | $A \subseteq P$ , say $p = \bigwedge A$ and $q = \bigwedge A$ with $p \neq q$ , which does not make sense if              |
| fong-spivak-invitation_F00375 | 38   | ■ 1.000 | $q = \bigwedge A$                                                           | $q = \bigwedge A$                                                           | $A \subseteq P$ , say $p = \bigwedge A$ and $q = \bigwedge A$ with $p \neq q$ , which does not make sense if              |
| fong-spivak-invitation_F00376 | 38   | ■ 1.000 | $p, q$                                                                      | $p, q$                                                                      | meets $p, q$ are automatically isomorphic: the very definition of meet forces both $p \leq$                               |
| fong-spivak-invitation_F00377 | 38   | ■ 1.000 | $p \leq q$                                                                  | $p \leq q$                                                                  | meets $p, q$ are automatically isomorphic: the very definition of meet forces both $p \leq$                               |
| fong-spivak-invitation_F00378 | 38   | ■ 1.000 | $p \cong q$                                                                 | $p \cong q$                                                                 | $q$ and $q \leq p$ , and thus we have $p \cong q$ . So, for any $x \in P$ , we have $p \leq x$ iff                        |
| fong-spivak-invitation_F00379 | 38   | ■ 1.000 | $x \in P$                                                                   | $x \in P$                                                                   | $q$ and $q \leq p$ , and thus we have $p \cong q$ . So, for any $x \in P$ , we have $p \leq x$ iff                        |
| fong-spivak-invitation_F00380 | 38   | ■ 1.000 | $p \leq x$                                                                  | $p \leq x$                                                                  | $q$ and $q \leq p$ , and thus we have $p \cong q$ . So, for any $x \in P$ , we have $p \leq x$ iff                        |

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| fong-spivak-invitation_F00381 | 38   | ■ 1.000 | $q \leq x$                  | $q \leq x$                  | $q \leq x$ and $x \leq p$ iff $x \leq q$ . Thus as long as we are only interested in elements of $P$             |
| fong-spivak-invitation_F00382 | 38   | ■ 1.000 | $x \leq p$                  | $x \leq p$                  | $q \leq x$ and $x \leq p$ iff $x \leq q$ . Thus as long as we are only interested in elements of $P$             |
| fong-spivak-invitation_F00383 | 38   | ■ 1.000 | $x \leq q$                  | $x \leq q$                  | $q \leq x$ and $x \leq p$ iff $x \leq q$ . Thus as long as we are only interested in elements of $P$             |
| fong-spivak-invitation_F00384 | 38   | ■ 1.000 | $P=\{p, q, r\}$             | $P = \{p, q, r\}$           | subset $A$ need not have a meet or a join. Consider the three-element set $P = \{p, q, r\}$                      |
| fong-spivak-invitation_F00385 | 38   | ■ 1.000 | $A=\{p, q\}$                | $A = \{p, q\}$              | with the discrete ordering. The set $A = \{p, q\}$ does not have a join in $P$ because if $x$                    |
| fong-spivak-invitation_F00386 | 39   | ■ 0.630 | $a, b\}$                    | $a, b\}$                    | Let $A$ be the subset $\{a, b\}$ in the preorder specified by this Hasse diagram. Then both $c$                  |
| fong-spivak-invitation_F00387 | 39   | ■ 0.992 | $c \leq d$                  | $c \leq d$                  | less than $d$ . Note that, as in Remark 1.77, $c \leq d$ and $d \leq c$ , so $c \cong d$ . Such will always      |
| fong-spivak-invitation_F00388 | 39   | ■ 0.992 | $d \leq c$                  | $d \leq c$                  | less than $d$ . Note that, as in Remark 1.77, $c \leq d$ and $d \leq c$ , so $c \cong d$ . Such will always      |
| fong-spivak-invitation_F00389 | 39   | ■ 0.992 | $c \cong d$                 | $c \cong d$                 | less than $d$ . Note that, as in Remark 1.77, $c \leq d$ and $d \leq c$ , so $c \cong d$ . Such will always      |
| fong-spivak-invitation_F00390 | 39   | ■ 1.000 | $A=\{p\}$                   | $A = \{p\}$                 | $A = \{p\}$ with one element.                                                                                    |
| fong-spivak-invitation_F00391 | 39   | ■ 1.000 | $\wedge A \cong p$          | $\wedge A \cong p$          | 1. Show that $\wedge A \cong p$ .                                                                                |
| fong-spivak-invitation_F00392 | 39   | ■ 1.000 | $\wedge A = p$              | $\wedge A = p$              | 2. Show that if $P$ is in fact a partial order, then $\wedge A = p$ .                                            |
| fong-spivak-invitation_F00393 | 39   | ■ 0.994 | $\wedge$                    | $\wedge$                    | 3. Are the analogous facts true when $\wedge$ is replaced by $\vee$ ?                                            |
| fong-spivak-invitation_F00394 | 39   | ■ 0.994 | $\vee$                      | $\vee$                      | 3. Are the analogous facts true when $\wedge$ is replaced by $\vee$ ?                                            |
| fong-spivak-invitation_F00395 | 39   | ■ 1.000 | $p \vee p = p \wedge p = p$ | $p \vee p = p \wedge p = p$ | <b>Example 1.81.</b> In any partial order $P$ , we have $p \vee p = p \wedge p = p$ . The reason is              |
| fong-spivak-invitation_F00396 | 39   | ■ 0.996 | $p \vee p$                  | $p \vee p$                  | that our notation says $p \vee p$ means $\vee\{p, p\}$ . But $\{p, p\} = \{p\}$ (see Section 1.2.1), so          |
| fong-spivak-invitation_F00397 | 39   | ■ 0.996 | $\bigvee\{p, p\}$           | $\bigvee\{p, p\}$           | that our notation says $p \vee p$ means $\vee\{p, p\}$ . But $\{p, p\} = \{p\}$ (see Section 1.2.1), so          |
| fong-spivak-invitation_F00398 | 39   | ■ 0.996 | $\{p, p\}=\{p\}$            | $\{p, p\} = \{p\}$          | that our notation says $p \vee p$ means $\vee\{p, p\}$ . But $\{p, p\} = \{p\}$ (see Section 1.2.1), so          |
| fong-spivak-invitation_F00399 | 39   | ■ 1.000 | $p \vee p = p$              | $p \vee p = p$              | $p \vee p = p$ by Exercise 1.80.                                                                                 |
| fong-spivak-invitation_F00400 | 39   | ■ 1.000 | $A, B \subseteq X$          | $A, B \subseteq X$          | <b>Example 1.82.</b> In a power set $P(X)$ , the meet of a collection of subsets, say $A, B \subseteq X$ ,       |
| fong-spivak-invitation_F00401 | 39   | ■ 1.000 | $A \wedge B = A \cap B$     | $A \wedge B = A \cap B$     | is their intersection $A \wedge B = A \cap B$ , while the join is their union, $A \vee B =$                      |
| fong-spivak-invitation_F00402 | 39   | ■ 1.000 | $A \vee B =$                | $A \vee B =$                | is their intersection $A \wedge B = A \cap B$ , while the join is their union, $A \vee B =$                      |
| fong-spivak-invitation_F00403 | 40   | ■ 1.000 | $n \mid m$                  | $n \mid m$                  | <b>Exercise 1.85.</b> Recall the division ordering on $\mathbb{N}$ from Example 1.40: we write $n \mid m$ if     |
| fong-spivak-invitation_F00404 | 40   | ■ 0.924 | $A \subseteq B \subseteq P$ | $A \subseteq B \subseteq P$ | <b>Proposition 1.86.</b> Suppose $(P, \leq)$ is a preorder and $A \subseteq B \subseteq P$ are subsets that have |
| fong-spivak-invitation_F00405 | 40   | ■ 1.000 | $\wedge B \leq \wedge A$    | $\wedge B \leq \wedge A$    | meets. Then $\wedge B \leq \wedge A$ .                                                                           |
| fong-spivak-invitation_F00406 | 40   | ■ 1.000 | $\bigvee A \leq \bigvee B$  | $\bigvee A \leq \bigvee B$  | Similarly, if $A$ and $B$ have joins, then $\bigvee A \leq \bigvee B$ .                                          |
| fong-spivak-invitation_F00407 | 40   | ■ 1.000 | $m=\wedge A$                | $m = \wedge A$              | <i>Proof.</i> Let $m = \wedge A$ and $n = \wedge B$ . Then for any $a \in A$ we also have $a \in B$ , so         |

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| fong-spivak-invitation_F00408 | 40   | ■ 1.000 | $n = \wedge B$                                    | $n = \wedge B$                                    | <i>Proof.</i> Let $m = \bigwedge A$ and $n = \bigwedge B$ . Then for any $a \in A$ we also have $a \in B$ , so                                                                 |
| fong-spivak-invitation_F00409 | 40   | ■ 1.000 | $a \in B$                                         | $a \in B$                                         | <i>Proof.</i> Let $m = \bigwedge A$ and $n = \bigwedge B$ . Then for any $a \in A$ we also have $a \in B$ , so                                                                 |
| fong-spivak-invitation_F00410 | 40   | ■ 1.000 | $n \leq a$                                        | $n \leq a$                                        | $n \leq a$ because $n$ is a lower bound for $B$ . Thus $n$ is also a lower bound for $A$ and                                                                                   |
| fong-spivak-invitation_F00411 | 40   | ■ 1.000 | $\Phi: P \rightarrow Q$                           | $\Phi: P \rightarrow Q$                           | $\Phi: P \rightarrow Q$ is a phenomenon (we might say “feature”) of $P$ as observed by $Q$ . Adam                                                                              |
| fong-spivak-invitation_F00412 | 40   | ■ 1.000 | $f(a \wedge$                                      | $f(a \wedge$                                      | <b>Definition 1.87.</b> We say that a monotone map $f: P \rightarrow Q$ <i>preserves meets</i> if $f(a \wedge$                                                                 |
| fong-spivak-invitation_F00413 | 40   | ■ 1.000 | $b) \cong f(a) \wedge f(b)$                       | $b) \cong f(a) \wedge f(b)$                       | $b) \cong f(a) \wedge f(b)$ for all $a, b \in P$ . We similarly say $f$ <i>preserves joins</i> if $f(a \vee b) \cong$                                                          |
| fong-spivak-invitation_F00414 | 40   | ■ 1.000 | $a, b \in P$                                      | $a, b \in P$                                      | $b) \cong f(a) \wedge f(b)$ for all $a, b \in P$ . We similarly say $f$ <i>preserves joins</i> if $f(a \vee b) \cong$                                                          |
| fong-spivak-invitation_F00415 | 40   | ■ 1.000 | $f(a \vee b) \cong$                               | $f(a \vee b) \cong$                               | $b) \cong f(a) \wedge f(b)$ for all $a, b \in P$ . We similarly say $f$ <i>preserves joins</i> if $f(a \vee b) \cong$                                                          |
| fong-spivak-invitation_F00416 | 40   | ■ 1.000 | $f(a) \vee f(b)$                                  | $f(a) \vee f(b)$                                  | $f(a) \vee f(b)$ for all $a, b \in P$ .                                                                                                                                        |
| fong-spivak-invitation_F00417 | 40   | ■ 1.000 | $f(a \vee b)$                                     | $f(a \vee b)$                                     | side $f(a \vee b)$ is the observation of the combined system $a \vee b$ . The inequality implies                                                                               |
| fong-spivak-invitation_F00418 | 40   | ■ 1.000 | $a \vee b$                                        | $a \vee b$                                        | side $f(a \vee b)$ is the observation of the combined system $a \vee b$ . The inequality implies                                                                               |
| fong-spivak-invitation_F00419 | 41   | ■ 1.000 | $f(a \vee$                                        | $f(a \vee$                                        | <b>Exercise 1.89.</b> In Definition 1.88, we defined generativity of $f$ as the inequality $f(a \vee$                                                                          |
| fong-spivak-invitation_F00420 | 41   | ■ 1.000 | $b) \neq f(a) \vee f(b)$                          | $b) \neq f(a) \vee f(b)$                          | $b) \neq f(a) \vee f(b)$ , but in the subsequent text we seemed to imply there would be not                                                                                    |
| fong-spivak-invitation_F00421 | 41   | ■ 1.000 | $f(a), f(b) \in Q$                                | $f(a), f(b) \in Q$                                | $f(a), f(b) \in Q$ have a join, then indeed $f(a) \vee f(b) \leq f(a \vee b)$ .                                                                                                |
| fong-spivak-invitation_F00422 | 41   | ■ 1.000 | $f(a) \vee f(b) \leq f(a \vee b)$                 | $f(a) \vee f(b) \leq f(a \vee b)$                 | $f(a), f(b) \in Q$ have a join, then indeed $f(a) \vee f(b) < f(a \vee b)$ .                                                                                                   |
| fong-spivak-invitation_F00423 | 41   | ■ 0.927 | $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ | $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ | <b>Example 1.91.</b> Consider the map $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ which sends $x \in \mathbb{Z}$ to $3x$ , which                                         |
| fong-spivak-invitation_F00424 | 41   | ■ 0.927 | $x \in \mathbb{Z}$                                | $x \in \mathbb{Z}$                                | <b>Example 1.91.</b> Consider the map $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ which sends $x \in \mathbb{Z}$ to $3x$ , which                                         |
| fong-spivak-invitation_F00425 | 41   | ■ 0.927 | $3x$                                              | $3x$                                              | <b>Example 1.91.</b> Consider the map $(3 \times -): \mathbb{Z} \rightarrow \mathbb{R}$ which sends $x \in \mathbb{Z}$ to $3x$ , which                                         |
| fong-spivak-invitation_F00426 | 41   | ■ 1.000 | $3x \in \mathbb{Z} \subseteq \mathbb{R}$          | $3x \in \mathbb{Z} \subseteq \mathbb{R}$          | we can consider as a real number $3x \in \mathbb{Z} \subseteq \mathbb{R}$ . Let’s find a left adjoint for the map                                                              |
| fong-spivak-invitation_F00427 | 42   | ■ 0.509 | $\lceil z \rceil$                                 | $\lceil z \rceil$                                 | Write $\lceil z \rceil$ for the smallest natural number above $z \in \mathbb{R}$ , and write $\lfloor z \rfloor$ for the largest                                               |
| fong-spivak-invitation_F00428 | 42   | ■ 0.509 | $z \in \mathbb{R}$                                | $z \in \mathbb{R}$                                | Write $\lceil z \rceil$ for the smallest natural number above $z \in \mathbb{R}$ , and write $\lfloor z \rfloor$ for the largest                                               |
| fong-spivak-invitation_F00429 | 42   | ■ 0.509 | $\lfloor z \rfloor$                               | $\lfloor z \rfloor$                               | Write $\lceil z \rceil$ for the smallest natural number above $z \in \mathbb{R}$ , and write $\lfloor z \rfloor$ for the largest                                               |
| fong-spivak-invitation_F00430 | 42   | ■ 0.937 | $z \in \mathbb{R}$                                | $z \in \mathbb{R}$                                | integer below $z \in \mathbb{R}$ , e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$ . <sup>9</sup> As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$ , let’s |
| fong-spivak-invitation_F00431 | 42   | ■ 0.937 | $\lceil 3.14 \rceil = 4$                          | $\lceil 3.14 \rceil = 4$                          | integer below $z \in \mathbb{R}$ , e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$ . <sup>9</sup> As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$ , let’s |
| fong-spivak-invitation_F00432 | 42   | ■ 0.937 | $\lfloor 3.14 \rfloor = 3$                        | $\lfloor 3.14 \rfloor = 3$                        | integer below $z \in \mathbb{R}$ , e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$ . <sup>9</sup> As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$ , let’s |
| fong-spivak-invitation_F00433 | 42   | ■ 0.937 | $\mathbb{R} \rightarrow \mathbb{Z}$               | $\mathbb{R} \rightarrow \mathbb{Z}$               | integer below $z \in \mathbb{R}$ , e.g. $\lceil 3.14 \rceil = 4$ and $\lfloor 3.14 \rfloor = 3$ . <sup>9</sup> As the left adjoint $\mathbb{R} \rightarrow \mathbb{Z}$ , let’s |
| fong-spivak-invitation_F00434 | 42   | ■ 0.994 | $\lceil -/3 \rceil$                               | $\lceil -/3 \rceil$                               | see if $\lceil -/3 \rceil$ works.                                                                                                                                              |

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| fong-spivak-invitation_F00435 | 42   | 0.651 | $(3 \times -)$                                          | $(3 \times -)$                                          | Success! Thus we have a Galois connection between $\lceil -/3 \rceil$ and $(3 \times -)$ .                                  |
| fong-spivak-invitation_F00436 | 42   | 1.000 | $P=Q=\underline{3}$                                     | $P = Q = \underline{3}$                                 | <b>Exercise 1.93.</b> Consider the preorder $P = Q = \underline{3}$ .                                                       |
| fong-spivak-invitation_F00437 | 42   | 1.000 | $f, g$                                                  | $f, g$                                                  | 1. Let $f, g$ be the monotone maps shown below:                                                                             |
| fong-spivak-invitation_F00438 | 42   | 1.000 | $1 \leq p, q \leq 3$                                    | $1 \leq p, q \leq 3$                                    | Is it the case that $f$ is left adjoint to $g$ ? Check that for each $1 \leq p, q \leq 3$ , one has                         |
| fong-spivak-invitation_F00439 | 42   | 1.000 | $f(p) \leq q$                                           | $f(p) \leq q$                                           | $f(p) \leq q$ iff $p \leq g(q)$ .                                                                                           |
| fong-spivak-invitation_F00440 | 42   | 1.000 | $p \leq g(q)$                                           | $p \leq g(q)$                                           | $f(p) \leq q$ iff $p \leq g(q)$ .                                                                                           |
| fong-spivak-invitation_F00441 | 42   | 1.000 | $L: \mathbb{Z} \rightarrow \mathbb{R}$                  | $L: \mathbb{Z} \rightarrow \mathbb{R}$                  | 1. Does $\lceil -/3 \rceil$ have a left adjoint $L: \mathbb{Z} \rightarrow \mathbb{R}$ ?                                    |
| fong-spivak-invitation_F00442 | 42   | 1.000 | $\lfloor 3 \rfloor = \lceil 3 \rceil$                   | $\lfloor 3 \rfloor = 3 = \lceil 3 \rceil$               | mouthful. Anyway, $\lfloor 3 \rfloor = 3 = \lceil 3 \rceil$ .                                                               |
| fong-spivak-invitation_F00443 | 43   | 1.000 | $\text{Prt}(S)$                                         | $\text{Prt}(S)$                                         | Recall from Example 1.47 that we can understand the set $\text{Prt}(S)$ of partitions on a set $S$                          |
| fong-spivak-invitation_F00444 | 43   | 1.000 | $g: S \rightarrow T$                                    | $g: S \rightarrow T$                                    | Suppose we are given any function $g: S \rightarrow T$ . We will show that this function                                    |
| fong-spivak-invitation_F00445 | 43   | 0.882 | $g!: \text{Prt}(S) \rightleftarrows \text{Prt}(T): g^*$ | $g!: \text{Prt}(S) \rightleftarrows \text{Prt}(T): g^*$ | $g$ induces a Galois connection $g!: \text{Prt}(S) \rightleftarrows \text{Prt}(T): g^*$ between the preorder of             |
| fong-spivak-invitation_F00446 | 43   | 0.993 | $g!: \text{Prt}(S) \rightarrow \text{Prt}(T)$           | $g!: \text{Prt}(S) \rightarrow \text{Prt}(T)$           | Start with $g: S \rightarrow T$ ; we first want to understand $g!: \text{Prt}(S) \rightarrow \text{Prt}(T)$ . So start with |
| fong-spivak-invitation_F00447 | 43   | 1.000 | $\sim_S$                                                | $\sim_S$                                                | a partition $\sim_S$ of $S$ . To begin the process of obtaining a partition $\sim_T$ on $T$ , say that                      |
| fong-spivak-invitation_F00448 | 43   | 1.000 | $\sim_T$                                                | $\sim_T$                                                | a partition $\sim_S$ of $S$ . To begin the process of obtaining a partition $\sim_T$ on $T$ , say that                      |
| fong-spivak-invitation_F00449 | 43   | 0.983 | $t_1, t_2 \in T$                                        | $t_1, t_2 \in T$                                        | two elements $t_1, t_2 \in T$ are in the same part, $t_1 \sim_T t_2$ , if there exist $s_1, s_2 \in S$ such                 |
| fong-spivak-invitation_F00450 | 43   | 0.983 | $t_1 \sim_T t_2$                                        | $t_1 \sim_T t_2$                                        | two elements $t_1, t_2 \in T$ are in the same part, $t_1 \sim_T t_2$ , if there exist $s_1, s_2 \in S$ such                 |
| fong-spivak-invitation_F00451 | 43   | 0.997 | $s_1 \sim_S s_2$                                        | $s_1 \sim_S s_2$                                        | that $s_1 \sim_S s_2$ and $g(s_1) = t_1$ and $g(s_2) = t_2$ . However, the result of doing so will                          |
| fong-spivak-invitation_F00452 | 43   | 0.997 | $g(s_1) = t_1$                                          | $g(s_1) = t_1$                                          | that $s_1 \sim_S s_2$ and $g(s_1) = t_1$ and $g(s_2) = t_2$ . However, the result of doing so will                          |
| fong-spivak-invitation_F00453 | 43   | 0.997 | $g(s_2) = t_2$                                          | $g(s_2) = t_2$                                          | that $s_1 \sim_S s_2$ and $g(s_1) = t_1$ and $g(s_2) = t_2$ . However, the result of doing so will                          |
| fong-spivak-invitation_F00454 | 43   | 0.769 | $t_2 \sim_T t_3$                                        | $t_2 \sim_T t_3$                                        | not necessarily be transitive – you may get $t_1 \sim_T t_2$ and $t_2 \sim_T t_3$ without $t_1 \stackrel{?}{\sim}_T t_3$ –  |
| fong-spivak-invitation_F00455 | 43   | 0.769 | $t_1 \stackrel{?}{\sim}_T t_3$                          | $t_1 \stackrel{?}{\sim}_T t_3$                          | not necessarily be transitive – you may get $t_1 \sim_T t_2$ and $t_2 \sim_T t_3$ without $t_1 \stackrel{?}{\sim}_T t_3$ –  |
| fong-spivak-invitation_F00456 | 43   | 1.000 | $g!(\sim_S) := \sim_T$                                  | $g!(\sim_S) := \sim_T$                                  | pieces (take the transitive closure). The result is $g!(\sim_S) := \sim_T$ .                                                |
| fong-spivak-invitation_F00457 | 43   | 0.998 | $g^*$                                                   | $g^*$                                                   | Again starting with $g$ , we want to get the right adjoint $g^*: \text{Prt}(T) \rightarrow \text{Prt}(S)$ . So start        |
| fong-spivak-invitation_F00458 | 43   | 0.998 | $\text{Prt}(T) \rightarrow \text{Prt}(S)$               | $\text{Prt}(T) \rightarrow \text{Prt}(S)$               | Again starting with $g$ , we want to get the right adjoint $g^*: \text{Prt}(T) \rightarrow \text{Prt}(S)$ . So start        |
| fong-spivak-invitation_F00459 | 43   | 1.000 | $g(s_1) \sim_T$                                         | $g(s_1) \sim_T$                                         | with a partition $\sim_T$ of $T$ . Get a partition $\sim_S$ on $S$ by saying that $s_1 \sim_S s_2$ iff $g(s_1) \sim_T$      |
| fong-spivak-invitation_F00460 | 43   | 1.000 | $g(s_2)$                                                | $g(s_2)$                                                | $g(s_2)$ . The result is $g^*(\sim_T) := \sim_S$ .                                                                          |
| fong-spivak-invitation_F00461 | 43   | 1.000 | $g^*(\sim_T) := \sim_S$                                 | $g^*(\sim_T) := \sim_S$                                 | $g(s_2)$ . The result is $g^*(\sim_T) := \sim_S$ .                                                                          |

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| fong-spivak-invitation_F00462 | 43   | ■ 1.000 | $S=\{1,2,3,4\}, T=\{12,3,4\}$                                        | $S = \{1, 2, 3, 4\}, T = \{12, 3, 4\}$                         | <b>Example 1.96.</b> Let $S = \{1, 2, 3, 4\}, T = \{12, 3, 4\}$ , and define $g: S \rightarrow T$ by $g(1) :=$                     |
| fong-spivak-invitation_F00463 | 43   | ■ 1.000 | $g(1) :=$                                                            | $g(1) :=$                                                      | <b>Example 1.96.</b> Let $S = \{1, 2, 3, 4\}, T = \{12, 3, 4\}$ , and define $g: S \rightarrow T$ by $g(1) :=$                     |
| fong-spivak-invitation_F00464 | 43   | ■ 1.000 | $g(2) := 12, g(3) := 3$                                              | $g(2) := 12, g(3) := 3$                                        | $g(2) := 12, g(3) := 3$ , and $g(4) := 4$ . The partition shown left below is translated by $g$ :                                  |
| fong-spivak-invitation_F00465 | 43   | ■ 1.000 | $g(4) := 4$                                                          | $g(4) := 4$                                                    | $g(2) := 12, g(3) := 3$ , and $g(4) := 4$ . The partition shown left below is translated by $g$ :                                  |
| fong-spivak-invitation_F00466 | 44   | ■ 0.872 | $c: S \rightarrowtail P$                                             | $c: S \rightarrow P$                                           | four different ones and, for each one, call it $c: S \rightarrowtail P$ , find $g_!(c)$ , where $S, T$ , and                       |
| fong-spivak-invitation_F00467 | 44   | ■ 0.872 | $g_!{}(c)$                                                           | $g_!(c)$                                                       | four different ones and, for each one, call it $c: S \rightarrowtail P$ , find $g_!(c)$ , where $S, T$ , and                       |
| fong-spivak-invitation_F00468 | 44   | ■ 0.872 | $S, T$                                                               | $S, T$                                                         | four different ones and, for each one, call it $c: S \rightarrowtail P$ , find $g_!(c)$ , where $S, T$ , and                       |
| fong-spivak-invitation_F00469 | 44   | ■ 1.000 | $T =$                                                                | $T =$                                                          | <b>Exercise 1.99.</b> There are five partitions possible on a set with three elements, say $T =$                                   |
| fong-spivak-invitation_F00470 | 44   | ■ 0.746 | $\{12, 3, 4\}$                                                       | $\{12, 3, 4\}$                                                 | $\{12, 3, 4\}$ . Using the same $S$ and $g: S \rightarrow T$ as in Example 1.96, determine the partition                           |
| fong-spivak-invitation_F00471 | 44   | ■ 1.000 | $g^*{}(c)$                                                           | $g^*(c)$                                                       | $g^*(c)$ on $S$ for each of the five partitions $c: T \rightarrowtail P$ .                                                         |
| fong-spivak-invitation_F00472 | 44   | ■ 1.000 | $c: T \rightarrowtail P$                                             | $c: T \rightarrow P$                                           | $g^*(c)$ on $S$ for each of the five partitions $c: T \rightarrowtail P$ .                                                         |
| fong-spivak-invitation_F00473 | 44   | ■ 1.000 | $g^*{}: \operatorname{Prt}(T) \rightarrowtail \operatorname{Prt}(S)$ | $g^*: \operatorname{Prt}(T) \rightarrow \operatorname{Prt}(S)$ | really is left adjoint to $g^*: \operatorname{Prt}(T) \rightarrow \operatorname{Prt}(S)$ would take too much time for this sketch. |
| fong-spivak-invitation_F00474 | 44   | ■ 1.000 | $d: T \rightarrowtail P'$                                            | $d: T \rightarrow P'$                                          | 2. Choose any coarser partition $d: T \rightarrowtail P'$ , i.e. where $g_!(c) \leq d$ .                                           |
| fong-spivak-invitation_F00475 | 44   | ■ 1.000 | $g_!{}(c) \leq d$                                                    | $g_!(c) \leq d$                                                | 2. Choose any coarser partition $d: T \rightarrowtail P'$ , i.e. where $g_!(c) \leq d$ .                                           |
| fong-spivak-invitation_F00476 | 44   | ■ 0.999 | $e: T \rightarrowtail Q$                                             | $e: T \rightarrow Q$                                           | 3. Choose any non-coarser partition $e: T \rightarrowtail Q$ , i.e. where $g_!(c) \not\leq e$ . (If you can't do                   |
| fong-spivak-invitation_F00477 | 44   | ■ 0.999 | $g_!{}(c) \not\leq e$                                                | $g_!(c) \not\leq e$                                            | 3. Choose any non-coarser partition $e: T \rightarrowtail Q$ , i.e. where $g_!(c) \not\leq e$ . (If you can't do                   |
| fong-spivak-invitation_F00478 | 44   | ■ 1.000 | $g^*{}(d)$                                                           | $g^*(d)$                                                       | 4. Find $g^*(d)$ and $g^*(e)$ .                                                                                                    |
| fong-spivak-invitation_F00479 | 44   | ■ 1.000 | $g^*{}(e)$                                                           | $g^*(e)$                                                       | 4. Find $g^*(d)$ and $g^*(e)$ .                                                                                                    |
| fong-spivak-invitation_F00480 | 44   | ■ 1.000 | $c \leq g^*{}(d)$                                                    | $c \leq g^*(d)$                                                | we should have $c \leq g^*(d)$ and $c \not\leq g^*(e)$ . Show that this is true.                                                   |
| fong-spivak-invitation_F00481 | 44   | ■ 1.000 | $c \not\leq g^*{}(e)$                                                | $c \not\leq g^*(e)$                                            | we should have $c \leq g^*(d)$ and $c \not\leq g^*(e)$ . Show that this is true.                                                   |
| fong-spivak-invitation_F00482 | 45   | ■ 1.000 | $q := f(p)$                                                          | $q := f(p)$                                                    | <i>Proof.</i> Suppose $f$ is left adjoint to $g$ . Take any $p \in P$ , and let $q := f(p)$ . By reflexiv-                         |
| fong-spivak-invitation_F00483 | 45   | ■ 1.000 | $p \leq g(f(p))$                                                     | $p \leq g(f(p))$                                               | but this means $p \leq g(f(p))$ . The proof that $f(g(q)) \leq q$ is similar.                                                      |
| fong-spivak-invitation_F00484 | 45   | ■ 1.000 | $f(g(q)) \leq q$                                                     | $f(g(q)) \leq q$                                               | but this means $p \leq g(f(p))$ . The proof that $f(g(q)) \leq q$ is similar.                                                      |
| fong-spivak-invitation_F00485 | 45   | ■ 1.000 | $g(f(p)) \leq$                                                       | $g(f(p)) \leq$                                                 | $f(p) \leq q$ iff $p \leq g(q)$ . Suppose $f(p) \leq q$ ; then since $g$ is monotonic, $g(f(p)) \leq$                              |
| fong-spivak-invitation_F00486 | 45   | ■ 1.000 | $g(q)$                                                               | $g(q)$                                                         | $g(q)$ , but $p \leq g(f(p))$ so $p \leq g(q)$ . The proof that $p \leq g(q)$ implies $f(p) \leq q$ is                             |
| fong-spivak-invitation_F00487 | 45   | ■ 1.000 | $g^{\prime}{}(q)$                                                    | $g'$                                                           | That means, for any other right adjoint $g'$ , we have $g(q) \cong g'(q)$ for all $q \in Q$ .                                      |
| fong-spivak-invitation_F00488 | 45   | ■ 1.000 | $g(q) \cong g^{\prime}{}(q)$                                         | $g(q) \cong g'(q)$                                             | That means, for any other right adjoint $g'$ , we have $g(q) \cong g'(q)$ for all $q \in Q$ .                                      |

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| fong-spivak-invitation_F00489 | 45   | ■ 1.000 | $A \subseteq Q$                                                                                                                   | $A \subseteq Q$                                                                                                                   | $g: Q \rightarrow P$ . Suppose that $A \subseteq Q$ is any subset, and let $g(A) := \{g(a) \mid a \in A\}$ be                                                                                                                                                                                    |
| fong-spivak-invitation_F00490 | 45   | ■ 1.000 | $g(A) := \{g(a) \mid a \in A\}$                                                                                                   | $g(A) := \{g(a) \mid a \in A\}$                                                                                                   | $g: Q \rightarrow P$ . Suppose that $A \subseteq Q$ is any subset, and let $g(A) := \{g(a) \mid a \in A\}$ be                                                                                                                                                                                    |
| fong-spivak-invitation_F00491 | 45   | ■ 1.000 | $\bigwedge A \in Q$                                                                                                               | $\bigwedge A \in Q$                                                                                                               | its image. Then if $A$ has a meet $\bigwedge A \in Q$ then $g(A)$ has a meet $\bigwedge g(A)$ in $P$ , and we                                                                                                                                                                                    |
| fong-spivak-invitation_F00492 | 45   | ■ 1.000 | $g(A)$                                                                                                                            | $g(A)$                                                                                                                            | its image. Then if $A$ has a meet $\bigwedge A \in Q$ then $g(A)$ has a meet $\bigwedge g(A)$ in $P$ , and we                                                                                                                                                                                    |
| fong-spivak-invitation_F00493 | 45   | ■ 1.000 | $\bigwedge g(A)$                                                                                                                  | $\bigwedge g(A)$                                                                                                                  | its image. Then if $A$ has a meet $\bigwedge A \in Q$ then $g(A)$ has a meet $\bigwedge g(A)$ in $P$ , and we                                                                                                                                                                                    |
| fong-spivak-invitation_F00494 | 45   | ■ 1.000 | $\bigvee A \in P$                                                                                                                 | $\bigvee A \in P$                                                                                                                 | is any subset that has a join $\bigvee A \in P$ , then $f(A)$ has a join $\bigvee f(A)$ in $Q$ , and we                                                                                                                                                                                          |
| fong-spivak-invitation_F00495 | 45   | ■ 1.000 | $f(A)$                                                                                                                            | $f(A)$                                                                                                                            | is any subset that has a join $\bigvee A \in P$ , then $f(A)$ has a join $\bigvee f(A)$ in $Q$ , and we                                                                                                                                                                                          |
| fong-spivak-invitation_F00496 | 45   | ■ 1.000 | $\bigvee f(A)$                                                                                                                    | $\bigvee f(A)$                                                                                                                    | is any subset that has a join $\bigvee A \in P$ , then $f(A)$ has a join $\bigvee f(A)$ in $Q$ , and we                                                                                                                                                                                          |
| fong-spivak-invitation_F00497 | 45   | ■ 1.000 | $m := \bigwedge A$                                                                                                                | $m := \bigwedge A$                                                                                                                | to $f$ . Let $A \subseteq Q$ be any subset and let $m := \bigwedge A$ be its meet. Then since $g$ is monotone                                                                                                                                                                                    |
| fong-spivak-invitation_F00498 | 45   | ■ 1.000 | $g(m) \leq g(a)$                                                                                                                  | $g(m) \leq g(a)$                                                                                                                  | $g(m) \leq g(a)$ for all $a \in A$ , so $g(m)$ is a lower bound for the set $g(A)$ . We will be done                                                                                                                                                                                             |
| fong-spivak-invitation_F00499 | 45   | ■ 1.000 | $g(m)$                                                                                                                            | $g(m)$                                                                                                                            | $g(m) \leq g(a)$ for all $a \in A$ , so $g(m)$ is a lower bound for the set $g(A)$ . We will be done                                                                                                                                                                                             |
| fong-spivak-invitation_F00500 | 45   | ■ 1.000 | $b \leq g(a)$                                                                                                                     | $b \leq g(a)$                                                                                                                     | have $b \leq g(a)$ and we want to show that $b \leq g(m)$ . Then by definition of $g$ being a                                                                                                                                                                                                    |
| fong-spivak-invitation_F00501 | 45   | ■ 1.000 | $b \leq g(m)$                                                                                                                     | $b \leq g(m)$                                                                                                                     | have $b \leq g(a)$ and we want to show that $b \leq g(m)$ . Then by definition of $g$ being a                                                                                                                                                                                                    |
| fong-spivak-invitation_F00502 | 45   | ■ 0.999 | $f(b) \leq a$                                                                                                                     | $f(b) \leq a$                                                                                                                     | right adjoint (Definition 1.90), we also have $f(b) \leq a$ . This means that $f(b)$ is a lower                                                                                                                                                                                                  |
| fong-spivak-invitation_F00503 | 45   | ■ 0.999 | $f(b)$                                                                                                                            | $f(b)$                                                                                                                            | right adjoint (Definition 1.90), we also have $f(b) \leq a$ . This means that $f(b)$ is a lower                                                                                                                                                                                                  |
| fong-spivak-invitation_F00504 | 45   | ■ 1.000 | $f(b) \leq m$                                                                                                                     | $f(b) \leq m$                                                                                                                     | bound for $A$ in $Q$ . Since the meet $m$ is the greatest lower bound, we have $f(b) \leq m$ .                                                                                                                                                                                                   |
| fong-spivak-invitation_F00505 | 46   | ■ 1.000 | $f(3.9) := 4$                                                                                                                     | $f(3.9) := 4$                                                                                                                     | far as possible but with $f(3.9) := 4$ . Both $f$ and $g$ are monotonic, and one can check                                                                                                                                                                                                       |
| fong-spivak-invitation_F00506 | 46   | ■ 1.000 | $1 \vee 2 = 4$                                                                                                                    | $1 \vee 2 = 4$                                                                                                                    | $1 \vee 2 = 4$ holds in $Q$ , whereas $g(1) \vee g(2) = 1 \vee 2 = 3.9 \neq 4 = g(4)$ in $P$ .                                                                                                                                                                                                   |
| fong-spivak-invitation_F00507 | 46   | ■ 1.000 | $g(1) \vee g(2) = 1 \vee 2 = 3.9 \neq 4 = g(4)$                                                                                   | $g(1) \vee g(2) = 1 \vee 2 = 3.9 \neq 4 = g(4)$                                                                                   | $1 \vee 2 = 4$ holds in $Q$ , whereas $g(1) \vee g(2) = 1 \vee 2 = 3.9 \neq 4 = g(4)$ in $P$ .                                                                                                                                                                                                   |
| fong-spivak-invitation_F00508 | 47   | ■ 1.000 | $\left\{q^{\prime} \in Q \mid p^{\prime} \leq g\left(q^{\prime}\right)\right\} \subseteq \left\{q \in Q \mid p \leq g(q)\right\}$ | $\left\{q^{\prime} \in Q \mid p^{\prime} \leq g\left(q^{\prime}\right)\right\} \subseteq \left\{q \in Q \mid p \leq g(q)\right\}$ | need to show it is monotone. So suppose that $p \leq p^{\prime}$ . Then $\left\{q^{\prime} \in Q \mid p^{\prime} \leq g\left(q^{\prime}\right)\right\} \subseteq \left\{q \in Q \mid p \leq g(q)\right\}$ . By Proposition 1.86, this implies $f(p) \leq f\left(p^{\prime}\right)$ . Thus $f$ is |
| fong-spivak-invitation_F00509 | 47   | ■ 1.000 | $\left\{q \in Q \mid p \leq g(q)\right\}$                                                                                         | $\left\{q \in Q \mid p \leq g(q)\right\}$                                                                                         | $\left\{q \in Q \mid p \leq g(q)\right\}$ . By Proposition 1.86, this implies $f(p) \leq f\left(p^{\prime}\right)$ . Thus $f$ is                                                                                                                                                                 |
| fong-spivak-invitation_F00510 | 47   | ■ 1.000 | $f(p) \leq f\left(p^{\prime}\right)$                                                                                              | $f(p) \leq f\left(p^{\prime}\right)$                                                                                              | $\left\{q \in Q \mid p \leq g(q)\right\}$ . By Proposition 1.86, this implies $f(p) \leq f\left(p^{\prime}\right)$ . Thus $f$ is                                                                                                                                                                 |
| fong-spivak-invitation_F00511 | 47   | ■ 1.000 | $p_{\{0\}} \leq g\left(f\left(p_{\{0\}}\right)\right)$                                                                            | $p_0 \leq g\left(f\left(p_0\right)\right)$                                                                                        | By Proposition 1.104, it suffices to show that $p_0 \leq g\left(f\left(p_0\right)\right)$ and that $f\left(g\left(q_0\right)\right) \leq q_0$                                                                                                                                                    |
| fong-spivak-invitation_F00512 | 47   | ■ 1.000 | $f\left(g\left(q_{\{0\}}\right)\right) \leq q_{\{0\}}$                                                                            | $f\left(g\left(q_0\right)\right) \leq q_0$                                                                                        | By Proposition 1.104, it suffices to show that $p_0 \leq g\left(f\left(p_0\right)\right)$ and that $f\left(g\left(q_0\right)\right) \leq q_0$                                                                                                                                                    |
| fong-spivak-invitation_F00513 | 47   | ■ 1.000 | $p_{\{0\}} \in P$                                                                                                                 | $p_0 \in P$                                                                                                                       | for all $p_0 \in P$ and $q_0 \in Q$ . For the first, we have                                                                                                                                                                                                                                     |
| fong-spivak-invitation_F00514 | 47   | ■ 1.000 | $q_{\{0\}} \in Q$                                                                                                                 | $q_0 \in Q$                                                                                                                       | for all $p_0 \in P$ and $q_0 \in Q$ . For the first, we have                                                                                                                                                                                                                                     |
| fong-spivak-invitation_F00515 | 47   | ■ 1.000 | $p_{\{0\}}$                                                                                                                       | $p_0$                                                                                                                             | where the first inequality follows from the fact that if $p_0$ is below every element of a set,                                                                                                                                                                                                  |

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| fong-spivak-invitation_F00516 | 47   | ■ 1.000 | $\left\{q_{\{0\}}\right\} \subseteq \left\{q \in Q \mid g\left(q_{\{0\}}\right) \leq \left(q_{\{0\}}\right)\right\}.$ | $\left\{q_0\right\} \subseteq\left\{q \in Q \mid g\left(q_0\right) \leq\right.$ | where the first inequality follows from Proposition 1.86 since $\boxed{\{q_0\} \subseteq \{q \in Q \mid g(q_0) \preceq$                   |
| fong-spivak-invitation_F00517 | 47   | ■ 0.990 | $g(q)\backslash\}$                                                                                                    | $g(q)\}$                                                                        | $g(q)\}$ , and the fact that $\bigwedge\left\{q_0\right\}=q_0$ .                                                                          |
| fong-spivak-invitation_F00518 | 47   | ■ 0.990 | $\wedge\left\{q_{\{0\}}\right\}=q_{\{0\}}$                                                                            | $\wedge\left\{q_0\right\}=q_0$                                                  | $g(q)\}$ , and the fact that $\bigwedge\left\{q_0\right\}=q_0$ .                                                                          |
| fong-spivak-invitation_F00519 | 47   | ■ 0.999 | $f^{\ast}: \mathrm{P}(B) \rightarrow \mathrm{P}(A)$                                                                   | $f^{\ast}: \mathrm{P}(B) \rightarrow \mathrm{P}(A)$                             | Then we have the monotone map $f^{\ast}: \mathrm{P}(B) \rightarrow \mathrm{P}(A)$ that category theorists call “pull-                     |
| fong-spivak-invitation_F00520 | 47   | ■ 1.000 | $B^{\prime} \subseteq B$                                                                                              | $B^{\prime} \subseteq B$                                                        | back along $f$ .” This map takes a subset $B^{\prime} \subseteq B$ to its preimage $f^{-1}\left(B^{\prime}\right) \subseteq A$ ; that is, |
| fong-spivak-invitation_F00521 | 47   | ■ 1.000 | $f^{-1}\left(B^{\prime}\right) \subseteq A$                                                                           | $f^{-1}\left(B^{\prime}\right) \subseteq A$                                     | back along $f$ .” This map takes a subset $B^{\prime} \subseteq B$ to its preimage $f^{-1}\left(B^{\prime}\right) \subseteq A$ ; that is, |
| fong-spivak-invitation_F00522 | 47   | ■ 1.000 | $B^{\prime}$                                                                                                          | $B^{\prime}$                                                                    | it takes a collection $B^{\prime}$ of buckets, and tells you all the apples that they contain in total.                                   |
| fong-spivak-invitation_F00523 | 47   | ■ 1.000 | $f_!(A)$                                                                                                              | $f_!(A)$                                                                        | The left adjoint $f_!(A)$ is given by the direct image: it maps a subset $A^{\prime} \subseteq A$ to                                      |
| fong-spivak-invitation_F00524 | 47   | ■ 1.000 | $A^{\prime} \subseteq A$                                                                                              | $A^{\prime} \subseteq A$                                                        | The left adjoint $f_!(A)$ is given by the direct image: it maps a subset $A^{\prime} \subseteq A$ to                                      |
| fong-spivak-invitation_F00525 | 47   | ■ 1.000 | $A^{\prime}$                                                                                                          | $A^{\prime}$                                                                    | This map takes a set $A^{\prime}$ of apples, and tells you all the buckets that contain at least one                                      |
| fong-spivak-invitation_F00526 | 47   | ■ 1.000 | $f_{\ast}$                                                                                                            | $f_{\ast}$                                                                      | The right adjoint $f_{\ast}$ maps a subset $A^{\prime} \subseteq A$ to                                                                    |
| fong-spivak-invitation_F00527 | 47   | ■ 0.923 | $f^{\ast}, f_!,$                                                                                                      | $f^{\ast}, f_!$                                                                 | whose only apples are from $A^{\prime}$ . But we did not invent these mappings $f^{\ast}, f_!$ , and                                      |
| fong-spivak-invitation_F00528 | 48   | ■ 1.000 | $B_{\{1\}}, B_{\{2\}} \subseteq Y$                                                                                    | $B_1, B_2 \subseteq Y$                                                          | 1. Choose two different subsets $B_1, B_2 \subseteq Y$ and find $f^{\ast}\left(B_1\right)$ and $f^{\ast}\left(B_2\right)$ .               |
| fong-spivak-invitation_F00529 | 48   | ■ 1.000 | $f^{\ast}\left(B_{\{1\}}\right)$                                                                                      | $f^{\ast}\left(B_1\right)$                                                      | 1. Choose two different subsets $B_1, B_2 \subseteq Y$ and find $f^{\ast}\left(B_1\right)$ and $f^{\ast}\left(B_2\right)$ .               |
| fong-spivak-invitation_F00530 | 48   | ■ 1.000 | $f^{\ast}\left(B_{\{2\}}\right)$                                                                                      | $f^{\ast}\left(B_2\right)$                                                      | 1. Choose two different subsets $B_1, B_2 \subseteq Y$ and find $f^{\ast}\left(B_1\right)$ and $f^{\ast}\left(B_2\right)$ .               |
| fong-spivak-invitation_F00531 | 48   | ■ 1.000 | $A_{\{1\}}, A_{\{2\}} \subseteq X$                                                                                    | $A_1, A_2 \subseteq X$                                                          | 2. Choose two different subsets $A_1, A_2 \subseteq X$ and find $f_!\left(A_1\right)$ and $f_!\left(A_2\right)$ .                         |
| fong-spivak-invitation_F00532 | 48   | ■ 1.000 | $f_!\left(A_{\{1\}}\right)$                                                                                           | $f_!\left(A_1\right)$                                                           | 2. Choose two different subsets $A_1, A_2 \subseteq X$ and find $f_!\left(A_1\right)$ and $f_!\left(A_2\right)$ .                         |
| fong-spivak-invitation_F00533 | 48   | ■ 1.000 | $f_!\left(A_{\{2\}}\right)$                                                                                           | $f_!\left(A_2\right)$                                                           | 2. Choose two different subsets $A_1, A_2 \subseteq X$ and find $f_!\left(A_1\right)$ and $f_!\left(A_2\right)$ .                         |
| fong-spivak-invitation_F00534 | 48   | ■ 1.000 | $f_{\ast}\left(A_{\{1\}}\right)$                                                                                      | $f_{\ast}\left(A_1\right)$                                                      | 3. With the same $A_1, A_2 \subseteq X$ , find $f_{\ast}\left(A_1\right)$ and $f_{\ast}\left(A_2\right)$ .                                |
| fong-spivak-invitation_F00535 | 48   | ■ 1.000 | $f_{\ast}\left(A_{\{2\}}\right)$                                                                                      | $f_{\ast}\left(A_2\right)$                                                      | 3. With the same $A_1, A_2 \subseteq X$ , find $f_{\ast}\left(A_1\right)$ and $f_{\ast}\left(A_2\right)$ .                                |
| fong-spivak-invitation_F00536 | 48   | ■ 1.000 | $f ; g: P \rightarrow P$                                                                                              | $f ; g: P \rightarrow P$                                                        | $f$ and $g$ to arrive at a monotone map $f \circ g: P \rightarrow P$ from preorder $P$ to itself. This                                    |
| fong-spivak-invitation_F00537 | 48   | ■ 0.999 | $p \leq(f \circ g)(p)$                                                                                                | $p \leq(f \circ g)(p)$                                                          | monotone map has the property that $p \leq(f \circ g)(p)$ , and that $(f \circ g \circ f \circ g)(p) \cong(f \circ g)(p)$                 |
| fong-spivak-invitation_F00538 | 48   | ■ 0.999 | $(f \circ g \circ f \circ g)(p) \cong(f \circ g)(p)$                                                                  | $(f \circ g \circ f \circ g)(p) \cong(f \circ g)(p)$                            | monotone map has the property that $p \leq(f \circ g)(p)$ , and that $(f \circ g \circ f \circ g)(p) \cong(f \circ g)(p)$                 |
| fong-spivak-invitation_F00539 | 48   | ■ 1.000 | $(f ; g ; f ; g)(p) \cong(f ; g)(p)$                                                                                  | $(f ; g ; f ; g)(p) \cong(f ; g)(p)$                                            | 2. $(f \circ g \circ f \circ g)(p) \cong(f \circ g)(p)$ . To prove this, show inequalities in both directions, $\leq$                     |
| fong-spivak-invitation_F00540 | 48   | ■ 1.000 | $\geq$                                                                                                                | $\geq$                                                                          | and $\geq$ .                                                                                                                              |
| fong-spivak-invitation_F00541 | 48   | ■ 1.000 | $j: P \rightarrow P$                                                                                                  | $j: P \rightarrow P$                                                            | <b>Definition 1.112.</b> A closure operator $j: P \rightarrow P$ on a preorder $P$ is a monotone map                                      |
| fong-spivak-invitation_F00542 | 48   | ■ 0.999 | $p \leq j(p)$                                                                                                         | $p \leq j(p)$                                                                   | (a) $p \leq j(p)$ ,                                                                                                                       |

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| fong-spivak-invitation_F00543 | 48   | ■ 1.000 | $j(j(p)) \backslash \text{cong } j(p)$                                                                                                        | $j(j(p)) \cong j(p)$                                  | (b) $j(j(p)) \cong j(p)$ .                                                                                                                                     |
| fong-spivak-invitation_F00544 | 48   | ■ 1.000 | $7+2+3$                                                                                                                                       | $7 + 2 + 3$                                           | expressions. For example, a computer can rewrite the expression $7 + 2 + 3$ as the expres-                                                                     |
| fong-spivak-invitation_F00545 | 48   | ■ 0.936 | $j: \text{exp} \backslash \text{rightarrow exp}$                                                                                              | $j : \text{exp} \rightarrow \text{exp}$               | and reducing it to another expression. So it is a map $j: \text{exp} \rightarrow \text{exp}$ . It furthermore                                                  |
| fong-spivak-invitation_F00546 | 48   | ■ 1.000 | $j(x)$                                                                                                                                        | $j(x)$                                                | means that if an expression $x$ can be rewritten into expression $y$ , then the reduction $j(x)$                                                               |
| fong-spivak-invitation_F00547 | 48   | ■ 1.000 | $j(y)$                                                                                                                                        | $j(y)$                                                | can be rewritten into $j(y)$ . Moreover, the requirement $x \leq j(x)$ implies that $j$ can only                                                               |
| fong-spivak-invitation_F00548 | 48   | ■ 1.000 | $x \backslash \text{leq } j(x)$                                                                                                               | $x \leq j(x)$                                         | can be rewritten into $j(y)$ . Moreover, the requirement $x \leq j(x)$ implies that $j$ can only                                                               |
| fong-spivak-invitation_F00549 | 48   | ■ 1.000 | $j$                                                                                                                                           | $j$                                                   | can be rewritten into $j(y)$ . Moreover, the requirement $x \leq j(x)$ implies that $j$ can only                                                               |
| fong-spivak-invitation_F00550 | 48   | ■ 1.000 | $j(j(x))=j(x)$                                                                                                                                | $j(j(x)) = j(x)$                                      | $j(j(x)) = j(x)$ implies that if you try to reduce an expression that has already been                                                                         |
| fong-spivak-invitation_F00551 | 48   | ■ 0.906 | $(g \backslash \text{circ } f)(q) \backslash \text{leq } q$                                                                                   | $(g \circ f)(q) \leq q$                               | <sup>10</sup> The other composite $g \circ f$ satisfies the dual properties: $(g \circ f)(q) \leq q$ and $(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$ |
| fong-spivak-invitation_F00552 | 48   | ■ 0.906 | $(g \backslash \text{circ } f \backslash \text{circ } g \backslash \text{circ } f)(q) \backslash \text{cong}(g \backslash \text{circ } f)(q)$ | $(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$ | <sup>10</sup> The other composite $g \circ f$ satisfies the dual properties: $(g \circ f)(q) \leq q$ and $(g \circ f \circ g \circ f)(q) \cong (g \circ f)(q)$ |
| fong-spivak-invitation_F00553 | 49   | ■ 0.673 | $\_{{j}}$                                                                                                                                     | $j$                                                   | $\text{fix}_j$ to have elements the fixed points of $j$ ; that is,                                                                                             |
| fong-spivak-invitation_F00554 | 49   | ■ 0.506 | $\mathrm{fix}\_{{j}}$                                                                                                                         | $\text{fix}_j$                                        | This is a subset of $P$ and inherits an order as a result; hence $\text{fix}_j$ is a sub-preorder of $P$ .                                                     |
| fong-spivak-invitation_F00555 | 49   | ■ 1.000 | $j(p)$                                                                                                                                        | $j(p)$                                                | Note that $j(p)$ is a fixed point for all $p \in P$ , since $j(j(p)) \cong j(p)$ .                                                                             |
| fong-spivak-invitation_F00556 | 49   | ■ 1.000 | $j: P \backslash \text{rightarrow } \mathrm{fix}\_{{j}}$                                                                                      | $j : P \rightarrow \text{fix}_j$                      | We define an adjunction with left adjoint $j: P \rightarrow \text{fix}_j$ sending $p$ to $j(p)$ , and right                                                    |
| fong-spivak-invitation_F00557 | 49   | ■ 0.888 | $g:$                                                                                                                                          | $g:$                                                  | adjoint $g: \text{fix}_j \rightarrow P$ simply the inclusion of the sub-preorder. To see it’s really an                                                        |
| fong-spivak-invitation_F00558 | 49   | ■ 0.888 | $\_{{j}} \backslash \text{rightarrow } P$                                                                                                     | $j \rightarrow P$                                     | adjoint $g: \text{fix}_j \rightarrow P$ simply the inclusion of the sub-preorder. To see it’s really an                                                        |
| fong-spivak-invitation_F00559 | 49   | ■ 0.979 | $q \in \operatorname{fix}\_{{j}}$                                                                                                             | $q \in \text{fix}_j$                                  | adjunction, we need to see that for any $p \in P$ and $q \in \text{fix}_j$ , we have $j(p) \leq q$                                                             |
| fong-spivak-invitation_F00560 | 49   | ■ 0.979 | $j(p) \backslash \text{leq } q$                                                                                                               | $j(p) \leq q$                                         | adjunction, we need to see that for any $p \in P$ and $q \in \text{fix}_j$ , we have $j(p) \leq q$                                                             |
| fong-spivak-invitation_F00561 | 49   | ■ 1.000 | $j(p) \backslash \text{leq } j(q) \backslash \text{cong } q$                                                                                  | $j(p) \leq j(q) \cong q$                              | $j(p) \leq j(q) \cong q$ .                                                                                                                                     |
| fong-spivak-invitation_F00562 | 49   | ■ 1.000 | $p \backslash \text{rightarrow } q$                                                                                                           | $p \Rightarrow q$                                     | $p \Rightarrow q$ . Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$ , we also have $p \Rightarrow r$ , this                 |
| fong-spivak-invitation_F00563 | 49   | ■ 1.000 | $p \backslash \text{rightarrow } p$                                                                                                           | $p \Rightarrow p$                                     | $p \Rightarrow q$ . Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$ , we also have $p \Rightarrow r$ , this                 |
| fong-spivak-invitation_F00564 | 49   | ■ 1.000 | $q \backslash \text{rightarrow } r$                                                                                                           | $q \Rightarrow r$                                     | $p \Rightarrow q$ . Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$ , we also have $p \Rightarrow r$ , this                 |
| fong-spivak-invitation_F00565 | 49   | ■ 1.000 | $p \backslash \text{rightarrow } r$                                                                                                           | $p \Rightarrow r$                                     | $p \Rightarrow q$ . Since $p \Rightarrow p$ and since whenever $p \Rightarrow q$ and $q \Rightarrow r$ , we also have $p \Rightarrow r$ , this                 |
| fong-spivak-invitation_F00566 | 49   | ■ 1.000 | $p \backslash \text{rightarrow } j(p)$                                                                                                        | $p \Rightarrow j(p)$                                  | propositions to propositions, for which $p \Rightarrow j(p)$ and $j(j(p)) = j(p)$ . An example                                                                 |
| fong-spivak-invitation_F00567 | 49   | ■ 1.000 | $j(j(p))=j(p)$                                                                                                                                | $j(j(p)) = j(p)$                                      | propositions to propositions, for which $p \Rightarrow j(p)$ and $j(j(p)) = j(p)$ . An example                                                                 |
| fong-spivak-invitation_F00568 | 49   | ■ 1.000 | $B \backslash \text{rightarrow } p$                                                                                                           | $B \Rightarrow p$                                     | “assuming Bob is in San Diego, $p$ ” means $B \Rightarrow p$ . Let’s see why $B \Rightarrow -$ is a closure                                                    |
| fong-spivak-invitation_F00569 | 49   | ■ 1.000 | $B \backslash \text{rightarrow -}$                                                                                                            | $B \Rightarrow -$                                     | “assuming Bob is in San Diego, $p$ ” means $B \Rightarrow p$ . Let’s see why $B \Rightarrow -$ is a closure                                                    |

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| fong-spivak-invitation_F00570 | 50   | ■ 0.951 | $\mathbf{Rel}(S)$                                                  | $\mathbf{Rel}(S)$                                                  | Given any set $S$ , there is a set $\mathbf{Rel}(S)$ of binary relations on $S$ . An element $R \in \mathbf{Rel}(S)$         |
| fong-spivak-invitation_F00571 | 50   | ■ 0.951 | $R \in \mathbf{Rel}(S)$                                            | $R \in \mathbf{Rel}(S)$                                            | Given any set $S$ , there is a set $\mathbf{Rel}(S)$ of binary relations on $S$ . An element $R \in \mathbf{Rel}(S)$         |
| fong-spivak-invitation_F00572 | 50   | ■ 0.958 | $R \subseteq S \times S$                                           | $R \subseteq S \times S$                                           | is formally a subset $R \subseteq S \times S$ . The set $\mathbf{Rel}(S)$ can be given an order via the subset               |
| fong-spivak-invitation_F00573 | 50   | ■ 0.958 | $\mathbf{Rel}(S)$                                                  | $\mathbf{Rel}(S)$                                                  | is formally a subset $R \subseteq S \times S$ . The set $\mathbf{Rel}(S)$ can be given an order via the subset               |
| fong-spivak-invitation_F00574 | 50   | ■ 1.000 | $R \subseteq R'$                                                   | $R \subseteq R'$                                                   | relation, $R \subseteq R'$ , i.e. if whenever $R(s_1, s_2)$ holds then so does $R'(s_1, s_2)$ .                              |
| fong-spivak-invitation_F00575 | 50   | ■ 1.000 | $R(s_1, s_2)$                                                      | $R(s_1, s_2)$                                                      | relation, $R \subseteq R'$ , i.e. if whenever $R(s_1, s_2)$ holds then so does $R'(s_1, s_2)$ .                              |
| fong-spivak-invitation_F00576 | 50   | ■ 1.000 | $R'(s_1, s_2)$                                                     | $R'(s_1, s_2)$                                                     | relation, $R \subseteq R'$ , i.e. if whenever $R(s_1, s_2)$ holds then so does $R'(s_1, s_2)$ .                              |
| fong-spivak-invitation_F00577 | 50   | ■ 0.999 | $\mathbf{Rel}(\{1\})$                                              | $\mathbf{Rel}(\{1\})$                                              | For example, the Hasse diagram for $\mathbf{Rel}(\{1\})$ is:                                                                 |
| fong-spivak-invitation_F00578 | 50   | ■ 0.948 | $\mathbf{Rel}(\{1, 2\})$                                           | $\mathbf{Rel}(\{1, 2\})$                                           | <b>Exercise 1.116.</b> Draw the Hasse diagram for the preorder $\mathbf{Rel}(\{1, 2\})$ of all binary                        |
| fong-spivak-invitation_F00579 | 50   | ■ 1.000 | $\mathbf{Pos}(S)$                                                  | $\mathbf{Pos}(S)$                                                  | For any set $S$ , there is also a set $\mathbf{Pos}(S)$ , consisting of all the preorder relations on $S$ .                  |
| fong-spivak-invitation_F00580 | 50   | ■ 0.936 | $\sqsubseteq$                                                      | $\sqsubseteq$                                                      | In fact there is a preorder structure $\sqsubseteq$ on $\mathbf{Pos}(S)$ , again given by inclusion: $\leq$ is below $\leq'$ |
| fong-spivak-invitation_F00581 | 50   | ■ 0.936 | $\leq'$                                                            | $\leq'$                                                            | In fact there is a preorder structure $\sqsubseteq$ on $\mathbf{Pos}(S)$ , again given by inclusion: $\leq$ is below $\leq'$ |
| fong-spivak-invitation_F00582 | 50   | ■ 0.989 | $\leq \sqsubseteq \leq'$                                           | $\leq \sqsubseteq \leq'$                                           | (we'll write $\leq \sqsubseteq \leq'$ ) if $a \leq b$ implies $a \leq' b$ for every $a, b \in S$ . A preorder of preorder    |
| fong-spivak-invitation_F00583 | 50   | ■ 0.989 | $a \leq' b$                                                        | $a \leq' b$                                                        | (we'll write $\leq \sqsubseteq \leq'$ ) if $a \leq b$ implies $a \leq' b$ for every $a, b \in S$ . A preorder of preorder    |
| fong-spivak-invitation_F00584 | 50   | ■ 0.989 | $a, b \in S$                                                       | $a, b \in S$                                                       | (we'll write $\leq \sqsubseteq \leq'$ ) if $a \leq b$ implies $a \leq' b$ for every $a, b \in S$ . A preorder of preorder    |
| fong-spivak-invitation_F00585 | 50   | ■ 1.000 | $\mathbf{Pos}(S) \rightarrow \mathbf{Rel}(S)$                      | $\mathbf{Pos}(S) \rightarrow \mathbf{Rel}(S)$                      | $\mathbf{Pos}(S) \rightarrow \mathbf{Rel}(S)$ . This is the right adjoint of a Galois connection. Its left adjoint is        |
| fong-spivak-invitation_F00586 | 50   | ■ 0.995 | $\mathrm{Cl}: \mathbf{Rel}(S) \rightarrow \mathbf{Pos}(S)$         | $\mathrm{Cl}: \mathbf{Rel}(S) \rightarrow \mathbf{Pos}(S)$         | a monotone map $\mathrm{Cl}: \mathbf{Rel}(S) \rightarrow \mathbf{Pos}(S)$ given by taking any relation $R$ , writing it in   |
| fong-spivak-invitation_F00587 | 50   | ■ 1.000 | $s \leq s$                                                         | $s \leq s$                                                         | infix notation using $\leq$ , and taking the reflexive and transitive closure, i.e. adding $s \leq s$                        |
| fong-spivak-invitation_F00588 | 50   | ■ 1.000 | $s \leq u$                                                         | $s \leq u$                                                         | for every $s$ and adding $s \leq u$ whenever $s \leq t$ and $t \leq u$ .                                                     |
| fong-spivak-invitation_F00589 | 50   | ■ 1.000 | $s \leq t$                                                         | $s \leq t$                                                         | for every $s$ and adding $s \leq u$ whenever $s \leq t$ and $t \leq u$ .                                                     |
| fong-spivak-invitation_F00590 | 50   | ■ 1.000 | $t \leq u$                                                         | $t \leq u$                                                         | for every $s$ and adding $s \leq u$ whenever $s \leq t$ and $t \leq u$ .                                                     |
| fong-spivak-invitation_F00591 | 50   | ■ 1.000 | $S = \{1, 2, 3\}$                                                  | $S = \{1, 2, 3\}$                                                  | <b>Exercise 1.117.</b> Let $S = \{1, 2, 3\}$ . Let's try to understand the adjunction discussed                              |
| fong-spivak-invitation_F00592 | 50   | ■ 1.000 | $U(\leq)$                                                          | $U(\leq)$                                                          | 1. Come up with any preorder relation $\leq$ on $S$ , and define $U(\leq)$ to be the subset                                  |
| fong-spivak-invitation_F00593 | 50   | ■ 1.000 | $U(\leq) := \{(s_1, s_2) \mid s_1 \leq s_2\} \subseteq S \times S$ | $U(\leq) := \{(s_1, s_2) \mid s_1 \leq s_2\} \subseteq S \times S$ | $U(\leq) := \{(s_1, s_2) \mid s_1 \leq s_2\} \subseteq S \times S$ , i.e. $U(\leq)$ is the image of $\leq$ under the         |
| fong-spivak-invitation_F00594 | 50   | ■ 1.000 | $Q \subseteq S \times S$                                           | $Q \subseteq S \times S$                                           | 2. Come up with any two binary relations $Q \subseteq S \times S$ and $Q' \subseteq S \times S$ such that                    |
| fong-spivak-invitation_F00595 | 50   | ■ 1.000 | $Q' \subseteq S \times S$                                          | $Q' \subseteq S \times S$                                          | 2. Come up with any two binary relations $Q \subseteq S \times S$ and $Q' \subseteq S \times S$ such that                    |
| fong-spivak-invitation_F00596 | 50   | ■ 1.000 | $Q \subseteq U(\leq)$                                              | $Q \subseteq U(\leq)$                                              | $Q \subseteq U(\leq)$ but $Q' \not\subseteq U(\leq)$ . Note that your choice of $Q, Q'$ do not have to come                  |

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| fong-spivak-invitation_F00597 | 50   | ■ 1.000 | $Q^{\prime} \not\subseteq U(\leq)$                  | $Q' \not\subseteq U(\leq)$                          | $Q \subseteq U(\leq)$ but $Q' \not\subseteq U(\leq)$ . Note that your choice of $Q, Q'$ do not have to come                     |
| fong-spivak-invitation_F00598 | 50   | ■ 1.000 | $Q, Q^{\prime}$                                     | $Q, Q'$                                             | $Q \subseteq U(\leq)$ but $Q' \not\subseteq U(\leq)$ . Note that your choice of $Q, Q'$ do not have to come                     |
| fong-spivak-invitation_F00599 | 50   | ■ 0.825 | $\mathrm{Cl}(Q) \sqsubseteq \leq$                   | $\mathrm{Cl}(Q) \sqsubseteq \leq$                   | that $\mathrm{Cl}(Q) \sqsubseteq \leq$ , where $\sqsubseteq$ is the order on $\mathbf{Pos}(S)$ , defined immediately above this |
| fong-spivak-invitation_F00600 | 50   | ■ 0.825 | $\operatorname{Pos}(S)$                             | $\operatorname{Pos}(S)$                             | that $\mathrm{Cl}(Q) \sqsubseteq \leq$ , where $\sqsubseteq$ is the order on $\mathbf{Pos}(S)$ , defined immediately above this |
| fong-spivak-invitation_F00601 | 50   | ■ 0.508 | $\mathrm{Cl}(Q^{\prime}) \not\subseteq \leq$        | $\mathrm{Cl}(Q') \not\subseteq \leq$                | 4. Concretely show that $\mathrm{Cl}(Q') \not\subseteq \leq$ .                                                                  |
| fong-spivak-invitation_F00602 | 51   | ■ 1.000 | $a \otimes b$                                       | $a \otimes b$                                       | a monoidal structure on preorders. This allows us to construct an element $a \otimes b$ of a                                    |
| fong-spivak-invitation_F00603 | 53   | ■ 1.000 | $\boldsymbol{a}$                                    | $\boldsymbol{a}$                                    | <b>2.1 Getting from <math>a</math> to <math>b</math></b>                                                                        |
| fong-spivak-invitation_F00604 | 53   | ■ 1.000 | $\boldsymbol{b}$                                    | $\boldsymbol{b}$                                    | <b>2.1 Getting from <math>a</math> to <math>b</math></b>                                                                        |
| fong-spivak-invitation_F00605 | 53   | ■ 1.000 | $d \in [0, \infty]$                                 | $d \in [0, \infty]$                                 | the question of getting from $a$ to $b$ has an answer $d \in [0, \infty]$ , a cost.                                             |
| fong-spivak-invitation_F00606 | 54   | ■ 0.994 | $(X, \leq)$                                         | $(X, \leq)$                                         | In Section 1.2.2 we introduced preorders. The notation for a preorder, namely $(X, \leq)$ ,                                     |
| fong-spivak-invitation_F00607 | 54   | ■ 1.000 | $I \in X$                                           | $I \in X$                                           | (i) an element $I \in X$ , called the <i>monoidal unit</i> ,                                                                    |
| fong-spivak-invitation_F00608 | 54   | ■ 1.000 | $\otimes: X \times X \rightarrow X$                 | $\otimes: X \times X \rightarrow X$                 | (ii) a function $\otimes: X \times X \rightarrow X$ , called the <i>monoidal product</i> .                                      |
| fong-spivak-invitation_F00609 | 54   | ■ 0.974 | $\otimes(x_1, x_2) =$                               | $\otimes(x_1, x_2) =$                               | These constituents must satisfy the following properties, where we write $\otimes(x_1, x_2) =$                                  |
| fong-spivak-invitation_F00610 | 54   | ■ 1.000 | $x_1 \otimes x_2$                                   | $x_1 \otimes x_2$                                   | $x_1 \otimes x_2$ :                                                                                                             |
| fong-spivak-invitation_F00611 | 54   | ■ 1.000 | $x_1, x_2, y_1, y_2 \in X$                          | $x_1, x_2, y_1, y_2 \in X$                          | (a) for all $x_1, x_2, y_1, y_2 \in X$ if $x_1 \leq y_1$ and $x_2 \leq y_2$ , then $x_1 \otimes x_2 \leq y_1 \otimes y_2$ ,     |
| fong-spivak-invitation_F00612 | 54   | ■ 1.000 | $x_1 \leq y_1$                                      | $x_1 \leq y_1$                                      | (a) for all $x_1, x_2, y_1, y_2 \in X$ , if $x_1 \leq y_1$ and $x_2 \leq y_2$ , then $x_1 \otimes x_2 \leq y_1 \otimes y_2$ ,   |
| fong-spivak-invitation_F00613 | 54   | ■ 1.000 | $x_2 \leq y_2$                                      | $x_2 \leq y_2$                                      | (a) for all $x_1, x_2, y_1, y_2 \in X$ , if $x_1 \leq y_1$ and $x_2 \leq y_2$ , then $x_1 \otimes x_2 \leq y_1 \otimes y_2$ ,   |
| fong-spivak-invitation_F00614 | 54   | ■ 1.000 | $x_1 \otimes x_2 \leq y_1 \otimes y_2$              | $x_1 \otimes x_2 \leq y_1 \otimes y_2$              | (a) for all $x_1, x_2, y_1, y_2 \in X$ , if $x_1 \leq y_1$ and $x_2 \leq y_2$ , then $x_1 \otimes x_2 \leq y_1 \otimes y_2$ ,   |
| fong-spivak-invitation_F00615 | 54   | ■ 1.000 | $I \otimes x = x$                                   | $I \otimes x = x$                                   | (b) for all $x \in X$ , the equations $I \otimes x = x$ and $x \otimes I = x$ hold,                                             |
| fong-spivak-invitation_F00616 | 54   | ■ 1.000 | $x \otimes I = x$                                   | $x \otimes I = x$                                   | (b) for all $x \in X$ , the equations $I \otimes x = x$ and $x \otimes I = x$ hold,                                             |
| fong-spivak-invitation_F00617 | 54   | ■ 1.000 | $x, y, z \in X$                                     | $x, y, z \in X$                                     | (c) for all $x, y, z \in X$ , the equation $(x \otimes y) \otimes z = x \otimes (y \otimes z)$ holds,                           |
| fong-spivak-invitation_F00618 | 54   | ■ 1.000 | $(x \otimes y) \otimes z = x \otimes (y \otimes z)$ | $(x \otimes y) \otimes z = x \otimes (y \otimes z)$ | (c) for all $x, y, z \in X$ , the equation $(x \otimes y) \otimes z = x \otimes (y \otimes z)$ holds,                           |
| fong-spivak-invitation_F00619 | 54   | ■ 1.000 | $x \otimes y = y \otimes x$                         | $x \otimes y = y \otimes x$                         | (d) for all $x, y \in X$ , the equation $x \otimes y = y \otimes x$ holds.                                                      |
| fong-spivak-invitation_F00620 | 54   | ■ 0.828 | $X, \leq, I, \otimes$                               | $X, \leq, I, \otimes$                               | tively. A preorder equipped with a symmetric monoidal structure, $(X, \leq, I, \otimes)$ , is called                            |
| fong-spivak-invitation_F00621 | 55   | ■ 1.000 | $I$                                                 | $I$                                                 | Anyone can propose a set $X$ , an order $\leq$ on $X$ , an element $I$ in $X$ , and a binary                                    |
| fong-spivak-invitation_F00622 | 55   | ■ 0.487 | $\cong$                                             | $\cong$                                             | <b>Remark 2.2.</b> It is often useful to replace $=$ with $\cong$ throughout Definition 2.1. The                                |
| fong-spivak-invitation_F00623 | 55   | ■ 0.962 | $\{*\}$                                             | $\{*\}$                                             | units we have seen include $I$ (as in the definition), $0, 1, \text{true}, \text{false}, \{*\}$ , and more.                     |

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| fong-spivak-invitation_F00624 | 55   | 0.572 | $+, *, \wedge, \vee$                                     | $+, *, \wedge, \vee$                                     | Monoidal products we have seen include $\otimes$ (as in the definition), $+$ , $*$ , $\wedge$ , $\vee$ , and $\times$ . The          |
| fong-spivak-invitation_F00625 | 55   | 0.572 | $\times$                                                 | $\times$                                                 | Monoidal products we have seen include $\otimes$ (as in the definition), $+$ , $*$ , $\wedge$ , $\vee$ , and $\times$ . The          |
| fong-spivak-invitation_F00626 | 55   | 1.000 | $-5 \leq \sqrt{2}$                                       | $-5 \leq \sqrt{2}$                                       | numbers; e.g. $-5 < \sqrt{2}$ . We propose 0 as a monoidal unit and $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ as a    |
| fong-spivak-invitation_F00627 | 55   | 1.000 | $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ | $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ | numbers; e.g. $-5 \leq \sqrt{2}$ . We propose 0 as a monoidal unit and $+: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ as a |
| fong-spivak-invitation_F00628 | 55   | 0.887 | $\mathbb{R}, \leq, 0, +$                                 | $\mathbb{R}, \leq, 0, +$                                 | monoidal product. Does $(\mathbb{R}, \leq, 0, +)$ satisfy the conditions of Definition 2.1?                                          |
| fong-spivak-invitation_F00629 | 55   | 1.000 | $x_1 + x_2 \leq y_1 + y_2$                               | $x_1 + x_2 \leq y_1 + y_2$                               | If $x_1 \leq y_1$ and $x_2 \leq y_2$ , it is true that $x_1 + x_2 \leq y_1 + y_2$ . It is also true that $0 + x = x$                 |
| fong-spivak-invitation_F00630 | 55   | 1.000 | $0 + x = x$                                              | $0 + x = x$                                              | If $x_1 \leq y_1$ and $x_2 \leq y_2$ , it is true that $x_1 + x_2 \leq y_1 + y_2$ . It is also true that $0 + x = x$                 |
| fong-spivak-invitation_F00631 | 55   | 0.988 | $x + 0 = x$                                              | $x + 0 = x$                                              | and $x + 0 = x$ , that $(x + y) + z = x + (y + z)$ , and that $x + y = y + x$ . Thus $(\mathbb{R}, \leq, 0, +)$                      |
| fong-spivak-invitation_F00632 | 55   | 0.988 | $(x + y) + z = x + (y + z)$                              | $(x + y) + z = x + (y + z)$                              | and $x + 0 = x$ , that $x + y + z = x + (y + z)$ , and that $x + y = y + x$ . Thus $(\mathbb{R}, \leq, 0, +)$                        |
| fong-spivak-invitation_F00633 | 55   | 0.988 | $x + y = y + x$                                          | $x + y = y + x$                                          | and $x + 0 = x$ , that $(x + y) + z = x + (y + z)$ , and that $x + y = y + x$ . Thus $(\mathbb{R}, \leq, 0, +)$                      |
| fong-spivak-invitation_F00634 | 55   | 0.988 | $(\mathbb{R}, \leq, 0, +)$                               | $(\mathbb{R}, \leq, 0, +)$                               | and $x + 0 = x$ , that $(x + y) + z = x + (y + z)$ , and that $x + y = y + x$ . Thus $(\mathbb{R}, \leq, 0, +)$                      |
| fong-spivak-invitation_F00635 | 55   | 0.842 | $\mathbb{R}, \leq$                                       | $\mathbb{R}, \leq$                                       | <b>Exercise 2.4.</b> Consider again the preorder $(\mathbb{R}, \leq)$ from Example 2.3. Someone pro-                                 |
| fong-spivak-invitation_F00636 | 55   | 0.975 | $M$                                                      | $M$                                                      | <b>Example 2.5.</b> A <i>monoid</i> consists of a set $M$ , a function $*$ : $M \times M \rightarrow M$ called the                   |
| fong-spivak-invitation_F00637 | 55   | 0.975 | $*$ : $M \times M \rightarrow M$                         | $*$ : $M \times M \rightarrow M$                         | <b>Example 2.5.</b> A <i>monoid</i> consists of a set $M$ , a function $*$ : $M \times M \rightarrow M$ called the                   |
| fong-spivak-invitation_F00638 | 55   | 1.000 | $e \in M$                                                | $e \in M$                                                | <i>monoid multiplication</i> , and an element $e \in M$ called the <i>monoid unit</i> , such that, when                              |
| fong-spivak-invitation_F00639 | 55   | 1.000 | $*(m, n)$                                                | $*(m, n)$                                                | you write $*(m, n)$ as $m * n$ , i.e. using infix notation, the equations                                                            |
| fong-spivak-invitation_F00640 | 55   | 1.000 | $m * n$                                                  | $m * n$                                                  | you write $*(m, n)$ as $m * n$ , i.e. using infix notation, the equations                                                            |
| fong-spivak-invitation_F00641 | 55   | 1.000 | $m, n, p \in M$                                          | $m, n, p \in M$                                          | hold for all $m, n, p \in M$ . It is called <i>commutative</i> if also $m * n = n * m$ .                                             |
| fong-spivak-invitation_F00642 | 55   | 1.000 | $m * n = n * m$                                          | $m * n = n * m$                                          | hold for all $m, n, p \in M$ . It is called <i>commutative</i> if also $m * n = n * m$ .                                             |
| fong-spivak-invitation_F00643 | 55   | 0.481 | $_S$                                                     | $S$                                                      | Every set $S$ determines a discrete preorder <b>Disc</b> <sub><math>S</math></sub> (where $m \leq n$ iff $m = n$ ; see               |
| fong-spivak-invitation_F00644 | 55   | 0.481 | $m = n$                                                  | $m = n$                                                  | Every set $S$ determines a discrete preorder <b>Disc</b> <sub><math>S</math></sub> (where $m \leq n$ iff $m = n$ ; see               |
| fong-spivak-invitation_F00645 | 55   | 0.769 | $(M, e, *)$                                              | $(M, e, *)$                                              | Example 1.27), and it is easy to check that if $(M, e, *)$ is a commutative monoid then                                              |
| fong-spivak-invitation_F00646 | 55   | 0.570 | $(\text{Disc}_M, =, e, *)$                               | $(\text{Disc}_M, =, e, *)$                               | $(\text{Disc}_M, =, e, *)$ is a symmetric monoidal preorder.                                                                         |
| fong-spivak-invitation_F00647 | 55   | 0.570 | $(\text{Disc}_M, =, e, *)$                               | $(\text{Disc}_M, =, e, *)$                               | $(\text{Disc}_M, =, e, *)$ is a symmetric monoidal preorder.                                                                         |
| fong-spivak-invitation_F00648 | 55   | 0.932 | $(M, *, e)$                                              | $(M, *, e)$                                              | <b>Exercise 2.6.</b> We said it was easy to check that if $(M, *, e)$ is a commutative monoid                                        |
| fong-spivak-invitation_F00649 | 55   | 0.366 | $(\text{Disc}_M, =, *, e)$                               | $(\text{Disc}_M, =, *, e)$                               | then $(\text{Disc}_M, =, *, e)$ is a symmetric monoidal preorder. Were we telling the truth? $\diamond$                              |
| fong-spivak-invitation_F00650 | 55   | 1.000 | $H$                                                      | $H$                                                      | Let $H$ be the set of all poker hands, where a hand means a choice of five cards                                                     |

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| fong-spivak-invitation_F00651 | 56   | ■ 1.000 | $h \leq h^{\prime}$                                                                                                                            | $h \leq h'$                                                                                                                                    | from the standard 52-card deck. As an order, put $h \leq h'$ if $h'$ beats or equals $h$ in                                                                                                |
| fong-spivak-invitation_F00652 | 56   | ■ 1.000 | $h^{\prime}$                                                                                                                                   | $h'$                                                                                                                                           | from the standard 52-card deck. As an order, put $h \leq h'$ if $h'$ beats or equals $h$ in                                                                                                |
| fong-spivak-invitation_F00653 | 56   | ■ 1.000 | $\otimes: H \times H \rightarrow H$                                                                                                            | $\otimes: H \times H \rightarrow H$                                                                                                            | One could propose a monoidal product $\otimes: H \times H \rightarrow H$ by assigning $h_1 \otimes h_2$ to be                                                                              |
| fong-spivak-invitation_F00654 | 56   | ■ 1.000 | $h_{\{1\}} \otimes h_{\{2\}}$                                                                                                                  | $h_1 \otimes h_2$                                                                                                                              | One could propose a monoidal product $\otimes: H \times H \rightarrow H$ by assigning $h_1 \otimes h_2$ to be                                                                              |
| fong-spivak-invitation_F00655 | 56   | ■ 1.000 | $h_{\{1\}}$                                                                                                                                    | $h_1$                                                                                                                                          | “the best hand one can form out of the ten cards in $h_1$ and $h_2$ .” If some cards are in both                                                                                           |
| fong-spivak-invitation_F00656 | 56   | ■ 1.000 | $h_{\{2\}}$                                                                                                                                    | $h_2$                                                                                                                                          | “the best hand one can form out of the ten cards in $h_1$ and $h_2$ .” If some cards are in both                                                                                           |
| fong-spivak-invitation_F00657 | 56   | ■ 0.179 | $\{2 \odot, 5 \curvearrowright, 6 \odot, 6 \infty, 7 \rightarrow\} = \{2 \odot, 3 \odot, 4 \curvearrowright, 5 \curvearrowright, 6 \bigodot\}$ | $\{2 \odot, 5 \curvearrowright, 6 \odot, 6 \infty, 7 \rightarrow\} = \{2 \odot, 3 \odot, 4 \curvearrowright, 5 \curvearrowright, 6 \bigodot\}$ | $\{2 \heartsuit, 5 \heartsuit, 6 \heartsuit, 6 \clubsuit, 7 \clubsuit\} = \{2 \heartsuit, 3 \heartsuit, 4 \heartsuit, 5 \heartsuit, 6 \heartsuit\}$ because the latter is the best         |
| fong-spivak-invitation_F00658 | 56   | ■ 1.000 | $h_{\{1\}} \leq i_{\{1\}}$                                                                                                                     | $h_1 \leq i_1$                                                                                                                                 | could be the case that $h_1 \leq i_1$ and $h_2 \leq i_2$ , and yet <i>not</i> be the case that $h_1 \otimes h_2 \leq i_1 \otimes i_2$ .                                                    |
| fong-spivak-invitation_F00659 | 56   | ■ 1.000 | $h_{\{2\}} \leq i_{\{2\}}$                                                                                                                     | $h_2 \leq i_2$                                                                                                                                 | could be the case that $h_1 \leq i_1$ and $h_2 \leq i_2$ , and yet <i>not</i> be the case that $h_1 \otimes h_2 \leq i_1 \otimes i_2$ .                                                    |
| fong-spivak-invitation_F00660 | 56   | ■ 1.000 | $h_{\{1\}} \otimes h_{\{2\}} \leq i_{\{1\}} \otimes i_{\{2\}}$                                                                                 | $h_1 \otimes h_2 \leq i_1 \otimes i_2$                                                                                                         | could be the case that $h_1 \leq i_1$ and $h_2 \leq i_2$ , and yet <i>not</i> be the case that $h_1 \otimes h_2 \leq i_1 \otimes i_2$ .                                                    |
| fong-spivak-invitation_F00661 | 56   | ■ 0.201 | $h_{\{1\}} \otimes h_{\{2\}} = \{10 \wedge, \mathrm{~J} \in, \mathrm{Q}, \mathrm{K}, \mathrm{A}\}$                                             | $h_1 \otimes h_2 = \{10 \wedge, \mathrm{~J} \in, \mathrm{Q}, \mathrm{K}, \mathrm{A}\}$                                                         | Here, $h_1 \leq i_1$ and $h_2 \leq i_2$ , but $h_1 \otimes h_2 = \{10 \clubsuit, \mathrm{~J} \clubsuit, \mathrm{~O} \clubsuit, \mathrm{~K} \clubsuit, \mathrm{~A} \clubsuit\}$ is the best |
| fong-spivak-invitation_F00662 | 56   | ■ 0.409 | $i_{\{1\}} \otimes i_{\{2\}} = \{5 \bullet, 5 \oslash, 6 \circ, 6 \diamond, \mathrm{Q} \diamond\}$                                             | $i_1 \otimes i_2 = \{5 \bullet, 5 \oslash, 6 \circ, 6 \diamond, \mathrm{Q} \diamond\}$                                                         | possible hand and beats $i_1 \otimes i_2 = \{5 \clubsuit, 5 \heartsuit, 6 \heartsuit, 6 \diamondsuit, \mathrm{Q} \diamondsuit\}$ .                                                         |
| fong-spivak-invitation_F00663 | 57   | ■ 0.689 | $(x \in X)$                                                                                                                                    | $(x \in X)$                                                                                                                                    | $(x \in X)$ , relationship ( $\leq$ ), and combination (using transitivity and $\otimes$ ), and so do wiring                                                                               |
| fong-spivak-invitation_F00664 | 57   | ■ 0.689 | $(\leq)$                                                                                                                                       | $(\leq)$                                                                                                                                       | $(x \in X)$ , relationship ( $\leq$ ), and combination (using transitivity and $\otimes$ ), and so do wiring                                                                               |
| fong-spivak-invitation_F00665 | 57   | ■ 0.921 | $(X, \leq I, \otimes)$                                                                                                                         | $(X, \leq I, \otimes)$                                                                                                                         | $(X, \leq, I, \otimes)$ as in Definition 2.1. Wiring diagrams have wires on the left and the right.                                                                                        |
| fong-spivak-invitation_F00666 | 57   | ■ 1.000 | $x \otimes y$                                                                                                                                  | $x \otimes y$                                                                                                                                  | draw one wire labeled $x \otimes y$ .                                                                                                                                                      |
| fong-spivak-invitation_F00667 | 58   | ■ 0.999 | $x_{\{1\}} \otimes x_{\{2\}} \leq y_{\{1\}} \otimes y_{\{2\}} \otimes y_{\{3\}}$                                                               | $x_1 \otimes x_2 \leq y_1 \otimes y_2 \otimes y_3$                                                                                             | The validity of the second box corresponds to the inequality $x_1 \otimes x_2 \leq y_1 \otimes y_2 \otimes y_3$ .                                                                          |
| fong-spivak-invitation_F00668 | 58   | ■ 0.933 | $4 \leq 7$                                                                                                                                     | $4 \leq 7$                                                                                                                                     | And here we express a couple of facts about $(\mathbb{R}, \leq, 0, +)$ in this language: $4 \leq 7$ and                                                                                    |
| fong-spivak-invitation_F00669 | 58   | ■ 1.000 | $2+5 \leq -1+5+3$                                                                                                                              | $2+5 \leq -1+5+3$                                                                                                                              | $2+5 \leq -1+5+3$ .                                                                                                                                                                        |
| fong-spivak-invitation_F00670 | 58   | ■ 1.000 | $x_{\{1\}} \otimes x_{\{2\}} \leq$                                                                                                             | $x_1 \otimes x_2 \leq$                                                                                                                         | structure (Definition 2.1). Property (a) says that if $x_1 \leq y_1$ and $x_2 \leq y_2$ then $x_1 \otimes x_2 \leq$                                                                        |
| fong-spivak-invitation_F00671 | 58   | ■ 1.000 | $y_{\{1\}} \otimes y_{\{2\}}$                                                                                                                  | $y_1 \otimes y_2$                                                                                                                              | $y_1 \otimes y_2$ . This corresponds to the idea that stacking any two valid boxes in parallel is still                                                                                    |
| fong-spivak-invitation_F00672 | 59   | ■ 0.977 | $\mathcal{X} = (X, \leq, I, \otimes)$                                                                                                          | $\mathcal{X} = (X, \leq, I, \otimes)$                                                                                                          | Given a monoidal preorder $\mathcal{X} = (X, \leq, I, \otimes)$ , a wiring diagram is a graphical proof of                                                                                 |
| fong-spivak-invitation_F00673 | 59   | ■ 0.973 | $\mathcal{X}$                                                                                                                                  | $\mathcal{X}$                                                                                                                                  | something about $\mathcal{X}$ . Each box in the diagram has a left side and a right side, say $x$ and                                                                                      |
| fong-spivak-invitation_F00674 | 60   | ■ 1.000 | $v+w+u$                                                                                                                                        | $v+w+u$                                                                                                                                        | technically there is no such thing as $v+w+u$ , for example. Instead, there is $(v+w)+u$                                                                                                   |
| fong-spivak-invitation_F00675 | 60   | ■ 1.000 | $(v+w)+u$                                                                                                                                      | $(v+w)+u$                                                                                                                                      | technically there is no such thing as $v+w+u$ , for example. Instead, there is $(v+w)+u$                                                                                                   |
| fong-spivak-invitation_F00676 | 60   | ■ 1.000 | $v+(w+u)$                                                                                                                                      | $v+(w+u)$                                                                                                                                      | and $v+(w+u)$ , and so on.                                                                                                                                                                 |
| fong-spivak-invitation_F00677 | 61   | ■ 1.000 | $2 \mathrm{~H}_{\{2\}} + 2 \mathrm{~Na}$                                                                                                       | $2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na}$                                                                                                       | The collection on the left, $2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na}$ , is called the <i>reactant</i> , and the collection on the                                                           |

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| fong-spivak-invitation_F00678 | 61   | ■ 1.000 | $2 \mathrm{NaOH} + \mathrm{H}_2$                            | $2\mathrm{NaOH} + \mathrm{H}_2$                      | right, $2\mathrm{NaOH} + \mathrm{H}_2$ , is called the <i>product</i> .                                                         |
| fong-spivak-invitation_F00679 | 61   | ■ 0.688 | $\mathrm{NaCl} \in \mathrm{Mat}$                            | $\mathrm{NaCl} \in \mathrm{Mat}$                     | tions of atoms and molecules, sometimes called <i>materials</i> . So we have $\mathrm{NaCl} \in \mathrm{Mat}$ and               |
| fong-spivak-invitation_F00680 | 61   | ■ 0.484 | $4 \mathrm{H}_2\mathrm{O} + 6 \mathrm{Ne} \in \mathrm{Mat}$ | $4\mathrm{H}_2\mathrm{O} + 6\mathrm{Ne} \in$         | <del><math>4\mathrm{H}_2\mathrm{O} + 6\mathrm{Ne} \in</math></del> Mat.                                                         |
| fong-spivak-invitation_F00681 | 62   | ■ 0.725 | $\otimes$                                                   | $\otimes$                                            | in the chemistry setting, taking the place of $\otimes$ . While it does not come up in practice,                                |
| fong-spivak-invitation_F00682 | 62   | ■ 1.000 | $k$                                                         | $k$                                                  | Here $k$ is the catalyst because it is found in both the reactant and the product of the                                        |
| fong-spivak-invitation_F00683 | 62   | ■ 1.000 | $x + y \rightarrow z$                                       | $x + y \rightarrow z$                                | reaction. It is said to catalyze the reaction $x + y \rightarrow z$ . The idea is that the reaction                             |
| fong-spivak-invitation_F00684 | 62   | ■ 1.000 | $x + y + k \rightarrow z + k$                               | $x + y + k \rightarrow z + k$                        | corresponds to the composite reaction <del><math>x + y + k \rightarrow z + k</math></del> .                                     |
| fong-spivak-invitation_F00685 | 63   | ■ 1.000 | $x \leq I$                                                  | $x \leq I$                                           | (e) $x \leq I$ for all $x \in X$ .                                                                                              |
| fong-spivak-invitation_F00686 | 63   | ■ 1.000 | $x \rightarrow 0$                                           | $x \rightarrow 0$                                    | chemistry section, we would write instead $x \rightarrow 0$ : any $x$ yields nothing. But this is                               |
| fong-spivak-invitation_F00687 | 63   | ■ 1.000 | $10^{23}$                                                   | $10^{23}$                                            | manipulate – each item produced. In chemistry, when you have $10^{23}$ of substance $A$                                         |
| fong-spivak-invitation_F00688 | 64   | ■ 1.000 | $x \leq x + x$                                              | $x \leq x + x$                                       | (f) <del><math>x \leq x + x</math></del> for all $x \in X$                                                                      |
| fong-spivak-invitation_F00689 | 65   | ■ 0.953 | $=0$                                                        | $= 0$                                                | $\text{false} = 0$ and $\text{true} = 1$ , then $\wedge$ corresponds to the usual multiplication operation                      |
| fong-spivak-invitation_F00690 | 65   | ■ 0.953 | $\operatorname{true}=1$                                     | $\text{true} = 1$                                    | $\text{false} = 0$ and $\text{true} = 1$ , then $\wedge$ corresponds to the usual multiplication operation                      |
| fong-spivak-invitation_F00691 | 65   | ■ 0.905 | $= (\mathbb{B}, \leq$                                       | $= (\mathbb{B}, \leq$                                | preorder which we denote $\mathbf{Bool} := (\mathbb{B}, \leq, \text{true}, \wedge)$ .                                           |
| fong-spivak-invitation_F00692 | 65   | ■ 0.905 | $\wedge)$                                                   | $\wedge)$                                            | preorder which we denote $\mathbf{Bool} := (\mathbb{B}, \leq, \text{true}, \wedge)$ .                                           |
| fong-spivak-invitation_F00693 | 65   | ■ 0.975 | $\mathcal{V} = (V, \leq, I, \otimes)$                       | $\mathcal{V} = (V, \leq, I, \otimes)$                | monoidal preorder $\mathcal{V} = (V, \leq, I, \otimes)$ means “letting $\mathcal{V}$ structure the question of get-             |
| fong-spivak-invitation_F00694 | 66   | ■ 0.921 | $\mathbb{B}, \leq$                                          | $\mathbb{B}, \leq$                                   | <b>Exercise 2.13.</b> Let $(\mathbb{B}, \leq)$ be as above, but now consider the monoidal product to be $\vee$                  |
| fong-spivak-invitation_F00695 | 66   | ■ 0.850 | $(\mathbb{N}, \leq)$                                        | $(\mathbb{N}, \leq)$                                 | preorder $(\mathbb{N}, \leq)$ of natural numbers, where $\leq$ is the usual ordering ( $0 \leq 1$ and $5 \leq 16$ ).            |
| fong-spivak-invitation_F00696 | 66   | ■ 0.850 | $5 \leq 16$                                                 | $5 \leq 16$                                          | preorder $(\mathbb{N}, \leq)$ of natural numbers, where $\leq$ is the usual ordering ( $0 \leq 1$ and $5 \leq 16$ ).            |
| fong-spivak-invitation_F00697 | 66   | ■ 0.999 | $6 + 4 = 10$                                                | $6 + 4 = 10$                                         | where the monoidal unit is 0 and the monoidal product is $+$ , i.e. $6 + 4 = 10$ . It is easy                                   |
| fong-spivak-invitation_F00698 | 66   | ■ 1.000 | $6 * 4 = 24$                                                | $6 * 4 = 24$                                         | product is $*$ , i.e. <del><math>6 * 4 = 24</math></del> . What should the monoidal unit be?                                    |
| fong-spivak-invitation_F00699 | 66   | ■ 1.000 | $m \mid n$                                                  | $m \mid n$                                           | “divisibility” order on $\mathbb{N}$ : we write $m \mid n$ to mean that $m$ divides into $n$ without remainder.                 |
| fong-spivak-invitation_F00700 | 66   | ■ 0.999 | $1 \mid m$                                                  | $1 \mid m$                                           | So $1 \mid m$ for all $m$ and $4 \mid 12$ .                                                                                     |
| fong-spivak-invitation_F00701 | 66   | ■ 0.999 | $4 \mid 12$                                                 | $4 \mid 12$                                          | So $1 \mid m$ for all $m$ and $4 \mid 12$ .                                                                                     |
| fong-spivak-invitation_F00702 | 66   | ■ 0.970 | $(\mathbb{N}, \mid)$                                        | $(\mathbb{N}, \mid)$                                 | There is a monoidal structure on $(\mathbb{N}, \mid)$ , where the monoidal unit is 1 and the                                    |
| fong-spivak-invitation_F00703 | 66   | ■ 1.000 | $x_1 \mid y_1$                                              | $x_1 \mid y_1$                                       | monoidal product is $*$ , i.e. $6 * 4 = 24$ . Then if $x_1 \mid y_1$ and $x_2 \mid y_2$ , then $(x_1 * x_2) \mid (y_1 * y_2)$ . |
| fong-spivak-invitation_F00704 | 66   | ■ 1.000 | $x_2 \mid y_2$                                              | $x_2 \mid y_2$                                       | monoidal product is $*$ , i.e. $6 * 4 = 24$ . Then if $x_1 \mid y_1$ and $x_2 \mid y_2$ , then $(x_1 * x_2) \mid (y_1 * y_2)$ . |
| fong-spivak-invitation_F00705 | 66   | ■ 1.000 | $\left(x_1 * x_2\right) \mid \left(y_1 * y_2\right)$        | $\left(x_1 * x_2\right) \mid \left(y_1 * y_2\right)$ | monoidal product is $*$ , i.e. $6 * 4 = 24$ . Then if $x_1 \mid y_1$ and $x_2 \mid y_2$ , then $(x_1 * x_2) \mid (y_1 * y_2)$ . |

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| fong-spivak-invitation_F00706 | 66   | ■ 1.000 | $p_{\{1\}}, p_{\{2\}} \in \mathbb{N}$                                                             | $p_1, p_2 \in \mathbb{N}$               | Indeed, if there is some $p_1, p_2 \in \mathbb{N}$ such that $x_1 * p_1 = y_1$ and $x_2 * p_2 = y_2$ , then                     |
| fong-spivak-invitation_F00707 | 66   | ■ 1.000 | $x_{\{1\}} * p_{\{1\}} = y_{\{1\}}$                                                               | $x_1 * p_1 = y_1$                       | Indeed, if there is some $p_1, p_2 \in \mathbb{N}$ such that $x_1 * p_1 = y_1$ and $x_2 * p_2 = y_2$ , then                     |
| fong-spivak-invitation_F00708 | 66   | ■ 1.000 | $x_{\{2\}} * p_{\{2\}} = y_{\{2\}}$                                                               | $x_2 * p_2 = y_2$                       | Indeed, if there is some $p_1, p_2 \in \mathbb{N}$ such that $x_1 * p_1 = y_1$ and $x_2 * p_2 = y_2$ , then                     |
| fong-spivak-invitation_F00709 | 66   | ■ 1.000 | $\left(p_{\{1\}} * p_{\{2\}}\right) * \left(x_{\{1\}} * x_{\{2\}}\right) = y_{\{1\}} * y_{\{2\}}$ | $(p_1 * p_2) * (x_1 * x_2) = y_1 * y_2$ | $(p_1 * p_2) * (x_1 * x_2) = y_1 * y_2$ .                                                                                       |
| fong-spivak-invitation_F00710 | 67   | ■ 1.000 | $\mathrm{P}(S)$                                                                                   | $P(S)$                                  | <b>Exercise 2.19.</b> Let $S$ be a set and let $P(S)$ be its power set, the set of all subsets of                               |
| fong-spivak-invitation_F00711 | 67   | ■ 1.000 | $\varnothing \subseteq S$                                                                         | $\emptyset \subseteq S$                 | $S$ , including the empty subset, $\emptyset \subseteq S$ , and the “everything” subset, $S \subseteq S$ . We can               |
| fong-spivak-invitation_F00712 | 67   | ■ 1.000 | $S \subseteq S$                                                                                   | $S \subseteq S$                         | $S$ , including the empty subset, $\emptyset \subseteq S$ , and the “everything” subset, $S \subseteq S$ . We can               |
| fong-spivak-invitation_F00713 | 67   | ■ 1.000 | $A \subseteq B$                                                                                   | $A \subseteq B$                         | give $P(S)$ an order: $A \leq B$ is given by the subset relation $A \subseteq B$ , as discussed in                              |
| fong-spivak-invitation_F00714 | 67   | ■ 1.000 | $A \cap B$                                                                                        | $A \cap B$                              | $S$ and monoidal product given by intersection $A \cap B$ .                                                                     |
| fong-spivak-invitation_F00715 | 67   | ■ 0.958 | $\{\}^{\mathbb{N}}$                                                                               | $\mathbb{N}$                            | <b>Exercise 2.20.</b> Let $\mathrm{Prop}^{\mathbb{N}}$ denote the set of all mathematical statements one can make               |
| fong-spivak-invitation_F00716 | 67   | ■ 0.422 | $\operatorname{Prop}^{\mathbb{N}}$                                                                | $\mathrm{Prop}^{\mathbb{N}}$            | if and only if the other is true. For example “ $n$ is prime” is an element of $\mathrm{Prop}^{\mathbb{N}}$ , and               |
| fong-spivak-invitation_F00717 | 67   | ■ 1.000 | $n=2$                                                                                             | $n = 2$                                 | so are “ $n = 2$ ” and “ $n \geq 11$ .” The statements “ $n + 2 = 5$ ” and “ $n$ is the least odd                               |
| fong-spivak-invitation_F00718 | 67   | ■ 1.000 | $n \geq 11$                                                                                       | $n \geq 11$                             | so are “ $n = 2$ ” and “ $n \geq 11$ .” The statements “ $n + 2 = 5$ ” and “ $n$ is the least odd                               |
| fong-spivak-invitation_F00719 | 67   | ■ 1.000 | $n+2=5$                                                                                           | $n + 2 = 5$                             | so are “ $n = 2$ ” and “ $n \geq 11$ .” The statements “ $n + 2 = 5$ ” and “ $n$ is the least odd                               |
| fong-spivak-invitation_F00720 | 67   | ■ 0.998 | $P, Q \in \operatorname{Prop}^{\mathbb{N}}$                                                       | $P, Q \in \mathrm{Prop}^{\mathbb{N}}$   | prime” are considered the same. Given $P, Q \in \mathrm{Prop}^{\mathbb{N}}$ , we say $P \leq Q$ if for all $n \in \mathbb{N}$ , |
| fong-spivak-invitation_F00721 | 67   | ■ 0.998 | $P \leq Q$                                                                                        | $P \leq Q$                              | prime” are considered the same. Given $P, Q \in \mathrm{Prop}^{\mathbb{N}}$ , we say $P \leq Q$ if for all $n \in \mathbb{N}$ , |
| fong-spivak-invitation_F00722 | 67   | ■ 1.000 | $P(n)$                                                                                            | $P(n)$                                  | whenever $P(n)$ is true, so is $Q(n)$ .                                                                                         |
| fong-spivak-invitation_F00723 | 67   | ■ 1.000 | $Q(n)$                                                                                            | $Q(n)$                                  | whenever $P(n)$ is true, so is $Q(n)$ .                                                                                         |
| fong-spivak-invitation_F00724 | 67   | ■ 1.000 | $[0, \infty]$                                                                                     | $[0, \infty]$                           | <b>Example 2.21</b> (Lawvere’s monoidal preorder, <b>Cost</b> ). Let $[0, \infty]$ denote the set of nonneg-                    |
| fong-spivak-invitation_F00725 | 67   | ■ 0.820 | $0, 1, 15.33 \overline{3}$                                                                        | $0, 1, 15.33\overline{3}$               | ative real numbers – such as $0, 1, 15.33\overline{3}$ , and $2\pi$ – together with $\infty$ . Consider the                     |
| fong-spivak-invitation_F00726 | 67   | ■ 0.820 | $2 \pi$                                                                                           | $2\pi$                                  | ative real numbers – such as $0, 1, 15.33\overline{3}$ , and $2\pi$ – together with $\infty$ . Consider the                     |
| fong-spivak-invitation_F00727 | 67   | ■ 0.820 | $\infty$                                                                                          | $\infty$                                | ative real numbers – such as $0, 1, 15.33\overline{3}$ , and $2\pi$ – together with $\infty$ . Consider the                     |
| fong-spivak-invitation_F00728 | 67   | ■ 0.972 | $[0, \infty], \geq$                                                                               | $[0, \infty], \geq$                     | preorder $([0, \infty], \geq)$ , with the usual notion of $\geq$ , where of course $\infty \geq x$ for all                      |
| fong-spivak-invitation_F00729 | 67   | ■ 0.972 | $\infty \geq x$                                                                                   | $\infty \geq x$                         | preorder $([0, \infty], \geq)$ , with the usual notion of $\geq$ , where of course $\infty \geq x$ for all                      |
| fong-spivak-invitation_F00730 | 67   | ■ 1.000 | $x \in [0, \infty]$                                                                               | $x \in [0, \infty]$                     | $x \in [0, \infty]$ .                                                                                                           |
| fong-spivak-invitation_F00731 | 67   | ■ 0.993 | $x + \infty = \infty$                                                                             | $x + \infty = \infty$                   | monoidal product is $+$ . In particular, $x + \infty = \infty$ for any $x \in [0, \infty]$ . Let’s call this                    |
| fong-spivak-invitation_F00732 | 67   | ■ 1.000 | $(X, \leq)^{\mathrm{op}} := (X, \geq)$                                                            | $(X, \leq)^{\mathrm{op}} := (X, \geq)$  | One can take the opposite of any preorder, just flip the order: $(X, \leq)^{\mathrm{op}} := (X, \geq)$ ; see                    |

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| fong-spivak-invitation_F00733 | 68   | ■ 0.514 | $\mathcal{X}=(X, \leq)$                                                                                                          | $\mathcal{X} = (X, \leq)$                                          | <b>Proposition 2.22.</b> Suppose $\mathcal{X} = (X, \leq)$ is a preorder and $\mathcal{X}^{\text{op}} = (X, \geq)$ is its                                      |
| fong-spivak-invitation_F00734 | 68   | ■ 0.514 | $X^{\{\mathrm{op}\}}=(X, \geq)$                                                                                                  | $\mathcal{X}^{\text{op}} = (X, \geq)$                              | <b>Proposition 2.22.</b> Suppose $\mathcal{X} = (X, \leq)$ is a preorder and $\mathcal{X}^{\text{op}} = (X, \geq)$ is its                                      |
| fong-spivak-invitation_F00735 | 68   | ■ 0.840 | $(X, \leq, I, \otimes)$                                                                                                          | $(X, \leq, I, \otimes)$                                            | opposite. If $(X, \leq, I, \otimes)$ is a symmetric monoidal preorder then so is its opposite,                                                                 |
| fong-spivak-invitation_F00736 | 68   | ■ 1.000 | $(X, \geq, I, \otimes)$                                                                                                          | $(X, \geq, I, \otimes)$                                            | $(X, \geq, I, \otimes)$ .                                                                                                                                      |
| fong-spivak-invitation_F00737 | 68   | ■ 0.481 | $x_{\{1\}} \geq y_{\{1\}}$                                                                                                       | $x_1 \geq y_1$                                                     | <i>Proof.</i> Let's first check monotonicity. Suppose $x_1 \geq y_1$ and $x_2 \geq y_2$ in $\mathcal{X}^{\text{op}}$ ; we need                                 |
| fong-spivak-invitation_F00738 | 68   | ■ 0.481 | $x_{\{2\}} \geq y_{\{2\}}$                                                                                                       | $x_2 \geq y_2$                                                     | <i>Proof.</i> Let's first check monotonicity. Suppose $x_1 \geq y_1$ and $x_2 \geq y_2$ in $\mathcal{X}^{\text{op}}$ ; we need                                 |
| fong-spivak-invitation_F00739 | 68   | ■ 0.481 | $\mathcal{X}^{\{\text{op}\}}$                                                                                                    | $\mathcal{X}^{\text{op}}$                                          | <i>Proof.</i> Let's first check monotonicity. Suppose $x_1 \geq y_1$ and $x_2 \geq y_2$ in $\mathcal{X}^{\text{op}}$ ; we need                                 |
| fong-spivak-invitation_F00740 | 68   | ■ 1.000 | $x_{\{1\}} \otimes x_{\{2\}} \geq y_{\{1\}} \otimes y_{\{2\}}$                                                                   | $x_1 \otimes x_2 \geq y_1 \otimes y_2$                             | to show that $x_1 \otimes x_2 \geq y_1 \otimes y_2$ . But by definition of opposite order, we have $y_1 \leq x_1$                                              |
| fong-spivak-invitation_F00741 | 68   | ■ 1.000 | $y_{\{1\}} \leq x_{\{1\}}$                                                                                                       | $y_1 \leq x_1$                                                     | to show that $x_1 \otimes x_2 \geq y_1 \otimes y_2$ . But by definition of opposite order, we have $y_1 \leq x_1$                                              |
| fong-spivak-invitation_F00742 | 68   | ■ 0.925 | $y_{\{2\}} \leq x_{\{2\}}$                                                                                                       | $y_2 \leq x_2$                                                     | and $y_2 \leq x_2$ in $\mathcal{X}$ , and thus $y_1 \otimes y_2 \leq x_1 \otimes x_2$ in $\mathcal{X}$ . Thus indeed $x_1 \otimes x_2 \geq y_1 \otimes y_2$ in |
| fong-spivak-invitation_F00743 | 68   | ■ 0.925 | $y_{\{1\}} \otimes y_{\{2\}} \leq x_{\{1\}} \otimes x_{\{2\}}$                                                                   | $y_1 \otimes y_2 \leq x_1 \otimes x_2$                             | and $y_2 \leq x_2$ in $\mathcal{X}$ , and thus $y_1 \otimes y_2 \leq x_1 \otimes x_2$ in $\mathcal{X}$ . Thus indeed $x_1 \otimes x_2 \geq y_1 \otimes y_2$ in |
| fong-spivak-invitation_F00744 | 68   | ■ 0.989 | $\{\}^{\{\mathrm{op}\}}$                                                                                                         | $^{\text{op}}$                                                     | <b>Cost</b> <sup>op</sup> is too.                                                                                                                              |
| fong-spivak-invitation_F00745 | 68   | ■ 1.000 | $x \cong x^{\{\mathrm{prime}\}}$                                                                                                 | $x \cong x'$                                                       | relation $\cong$ on $X$ , where $x \cong x'$ iff both $x \leq x'$ and $x' \leq x$ .                                                                            |
| fong-spivak-invitation_F00746 | 68   | ■ 1.000 | $x \leq x^{\{\mathrm{prime}\}}$                                                                                                  | $x \leq x'$                                                        | relation $\cong$ on $X$ , where $x \cong x'$ iff both $x \leq x'$ and $x' \leq x$ .                                                                            |
| fong-spivak-invitation_F00747 | 68   | ■ 1.000 | $x^{\{\mathrm{prime}\}} \leq x$                                                                                                  | $x' \leq x$                                                        | relation $\cong$ on $X$ , where $x \cong x'$ iff both $x \leq x'$ and $x' \leq x$ .                                                                            |
| fong-spivak-invitation_F00748 | 68   | ■ 0.904 | $\mathcal{P}=\left(P, \leq_{\{P\}}, I_{\{P\}}, \otimes_{\{P\}}\right)$                                                           | $\mathcal{P} = (P, \leq_P, I_P, \otimes_P)$                        | <b>Definition 2.25.</b> Let $\mathcal{P} = (P, \leq_P, I_P, \otimes_P)$ and $\mathcal{Q} = (Q, \leq_Q, I_Q, \otimes_Q)$ be monoidal                            |
| fong-spivak-invitation_F00749 | 68   | ■ 0.904 | $\mathcal{Q}=\left(Q, \leq_{\{Q\}}, I_{\{Q\}}, \otimes_{\{Q\}}\right)$                                                           | $\mathcal{Q} = (Q, \leq_Q, I_Q, \otimes_Q)$                        | <b>Definition 2.25.</b> Let $\mathcal{P} = (P, \leq_P, I_P, \otimes_P)$ and $\mathcal{Q} = (Q, \leq_Q, I_Q, \otimes_Q)$ be monoidal                            |
| fong-spivak-invitation_F00750 | 68   | ■ 0.945 | $\mathcal{P}$                                                                                                                    | $\mathcal{P}$                                                      | preorders. A <i>monoidal monotone</i> from $\mathcal{P}$ to $\mathcal{Q}$ is a monotone map $f: (P, \leq_P) \rightarrow$                                       |
| fong-spivak-invitation_F00751 | 68   | ■ 0.945 | $f:\left(P, \leq_{\{P\}}\right) \rightarrow$                                                                                     | $f: (P, \leq_P) \rightarrow$                                       | preorders. A <i>monoidal monotone</i> from $\mathcal{P}$ to $\mathcal{Q}$ is a monotone map $f: (P, \leq_P) \rightarrow$                                       |
| fong-spivak-invitation_F00752 | 68   | ■ 1.000 | $I_{\{Q\}} \leq_{\{Q\}} f\left(I_{\{P\}}\right)$                                                                                 | $I_Q \leq_Q f(I_P)$                                                | (a) $I_Q \leq_Q f(I_P)$ ,                                                                                                                                      |
| fong-spivak-invitation_F00753 | 68   | ■ 1.000 | $f\left(p_{\{1\}}\right) \otimes_{\{Q\}} f\left(p_{\{2\}}\right) \leq_{\{Q\}} f\left(p_{\{1\}} \otimes_{\{P\}} p_{\{2\}}\right)$ | $f(p_1) \otimes_Q f(p_2) \leq_Q f(p_1 \otimes_P p_2)$              | (b) $f(p_1) \otimes_Q f(p_2) \leq_Q f(p_1 \otimes_P p_2)$ ,                                                                                                    |
| fong-spivak-invitation_F00754 | 68   | ■ 1.000 | $p_{\{1\}}, p_{\{2\}} \in P$                                                                                                     | $p_1, p_2 \in P$                                                   | for all $p_1, p_2 \in P$ .                                                                                                                                     |
| fong-spivak-invitation_F00755 | 68   | ■ 0.970 | $I_{\{Q\}} \cong f\left(I_{\{P\}}\right)$                                                                                        | $I_Q \cong f(I_P)$                                                 | (a') $I_Q \cong f(I_P)$ ,                                                                                                                                      |
| fong-spivak-invitation_F00756 | 68   | ■ 1.000 | $f\left(p_{\{1\}}\right) \otimes_{\{Q\}} f\left(p_{\{2\}}\right) \cong f\left(p_{\{1\}} \otimes_{\{P\}} p_{\{2\}}\right)$        | $f(p_1) \otimes_Q f(p_2) \cong f(p_1 \otimes_P p_2)$               | (b') $f(p_1) \otimes_Q f(p_2) \cong f(p_1 \otimes_P p_2)$ ;                                                                                                    |
| fong-spivak-invitation_F00757 | 68   | ■ 0.507 | $\left(\mathrm{a}^{\{\mathrm{prime}\}}\right) I_{\{Q\}}=f\left(I_{\{P\}}\right)$                                                 | (a'') $I_Q = f(I_P)$                                               | (a'') $I_Q = f(I_P)$ ,                                                                                                                                         |
| fong-spivak-invitation_F00758 | 68   | ■ 0.998 | $f\left(p_{\{1\}}\right) \otimes_{\{Q\}} f\left(p_{\{2\}}\right)=f\left(p_{\{1\}} \otimes_{\{P\}} p_{\{2\}}\right)$              | $f(p_1) \otimes_Q f(p_2) = f(p_1 \otimes_P p_2)$                   | (b'') $f(p_1) \otimes_Q f(p_2) = f(p_1 \otimes_P p_2)$ .                                                                                                       |
| fong-spivak-invitation_F00759 | 69   | ■ 1.000 | $i:\left(\mathbb{Z}, \leq, 0,+\right) \rightarrow\left(\mathbb{R}, \leq, 0,+\right)$                                             | $i: (\mathbb{Z}, \leq, 0, +) \rightarrow (\mathbb{R}, \leq, 0, +)$ | <b>Example 2.26.</b> There is a monoidal monotone $i: (\mathbb{Z}, \leq, 0, +) \rightarrow (\mathbb{R}, \leq, 0, +)$ , where                                   |

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| fong-spivak-invitation_F00760 | 69   | ■ 1.000 | $i(n)=n$                                                           | $i(n) = n$                                                         | $i(n) = n$ for all $n \in \mathbb{Z}$ . It is clearly monotonic, $m \leq n$ implies $i(m) \leq i(n)$ . It is even                                 |
| fong-spivak-invitation_F00761 | 69   | ■ 1.000 | $n \in \mathbb{Z}$                                                 | $n \in \mathbb{Z}$                                                 | $i(n) = n$ for all $n \in \mathbb{Z}$ . It is clearly monotonic, $m \leq n$ implies $i(m) \leq i(n)$ . It is even                                 |
| fong-spivak-invitation_F00762 | 69   | ■ 1.000 | $i(m) \leq i(n)$                                                   | $i(m) \leq i(n)$                                                   | $i(n) = n$ for all $n \in \mathbb{Z}$ . It is clearly monotonic, $m \leq n$ implies $i(m) \leq i(n)$ . It is even                                 |
| fong-spivak-invitation_F00763 | 69   | ■ 0.997 | $i(0)=0$                                                           | $i(0) = 0$                                                         | strict monoidal because $i(0) = 0$ and $i(m+n) = i(m) + i(n)$ .                                                                                   |
| fong-spivak-invitation_F00764 | 69   | ■ 0.997 | $i(m+n)=i(m)+i(n)$                                                 | $i(m+n) = i(m) + i(n)$                                             | strict monoidal because $i(0) = 0$ and $i(m+n) = i(m) + i(n)$ .                                                                                   |
| fong-spivak-invitation_F00765 | 69   | ■ 1.000 | $f: (\mathbb{R}, \leq, 0, +) \rightarrow (\mathbb{Z}, \leq, 0, +)$ | $f: (\mathbb{R}, \leq, 0, +) \rightarrow (\mathbb{Z}, \leq, 0, +)$ | There is also a monoidal monotone $f: (\mathbb{R}, \leq, 0, +) \rightarrow (\mathbb{Z}, \leq, 0, +)$ going the other                              |
| fong-spivak-invitation_F00766 | 69   | ■ 1.000 | $f(x) := \lfloor x \rfloor$                                        | $f(x) := \lfloor x \rfloor$                                        | way. Here $f(x) := \lfloor x \rfloor$ is the floor function, e.g. $f(3.14) = 3$ . It is monotonic because                                         |
| fong-spivak-invitation_F00767 | 69   | ■ 1.000 | $f(3.14)=3$                                                        | $f(3.14) = 3$                                                      | way. Here $f(x) := \lfloor x \rfloor$ is the floor function, e.g. $f(3.14) = 3$ . It is monotonic because                                         |
| fong-spivak-invitation_F00768 | 69   | ■ 1.000 | $f(0)=0$                                                           | $f(0) = 0$                                                         | $x \leq y$ implies $f(x) \leq f(y)$ . Also $f(0) = 0$ and $f(x) + f(y) \leq f(x+y)$ , so it is                                                    |
| fong-spivak-invitation_F00769 | 69   | ■ 1.000 | $f(x)+f(y) \leq f(x+y)$                                            | $f(x) + f(y) \leq f(x+y)$                                          | $x \leq y$ implies $f(x) \leq f(y)$ . Also $f(0) = 0$ and $f(x) + f(y) \leq f(x+y)$ , so it is                                                    |
| fong-spivak-invitation_F00770 | 69   | ■ 1.000 | $f(0.5)+f(0.5) \neq$                                               | $f(0.5) + f(0.5) \neq$                                             | a monoidal monotone. But it is not strict or even strong because $f(0.5) + f(0.5) \neq$                                                           |
| fong-spivak-invitation_F00771 | 69   | ■ 1.000 | $f(0.5+0.5)$                                                       | $f(0.5 + 0.5)$                                                     | $f(0.5 + 0.5)$ .                                                                                                                                  |
| fong-spivak-invitation_F00772 | 69   | ■ 0.970 | $=([0, \infty], \geq, 0, +)$                                       | $= ([0, \infty], \geq, 0, +)$                                      | Recall <b>Bool</b> = $(\mathbb{B}, \leq, \text{true}, \wedge)$ from Example 2.12 and <b>Cost</b> = $([0, \infty], \geq, 0, +)$                    |
| fong-spivak-invitation_F00773 | 69   | ■ 0.835 | $:=\infty$                                                         | $:= \infty$                                                        | $g(\text{false}) := \infty$ and $g(\text{true}) := 0$ .                                                                                           |
| fong-spivak-invitation_F00774 | 69   | ■ 0.835 | $:=0$                                                              | $:= 0$                                                             | $g(\text{false}) := \infty$ and $g(\text{true}) := 0$ .                                                                                           |
| fong-spivak-invitation_F00775 | 69   | ■ 0.483 | $g: (\mathbb{B}, \leq$                                             | $g: (\mathbb{B}, \leq$                                             | 1. Check that the map $g: (\mathbb{B}, \leq, \text{true}, \wedge) \rightarrow ([0, \infty], \geq, 0, +)$ presented above                          |
| fong-spivak-invitation_F00776 | 69   | ■ 0.483 | $\wedge) \rightarrow ([0, \infty], \geq, 0, +)$                    | $\wedge) \rightarrow ([0, \infty], \geq, 0, +)$                    | 1. Check that the map $g: (\mathbb{B}, \leq, \text{true}, \wedge) \rightarrow ([0, \infty], \geq, 0, +)$ presented above                          |
| fong-spivak-invitation_F00777 | 69   | ■ 1.000 | $d, u: [0, \infty] \rightarrow \mathbb{B}$                         | $d, u: [0, \infty] \rightarrow \mathbb{B}$                         | functions $d, u: [0, \infty] \rightarrow \mathbb{B}$ defined as follows:                                                                          |
| fong-spivak-invitation_F00778 | 69   | ■ 1.000 | $u$                                                                | $u$                                                                | 4. Is $u$ monotonic?                                                                                                                              |
| fong-spivak-invitation_F00779 | 69   | ■ 0.860 | $(\mathbb{N}, \leq, 1, *)$                                         | $(\mathbb{N}, \leq, 1, *)$                                         | 1. Is $(\mathbb{N}, \leq, 1, *)$ a monoidal preorder, where $*$ is the usual multiplication of natural                                            |
| fong-spivak-invitation_F00780 | 69   | ■ 1.000 | $(\mathbb{N}, \leq, 0, +) \rightarrow$                             | $(\mathbb{N}, \leq, 0, +) \rightarrow$                             | 2. If not, why not? If so, does there exist a monoidal monotone $(\mathbb{N}, \leq, 0, +) \rightarrow$                                            |
| fong-spivak-invitation_F00781 | 69   | ■ 0.988 | $\mathbb{Z}, \leq, 1, *$                                           | $\mathbb{Z}, \leq, 1, *$                                           | 3. Is $(\mathbb{Z}, \leq, 1, *)$ a monoidal preorder?                                                                                             |
| fong-spivak-invitation_F00782 | 70   | ■ 0.763 | $\text{Ob}(\mathcal{X})$                                           | $\text{Ob}(\mathcal{X})$                                           | (i) one specifies a set $\text{Ob}(\mathcal{X})$ , elements of which are called <i>objects</i> ;                                                  |
| fong-spivak-invitation_F00783 | 70   | ■ 1.000 | $\mathcal{X}(x, y) \in V$                                          | $\mathcal{X}(x, y) \in V$                                          | (ii) for every two objects $x, y$ , one specifies an element $\mathcal{X}(x, y) \in V$ , called the <i>hom-</i>                                   |
| fong-spivak-invitation_F00784 | 70   | ■ 0.622 | $x \in \text{Ob}(\mathcal{X})$                                     | $x \in \text{Ob}(\mathcal{X})$                                     | (a) for every object $x \in \text{Ob}(\mathcal{X})$ we have $I \leq \mathcal{X}(x, x)$ ,                                                          |
| fong-spivak-invitation_F00785 | 70   | ■ 0.622 | $I \leq \mathcal{X}(x, x)$                                         | $I \leq \mathcal{X}(x, x)$                                         | (a) for every object $x \in \text{Ob}(\mathcal{X})$ we have $I \leq \mathcal{X}(x, x)$ ,                                                          |
| fong-spivak-invitation_F00786 | 70   | ■ 0.288 | $x, y, z \in \text{Ob}(\mathcal{X})$                               | $x, y, z \in \text{Ob}(\mathcal{X})$                               | (b) for every three objects $x, y, z \in \text{Ob}(\mathcal{X})$ , we have $\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$ . |

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| fong-spivak-invitation_F00787 | 70   | ■ 0.288 | $\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$ | $\mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \mathfrak{X}(x, z)$ | (b) for every three objects $x, y, z \in \text{Ob}(\mathcal{X})$ , we have $\mathfrak{X}(x, y) \otimes \mathfrak{X}(y, z) \leq \mathfrak{X}(x, z)$ . |
| fong-spivak-invitation_F00788 | 70   | ■ 0.848 | $\mathfrak{X}$                                                       | $\mathfrak{X}$                                                        | How does this correspond to a <b>Bool</b> -category $\mathfrak{X}$ ? Well, the objects of $\mathfrak{X}$ are simply                                  |
| fong-spivak-invitation_F00789 | 70   | ■ 0.751 | $\text{Ob}(\mathcal{X}) = \{p, q, r, s, t\}$                         | $\text{Ob}(\mathcal{X}) = \{p, q, r, s, t\}$                          | the elements of the preorder, i.e. $\text{Ob}(\mathfrak{X}) = \{p, q, r, s, t\}$ . Next, for every pair of objects                                   |
| fong-spivak-invitation_F00790 | 70   | ■ 0.940 | $\mathcal{X}(x, y)$                                                  | $\mathcal{X}(x, y)$                                                   | name for $\mathfrak{X}(x, y)$ might be <i>mapping object</i> , but we use “hom” mainly because it is an important jargon                             |
| fong-spivak-invitation_F00791 | 71   | ■ 0.998 | $t \not\leq s$                                                       | $t \not\leq s$                                                        | false otherwise. So, for example, since $s \leq t$ and $t \not\leq s$ , we have $\mathfrak{X}(s, t) = \text{true}$                                   |
| fong-spivak-invitation_F00792 | 71   | ■ 0.998 | $\mathcal{X}(s, t) =$                                                | $\mathcal{X}(s, t) =$                                                 | false otherwise. So, for example, since $s \leq t$ and $t \not\leq s$ , we have $\mathfrak{X}(s, t) = \text{true}$                                   |
| fong-spivak-invitation_F00793 | 71   | ■ 0.998 | $\mathcal{X}(t, s) =$                                                | $\mathcal{X}(t, s) =$                                                 | and $\mathfrak{X}(t, s) = \text{false}$ . Recalling from Example 2.12 that the monoidal unit $I$ of <b>Bool</b>                                      |
| fong-spivak-invitation_F00794 | 132  | ■ 1.000 | $r$                                                                  | $r$                                                                   | infeasible – for that pair. In other words, “yes I can provide $p$ given $r$ ” or “no, I cannot                                                      |
| fong-spivak-invitation_F00795 | 71   | ■ 0.538 | $\text{Ob}(\mathfrak{X})$                                            | $\text{Ob}(\mathfrak{X})$                                             | (i) a set $\text{Ob}(\mathfrak{X})$ ,                                                                                                                |
| fong-spivak-invitation_F00796 | 71   | ■ 0.656 | $x, y \in \text{Ob}(\mathfrak{X})$                                   | $x, y \in \text{Ob}(\mathfrak{X})$                                    | (ii) for every $x, y \in \text{Ob}(\mathfrak{X})$ an element $\mathfrak{X}(x, y) \in \mathbb{B}$ , i.e. the element $\mathfrak{X}(x, y)$ is either   |
| fong-spivak-invitation_F00797 | 71   | ■ 0.656 | $\mathcal{X}(x, y) \in \mathbb{B}$                                   | $\mathcal{X}(x, y) \in \mathbb{B}$                                    | (ii) for every $x, y \in \text{Ob}(\mathfrak{X})$ an element $\mathfrak{X}(x, y) \in \mathbb{B}$ , i.e. the element $\mathfrak{X}(x, y)$ is either   |
| fong-spivak-invitation_F00798 | 71   | ■ 0.510 | $X := \text{Ob}(\mathfrak{X})$                                       | $X := \text{Ob}(\mathfrak{X})$                                        | of $\mathfrak{X}$ . So let’s call the preorder $(X, \leq)$ , and let $X := \text{Ob}(\mathfrak{X})$ . For the $\leq$ relation, let’s                 |
| fong-spivak-invitation_F00799 | 71   | ■ 0.932 | $\mathcal{X}(x, y) =$                                                | $\mathcal{X}(x, y) =$                                                 | declare $x \leq y$ iff $\mathfrak{X}(x, y) = \text{true}$ . We have the makings of a preorder, but for it to                                         |
| fong-spivak-invitation_F00800 | 72   | ■ 0.767 | $\leq \mathcal{X}(x, x)$                                             | $\leq \mathcal{X}(x, x)$                                              | (a) for every element $x \in X$ we have $\text{true} \leq \mathfrak{X}(x, x)$ ,                                                                      |
| fong-spivak-invitation_F00801 | 72   | ■ 0.835 | $\mathcal{X}(x, y) \wedge \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$  | $\mathcal{X}(x, y) \wedge \mathcal{X}(y, z) \leq \mathcal{X}(x, z)$   | (b) for every three elements $x, y, z \in X$ we have $\mathfrak{X}(x, y) \wedge \mathfrak{X}(y, z) \leq \mathfrak{X}(x, z)$ .                        |
| fong-spivak-invitation_F00802 | 72   | ■ 0.943 | $b \in$                                                              | $b \in$                                                               | For $b \in \mathbf{Bool}$ , if $\text{true} \leq b$ then $b = \text{true}$ , so the first statement says $\mathfrak{X}(x, x) =$                      |
| fong-spivak-invitation_F00803 | 72   | ■ 0.943 | $\leq b$                                                             | $\leq b$                                                              | For $b \in \mathbf{Bool}$ , if $\text{true} \leq b$ then $b = \text{true}$ , so the first statement says $\mathfrak{X}(x, x) =$                      |
| fong-spivak-invitation_F00804 | 72   | ■ 0.943 | $b =$                                                                | $b =$                                                                 | For $b \in \mathbf{Bool}$ , if $\text{true} \leq b$ then $b = \text{true}$ , so the first statement says $\mathfrak{X}(x, x) =$                      |
| fong-spivak-invitation_F00805 | 72   | ■ 0.943 | $\mathcal{X}(x, x) =$                                                | $\mathcal{X}(x, x) =$                                                 | For $b \in \mathbf{Bool}$ , if $\text{true} \leq b$ then $b = \text{true}$ , so the first statement says $\mathfrak{X}(x, x) =$                      |
| fong-spivak-invitation_F00806 | 72   | ■ 0.555 | $b \in \mathbb{B}$                                                   | $b \in \mathbb{B}$                                                    | $\text{false} \leq b$ for all $b \in \mathbb{B}$ , the only way statement (b) has any force is if $\mathfrak{X}(x, y) = \text{true}$                 |
| fong-spivak-invitation_F00807 | 72   | ■ 0.555 | $\mathcal{X}(x, y) = \text{tr}$                                      | $\mathcal{X}(x, y) = \text{tr}$                                       | $\text{false} \leq b$ for all $b \in \mathbb{B}$ , the only way statement (b) has any force is if $\mathfrak{X}(x, y) = \text{true}$                 |
| fong-spivak-invitation_F00808 | 72   | ■ 0.994 | $\mathcal{X}(y, z) =$                                                | $\mathcal{X}(y, z) =$                                                 | and $\mathfrak{X}(y, z) = \text{true}$ , in which case it forces $\mathfrak{X}(x, z) = \text{true}$ . This condition exactly                         |
| fong-spivak-invitation_F00809 | 72   | ■ 0.994 | $\mathcal{X}(x, z) =$                                                | $\mathcal{X}(x, z) =$                                                 | and $\mathfrak{X}(y, z) = \text{true}$ , in which case it forces $\mathfrak{X}(x, z) = \text{true}$ . This condition exactly                         |
| fong-spivak-invitation_F00810 | 72   | ■ 0.777 | $X, d$                                                               | $X, d$                                                                | <b>Definition 2.34.</b> A metric space $(X, d)$ consists of:                                                                                         |
| fong-spivak-invitation_F00811 | 72   | ■ 1.000 | $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$                      | $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$                       | (ii) a function $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$ , where $d(x, y)$ is called the <i>distance between</i> $x$                          |
| fong-spivak-invitation_F00812 | 72   | ■ 1.000 | $d(x, y)$                                                            | $d(x, y)$                                                             | (ii) a function $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$ , where $d(x, y)$ is called the <i>distance between</i> $x$                          |
| fong-spivak-invitation_F00813 | 72   | ■ 1.000 | $d(x, x) = 0$                                                        | $d(x, x) = 0$                                                         | (a) for every $x \in X$ , we have $d(x, x) = 0$ ,                                                                                                    |

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| fong-spivak-invitation_F00814 | 72   | ■ 1.000 | $d(x, y)=0$                                                                   | $d(x, y) = 0$                                                                 | (b) for every $x, y \in X$ , if $d(x, y) = 0$ then $x = y$ ,                                                                        |
| fong-spivak-invitation_F00815 | 72   | ■ 1.000 | $d(x, y)=d(y, x)$                                                             | $d(x, y) = d(y, x)$                                                           | (c) for every $x, y \in X$ , we have $d(x, y) = d(y, x)$ ,                                                                          |
| fong-spivak-invitation_F00816 | 72   | ■ 1.000 | $d(x, y)+d(y, z) \geq d(x, z)$                                                | $d(x, y) + d(y, z) \geq d(x, z)$                                              | (d) for every $x, y, z \in X$ , we have $d(x, y) + d(y, z) \geq d(x, z)$ .                                                          |
| fong-spivak-invitation_F00817 | 72   | ■ 1.000 | $d: X \times X \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$ | $d: X \times X \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$ | If we ask instead in (ii) for a function $d: X \times X \rightarrow [0, \infty] = \mathbb{R}_{\geq 0} \cup \{\infty\}$ we call      |
| fong-spivak-invitation_F00818 | 72   | ■ 0.672 | $(X, d)$                                                                      | $(X, d)$                                                                      | $(X, d)$ an <i>extended metric space</i> .                                                                                          |
| fong-spivak-invitation_F00819 | 73   | ■ 1.000 | $z$                                                                           | $z$                                                                           | The triangle inequality says that when plotting a route from $x$ to $z$ , the distance is                                           |
| fong-spivak-invitation_F00820 | 73   | ■ 1.000 | $x \rightarrow y \rightarrow z$                                               | $x \rightarrow y \rightarrow z$                                               | always at most what you get by choosing an intermediate point $y$ and going $x \rightarrow y \rightarrow z$ .                       |
| fong-spivak-invitation_F00821 | 73   | ■ 1.000 | $3+5 \geq 7.2$                                                                | $3 + 5 \geq 7.2$                                                              | It can be invoked in three different ways in the above picture: $3 + 5 \geq 7.2$ , but also                                         |
| fong-spivak-invitation_F00822 | 73   | ■ 0.997 | $5+7.2 \geq 3$                                                                | $5 + 7.2 \geq 3$                                                              | $5 + 7.2 \geq 3$ , and $3 + 7.2 \geq 5$ . Oh yeah, and $5 + 3 \geq 7.2$ , $7.2 + 5 \geq 3$ , and $7.2 + 3 \geq 5$ .                 |
| fong-spivak-invitation_F00823 | 73   | ■ 0.997 | $3+7.2 \geq 5$                                                                | $3 + 7.2 \geq 5$                                                              | $5 + 7.2 \geq 3$ , and $3 + 7.2 \geq 5$ . Oh yeah, and $5 + 3 \geq 7.2$ , $7.2 + 5 \geq 3$ , and $7.2 + 3 \geq 5$ .                 |
| fong-spivak-invitation_F00824 | 73   | ■ 0.997 | $5+3 \geq 7.2, 7.2+5 \geq 3$                                                  | $5 + 3 \geq 7.2, 7.2 + 5 \geq 3$                                              | $5 + 7.2 \geq 3$ , and $3 + 7.2 \geq 5$ . Oh yeah, and $5 + 3 \geq 7.2, 7.2 + 5 \geq 3$ , and $7.2 + 3 \geq 5$ .                    |
| fong-spivak-invitation_F00825 | 73   | ■ 0.997 | $7.2+3 \geq 5$                                                                | $7.2 + 3 \geq 5$                                                              | $5 + 7.2 \geq 3$ , and $3 + 7.2 \geq 5$ . Oh yeah, and $5 + 3 \geq 7.2, 7.2 + 5 \geq 3$ , and $7.2 + 3 \geq 5$ .                    |
| fong-spivak-invitation_F00826 | 73   | ■ 0.528 | $d(x, y)=? d(y, x)$                                                           | $d(x, y) =? d(y, x)$                                                          | are any hills, the symmetry axiom, $d(x, y) =? d(y, x)$ , fails: it's easier to get from $x$                                        |
| fong-spivak-invitation_F00827 | 73   | ■ 0.358 | $d($                                                                          | $d($                                                                          | It also violates another condition, because $d(\text{Boston}, \text{US}) = 0$ . No matter where you                                 |
| fong-spivak-invitation_F00828 | 73   | ■ 0.374 | $) \neq 0$                                                                    | $) \neq 0$                                                                    | $d(\text{US}, \text{Boston}) \neq 0$ .                                                                                              |
| fong-spivak-invitation_F00829 | 73   | ■ 0.982 | $U \subseteq X$                                                               | $U \subseteq X$                                                               | <sup>3</sup> The Hausdorff distance gives a metric on the set of all subsets $U \subseteq X$ of a given metric space $(X, d)$ . One |
| fong-spivak-invitation_F00830 | 73   | ■ 1.000 | $d(U, V) := \max(d_L(U, V), d_L(V, U))$                                       | $d(U, V) := \max(d_L(U, V), d_L(V, U))$                                       | formula and symmetrize it: $d(U, V) := \max(d_L(U, V), d_L(V, U))$ . We happen to see the unsymmetrized                             |
| fong-spivak-invitation_F00831 | 74   | ■ 0.958 | $\{r \in \mathbb{R} \mid r < 0\}$                                             | $\{r \in \mathbb{R} \mid r < 0\}$                                             | two regions is often infinite, e.g. the distance between $\{r \in \mathbb{R} \mid r < 0\}$ and $\{0\}$ as subsets                   |
| fong-spivak-invitation_F00832 | 74   | ■ 0.958 | $\{0\}$                                                                       | $\{0\}$                                                                       | two regions is often infinite, e.g. the distance between $\{r \in \mathbb{R} \mid r < 0\}$ and $\{0\}$ as subsets                   |
| fong-spivak-invitation_F00833 | 74   | ■ 0.725 | $(\mathbb{R}, d)$                                                             | $(\mathbb{R}, d)$                                                             | of $(\mathbb{R}, d)$ is infinite.                                                                                                   |
| fong-spivak-invitation_F00834 | 74   | ■ 0.578 | $x, y \in \text{Ob}(\mathcal{X})$                                             | $x, y \in \text{Ob}(\mathcal{X})$                                             | (ii) for every $x, y \in \text{Ob}(\mathcal{X})$ an element $\mathcal{X}(x, y) \in [0, \infty]$ .                                   |
| fong-spivak-invitation_F00835 | 74   | ■ 0.578 | $\mathcal{X}(x, y) \in [0, \infty]$                                           | $\mathcal{X}(x, y) \in [0, \infty]$                                           | (ii) for every $x, y \in \text{Ob}(\mathcal{X})$ an element $\mathcal{X}(x, y) \in [0, \infty]$ .                                   |
| fong-spivak-invitation_F00836 | 74   | ■ 1.000 | $0 \geq d(x, x)$                                                              | $0 \geq d(x, x)$                                                              | (a) $0 \geq d(x, x)$ for all $x \in X$ ,                                                                                            |
| fong-spivak-invitation_F00837 | 74   | ■ 1.000 | $d(x, x) \in [0, \infty]$                                                     | $d(x, x) \in [0, \infty]$                                                     | Since $d(x, x) \in [0, \infty]$ , if $0 \geq d(x, x)$ then $d(x, x) = 0$ . So the first condition is                                |
| fong-spivak-invitation_F00838 | 74   | ■ 1.000 | $d(x, y) :=  y - x $                                                          | $d(x, y) :=  y - x $                                                          | hence a Lawvere metric space structure. Namely $d(x, y) :=  y - x $ , the absolute value                                            |
| fong-spivak-invitation_F00839 | 74   | ■ 1.000 | $d(3, 7)=4$                                                                   | $d(3, 7) = 4$                                                                 | of the difference. So $d(3, 7) = 4$ .                                                                                               |
| fong-spivak-invitation_F00840 | 74   | ■ 0.953 | $\mathbb{R}_{\geq 0}, \geq, 0, +$                                             | $\mathbb{R}_{\geq 0}, \geq, 0, +$                                             | <b>Exercise 2.38.</b> Consider the symmetric monoidal preorder $(\mathbb{R}_{>0}, \geq, 0, +)$ , which is                           |

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| fong-spivak-invitation_F00841 | 74   | ■ 0.994 | $\left(\mathbb{R}_{\geq 0}, \geq, 0, +\right)$                    | $(\mathbb{R}_{\geq 0}, \geq, 0, +)$                               | difference between a Lawvere metric space and a $(\mathbb{R}_{\geq 0}, \geq, 0, +)$ -category in the sense                                     |
| fong-spivak-invitation_F00842 | 74   | ■ 0.998 | $w \geq 0$                                                        | $w \geq 0$                                                        | graph – a graph whose edges are labeled with numbers $w \geq 0$ – into a Lawvere metric                                                        |
| fong-spivak-invitation_F00843 | 75   | ■ 1.000 | $d_{\{X\}}$                                                       | $d_X$                                                             | Given a weighted graph, one forms a metric $d_X$ on its set $X$ of vertices by setting $d(p, q)$                                               |
| fong-spivak-invitation_F00844 | 75   | ■ 1.000 | $d(p, q)$                                                         | $d(p, q)$                                                         | Given a weighted graph, one forms a metric $d_X$ on its set $X$ of vertices by setting $d(p, q)$                                               |
| fong-spivak-invitation_F00846 | 106  | ■ 1.000 | $D$                                                               | $D$                                                               | $C$ , and $D'$ to $D$ . That is, once we have made this decision about how $F$ acts on objects,                                                |
| fong-spivak-invitation_F00847 | 75   | ■ 1.000 | $M_{\{G\}}$                                                       | $M_G$                                                             | $G$ and getting a square matrix $M_G$ , whose rows and columns are indexed by the vertices                                                     |
| fong-spivak-invitation_F00848 | 76   | ■ 1.000 | $d_{\{Y\}}$                                                       | $d_Y$                                                             | that the more difficult “distance matrices” $d_Y$ , such as (2.19), can be obtained from                                                       |
| fong-spivak-invitation_F00849 | 76   | ■ 1.000 | $M_{\{Y\}}$                                                       | $M_Y$                                                             | the easy graph matrices $M_Y$ , such as (2.20), by repeating a certain sort of “matrix                                                         |
| fong-spivak-invitation_F00850 | 76   | ■ 1.000 | $M_{\{X\}}$                                                       | $M_X$                                                             | <b>Exercise 2.40.</b> Copy and complete the matrix $M_X$ associated to the graph $X$ in                                                        |
| fong-spivak-invitation_F00852 | 76   | ■ 0.731 | $\mathcal{M} := (\mathrm{P}(M), \subseteq, M, \cap)$              | $\mathcal{M} := (\mathrm{P}(M), \subseteq, M, \cap)$              | <b>Exercise 2.42.</b> Let $M$ be a set and let $\mathcal{M} := (\mathrm{P}(M), \subseteq, M, \cap)$ be the monoidal preorder                   |
| fong-spivak-invitation_F00853 | 76   | ■ 1.000 | $\mathcal{M}$                                                     | $\mathcal{M}$                                                     | of modes of transportation (e.g. car, boat, foot). Then an $\mathcal{M}$ -category $\mathcal{X}$ tells you                                     |
| fong-spivak-invitation_F00854 | 76   | ■ 1.000 | $a, b \in$                                                        | $a, b \in$                                                        | all the modes that will get you from $a$ all the way to $b$ , for any two points $a, b \in$                                                    |
| fong-spivak-invitation_F00855 | 77   | ■ 0.981 | $M =$                                                             | $M =$                                                             | $M = \{\text{car, boat, foot}\}.$                                                                                                              |
| fong-spivak-invitation_F00856 | 77   | ■ 0.925 | $\mathcal{W} := (\mathbb{N} \cup \{\infty\}, \leq, \infty, \min)$ | $\mathcal{W} := (\mathbb{N} \cup \{\infty\}, \leq, \infty, \min)$ | <b>Exercise 2.43.</b> Consider the monoidal preorder $\mathcal{W} := (\mathbb{N} \cup \{\infty\}, \leq, \infty, \min)$ .                       |
| fong-spivak-invitation_F00857 | 77   | ■ 1.000 | $\mathbb{N} \cup \{\infty\}$                                      | $\mathbb{N} \cup \{\infty\}$                                      | 1. Draw a small graph labeled by elements of $\mathbb{N} \cup \{\infty\}$ .                                                                    |
| fong-spivak-invitation_F00858 | 77   | ■ 1.000 | $\mathcal{W}$                                                     | $\mathcal{W}$                                                     | 3. Prove that this matrix is the matrix of hom-objects for a $\mathcal{W}$ -category. This will give                                           |
| fong-spivak-invitation_F00859 | 77   | ■ 0.998 | $f: \mathcal{V} \rightarrow \mathcal{W}$                          | $f: \mathcal{V} \rightarrow \mathcal{W}$                          | base. This allows us to use a monoidal monotone $f: \mathcal{V} \rightarrow \mathcal{W}$ to construct $\mathcal{W}$ -categories                |
| fong-spivak-invitation_F00860 | 77   | ■ 1.000 | $\mathcal{V} \rightarrow \mathcal{W}$                             | $\mathcal{V} \rightarrow \mathcal{W}$                             | Any monoidal monotone $\mathcal{V} \rightarrow \mathcal{W}$ between symmetric monoidal preorders lets us                                       |
| fong-spivak-invitation_F00861 | 77   | ■ 1.000 | $\mathcal{C}$                                                     | $\mathcal{C}$                                                     | <b>Construction 2.44.</b> Let $f: \mathcal{V} \rightarrow \mathcal{W}$ be a monoidal monotone. Given a $\mathcal{V}$ -category $\mathcal{C}$ , |
| fong-spivak-invitation_F00862 | 77   | ■ 1.000 | $\mathcal{C}_f$                                                   | $\mathcal{C}_f$                                                   | one forms the associated $\mathcal{W}$ -category, say $\mathcal{C}_f$ , as follows.                                                            |
| fong-spivak-invitation_F00863 | 77   | ■ 0.999 | $\mathrm{Ob}(\mathcal{C}_f) := \mathrm{Ob}(\mathcal{C})$          | $\mathrm{Ob}(\mathcal{C}_f) := \mathrm{Ob}(\mathcal{C})$          | (i) We take the same objects: $\mathrm{Ob}(\mathcal{C}_f) := \mathrm{Ob}(\mathcal{C})$ .                                                       |
| fong-spivak-invitation_F00864 | 77   | ■ 0.979 | $c, d \in \mathrm{Ob}(\mathcal{C})$                               | $c, d \in \mathrm{Ob}(\mathcal{C})$                               | (ii) For any $c, d \in \mathrm{Ob}(\mathcal{C})$ , put $\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$ .                                         |
| fong-spivak-invitation_F00865 | 77   | ■ 0.979 | $\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$                     | $\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$                     | (ii) For any $c, d \in \mathrm{Ob}(\mathcal{C})$ , put $\mathcal{C}_f(c, d) := f(\mathcal{C}(c, d))$ .                                         |
| fong-spivak-invitation_F00866 | 78   | ■ 1.000 | $c \in \mathcal{C}$                                               | $c \in \mathcal{C}$                                               | (a) for every $c \in \mathcal{C}$ , we have                                                                                                    |
| fong-spivak-invitation_F00867 | 78   | ■ 0.945 | $c, d, e \in \mathrm{Ob}(\mathcal{C})$                            | $c, d, e \in \mathrm{Ob}(\mathcal{C})$                            | (b) for every $c, d, e \in \mathrm{Ob}(\mathcal{C})$ , we have                                                                                 |
| fong-spivak-invitation_F00868 | 78   | ■ 0.998 | $f: [0, \infty] \rightarrow \{\text{true, false}\}$               | $f: [0, \infty] \rightarrow \{\text{true, false}\}$               | <b>Example 2.45.</b> As an example, consider the function $f: [0, \infty] \rightarrow \{\text{true, false}\}$                                  |
| fong-spivak-invitation_F00869 | 78   | ■ 0.831 | $x_f \neq x_g$                                                    | $x_f \neq x_g$                                                    | $\mathcal{X}_f \neq \mathcal{X}_g$ .                                                                                                           |

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| fong-spivak-invitation_F00870 | 79   | 0.703 | $\mathcal{Y}$                                                                                                | $\mathcal{Y}$                                                                                                | $\mathcal{V}$ -categories. A $\mathcal{V}$ -functor from $\mathcal{X}$ to $\mathcal{Y}$ , denoted $F: \mathcal{X} \rightarrow \mathcal{Y}$ , consists of one constituent   |
| fong-spivak-invitation_F00871 | 79   | 0.703 | $F: \mathcal{X} \rightarrow \mathcal{Y}$                                                                     | $F: \mathfrak{X} \rightarrow y$                                                                              | $\mathcal{V}$ -categories. A $\mathcal{V}$ -functor from $\mathcal{X}$ to $\mathcal{Y}$ , denoted $F: \mathcal{X} \rightarrow \mathcal{Y}$ , consists of one constituent   |
| fong-spivak-invitation_F00872 | 79   | 0.752 | $F: \mathcal{O}b(\mathcal{X}) \rightarrow \mathcal{O}b(\mathcal{Y})$                                         | $F: \mathcal{O}b(\mathcal{X}) \rightarrow \mathcal{O}b(\mathcal{Y})$                                         | (i) a function $F: \mathcal{O}b(\mathcal{X}) \rightarrow \mathcal{O}b(\mathcal{Y})$                                                                                        |
| fong-spivak-invitation_F00873 | 79   | 0.907 | $x_1, x_2 \in \mathcal{O}b(\mathcal{X})$                                                                     | $x_1, x_2 \in \mathcal{O}b(\mathcal{X})$                                                                     | (a) for all $x_1, x_2 \in \mathcal{O}b(\mathcal{X})$ , one has $\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$ .                                                  |
| fong-spivak-invitation_F00874 | 79   | 0.907 | $\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$                                                     | $\mathcal{X}(x_1, x_2) \leq y(F(x_1), F(x_2))$                                                               | (a) for all $x_1, x_2 \in \mathcal{O}b(\mathcal{X})$ , one has $\mathcal{X}(x_1, x_2) \leq \mathcal{Y}(F(x_1), F(x_2))$ .                                                  |
| fong-spivak-invitation_F00875 | 79   | 0.931 | $\mathcal{X}(x_1, x_2) =$                                                                                    | $\mathcal{X}(x_1, x_2) =$                                                                                    | preorders are <b>Bool</b> -categories, where $\mathcal{X}(x_1, x_2) = \text{true}$ is denoted $x_1 \leq x_2$ . One                                                         |
| fong-spivak-invitation_F00876 | 79   | 0.931 | $x_1 \leq x_2$                                                                                               | $x_1 \leq x_2$                                                                                               | preorders are <b>Bool</b> -categories, where $\mathcal{X}(x_1, x_2) = \text{true}$ is denoted $x_1 \leq x_2$ . One                                                         |
| fong-spivak-invitation_F00877 | 79   | 0.968 | $(X, \leq_X) \rightarrow (Y, \leq_Y)$                                                                        | $(X, \leq_X) \rightarrow (Y, \leq_Y)$                                                                        | functors, and that's true. A monotone map $(X, \leq_X) \rightarrow (Y, \leq_Y)$ is a function $F: X \rightarrow Y$                                                         |
| fong-spivak-invitation_F00878 | 79   | 0.998 | $x_1, x_2 \in X$                                                                                             | $x_1, x_2 \in X$                                                                                             | such that, for every $x_1, x_2 \in X$ , if $x_1 \leq_X x_2$ then $F(x_1) \leq_Y F(x_2)$ . In other words, we                                                               |
| fong-spivak-invitation_F00879 | 79   | 0.998 | $x_1 \leq_X x_2$                                                                                             | $x_1 \leq_X x_2$                                                                                             | such that, for every $x_1, x_2 \in X$ , if $x_1 \leq_X x_2$ then $F(x_1) \leq_Y F(x_2)$ . In other words, we                                                               |
| fong-spivak-invitation_F00880 | 79   | 0.998 | $F(x_1) \leq_Y F(x_2)$                                                                                       | $F(x_1) \leq_Y F(x_2)$                                                                                       | such that, for every $x_1, x_2 \in X$ , if $x_1 \leq_X x_2$ then $F(x_1) \leq_Y F(x_2)$ . In other words, we                                                               |
| fong-spivak-invitation_F00881 | 79   | 0.803 | $\mathcal{V} = \mathbf{Bool}$                                                                                | $\mathcal{V} = \mathbf{Bool}$                                                                                | where the above $\leq$ takes place in the enriching category $\mathcal{V} = \mathbf{Bool}$ ; this is exactly the                                                           |
| fong-spivak-invitation_F00882 | 79   | 0.999 | $(X, d_X)$                                                                                                   | $(X, d_X)$                                                                                                   | points does not increase. That is, given Lawvere metric spaces $(X, d_X)$ and $(Y, d_Y)$ , a                                                                               |
| fong-spivak-invitation_F00883 | 79   | 0.999 | $(Y, d_Y)$                                                                                                   | $(Y, d_Y)$                                                                                                   | points does not increase. That is, given Lawvere metric spaces $(X, d_X)$ and $(Y, d_Y)$ , a                                                                               |
| fong-spivak-invitation_F00884 | 79   | 0.999 | $d_X(x_1, x_2) \geq d_Y(F(x_1), F(x_2))$                                                                     | $d_X(x_1, x_2) \geq d_Y(F(x_1), F(x_2))$                                                                     | have $d_Y(x_1, x_2) > d_Y(F(x_1), F(x_2))$ .                                                                                                                               |
| fong-spivak-invitation_F00885 | 79   | 0.888 | $\mathcal{X}^{\text{op}}$                                                                                    | $\mathcal{X}^{\text{op}}$                                                                                    | category $\mathcal{X}$ is denoted $\mathcal{X}^{\text{op}}$ and is defined by                                                                                              |
| fong-spivak-invitation_F00886 | 79   | 0.625 | $\mathcal{O}b(\mathcal{X}^{\text{op}}) := \mathcal{O}b(\mathcal{X})$                                         | $\mathcal{O}b(\mathcal{X}^{\text{op}}) := \mathcal{O}b(\mathcal{X})$                                         | (i) $\mathcal{O}b(\mathcal{X}^{\text{op}}) := \mathcal{O}b(\mathcal{X})$ , and                                                                                             |
| fong-spivak-invitation_F00887 | 79   | 0.688 | $x, y \in \mathcal{X}$                                                                                       | $x, y \in \mathcal{X}$                                                                                       | (ii) for all $x, y \in \mathcal{X}$ , we have $\mathcal{X}^{\text{op}}(x, y) := \mathcal{X}(y, x)$ .                                                                       |
| fong-spivak-invitation_F00888 | 79   | 0.688 | $\mathcal{X}^{\text{op}}(x, y) := \mathcal{X}(y, x)$                                                         | $\mathcal{X}^{\text{op}}(x, y) := \mathcal{X}(y, x)$                                                         | (ii) for all $x, y \in \mathcal{X}$ , we have $\mathcal{X}^{\text{op}}(x, y) := \mathcal{X}(y, x)$ .                                                                       |
| fong-spivak-invitation_F00889 | 79   | 1.000 | $\dagger: \mathfrak{X} \rightarrow$                                                                          | $\dagger: \mathfrak{X} \rightarrow$                                                                          | A $\mathcal{V}$ -category $\mathcal{X}$ is a <i>dagger</i> $\mathcal{V}$ -category if the identity function is a $\mathcal{V}$ -functor $\dagger: \mathcal{X} \rightarrow$ |
| fong-spivak-invitation_F00890 | 79   | 0.154 | $\mathcal{X}^{\text{Op}}$                                                                                    | $\mathcal{X}^{\text{Op}}$                                                                                    | $\mathcal{X}^{\text{op}}$ . And a <i>skeletal</i> $\mathcal{V}$ -category is one in which if $I \leq \mathcal{X}(x, y)$ and $I \leq \mathcal{X}(y, x)$ , then              |
| fong-spivak-invitation_F00891 | 79   | 0.154 | $I \leq \mathcal{X}(x, y)$                                                                                   | $I \leq \mathfrak{X}(x, y)$                                                                                  | $\mathcal{X}^{\text{op}}$ . And a <i>skeletal</i> $\mathcal{V}$ -category is one in which if $I \leq \mathcal{X}(x, y)$ and $I \leq \mathcal{X}(y, x)$ , then              |
| fong-spivak-invitation_F00892 | 79   | 0.154 | $I \leq \mathcal{X}(y, x)$                                                                                   | $I \leq \mathfrak{X}(y, x)$                                                                                  | $\mathcal{X}^{\text{op}}$ . And a <i>skeletal</i> $\mathcal{V}$ -category is one in which if $I \leq \mathcal{X}(x, y)$ and $I \leq \mathcal{X}(y, x)$ , then              |
| fong-spivak-invitation_F00893 | 80   | 0.797 | $\mathcal{X} \times \mathcal{Y}$                                                                             | $\mathcal{X} \times \mathcal{Y}$                                                                             | <i>product</i> , to be the $\mathcal{V}$ -category $\mathcal{X} \times \mathcal{Y}$ with                                                                                   |
| fong-spivak-invitation_F00894 | 80   | 0.435 | $\mathcal{O}b(\mathcal{X} \times \mathcal{Y}) := \mathcal{O}b(\mathcal{X}) \times \mathcal{O}b(\mathcal{Y})$ | $\mathcal{O}b(\mathcal{X} \times \mathcal{Y}) := \mathcal{O}b(\mathcal{X}) \times \mathcal{O}b(\mathcal{Y})$ | (i) $\mathcal{O}b(\mathcal{X} \times \mathcal{Y}) := \mathcal{O}b(\mathcal{X}) \times \mathcal{O}b(\mathcal{Y})$                                                           |
| fong-spivak-invitation_F00895 | 80   | 0.978 | $(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$        | $(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$        | (ii) $(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$ .                                                               |

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| fong-spivak-invitation_F00896 | 80   | ■ 0.977 | $\left(x^{\wedge\{\text{prime}\}}, y^{\wedge\{\text{prime}\}}\right)$                                                        | $(x', y')$                                                | for two objects $(x, y)$ and $(x', y')$ in $\text{Ob}(\mathcal{X} \times \mathcal{Y})$ .                                                                  |
| fong-spivak-invitation_F00897 | 80   | ■ 0.977 | $\operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$                                                                          | $\text{Ob}(\mathcal{X} \times \mathcal{Y})$               | for two objects $(x, y)$ and $(x', y')$ in $\text{Ob}(\mathcal{X} \times \mathcal{Y})$ .                                                                  |
| fong-spivak-invitation_F00898 | 80   | ■ 0.722 | $(x, y) \in \operatorname{Ob}(\mathcal{X} \times \mathcal{Y})$                                                               | $(x, y) \in \text{Ob}(\mathcal{X} \times \mathcal{Y})$    | 1. Check that for every object $(x, y) \in \text{Ob}(\mathcal{X} \times \mathcal{Y})$ we have $I \leq (\mathcal{X} \times \mathcal{Y})((x, y), (x, y))$ . |
| fong-spivak-invitation_F00899 | 80   | ■ 0.722 | $I \leq (\mathcal{X} \times \mathcal{Y})((x, y), (x, y))$                                                                    | $I \leq (\mathcal{X} \times \mathcal{Y})((x, y), (x, y))$ | 1. Check that for every object $(x, y) \in \text{Ob}(\mathcal{X} \times \mathcal{Y})$ we have $I \leq (\mathcal{X} \times \mathcal{Y})((x, y), (x, y))$ . |
| fong-spivak-invitation_F00900 | 80   | ■ 1.000 | $\left(x_1, y_1\right), \left(x_2, y_2\right)$                                                                               | $(x_1, y_1), (x_2, y_2)$                                  | 2. Check that for every three objects $(x_1, y_1), (x_2, y_2)$ , and $(x_3, y_3)$ , we have                                                               |
| fong-spivak-invitation_F00901 | 80   | ■ 1.000 | $\left(x_3, y_3\right)$                                                                                                      | $(x_3, y_3)$                                              | 2. Check that for every three objects $(x_1, y_1), (x_2, y_2)$ , and $(x_3, y_3)$ , we have                                                               |
| fong-spivak-invitation_F00902 | 80   | ■ 1.000 | $\left(P, \leq_P\right) \times \left(Q, \leq_Q\right)$                                                                       | $(P, \leq_P) \times (Q, \leq_Q)$                          | When taking the product of two preorders $(P, \leq_P) \times (Q, \leq_Q)$ , as first described in                                                         |
| fong-spivak-invitation_F00903 | 80   | ■ 0.879 | $\left(p_1, q_1\right) \leq \left(p_2, q_2\right)$                                                                           | $(p_1, q_1) \leq (p_2, q_2)$                              | Example 1.51, we say that $(p_1, q_1) \leq (p_2, q_2)$ iff both $p_1 \leq p_2$ AND $q_1 \leq q_2$ ; the AND                                               |
| fong-spivak-invitation_F00904 | 80   | ■ 0.879 | $q_1 \leq q_2$                                                                                                               | $q_1 \leq q_2$                                            | Example 1.51, we say that $(p_1, q_1) \leq (p_2, q_2)$ iff both $p_1 \leq p_2$ AND $q_1 \leq q_2$ ; the AND                                               |
| fong-spivak-invitation_F00905 | 81   | ■ 0.996 | $d_{\mathcal{X} \times \mathcal{Y}}\left((x, y), \left(x^{\wedge\{\text{prime}\}}, y^{\wedge\{\text{prime}\}}\right)\right)$ | $d_{\mathcal{X} \times \mathcal{Y}}((x, y), (x', y'))$    | objects in $\mathcal{X} \times \mathcal{Y}$ . And the distance $d_{\mathcal{X} \times \mathcal{Y}}((x, y), (x', y'))$ between any two points is           |
| fong-spivak-invitation_F00906 | 81   | ■ 1.000 | $d_{\mathcal{X}}\left(x, x^{\wedge\{\text{prime}\}}\right)+d_{\mathcal{Y}}\left(y, y^{\wedge\{\text{prime}\}}\right)$        | $d_X(x, x') + d_Y(y, y')$                                 | given by the sum $d_X(x, x') + d_Y(y, y')$ .                                                                                                              |
| fong-spivak-invitation_F00907 | 81   | ■ 0.999 | $\mathbb{R} \times \mathbb{R}$                                                                                               | $\mathbb{R} \times \mathbb{R}$                            | Example 2.37). Form the <b>Cost</b> -product $\mathbb{R} \times \mathbb{R}$ . What is the distance from $(5, 6)$ to                                       |
| fong-spivak-invitation_F00908 | 81   | ■ 0.999 | $(5, 6)$                                                                                                                     | $(5, 6)$                                                  | Example 2.37). Form the <b>Cost</b> -product $\mathbb{R} \times \mathbb{R}$ . What is the distance from $(5, 6)$ to                                       |
| fong-spivak-invitation_F00909 | 81   | ■ 0.991 | $(-1, 4)$                                                                                                                    | $(-1, 4)$                                                 | $(-1, 4)$ ? Hint: apply Definition 2.53; the answer is not $\sqrt{40}$ .                                                                                  |
| fong-spivak-invitation_F00910 | 81   | ■ 0.991 | $\sqrt{40}$                                                                                                                  | $\sqrt{40}$                                               | $(-1, 4)$ ? Hint: apply Definition 2.53; the answer is not $\sqrt{40}$ .                                                                                  |
| fong-spivak-invitation_F00911 | 125  | ■ 1.000 | $x \times y$                                                                                                                 | $x \times y$                                              | corresponding category. Show that the product $x \times y$ in $\mathcal{P}$ agrees with their meet $x \wedge y$                                           |
| fong-spivak-invitation_F00921 | 82   | ■ 0.788 | $\mathcal{V}=(V, \leq I, \otimes)$                                                                                           | $\mathcal{V} = (V, \leq I, \otimes)$                      | <b>Definition 2.57.</b> A symmetric monoidal preorder $\mathcal{V} = (V, \leq, I, \otimes)$ is called <i>symmetric</i>                                    |
| fong-spivak-invitation_F00922 | 82   | ■ 1.000 | $v, w \in V$                                                                                                                 | $v, w \in V$                                              | <i>monoidal closed</i> (or just <i>closed</i> ) if, for every two elements $v, w \in V$ , there is an element                                             |
| fong-spivak-invitation_F00923 | 82   | ■ 0.984 | $v \multimap w$                                                                                                              | $v \multimap w$                                           | $v \multimap w$ in $\mathcal{V}$ , called the <i>hom-element</i> , with the property                                                                      |
| fong-spivak-invitation_F00924 | 82   | ■ 1.000 | $a, v, w \in V$                                                                                                              | $a, v, w \in V$                                           | for all $a, v, w \in V$ .                                                                                                                                 |
| fong-spivak-invitation_F00925 | 82   | ■ 1.000 | $v \in V$                                                                                                                    | $v \in V$                                                 | preorder is monoidal closed iff, given any $v \in V$ , the map $(- \otimes v): V \rightarrow V$ given by                                                  |
| fong-spivak-invitation_F00926 | 82   | ■ 1.000 | $(- \otimes v): V \rightarrow V$                                                                                             | $(- \otimes v): V \rightarrow V$                          | preorder is monoidal closed iff, given any $v \in V$ , the map $(- \otimes v): V \rightarrow V$ given by                                                  |
| fong-spivak-invitation_F00927 | 82   | ■ 0.756 | $(v \multimap -): V \rightarrow V$                                                                                           | $(v \multimap -): V \rightarrow V$                        | multiplying with $v$ has a right adjoint. We write this right adjoint $(v \multimap -): V \rightarrow V$ .                                                |
| fong-spivak-invitation_F00928 | 82   | ■ 1.000 | $(- \otimes v)$                                                                                                              | $(- \otimes v)$                                           | 1. Using Definition 2.1, show that $(- \otimes v)$ is monotone.                                                                                           |
| fong-spivak-invitation_F00929 | 82   | ■ 1.000 | $((v \multimap w) \otimes v) \leq w$                                                                                         | $((v \multimap w) \otimes v) \leq w$                      | 2. Supposing that $\mathcal{V}$ is closed, show that for all $v, w \in V$ we have $((v \multimap w) \otimes v) \leq w$ .                                  |
| fong-spivak-invitation_F00930 | 82   | ■ 0.771 | $(v \multimap -)$                                                                                                            | $(v \multimap -)$                                         | 3. Using part 2, show that $(v \multimap -)$ is monotone.                                                                                                 |
| fong-spivak-invitation_F00931 | 83   | ■ 1.000 | $x, y \in[0, \infty]$                                                                                                        | $x, y \in[0, \infty]$                                     | Indeed, for any $x, y \in[0, \infty]$ , define $x \multimap y:=\max (0, y-x)$ . Then, for any $a, x, y \in$                                               |

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| fong-spivak-invitation_F00932 | 83   | ■ 1.000 | $x \multimap y := \max(0, y - x)$                                           | $x \multimap y := \max(0, y - x)$                                        | Indeed, for any $x, y \in [0, \infty]$ , define $x \multimap y := \max(0, y - x)$ . Then, for any $a, x, y \in$                  |
| fong-spivak-invitation_F00933 | 83   | ■ 1.000 | $a, x, y \in$                                                               | $a, x, y \in$                                                            | Indeed, for any $x, y \in [0, \infty]$ , define $x \multimap y := \max(0, y - x)$ . Then, for any $a, x, y \in$                  |
| fong-spivak-invitation_F00934 | 83   | ■ 0.543 | $(\mathbb{B}, \leq$                                                         | $(\mathbb{B}, \leq$                                                      | <b>Example 2.62.</b> A non-example is $(\mathbb{B}, \leq, \text{false}, \vee)$ . Indeed, suppose we had a $\multimap$ oper-      |
| fong-spivak-invitation_F00935 | 83   | ■ 0.543 | $\vee)$                                                                     | $\vee)$                                                                  | <b>Example 2.62.</b> A non-example is $(\mathbb{B}, \leq, \text{false}, \vee)$ . Indeed, suppose we had a $\multimap$ oper-      |
| fong-spivak-invitation_F00936 | 83   | ■ 0.812 | $\leq p \multimap q$                                                        | $\leq p \multimap q$                                                     | ator as in Definition 2.57. Note that $\text{false} \leq p \multimap q$ , for any $p, q$ no matter what $\multimap$              |
| fong-spivak-invitation_F00937 | 83   | ■ 0.812 | $\longrightarrow$                                                           | $\longrightarrow$                                                        | ator as in Definition 2.57. Note that $\text{false} \leq p \multimap q$ , for any $p, q$ no matter what $\multimap$              |
| fong-spivak-invitation_F00938 | 83   | ■ 1.000 | $a =$                                                                       | $a =$                                                                    | is, because $\text{false}$ is less than everything. But using $a = \text{false}$ , $p = \text{true}$ , and                       |
| fong-spivak-invitation_F00939 | 83   | ■ 1.000 | $p =$                                                                       | $p =$                                                                    | is, because $\text{false}$ is less than everything. But using $a = \text{false}$ , $p = \text{true}$ , and                       |
| fong-spivak-invitation_F00940 | 83   | ■ 0.927 | $q =$                                                                       | $q =$                                                                    | $q = \text{false}$ , we then get a contradiction: $(a \vee p) \not\leq q$ and yet $a \leq (p \multimap q)$ .                     |
| fong-spivak-invitation_F00941 | 83   | ■ 0.927 | $(a \vee p) \not\leq q$                                                     | $(a \vee p) \not\leq q$                                                  | $q = \text{false}$ , we then get a contradiction: $(a \vee p) \not\leq q$ and yet $a \leq (p \multimap q)$ .                     |
| fong-spivak-invitation_F00942 | 83   | ■ 0.927 | $a \leq (p \multimap q)$                                                    | $a \leq (p \multimap q)$                                                 | $q = \text{false}$ , we then get a contradiction: $(a \vee p) \not\leq q$ and yet $a \leq (p \multimap q)$ .                     |
| fong-spivak-invitation_F00943 | 83   | ■ 0.969 | $c, d$                                                                      | $c, d$                                                                   | that for every two material collections $c, d$ one can form a material collection $c \multimap d$                                |
| fong-spivak-invitation_F00944 | 83   | ■ 0.969 | $c \longrightarrow d$                                                       | $c \rightarrow d$                                                        | that for every two material collections $c, d$ one can form a material collection $c \multimap d$                                |
| fong-spivak-invitation_F00945 | 83   | ■ 1.000 | $2 \text{H}_2\text{O} + 2 \text{Na} \rightarrow 2 \text{NaOH} + \text{H}_2$ | $2\text{H}_2\text{O} + 2\text{Na} \rightarrow 2\text{NaOH} + \text{H}_2$ | Or, more down to earth, since we have the reaction $2\text{H}_2\text{O} + 2\text{Na} \rightarrow 2\text{NaOH} + \text{H}_2$ , we |
| fong-spivak-invitation_F00946 | 83   | ■ 0.975 | $2 \text{Na} \multimap (2 \text{NaOH} + \text{H}_2)$                        | $2\text{Na} \multimap (2\text{NaOH} + \text{H}_2)$                       | But it is not so far-fetched: this hypothetical new substance $(2\text{Na} \multimap (2\text{NaOH} + \text{H}_2))$               |
| fong-spivak-invitation_F00947 | 83   | ■ 0.977 | $\mathcal{V} = (V, \leq, I, \otimes, \multimap)$                            | $\mathcal{V} = (V, \leq, I, \otimes, \multimap)$                         | <b>Proposition 2.64.</b> Suppose $\mathcal{V} = (V, <, I, \otimes, \multimap)$ is a symmetric monoidal preorder                  |
| fong-spivak-invitation_F00948 | 83   | ■ 1.000 | $\multimap v : (V, \leq) \rightarrow (V, \leq)$                             | $\multimap v : (V, \leq) \rightarrow (V, \leq)$                          | (a) For every $v \in V$ , the monotone map $\multimap v : (V, \leq) \rightarrow (V, \leq)$ is left adjoint to                    |
| fong-spivak-invitation_F00949 | 83   | ■ 0.779 | $v \multimap - : (V, \leq) \rightarrow (V, \leq)$                           | $v \multimap - : (V, \leq) \rightarrow (V, \leq)$                        | $v \multimap - : (V, <) \rightarrow (V, <)$ .                                                                                    |
| fong-spivak-invitation_F00950 | 84   | ■ 1.000 | $A \subseteq V$                                                             | $A \subseteq V$                                                          | (b) For any element $v \in V$ and set of elements $A \subseteq V$ , if the join $\bigvee_{a \in A} a$ exists then                |
| fong-spivak-invitation_F00951 | 84   | ■ 1.000 | $\bigvee_{a \in A} a$                                                       | $\bigvee_{a \in A} a$                                                    | (b) For any element $v \in V$ and set of elements $A \subseteq V$ , if the join $\bigvee_{a \in A} a$ exists then                |
| fong-spivak-invitation_F00952 | 84   | ■ 1.000 | $\bigvee_{a \in A} v \otimes a$                                             | $\bigvee_{a \in A} v \otimes a$                                          | so does $\bigvee_{a \in A} v \otimes a$ and we have                                                                              |
| fong-spivak-invitation_F00953 | 84   | ■ 1.000 | $v \otimes (v \multimap w) \leq w$                                          | $v \otimes (v \multimap w) \leq w$                                       | (c) For any $v, w \in V$ , we have $v \otimes (v \multimap w) \leq w$ .                                                          |
| fong-spivak-invitation_F00954 | 84   | ■ 1.000 | $v \cong (I \multimap v)$                                                   | $v \cong (I \multimap v)$                                                | (d) For any $v \in V$ , we have $v \cong (I \multimap v)$ .                                                                      |
| fong-spivak-invitation_F00955 | 84   | ■ 1.000 | $u, v, w \in V$                                                             | $u, v, w \in V$                                                          | (e) For any $u, v, w \in V$ , we have $(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$ .                           |
| fong-spivak-invitation_F00956 | 84   | ■ 1.000 | $(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$              | $(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$           | (e) For any $u, v, w \in V$ , we have $(u \multimap v) \otimes (v \multimap w) \leq (u \multimap w)$ .                           |
| fong-spivak-invitation_F00957 | 84   | ■ 1.000 | $(v \multimap w) \leq (v \multimap w)$                                      | $(v \multimap w) \leq (v \multimap w)$                                   | Proposition 1.101. More concretely, from reflexivity $(v \multimap w) \leq (v \multimap w)$ , we                                 |
| fong-spivak-invitation_F00958 | 84   | ■ 1.000 | $(v \multimap w) \otimes v \leq w$                                          | $(v \multimap w) \otimes v \leq w$                                       | obtain $(v \multimap w) \otimes v \leq w$ Eq. (2.23), and we are done by symmetry, which says                                    |

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| fong-spivak-invitation_F00959 | 84   | ■ 1.000 | $v \otimes (v \multimap w) = (v \multimap w) \otimes v$    | $v \otimes (v \multimap w) = (v \multimap w) \otimes v$    | $v \otimes (v \multimap w) = (v \multimap w) \otimes v$ .                                                                                      |
| fong-spivak-invitation_F00960 | 84   | ■ 1.000 | $v \otimes I = v \leq v$                                   | $v \otimes I = v \leq v$                                   | (d) Since $v \otimes I = v \leq v$ , Eq. (2.23) says $v \leq (I \multimap v)$ . For the other direction, we have                               |
| fong-spivak-invitation_F00961 | 84   | ■ 1.000 | $v \leq (I \multimap v)$                                   | $v \leq (I \multimap v)$                                   | (d) Since $v \otimes I = v \leq v$ , Eq. (2.23) says $v \leq (I \multimap v)$ . For the other direction, we have                               |
| fong-spivak-invitation_F00962 | 84   | ■ 1.000 | $(I \multimap v) = I \otimes (I \multimap v) \leq v$       | $(I \multimap v) = I \otimes (I \multimap v) \leq v$       | $(I \multimap v) = I \otimes (I \multimap v) < v$ by (c).                                                                                      |
| fong-spivak-invitation_F00963 | 84   | ■ 1.000 | $u \otimes (u \multimap v) \otimes (v \multimap w) \leq w$ | $u \otimes (u \multimap v) \otimes (v \multimap w) \leq w$ | (e) To obtain this inequality, we just need $u \otimes (u \multimap v) \otimes (v \multimap w) < w$ . But this                                 |
| fong-spivak-invitation_F00964 | 84   | ■ 1.000 | $v, w \in$                                                 | $v, w \in$                                                 | <b>Remark 2.65.</b> We can consider $\mathcal{V}$ to be enriched in itself. That is, for every $v, w \in$                                      |
| fong-spivak-invitation_F00965 | 84   | ■ 0.999 | $\operatorname{Ob}(\mathcal{V})$                           | $\operatorname{Ob}(\mathcal{V})$                           | $\operatorname{Ob}(\mathcal{V})$ , we can define $\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$ . For this to really be an enrichment, |
| fong-spivak-invitation_F00966 | 84   | ■ 0.999 | $\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$     | $\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$     | $\operatorname{Ob}(\mathcal{V})$ , we can define $\mathcal{V}(v, w) := (v \multimap w) \in \mathcal{V}$ . For this to really be an enrichment, |
| fong-spivak-invitation_F00967 | 84   | ■ 1.000 | $I \leq$                                                   | $I \leq$                                                   | we just need to check the two conditions of Definition 2.30. The first condition $I \leq$                                                      |
| fong-spivak-invitation_F00968 | 84   | ■ 0.688 | $\mathcal{X}(x, x) = (x \multimap x)$                      | $\mathcal{X}(x, x) = (x \multimap x)$                      | $\mathcal{X}(x, x) = (x \multimap x)$ is satisfied because $I \otimes x \leq x$ . The second condition is satisfied by                         |
| fong-spivak-invitation_F00969 | 84   | ■ 0.688 | $I \otimes x \leq x$                                       | $I \otimes x \leq x$                                       | $\mathcal{X}(x, x) = (x \multimap x)$ is satisfied because $I \otimes x < x$ . The second condition is satisfied by                            |
| fong-spivak-invitation_F00970 | 84   | ■ 0.901 | $\mathcal{V} = (V, \leq, I, \otimes, \multimap)$           | $\mathcal{V} = (V, \leq, I, \otimes, \multimap)$           | order $\mathcal{V} = (V, \leq, I, \otimes, \multimap)$ that has all joins: $\bigvee A$ exists for every $A \subseteq V$ . In particular,       |
| fong-spivak-invitation_F00971 | 84   | ■ 0.901 | $\bigvee A$                                                | $\bigvee A$                                                | order $\mathcal{V} = (V, \leq, I, \otimes, \multimap)$ that has all joins: $\bigvee A$ exists for every $A \subseteq V$ . In particular,       |
| fong-spivak-invitation_F00972 | 84   | ■ 0.928 | $0 := \bigvee \varnothing$                                 | $0 := \bigvee \varnothing$                                 | we often denote the empty join by $0 := \bigvee \varnothing$ .                                                                                 |
| fong-spivak-invitation_F00973 | 85   | ■ 1.000 | $A \subseteq [0, \infty]$                                  | $A \subseteq [0, \infty]$                                  | <b>Cost</b> is a quantale, we take an arbitrary set of elements $A \subseteq [0, \infty]$ and ask if it has a                                  |
| fong-spivak-invitation_F00974 | 85   | ■ 0.999 | $a \geq \bigvee A$                                         | $a \geq \bigvee A$                                         | a. $a \geq \bigvee A$ for all $a \in A$ , and                                                                                                  |
| fong-spivak-invitation_F00975 | 85   | ■ 1.000 | $b \in [0, \infty]$                                        | $b \in [0, \infty]$                                        | b. if $b \in [0, \infty]$ is any element such that $a \geq b$ for all $a \in A$ , then $\bigvee A \geq b$ .                                    |
| fong-spivak-invitation_F00976 | 85   | ■ 1.000 | $a \geq b$                                                 | $a \geq b$                                                 | b. if $b \in [0, \infty]$ is any element such that $a \geq b$ for all $a \in A$ , then $\bigvee A \geq b$ .                                    |
| fong-spivak-invitation_F00977 | 85   | ■ 1.000 | $\bigvee A \geq b$                                         | $\bigvee A \geq b$                                         | b. if $b \in [0, \infty]$ is any element such that $a \geq b$ for all $a \in A$ , then $\bigvee A \geq b$ .                                    |
| fong-spivak-invitation_F00978 | 85   | ■ 0.544 | $A \cdot \{ \}^{\{5\}}$                                    | $A \cdot \{ \}^{\{5\}}$                                    | bound, of $A \cdot \{ \}^{\{5\}}$ For example, if $A = \{2, 3\}$ then $\bigvee A = 2$ . We have joins for infinite                             |
| fong-spivak-invitation_F00979 | 85   | ■ 0.544 | $A = \{2, 3\}$                                             | $A = \{2, 3\}$                                             | bound, of $A \cdot \{ \}^{\{5\}}$ For example, if $A = \{2, 3\}$ then $\bigvee A = 2$ . We have joins for infinite                             |
| fong-spivak-invitation_F00980 | 85   | ■ 0.544 | $\bigvee A = 2$                                            | $\bigvee A = 2$                                            | bound, of $A \cdot \{ \}^{\{5\}}$ For example, if $A = \{2, 3\}$ then $\bigvee A = 2$ . We have joins for infinite                             |
| fong-spivak-invitation_F00981 | 85   | ■ 0.996 | $B = \{2.5, 2.05, 2.005, \dots\}$                          | $B = \{2.5, 2.05, 2.005, \dots\}$                          | sets too: if $B = \{2.5, 2.05, 2.005, \dots\}$ , its infimum is 2. Finally, in order to say that                                               |
| fong-spivak-invitation_F00982 | 85   | ■ 0.997 | $([0, \infty], \geq)$                                      | $([0, \infty], \geq)$                                      | $([0, \infty], \geq)$ has all joins, we need a join to exist for the empty set $A = \varnothing$ too. The first                                |
| fong-spivak-invitation_F00983 | 85   | ■ 0.997 | $A = \varnothing$                                          | $A = \varnothing$                                          | $([0, \infty], \geq)$ has all joins, we need a join to exist for the empty set $A = \varnothing$ too. The first                                |
| fong-spivak-invitation_F00984 | 85   | ■ 0.999 | $\bigvee \varnothing \geq b$                               | $\bigvee \varnothing \geq b$                               | for any $b \in [0, \infty]$ we have $\bigvee \varnothing \geq b$ ; this means $\bigvee \varnothing = \infty$ .                                 |
| fong-spivak-invitation_F00985 | 85   | ■ 0.999 | $\bigvee \varnothing = \infty$                             | $\bigvee \varnothing = \infty$                             | for any $b \in [0, \infty]$ we have $\bigvee \varnothing \geq b$ ; this means $\bigvee \varnothing = \infty$ .                                 |

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| fong-spivak-invitation_F00986 | 85   | ■ 0.934 | $\bigvee \varnothing$                                     | $\bigvee \emptyset$                                       | 1. What is $\bigvee \emptyset$ , which we generally denote 0, in the case                                                            |
| fong-spivak-invitation_F00987 | 85   | ■ 0.921 | $\mathcal{V} = \mathbf{Bool} = (\mathbb{B}, \leq)$        | $\mathcal{V} = \mathbf{Bool} = (\mathbb{B}, \leq)$        | a. $\mathcal{V} = \mathbf{Bool} = (\mathbb{B}, \leq, \text{true}, \wedge)$ ?                                                         |
| fong-spivak-invitation_F00988 | 85   | ■ 0.735 | $\mathcal{V} = \mathbf{Cost} = ([0, \infty], \geq, 0, +)$ | $\mathcal{V} = \mathbf{Cost} = ([0, \infty], \geq, 0, +)$ | b. $\mathcal{V} = \mathbf{Cost} = ([0, \infty], \geq, 0, +)$ ?                                                                       |
| fong-spivak-invitation_F00989 | 85   | ■ 1.000 | $x \vee y$                                                | $x \vee y$                                                | 2. What is the join $x \vee y$ in the case                                                                                           |
| fong-spivak-invitation_F00990 | 85   | ■ 0.802 | $\mathcal{V} =$                                           | $\mathcal{V} =$                                           | a. $\mathcal{V} = \mathbf{Bool}$ , and $x, y \in \mathbb{B}$ are booleans?                                                           |
| fong-spivak-invitation_F00991 | 85   | ■ 0.802 | $x, y \in \mathbb{B}$                                     | $x, y \in \mathbb{B}$                                     | a. $\mathcal{V} = \mathbf{Bool}$ , and $x, y \in \mathbb{B}$ are booleans?                                                           |
| fong-spivak-invitation_F00992 | 85   | ■ 0.422 | $\mathbf{Bool} = (\mathbb{B}, \leq)$                      | $\mathbf{Bool} = (\mathbb{B}, \leq)$                      | <b>Exercise 2.69.</b> Show that $\mathbf{Bool} = (\mathbb{B}, \leq, \text{true}, \wedge)$ is a quantale.                             |
| fong-spivak-invitation_F00993 | 85   | ■ 0.737 | $P(S), \subseteq, S, \cap$                                | $P(S), \subseteq, S, \cap$                                | <b>Exercise 2.70.</b> Let $S$ be a set and recall the power set monoidal preorder $(P(S), \subseteq, S, \cap)$                       |
| fong-spivak-invitation_F00994 | 85   | ■ 0.720 | $\{3.1, 3.01, 3.001, \dots\}$                             | $\{3.1, 3.01, 3.001, \dots\}$                             | number that is $\leq$ everything in $A$ . For example, the infimum of $\{3.1, 3.01, 3.001, \dots\}$ is 3. But note that              |
| fong-spivak-invitation_F00995 | 86   | ■ 1.000 | $\mathcal{P} = (P, \leq)$                                 | $\mathcal{P} = (P, \leq)$                                 | <b>Proposition 2.72.</b> Let $\mathcal{P} = (P, \leq)$ be a preorder. It has all joins iff it has all meets.                         |
| fong-spivak-invitation_F00996 | 86   | ■ 0.931 | $\mathcal{P}^{\text{op}}$                                 | $\mathcal{P}^{\text{op}}$                                 | <i>Proof.</i> The joins (resp. meets) in $\mathcal{P}$ are the meets (resp. joins) in $\mathcal{P}^{\text{op}}$ , so the two claims  |
| fong-spivak-invitation_F00997 | 86   | ■ 1.000 | $A \subseteq \mathcal{P}$                                 | $A \subseteq \mathcal{P}$                                 | Suppose $\mathcal{P}$ has all joins and suppose that $A \subseteq \mathcal{P}$ is a subset for which we want the                     |
| fong-spivak-invitation_F00998 | 86   | ■ 1.000 | $M_A := \{p \in P \mid p \leq a\}$                        | $M_A := \{p \in P \mid p \leq a\}$                        | meet. Consider the set $M_A := \{p \in P \mid p \leq a \text{ for all } a \in A\}$ of elements below                                 |
| fong-spivak-invitation_F00999 | 86   | ■ 1.000 | $a \in A$                                                 | $a \in A$                                                 | meet. Consider the set $M_A := \{p \in P \mid p \leq a \text{ for all } a \in A\}$ of elements below                                 |
| fong-spivak-invitation_F01000 | 86   | ■ 1.000 | $m_A := \bigvee_{p \in M_A} p$                            | $m_A := \bigvee_{p \in M_A} p$                            | everything in $A$ . Let $m_A := \bigvee_{p \in M_A} p$ be their join. We claim that $m_A$ is a meet for $A$ .                        |
| fong-spivak-invitation_F01001 | 86   | ■ 1.000 | $m_A$                                                     | $m_A$                                                     | everything in $A$ . Let $m_A := \bigvee_{p \in M_A} p$ be their join. We claim that $m_A$ is a meet for $A$ .                        |
| fong-spivak-invitation_F01002 | 86   | ■ 1.000 | $m_A \leq a$                                              | $m_A \leq a$                                              | We first need to know that for any $a \in A$ we have $m_A \leq a$ , but this is by definition                                        |
| fong-spivak-invitation_F01003 | 86   | ■ 1.000 | $p \in M_A$                                               | $p \in M_A$                                               | of join: since all $p \in M_A$ satisfy $p \leq a$ , so does their join $m_A \leq a$ . We second need                                 |
| fong-spivak-invitation_F01004 | 86   | ■ 1.000 | $m' \in P$                                                | $m' \in P$                                                | to know that for any $m' \in P$ with $m' \leq a$ for all $a \in A$ , we have $m' \leq m$ . But every                                 |
| fong-spivak-invitation_F01005 | 86   | ■ 1.000 | $m' \leq a$                                               | $m' \leq a$                                               | to know that for any $m' \in P$ with $m' \leq a$ for all $a \in A$ , we have $m' \leq m$ . But every                                 |
| fong-spivak-invitation_F01006 | 86   | ■ 1.000 | $m' \leq m$                                               | $m' \leq m$                                               | to know that for any $m' \in P$ with $m' \leq a$ for all $a \in A$ , we have $m' \leq m$ . But every                                 |
| fong-spivak-invitation_F01007 | 86   | ■ 1.000 | $m'$                                                      | $m'$                                                      | such $m'$ is actually an element of $M_A$ and $m$ is their join, so $m' \leq m$ . This completes the                                 |
| fong-spivak-invitation_F01008 | 86   | ■ 1.000 | $M_A$                                                     | $M_A$                                                     | such $m'$ is actually an element of $M_A$ and $m$ is their join, so $m' \leq m$ . This completes the                                 |
| fong-spivak-invitation_F01009 | 86   | ■ 1.000 | $V \subseteq X$                                           | $V \subseteq X$                                           | and $V \subseteq X$ , we can generalize the usual Hausdorff distance, on the left below, to the                                      |
| fong-spivak-invitation_F01010 | 86   | ■ 0.877 | $u \in U$                                                 | $u \in U$                                                 | answers the question “can I get into $V$ from every $u \in U$ ,” i.e. $\forall u \in U. \exists v \in V. u \leq v$ . <sup>6</sup> Or |
| fong-spivak-invitation_F01011 | 86   | ■ 0.877 | $\forall u \in U. \exists v \in V. u \leq v$              | $\forall u \in U. \exists v \in V. u \leq v$              | answers the question “can I get into $V$ from every $u \in U$ ,” i.e. $\forall u \in U. \exists v \in V. u \leq v$ . <sup>6</sup> Or |
| fong-spivak-invitation_F01012 | 86   | ■ 1.000 | $\mathcal{V} = P(M)$                                      | $\mathcal{V} = P(M)$                                      | for another example, use $\mathcal{V} = P(M)$ with its interpretation as modes of transportation,                                    |

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| fong-spivak-invitation_F01013 | 86   | ■ 1.000 | $d(U, V) \in \mathrm{P}(M)$                                | $d(U, V) \in \mathrm{P}(M)$                                | as in Exercise 2.42. Then the Hausdorff distance $d(U, V) \in \mathrm{P}(M)$ tells us those modes                                                                     |
| fong-spivak-invitation_F01014 | 86   | ■ 1.000 | $-\otimes v: V \rightarrow V$                              | $-\otimes v: V \rightarrow V$                              | Eq. (2.24) holds. We need to show that if Eq. (2.24) holds then $-\otimes v: V \rightarrow V$ has a                                                                   |
| fong-spivak-invitation_F01015 | 86   | ■ 0.999 | $v \multimap -$                                            | $v \multimap -$                                            | right adjoint $v \multimap -$ . This is just the adjoint functor theorem, Theorem 1.108. It says                                                                      |
| fong-spivak-invitation_F01016 | 86   | ■ 1.000 | $\forall$                                                  | $\forall$                                                  | <sup>6</sup> Here $\forall$ is the universal quantifier, “for all,” and $\exists$ is the existential quantifier, “there exists.”                                      |
| fong-spivak-invitation_F01017 | 86   | ■ 1.000 | $\exists$                                                  | $\exists$                                                  | <sup>6</sup> Here $\forall$ is the universal quantifier, “for all,” and $\exists$ is the existential quantifier, “there exists.”                                      |
| fong-spivak-invitation_F01018 | 87   | ■ 0.959 | $\sum$                                                     | $\sum$                                                     | We will get a formula where joins stand in for the sum operation $\sum$ , and $\otimes$ stands in for                                                                 |
| fong-spivak-invitation_F01019 | 87   | ■ 1.000 | $\mathcal{V}=(V, \leq, \otimes, I)$                        | $\mathcal{V}=(V, \leq, \otimes, I)$                        | <b>Definition 2.75.</b> Let $\mathcal{V}=(V, <, \otimes, I)$ be a quantale. Given sets $X$ and $Y$ , a <i>matrix with</i>                                             |
| fong-spivak-invitation_F01020 | 87   | ■ 1.000 | $M: X \times Y \rightarrow V$                              | $M: X \times Y \rightarrow V$                              | <i>entries in <math>\mathcal{V}</math></i> , or simply a <i><math>\mathcal{V}</math>-matrix</i> , is a function $M: X \times Y \rightarrow V$ . For any $x \in X$ and |
| fong-spivak-invitation_F01021 | 87   | ■ 0.991 | $M(x, y)$                                                  | $M(x, y)$                                                  | $y \in Y$ , we call $M(x, y)$ the <i><math>(x, y)</math>-entry</i> .                                                                                                  |
| fong-spivak-invitation_F01022 | 87   | ■ 1.000 | $N: Y \times Z \rightarrow V$                              | $N: Y \times Z \rightarrow V$                              | Here is how you multiply $\mathcal{V}$ -matrices $M: X \times Y \rightarrow V$ and $N: Y \times Z \rightarrow V$ . Their                                              |
| fong-spivak-invitation_F01023 | 87   | ■ 1.000 | $(M * N): X \times Z \rightarrow V$                        | $(M * N): X \times Z \rightarrow V$                        | product is defined to be the matrix $(M * N): X \times Z \rightarrow V$ , whose entries are given by                                                                  |
| fong-spivak-invitation_F01024 | 87   | ■ 0.999 | $M * N$                                                    | $M * N$                                                    | <b>Example 2.76.</b> Let $\mathcal{V} = \mathbf{Bool}$ . Here is an example of matrix multiplication $M * N$ .                                                        |
| fong-spivak-invitation_F01025 | 87   | ■ 1.000 | $X=\{1,2,3\}, Y=\{1,2\}$                                   | $X=\{1,2,3\}, Y=\{1,2\}$                                   | Here $X=\{1,2,3\}$ , $Y=\{1,2\}$ , and $Z=\{1,2,3\}$ , matrices $M: X \times Y \rightarrow \mathbb{B}$                                                                |
| fong-spivak-invitation_F01026 | 87   | ■ 1.000 | $Z=\{1,2,3\}$                                              | $Z=\{1,2,3\}$                                              | Here $X=\{1,2,3\}$ , $Y=\{1,2\}$ , and $Z=\{1,2,3\}$ , matrices $M: X \times Y \rightarrow \mathbb{B}$                                                                |
| fong-spivak-invitation_F01027 | 87   | ■ 1.000 | $M: X \times Y \rightarrow \mathbb{B}$                     | $M: X \times Y \rightarrow \mathbb{B}$                     | Here $X=\{1,2,3\}$ , $Y=\{1,2\}$ , and $Z=\{1,2,3\}$ , matrices $M: X \times Y \rightarrow \mathbb{B}$                                                                |
| fong-spivak-invitation_F01028 | 87   | ■ 1.000 | $N: Y \times Z \rightarrow \mathbb{B}$                     | $N: Y \times Z \rightarrow \mathbb{B}$                     | and $N: Y \times Z \rightarrow \mathbb{B}$ are shown to the left below, and their product is shown to the                                                             |
| fong-spivak-invitation_F01029 | 87   | ■ 1.000 | $I_X: X \times X \rightarrow V$                            | $I_X: X \times X \rightarrow V$                            | The identity $V$ -matrix on a set $X$ is $I_X: X \times X \rightarrow V$ given by                                                                                     |
| fong-spivak-invitation_F01030 | 87   | ■ 0.984 | $(2 \times 2)$                                             | $(2 \times 2)$                                             | <b>Exercise 2.77.</b> Write down the $(2 \times 2)$ -identity matrix for each of the quantales                                                                        |
| fong-spivak-invitation_F01031 | 88   | ■ 1.000 | $I_X * M = M$                                              | $I_X * M = M$                                              | $I_X * M = M$ .                                                                                                                                                       |
| fong-spivak-invitation_F01032 | 88   | ■ 1.000 | $M: W \times X \rightarrow V, N: X \times Y \rightarrow V$ | $M: W \times X \rightarrow V, N: X \times Y \rightarrow V$ | 2. Prove the <i>associative law</i> : for any matrices $M: W \times X \rightarrow V, N: X \times Y \rightarrow V$ , and                                               |
| fong-spivak-invitation_F01033 | 88   | ■ 1.000 | $P: Y \times Z \rightarrow V$                              | $P: Y \times Z \rightarrow V$                              | $P: Y \times Z \rightarrow V$ , one has $(M * N) * P = M * (N * P)$ .                                                                                                 |
| fong-spivak-invitation_F01034 | 88   | ■ 1.000 | $(M * N) * P = M * (N * P)$                                | $(M * N) * P = M * (N * P)$                                | $P: Y \times Z \rightarrow V$ , one has $(M * N) * P = M * (N * P)$ .                                                                                                 |
| fong-spivak-invitation_F01035 | 88   | ■ 1.000 | $M_Y^2$                                                    | $M_Y^2$                                                    | When we multiply $M_Y$ by itself using the formula Eq. (2.26), the result $M_Y^2$ tells us                                                                            |
| fong-spivak-invitation_F01036 | 88   | ■ 1.000 | $M_Y^3$                                                    | $M_Y^3$                                                    | the length of the shortest path traversing two edges or fewer. Similarly $M_Y^3$ tells us about                                                                       |
| fong-spivak-invitation_F01039 | 88   | ■ 1.000 | $M_Y^2 = M_Y^3$                                            | $M_Y^2 = M_Y^3$                                            | One sees that the powers stabilize: $M_Y^2 = M_Y^3$ ; as soon as that happens one has the                                                                             |
| fong-spivak-invitation_F01040 | 88   | ■ 0.999 | $M_Y^n$                                                    | $M_Y^n$                                                    | matrix of distances, $d_Y$ . Indeed $M_Y^n$ records the lengths of the shortest path traverse $n$                                                                     |
| fong-spivak-invitation_F01041 | 88   | ■ 1.000 | $M_X^2, M_X^3$                                             | $M_X^2, M_X^3$                                             | culate $M_X^2, M_X^3$ , and $M_X^4$ . Check that $M_X^4$ is what you got for the distance matrix in                                                                   |

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| fong-spivak-invitation_F01042 | 88   | ■ 1.000 | $M_X^4$                                          | $M_X^4$                                          | culate $M_X^2$ , $M_X^3$ , and $M_X^4$ . Check that $M_X^4$ is what you got for the distance matrix in                                                           |
| fong-spivak-invitation_F01043 | 91   | ■ 1.000 | $\$ 2.5$                                         | \$2.5                                            | ogy) budgets, and that the market for data integration software was \$2.5 billion in 2007                                                                        |
| fong-spivak-invitation_F01044 | 93   | ■ 1.000 | $\mathcal{C} \rightarrow$                        | $\mathcal{C} \rightarrow$                        | given by a “set-valued” functor $\mathcal{C} \rightarrow \mathbf{Set}$ , and that databases can be mapped to each                                                |
| fong-spivak-invitation_F01045 | 93   | ■ 0.999 | $\mathcal{C} \rightarrow \mathcal{D}$            | $\mathcal{C} \rightarrow \mathcal{D}$            | other via functors $\mathcal{C} \rightarrow \mathcal{D}$ . In other words, there is a relatively large overlap between                                           |
| fong-spivak-invitation_F01046 | 95   | ■ 1.000 | $\{ \}^2 \mathrm{Ob}(\mathcal{C})$               | ${}^2\mathrm{Ob}(\mathcal{C})$                   | (i) one specifies a collection <sup>2</sup> $\mathrm{Ob}(\mathcal{C})$ , elements of which are called <i>objects</i> ;                                           |
| fong-spivak-invitation_F01047 | 95   | ■ 0.959 | $\mathcal{C}(c, d), \{ \}^3$                     | $\mathcal{C}(c, d), {}^3$                        | (ii) for every two objects $c, d$ , one specifies a set $\mathcal{C}(c, d)$ , <sup>3</sup> elements of which are called                                          |
| fong-spivak-invitation_F01048 | 95   | ■ 0.901 | $c \in \mathrm{Ob}(\mathcal{C})$                 | $c \in \mathrm{Ob}(\mathcal{C})$                 | (iii) for every object $c \in \mathrm{Ob}(\mathcal{C})$ , one specifies a morphism $\mathrm{id}_c \in \mathcal{C}(c, c)$ , called the                            |
| fong-spivak-invitation_F01049 | 95   | ■ 0.901 | $\mathrm{id}_c \in \mathcal{C}(c, c)$            | $\mathrm{id}_c \in \mathcal{C}(c, c)$            | (iii) for every object $c \in \mathrm{Ob}(\mathcal{C})$ , one specifies a morphism $\mathrm{id}_c \in \mathcal{C}(c, c)$ , called the                            |
| fong-spivak-invitation_F01050 | 95   | ■ 0.791 | $f \in \mathcal{C}(c, d)$                        | $f \in \mathcal{C}(c, d)$                        | (iv) for every three objects $c, d, e \in \mathrm{Ob}(\mathcal{C})$ and morphisms $f \in \mathcal{C}(c, d)$ and $g \in \mathcal{C}(d, e)$ ,                      |
| fong-spivak-invitation_F01051 | 95   | ■ 0.791 | $g \in \mathcal{C}(d, e)$                        | $g \in \mathcal{C}(d, e)$                        | (iv) for every three objects $c, d, e \in \mathrm{Ob}(\mathcal{C})$ and morphisms $f \in \mathcal{C}(c, d)$ and $g \in \mathcal{C}(d, e)$ ,                      |
| fong-spivak-invitation_F01052 | 95   | ■ 0.983 | $f ; g \in \mathcal{C}(c, e)$                    | $f ; g \in \mathcal{C}(c, e)$                    | one specifies a morphism $f \circ g \in \mathcal{C}(c, e)$ , called <i>the composite of f and g</i> .                                                            |
| fong-spivak-invitation_F01053 | 95   | ■ 0.843 | $c \in \operatorname{Ob}(\mathcal{C})$           | $c \in \mathrm{Ob}(\mathcal{C})$                 | We will sometimes write an object $c \in \mathcal{C}$ , instead of $c \in \mathrm{Ob}(\mathcal{C})$ . It will also be conve-                                     |
| fong-spivak-invitation_F01054 | 95   | ■ 0.874 | $f: c \rightarrow d$                             | $f: c \rightarrow d$                             | nient to denote elements $f \in \mathcal{C}(c, d)$ as $f: c \rightarrow d$ . Here, $c$ is called the <i>domain</i> of $f$ ,                                      |
| fong-spivak-invitation_F01055 | 95   | ■ 0.664 | $\mathrm{id}_c ; f = f$                          | $\mathrm{id}_c; f = f$                           | nothing: $\mathrm{id}_c \circ f = f$ and $f \circ \mathrm{id}_d = f$ ;                                                                                           |
| fong-spivak-invitation_F01056 | 95   | ■ 0.664 | $f ; \mathrm{id}_d = f$                          | $f; \mathrm{id}_d = f$                           | nothing: $\mathrm{id}_c \circ f = f$ and $f \circ \mathrm{id}_d = f$ ;                                                                                           |
| fong-spivak-invitation_F01057 | 95   | ■ 1.000 | $f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$ | $f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$ | (b) <i>associativity</i> : for any three morphisms $f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$ , and $h: c_2 \rightarrow c_3$ ,                             |
| fong-spivak-invitation_F01058 | 95   | ■ 1.000 | $h: c_2 \rightarrow c_3$                         | $h: c_2 \rightarrow c_3$                         | (b) <i>associativity</i> : for any three morphisms $f: c_0 \rightarrow c_1, g: c_1 \rightarrow c_2$ , and $h: c_2 \rightarrow c_3$ ,                             |
| fong-spivak-invitation_F01059 | 95   | ■ 0.998 | $(f ; g) ; h = f ; (g ; h)$                      | $(f;g);h = f;(g;h)$                              | the following are equal: $(f \circ g) \circ h = f \circ (g \circ h)$ . We write this composite simply as                                                         |
| fong-spivak-invitation_F01060 | 95   | ■ 0.957 | $\mathcal{C}(c, d)$                              | $\mathcal{C}(c, d)$                              | <sup>3</sup> This set $\mathcal{C}(c, d)$ is often denoted $\mathrm{Hom}_{\mathcal{C}}(c, d)$ , and called the “hom-set from $c$ to $d$ .” The word “hom” stands |
| fong-spivak-invitation_F01061 | 95   | ■ 0.957 | $\operatorname{Hom}_{\mathcal{C}}(c, d)$         | $\mathrm{Hom}_{\mathcal{C}}(c, d)$               | <sup>3</sup> This set $\mathcal{C}(c, d)$ is often denoted $\mathrm{Hom}_{\mathcal{C}}(c, d)$ , and called the “hom-set from $c$ to $d$ .” The word “hom” stands |
| fong-spivak-invitation_F01062 | 96   | ■ 1.000 | $p: v \rightarrow w$                             | $p: v \rightarrow w$                             | $p: v \rightarrow w$ . To every vertex $v$ , there is a trivial path, containing no arrows, starting                                                             |
| fong-spivak-invitation_F01063 | 96   | ■ 0.999 | $\mathrm{id}_v$                                  | $\mathrm{id}_v$                                  | and ending at $v$ ; we often denote it by $\mathrm{id}_v$ or simply by $v$ . We may also concatenate                                                             |
| fong-spivak-invitation_F01064 | 96   | ■ 1.000 | $q: w \rightarrow x$                             | $q: w \rightarrow x$                             | paths: given $p: v \rightarrow w$ and $q: w \rightarrow x$ , their concatenation is denoted $p \circ q$ , and it                                                 |
| fong-spivak-invitation_F01065 | 96   | ■ 1.000 | $p ; q$                                          | $p; q$                                           | paths: given $p: v \rightarrow w$ and $q: w \rightarrow x$ , their concatenation is denoted $p \circ q$ , and it                                                 |
| fong-spivak-invitation_F01066 | 96   | ■ 1.000 | $v \rightarrow x$                                | $v \rightarrow x$                                | goes $\Rightarrow$ .                                                                                                                                             |
| fong-spivak-invitation_F01067 | 96   | ■ 0.658 | $(V, \leq)$                                      | $(V, \leq)$                                      | In Chapter 1, we used graphs to depict preorders $(V, \leq)$ : the vertices form the ele-                                                                        |
| fong-spivak-invitation_F01068 | 96   | ■ 0.977 | $\operatorname{Free}(G)$                         | $\mathrm{Free}(G)$                               | <b>Definition 3.3.</b> For any graph $G = (V, A, s, t)$ , we can define a category $\mathbf{Free}(G)$ ,                                                          |

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|-------------------------------|------|-------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fong-spivak-invitation_F01069 | 96   | 0.651 | $v_{\{1\}}$                                                                                                 | $v_1$                                                                | It has two objects $v_1$ and $v_2$ , and three morphisms: $\mathrm{id}_{v_1} : v_1 \rightarrow v_1$ , $f_1 : v_1 \rightarrow v_2$ , and                                       |
| fong-spivak-invitation_F01070 | 96   | 0.651 | $v_{\{2\}}$                                                                                                 | $v_2$                                                                | It has two objects $v_1$ and $v_2$ , and three morphisms: $\mathrm{id}_{v_1} : v_1 \rightarrow v_1$ , $f_1 : v_1 \rightarrow v_2$ , and                                       |
| fong-spivak-invitation_F01071 | 96   | 0.651 | $\mathrm{id}_{v_{\{1\}}} : v_{\{1\}} \rightarrow v_{\{1\}}$ , $f_{\{1\}} : v_{\{1\}} \rightarrow v_{\{2\}}$ | $\mathrm{id}_{v_1} : v_1 \rightarrow v_1, f_1 : v_1 \rightarrow v_2$ | It has two objects $v_1$ and $v_2$ , and three morphisms: $\mathrm{id}_{v_1} : v_1 \rightarrow v_1$ , $f_1 : v_1 \rightarrow v_2$ , and                                       |
| fong-spivak-invitation_F01072 | 96   | 0.992 | $\mathrm{id}_{v_{\{2\}}} : v_{\{2\}} \rightarrow v_{\{2\}}$                                                 | $\mathrm{id}_{v_2} : v_2 \rightarrow v_2$                            | $\mathrm{id}_{v_2} : v_2 \rightarrow v_2$ . Here $\mathrm{id}_{v_1}$ is the path of length 0 starting and ending at $v_1$ , $f_1$ is the path                                 |
| fong-spivak-invitation_F01073 | 96   | 0.992 | $\mathrm{id}_{v_{\{1\}}}$                                                                                   | $\mathrm{id}_{v_1}$                                                  | $\mathrm{id}_{v_2} : v_2 \rightarrow v_2$ . Here $\mathrm{id}_{v_1}$ is the path of length 0 starting and ending at $v_1$ , $f_1$ is the path                                 |
| fong-spivak-invitation_F01074 | 96   | 0.992 | $v_{\{1\}}$ , $f_{\{1\}}$                                                                                   | $v_1, f_1$                                                           | $\mathrm{id}_{v_2} : v_2 \rightarrow v_2$ . Here $\mathrm{id}_{v_1}$ is the path of length 0 starting and ending at $v_1$ , $f_1$ is the path                                 |
| fong-spivak-invitation_F01075 | 96   | 0.999 | $f_{\{1\}}$                                                                                                 | $f_1$                                                                | of length 1 consisting of just the arrow $f_1$ , and $\mathrm{id}_{v_2}$ is the length 0 path at $v_2$ . As our                                                               |
| fong-spivak-invitation_F01076 | 96   | 0.999 | $\mathrm{id}_{v_{\{2\}}}$                                                                                   | $\mathrm{id}_{v_2}$                                                  | of length 1 consisting of just the arrow $f_1$ , and $\mathrm{id}_{v_2}$ is the length 0 path at $v_2$ . As our                                                               |
| fong-spivak-invitation_F01077 | 96   | 0.424 | $f_{\{1\}}=f_{\{1\}}$                                                                                       | $f_1 = f_1$                                                          | $v_2$ . As composition is given by concatenation, we have, for example, $\mathrm{id}_{v_1} \circ f_1 = f_1$ ,                                                                 |
| fong-spivak-invitation_F01078 | 96   | 0.596 | $\mathrm{id}_{v_{\{2\}}} \circ \mathrm{id}_{v_{\{2\}}} = \mathrm{id}_{v_{\{2\}}}$                           | $\mathrm{id}_{v_2} \circ \mathrm{id}_{v_2} = \mathrm{id}_{v_2}$      | $\mathrm{id}_{v_2} \circ \mathrm{id}_{v_2} = \mathrm{id}_{v_2}$ , and so on.                                                                                                  |
| fong-spivak-invitation_F01079 | 96   | 0.821 | $\mathrm{Free}(G)$                                                                                          | $\mathbf{Free}(G)$                                                   | free category $\mathbf{Free}(G)$ , at least when the one we mean is clear enough from context.                                                                                |
| fong-spivak-invitation_F01080 | 97   | 0.850 | $\mathbf{0}$                                                                                                | $\mathbf{0}$                                                         | 2. What is the category $\mathbf{0}$ ?                                                                                                                                        |
| fong-spivak-invitation_F01081 | 97   | 0.994 | $\mathbf{n}$                                                                                                | $\mathbf{n}$                                                         | 3. What is the formula for the number of morphisms in $\mathbf{n}$ for arbitrary $n \in \mathbb{N}$ ?                                                                         |
| fong-spivak-invitation_F01082 | 97   | 0.337 | $\mathrm{Path}(z, s) = \{z, s, (s \circ s_9) \circ \dots\}$                                                 | $= \{z, s, (s_9 \circ s) \circ \dots\}$                              | a unique path of length $n$ for every natural number $n \in \mathbb{N}$ . That is, $\mathrm{Path}(z, s) = \{z, s, (s \circ s_9) \circ \dots\}$                                |
| fong-spivak-invitation_F01083 | 97   | 0.999 | $\mathrm{id}_z$                                                                                             | $\mathrm{id}_z$                                                      | $\mathrm{id}_z$ . There is a one-to-one correspondence between $\mathrm{Path}(z, s)$ and the natural numbers, $\mathbb{N} = \{0, 1, 2, 3, \dots\}$                            |
| fong-spivak-invitation_F01084 | 97   | 0.999 | $\mathbb{N}$                                                                                                | $\mathbb{N}$                                                         | $\mathrm{id}_z$ . There is a one-to-one correspondence between $\mathrm{Path}(z, s)$ and the natural numbers, $\mathbb{N} = \{0, 1, 2, 3, \dots\}$                            |
| fong-spivak-invitation_F01085 | 97   | 0.567 | $\{0, 1, 2, 3, \dots\}$                                                                                     | $\{0, 1, 2, 3, \dots\}$                                              | $\{0, 1, 2, 3, \dots\}$                                                                                                                                                       |
| fong-spivak-invitation_F01086 | 97   | 1.000 | $m * 1 = m = 1 * m$                                                                                         | $m * 1 = m = 1 * m$                                                  | $m * 1 = m = 1 * m$ and $(m * n) * p = m * (n * p)$ .                                                                                                                         |
| fong-spivak-invitation_F01087 | 97   | 1.000 | $(m * n) * p = m * (n * p)$                                                                                 | $(m * n) * p = m * (n * p)$                                          | $m * 1 = m = 1 * m$ and $(m * n) * p = m * (n * p)$ .                                                                                                                         |
| fong-spivak-invitation_F01088 | 97   | 0.996 | $M := \mathcal{C}(\bullet, \bullet)$                                                                        | $M := \mathcal{C}(\bullet, \bullet)$                                 | nection. Given a category $\mathcal{C}$ with one object, say $\bullet$ , let $M := \mathcal{C}(\bullet, \bullet)$ , let $e = \mathrm{id}_\bullet$ ,                           |
| fong-spivak-invitation_F01089 | 97   | 0.996 | $e = \mathrm{id}_\bullet$                                                                                   | $e = \mathrm{id}_\bullet$                                            | nection. Given a category $\mathcal{C}$ with one object, say $\bullet$ , let $M := \mathcal{C}(\bullet, \bullet)$ , let $e = \mathrm{id}_\bullet$ ,                           |
| fong-spivak-invitation_F01090 | 97   | 0.896 | $*$                                                                                                         | $*$                                                                  | and let $*$ : $\mathcal{C}(\bullet, \bullet) \times \mathcal{C}(\bullet, \bullet) \rightarrow \mathcal{C}(\bullet, \bullet)$ be the composition operation $*$ = $\circ$ . The |
| fong-spivak-invitation_F01091 | 97   | 0.896 | $*$                                                                                                         | $*$                                                                  | and let $*$ : $\mathcal{C}(\bullet, \bullet) \times \mathcal{C}(\bullet, \bullet) \rightarrow \mathcal{C}(\bullet, \bullet)$ be the composition operation $*$ = $\circ$ . The |
| fong-spivak-invitation_F01092 | 97   | 1.000 | $m, n \in \mathbb{N}$                                                                                       | $m, n \in \mathbb{N}$                                                | numbers $n \in \mathbb{N}$ . Paths can be concatenated. Given numbers $m, n \in \mathbb{N}$ , what number                                                                     |
| fong-spivak-invitation_F01093 | 98   | 0.170 | $i$                                                                                                         | $i$                                                                  | because $f \circ h$ and $g \circ i$ are made equal. This category is called the “commutative square.”                                                                         |
| fong-spivak-invitation_F01094 | 98   | 0.251 | $g \circ i$                                                                                                 | $g \circ i$                                                          | One might say “the missing one is $g \circ i$ ,” but that is not quite right: $g \circ i$ is there too,                                                                       |
| fong-spivak-invitation_F01095 | 98   | 0.251 | $g \circ i$                                                                                                 | $g \circ i$                                                          | One might say “the missing one is $g \circ i$ ,” but that is not quite right: $g \circ i$ is there too,                                                                       |

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| fong-spivak-invitation_F01096 | 98   | 0.696 | $f ; h$                                                | $f;h$                                               | because it is equal to $f \circ h$ . As usual, $A$ denotes $\text{id}_A$ , etc.                                                    |
| fong-spivak-invitation_F01097 | 98   | 0.696 | $\operatorname{id}_{\{A\}}$                            | $\text{id}_A$                                       | because it is equal to $f \circ h$ . As usual, $A$ denotes $\text{id}_A$ , etc.                                                    |
| fong-spivak-invitation_F01098 | 99   | 0.255 | $s ; s=z$                                              | $s; s = z$                                          | In $\mathcal{C}$ we have the equation $s \circ s = z$ . But this implies $s \circ s \circ s = z \circ s = s$ ! And similarly       |
| fong-spivak-invitation_F01099 | 99   | 0.255 | $s ; s ; s=z$                                          | $s; s; s = z$                                       | In $\mathcal{C}$ we have the equation $s \circ s = z$ . But this implies $s \circ s \circ s = z \circ s = s$ ! And similarly       |
| fong-spivak-invitation_F01100 | 99   | 0.153 | $s=z$                                                  | $s = z$                                             | we have $s \circ s \circ s \circ s = z \circ z = z$ . The set of morphisms in $\mathcal{C}$ is in fact merely $\{z, s\}$ , with    |
| fong-spivak-invitation_F01101 | 99   | 0.153 | $z=z$                                                  | $z = z$                                             | we have $s \circ s \circ s \circ s = z \circ z = z$ . The set of morphisms in $\mathcal{C}$ is in fact merely $\{z, s\}$ , with    |
| fong-spivak-invitation_F01102 | 99   | 0.153 | $\{z, s\}$                                             | $\{z, s\}$                                          | we have $s \circ s \circ s \circ s = z \circ z = z$ . The set of morphisms in $\mathcal{C}$ is in fact merely $\{z, s\}$ , with    |
| fong-spivak-invitation_F01103 | 99   | 0.217 | $=z$                                                   | $= z$                                               | composition described by $s \circ s = z \circ z = z$ , and $z \circ s = s \circ z = s$ . In group theory, one                      |
| fong-spivak-invitation_F01104 | 99   | 0.217 | $s=s$                                                  | $s = s$                                             | composition described by $s \circ s = z \circ z = z$ , and $z \circ s = s \circ z = s$ . In group theory, one                      |
| fong-spivak-invitation_F01105 | 99   | 0.217 | $z=s$                                                  | $z = s$                                             | composition described by $s \circ s = z \circ z = z$ , and $z \circ s = s \circ z = s$ . In group theory, one                      |
| fong-spivak-invitation_F01106 | 99   | 1.000 | $\mathbb{Z} / 2 \mathbb{Z}$                            | $\mathbb{Z}/2\mathbb{Z}$                            | would speak of a group called $\mathbb{Z}/2\mathbb{Z}$ .                                                                           |
| fong-spivak-invitation_F01107 | 99   | 1.000 | $\mathcal{D}$                                          | $\mathcal{D}$                                       | <b>Exercise 3.12.</b> Write down all the morphisms in the category $\mathcal{D}$ from Example 3.11. $\diamond$                     |
| fong-spivak-invitation_F01108 | 99   | 0.991 | $\operatorname{Ob}(\mathcal{P})=P$                     | $\text{Ob}(\mathcal{P}) = P$                        | are precisely the elements of $P$ ; that is, $\text{Ob}(\mathcal{P}) = P$ . As for morphisms, $\mathcal{P}$ has exactly            |
| fong-spivak-invitation_F01109 | 99   | 1.000 | $p \rightarrow q$                                      | $p \rightarrow q$                                   | one morphism $p \rightarrow q$ if $p \leq q$ and no morphisms $p \rightarrow q$ if $p \not\leq q$ . The fact that                  |
| fong-spivak-invitation_F01110 | 99   | 1.000 | $p \not \leq q$                                        | $p \not\leq q$                                      | one morphism $p \rightarrow q$ if $p \leq q$ and no morphisms $p \rightarrow q$ if $p \not\leq q$ . The fact that                  |
| fong-spivak-invitation_F01111 | 100  | 0.921 | $(C, \leq)$                                            | $(C, \leq)$                                         | Given any category $\mathcal{C}$ , one can obtain a preorder $(C, \leq)$ from it by destroying the dis-                            |
| fong-spivak-invitation_F01112 | 100  | 0.931 | $C:=\operatorname{Ob}(\mathcal{C})$                    | $C := \text{Ob}(\mathcal{C})$                       | tinction between any two parallel morphisms. That is, let $C := \text{Ob}(\mathcal{C})$ , and put $c_1 \leq c_2$                   |
| fong-spivak-invitation_F01113 | 100  | 0.931 | $c_{\{1\}} \leq c_{\{2\}}$                             | $c_1 \leq c_2$                                      | tinction between any two parallel morphisms. That is, let $C := \text{Ob}(\mathcal{C})$ , and put $c_1 \leq c_2$                   |
| fong-spivak-invitation_F01114 | 100  | 0.999 | $\mathcal{C}(c_{\{1\}}, c_{\{2\}}) \neq \varnothing$   | $\mathcal{C}(c_1, c_2) \neq \emptyset$              | iff $\mathcal{C}(c_1, c_2) \neq \emptyset$ . If there is one, or two, or fifty, or infinitely many morphisms $c_1 \rightarrow c_2$ |
| fong-spivak-invitation_F01115 | 100  | 0.999 | $c_{\{1\}} \rightarrow c_{\{2\}}$                      | $c_1 \rightarrow c_2$                               | iff $\mathcal{C}(c_1, c_2) \neq \emptyset$ . If there is one, or two, or fifty, or infinitely many morphisms $c_1 \rightarrow c_2$ |
| fong-spivak-invitation_F01116 | 101  | 0.878 | $\operatorname{Set}(S, T)=\{f: S \rightarrow T \mid f$ | $\mathbf{Set}(S, T) = \{f : S \rightarrow T \mid f$ | (ii) If $S$ and $T$ are sets, then $\mathbf{Set}(S, T) = \{f : S \rightarrow T \mid f \text{ is a function}\}$ .                   |
| fong-spivak-invitation_F01117 | 101  | 1.000 | $\operatorname{id}_{\{S\}}: S \rightarrow S$           | $\text{id}_S : S \rightarrow S$                     | (iii) For each set $S$ , the identity morphism is the function $\text{id}_S : S \rightarrow S$ given by                            |
| fong-spivak-invitation_F01118 | 101  | 0.979 | $\operatorname{id}_{\{S\}}(s):=s$                      | $\text{id}_S(s) := s$                               | $\text{id}_S(s) := s$ for each $s \in S$ .                                                                                         |
| fong-spivak-invitation_F01119 | 101  | 1.000 | $f: S \rightarrow T$                                   | $f : S \rightarrow T$                               | (iv) Given $f : S \rightarrow T$ and $g : T \rightarrow U$ , their composite is the function $f \circ g : S \rightarrow U$         |
| fong-spivak-invitation_F01120 | 101  | 1.000 | $g: T \rightarrow U$                                   | $g : T \rightarrow U$                               | (iv) Given $f : S \rightarrow T$ and $g : T \rightarrow U$ , their composite is the function $f \circ g : S \rightarrow U$         |
| fong-spivak-invitation_F01121 | 101  | 1.000 | $f ; g: S \rightarrow U$                               | $f; g : S \rightarrow U$                            | (iv) Given $f : S \rightarrow T$ and $g : T \rightarrow U$ , their composite is the function $f \circ g : S \rightarrow U$         |
| fong-spivak-invitation_F01122 | 101  | 1.000 | $(f ; g)(s):=g(f(s))$                                  | $(f; g)(s) := g(f(s))$                              | given by $(f \circ g)(s) := g(f(s))$ .                                                                                             |

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| fong-spivak-invitation_F01123 | 101  | ■ 1.000 | $\underline{2}=\{1,2\}$                                                          | $\underline{2} = \{1, 2\}$                                                       | <b>Exercise 3.18.</b> Let $\underline{2} = \{1, 2\}$ and $\underline{3} = \{1, 2, 3\}$ . These are objects in the category                                                   |
| fong-spivak-invitation_F01124 | 101  | ■ 1.000 | $\underline{3}=\{1,2,3\}$                                                        | $\underline{3} = \{1, 2, 3\}$                                                    | <b>Exercise 3.18.</b> Let $\underline{2} = \{1, 2\}$ and $\underline{3} = \{1, 2, 3\}$ . These are objects in the category                                                   |
| fong-spivak-invitation_F01125 | 102  | ■ 0.881 | $\mathcal{C}^{\mathrm{op}}$                                                      | $\mathcal{C}^{\mathrm{op}}$                                                      | <b>Example 3.20.</b> Let $\mathcal{C}$ be a category. Its <i>opposite</i> , denoted $\mathcal{C}^{\mathrm{op}}$ , is the category with the                                   |
| fong-spivak-invitation_F01126 | 102  | ■ 0.488 | $\operatorname{Ob}(\mathcal{C}^{\mathrm{op}}) := \operatorname{Ob}(\mathcal{C})$ | $\operatorname{Ob}(\mathcal{C}^{\mathrm{op}}) := \operatorname{Ob}(\mathcal{C})$ | same objects, $\operatorname{Ob}(\mathcal{C}^{\mathrm{op}}) := \operatorname{Ob}(\mathcal{C})$ , and for any two objects $c, d \in \operatorname{Ob}(\mathcal{C})$ , one has |
| fong-spivak-invitation_F01127 | 102  | ■ 0.968 | $\mathcal{C}^{\mathrm{op}}(c, d) := \mathcal{C}(d, c)$                           | $\mathcal{C}^{\mathrm{op}}(c, d) := \mathcal{C}(d, c)$                           | $\mathcal{C}^{\mathrm{op}}(c, d) := \mathcal{C}(d, c)$ . Identities and composition are as in $\mathcal{C}$ .                                                                |
| fong-spivak-invitation_F01128 | 102  | ■ 0.107 | $\{\boxed{\phantom{0}}, \sim\} \square$                                          | $\{\boxed{\phantom{0}}, \square\}$                                               | in the category <b>Set</b> , one can exchange the set $\{\blacksquare, \square\}$ for the set $\{0, 1\}$ and everything                                                      |
| fong-spivak-invitation_F01129 | 102  | ■ 1.000 | $a \cong b$                                                                      | $a \cong b$                                                                      | with elements $a, b$ , such that $a \leq b$ and $b \leq a$ , i.e. $a \cong b$ , then $a$ and $b$ are essentially                                                             |
| fong-spivak-invitation_F01130 | 102  | ■ 1.000 | $c \leq a$                                                                       | $c \leq a$                                                                       | $c \leq a$ iff $c \leq b$ , and $c \geq a$ iff $c \geq b$ . The notion of isomorphism formalizes this notion                                                                 |
| fong-spivak-invitation_F01131 | 102  | ■ 1.000 | $c \leq b$                                                                       | $c \leq b$                                                                       | $c \leq a$ iff $c \leq b$ , and $c \geq a$ iff $c \geq b$ . The notion of isomorphism formalizes this notion                                                                 |
| fong-spivak-invitation_F01132 | 102  | ■ 1.000 | $c \geq a$                                                                       | $c \geq a$                                                                       | $c \leq a$ iff $c \leq b$ , and $c \geq a$ iff $c \geq b$ . The notion of isomorphism formalizes this notion                                                                 |
| fong-spivak-invitation_F01133 | 102  | ■ 1.000 | $c \geq b$                                                                       | $c \geq b$                                                                       | $c \leq a$ iff $c \leq b$ , and $c \geq a$ iff $c \geq b$ . The notion of isomorphism formalizes this notion                                                                 |
| fong-spivak-invitation_F01134 | 102  | ■ 0.456 | $g: B \rightarrow A$                                                             | $g: B \rightarrow A$                                                             | morphism $g: B \rightarrow A$ satisfying $f \circ g = \operatorname{id}_A$ and $g \circ f = \operatorname{id}_B$ . In this case we call $f$                                  |
| fong-spivak-invitation_F01135 | 102  | ■ 0.456 | $f \circ g = \operatorname{id}_A$                                                | $f \circ g = \operatorname{id}_A$                                                | morphism $g: B \rightarrow A$ satisfying $f \circ g = \operatorname{id}_A$ and $g \circ f = \operatorname{id}_B$ . In this case we call $f$                                  |
| fong-spivak-invitation_F01136 | 102  | ■ 0.456 | $g \circ f = \operatorname{id}_B$                                                | $g \circ f = \operatorname{id}_B$                                                | morphism $g: B \rightarrow A$ satisfying $f \circ g = \operatorname{id}_A$ and $g \circ f = \operatorname{id}_B$ . In this case we call $f$                                  |
| fong-spivak-invitation_F01137 | 102  | ■ 1.000 | $g = f^{-1}$                                                                     | $g = f^{-1}$                                                                     | and $g$ <i>inverses</i> , and we often write $g = f^{-1}$ , or equivalently $f = g^{-1}$ . We also say that                                                                  |
| fong-spivak-invitation_F01138 | 102  | ■ 1.000 | $f = g^{-1}$                                                                     | $f = g^{-1}$                                                                     | and $g$ <i>inverses</i> , and we often write $g = f^{-1}$ , or equivalently $f = g^{-1}$ . We also say that                                                                  |
| fong-spivak-invitation_F01139 | 102  | ■ 1.000 | $A := \{a, b, c\}$                                                               | $A := \{a, b, c\}$                                                               | <b>Example 3.22.</b> The set $A := \{a, b, c\}$ and the set $\underline{3} = \{1, 2, 3\}$ are isomorphic, i.e. there                                                         |
| fong-spivak-invitation_F01140 | 102  | ■ 1.000 | $f: A \rightarrow \underline{3}$                                                 | $f: A \rightarrow \underline{3}$                                                 | exists an isomorphism $f: A \rightarrow \underline{3}$ given by $f(a) = 2, f(b) = 1, f(c) = 3$ . The                                                                         |
| fong-spivak-invitation_F01141 | 102  | ■ 1.000 | $f(a)=2, f(b)=1, f(c)=3$                                                         | $f(a) = 2, f(b) = 1, f(c) = 3$                                                   | exists an isomorphism $f: A \rightarrow \underline{3}$ given by $f(a) = 2, f(b) = 1, f(c) = 3$ . The                                                                         |
| fong-spivak-invitation_F01142 | 103  | ■ 0.979 | $A \in$                                                                          | $A \in$                                                                          | understood in terms of isomorphisms in <b>FinSet</b> . Namely, for any finite set $A \in \mathbf{FinSet}$ ,                                                                  |
| fong-spivak-invitation_F01143 | 103  | ■ 1.000 | $A \cong \underline{n}$                                                          | $A \cong \underline{n}$                                                          | its cardinality is the number $n \in \mathbb{N}$ such that there exists an isomorphism $A \cong \underline{n}$ . Georg                                                       |
| fong-spivak-invitation_F01144 | 103  | ■ 1.000 | $f^{-1}: \underline{3} \rightarrow A$                                            | $f^{-1}: \underline{3} \rightarrow A$                                            | 1. What is the inverse $f^{-1}: \underline{3} \rightarrow A$ of the function $f$ given in Example 3.22?                                                                      |
| fong-spivak-invitation_F01145 | 103  | ■ 1.000 | $A \rightarrow \underline{3}$                                                    | $A \rightarrow \underline{3}$                                                    | 2. How many distinct isomorphisms are there $A \rightarrow \underline{3}$ ?                                                                                                  |
| fong-spivak-invitation_F01146 | 103  | ■ 0.998 | $\operatorname{id}_c$                                                            | $\operatorname{id}_c$                                                            | identity $\operatorname{id}_c$ is an isomorphism.                                                                                                                            |
| fong-spivak-invitation_F01147 | 103  | ■ 1.000 | $f: \underline{2} \rightarrow \underline{3}$                                     | $f: \underline{2} \rightarrow \underline{3}$                                     | Consider the functions $f: \underline{2} \rightarrow \underline{3}$ and $g: \underline{3} \rightarrow \underline{2}$ drawn below:                                            |
| fong-spivak-invitation_F01148 | 103  | ■ 1.000 | $g: \underline{3} \rightarrow \underline{2}$                                     | $g: \underline{3} \rightarrow \underline{2}$                                     | Consider the functions $f: \underline{2} \rightarrow \underline{3}$ and $g: \underline{3} \rightarrow \underline{2}$ drawn below:                                            |
| fong-spivak-invitation_F01149 | 103  | ■ 0.180 | $f \circ g = \operatorname{id}_{\underline{2}}$                                  | $f \circ g = \operatorname{id}_{\underline{2}}$                                  | Then the reader should be able to check instantly that $f \circ g = \operatorname{id}_{\underline{2}}$ but $g \circ f \neq \operatorname{id}_{\underline{3}}$ . Thus         |

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| fong-spivak-invitation_F01150 | 103  | ■ 0.180 | $g: f \not\equiv \mathrm{id}_{\underline{3}}$     | $g: f \neq \mathrm{id}_3$                         | Then the reader should be able to check instantly that $f \circ g = \mathrm{id}_2$ but $g \circ f \neq \mathrm{id}_3$ . Thus                                           |
| fong-spivak-invitation_F01151 | 105  | ■ 1.000 | $F: \mathcal{C} \rightarrow \mathcal{D}$          | $F: \mathcal{C} \rightarrow \mathcal{D}$          | $F: \mathcal{C} \rightarrow \mathcal{D}$ ,                                                                                                                             |
| fong-spivak-invitation_F01152 | 105  | ■ 0.802 | $F(c) \in \mathrm{Ob}(\mathcal{D})$               | $F(c) \in \mathrm{Ob}(\mathcal{D})$               | (i) for every object $c \in \mathrm{Ob}(\mathcal{C})$ , one specifies an object $F(c) \in \mathrm{Ob}(\mathcal{D})$ ;                                                  |
| fong-spivak-invitation_F01153 | 105  | ■ 1.000 | $f: c_1 \rightarrow c_2$                          | $f: c_1 \rightarrow c_2$                          | (ii) for every morphism $f: c_1 \rightarrow c_2$ in $\mathcal{C}$ , one specifies a morphism $F(f): F(c_1) \rightarrow$                                                |
| fong-spivak-invitation_F01154 | 105  | ■ 1.000 | $F(f): F(c_1) \rightarrow$                        | $F(f): F(c_1) \rightarrow$                        | (ii) for every morphism $f: c_1 \rightarrow c_2$ in $\mathcal{C}$ , one specifies a morphism $F(f): F(c_1) \rightarrow$                                                |
| fong-spivak-invitation_F01155 | 105  | ■ 1.000 | $F(c_2)$                                          | $F(c_2)$                                          | $F(c_2)$ in $\mathcal{D}$ .                                                                                                                                            |
| fong-spivak-invitation_F01156 | 105  | ■ 0.306 | $F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$           | $F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$           | (a) for every object $c \in \mathrm{Ob}(\mathcal{C})$ , we have $F(\mathrm{id}_c) = \mathrm{id}_{F(c)}$ ;                                                              |
| fong-spivak-invitation_F01157 | 105  | ■ 0.620 | $c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$      | $c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$      | (b) for every three objects $c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$ and two morphisms $f \in \mathcal{C}(c_1, c_2), g \in$                                        |
| fong-spivak-invitation_F01158 | 105  | ■ 0.620 | $f \in \mathcal{C}(c_1, c_2), g \in$              | $f \in \mathcal{C}(c_1, c_2), g \in$              | (b) for every three objects $c_1, c_2, c_3 \in \mathrm{Ob}(\mathcal{C})$ and two morphisms $f \in \mathcal{C}(c_1, c_2), g \in$                                        |
| fong-spivak-invitation_F01159 | 105  | ■ 1.000 | $\mathcal{C}(c_2, c_3)$                           | $\mathcal{C}(c_2, c_3)$                           | $\mathcal{C}(c_2, c_3)$ , the equation $F(f \circ g) = F(f) \circ F(g)$ holds in $\mathcal{D}$ .                                                                       |
| fong-spivak-invitation_F01160 | 105  | ■ 1.000 | $F(f \circ g) = F(f) \circ F(g)$                  | $F(f \circ g) = F(f) \circ F(g)$                  | $\mathcal{C}(c_2, c_3)$ , the equation $F(f \circ g) = F(f) \circ F(g)$ holds in $\mathcal{D}$ .                                                                       |
| fong-spivak-invitation_F01161 | 105  | ■ 0.979 | $F: \mathbf{2} \rightarrow \mathbf{3}$            | $F: \mathbf{2} \rightarrow \mathbf{3}$            | <b>Example 3.29.</b> For example, here we draw three functors $F: \mathbf{2} \rightarrow \mathbf{3}$ :                                                                 |
| fong-spivak-invitation_F01162 | 106  | ■ 1.000 | $n_0$                                             | $n_0$                                             | path to the trivial path, i.e. the identity on $n_0$ . In the middle diagram $F(m_0) = n_0$ ,                                                                          |
| fong-spivak-invitation_F01163 | 106  | ■ 1.000 | $F(m_0) = n_0$                                    | $F(m_0) = n_0$                                    | path to the trivial path, i.e. the identity on $n_0$ . In the middle diagram $F(m_0) = n_0$ ,                                                                          |
| fong-spivak-invitation_F01164 | 106  | ■ 1.000 | $F(f_1) = g_1$                                    | $F(f_1) = g_1$                                    | $F(f_1) = g_1$ , and $F(m_1) = n_1$ . In the right-hand diagram, $F(m_0) = n_0, F(m_1) = n_2$ ,                                                                        |
| fong-spivak-invitation_F01165 | 106  | ■ 1.000 | $F(m_1) = n_1$                                    | $F(m_1) = n_1$                                    | $F(f_1) = g_1$ , and $F(m_1) = n_1$ . In the right-hand diagram, $F(m_0) = n_0, F(m_1) = n_2$ ,                                                                        |
| fong-spivak-invitation_F01166 | 106  | ■ 1.000 | $F(m_0) = n_0, F(m_1) = n_2$                      | $F(m_0) = n_0, F(m_1) = n_2$                      | $F(f_1) = g_1$ , and $F(m_1) = n_1$ . In the right-hand diagram, $F(m_0) = n_0, F(m_1) = n_2$ ,                                                                        |
| fong-spivak-invitation_F01167 | 106  | ■ 0.917 | $g_2$                                             | $g_2$                                             | and $F(f_1) = g_1 \circ g_2$ .                                                                                                                                         |
| fong-spivak-invitation_F01168 | 106  | ■ 0.974 | $2 \rightarrow 3$                                 | $2 \rightarrow 3$                                 | <b>Exercise 3.30.</b> Above we wrote down three functors $2 \rightarrow 3$ . Find and write down all                                                                   |
| fong-spivak-invitation_F01169 | 106  | ■ 0.915 | $\mathbf{2} \rightarrow \mathbf{3}$               | $\mathbf{2} \rightarrow \mathbf{3}$               | the remaining functors $2 \rightarrow 3$ .                                                                                                                             |
| fong-spivak-invitation_F01170 | 106  | ■ 1.000 | $\mathcal{F}$                                     | $\mathcal{F}$                                     | There are lots of functors from the free square category (let's call it $\mathcal{F}$ ) to the                                                                         |
| fong-spivak-invitation_F01171 | 106  | ■ 1.000 | $F: \mathcal{F} \rightarrow \mathcal{C}$          | $F: \mathcal{F} \rightarrow \mathcal{C}$          | However, there is exactly one functor $F: \mathcal{F} \rightarrow \mathcal{C}$ that sends $A'$ to $A$ , $B'$ to $B$ , $C'$ to                                          |
| fong-spivak-invitation_F01172 | 106  | ■ 1.000 | $A, B'$                                           | $A, B'$                                           | However, there is exactly one functor $F: \mathcal{F} \rightarrow \mathcal{C}$ that sends $A'$ to $A$ , $B'$ to $B$ , $C'$ to                                          |
| fong-spivak-invitation_F01173 | 106  | ■ 1.000 | $B, C'$                                           | $B, C'$                                           | However, there is exactly one functor $F: \mathcal{F} \rightarrow \mathcal{C}$ that sends $A'$ to $A$ , $B'$ to $B$ , $C'$ to                                          |
| fong-spivak-invitation_F01174 | 106  | ■ 1.000 | $D'$                                              | $D'$                                              | $C$ , and $D'$ to $D$ . That is, once we have made this decision about how $F$ acts on objects,                                                                        |
| fong-spivak-invitation_F01175 | 106  | ■ 0.851 | $\mathcal{C} = \bullet \rightarrow \bullet$       | $\mathcal{C} = \bullet \rightarrow \bullet$       | <b>Exercise 3.33.</b> Consider the free categories $\mathcal{C} = \boxed{\bullet \rightarrow \bullet}$ and $\mathcal{D} = \boxed{\bullet \rightrightarrows \bullet}$ . |
| fong-spivak-invitation_F01176 | 106  | ■ 0.851 | $\mathcal{D} = \bullet \rightrightarrows \bullet$ | $\mathcal{D} = \bullet \rightrightarrows \bullet$ | <b>Exercise 3.33.</b> Consider the free categories $\mathcal{C} = \boxed{\bullet \rightarrow \bullet}$ and $\mathcal{D} = \boxed{\bullet \rightrightarrows \bullet}$ . |

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| fong-spivak-invitation_F01177 | 106  | ■ 1.000 | $F, G: \mathcal{C} \rightarrow \mathcal{D}$                            | $F, G: \mathcal{C} \rightarrow \mathcal{D}$                            | Give two functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$ that act the same on objects but differently on                                                           |
| fong-spivak-invitation_F01178 | 106  | ■ 1.000 | $A^{\prime}, B$                                                        | $A', B$                                                                | gory $\mathcal{C}$ to the free square category $\mathcal{F}$ , but <i>none</i> that sends $A$ to $A'$ , $B$ to $B'$ , $C$ to $C'$ ,                                     |
| fong-spivak-invitation_F01179 | 106  | ■ 1.000 | $B^{\prime}, C$                                                        | $B', C$                                                                | gory $\mathcal{C}$ to the free square category $\mathcal{F}$ , but <i>none</i> that sends $A$ to $A'$ , $B$ to $B'$ , $C$ to $C'$ ,                                     |
| fong-spivak-invitation_F01180 | 106  | ■ 1.000 | $C^{\prime}$                                                           | $C'$                                                                   | gory $\mathcal{C}$ to the free square category $\mathcal{F}$ , but <i>none</i> that sends $A$ to $A'$ , $B$ to $B'$ , $C$ to $C'$ ,                                     |
| fong-spivak-invitation_F01181 | 106  | ■ 0.961 | $f; h = g; i$                                                          | $f; h = g; i$                                                          | and $D$ to $D'$ . The reason is that if $F$ were such a functor, then since $f \circ h = g \circ i$ in $\mathcal{C}$ , we                                               |
| fong-spivak-invitation_F01182 | 107  | ■ 1.000 | $F(f; h) = F(g; i)$                                                    | $F(f; h) = F(g; i)$                                                    | would have $F(f \circ h) = F(g \circ i)$ , but then the rules of functors would let us reason as                                                                        |
| fong-spivak-invitation_F01183 | 107  | ■ 0.600 | $f^{\prime}; h^{\prime} = g^{\prime}; i^{\prime}$                      | $f'; h' = g'; i'$                                                      | The resulting equation, $f' \circ h' = g' \circ i'$ does not hold in $\mathcal{F}$ because it is a free category                                                        |
| fong-spivak-invitation_F01184 | 107  | ■ 0.998 | $\mathcal{N}$                                                          | $\mathcal{N}$                                                          | For example, consider the preorder $(\mathbb{N}, \leq)$ considered as a category $\mathcal{N}$ with objects                                                             |
| fong-spivak-invitation_F01185 | 107  | ■ 0.998 | $\operatorname{Ob}(\mathcal{N}) = \mathbb{N}$                          | $\operatorname{Ob}(\mathcal{N}) = \mathbb{N}$                          | $\operatorname{Ob}(\mathcal{N}) = \mathbb{N}$ and a unique morphism $m \rightarrow n$ iff $m \leq n$ . A functor $F: \mathcal{N} \rightarrow \mathcal{N}$               |
| fong-spivak-invitation_F01186 | 107  | ■ 0.998 | $m \rightarrow n$                                                      | $m \rightarrow n$                                                      | $\operatorname{Ob}(\mathcal{N}) = \mathbb{N}$ and a unique morphism $m \rightarrow n$ iff $m \leq n$ . A functor $F: \mathcal{N} \rightarrow \mathcal{N}$               |
| fong-spivak-invitation_F01187 | 107  | ■ 0.998 | $F: \mathcal{N} \rightarrow \mathcal{N}$                               | $F: \mathcal{N} \rightarrow \mathcal{N}$                               | $\operatorname{Ob}(\mathcal{N}) = \mathbb{N}$ and a unique morphism $m \rightarrow n$ iff $m \leq n$ . A functor $F: \mathcal{N} \rightarrow \mathcal{N}$               |
| fong-spivak-invitation_F01188 | 107  | ■ 1.000 | $F(n) \in \mathbb{N}$                                                  | $F(n) \in \mathbb{N}$                                                  | sends each object $n \in \mathbb{N}$ to an object $F(n) \in \mathbb{N}$ . It must send morphisms in $\mathcal{N}$ to                                                    |
| fong-spivak-invitation_F01189 | 107  | ■ 1.000 | $F(m) \rightarrow F(n)$                                                | $F(m) \rightarrow F(n)$                                                | also be a morphism $F(m) \rightarrow F(n)$ . In other words, if $m \leq n$ , then we must also have                                                                     |
| fong-spivak-invitation_F01190 | 107  | ■ 1.000 | $F(m) \leq F(n)$                                                       | $F(m) \leq F(n)$                                                       | $F(m) \leq F(n)$ . But as long as $m \leq n$ implies $F(m) \leq F(n)$ , we have a functor.                                                                              |
| fong-spivak-invitation_F01191 | 107  | ■ 1.000 | $\mathbb{N} \rightarrow \mathbb{N}$                                    | $\mathbb{N} \rightarrow \mathbb{N}$                                    | Thus a functor $F: \mathcal{N} \rightarrow \mathcal{N}$ and a monotone map $\mathbb{N} \rightarrow \mathbb{N}$ are the same thing.                                      |
| fong-spivak-invitation_F01192 | 107  | ■ 0.908 | $\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$ | $\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$ | 1. Given any category $\mathcal{C}$ , show that there exists a functor $\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$ , known as the            |
| fong-spivak-invitation_F01193 | 107  | ■ 1.000 | $\operatorname{Ob}(\mathcal{C})$                                       | $\operatorname{Ob}(\mathcal{C})$                                       | Note that a functor $\mathcal{C} \rightarrow \mathcal{D}$ consists of a function from $\operatorname{Ob}(\mathcal{C})$ to $\operatorname{Ob}(\mathcal{D})$ and for each |
| fong-spivak-invitation_F01194 | 107  | ■ 1.000 | $\operatorname{Ob}(\mathcal{D})$                                       | $\operatorname{Ob}(\mathcal{D})$                                       | Note that a functor $\mathcal{C} \rightarrow \mathcal{D}$ consists of a function from $\operatorname{Ob}(\mathcal{C})$ to $\operatorname{Ob}(\mathcal{D})$ and for each |
| fong-spivak-invitation_F01195 | 107  | ■ 0.965 | $c_1, c_2 \in \mathcal{C}$                                             | $c_1, c_2 \in \mathcal{C}$                                             | pair of objects $c_1, c_2 \in \mathcal{C}$ a function from $\mathcal{C}(c_1, c_2)$ to $\mathcal{D}(F(c_1), F(c_2))$ .                                                   |
| fong-spivak-invitation_F01196 | 107  | ■ 0.965 | $\mathcal{C}(c_1, c_2)$                                                | $\mathcal{C}(c_1, c_2)$                                                | pair of objects $c_1, c_2 \in \mathcal{C}$ a function from $\mathcal{C}(c_1, c_2)$ to $\mathcal{D}(F(c_1), F(c_2))$ .                                                   |
| fong-spivak-invitation_F01197 | 107  | ■ 0.965 | $\mathcal{D}(F(c_1), F(c_2))$                                          | $\mathcal{D}(F(c_1), F(c_2))$                                          | pair of objects $c_1, c_2 \in \mathcal{C}$ a function from $\mathcal{C}(c_1, c_2)$ to $\mathcal{D}(F(c_1), F(c_2))$ .                                                   |
| fong-spivak-invitation_F01198 | 107  | ■ 0.907 | $G: \mathcal{D} \rightarrow \mathcal{E}$                               | $G: \mathcal{D} \rightarrow \mathcal{E}$                               | 2. Show that given $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{E}$ , we can define a new functor $(F \circ$                       |
| fong-spivak-invitation_F01199 | 107  | ■ 1.000 | $G): \mathcal{C} \rightarrow \mathcal{E}$                              | $G): \mathcal{C} \rightarrow \mathcal{E}$                              | $G): \mathcal{C} \rightarrow \mathcal{E}$ just by composing functions.                                                                                                  |
| fong-spivak-invitation_F01200 | 107  | ■ 1.000 | $F: \mathcal{C} \rightarrow$                                           | $F: \mathcal{C} \rightarrow$                                           | Let $\mathcal{C}$ be a category, and recall the category <b>Set</b> from Definition 3.17. A functor $F: \mathcal{C} \rightarrow$                                        |
| fong-spivak-invitation_F01201 | 107  | ■ 0.817 | $I: \mathcal{C} \rightarrow$                                           | $I: \mathcal{C} \rightarrow$                                           | the database – is given by a set-valued functor $I: \mathcal{C} \rightarrow \mathbf{Set}$ . The only additional detail                                                  |
| fong-spivak-invitation_F01202 | 108  | ■ 1.000 | $c = \stackrel{\text{string}}{\circ}$                                  | $c = \stackrel{\text{string}}{\circ}$                                  | is that for any white node, such as $c = \stackrel{\text{string}}{\circ}$ , we want to force $I$ to map to the set of                                                   |
| fong-spivak-invitation_F01203 | 108  | ■ 0.694 | $\{5\}$                                                                | $\{5\}$                                                                | functor $I: \mathcal{C} \rightarrow \mathbf{Set}$ . <sup>5</sup>                                                                                                        |

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| fong-spivak-invitation_F01204 | 108  | ■ 0.991 | $\mathrm{id}_{\mathbf{1}}$                  | $\mathrm{id}_{\mathbf{1}}$                  | phism $\mathrm{id}_{\mathbf{1}}$ , and no other morphisms. For any functor $F: \mathbf{1} \rightarrow \mathbf{Set}$ one can extract a set             |
| fong-spivak-invitation_F01205 | 108  | ■ 0.991 | $F: \mathbf{1} \rightarrow \mathbf{Set}$    | $F: \mathbf{1} \rightarrow \mathbf{Set}$    | phism $\mathrm{id}_{\mathbf{1}}$ , and no other morphisms. For any functor $F: \mathbf{1} \rightarrow \mathbf{Set}$ one can extract a set             |
| fong-spivak-invitation_F01206 | 108  | ■ 0.941 | $F(1)$                                      | $F(1)$                                      | $F(1)$ . Show that, for any set $S$ , there is a functor $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ such that $F_S(1) = S$ . $\diamond$               |
| fong-spivak-invitation_F01207 | 108  | ■ 0.941 | $F_S: \mathbf{1} \rightarrow \mathbf{Set}$  | $F_S: \mathbf{1} \rightarrow \mathbf{Set}$  | $F(1)$ . Show that, for any set $S$ , there is a functor $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ such that $F_S(1) = S$ . $\diamond$               |
| fong-spivak-invitation_F01208 | 108  | ■ 0.941 | $F_S(1) = S$                                | $F_S(1) = S$                                | $F(1)$ . Show that, for any set $S$ , there is a functor $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ such that $F_S(1) = S$ . $\diamond$               |
| fong-spivak-invitation_F01209 | 108  | ■ 0.998 | $Z := F(z)$                                 | $Z := F(z)$                                 | What is a set-valued functor $F: \mathcal{C} \rightarrow \mathbf{Set}$ ? It will consist of a set $Z := F(z)$ and a                                   |
| fong-spivak-invitation_F01210 | 108  | ■ 0.993 | $S := F(s): Z \rightarrow Z$                | $S := F(s): Z \rightarrow Z$                | function $S := F(s): Z \rightarrow Z$ , subject to the requirement that $S \circ S = S$ . Here are some                                               |
| fong-spivak-invitation_F01211 | 108  | ■ 0.993 | $S; S = S$                                  | $S; S = S$                                  | function $S := F(s): Z \rightarrow Z$ , subject to the requirement that $S \circ S = S$ . Here are some                                               |
| fong-spivak-invitation_F01212 | 108  | ■ 0.934 | $Z = \mathbb{N}$                            | $Z = \mathbb{N}$                            | • $Z = \mathbb{N}$ is the set of natural numbers and $S$ sends each number to 0. In particular, 0                                                     |
| fong-spivak-invitation_F01213 | 108  | ■ 0.255 | $13 + (2 * 4)$                              | $13 + (2 * 4)$                              | • $Z$ is the set of all well-formed arithmetic expressions, such as $13 + (2 * 4)$ or $-5$ , that                                                     |
| fong-spivak-invitation_F01214 | 108  | ■ 1.000 | $Z = \mathbb{N}_{\geq 2}$                   | $Z = \mathbb{N}_{\geq 2}$                   | • $Z = \mathbb{N}_{\geq 2}$ , and $S$ sends $n$ to its smallest prime factor. The smallest prime factor of a                                          |
| fong-spivak-invitation_F01215 | 108  | ■ 1.000 | $r \in I(c)$                                | $r \in I(c)$                                | row $r \in I(c)$ .                                                                                                                                    |
| fong-spivak-invitation_F01217 | 110  | ■ 0.997 | $\alpha: F \Rightarrow G$                   | $\alpha: F \Rightarrow G$                   | specify a natural transformation $\alpha: F \Rightarrow G$ .                                                                                          |
| fong-spivak-invitation_F01218 | 110  | ■ 0.999 | $\alpha_c: F(c) \rightarrow G(c)$           | $\alpha_c: F(c) \rightarrow G(c)$           | (i) for each object $c \in \mathcal{C}$ , one specifies a morphism $\alpha_c: F(c) \rightarrow G(c)$ in $\mathcal{D}$ , called the                    |
| fong-spivak-invitation_F01219 | 110  | ■ 1.000 | $\alpha$                                    | $\alpha$                                    | $c$ -component of $\alpha$ .                                                                                                                          |
| fong-spivak-invitation_F01220 | 110  | ■ 1.000 | $\alpha: F \rightarrow G$                   | $\alpha: F \rightarrow G$                   | A natural transformation $\alpha: F \rightarrow G$ is called a natural isomorphism if each                                                            |
| fong-spivak-invitation_F01221 | 110  | ■ 0.986 | $\alpha_c$                                  | $\alpha_c$                                  | component $\alpha_c$ is an isomorphism in $\mathcal{D}$ .                                                                                             |
| fong-spivak-invitation_F01222 | 110  | ■ 0.854 | $D: \mathcal{J} \rightarrow \mathcal{C}$    | $D: \mathcal{J} \rightarrow \mathcal{C}$    | <b>Definition 3.42.</b> A diagram $D$ in $\mathcal{C}$ is a functor $D: \mathcal{J} \rightarrow \mathcal{C}$ from any category $\mathcal{J}$ , called |
| fong-spivak-invitation_F01223 | 110  | ■ 0.854 | $\mathcal{J}$                               | $\mathcal{J}$                               | <b>Definition 3.42.</b> A diagram $D$ in $\mathcal{C}$ is a functor $D: \mathcal{J} \rightarrow \mathcal{C}$ from any category $\mathcal{J}$ , called |
| fong-spivak-invitation_F01224 | 110  | ■ 1.000 | $D(f) = D(f')$                              | $D(f) = D(f')$                              | the indexing category of the diagram $D$ . We say that $D$ commutes if $D(f) = D(f')$                                                                 |
| fong-spivak-invitation_F01225 | 110  | ■ 0.961 | $f, f': a \rightarrow b$                    | $f, f': a \rightarrow b$                    | holds for every parallel pair of morphisms $f, f': a \rightarrow b$ in $\mathcal{J}$ .                                                                |
| fong-spivak-invitation_F01226 | 110  | ■ 0.961 | $\mathcal{J}^7$                             | $\mathcal{J}^7$                             | holds for every parallel pair of morphisms $f, f': a \rightarrow b$ in $\mathcal{J}$ .                                                                |
| fong-spivak-invitation_F01227 | 110  | ■ 1.000 | $F(c) \Rightarrow G(d)$                     | $F(c) \Rightarrow G(d)$                     | In terms of Eq. (3.9), the only case of two parallel morphisms is that of $F(c) \Rightarrow G(d)$ ,                                                   |
| fong-spivak-invitation_F01228 | 110  | ■ 0.920 | $F(f) \circ \alpha_d = \alpha_c \circ G(f)$ | $F(f) \circ \alpha_d = \alpha_c \circ G(f)$ | so to say that the diagram commutes is to say that $F(f) \circ \alpha_d = \alpha_c \circ G(f)$ . This is exactly                                      |
| fong-spivak-invitation_F01229 | 111  | ■ 1.000 | $\alpha_1: v \rightarrow x$                 | $\alpha_1: v \rightarrow x$                 | ural transformation $\alpha: F \Rightarrow G$ is given by choosing components $\alpha_1: v \rightarrow x$ and                                         |
| fong-spivak-invitation_F01230 | 111  | ■ 1.000 | $\alpha_2: w \rightarrow y$                 | $\alpha_2: w \rightarrow y$                 | $\alpha_2: w \rightarrow y$ . We have highlighted the only choice for each in green: namely, $\alpha_1 = c$ and                                       |
| fong-spivak-invitation_F01231 | 111  | ■ 1.000 | $\alpha_1 = c$                              | $\alpha_1 = c$                              | $\alpha_2: w \rightarrow y$ . We have highlighted the only choice for each in green: namely, $\alpha_1 = c$ and                                       |

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| fong-spivak-invitation_F01232 | 111  | ■ 0.998 | $\alpha_2 = g$                               | $\alpha_2 = g$                               | $\alpha_2 = g.$                                                                                                                             |
| fong-spivak-invitation_F01233 | 111  | ■ 0.476 | $\mathcal{C} \xrightarrow[G]{F} \mathcal{D}$ | $\mathcal{C} \xrightarrow[G]{F} \mathcal{D}$ | notation $\mathcal{C} \begin{array}{c} \xrightarrow{F} \\ \alpha \Downarrow \\ \xrightarrow{G} \end{array} \mathcal{D}?$                    |
| fong-spivak-invitation_F01234 | 111  | ■ 1.000 | $A, B: \mathbf{1} \rightarrow$               | $A, B: \mathbf{1} \rightarrow$               | fied with a set. So suppose $A$ and $B$ are sets considered as functors $A, B: \mathbf{1} \rightarrow$                                      |
| fong-spivak-invitation_F01235 | 111  | ■ 0.852 | $\mathcal{D}^{\mathcal{C}}$                  | $\mathcal{D}^{\mathcal{C}}$                  | <b>Definition 3.45.</b> Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We denote by $\mathcal{D}^{\mathcal{C}}$ the category whose      |
| fong-spivak-invitation_F01236 | 111  | ■ 0.997 | $\mathcal{D}^{\mathcal{C}}(F, G)$            | $\mathcal{D}^{\mathcal{C}}(F, G)$            | objects are functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and whose morphisms $\mathcal{D}^{\mathcal{C}}(F, G)$ are the natural trans-  |
| fong-spivak-invitation_F01237 | 111  | ■ 0.998 | $F \in \mathcal{D}^{\mathcal{P}}$            | $F \in \mathcal{D}^{\mathcal{P}}$            | 2. Propose an identity natural transformation on any object $F \in \mathcal{D}^{\mathcal{C}}$ , and check that it                           |
| fong-spivak-invitation_F01238 | 111  | ■ 0.748 | $\mathbf{1}$                                 | $\mathbf{1}$                                 | <b>Example 3.47.</b> In our new language, Example 3.44 says that $\mathbf{Set}^{\mathbf{1}}$ is isomorphic to $\mathbf{Set}$ .              |
| fong-spivak-invitation_F01239 | 111  | ■ 0.701 | $F_0, F_1, F_2, \dots$                       | $F_0, F_1, F_2, \dots$                       | sequence $(F_0, F_1, F_2, \dots)$ of natural numbers, i.e. $F_0 \leq F_1 \leq F_2 \leq \dots$ . If $G$ is another                           |
| fong-spivak-invitation_F01240 | 111  | ■ 0.701 | $F_0 \leq F_1 \leq F_2 \leq \dots$           | $F_0 \leq F_1 \leq F_2 \leq \dots$           | sequence $(F_0, F_1, F_2, \dots)$ of natural numbers, i.e. $F_0 < F_1 < F_2 < \dots$ . If $G$ is another                                    |
| fong-spivak-invitation_F01241 | 111  | ■ 1.000 | $\alpha_n: F_n \rightarrow G_n$              | $\alpha_n: F_n \rightarrow G_n$              | nent $\alpha_n: F_n \rightarrow G_n$ has no data, it just tells us a fact: that $F_n \leq G_n$ . And the naturality                         |
| fong-spivak-invitation_F01242 | 111  | ■ 1.000 | $F_n \leq G_n$                               | $F_n \leq G_n$                               | nent $\alpha_n: F_n \rightarrow G_n$ has no data, it just tells us a fact: that $F_n \leq G_n$ . And the naturality                         |
| fong-spivak-invitation_F01243 | 112  | ■ 1.000 | $F \Rightarrow G$                            | $F \Rightarrow G$                            | $F \Rightarrow G$ are the same. In other words, the category $\mathcal{N}^{\mathcal{N}}$ is itself a preorder, namely the                   |
| fong-spivak-invitation_F01244 | 112  | ■ 1.000 | $\mathcal{N}^{\mathcal{N}}$                  | $\mathcal{N}^{\mathcal{N}}$                  | $F \Rightarrow G$ are the same. In other words, the category $\mathcal{N}^{\mathcal{N}}$ is itself a preorder, namely the                   |
| fong-spivak-invitation_F01245 | 112  | ■ 1.000 | $F, G: \mathcal{C} \rightarrow \mathcal{P}$  | $F, G: \mathcal{C} \rightarrow \mathcal{P}$  | 1. For any two functors $F, G: \mathcal{C} \rightarrow \mathcal{P}$ , there is at most one natural transformation                           |
| fong-spivak-invitation_F01246 | 112  | ■ 1.000 | $F \rightarrow G$                            | $F \rightarrow G$                            | $F \Rightarrow G$ .                                                                                                                         |
| fong-spivak-invitation_F01247 | 112  | ■ 1.000 | $F, G: \mathcal{P} \rightarrow \mathcal{C}$  | $F, G: \mathcal{P} \rightarrow \mathcal{C}$  | 2. For any two functors $F, G: \mathcal{P} \rightarrow \mathcal{C}$ , there is at most one natural transformation                           |
| fong-spivak-invitation_F01248 | 112  | ■ 0.689 | $F:$                                         | $F:$                                         | these two categories equivalent because there exist functors $F: \mathbf{PrO} \rightarrow \mathbf{Bool-Cat}$ and                            |
| fong-spivak-invitation_F01249 | 112  | ■ 0.795 | $G \cong \mathrm{id}_{\text{Pro}}$           | $G \cong \mathrm{id}_{\text{Pro}}$           | $G: \mathbf{Bool-Cat} \rightarrow \mathbf{PrO}$ such that there exist natural isomorphisms $F \circ G \cong \mathrm{id}_{\mathbf{PrO}}$ and |
| fong-spivak-invitation_F01250 | 112  | ■ 0.563 | $G: F \cong \mathrm{id}_{\text{Bool-Cat}}$   | $G: F \cong \mathrm{id}_{\text{Bool-Cat}}$   | $G \circ F \cong \mathrm{id}_{\mathbf{Bool-Cat}}$ in the sense of Definition 3.41.                                                          |
| fong-spivak-invitation_F01251 | 112  | ■ 1.000 | $I, J: \mathcal{C} \rightarrow$              | $I, J: \mathcal{C} \rightarrow$              | <b>Definition 3.51.</b> Suppose that $\mathcal{C}$ is a database schema and $I, J: \mathcal{C} \rightarrow \mathbf{Set}$ are                |
| fong-spivak-invitation_F01252 | 112  | ■ 0.949 | $\alpha: I \rightarrow J$                    | $\alpha: I \rightarrow J$                    | tion $\alpha: I \rightarrow J$ . Write $\mathcal{C}\text{-Inst} := \mathbf{Set}^{\mathcal{C}}$ to denote the functor category as defined in |
| fong-spivak-invitation_F01253 | 112  | ■ 0.949 | $:=$                                         | $:=$                                         | tion $\alpha: I \rightarrow J$ . Write $\mathcal{C}\text{-Inst} := \mathbf{Set}^{\mathcal{C}}$ to denote the functor category as defined in |
| fong-spivak-invitation_F01254 | 112  | ■ 0.949 | $\mathcal{C}$                                | $\mathcal{C}$                                | tion $\alpha: I \rightarrow J$ . Write $\mathcal{C}\text{-Inst} := \mathbf{Set}^{\mathcal{C}}$ to denote the functor category as defined in |
| fong-spivak-invitation_F01255 | 113  | ■ 0.993 | $I: \mathbf{Gr} \rightarrow$                 | $I: \mathbf{Gr} \rightarrow$                 | Here is an example of a $\mathbf{Gr}$ -instance, i.e. a set-valued functor $I: \mathbf{Gr} \rightarrow \mathbf{Set}$ , in table             |
| fong-spivak-invitation_F01256 | 113  | ■ 0.508 | $=\{a, b, c, d, e\}$                         | $=\{a, b, c, d, e\}$                         | Here $I(\text{Arrow}) = \{a, b, c, d, e\}$ , and $I(\text{Vertex}) = \{1, 2, 3, 4\}$ . One can draw the                                     |
| fong-spivak-invitation_F01257 | 113  | ■ 0.508 | $=\{1, 2, 3, 4\}$                            | $=\{1, 2, 3, 4\}$                            | Here $I(\text{Arrow}) = \{a, b, c, d, e\}$ , and $I(\text{Vertex}) = \{1, 2, 3, 4\}$ . One can draw the                                     |

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| fong-spivak-invitation_F01258 | 113  | ■ 0.987 | $G, H: \mathbf{Gr} \rightarrow \mathbf{Set}$ | $G, H: \mathbf{Gr} \rightarrow \mathbf{Set}$ | are called <i>graph homomorphisms</i> . Let’s unwind this. Suppose that $G, H: \mathbf{Gr} \rightarrow \mathbf{Set}$                                                              |
| fong-spivak-invitation_F01259 | 113  | ■ 0.990 | $G, H \in \mathbf{Gr}$                       | $G, H \in \mathbf{Gr}$                       | are functors (i.e. <b>Gr</b> -instances); that is, they are objects $G, H \in \mathbf{Gr-Inst}$ . A mor-                                                                          |
| fong-spivak-invitation_F01260 | 113  | ■ 1.000 | $G \rightarrow H$                            | $G \rightarrow H$                            | phism $G \rightarrow H$ is a natural transformation $\alpha: G \rightarrow H$ between them; what does that                                                                        |
| fong-spivak-invitation_F01261 | 113  | ■ 1.000 | $\alpha: G \rightarrow H$                    | $\alpha: G \rightarrow H$                    | phism $G \rightarrow H$ is a natural transformation $\alpha: G \rightarrow H$ between them; what does that                                                                        |
| fong-spivak-invitation_F01262 | 113  | ■ 0.981 | $\mathbf{Gr}$                                | <b>Gr</b>                                    | By Definition 3.41, since <b>Gr</b> has two objects, $\alpha$ consists of two components,                                                                                         |
| fong-spivak-invitation_F01263 | 114  | ■ 1.000 | $\alpha_{\text{Vertex}}$                     | $\alpha_{\text{Vertex}}$                     | tions $\alpha_{\text{Vertex}}$ and $\alpha_{\text{Arrow}}$ to respect source and targets in $G$ and $H$ . The left diagram                                                        |
| fong-spivak-invitation_F01264 | 114  | ■ 1.000 | $\alpha_{\text{Arrow}}$                      | $\alpha_{\text{Arrow}}$                      | tions $\alpha_{\text{Vertex}}$ and $\alpha_{\text{Arrow}}$ to respect source and targets in $G$ and $H$ . The left diagram                                                        |
| fong-spivak-invitation_F01265 | 114  | ■ 1.000 | $G, H$                                       | $G, H$                                       | <b>Exercise 3.54.</b> We claim that – with $G, H$ as in Example 3.53 – there is exactly one                                                                                       |
| fong-spivak-invitation_F01266 | 114  | ■ 0.991 | $\alpha_{\text{Arrow}}(a) = d$               | $\alpha_{\text{Arrow}}(a) = d$               | graph homomorphism $\alpha: G \rightarrow H$ such that $\alpha_{\text{Arrow}}(a) = d$ .                                                                                           |
| fong-spivak-invitation_F01267 | 115  | ■ 0.994 | $F: \mathbf{Gr} \rightarrow$                 | $F: \mathbf{Gr} \rightarrow$                 | The reader can create any functor she likes, but we will use a specific functor $F: \mathbf{Gr} \rightarrow$                                                                      |
| fong-spivak-invitation_F01268 | 116  | ■ 0.156 | $I:$                                         | $I:$                                         | $I: \mathbf{Gr} \rightarrow \mathbf{Set}$ . Both $F$ and $I$ respect identities and composition, so $F \circ I$ does too. Thus                                                    |
| fong-spivak-invitation_F01269 | 116  | ■ 0.156 | $\rightarrow$                                | $\rightarrow$                                | $I: \mathbf{Gr} \rightarrow \mathbf{Set}$ . Both $F$ and $I$ respect identities and composition, so $F \circ I$ does too. Thus                                                    |
| fong-spivak-invitation_F01270 | 116  | ■ 0.156 | $F \circ I$                                  | $F \circ I$                                  | $I: \mathbf{Gr} \rightarrow \mathbf{Set}$ . Both $F$ and $I$ respect identities and composition, so $F \circ I$ does too. Thus                                                    |
| fong-spivak-invitation_F01271 | 117  | ■ 0.649 | $G: \mathbf{Gr} \rightarrow \mathbf{DDS}$    | $G: \mathbf{Gr} \rightarrow \mathbf{DDS}$    | <b>Exercise 3.55.</b> Consider the functor $G: \mathbf{Gr} \rightarrow \mathbf{DDS}$ given by sending “source” to                                                                 |
| fong-spivak-invitation_F01272 | 117  | ■ 0.858 | $I: \mathcal{D} \rightarrow$                 | $I: \mathcal{D} \rightarrow$                 | set-valued functor $I: \mathcal{D} \rightarrow \mathbf{Set}$ , we refer to the composite functor $F \circ I: \mathcal{C} \rightarrow \mathbf{Set}$ as                             |
| fong-spivak-invitation_F01273 | 117  | ■ 0.858 | $F: I: \mathcal{C} \rightarrow$              | $F: I: \mathcal{C} \rightarrow$              | set-valued functor $I: \mathcal{D} \rightarrow \mathbf{Set}$ , we refer to the composite functor $F \circ I: \mathcal{C} \rightarrow \mathbf{Set}$ as                             |
| fong-spivak-invitation_F01274 | 117  | ■ 0.957 | $\alpha: I \Rightarrow J$                    | $\alpha: I \Rightarrow J$                    | the <i>pullback of <math>I</math> along <math>F</math></i> . Given a natural transformation $\alpha: I \Rightarrow J$ , there is a natural                                        |
| fong-spivak-invitation_F01275 | 117  | ■ 0.587 | $\alpha_F: F; I \Rightarrow F; J$            | $\alpha_F: F; I \Rightarrow F; J$            | transformation $\alpha_F: F \circ I \Rightarrow F \circ J$ , whose component $(F \circ I)(c) \rightarrow (F \circ J)(c)$ for any                                                  |
| fong-spivak-invitation_F01276 | 117  | ■ 0.587 | $(F; I)(c) \rightarrow (F; J)(c)$            | $(F; I)(c) \rightarrow (F; J)(c)$            | transformation $\alpha_F: F \circ I \Rightarrow F \circ J$ , whose component $(F \circ I)(c) \rightarrow (F \circ J)(c)$ for any                                                  |
| fong-spivak-invitation_F01277 | 117  | ■ 0.677 | $(\alpha_F)_c := \alpha_{Fc}$                | $(\alpha_F)_c := \alpha_{Fc}$                | $c \in \text{Ob}(\mathcal{C})$ is given by $(\alpha_F)_c := \alpha_{Fc}$ .                                                                                                        |
| fong-spivak-invitation_F01278 | 117  | ■ 0.970 | $\Delta_F: \mathcal{D}$                      | $\Delta_F: \mathcal{D}$                      | This uses the data of $F$ to define a functor $\Delta_F: \mathcal{D-Inst} \rightarrow \mathcal{C-Inst}$ .                                                                         |
| fong-spivak-invitation_F01279 | 117  | ■ 1.000 | $\Delta$                                     | $\Delta$                                     | in general. The relevance to databases is that the data migration functor $\Delta$ from Defi-                                                                                     |
| fong-spivak-invitation_F01280 | 117  | ■ 1.000 | $\Sigma$                                     | $\Sigma$                                     | nition 3.56 always has two adjoints of its own: a left adjoint which we denote $\Sigma$ and a                                                                                     |
| fong-spivak-invitation_F01281 | 117  | ■ 1.000 | $\Pi$                                        | $\Pi$                                        | right adjoint which we denote $\Pi$ .                                                                                                                                             |
| fong-spivak-invitation_F01282 | 117  | ■ 1.000 | $P(a, b)$                                    | $P(a, b)$                                    | Recall from Section 3.2.3 that in a preorder $P$ , a hom-set $P(a, b)$ has one element                                                                                            |
| fong-spivak-invitation_F01283 | 117  | ■ 1.000 | $Q(f(p), q) \cong P(p, g(q))$                | $Q(f(p), q) \cong P(p, g(q))$                | isomorphism of sets $Q(f(p), q) \cong P(p, g(q))$ : either both are one-element sets                                                                                              |
| fong-spivak-invitation_F01284 | 118  | ■ 1.000 | $L: \mathcal{C} \rightarrow \mathcal{D}$     | $L: \mathcal{C} \rightarrow \mathcal{D}$     | <b>Definition 3.57.</b> Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{D} \rightarrow \mathcal{C}$ be functors. |

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| fong-spivak-invitation_F01285 | 118  | ■ 1.000 | $R: \mathcal{D} \rightarrow \mathcal{C}$                   | $R: \mathcal{D} \rightarrow \mathcal{C}$                   | <b>Definition 3.57.</b> Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{D} \rightarrow \mathcal{C}$ be functors.                    |
| fong-spivak-invitation_F01286 | 118  | ■ 1.000 | $L$                                                        | $L$                                                        | We say that $L$ is <i>left adjoint to <math>R</math></i> (and that $R$ is <i>right adjoint to <math>L</math></i> ) if, for any $c \in \mathcal{C}$ and                                               |
| fong-spivak-invitation_F01287 | 118  | ■ 0.990 | $d \in \mathcal{D}$                                        | $d \in \mathcal{D}$                                        | $d \in \mathcal{D}$ , there is an isomorphism of hom-sets                                                                                                                                            |
| fong-spivak-invitation_F01288 | 118  | ■ 0.798 | $d.^8$                                                     | $d.^8$                                                     | that is natural in $c$ and $d$ . <sup>8</sup>                                                                                                                                                        |
| fong-spivak-invitation_F01289 | 118  | ■ 1.000 | $f: c \rightarrow R(d)$                                    | $f: c \rightarrow R(d)$                                    | Given a morphism $f: c \rightarrow R(d)$ in $\mathcal{C}$ , its image $g := \alpha_{c,d}(f)$ is called its <i>mate</i> .                                                                             |
| fong-spivak-invitation_F01290 | 118  | ■ 1.000 | $g := \alpha_{c,d}(f)$                                     | $g := \alpha_{c,d}(f)$                                     | Given a morphism $f: c \rightarrow R(d)$ in $\mathcal{C}$ , its image $g := \alpha_{c,d}(f)$ is called its <i>mate</i> .                                                                             |
| fong-spivak-invitation_F01291 | 118  | ■ 1.000 | $g: L(c) \rightarrow d$                                    | $g: L(c) \rightarrow d$                                    | Similarly, the mate of $g: L(c) \rightarrow d$ is $f$ .                                                                                                                                              |
| fong-spivak-invitation_F01292 | 118  | ■ 1.000 | $L \dashv R$                                               | $L \dashv R$                                               | To denote an adjunction we write $L \dashv R$ , or in diagrams                                                                                                                                       |
| fong-spivak-invitation_F01293 | 118  | ■ 0.528 | $B \in \operatorname{Ob}$                                  | $B \in \operatorname{Ob}$                                  | <b>Example 3.59.</b> Let $B \in \operatorname{Ob}(\mathbf{Set})$ be any set. There is an adjunction called “currying $B$ .”                                                                          |
| fong-spivak-invitation_F01294 | 118  | ■ 1.000 | $A, C$                                                     | $A, C$                                                     | Abstractly we write it as on the left, but what this means is that, for any sets $A, C$ , there                                                                                                      |
| fong-spivak-invitation_F01295 | 118  | ■ 0.997 | $B \rightarrow C$                                          | $B \rightarrow C$                                          | $C$ are sets. Then the set of functions $B \rightarrow C$ is also a set; let’s denote it $C^B$ . It’s written                                                                                        |
| fong-spivak-invitation_F01296 | 118  | ■ 0.997 | $C^B$                                                      | $C^B$                                                      | $C$ are sets. Then the set of functions $B \rightarrow C$ is also a set; let’s denote it $C^B$ . It’s written                                                                                        |
| fong-spivak-invitation_F01297 | 118  | ■ 1.000 | $10^3$                                                     | $10^3$                                                     | this way because if $C$ has 10 elements and $B$ has 3 elements then $C^B$ has $10^3$ elements,                                                                                                       |
| fong-spivak-invitation_F01298 | 118  | ■ 1.000 | $ \left C^B\right  = C ^{ B }$                             | $ C^B  =  C ^{ B }$                                        | and more generally for any two finite sets $ C^B  =  C ^{ B }$ .                                                                                                                                     |
| fong-spivak-invitation_F01299 | 118  | ■ 1.000 | $(A \times B) \rightarrow C$                               | $(A \times B) \rightarrow C$                               | dence between functions $(A \times B) \rightarrow C$ and functions $A \rightarrow C^B$ . Intuitively, if I have a                                                                                    |
| fong-spivak-invitation_F01300 | 118  | ■ 1.000 | $A \rightarrow C^B$                                        | $A \rightarrow C^B$                                        | dence between functions $(A \times B) \rightarrow C$ and functions $A \rightarrow C^B$ . Intuitively, if I have a                                                                                    |
| fong-spivak-invitation_F01301 | 118  | ■ 0.522 | $\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \rightarrow$ | $\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \rightarrow$ | <sup>8</sup> This naturality is between functors $\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \rightarrow \mathbf{Set}$ . It says that for any morphisms $f: c' \rightarrow c$ in $\mathcal{C}$ and |
| fong-spivak-invitation_F01302 | 118  | ■ 0.522 | $f: c^{\prime} \rightarrow c$                              | $f: c' \rightarrow c$                                      | <sup>8</sup> This naturality is between functors $\mathcal{C}^{\mathrm{op}} \times \mathcal{D} \rightarrow \mathbf{Set}$ . It says that for any morphisms $f: c' \rightarrow c$ in $\mathcal{C}$ and |
| fong-spivak-invitation_F01303 | 118  | ■ 1.000 | $g: d \rightarrow d^{\prime}$                              | $g: d \rightarrow d'$                                      | $g: d \rightarrow d'$ in $\mathcal{D}$ , the following diagram commutes:                                                                                                                             |
| fong-spivak-invitation_F01304 | 119  | ■ 1.000 | $b \multimap c$                                            | $b \multimap c$                                            | objects and what we called hom-elements $b \multimap c$ in Definition 2.57.                                                                                                                          |
| fong-spivak-invitation_F01305 | 119  | ■ 1.000 | $- \times B$                                               | $- \times B$                                               | <b>Exercise 3.60.</b> In Example 3.59, we discussed an adjunction between functors $- \times B$                                                                                                      |
| fong-spivak-invitation_F01306 | 119  | ■ 1.000 | $(-)^B$                                                    | $(-)^B$                                                    | and $(-)^B$ . But we only said how these functors worked on objects: for an arbitrary set                                                                                                            |
| fong-spivak-invitation_F01307 | 119  | ■ 1.000 | $X \times B$                                               | $X \times B$                                               | $X$ , they return sets $X \times B$ and $X^B$ respectively.                                                                                                                                          |
| fong-spivak-invitation_F01308 | 119  | ■ 1.000 | $X^B$                                                      | $X^B$                                                      | $X$ , they return sets $X \times B$ and $X^B$ respectively.                                                                                                                                          |
| fong-spivak-invitation_F01309 | 119  | ■ 1.000 | $- \times B: X \times B \rightarrow Y \times B$            | $- \times B: X \times B \rightarrow Y \times B$            | 1. Given a morphism $f: X \rightarrow Y$ , what morphism should $- \times B: X \times B \rightarrow Y \times B$                                                                                      |
| fong-spivak-invitation_F01310 | 119  | ■ 1.000 | $(-)^B: X^B \rightarrow Y^B$                               | $(-)^B: X^B \rightarrow Y^B$                               | 2. Given a morphism $f: X \rightarrow Y$ , what morphism should $(-)^B: X^B \rightarrow Y^B$ return?                                                                                                 |
| fong-spivak-invitation_F01311 | 119  | ■ 1.000 | $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$   | $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$   | 3. Consider the function $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ , which sends $(a, b) \mapsto a + b$ . Currying $+$ ,                                                              |

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| fong-spivak-invitation_F01312 | 119  | ■ 1.000 | (a, b) \mapsto a+b                                | $(a, b) \mapsto a + b$                              | 3. Consider the function $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ , which sends $(a, b) \mapsto a + b$ . Currying $+$ ,                                        |
| fong-spivak-invitation_F01313 | 119  | ■ 1.000 | p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}} | $p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$ | we get a certain function $p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$ . What is $p(3)$ ?                                                                               |
| fong-spivak-invitation_F01314 | 119  | ■ 1.000 | p(3)                                              | $p(3)$                                              | we get a certain function $p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$ . What is $p(3)$ ?                                                                               |
| fong-spivak-invitation_F01315 | 119  | ■ 1.000 | \Delta_{\mathcal{F}}                              | $\Delta_F$                                          | Given $F: \mathcal{C} \rightarrow \mathcal{D}$ , the data migration functor $\Delta_F$ turns $\mathcal{D}$ -instances into $\mathcal{C}$ -instances.                           |
| fong-spivak-invitation_F01316 | 120  | ■ 1.000 | \Sigma_{\mathcal{F}}                              | $\Sigma_F$                                          | <b>DDS</b> and get a graph, the migration functors $\Sigma_F$ and $\Pi_F$ can be used to turn any graph                                                                        |
| fong-spivak-invitation_F01317 | 120  | ■ 1.000 | \Pi_{\mathcal{F}}                                 | $\Pi_F$                                             | <b>DDS</b> and get a graph, the migration functors $\Sigma_F$ and $\Pi_F$ can be used to turn any graph                                                                        |
| fong-spivak-invitation_F01318 | 120  | ■ 1.000 | J: \mathbf{Gr} \rightarrow                        | $J: \mathbf{Gr} \rightarrow$                        | into a discrete dynamical system. That is, given an instance $J: \mathbf{Gr} \rightarrow \mathbf{Set}$ , we can get                                                            |
| fong-spivak-invitation_F01319 | 120  | ■ 1.000 | \Sigma_{\mathcal{F}}(J)                           | $\Sigma_F(J)$                                       | instances $\Sigma_F(J)$ and $\Pi_F(J)$ on <b>DDS</b> . This, however, is quite technical, and we leave                                                                         |
| fong-spivak-invitation_F01320 | 120  | ■ 1.000 | \Pi_{\mathcal{F}}(J)                              | $\Pi_F(J)$                                          | instances $\Sigma_F(J)$ and $\Pi_F(J)$ on <b>DDS</b> . This, however, is quite technical, and we leave                                                                         |
| fong-spivak-invitation_F01321 | 120  | ■ 0.999 | (e, f)                                            | $(e, f)$                                            | The functor $\Pi_F$ puts into Airline Seat the set of those pairs $(e, f)$ where $e$ is an Econ-                                                                               |
| fong-spivak-invitation_F01322 | 121  | ■ 1.000 | \Pi_{\mathcal{F}}(I)                              | $\Pi_F(I)$                                          | $I$ , in which case $\Pi_F(I)(\text{Airline Seat})$ would be empty. But in other uses of these same                                                                            |
| fong-spivak-invitation_F01323 | 121  | ■ 1.000 | \Delta_{\mathcal{F}}, \Sigma_{\mathcal{F}}        | $\Delta_F, \Sigma_F$                                | $F$ between schemas and using the “induced” functors $\Delta_F$ , $\Sigma_F$ , and $\Pi_F$ . Indeed, in                                                                        |
| fong-spivak-invitation_F01324 | 121  | ■ 0.989 | \mathcal{C} \rightarrow \mathbf{1}                | $\mathcal{C} \rightarrow \mathbf{1}$                | exactly one functor $\mathcal{C} \Rightarrow \mathbf{1}$ for any $\mathcal{C}$ ; let’s denote it                                                                               |
| fong-spivak-invitation_F01325 | 121  | ■ 0.856 | \Sigma_{\mathcal{I}}                              | $\Sigma_{\mathcal{I}}$                              | We want to consider the data migration functors $\Sigma_{\mathcal{I}}: \mathcal{C}\text{-}\mathbf{Inst} \rightarrow \mathbf{1}\text{-}\mathbf{Inst}$ and                       |
| fong-spivak-invitation_F01326 | 121  | ■ 0.451 | \Pi_{\mathcal{I}}: \mathcal{C}                    | $\Pi_{\mathcal{I}}: \mathcal{C}$                    | $\Pi_{\mathcal{I}}: \mathcal{C}\text{-}\mathbf{Inst} \rightarrow \mathbf{1}\text{-}\mathbf{Inst}$ . In Example 3.44, we saw that an instance on $\mathbf{1}$ is the same thing |
| fong-spivak-invitation_F01327 | 121  | ■ 0.772 | \Sigma_{\mathcal{I}}(I)                           | $\Sigma_{\mathcal{I}}(I)$                           | Given any schema $\mathcal{C}$ and instance $I: \mathcal{C} \rightarrow \mathbf{Set}$ , we will get sets $\Sigma_{\mathcal{I}}(I)$ and $\Pi_{\mathcal{I}}(I)$ . Think-         |
| fong-spivak-invitation_F01328 | 121  | ■ 0.772 | \Pi_{\mathcal{I}}(I)                              | $\Pi_{\mathcal{I}}(I)$                              | Given any schema $\mathcal{C}$ and instance $I: \mathcal{C} \rightarrow \mathbf{Set}$ , we will get sets $\Sigma_{\mathcal{I}}(I)$ and $\Pi_{\mathcal{I}}(I)$ . Think-         |
| fong-spivak-invitation_F01329 | 121  | ■ 1.000 | I: \mathcal{G} \rightarrow                        | $I: \mathcal{G} \rightarrow$                        | Here’s a sample instance $I: \mathcal{G} \rightarrow \mathbf{Set}$ :                                                                                                           |
| fong-spivak-invitation_F01330 | 122  | ■ 0.969 | \mathcal{G}                                       | $\mathcal{G}$                                       | <b>Exercise 3.63.</b> Note that $\mathcal{G}$ from Eq. (3.15) is isomorphic to the schema $\mathbf{Gr}$ . In Sec-                                                              |
| fong-spivak-invitation_F01331 | 122  | ■ 0.952 | !: \mathcal{G} \rightarrow \mathbf{1}             | $!: \mathcal{G} \rightarrow \mathbf{1}$             | Now we have a unique functor $!: \mathcal{G} \rightarrow \mathbf{1}$ , and we want to say what $\Sigma_{\mathcal{I}}(I)$ and                                                   |
| fong-spivak-invitation_F01332 | 122  | ■ 0.989 | \Pi_{\mathcal{I}}                                 | $\Pi_{\mathcal{I}}$                                 | Where do these facts – that $\Pi_{\mathcal{I}}$ and $\Sigma_{\mathcal{I}}$ act the way we said – come from? Everything                                                         |
| fong-spivak-invitation_F01333 | 123  | ■ 0.578 | !: C \rightarrow Z                                | $!: C \rightarrow Z$                                | each object $C$ of $\mathcal{C}$ , there exists a unique morphism $!: C \rightarrow Z$ .                                                                                       |
| fong-spivak-invitation_F01334 | 123  | ■ 0.900 | C \rightarrow \{\bullet\}                         | $C \rightarrow \{\bullet\}$                         | one function $!: C \rightarrow \{\bullet\}$ . This unique function is the one that does the only thing that                                                                    |
| fong-spivak-invitation_F01335 | 123  | ■ 1.000 | c \in C                                           | $c \in C$                                           | can be done: it maps each element $c \in C$ to the element $\bullet \in \{\bullet\}$ .                                                                                         |
| fong-spivak-invitation_F01336 | 123  | ■ 1.000 | \bullet \in \{\bullet\}                           | $\bullet \in \{\bullet\}$                           | can be done: it maps each element $c \in C$ to the element $\bullet \in \{\bullet\}$ .                                                                                         |
| fong-spivak-invitation_F01337 | 123  | ■ 0.825 | z \in P                                           | $z \in P$                                           | <b>Exercise 3.66.</b> Let $(P, \leq)$ be a preorder, let $z \in P$ be an element, and let $\mathcal{P}$ be the                                                                 |
| fong-spivak-invitation_F01338 | 123  | ■ 1.000 | c \in P                                           | $c \in P$                                           | only if it is a <i>top element</i> in $P$ : that is, if and only if for all $c \in P$ we have $c \leq z$ .                                                                     |
| fong-spivak-invitation_F01339 | 123  | ■ 1.000 | c \leq z                                          | $c \leq z$                                          | only if it is a <i>top element</i> in $P$ : that is, if and only if for all $c \in P$ we have $c \leq z$ .                                                                     |

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| fong-spivak-invitation_F01340 | 123  | ■ 1.000 | $Z^{\{\prime\}}$                                      | $Z'$                                      | <i>Proof.</i> This is a simple, but powerful standard argument. Suppose $Z$ and $Z'$ are both                                              |
| fong-spivak-invitation_F01341 | 123  | ■ 1.000 | $a: Z \rightarrow Z^{\{\prime\}}$                     | $a: Z \rightarrow Z'$                     | terminal objects in some category $\mathcal{C}$ . Then there exist (unique) maps $a: Z \rightarrow Z'$ and                                 |
| fong-spivak-invitation_F01342 | 123  | ■ 1.000 | $b: Z^{\{\prime\}} \rightarrow Z$                     | $b: Z' \rightarrow Z$                     | $b: Z' \rightarrow Z$ . Composing these, we get a map $a \circ b: Z \rightarrow Z$ . Now since $Z$ is termi-                               |
| fong-spivak-invitation_F01343 | 123  | ■ 1.000 | $a; b: Z \rightarrow Z$                               | $a; b: Z \rightarrow Z$                   | $b: Z' \rightarrow Z$ . Composing these, we get a map $a \circ b: Z \rightarrow Z$ . Now since $Z$ is termi-                               |
| fong-spivak-invitation_F01344 | 123  | ■ 0.992 | $Z \rightarrow Z$                                     | $Z \rightarrow Z$                         | nal, this map $Z \rightarrow Z$ must be unique. But $\text{id}_Z$ is also such a map. So we must have                                      |
| fong-spivak-invitation_F01345 | 123  | ■ 0.992 | $\mathrm{id}_Z$                                       | $\text{id}_Z$                             | nal, this map $Z \rightarrow Z$ must be unique. But $\text{id}_Z$ is also such a map. So we must have                                      |
| fong-spivak-invitation_F01346 | 123  | ■ 0.999 | $a; b=\mathrm{id}_{\mathrm{Z}}$                       | $a; b = \text{id}_Z$                      | $a \circ b = \text{id}_Z$ . Similarly, we find that $b \circ a = \text{id}_{Z'}$ . Thus $a$ is an isomorphism, with                        |
| fong-spivak-invitation_F01347 | 123  | ■ 0.999 | $b; a=\mathrm{id}_{\mathrm{Z}^{\{\prime\}}}$          | $b; a = \text{id}_{Z'}$                   | $a \circ b = \text{id}_Z$ . Similarly, we find that $b \circ a = \text{id}_{Z'}$ . Thus $a$ is an isomorphism, with                        |
| fong-spivak-invitation_F01348 | 124  | ■ 1.000 | $X, Y$                                                | $X, Y$                                    | <b>Definition 3.71.</b> Let $\mathcal{C}$ be a category, and let $X, Y$ be objects in $\mathcal{C}$ . A <i>product</i> of $X$              |
| fong-spivak-invitation_F01349 | 124  | ■ 1.000 | $p_X: X \times Y \rightarrow X$                       | $p_X: X \times Y \rightarrow X$           | and $Y$ is an object, denoted $X \times Y$ , together with morphisms $p_X: X \times Y \rightarrow X$ and                                   |
| fong-spivak-invitation_F01350 | 124  | ■ 1.000 | $p_Y: X \times Y \rightarrow Y$                       | $p_Y: X \times Y \rightarrow Y$           | $p_Y: X \times Y \rightarrow Y$ such that for all objects $C$ together with morphisms $f: C \rightarrow X$ and                             |
| fong-spivak-invitation_F01351 | 124  | ■ 1.000 | $f: C \rightarrow X$                                  | $f: C \rightarrow X$                      | $p_Y: X \times Y \rightarrow Y$ such that for all objects $C$ together with morphisms $f: C \rightarrow X$ and                             |
| fong-spivak-invitation_F01352 | 124  | ■ 1.000 | $g: C \rightarrow Y$                                  | $g: C \rightarrow Y$                      | $g: C \rightarrow Y$ , there exists a unique morphism $C \rightarrow X \times Y$ , denoted $\langle f, g \rangle$ , for which the          |
| fong-spivak-invitation_F01353 | 124  | ■ 1.000 | $C \rightarrow X \times Y$                            | $C \rightarrow X \times Y$                | $g: C \rightarrow Y$ , there exists a unique morphism $C \rightarrow X \times Y$ , denoted $\langle f, g \rangle$ , for which the          |
| fong-spivak-invitation_F01354 | 124  | ■ 1.000 | $\langle f, g \rangle$                                | $\langle f, g \rangle$                    | $g: C \rightarrow Y$ , there exists a unique morphism $C \rightarrow X \times Y$ , denoted $\langle f, g \rangle$ , for which the          |
| fong-spivak-invitation_F01355 | 124  | ■ 1.000 | $p_X(x, y) := x$                                      | $p_X(x, y) := x$                          | $p_X(x, y) := x$ and $p_Y(x, y) := y$ .                                                                                                    |
| fong-spivak-invitation_F01356 | 124  | ■ 1.000 | $p_Y(x, y) := y$                                      | $p_Y(x, y) := y$                          | $p_X(x, y) := x$ and $p_Y(x, y) := y$ .                                                                                                    |
| fong-spivak-invitation_F01357 | 124  | ■ 1.000 | $\langle f, g \rangle(c) := (f(c), g(c))$             | $\langle f, g \rangle(c) := (f(c), g(c))$ | to $X \times Y$ such that the required diagram commutes is given by $\langle f, g \rangle(c) := (f(c), g(c))$ .                            |
| fong-spivak-invitation_F01358 | 124  | ■ 0.999 | $\underline{6} \times \underline{4}$                  | $\underline{6} \times \underline{4}$      | Here is a picture of the product $\underline{6} \times \underline{4}$ of sets $\underline{6}$ and $\underline{4}$ .                        |
| fong-spivak-invitation_F01359 | 124  | ■ 0.999 | $\underline{6}$                                       | $\underline{6}$                           | Here is a picture of the product $\underline{6} \times \underline{4}$ of sets $\underline{6}$ and $\underline{4}$ .                        |
| fong-spivak-invitation_F01360 | 124  | ■ 0.999 | $\underline{4}$                                       | $\underline{4}$                           | Here is a picture of the product $\underline{6} \times \underline{4}$ of sets $\underline{6}$ and $\underline{4}$ .                        |
| fong-spivak-invitation_F01361 | 125  | ■ 0.906 | $x, y \in P$                                          | $x, y \in P$                              | <b>Exercise 3.73.</b> Let $(P, \leq)$ be a preorder, let $x, y \in P$ be elements, and let $\mathcal{P}$ be the                            |
| fong-spivak-invitation_F01362 | 125  | ■ 1.000 | $x \wedge y$                                          | $x \wedge y$                              | corresponding category. Show that the product $x \times y$ in $\mathcal{P}$ agrees with their meet $x \wedge y$                            |
| fong-spivak-invitation_F01363 | 125  | ■ 1.000 | $\mathcal{C} \times \mathcal{D}$                      | $\mathcal{C} \times \mathcal{D}$          | <b>Example 3.74.</b> Given two categories $\mathcal{C}$ and $\mathcal{D}$ , their product $\mathcal{C} \times \mathcal{D}$ may be given as |
| fong-spivak-invitation_F01364 | 125  | ■ 0.953 | $(c, d)$                                              | $(c, d)$                                  | follows. The objects of this category are pairs $(c, d)$ , where $c$ is an object of $\mathcal{C}$ and $d$ is                              |
| fong-spivak-invitation_F01365 | 125  | ■ 0.999 | $(c, d) \rightarrow (c^{\{\prime\}}, d^{\{\prime\}})$ | $(c, d) \rightarrow (c', d')$             | an object of $\mathcal{D}$ . Similarly, morphisms $(c, d) \rightarrow (c', d')$ are pairs $(f, g)$ where $f: c \rightarrow c'$             |
| fong-spivak-invitation_F01366 | 125  | ■ 0.999 | $(f, g)$                                              | $(f, g)$                                  | an object of $\mathcal{D}$ . Similarly, morphisms $(c, d) \rightarrow (c', d')$ are pairs $(f, g)$ where $f: c \rightarrow c'$             |

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| fong-spivak-invitation_F01367 | 125  | ■ 0.999 | $f: c \rightarrow c^{\prime}$                                           | $f: c \rightarrow c'$                               | an object of $\mathcal{D}$ . Similarly, morphisms $(c, d) \rightarrow (c', d')$ are pairs $(f, g)$ where $f: c \rightarrow c'$                                        |
| fong-spivak-invitation_F01368 | 125  | ■ 0.487 | $(f, g); \left(f^{\prime}, g^{\prime}\right)=$                          | $(f, g); (f', g') =$                                | simply given by composing each entry in the pair separately, so $(f, g) \circ (f', g') =$                                                                             |
| fong-spivak-invitation_F01369 | 125  | ■ 0.807 | $\left(f \circ f^{\prime}, g \circ g^{\prime}\right)$                   | $(f \circ f', g \circ g')$                          | $f \circ f', g \circ g'$ .                                                                                                                                            |
| fong-spivak-invitation_F01370 | 125  | ■ 1.000 | $\mathbf{1} \times \mathbf{2}$                                          | $\mathbf{1} \times \mathbf{2}$                      | 3. What is the product category $\mathbf{1} \times \mathbf{2}$ ?                                                                                                      |
| fong-spivak-invitation_F01371 | 125  | ■ 0.937 | $\mathcal{P} \times \mathcal{Q}$                                        | $\mathcal{P} \times \mathcal{Q}$                    | 4. What is the product category $\mathcal{P} \times \mathcal{Q}$ when $\mathcal{P}$ and $\mathcal{Q}$ are preorders and $\mathcal{P}$ and $\mathcal{Q}$ are           |
| fong-spivak-invitation_F01372 | 125  | ■ 1.000 | $X \stackrel{f}{\leftarrow} C \stackrel{g}{\rightarrow} Y$              | $X \xleftarrow{f} C \xrightarrow{g} Y$              | pair of maps $X \xleftarrow{f} C \xrightarrow{g} Y$ , there exists a unique map $C \rightarrow X \times Y$ such that certain                                          |
| fong-spivak-invitation_F01373 | 125  | ■ 0.991 | $(X, Y)$                                                                | $(X, Y)$                                            | call <b>Cone</b> $(X, Y)$ , the category of cones over $X$ and $Y$ in $\mathcal{C}$ . We shall see in Exercise 3.76                                                   |
| fong-spivak-invitation_F01374 | 125  | ■ 0.644 | $X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$ | $X \xleftarrow{p_X} X \times Y \xrightarrow{p_Y} Y$ | that $X \xleftarrow{p_X} X \times Y \xrightarrow{p_Y} Y$ is a terminal object in <b>Cone</b> $(X, Y)$ .                                                               |
| fong-spivak-invitation_F01375 | 125  | ■ 0.644 | $\mathbf{Cone}(X, Y)$                                                   | <b>Cone</b> $(X, Y)$                                | that $X \xleftarrow{p_X} X \times Y \xrightarrow{p_Y} Y$ is a terminal object in <b>Cone</b> $(X, Y)$ .                                                               |
| fong-spivak-invitation_F01376 | 125  | ■ 0.849 | $X \stackrel{f'}{\leftarrow} C' \stackrel{g'}{\rightarrow} Y$           | $X \xleftarrow{f'} C' \xrightarrow{g'} Y$           | $X \xleftarrow{f} C \xrightarrow{g} Y$ to $X \xleftarrow{f'} C' \xrightarrow{g'} Y$ in <b>Cone</b> $(X, Y)$ is a morphism $a: C \rightarrow C'$ in $\mathcal{C}$ such |
| fong-spivak-invitation_F01377 | 125  | ■ 0.849 | $a: C \rightarrow C^{\prime}$                                           | $a: C \rightarrow C'$                               | $X \xleftarrow{f} C \xrightarrow{g} Y$ to $X \xleftarrow{f'} C' \xrightarrow{g'} Y$ in <b>Cone</b> $(X, Y)$ is a morphism $a: C \rightarrow C'$ in $\mathcal{C}$ such |
| fong-spivak-invitation_F01378 | 125  | ■ 0.622 | $X \stackrel{p_X}{\leftarrow} X \times Y \stackrel{p_Y}{\rightarrow} Y$ | $X \xleftarrow{p_X} X \times Y \xrightarrow{p_Y} Y$ | <b>Exercise 3.76.</b> Check that a product $X \xleftarrow{p_X} X \times Y \xrightarrow{p_Y} Y$ is exactly the same as a                                               |
| fong-spivak-invitation_F01379 | 126  | ■ 0.994 | $(C, c_*)$                                                              | $(C, c_*)$                                          | <b>Definition 3.77.</b> Let $D: \mathcal{J} \rightarrow \mathcal{C}$ be a diagram. A cone $(C, c_*)$ over $D$ consists of                                             |
| fong-spivak-invitation_F01380 | 126  | ■ 1.000 | $C \in \mathcal{C}$                                                     | $C \in \mathcal{C}$                                 | (i) an object $C \in \mathcal{C}$ ;                                                                                                                                   |
| fong-spivak-invitation_F01381 | 126  | ■ 0.999 | $j \in \mathcal{J}$                                                     | $j \in \mathcal{J}$                                 | (ii) for each object $j \in \mathcal{J}$ , a morphism $c_j: C \rightarrow D(j)$ .                                                                                     |
| fong-spivak-invitation_F01382 | 126  | ■ 0.999 | $c_j: C \rightarrow D(j)$                                               | $c_j: C \rightarrow D(j)$                           | (ii) for each object $j \in \mathcal{J}$ , a morphism $c_j: C \rightarrow D(j)$ .                                                                                     |
| fong-spivak-invitation_F01383 | 126  | ■ 0.812 | $f: j \rightarrow k$                                                    | $f: j \rightarrow k$                                | (a) for each $f: j \rightarrow k$ in $\mathcal{J}$ , we have $c_k = c_j \circ D(f)$ .                                                                                 |
| fong-spivak-invitation_F01384 | 126  | ■ 0.812 | $c_k = c_j$                                                             | $c_k = c_j$                                         | (a) for each $f: j \rightarrow k$ in $\mathcal{J}$ , we have $c_k = c_j \circ D(f)$ .                                                                                 |
| fong-spivak-invitation_F01385 | 126  | ■ 0.812 | $D(f)$                                                                  | $D(f)$                                              | (a) for each $f: j \rightarrow k$ in $\mathcal{J}$ , we have $c_k = c_j \circ D(f)$ .                                                                                 |
| fong-spivak-invitation_F01386 | 126  | ■ 1.000 | $(C, c_*) \rightarrow (C', c'_*)$                                       | $(C, c_*) \rightarrow (C', c'_*)$                   | A morphism of cones $(C, c_*) \rightarrow (C', c'_*)$ is a morphism $a: C \rightarrow C'$ in $\mathcal{C}$ such that                                                  |
| fong-spivak-invitation_F01387 | 126  | ■ 0.918 | $c_j = a \circ c'_j$                                                    | $c_j = a \circ c'_j$                                | for all $j \in \mathcal{J}$ we have $c_j = a \circ c'_j$ . Cones over $D$ , and their morphisms, form a category                                                      |
| fong-spivak-invitation_F01388 | 126  | ■ 0.997 | $(D)$                                                                   | $(D)$                                               | <b>Cone</b> $(D)$ .                                                                                                                                                   |
| fong-spivak-invitation_F01389 | 126  | ■ 0.480 | $\lim(D)$                                                               | $\lim(D)$                                           | The limit of $D$ , denoted $\lim(D)$ , is the terminal object in the category <b>Cone</b> $(D)$ . Say                                                                 |
| fong-spivak-invitation_F01390 | 126  | ■ 0.480 | <b>Cone</b> $(D)$                                                       | <b>Cone</b> $(D)$                                   | The limit of $D$ , denoted $\lim(D)$ , is the terminal object in the category <b>Cone</b> $(D)$ . Say                                                                 |
| fong-spivak-invitation_F01391 | 126  | ■ 0.991 | $\lim(D) = (C, c_*)$                                                    | $\lim(D) = (C, c_*)$                                | it is the cone $\lim(D) = (C, c_*)$ ; we refer to $C$ as the limit object and the map $c_j$ for any                                                                   |
| fong-spivak-invitation_F01392 | 126  | ■ 0.991 | $c_j$                                                                   | $c_j$                                               | it is the cone $\lim(D) = (C, c_*)$ ; we refer to $C$ as the limit object and the map $c_j$ for any                                                                   |

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| fong-spivak-invitation_F01393 | 126  | 0.854 | $j \in \mathcal{J}$                                              | $j \in \mathfrak{J}$                                                      | $j \in \mathcal{J}$ as the $j$ th projection map.                                                                                                                                                                                               |
| fong-spivak-invitation_F01394 | 126  | 1.000 | $D: \mathcal{J} \rightarrow$                                     | $D: \mathcal{J} \rightarrow$                                              | with five objects and five non-identity morphisms, then we may draw a diagram $D: \mathcal{J} \rightarrow$                                                                                                                                      |
| fong-spivak-invitation_F01395 | 126  | 0.950 | $\mathcal{J}=\varnothing$                                        | $\mathcal{J} = \varnothing$                                               | $\mathcal{J} = \varnothing$ .                                                                                                                                                                                                                   |
| fong-spivak-invitation_F01396 | 127  | 0.159 | $v, w$                                                           | $v, w$                                                                    | $v, w$ and no arrows, $\mathcal{J} = \left  \begin{array}{cc} v & w \\ \bullet & \bullet \end{array} \right $ .                                                                                                                                 |
| fong-spivak-invitation_F01397 | 127  | 0.159 | $\mathfrak{d}=\stackrel{v}{\bullet} \quad \stackrel{w}{\bullet}$ | $\mathfrak{d} = \begin{array}{cc} v & w \\ \bullet & \bullet \end{array}$ | $v, w$ and no arrows, $\mathcal{J} = \left  \begin{array}{cc} v & w \\ \bullet & \bullet \end{array} \right $ .                                                                                                                                 |
| fong-spivak-invitation_F01398 | 127  | 0.762 | $V, A, s, t$                                                     | $V, A, s, t$                                                              | <b>Theorem 3.80.</b> Let $\mathcal{J}$ be a category presented by the finite graph $(V, A, s, t)$ together                                                                                                                                      |
| fong-spivak-invitation_F01399 | 127  | 0.989 | $V=$                                                             | $V =$                                                                     | with some equations, and let $D: \mathcal{J} \rightarrow \mathbf{Set}$ be a set-valued functor. Write $V =$                                                                                                                                     |
| fong-spivak-invitation_F01400 | 127  | 1.000 | $\{v_1, \dots, v_n\}$                                            | $\{v_1, \dots, v_n\}$                                                     | $\{v_1, \dots, v_n\}$ . The set                                                                                                                                                                                                                 |
| fong-spivak-invitation_F01401 | 127  | 0.999 | $p_i: \left( \lim_{\mathcal{J}} D \right) \rightarrow D(v_i)$    | $p_i: \left( \lim_{\mathcal{J}} D \right) \rightarrow D(v_i)$             | together with the projection maps $p_i: (\lim_{\mathcal{J}} D) \rightarrow D(v_i)$ given by $p_i(d_1, \dots, d_n) :=$                                                                                                                           |
| fong-spivak-invitation_F01402 | 127  | 0.999 | $p_i(d_1, \dots, d_n) :=$                                        | $p_i(d_1, \dots, d_n) :=$                                                 | together with the projection maps $p_i: (\lim_{\mathcal{J}} D) \rightarrow D(v_i)$ given by $p_i(d_1, \dots, d_n) :=$                                                                                                                           |
| fong-spivak-invitation_F01403 | 127  | 1.000 | $d_i$                                                            | $d_i$                                                                     | $d_i$ is a limit of $D$ .                                                                                                                                                                                                                       |
| fong-spivak-invitation_F01404 | 127  | 0.958 | $J$                                                              | $J$                                                                       | <b>Example 3.81.</b> If $J$ is the empty graph $\square$ , then $n = 0$ : there are no vertices. There is                                                                                                                                       |
| fong-spivak-invitation_F01405 | 127  | 0.958 | $n=0$                                                            | $n = 0$                                                                   | <b>Example 3.81.</b> If $J$ is the empty graph $\square$ , then $n = 0$ : there are no vertices. There is                                                                                                                                       |
| fong-spivak-invitation_F01406 | 127  | 1.000 | $D: \mathbf{1} \rightarrow$                                      | $D: \mathbf{1} \rightarrow$                                               | <b>Exercise 3.83.</b> If $D: \mathbf{1} \rightarrow \mathbf{Set}$ is a functor, what is the limit of $D$ ? Compute it using                                                                                                                     |
| fong-spivak-invitation_F01407 | 127  | 0.997 | $(d_1, \dots, d_n)$                                              | $(d_1, \dots, d_n)$                                                       | In particular, the condition that the limit of $D: \mathcal{J} \rightarrow \mathbf{Set}$ selects tuples $(d_1, \dots, d_n)$ such                                                                                                                |
| fong-spivak-invitation_F01408 | 127  | 0.942 | $D(a)(d_i)=d_j$                                                  | $D(a)(d_i) = d_j$                                                         | that $D(a)(d_i) = d_j$ for each morphism $a: i \rightarrow j$ in $\mathcal{J}$ allows us to use limits to select                                                                                                                                |
| fong-spivak-invitation_F01409 | 127  | 0.942 | $a: i \rightarrow j$                                             | $a: i \rightarrow j$                                                      | that $D(a)(d_i) = d_j$ for each morphism $a: i \rightarrow j$ in $\mathcal{J}$ allows us to use limits to select                                                                                                                                |
| fong-spivak-invitation_F01410 | 127  | 0.942 | $\mathcal{J}$                                                    | $\mathfrak{J}$                                                            | that $D(a)(d_i) = d_j$ for each morphism $a: i \rightarrow j$ in $\mathcal{J}$ allows us to use limits to select                                                                                                                                |
| fong-spivak-invitation_F01411 | 128  | 0.282 | $\xrightarrow{x} \xrightarrow{a} \xrightarrow{g}$                | $\xrightarrow{x} \xrightarrow{a} \xrightarrow{g}$                         | <b>Example 3.84.</b> If $\mathcal{J}$ is presented by the <i>cospan</i> graph $\left  \begin{array}{ccc} x & \xrightarrow{f} & a \\ \bullet & & \bullet \\ & \xleftarrow{g} & y \\ & \bullet & \bullet \end{array} \right $ , then its limit is |
| fong-spivak-invitation_F01412 | 128  | 1.000 | $X \xrightarrow{f} A \xleftarrow{g} Y$                           | $X \xrightarrow{f} A \xleftarrow{g} Y$                                    | known as a <i>pullback</i> . Given the diagram $X \xrightarrow{f} A \xleftarrow{g} Y$ , the pullback is the cone shown                                                                                                                          |
| fong-spivak-invitation_F01413 | 128  | 1.000 | $c_a$                                                            | $c_a$                                                                     | The fact that the diagram commutes means that the diagonal arrow $c_a$ is in some                                                                                                                                                               |
| fong-spivak-invitation_F01414 | 128  | 0.492 | $X \times_A Y$                                                   | $X \times_A Y$                                                            | naming the cone point $X \times_A Y$ , and adding the $\lrcorner$ symbol, as shown to the right                                                                                                                                                 |
| fong-spivak-invitation_F01415 | 128  | 1.000 | $X=\underline{6}, Y=\underline{4}$                               | $X = \underline{6}, Y = \underline{4}$                                    | Here is a picture to help us unpack the definition in <b>Set</b> . We take $X = \underline{6}, Y = \underline{4}$ , and                                                                                                                         |
| fong-spivak-invitation_F01416 | 128  | 1.000 | $f: \underline{6} \rightarrow A$                                 | $f: \underline{6} \rightarrow A$                                          | The functions $f: \underline{6} \rightarrow A$ and $g: \underline{4} \rightarrow A$ are expressed in the coloring of the dots:                                                                                                                  |
| fong-spivak-invitation_F01417 | 128  | 1.000 | $g: \underline{4} \rightarrow A$                                 | $g: \underline{4} \rightarrow A$                                          | The functions $f: \underline{6} \rightarrow A$ and $g: \underline{4} \rightarrow A$ are expressed in the coloring of the dots:                                                                                                                  |
| fong-spivak-invitation_F01418 | 128  | 1.000 | $g(2)=g(4)=$                                                     | $g(2) = g(4) =$                                                           | for example, $g(2) = g(4) =$ red, while $f(5) =$ black. The pullback selects pairs                                                                                                                                                              |

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| fong-spivak-invitation_F01419 | 128  | ■ 1.000 | $f(5)=$                                                                                | $f(5) =$                                                                               | for example, $g(2) = g(4) = \text{red}$ , while $f(5) = \text{black}$ . The pullback selects pairs                                                                             |
| fong-spivak-invitation_F01420 | 128  | ■ 1.000 | $(i, j) \in \underline{6} \times \underline{4}$                                        | $(i, j) \in \underline{6} \times \underline{4}$                                        | $(i, j) \in \underline{6} \times \underline{4}$ such that $f(i)$ and $g(j)$ have the same color.                                                                               |
| fong-spivak-invitation_F01421 | 128  | ■ 1.000 | $f(i)$                                                                                 | $f(i)$                                                                                 | $(i, j) \in \underline{6} \times \underline{4}$ such that $f(i)$ and $g(j)$ have the same color.                                                                               |
| fong-spivak-invitation_F01422 | 128  | ■ 1.000 | $g(j)$                                                                                 | $g(j)$                                                                                 | $(i, j) \in \underline{6} \times \underline{4}$ such that $f(i)$ and $g(j)$ have the same color.                                                                               |
| fong-spivak-invitation_F01423 | 129  | ■ 0.989 | $F^{\text{op}}$                                                                        | $F^{\text{op}}$                                                                        | $F^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$ ? That is, how should $F^{\text{op}}$ act on objects, and how should it act on                    |
| fong-spivak-invitation_F01424 | 129  | ■ 0.989 | $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$                          | $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$                          | $F^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$ ? That is, how should $F^{\text{op}}$ act on objects, and how should it act on                    |
| fong-spivak-invitation_F01425 | 129  | ■ 0.999 | $\mathcal{D}^{\text{op}}: \mathcal{J}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$ | $\mathcal{D}^{\text{op}}: \mathcal{J}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$ | Given a diagram $D: \mathcal{J} \rightarrow \mathcal{C}$ , we may take the limit of the functor $D^{\text{op}}: \mathcal{J}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$ . |
| fong-spivak-invitation_F01426 | 130  | ■ 0.988 | $\Delta, \Sigma, \Pi$                                                                  | $\Delta, \Sigma, \Pi$                                                                  | three data migration functors $\Delta, \Sigma, \Pi$ ; see [Spi12; SW15b]. The version of this story                                                                            |
| fong-spivak-invitation_F01427 | 132  | ■ 0.699 | $\leq s$                                                                               | $\leq s$                                                                               | the $\leq s$ on each wire; they indicate that if box $A$ requires a resource that box $B$ produces,                                                                            |
| fong-spivak-invitation_F01428 | 132  | ■ 1.000 | $\$ 10 \leq \$ 20, 5 \text{ W} \leq 6 \text{ W}$                                       | $\$10 \leq \$20, 5 \text{ W} \leq 6 \text{ W}$                                         | $\$10 < \$20, 5\text{W} < 6\text{W}$ and so on.                                                                                                                                |
| fong-spivak-invitation_F01429 | 132  | ■ 1.000 | $(p, r) \in P \times R$                                                                | $(p, r) \in P \times R$                                                                | pair $(p, r) \in P \times R$ , where $P$ is the preorder of resources to be produced and $R$ is                                                                                |
| fong-spivak-invitation_F01430 | 132  | ■ 1.000 | $\Phi: P \times R \rightarrow$                                                         | $\Phi: P \times R \rightarrow$                                                         | Feasibility relations hence define a function $\Phi: P \times R \rightarrow \mathbf{Bool}$ . For a function                                                                    |
| fong-spivak-invitation_F01431 | 132  | ■ 1.000 | $\Phi(p, r)=$                                                                          | $\Phi(p, r) =$                                                                         | (a) If $\Phi(p, r) = \text{true}$ and $p' \leq p$ , then $\Phi(p', r) = \text{true}$ .                                                                                         |
| fong-spivak-invitation_F01432 | 132  | ■ 1.000 | $p^{\text{prime}} \leq p$                                                              | $p' \leq p$                                                                            | (a) If $\Phi(p, r) = \text{true}$ and $p' \leq p$ , then $\Phi(p', r) = \text{true}$ .                                                                                         |
| fong-spivak-invitation_F01433 | 132  | ■ 1.000 | $\Phi(p^{\text{prime}}, r)=$                                                           | $\Phi(p', r) =$                                                                        | (a) If $\Phi(p, r) = \text{true}$ and $p' \leq p$ , then $\Phi(p', r) = \text{true}$ .                                                                                         |
| fong-spivak-invitation_F01434 | 132  | ■ 0.631 | $r \leq r^{\text{prime}}$                                                              | $r \leq r'$                                                                            | (b) If $\Phi(p, r) = \text{true}$ and $r \leq r'$ then $\Phi(p, r') = \text{true}$ .                                                                                           |
| fong-spivak-invitation_F01435 | 132  | ■ 0.631 | $\Phi(p, r^{\text{prime}})=$                                                           | $\Phi(p, r') =$                                                                        | (b) If $\Phi(p, r) = \text{true}$ and $r \leq r'$ then $\Phi(p, r') = \text{true}$ .                                                                                           |
| fong-spivak-invitation_F01436 | 133  | ■ 1.000 | $r^{\text{prime}} \geq r$                                                              | $r' \geq r$                                                                            | (b) also produce $p$ given more resources $r' \geq r$ . We shall see that these two conditions                                                                                 |
| fong-spivak-invitation_F01437 | 133  | ■ 0.600 | $P^{\text{op}} \times R \rightarrow$                                                   | $P^{\text{op}} \times R \rightarrow$                                                   | are formalized by requiring $\Phi$ to be a monotone map $P^{\text{op}} \times R \rightarrow \mathbf{Bool}$ .                                                                   |
| fong-spivak-invitation_F01438 | 133  | ■ 0.606 | $\$$                                                                                   | $\$$                                                                                   | $\$ -$ is structured as a preorder. The order $x \leq y$ represents the <i>availability of <math>x</math> given <math>y</math></i> ,                                           |
| fong-spivak-invitation_F01439 | 133  | ■ 0.712 | $5 \text{ W} \leq 10 \text{ W}$                                                        | $5 \text{ W} \leq 10 \text{ W}$                                                        | if $5\text{W} \leq 10\text{W}$ , it means that whenever we are provided $10\text{W}$ , we implicitly also have                                                                 |
| fong-spivak-invitation_F01440 | 134  | ■ 0.792 | $\mathcal{X}^{\text{op}} = (X, \geq)$                                                  | $\mathcal{X}^{\text{op}} = (X, \geq)$                                                  | opposite $\mathcal{X}^{\text{op}} = (X, \geq)$ has $x \geq y$ iff $y \leq x$ .                                                                                                 |
| fong-spivak-invitation_F01441 | 134  | ■ 0.792 | $x \geq y$                                                                             | $x \geq y$                                                                             | opposite $\mathcal{X}^{\text{op}} = (X, \geq)$ has $x \geq y$ iff $y \leq x$ .                                                                                                 |
| fong-spivak-invitation_F01442 | 134  | ■ 0.940 | $\mathcal{X} = \left(X, \leq_X\right)$                                                 | $\mathcal{X} = (X, \leq_X)$                                                            | <b>Definition 4.1.</b> Let $\mathcal{X} = (X, \leq_X)$ and $\mathcal{Y} = (Y, \leq_Y)$ be preorders. A <i>feasibility relation</i>                                             |
| fong-spivak-invitation_F01443 | 134  | ■ 0.940 | $\mathcal{Y} = \left(Y, \leq_Y\right)$                                                 | $\mathcal{Y} = (Y, \leq_Y)$                                                            | <b>Definition 4.1.</b> Let $\mathcal{X} = (X, \leq_X)$ and $\mathcal{Y} = (Y, \leq_Y)$ be preorders. A <i>feasibility relation</i>                                             |
| fong-spivak-invitation_F01444 | 134  | ■ 0.616 | $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$                                            | $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$                                            | We denote this by $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$ .                                                                                                                |
| fong-spivak-invitation_F01445 | 134  | ■ 1.000 | $\Phi(x, y)=$                                                                          | $\Phi(x, y) =$                                                                         | Given $x \in X$ and $y \in Y$ , if $\Phi(x, y) = \text{true}$ we say <i><math>x</math> can be obtained given <math>y</math></i> .                                              |

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| fong-spivak-invitation_F01446 | 134  | 0.832 | $x^{\prime} \leq_X x$                                                          | $x' \leq_X x$                                                                  | $x' \leq_X x$ and $y \leq_Y y'$ then $\Phi(x, y) \leq_{\mathbf{Bool}} \Phi(x', y')$ . In other words, if $x$ can be                                                                                  |
| fong-spivak-invitation_F01447 | 134  | 0.832 | $y \leq_Y y^{\prime}$                                                          | $y \leq_Y y'$                                                                  | $x' \leq_X x$ and $y \leq_Y y'$ then $\Phi(x, y) \leq_{\mathbf{Bool}} \Phi(x', y')$ . In other words, if $x$ can be                                                                                  |
| fong-spivak-invitation_F01448 | 134  | 0.832 | $\Phi(x, y) \leq_{\text{Bool}} \Phi(x^{\prime}, y^{\prime})$                   | $\Phi(x, y) \leq_{\mathbf{Bool}} \Phi(x', y')$                                 | $x' \leq_X x$ and $y \leq_Y y'$ then $\Phi(x, y) \leq_{\mathbf{Bool}} \Phi(x', y')$ . In other words, if $x$ can be                                                                                  |
| fong-spivak-invitation_F01449 | 134  | 1.000 | $x^{\prime}$                                                                   | $x'$                                                                           | obtained given $y$ , and if $x'$ is available given $x$ , then $x'$ can be obtained given $y$ . And if                                                                                               |
| fong-spivak-invitation_F01450 | 134  | 1.000 | $y^{\prime}$                                                                   | $y'$                                                                           | furthermore $y$ is available given $y'$ , then $x'$ can also be obtained given $y'$ .                                                                                                                |
| fong-spivak-invitation_F01451 | 134  | 0.387 | $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$                                 | $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$                                 | 1. Draw the Hasse diagram for the preorder $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$ .                                                                                                          |
| fong-spivak-invitation_F01452 | 134  | 0.710 | $\Lambda: \mathcal{X} \rightarrow \mathcal{Y}$                                 | $\Lambda: \mathcal{X} \rightarrow \mathcal{Y}$                                 | 2. Write down a profunctor $\Lambda: \mathcal{X} \nrightarrow \mathcal{Y}$ and, reading $\Lambda(x, y) = \text{true}$ as “my aunt can                                                                |
| fong-spivak-invitation_F01453 | 134  | 0.710 | $\Lambda(x, y) =$                                                              | $\Lambda(x, y) =$                                                              | 2. Write down a profunctor $\Lambda: \mathcal{X} \nrightarrow \mathcal{Y}$ and, reading $\Lambda(x, y) = \text{true}$ as “my aunt can                                                                |
| fong-spivak-invitation_F01454 | 134  | 0.335 | $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$                                 | $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$                                 | forms an upper set in $\mathcal{X}^{\mathrm{op}} \times \mathcal{Y}$ .                                                                                                                               |
| fong-spivak-invitation_F01455 | 134  | 1.000 | $\Rightarrow: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$             | $\Rightarrow: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$             | and a hom-operation, which we’ll write $\Rightarrow: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ . By definition, this operation                                                            |
| fong-spivak-invitation_F01456 | 134  | 1.000 | $b, c, d \in \mathbb{B}$                                                       | $b, c, d \in \mathbb{B}$                                                       | satisfies the property that for all $b, c, d \in \mathbb{B}$ one has                                                                                                                                 |
| fong-spivak-invitation_F01458 | 135  | 0.311 | $\Phi: \mathcal{X} \mapsto \mathcal{Y}$                                        | $\Phi: \mathcal{X} \mapsto \mathcal{Y}$                                        | and $\mathcal{Y}$ be $\mathcal{V}$ -categories. A $\mathcal{V}$ -profunctor from $\mathcal{X}$ to $\mathcal{Y}$ , denoted $\Phi: \mathcal{X} \nrightarrow \mathcal{Y}$ , is a $\mathcal{V}$ -functor |
| fong-spivak-invitation_F01459 | 135  | 0.322 | $\Phi: \mathrm{Ob}(\mathcal{X}) \times \mathrm{Ob}(\mathcal{Y}) \rightarrow V$ | $\Phi: \mathrm{Ob}(\mathcal{X}) \times \mathrm{Ob}(\mathcal{Y}) \rightarrow V$ | $\Phi: \mathrm{Ob}(\mathcal{X}) \times \mathrm{Ob}(\mathcal{Y}) \rightarrow V$ such that for any $x, x' \in \mathcal{X}$ and $y, y' \in \mathcal{Y}$ the following                                   |
| fong-spivak-invitation_F01460 | 135  | 0.322 | $x, x^{\prime} \in \mathcal{X}$                                                | $x, x' \in \mathcal{X}$                                                        | $\Phi: \mathrm{Ob}(\mathcal{X}) \times \mathrm{Ob}(\mathcal{Y}) \rightarrow V$ such that for any $x, x' \in \mathcal{X}$ and $y, y' \in \mathcal{Y}$ the following                                   |
| fong-spivak-invitation_F01461 | 135  | 0.322 | $y, y^{\prime} \in \mathcal{Y}$                                                | $y, y' \in \mathcal{Y}$                                                        | $\Phi: \mathrm{Ob}(\mathcal{X}) \times \mathrm{Ob}(\mathcal{Y}) \rightarrow V$ such that for any $x, x' \in \mathcal{X}$ and $y, y' \in \mathcal{Y}$ the following                                   |
| fong-spivak-invitation_F01462 | 135  | 0.698 | $\Phi: X \rightarrow Y$                                                        | $\Phi: X \rightarrow Y$                                                        | tion 4.4 in the case $\mathcal{V} = \mathbf{Bool}$ . One way to imagine a <b>Bool</b> -profunctor $\Phi: X \nrightarrow Y$                                                                           |
| fong-spivak-invitation_F01463 | 136  | 1.000 | $W \leq N$                                                                     | $W \leq N$                                                                     | Both $X$ and $Y$ are preorders, e.g. with $W \leq N$ and $b \leq a$ . With bridges coming                                                                                                            |
| fong-spivak-invitation_F01464 | 136  | 0.837 | $N$                                                                            | $N$                                                                            | a path from $N$ to $e$ and $E$ to $a$ , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$ .                                                                                          |
| fong-spivak-invitation_F01465 | 136  | 0.837 | $E$                                                                            | $E$                                                                            | a path from $N$ to $e$ and $E$ to $a$ , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$ .                                                                                          |
| fong-spivak-invitation_F01466 | 136  | 0.837 | $\Phi(N, e) =$                                                                 | $\Phi(N, e) =$                                                                 | a path from $N$ to $e$ and $E$ to $a$ , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$ .                                                                                          |
| fong-spivak-invitation_F01467 | 136  | 0.837 | $\Phi(E, a) =$                                                                 | $\Phi(E, a) =$                                                                 | a path from $N$ to $e$ and $E$ to $a$ , we have $\Phi(N, e) = \text{true}$ and $\Phi(E, a) = \text{true}$ .                                                                                          |
| fong-spivak-invitation_F01468 | 136  | 1.000 | $W$                                                                            | $W$                                                                            | On the other hand, since there is no path from $W$ to $d$ , we have $\Phi(W, d) =$                                                                                                                   |
| fong-spivak-invitation_F01469 | 136  | 1.000 | $\Phi(W, d) =$                                                                 | $\Phi(W, d) =$                                                                 | On the other hand, since there is no path from $W$ to $d$ , we have $\Phi(W, d) =$                                                                                                                   |
| fong-spivak-invitation_F01470 | 136  | 1.000 | $W \leq N \leq c \leq a$                                                       | $W \leq N \leq c \leq a$                                                       | $W \leq N \leq c \leq a$ , etc. This is called the <i>collage</i> of $\Phi$ ; we’ll explore this in more detail                                                                                      |
| fong-spivak-invitation_F01471 | 136  | 1.000 | $\Phi(m, n) \in \mathbb{B}$                                                    | $\Phi(m, n) \in \mathbb{B}$                                                    | $\Phi(m, n) \in \mathbb{B}$ . Copy and complete the <b>Bool</b> -matrix:                                                                                                                             |
| fong-spivak-invitation_F01472 | 137  | 0.722 | $\Phi: X \mapsto Y$                                                            | $\Phi: X \mapsto Y$                                                            | Here is a depiction of a <b>Cost</b> -profunctor $\Phi: X \nrightarrow Y$ :                                                                                                                          |
| fong-spivak-invitation_F01474 | 137  | 1.000 | $d_X * M_{\Phi} * d_Y$                                                         | $d_X * M_{\Phi} * d_Y$                                                         | The matrix of distances for the profunctor $\Phi$ will be equal to $d_X * M_{\Phi} * d_Y$ . In fact, since                                                                                           |

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| fong-spivak-invitation_F01475 | 137  | ■ 1.000 | $\Phi = M_X^3 * M_\Phi * M_Y^2$                        | $\Phi = M_X^3 * M_\Phi * M_Y^2$                        | $X$ has four elements and $Y$ has three, we also know that $\Phi = M_X^3 * M_\Phi * M_Y^2$ .                                                                                                |
| fong-spivak-invitation_F01476 | 138  | ■ 1.000 | $M_X^3 * M_\Phi * M_Y^2$                               | $M_X^3 * M_\Phi * M_Y^2$                               | <b>Exercise 4.12.</b> Calculate $M_X^3 * M_\Phi * M_Y^2$ , remembering to do matrix multiplication                                                                                          |
| fong-spivak-invitation_F01477 | 138  | ■ 0.484 | $[0, \infty], \leq$                                    | $[0, \infty], \leq$                                    | ple, load might be the preorder $([0, \infty], \leq)$ , where an element $x \in [0, \infty]$ represents                                                                                     |
| fong-spivak-invitation_F01478 | 138  | —       | 100, more_than_\                                       | (not rendered)                                         | —                                                                                                                                                                                           |
| fong-spivak-invitation_F01479 | 138  | ■ 0.941 | $\Phi: (T \times E) \mapsto \$$                        | $\Phi: (T \times E) \mapsto \$$                        | obtained at each cost; as such, it is described by a feasibility relation $\Phi: (T \times E) \multimap \$$ .                                                                               |
| fong-spivak-invitation_F01480 | 138  | ■ 0.626 | $T, E$                                                 | $T, E$                                                 | where $T$ , $E$ , and $\$$ are the preorders drawn below:                                                                                                                                   |
| fong-spivak-invitation_F01481 | 139  | —       | 500K given \                                           | (not rendered)                                         | —                                                                                                                                                                                           |
| fong-spivak-invitation_F01482 | 139  | ■ 0.997 | $\$ 500 \text{ \textit{K}}$                            | $\$500 \text{ K}$                                      | movie given \$500K,” and “I can provide a mean and boring movie given a good-                                                                                                               |
| fong-spivak-invitation_F01483 | 139  | ■ 0.988 | $\mathcal{P}, Q$                                       | $\mathcal{P}, Q$                                       | posing two of them in series. Imagine you have cities $\mathcal{P}$ , $\mathcal{Q}$ , and $\mathcal{R}$ and you have                                                                        |
| fong-spivak-invitation_F01484 | 139  | ■ 0.988 | $\mathcal{R}$                                          | $\mathcal{R}$                                          | posing two of them in series. Imagine you have cities $\mathcal{P}$ , $\mathcal{Q}$ , and $\mathcal{R}$ and you have                                                                        |
| fong-spivak-invitation_F01485 | 139  | ■ 0.936 | $\Phi: \mathcal{P} \mapsto Q$                          | $\Phi: \mathcal{P} \mapsto Q$                          | bridges – and hence feasibility matrices – connecting these cities, say $\Phi: \mathcal{P} \multimap Q$ and                                                                                 |
| fong-spivak-invitation_F01486 | 139  | ■ 0.586 | $\Psi: Q \mapsto \mathcal{R}$                          | $\Psi: Q \mapsto \mathcal{R}$                          | $\Psi: Q \multimap \mathcal{R}$ .                                                                                                                                                           |
| fong-spivak-invitation_F01487 | 140  | ■ 1.000 | $\Psi$                                                 | $\Psi$                                                 | The feasibility matrices for $\Phi$ (in blue) and $\Psi$ (in red) are:                                                                                                                      |
| fong-spivak-invitation_F01488 | 140  | ■ 1.000 | $\Phi; \Psi$                                           | $\Phi; \Psi$                                           | trying to give a feasibility matrix $\Phi; \Psi$ for getting from $\mathcal{P}$ to $\mathcal{R}$ . How should this be                                                                       |
| fong-spivak-invitation_F01489 | 140  | ■ 0.844 | $p \in \mathcal{P}$                                    | $p \in \mathcal{P}$                                    | done? Basically, for every pair $p \in \mathcal{P}$ and $r \in \mathcal{R}$ , the navigator searches through $\mathcal{Q}$ for a                                                            |
| fong-spivak-invitation_F01490 | 140  | ■ 0.844 | $r \in \mathcal{R}$                                    | $r \in \mathcal{R}$                                    | done? Basically, for every pair $p \in \mathcal{P}$ and $r \in \mathcal{R}$ , the navigator searches through $\mathcal{Q}$ for a                                                            |
| fong-spivak-invitation_F01492 | 141  | ■ 0.776 | $\mathcal{X}, \mathcal{Y}$                             | $\mathcal{X}, \mathcal{Y}$                             | <b>Definition 4.14.</b> Let $\mathcal{V}$ be a quantale, let $\mathcal{X}, \mathcal{Y}$ , and $\mathcal{Z}$ be $\mathcal{V}$ -categories, and let $\Phi: \mathcal{X} \multimap \mathcal{Y}$ |
| fong-spivak-invitation_F01493 | 141  | ■ 0.776 | $\mathcal{Z}$                                          | $\mathcal{Z}$                                          | <b>Definition 4.14.</b> Let $\mathcal{V}$ be a quantale, let $\mathcal{X}, \mathcal{Y}$ , and $\mathcal{Z}$ be $\mathcal{V}$ -categories, and let $\Phi: \mathcal{X} \multimap \mathcal{Y}$ |
| fong-spivak-invitation_F01494 | 141  | ■ 0.521 | $\Psi: y \rightarrow z$                                | $\Psi: y \rightarrow z$                                | and $\Psi: \mathcal{Y} \multimap \mathcal{Z}$ be $\mathcal{V}$ -profunctors. We define their <i>composite</i> , denoted $\Phi; \Psi: \mathcal{X} \multimap \mathcal{Z}$                     |
| fong-spivak-invitation_F01495 | 141  | ■ 0.521 | $\Phi: \Psi: \mathcal{X} \mapsto z$                    | $\Phi: \Psi: \mathcal{X} \mapsto z$                    | and $\Psi: \mathcal{Y} \multimap \mathcal{Z}$ be $\mathcal{V}$ -profunctors. We define their <i>composite</i> , denoted $\Phi; \Psi: \mathcal{X} \multimap \mathcal{Z}$                     |
| fong-spivak-invitation_F01496 | 141  | ■ 0.496 | $_{\mathcal{V}}$                                       | $\mathcal{V}$                                          | <b>Theorem 4.16.</b> For any skeletal quantale $\mathcal{V}$ , there is a category <b>Prof</b> $_{\mathcal{V}}$ whose objects                                                               |
| fong-spivak-invitation_F01497 | 141  | ■ 0.411 | $\mathcal{X} \mapsto 1$                                | $\mathcal{X} \mapsto 1$                                | are $\mathcal{V}$ -categories $\mathcal{X}$ , whose morphisms are $\mathcal{V}$ -profunctors $\mathcal{X} \multimap \mathcal{Y}$ , and with composition                                     |
| fong-spivak-invitation_F01498 | 141  | ■ 0.877 | $_{\text{Bool}}$                                       | $\text{Bool}$                                          | <b>Definition 4.17.</b> We define <b>Feas</b> $:= \mathbf{Prof}_{\text{Bool}}$ .                                                                                                            |
| fong-spivak-invitation_F01499 | 142  | ■ 0.661 | $\mathrm{U}_X: \mathcal{X} \mapsto X$                  | $U_X: \mathcal{X} \mapsto X$                           | Define the <i>unit profunctor</i> $U_X: \mathcal{X} \multimap X$ on a $\mathcal{V}$ -category $\mathcal{X}$ by the formula                                                                  |
| fong-spivak-invitation_F01500 | 142  | ■ 0.798 | $\mathcal{X}(x, y) \in \mathcal{V}$                    | $\mathcal{X}(x, y) \in \mathcal{V}$                    | of elements $x, y \in \mathcal{X}$ a hom-object $\mathcal{X}(x, y) \in \mathcal{V}$ . The unit profunctor $U_X$ just assigns                                                                |
| fong-spivak-invitation_F01501 | 142  | ■ 0.798 | $\mathrm{U}_X$                                         | $U_X$                                                  | of elements $x, y \in \mathcal{X}$ a hom-object $\mathcal{X}(x, y) \in \mathcal{V}$ . The unit profunctor $U_X$ just assigns                                                                |
| fong-spivak-invitation_F01502 | 142  | ■ 0.582 | $U_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$ | $U_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$ | for the unit profunctor $U_X: \mathcal{X} \multimap X$ .                                                                                                                                    |

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| fong-spivak-invitation_F01503 | 142  | 0.752 | $\mathrm{U}_{\mathcal{P}}$                                                             | $U_{\mathcal{P}}$                                                                      | <b>Lemma 4.19.</b> Composing any profunctor $\Phi: \mathcal{P} \multimap \mathcal{Q}$ with either unit profunctor, $U_{\mathcal{P}}$                                                    |
| fong-spivak-invitation_F01504 | 142  | 0.412 | $U_{\mathcal{Q}}$                                                                      | $U_{\mathcal{Q}}$                                                                      | or $U_{\mathcal{Q}}$ , returns $\Phi$ :                                                                                                                                                 |
| fong-spivak-invitation_F01505 | 142  | 0.524 | $U_{\mathcal{P}}; \Phi = \Phi$                                                         | $U_{\mathcal{P}}; \Phi = \Phi$                                                         | <i>Proof.</i> We show that $U_{\mathcal{P}} \circ \Phi = \Phi$ holds; proving $\Phi = \Phi \circ U_{\mathcal{Q}}$ is similar. Fix $p \in P$                                             |
| fong-spivak-invitation_F01506 | 142  | 0.524 | $\Phi = \Phi; U_{\mathcal{Q}}$                                                         | $\Phi = \Phi; U_{\mathcal{Q}}$                                                         | <i>Proof.</i> We show that $U_{\mathcal{P}} \circ \Phi = \Phi$ holds; proving $\Phi = \Phi \circ U_{\mathcal{Q}}$ is similar. Fix $p \in P$                                             |
| fong-spivak-invitation_F01507 | 142  | 0.998 | $\Phi \leq \mathrm{U}_{\mathcal{P}}; \Phi$                                             | $\Phi \leq U_{\mathcal{P}}; \Phi$                                                      | and $q \in Q$ . Since $\mathcal{V}$ is skeletal, to prove the equality it's enough to show $\Phi \leq U_{\mathcal{P}} \circ \Phi$                                                       |
| fong-spivak-invitation_F01508 | 142  | 0.519 | $U_{\mathcal{P}}; \Phi \leq \Phi$                                                      | $U_{\mathcal{P}}; \Phi \leq \Phi$                                                      | and $U_{\mathcal{P}} \circ \Phi \leq \Phi$ . We have one direction:                                                                                                                     |
| fong-spivak-invitation_F01509 | 142  | 1.000 | $\bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) \leq \Phi(p, q)$       | $\bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) \leq \Phi(p, q)$       | For the other direction, we must show $\bigvee_{p_1 \in P} (\mathcal{P}(p, p_1) \otimes \Phi(p_1, q)) \leq \Phi(p, q)$ . But by                                                         |
| fong-spivak-invitation_F01510 | 142  | 1.000 | $\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \leq \Phi(p, q)$                             | $\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \leq \Phi(p, q)$                             | definition of join, this holds iff $\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) < \Phi(p, q)$ is true for each $p_1 \in P$ .                                                               |
| fong-spivak-invitation_F01511 | 142  | 1.000 | $p_1 \in P$                                                                            | $p_1 \in P$                                                                            | definition of join, this holds iff $\mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \leq \Phi(p, q)$ is true for each $p_1 \in P$ .                                                            |
| fong-spivak-invitation_F01512 | 143  | 0.676 | $(=, \leq, \leq, =)$                                                                   | $(=, \leq, \leq, =)$                                                                   | 1. Justify each of the four steps $\equiv, <, <, \equiv$ in Eq. (4.9).                                                                                                                  |
| fong-spivak-invitation_F01513 | 143  | 0.915 | $\Phi: \mathcal{P} \rightarrow \mathcal{Q}, \Psi: \mathcal{Q} \rightarrow \mathcal{R}$ | $\Phi: \mathcal{P} \rightarrow \mathcal{Q}, \Psi: \mathcal{Q} \rightarrow \mathcal{R}$ | tors $\Phi: \mathcal{P} \rightarrow \mathcal{Q}, \Psi: \mathcal{Q} \rightarrow \mathcal{R}$ , and $\Upsilon: \mathcal{R} \rightarrow \mathcal{S}$ , we have                             |
| fong-spivak-invitation_F01514 | 143  | 0.915 | $\Upsilon: \mathcal{R} \rightarrow \mathcal{S}$                                        | $\Upsilon: \mathcal{R} \rightarrow \mathcal{S}$                                        | tors $\Phi: \mathcal{P} \rightarrow \mathcal{Q}, \Psi: \mathcal{Q} \rightarrow \mathcal{R}$ , and $\Upsilon: \mathcal{R} \rightarrow \mathcal{S}$ , we have                             |
| fong-spivak-invitation_F01515 | 144  | 0.744 | $\mathrm{C}$                                                                           | $C$                                                                                    | when defining the category $\mathbf{Cospan}_{\mathcal{C}}$ in Definition 6.39. An alternative approach is to                                                                            |
| fong-spivak-invitation_F01516 | 144  | 1.000 | $F: \mathcal{P} \rightarrow \mathcal{Q}$                                               | $F: \mathcal{P} \rightarrow \mathcal{Q}$                                               | <b>Definition 4.24.</b> Let $F: \mathcal{P} \rightarrow \mathcal{Q}$ be a $\mathcal{V}$ -functor. The <i>companion of <math>F</math></i> , denoted                                      |
| fong-spivak-invitation_F01517 | 144  | 0.592 | $\widehat{F}: \mathcal{P} \mapsto \mathcal{Q}$                                         | $\widehat{F}: \mathcal{P} \mapsto \mathcal{Q}$                                         | $\widehat{F}: \mathcal{P} \multimap \mathcal{Q}$ , and the <i>conjoint of <math>F</math></i> , denoted $\check{F}: \mathcal{Q} \multimap \mathcal{P}$ , are defined to be the following |
| fong-spivak-invitation_F01518 | 144  | 0.592 | $\check{F}: \mathcal{Q} \rightarrow \mathcal{P}$                                       | $\check{F}: \mathcal{Q} \rightarrow \mathcal{P}$                                       | $\widehat{F}: \mathcal{P} \multimap \mathcal{Q}$ , and the <i>conjoint of <math>F</math></i> , denoted $\check{F}: \mathcal{Q} \multimap \mathcal{P}$ , are defined to be the following |
| fong-spivak-invitation_F01519 | 144  | 1.000 | $F(p) \in Q$                                                                           | $F(p) \in Q$                                                                           | $F(p) \in Q$ .                                                                                                                                                                          |
| fong-spivak-invitation_F01520 | 144  | 1.000 | $\widehat{F}$                                                                          | $\widehat{F}$                                                                          | picture in that light, we see the companion $\widehat{F}$ . But now mentally reverse every dotted                                                                                       |
| fong-spivak-invitation_F01521 | 144  | 0.866 | $Q \mapsto \mathcal{P}$                                                                | $Q \mapsto \mathcal{P}$                                                                | arrow, and the result would be bridges $\mathcal{Q}$ to $\mathcal{P}$ . This is a profunctor $\mathcal{Q} \multimap \mathcal{P}$ ! We call it $\check{F}$ .                             |
| fong-spivak-invitation_F01522 | 144  | 0.866 | $\check{F}$                                                                            | $\check{F}$                                                                            | arrow, and the result would be bridges $\mathcal{Q}$ to $\mathcal{P}$ . This is a profunctor $\mathcal{Q} \multimap \mathcal{P}$ ! We call it $\check{F}$ .                             |
| fong-spivak-invitation_F01523 | 144  | 0.999 | $\mathcal{P} \rightarrow \mathcal{P}$                                                  | $\mathcal{P} \rightarrow \mathcal{P}$                                                  | <b>Example 4.25.</b> For any preorder $\mathcal{P}$ , there is an identity functor $\mathrm{id}: \mathcal{P} \rightarrow \mathcal{P}$ . Its com-                                        |
| fong-spivak-invitation_F01524 | 144  | 0.993 | $\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}: \mathcal{P} \mapsto \mathcal{P}$     | $\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}: \mathcal{P} \mapsto \mathcal{P}$     | panion and conjoint agree $\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}: \mathcal{P} \multimap \mathcal{P}$ . The resulting profunctor is in fact the unit                           |
| fong-spivak-invitation_F01525 | 145  | 0.951 | $\widehat{\mathrm{id}}$                                                                | $\widehat{\mathrm{id}}$                                                                | <b>Exercise 4.26.</b> Check that the companion $\widehat{\mathrm{id}}$ of $\mathrm{id}: \mathcal{P} \rightarrow \mathcal{P}$ really has the unit profunctor                             |
| fong-spivak-invitation_F01526 | 145  | 0.895 | $+: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$             | $+: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$             | <b>Example 4.27.</b> Consider the function $+: \mathbb{R} \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ , sending a triple $(a, b, c)$                                    |
| fong-spivak-invitation_F01527 | 145  | 0.969 | $a+b+c \in \mathbb{R}$                                                                 | $a+b+c \in \mathbb{R}$                                                                 | of real numbers to $a+b+c \in \mathbb{R}$ . This function is monotonic, because if $(a, b, c) \leq$                                                                                     |
| fong-spivak-invitation_F01528 | 145  | 0.969 | $(a, b, c) \leq$                                                                       | $(a, b, c) \leq$                                                                       | of real numbers to $a+b+c \in \mathbb{R}$ . This function is monotonic, because if $(a, b, c) \leq$                                                                                     |

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| fong-spivak-invitation_F01529 | 145  | ■ 0.982 | $a^{\wedge\{\prime\}}, b^{\wedge\{\prime\}}, c^{\wedge\{\prime\}}$                            | $a', b', c'$                                                                                  | $(a', b', c')$ – i.e. if $a \leq a', b \leq b'$ , and $c \leq c'$ – then obviously $a + b + c \leq a' + b' + c'$ .                                                             |
| fong-spivak-invitation_F01530 | 145  | ■ 0.982 | $a \leq a^{\wedge\{\prime\}}, b \leq b^{\wedge\{\prime\}}$                                    | $a \leq a', b \leq b'$                                                                        | $(a', b', c')$ – i.e. if $\boxed{a \leq a', b \leq b'}$ , and $c \leq c'$ – then obviously $a + b + c \leq a' + b' + c'$ .                                                     |
| fong-spivak-invitation_F01531 | 145  | ■ 0.982 | $c \leq c^{\wedge\{\prime\}}$                                                                 | $c \leq c'$                                                                                   | $(a', b', c')$ – i.e. if $a \leq a', b \leq b'$ , and $c \leq c'$ – then obviously $a + b + c \leq a' + b' + c'$ .                                                             |
| fong-spivak-invitation_F01532 | 145  | ■ 0.982 | $a+b+c \leq a^{\wedge\{\prime\}}+b^{\wedge\{\prime\}}+c^{\wedge\{\prime\}}$                   | $a + b + c \leq a' + b' + c'$                                                                 | $(a', b', c')$ – i.e. if $a \leq a', b \leq b'$ , and $c \leq c'$ – then obviously $\boxed{a + b + c \leq a' + b' + c'}$ .                                                     |
| fong-spivak-invitation_F01533 | 145  | ■ 0.513 | $\widehat{+} : (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \dashv \mathbb{R}$            | $\hat{+} : (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \dashv \mathbb{R}$                | Its companion $\boxed{\hat{+} : (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \rightarrow \mathbb{R}}$ is the function that sends $(a, b, c, d)$ to <code>true</code>       |
| fong-spivak-invitation_F01534 | 145  | ■ 0.513 | $a, b, c, d$                                                                                  | $a, b, c, d$                                                                                  | Its companion $\hat{+} : (\mathbb{R} \times \mathbb{R} \times \mathbb{R}) \rightarrow \mathbb{R}$ is the function that sends $(a, b, c, d)$ to <code>true</code>               |
| fong-spivak-invitation_F01535 | 145  | ■ 1.000 | $a+b+c \leq d$                                                                                | $a + b + c \leq d$                                                                            | if $a + b + c \leq d$ and to <code>false</code> otherwise.                                                                                                                     |
| fong-spivak-invitation_F01536 | 145  | ■ 0.472 | $\check{+}$                                                                                   | $\check{+}$                                                                                   | joint $\check{+}$ ?                                                                                                                                                            |
| fong-spivak-invitation_F01537 | 145  | ■ 0.964 | $\mathcal{Q}$                                                                                 | $\mathcal{Q}$                                                                                 | tion between preorders $\mathcal{P}$ and $\mathcal{Q}$ . The definition of adjoint can be extended from the                                                                    |
| fong-spivak-invitation_F01538 | 145  | ■ 0.706 | $G: \mathcal{Q} \rightarrow \mathcal{P}$                                                      | $G: \mathcal{Q} \rightarrow \mathcal{P}$                                                      | $\mathcal{V}$ -functors $F: \mathcal{P} \rightarrow \mathcal{Q}$ and $G: \mathcal{Q} \rightarrow \mathcal{P}$ such that the following holds for all $p \in \mathcal{P}$ and    |
| fong-spivak-invitation_F01539 | 145  | ■ 1.000 | $\widehat{F} = \widetilde{G}$                                                                 | $\widehat{F} = \widetilde{G}$                                                                 | the former equals the conjoint of the latter: $\widehat{F} = \widetilde{G}$ .                                                                                                  |
| fong-spivak-invitation_F01540 | 145  | ■ 1.000 | $\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}$                                             | $\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}$                                             | 2. Use this to prove that $\widehat{\mathrm{id}} = \widetilde{\mathrm{id}}$ , as was stated in Example 4.25.                                                                   |
| fong-spivak-invitation_F01541 | 145  | ■ 0.611 | $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$                                                   | $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$                                                   | <b>Definition 4.31.</b> Let $\mathcal{V}$ be a quantale, let $\mathcal{X}$ and $\mathcal{Y}$ be $\mathcal{V}$ -categories, and let $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$ |
| fong-spivak-invitation_F01542 | 145  | ■ 0.969 | $\varnothing := \bigvee \varnothing$                                                          | $\varnothing := \bigvee \varnothing$                                                          | be a $\mathcal{V}$ -profunctor. Recall the notation $\varnothing := \bigvee \varnothing$ . The <i>collage of <math>\Phi</math></i> , denoted $\mathbf{Col}(\Phi)$ is           |
| fong-spivak-invitation_F01543 | 145  | ■ 0.969 | $\mathbf{Col}(\Phi)$                                                                          | $\mathbf{Col}(\Phi)$                                                                          | be a $\mathcal{V}$ -profunctor. Recall the notation $\varnothing := \bigvee \varnothing$ . The <i>collage of <math>\Phi</math></i> , denoted $\mathbf{Col}(\Phi)$ is           |
| fong-spivak-invitation_F01544 | 145  | ■ 0.778 | $\mathrm{Ob}(\mathbf{Col}(\Phi)) := \mathrm{Ob}(\mathcal{X}) \sqcup \mathrm{Ob}(\mathcal{Y})$ | $\mathrm{Ob}(\mathbf{Col}(\Phi)) := \mathrm{Ob}(\mathcal{X}) \sqcup \mathrm{Ob}(\mathcal{Y})$ | (i) $\boxed{\mathrm{Ob}(\mathbf{Col}(\Phi)) := \mathrm{Ob}(\mathcal{X}) \sqcup \mathrm{Ob}(\mathcal{Y})}$ .                                                                    |
| fong-spivak-invitation_F01545 | 145  | ■ 0.964 | $a, b \in \mathrm{Ob}(\mathbf{Col}(\Phi))$                                                    | $a, b \in \mathrm{Ob}(\mathbf{Col}(\Phi))$                                                    | (ii) For any $\boxed{a, b \in \mathrm{Ob}(\mathbf{Col}(\Phi))}$ , define $\mathbf{Col}(\Phi)(a, b) \in \mathcal{V}$ to be                                                      |
| fong-spivak-invitation_F01546 | 145  | ■ 0.964 | $\mathbf{Col}(\Phi)(a, b) \in \mathcal{V}$                                                    | $\mathbf{Col}(\Phi)(a, b) \in \mathcal{V}$                                                    | (ii) For any $a, b \in \mathrm{Ob}(\mathbf{Col}(\Phi))$ , define $\boxed{\mathbf{Col}(\Phi)(a, b) \in \mathcal{V}}$ to be                                                      |
| fong-spivak-invitation_F01547 | 146  | ■ 0.660 | $i_x: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$                                             | $i_x: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$                                             | There are obvious functors $i_{\mathcal{X}}: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$ and $i_{\mathcal{Y}}: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$ , sending each      |
| fong-spivak-invitation_F01548 | 146  | ■ 0.660 | $i_y: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$                                             | $i_y: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$                                             | There are obvious functors $i_{\mathcal{X}}: \mathcal{X} \rightarrow \mathbf{Col}(\Phi)$ and $i_{\mathcal{Y}}: \mathcal{Y} \rightarrow \mathbf{Col}(\Phi)$ , sending each      |
| fong-spivak-invitation_F01549 | 146  | ■ 0.311 | $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$                                                   | $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$                                                   | <b>Example 4.32.</b> Consider the following picture of a <b>Cost</b> -profunctor $\Phi: \mathcal{X} \rightarrow \mathcal{Y}$ :                                                 |
| fong-spivak-invitation_F01551 | 146  | ■ 0.999 | $\mathbf{Col}(\Phi)$                                                                          | $\mathbf{Col}(\Phi)$                                                                          | profunctor $\Phi$ , we draw the Hasse diagram for $\mathbf{Col}(\Phi)$ below left, and the <b>Cost</b> -matrix                                                                 |
| fong-spivak-invitation_F01553 | 147  | ■ 1.000 | $5+3=8$                                                                                       | $5 + 3 = 8$                                                                                   | that $5 + 3 = 8$ . One way to categorify $\mathbb{N}$ is to use the category <b>FinSet</b> of finite                                                                           |
| fong-spivak-invitation_F01554 | 147  | ■ 0.840 | $\overline{5} =$                                                                              | $\overline{5} =$                                                                              | sets of that many elements, say $\overline{5} = \{\text{apple, banana, cherry, dragonfruit, elephant}\}$ ,                                                                     |
| fong-spivak-invitation_F01555 | 147  | ■ 0.539 | $\overline{3} =$                                                                              | $\overline{3} =$                                                                              | $\overline{3} = \{\text{apple, tomato, cantaloupe}\}$ , and $\overline{8} = \{\text{Ali, Bob, Carl, Deb, Eli, Fritz, Gem, Helen}\}$                                            |
| fong-spivak-invitation_F01556 | 147  | ■ 0.539 | $\overline{8} = \{$                                                                           | $\overline{8} = \{$                                                                           | $\overline{3} = \{\text{apple, tomato, cantaloupe}\}$ , and $\overline{8} = \{\text{Ali, Bob, Carl, Deb, Eli, Fritz, Gem, Helen}\}$                                            |

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| fong-spivak-invitation_F01557 | 147  | ■ 1.000 | $\overline{5} \sqcup \overline{3} \cong \overline{8}$             | $\overline{5} \sqcup \overline{3} \cong \overline{8}$             | $\overline{5} \sqcup \overline{3} \cong \overline{8}$ .                                                                                  |
| fong-spivak-invitation_F01558 | 147  | ■ 1.000 | $\overline{5} \sqcup \overline{3}$                                | $\overline{5} \sqcup \overline{3}$                                | Thus it's not the case that $\overline{5} \sqcup \overline{3}$ is <i>equal</i> to our chosen eight-element set $\overline{8}$ , but more |
| fong-spivak-invitation_F01559 | 147  | ■ 1.000 | $\overline{8}$                                                    | $\overline{8}$                                                    | Thus it's not the case that $\overline{5} \sqcup \overline{3}$ is <i>equal</i> to our chosen eight-element set $\overline{8}$ , but more |
| fong-spivak-invitation_F01560 | 148  | ■ 0.964 | $\mathcal{P}(a, b) =$                                             | $\mathcal{P}(a, b) =$                                             | a morphism $a \rightarrow b$ ,” denoted $a \leq b$ or $\mathcal{P}(a, b) = \text{true}$ , we instead say “here is a set                  |
| fong-spivak-invitation_F01561 | 148  | ■ 1.000 | $x_{\{0\}} \leq x_{\{1\}}$                                        | $x_0 \leq x_1$                                                    | whenever $x_0 \leq x_1$ . Chaining these together, we could prove facts in our preorder. For                                             |
| fong-spivak-invitation_F01562 | 148  | ■ 1.000 | $x_{\{0\}} \leq x_{\{3\}}$                                        | $x_0 \leq x_3$                                                    | provides a proof that $x_0 \leq x_3$ (the exterior box) using three facts (the interior boxes),                                          |
| fong-spivak-invitation_F01563 | 148  | ■ 1.000 | $x_{\{0\}} \leq x_{\{1\}}, x_{\{1\}} \leq x_{\{2\}}$              | $x_0 \leq x_1, x_1 \leq x_2$                                      | $x_0 \leq x_1, x_1 \leq x_2$ , and $x_2 \leq x_3$ .                                                                                      |
| fong-spivak-invitation_F01564 | 148  | ■ 1.000 | $x_{\{2\}} \leq x_{\{3\}}$                                        | $x_2 \leq x_3$                                                    | $x_0 \leq x_1, x_1 \leq x_2$ , and $x_2 \leq x_3$ .                                                                                      |
| fong-spivak-invitation_F01565 | 148  | ■ 1.000 | $g: B \rightarrow C$                                              | $g: B \rightarrow C$                                              | Suppose we are given additional morphisms $g: B \rightarrow C$ , and $h: C \rightarrow D$ . Representing                                 |
| fong-spivak-invitation_F01566 | 148  | ■ 1.000 | $h: C \rightarrow D$                                              | $h: C \rightarrow D$                                              | Suppose we are given additional morphisms $g: B \rightarrow C$ , and $h: C \rightarrow D$ . Representing                                 |
| fong-spivak-invitation_F01567 | 148  | ■ 0.305 | $f; (g; h)$                                                       | $f; (g; h)$                                                       | We don't know which morphism it is. Is it $f \circ (g \circ h)$ ? Or $(f \circ g) \circ h$ ? It's good that                              |
| fong-spivak-invitation_F01568 | 148  | ■ 0.305 | $(f; g) \circ h$                                                  | $(f; g) \circ h$                                                  | We don't know which morphism it is. Is it $f \circ (g \circ h)$ ? Or $(f \circ g) \circ h$ ? It's good that                              |
| fong-spivak-invitation_F01569 | 149  | ■ 0.440 | $f; (g; h) = (f; g); h$                                           | $f; (g; h) = (f; g); h$                                           | Associativity says $f \circ (g \circ h) = (f \circ g) \circ h$ , so it doesn't matter which way we chose to                              |
| fong-spivak-invitation_F01570 | 149  | ■ 0.994 | $\mathrm{id}_x$                                                   | $\mathrm{id}_x$                                                   | shall see that it is not harmful to draw $\mathrm{id}_x$ in any of the following three ways:                                             |
| fong-spivak-invitation_F01571 | 149  | ■ 0.486 | $\mathrm{id}_x \circ f = f = f \circ \mathrm{id}_y$               | $\mathrm{id}_x \circ f = f = f \circ \mathrm{id}_y$               | property and the associativity property. The unitality says that $\mathrm{id}_x \circ f = f = f \circ \mathrm{id}_y$ for                 |
| fong-spivak-invitation_F01572 | 149  | ■ 1.000 | $f: x \rightarrow y$                                              | $f: x \rightarrow y$                                              | any $f: x \rightarrow y$ . In terms of diagrams this would say                                                                           |
| fong-spivak-invitation_F01573 | 150  | ■ 1.000 | $\{1\}, \times$                                                   | $\{1\}, \times$                                                   | We will soon meet the monoidal category $(\mathbf{Set}, \{1\}, \times)$ . The monoidal product will                                      |
| fong-spivak-invitation_F01574 | 150  | ■ 1.000 | $S \times T = \{(s, t) \mid s \in S, t \in T\}$                   | $S \times T = \{(s, t) \mid s \in S, t \in T\}$                   | take two sets, $S$ and $T$ , and return the set $S \times T = \{(s, t) \mid s \in S, t \in T\}$ . But whereas                            |
| fong-spivak-invitation_F01575 | 150  | ■ 0.997 | $(p \otimes q) \otimes r = p \otimes (q \otimes r)$               | $(p \otimes q) \otimes r = p \otimes (q \otimes r)$               | for monoidal preorders we had the brute associative property $(p \otimes q) \otimes r = p \otimes (q \otimes r)$ ,                       |
| fong-spivak-invitation_F01576 | 150  | ■ 1.000 | $T \times U$                                                      | $T \times U$                                                      | and an element in $T \times U$ , while the second contains pairs consisting of an element in                                             |
| fong-spivak-invitation_F01577 | 150  | ■ 1.000 | $S \times T$                                                      | $S \times T$                                                      | $S \times T$ and an element in $U$ . The sets are not equal, but they are clearly isomorphic, i.e.                                       |
| fong-spivak-invitation_F01578 | 150  | ■ 1.000 | $\alpha_{s, t, u}$                                                | $\alpha_{s, t, u}$                                                | example, above we replaced the associativity equation with an isomorphism $\alpha_{s, t, u}$ , but                                       |
| fong-spivak-invitation_F01579 | 150  | ■ 0.913 | $\alpha \mathrm{mathrm{s}}$                                       | $\alpha \mathrm{s}$                                               | now we need a more complex property to ensure that all these $\alpha$ s behave reasonably!                                               |
| fong-spivak-invitation_F01580 | 150  | ■ 0.907 | $I \in \mathrm{Ob}(\mathcal{C})$                                  | $I \in \mathrm{Ob}(\mathcal{C})$                                  | (i) an object $I \in \mathrm{Ob}(\mathcal{C})$ called the <i>monoidal unit</i> ,                                                         |
| fong-spivak-invitation_F01581 | 150  | ■ 1.000 | $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ | $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ | (ii) a functor $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ , called the <i>monoidal product</i>                    |
| fong-spivak-invitation_F01582 | 151  | ■ 0.863 | $\lambda_c: I \otimes c \cong c$                                  | $\lambda_c: I \otimes c \cong c$                                  | (a) $\lambda_c: I \otimes c \cong c$ for every $c \in \mathrm{Ob}(\mathcal{C})$ ,                                                        |
| fong-spivak-invitation_F01583 | 151  | ■ 0.997 | $\rho_c: c \otimes I \cong c$                                     | $\rho_c: c \otimes I \cong c$                                     | (b) $\rho_c: c \otimes I \cong c$ for every $c \in \mathrm{Ob}(\mathcal{C})$ ,                                                           |

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| fong-spivak-invitation_F01584 | 151  | ■ 0.984 | $\alpha_{c,d,e}:(c \otimes d) \otimes e \cong c \otimes (d \otimes e)$ | $\alpha_{c,d,e}:(c \otimes d) \otimes e \cong c \otimes (d \otimes e)$ | (c) $\alpha_{c,d,e}:(c \otimes d) \otimes e \cong c \otimes (d \otimes e)$ for every $c, d, e \in \text{Ob}(\mathcal{C})$ ,            |
| fong-spivak-invitation_F01585 | 151  | ■ 0.999 | $\sigma_{c,d}: c \otimes d \cong d \otimes c$                          | $\sigma_{c,d}: c \otimes d \cong d \otimes c$                          | (d) $\sigma_{c,d}: c \otimes d \cong d \otimes c$ for every $c, d \in \text{Ob}(\mathcal{C})$ , called the <i>swap map</i> , such that |
| fong-spivak-invitation_F01586 | 151  | ■ 0.999 | $c, d \in \text{Ob}(\mathcal{C})$                                      | $c, d \in \text{Ob}(\mathcal{C})$                                      | (d) $\sigma_{c,d}: c \otimes d \cong d \otimes c$ for every $c, d \in \text{Ob}(\mathcal{C})$ , called the <i>swap map</i> , such that |
| fong-spivak-invitation_F01587 | 151  | ■ 1.000 | $\sigma \circ \sigma = \text{id}$                                      | $\sigma \circ \sigma = \text{id}$                                      | $\sigma \circ \sigma = \text{id}$ .                                                                                                    |
| fong-spivak-invitation_F01588 | 151  | ■ 0.998 | $(\mathcal{P}, I, \otimes)$                                            | $(\mathcal{P}, I, \otimes)$                                            | a monoidal preorder is a monoidal category $(\mathcal{P}, I, \otimes)$ where, for every $p, q \in \mathcal{P}$ , the set               |
| fong-spivak-invitation_F01589 | 151  | ■ 0.998 | $p, q \in \mathcal{P}$                                                 | $p, q \in \mathcal{P}$                                                 | a monoidal preorder is a monoidal category $(\mathcal{P}, I, \otimes)$ where, for every $p, q \in \mathcal{P}$ , the set               |
| fong-spivak-invitation_F01590 | 151  | ■ 1.000 | $\mathcal{P}(p, q)$                                                    | $\mathcal{P}(p, q)$                                                    | $\mathcal{P}(p, q)$ has at most one element.                                                                                           |
| fong-spivak-invitation_F01591 | 151  | ■ 1.000 | $I := \{1\}$                                                           | $I := \{1\}$                                                           | monoidal unit is some choice of singleton set, say $I := \{1\}$ , and the monoidal product                                             |
| fong-spivak-invitation_F01592 | 151  | ■ 0.990 | $\otimes := \times$                                                    | $\otimes := \times$                                                    | is $\otimes := \times$ . What it means that $\times$ is a functor is that:                                                             |
| fong-spivak-invitation_F01593 | 151  | ■ 0.600 | $(S, T) \in \text{Ob}(\mathcal{C})$                                    | $(S, T) \in \text{Ob}(\mathcal{C})$                                    | • For any pair of objects, i.e. sets $(S, T) \in \text{Ob}(\mathbf{Set} \times \mathbf{Set})$ , one obtains a set $(S \times T) \in$   |
| fong-spivak-invitation_F01594 | 151  | ■ 0.600 | $)$                                                                    | $)$                                                                    | • For any pair of objects, i.e. sets $(S, T) \in \text{Ob}(\mathbf{Set} \times \mathbf{Set})$ , one obtains a set $(S \times T) \in$   |
| fong-spivak-invitation_F01595 | 151  | ■ 0.600 | $(S \times T) \in$                                                     | $(S \times T) \in$                                                     | • For any pair of objects, i.e. sets $(S, T) \in \text{Ob}(\mathbf{Set} \times \mathbf{Set})$ , one obtains a set $(S \times T) \in$   |
| fong-spivak-invitation_F01596 | 151  | ■ 0.992 | $\{(s, t) \mid s \in S, t \in T\}$                                     | $\{(s, t) \mid s \in S, t \in T\}$                                     | $\text{Ob}(\mathbf{Set})$ . We know what it is: the set of pairs $\{(s, t) \mid s \in S, t \in T\}$ .                                  |
| fong-spivak-invitation_F01597 | 151  | ■ 1.000 | $f: S \rightarrow S'$                                                  | $f: S \rightarrow S'$                                                  | • For any pair of morphisms, i.e. functions $f: S \rightarrow S'$ and $g: T \rightarrow T'$ , one obtains                              |
| fong-spivak-invitation_F01598 | 151  | ■ 1.000 | $g: T \rightarrow T'$                                                  | $g: T \rightarrow T'$                                                  | • For any pair of morphisms, i.e. functions $f: S \rightarrow S'$ and $g: T \rightarrow T'$ , one obtains                              |
| fong-spivak-invitation_F01599 | 151  | ■ 1.000 | $(f \times g): (S \times T) \rightarrow (S' \times T')$                | $(f \times g): (S \times T) \rightarrow (S' \times T')$                | a function $(f \times g): (S \times T) \rightarrow (S' \times T')$ . It works pointwise: $(f \times g)(s, t) :=$                       |
| fong-spivak-invitation_F01600 | 151  | ■ 1.000 | $(f \times g)(s, t) :=$                                                | $(f \times g)(s, t) :=$                                                | a function $(f \times g): (S \times T) \rightarrow (S' \times T')$ . It works pointwise: $(f \times g)(s, t) :=$                       |
| fong-spivak-invitation_F01601 | 151  | ■ 0.991 | $(f(s), g(t))$                                                         | $(f(s), g(t))$                                                         | $(f(s), g(t))$ .                                                                                                                       |
| fong-spivak-invitation_F01602 | 151  | ■ 0.975 | $\text{id}_S \times \text{id}_T = \text{id}_{S \times T}$              | $\text{id}_S \times \text{id}_T = \text{id}_{S \times T}$              | • These should preserve identities: $\text{id}_S \times \text{id}_T = \text{id}_{S \times T}$ for any sets $S, T$ .                    |
| fong-spivak-invitation_F01603 | 152  | ■ 1.000 | $S \xrightarrow{f} S' \xrightarrow{f'} S''$                            | $S \xrightarrow{f} S' \xrightarrow{f'} S''$                            | • These should preserve composition: for any functions $S \xrightarrow{f} S' \xrightarrow{f'} S''$ and $T \xrightarrow{g}$             |
| fong-spivak-invitation_F01604 | 152  | ■ 1.000 | $T \xrightarrow{g}$                                                    | $T \xrightarrow{g}$                                                    | • These should preserve composition: for any functions $S \xrightarrow{f} S' \xrightarrow{f'} S''$ and $T \xrightarrow{g}$             |
| fong-spivak-invitation_F01605 | 152  | ■ 1.000 | $T' \xrightarrow{g'} T''$                                              | $T' \xrightarrow{g'} T''$                                              | $T' \xrightarrow{g'} T''$ , one has                                                                                                    |
| fong-spivak-invitation_F01606 | 152  | ■ 1.000 | $\{1\} \times S \cong S$                                               | $\{1\} \times S \cong S$                                               | The four conditions, (a), (b), (c), and (d) give isomorphisms $\{1\} \times S \cong S$ , etc. These                                    |
| fong-spivak-invitation_F01607 | 152  | ■ 1.000 | $\{(1, s) \mid s \in S\} \rightarrow S$                                | $\{(1, s) \mid s \in S\} \rightarrow S$                                | maps are obvious in the case of $\mathbf{Set}$ , e.g. the function $\{(1, s) \mid s \in S\} \rightarrow S$ sending                     |
| fong-spivak-invitation_F01608 | 152  | ■ 1.000 | $(1, s)$                                                               | $(1, s)$                                                               | $(1, s)$ to $s$ . We have been calling such things bookkeeping.                                                                        |
| fong-spivak-invitation_F01609 | 152  | ■ 1.000 | $1, \times$                                                            | $1, \times$                                                            | <b>Exercise 4.39.</b> Consider the monoidal category $(\mathbf{Set}, 1, \times)$ , together with the diagram                           |
| fong-spivak-invitation_F01610 | 152  | ■ 0.740 | $A=B=C=D=F=G=\mathbb{Z}$                                               | $A = B = C = D = F = G = \mathbb{Z}$                                   | Suppose that $A = B = C = D = F = G = \mathbb{Z}$ and $E = \mathbb{B} = \{\text{true}, \text{false}\}$ , and                           |

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| fong-spivak-invitation_F01611 | 152  | 0.740 | $E=\mathbb{B}=\$                                                                                    | $E = \mathbb{B} =$                                                                 | Suppose that $A = B = C = D = F = G = \mathbb{Z}$ and $E = \mathbb{B} = \{\text{true}, \text{false}\}$ , and                                               |
| fong-spivak-invitation_F01612 | 152  | 0.766 | $f_{\{C\}}(a)= a $ , $f_{\{D\}}(a)=a * 5$ , $g_{\{E\}}(d, b)="\ d \leq b$ , " $g_{\{F\}}(d, b)=d-b$ | $f_C(a) =  a $ , $f_D(a) = a * 5$ , $g_E(d, b) = "d \leq b,$ " $g_F(d, b) = d - b$ | suppose that $f_C(a) =  a $ , $f_D(a) = a * 5$ , $g_F(d, b) = "d < b,"$ $g_F(d, b) = d - b$ , and                                                          |
| fong-spivak-invitation_F01613 | 152  | 0.999 | $h(c, e)=$                                                                                          | $h(c, e) =$                                                                        | $h(c, e) \equiv$ if $e$ then $c$ else $1 - c$ .                                                                                                            |
| fong-spivak-invitation_F01614 | 152  | 0.999 | $1-c$                                                                                               | $1 - c$                                                                            | $h(c, e) =$ if $e$ then $c$ else $1 - c$ .                                                                                                                 |
| fong-spivak-invitation_F01615 | 152  | 1.000 | $g_{\{E\}}(5,3)$                                                                                    | $g_E(5,3)$                                                                         | 1. What are $g_E(5, 3)$ and $g_F(5, 3)$ ?                                                                                                                  |
| fong-spivak-invitation_F01616 | 152  | 1.000 | $g_{\{F\}}(5,3)$                                                                                    | $g_F(5,3)$                                                                         | 1. What are $g_E(5, 3)$ and $g_F(5, 3)$ ?                                                                                                                  |
| fong-spivak-invitation_F01617 | 152  | 1.000 | $g_{\{E\}}(3,5)$                                                                                    | $g_E(3,5)$                                                                         | 2. What are $g_E(3, 5)$ and $g_F(3, 5)$ ?                                                                                                                  |
| fong-spivak-invitation_F01618 | 152  | 1.000 | $g_{\{F\}}(3,5)$                                                                                    | $g_F(3,5)$                                                                         | 2. What are $g_E(3, 5)$ and $g_F(3, 5)$ ?                                                                                                                  |
| fong-spivak-invitation_F01619 | 152  | 0.997 | $h(5$                                                                                               | $h(5$                                                                              | 3. What is $h(5, \text{true})$ ?                                                                                                                           |
| fong-spivak-invitation_F01620 | 152  | 0.930 | $h(-5$                                                                                              | $h(-5$                                                                             | 4. What is $h(-5, \text{true})$ ?                                                                                                                          |
| fong-spivak-invitation_F01621 | 152  | 1.000 | $A \times B \rightarrowtail G \times F$                                                             | $A \times B \rightarrow G \times F$                                                | The whole diagram now defines a function $A \times B \rightarrowtail G \times F$ ; call it $q$ .                                                           |
| fong-spivak-invitation_F01622 | 152  | 1.000 | $q_{\{G\}}(-2,3)$                                                                                   | $q_G(-2,3)$                                                                        | 6. What are $q_G(-2, 3)$ and $q_F(-2, 3)$ ?                                                                                                                |
| fong-spivak-invitation_F01623 | 152  | 1.000 | $q_{\{F\}}(-2,3)$                                                                                   | $q_F(-2,3)$                                                                        | 6. What are $q_G(-2, 3)$ and $q_F(-2, 3)$ ?                                                                                                                |
| fong-spivak-invitation_F01624 | 152  | 1.000 | $q_{\{G\}}(2,3)$                                                                                    | $q_G(2,3)$                                                                         | 7. What are $q_G(2, 3)$ and $q_F(2, 3)$ ?                                                                                                                  |
| fong-spivak-invitation_F01625 | 152  | 1.000 | $q_{\{F\}}(2,3)$                                                                                    | $q_F(2,3)$                                                                         | 7. What are $q_G(2, 3)$ and $q_F(2, 3)$ ?                                                                                                                  |
| fong-spivak-invitation_F01626 | 153  | 0.298 | $x, y \in \mathrm{Ob}(\mathcal{X})$                                                                 | $x, y \in \mathrm{Ob}(\mathcal{X})$                                                | (ii) for every pair $x, y \in \mathrm{Ob}(\mathcal{X})$ , one specifies an object $\mathcal{X}(x, y) \in \mathcal{V}$ , called the <i>hom-</i>             |
| fong-spivak-invitation_F01627 | 153  | 0.420 | $\mathrm{id}_x: I \rightarrowtail \mathcal{X}(x, x)$                                                | $\mathrm{id}_x : I \rightarrow \mathcal{X}(x, x)$                                  | (iii) for every $x \in \mathrm{Ob}(\mathcal{X})$ , one specifies a morphism $\mathrm{id}_x: I \rightarrow \mathcal{X}(x, x)$ in $\mathcal{V}$ , called the |
| fong-spivak-invitation_F01628 | 153  | 0.596 | $x, y, z \in \mathrm{operatorname{Ob}}(\mathcal{X})$                                                | $x, y, z \in \mathrm{Ob}(\mathcal{X})$                                             | (iv) for each $x, y, z \in \mathrm{Ob}(\mathcal{X})$ , one specifies a morphism $\mathbin{\circ}: \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \rightarrow$ |
| fong-spivak-invitation_F01629 | 153  | 0.596 | $\{ \}_{9}: \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \rightarrowtail$                            | $\mathbin{\circ}: \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \rightarrow$         | (iv) for each $x, y, z \in \mathrm{Ob}(\mathcal{X})$ , one specifies a morphism $\mathbin{\circ}: \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \rightarrow$ |
| fong-spivak-invitation_F01630 | 153  | 0.809 | $\mathcal{X}(x, z)$                                                                                 | $\mathcal{X}(x, z)$                                                                | $\mathcal{X}(x, z)$ , called the <i>composition morphism</i> .                                                                                             |
| fong-spivak-invitation_F01631 | 153  | 0.922 | $\mathcal{V}=($                                                                                     | $\mathcal{V} = ($                                                                  | <b>Exercise 4.41.</b> Recall from Example 4.38 that $\mathcal{V} = (\mathbf{Set}, \{1\}, \times)$ is a symmetric                                           |
| fong-spivak-invitation_F01632 | 153  | 0.922 | $, \{1\}, \times)$                                                                                  | $, \{1\}, \times)$                                                                 | <b>Exercise 4.41.</b> Recall from Example 4.38 that $\mathcal{V} = (\mathbf{Set}, \{1\}, \times)$ is a symmetric                                           |
| fong-spivak-invitation_F01633 | 153  | 0.496 | $\mathrm{operatorname{id}}_x: I \rightarrowtail \mathcal{X}(x, x)$                                  | $\mathrm{id}_x : I \rightarrow \mathcal{X}(x, x)$                                  | $\mathrm{id}_x: I \rightarrow \mathcal{X}(x, x)$ for the object $x$ simply becomes the requirement (a) that $I \leq \mathcal{X}(x, x)$                     |
| fong-spivak-invitation_F01634 | 155  | 0.650 | $(\mathcal{C}, I, \otimes)$                                                                         | $(\mathcal{C}, I, \otimes)$                                                        | <b>Definition 4.44.</b> Let $(\mathcal{C}, I, \otimes)$ be a symmetric monoidal category, and $c \in \mathrm{Ob}(\mathcal{C})$ an                          |
| fong-spivak-invitation_F01635 | 155  | 0.999 | $c^* \in \mathrm{operatorname{Ob}}(\mathcal{C})$                                                    | $c^* \in \mathrm{Ob}(\mathcal{C})$                                                 | (i) an object $c^* \in \mathrm{Ob}(\mathcal{C})$ , called the <i>dual of c</i> ,                                                                           |
| fong-spivak-invitation_F01636 | 155  | 1.000 | $\eta_c: I \rightarrowtail c^* \otimes c$                                                           | $\eta_c : I \rightarrow c^* \otimes c$                                             | (ii) a morphism $\eta_c: I \rightarrowtail c^* \otimes c$ , called the <i>unit for c</i> ,                                                                 |
| fong-spivak-invitation_F01637 | 155  | 1.000 | $\epsilon_c: c \otimes c^* \rightarrowtail I$                                                       | $\epsilon_c : c \otimes c^* \rightarrow I$                                         | (iii) a morphism $\epsilon_c: c \otimes c^* \rightarrowtail I$ , called the <i>counit for c</i> .                                                          |

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| fong-spivak-invitation_F01638 | 155  | 0.881 | $c^{\{*\}}$                                                       | $c^*$                                                             | If for every object $c \in \text{Ob}(\mathcal{C})$ there exists a dual $c^*$ for $c$ , then we say that $(\mathcal{C}, I, \otimes)$                               |
| fong-spivak-invitation_F01639 | 155  | 1.000 | $\xrightarrow{c}$                                                 | $\xrightarrow{c}$                                                 | labeled $c^*$ , i.e. $\xrightarrow{c}$ is the same as $\xleftarrow{c^*}$ . The cup and cap discussed above are in fact the                                        |
| fong-spivak-invitation_F01640 | 155  | 1.000 | $\xleftarrow{c^{\{*\}}}$                                          | $\xleftarrow{c^*}$                                                | labeled $c^*$ , i.e. $\xrightarrow{c}$ is the same as $\xleftarrow{c^*}$ . The cup and cap discussed above are in fact the                                        |
| fong-spivak-invitation_F01641 | 156  | 0.844 | $\epsilon_{\text{sound}}$                                         | $\epsilon_{\text{sound}}$                                         | Note that the pictures in Eq. (4.14) correspond to $\epsilon_{\text{sound}}$ and $\eta_{\text{sound}^*}$ .                                                        |
| fong-spivak-invitation_F01642 | 156  | 0.844 | $\eta_{\text{sound}^*}$                                           | $\eta_{\text{sound}^*}$                                           | Note that the pictures in Eq. (4.14) correspond to $\epsilon_{\text{sound}}$ and $\eta_{\text{sound}^*}$ .                                                        |
| fong-spivak-invitation_F01643 | 156  | 0.738 | $c \multimap d$                                                   | $c \multimap d$                                                   | object $c \multimap d$ and an isomorphism $\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$ , natural in $b$ . While we                           |
| fong-spivak-invitation_F01644 | 156  | 0.738 | $\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$ | $\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$ | object $c \multimap d$ and an isomorphism $\mathcal{C}(b \otimes c, d) \cong \mathcal{C}(b, c \multimap d)$ , natural in $b$ . While we                           |
| fong-spivak-invitation_F01645 | 156  | 1.000 | $c^{\{\prime\}}$                                                  | $c'$                                                              | 2. if $c^*$ and $c'$ are both duals to $c$ then there is an isomorphism $c^* \cong c'$ ; and                                                                      |
| fong-spivak-invitation_F01646 | 156  | 1.000 | $c^{\{*\}} \cong c^{\{\prime\}}$                                  | $c^* \cong c'$                                                    | 2. if $c^*$ and $c'$ are both duals to $c$ then there is an isomorphism $c^* \cong c'$ ; and                                                                      |
| fong-spivak-invitation_F01647 | 156  | 1.000 | $c \cong c^{**}$                                                  | $c \cong c^{**}$                                                  | 3. there is an isomorphism between $c$ and its double-dual, $c \cong c^{**}$ .                                                                                    |
| fong-spivak-invitation_F01648 | 156  | 0.668 | $c^{\{*\}} \otimes d$                                             | $c^* \otimes d$                                                   | To prove part 1, the key idea is that, for any $c$ and $d$ , the object $c \multimap d$ is given by $c^* \otimes d$ ,                                             |
| fong-spivak-invitation_F01649 | 156  | 0.959 | $\text{id}_b \otimes \eta_c$                                      | $\text{id}_b \otimes \eta_c$                                      | $\text{id}_b \otimes \eta_c$ .                                                                                                                                    |
| fong-spivak-invitation_F01650 | 157  | 1.000 | $\alpha: A \rightarrow B$                                         | $\alpha: A \rightarrow B$                                         | look at than to write down formally. <sup>2</sup> If in addition to the corelation $\alpha: A \rightarrow B$ above                                                |
| fong-spivak-invitation_F01651 | 157  | 1.000 | $\beta: B \rightarrow C$                                          | $\beta: B \rightarrow C$                                          | we have another corelation $\beta: B \rightarrow C$                                                                                                               |
| fong-spivak-invitation_F01652 | 157  | 0.999 | $\beta \circ \alpha$                                              | $\beta \circ \alpha$                                              | then the composite $\beta \circ \alpha$ of our two corelations is given by                                                                                        |
| fong-spivak-invitation_F01653 | 157  | 1.000 | $\beta$                                                           | $\beta$                                                           | one to the other staying within equivalence classes of either $\alpha$ or $\beta$ .                                                                               |
| fong-spivak-invitation_F01654 | 157  | 1.000 | $A \sqcup A :=$                                                   | $A \sqcup A :=$                                                   | Indeed, note that for any finite set $A$ there is an equivalence relation on $A \sqcup A :=$                                                                      |
| fong-spivak-invitation_F01655 | 157  | 0.998 | $\{(a, 1), (a, 2) \mid a \in A\}$                                 | $\{(a, 1), (a, 2) \mid a \in A\}$                                 | $\{(a, 1), (a, 2) \mid a \in A\}$ where each part simply consists of the two elements $(a, 1)$ and                                                                |
| fong-spivak-invitation_F01656 | 157  | 0.998 | $(a, 1)$                                                          | $(a, 1)$                                                          | $\{(a, 1), (a, 2) \mid a \in A\}$ where each part simply consists of the two elements $(a, 1)$ and                                                                |
| fong-spivak-invitation_F01657 | 157  | 1.000 | $(a, 2)$                                                          | $(a, 2)$                                                          | $(a, 2)$ for each $a \in A$ . The unit on a finite set $A$ is the corelation $\eta_A: \emptyset \rightarrow A \sqcup A$                                           |
| fong-spivak-invitation_F01658 | 157  | 1.000 | $\eta_A: \emptyset \rightarrow A \sqcup A$                        | $\eta_A: \emptyset \rightarrow A \sqcup A$                        | $(a, 2)$ for each $a \in A$ . The unit on a finite set $A$ is the corelation $\eta_A: \emptyset \rightarrow A \sqcup A$                                           |
| fong-spivak-invitation_F01659 | 157  | 1.000 | $\epsilon_A: A \sqcup A \rightarrow \emptyset$                    | $\epsilon_A: A \sqcup A \rightarrow \emptyset$                    | $\epsilon_A: A \sqcup A \rightarrow \emptyset$ specified by this same equivalence relation.                                                                       |
| fong-spivak-invitation_F01660 | 157  | 0.992 | $\emptyset \rightarrow \underline{3} \sqcup \underline{3}$        | $\emptyset \rightarrow \underline{3} \sqcup \underline{3}$        | 1. Draw a picture of the unit corelation $\emptyset \rightarrow \underline{3} \sqcup \underline{3}$ .                                                             |
| fong-spivak-invitation_F01661 | 157  | 0.989 | $\underline{3} \sqcup \underline{3} \rightarrow \emptyset$        | $\underline{3} \sqcup \underline{3} \rightarrow \emptyset$        | 2. Draw a picture of the counit corelation $\underline{3} \sqcup \underline{3} \rightarrow \emptyset$ .                                                           |
| fong-spivak-invitation_F01662 | 157  | 0.907 | $\alpha; \beta$                                                   | $\alpha; \beta$                                                   | <sup>2</sup> To compose corelations $\alpha: A \rightarrow B$ and $\beta: B \rightarrow C$ , we need to construct an equivalence relation $\alpha \circ \beta$ on |
| fong-spivak-invitation_F01663 | 157  | 1.000 | $A \sqcup C$                                                      | $A \sqcup C$                                                      | $A \sqcup C$ . To do so requires three steps: (i) consider $\alpha$ and $\beta$ as relations on $A \sqcup B \sqcup C$ , (ii) take the transitive                  |
| fong-spivak-invitation_F01664 | 157  | 1.000 | $A \sqcup B \sqcup C$                                             | $A \sqcup B \sqcup C$                                             | $A \sqcup C$ . To do so requires three steps: (i) consider $\alpha$ and $\beta$ as relations on $A \sqcup B \sqcup C$ , (ii) take the transitive                  |

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| fong-spivak-invitation_F01665 | 157  | ■ 1.000 | $\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$                                                                              | $\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$                                                                              | description. Note that, as binary relations, we have $\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$ , and $\beta \subseteq (B \sqcup C) \times (B \sqcup C)$ .               |
| fong-spivak-invitation_F01666 | 157  | ■ 1.000 | $\beta \subseteq (B \sqcup C) \times (B \sqcup C)$                                                                               | $\beta \subseteq (B \sqcup C) \times (B \sqcup C)$                                                                               | description. Note that, as binary relations, we have $\alpha \subseteq (A \sqcup B) \times (A \sqcup B)$ , and $\beta \subseteq (B \sqcup C) \times (B \sqcup C)$ .               |
| fong-spivak-invitation_F01667 | 157  | ■ 1.000 | $\iota_{A \sqcup B}: A \sqcup B \rightarrow A \sqcup B \sqcup C, \iota_{B \sqcup C}: B \sqcup C \rightarrow A \sqcup B \sqcup C$ | $\iota_{A \sqcup B}: A \sqcup B \rightarrow A \sqcup B \sqcup C, \iota_{B \sqcup C}: B \sqcup C \rightarrow A \sqcup B \sqcup C$ | We also have three inclusions: $\iota_{A \sqcup B}: A \sqcup B \rightarrow A \sqcup B \sqcup C, \iota_{B \sqcup C}: B \sqcup C \rightarrow A \sqcup B \sqcup C$ , and             |
| fong-spivak-invitation_F01668 | 157  | ■ 0.778 | $\{ \}^{\iota} A \sqcup C: A \sqcup C \rightarrow A \sqcup B \sqcup C$                                                           | $\iota_{A \sqcup C}: A \sqcup C \rightarrow A \sqcup B \sqcup C$                                                                 | $\iota_{A \sqcup C}: A \sqcup C \rightarrow A \sqcup B \sqcup C$ . <b>Recalling our notation from Section</b> 1.4, we define                                                      |
| fong-spivak-invitation_F01669 | 158  | ■ 0.544 | $\{ \}_{\text{Bool}}$                                                                                                            | Bool                                                                                                                             | Instead of just detailing this compact closed structure for <b>Feas</b> = <b>Prof</b> <sub>Bool</sub> , it's no                                                                   |
| fong-spivak-invitation_F01670 | 158  | ■ 0.992 | $(\mathcal{V}, I, \otimes)$                                                                                                      | $(\mathcal{V}, I, \otimes)$                                                                                                      | extra work to prove that for any skeletal (unital, commutative) quantale $(\mathcal{V}, I, \otimes)$ the                                                                          |
| fong-spivak-invitation_F01671 | 158  | ■ 0.529 | $\operatorname{Prof}_{\mathcal{V}}$                                                                                              | $\operatorname{Prof}_{\mathcal{V}}$                                                                                              | <b>Theorem 4.48.</b> Let $\mathcal{V}$ be a skeletal quantale. The category <b>Prof</b> <sub><math>\mathcal{V}</math></sub> can be given the                                      |
| fong-spivak-invitation_F01672 | 158  | ■ 0.583 | $\{ \}_{\mathcal{V}}$                                                                                                            | $\mathcal{V}$                                                                                                                    | But monoidal products need to be given on morphisms also, and the morphisms in <b>Prof</b> <sub><math>\mathcal{V}</math></sub>                                                    |
| fong-spivak-invitation_F01673 | 158  | ■ 0.843 | $\Phi: \mathcal{X}_1 \mapsto \mathcal{X}_2$                                                                                      | $\Phi: \mathcal{X}_1 \mapsto \mathcal{X}_2$                                                                                      | are $\mathcal{V}$ -profunctors. So, given $\mathcal{V}$ -profunctors $\Phi: \mathcal{X}_1 \mapsto \mathcal{X}_2$ and $\Psi: \mathcal{Y}_1 \mapsto \mathcal{Y}_2$ , one defines    |
| fong-spivak-invitation_F01674 | 158  | ■ 0.843 | $\Psi: \mathcal{Y}_1 \mapsto \mathcal{Y}_2$                                                                                      | $\Psi: \mathcal{Y}_1 \mapsto \mathcal{Y}_2$                                                                                      | are $\mathcal{V}$ -profunctors. So, given $\mathcal{V}$ -profunctors $\Phi: \mathcal{X}_1 \mapsto \mathcal{X}_2$ and $\Psi: \mathcal{Y}_1 \mapsto \mathcal{Y}_2$ , one defines    |
| fong-spivak-invitation_F01675 | 158  | ■ 0.787 | $(\Phi \times \Psi): \mathcal{X}_1 \times \mathcal{Y}_1 \mapsto \mathcal{X}_2 \times \mathcal{Y}_2$                              | $(\Phi \times \Psi): \mathcal{X}_1 \times \mathcal{Y}_1 \mapsto \mathcal{X}_2 \times \mathcal{Y}_2$                              | a $\mathcal{V}$ -profunctor $\Phi \times \Psi: \mathcal{X}_1 \times \mathcal{Y}_1 \mapsto \mathcal{X}_2 \times \mathcal{Y}_2$ by                                                  |
| fong-spivak-invitation_F01676 | 159  | ■ 0.977 | $\Phi \times \Psi$                                                                                                               | $\Phi \times \Psi$                                                                                                               | the monoidal product of resource preorders $\mathcal{X}$ and $\mathcal{Y}$ and why $\Phi \times \Psi$ makes sense as the                                                          |
| fong-spivak-invitation_F01677 | 159  | ■ 0.665 | $(1, 1) = I$                                                                                                                     | $(1, 1) = I$                                                                                                                     | 1, and $(1, 1) = I$ is the monoidal unit of $\mathcal{V}$ . We take <b>1</b> to be the monoidal unit of <b>Prof</b> <sub><math>\mathcal{V}</math></sub> .                         |
| fong-spivak-invitation_F01678 | 159  | ■ 0.400 | $X \times \mathbf{1} \mapsto X$                                                                                                  | $X \times \mathbf{1} \mapsto X$                                                                                                  | morphisms $X \times \mathbf{1} \mapsto X$ and $\mathbf{1} \times X \mapsto X$ in <b>Prof</b> <sub><math>\mathcal{V}</math></sub> , for any $\mathcal{V}$ -category $X$ . What are |
| fong-spivak-invitation_F01679 | 159  | ■ 0.400 | $\mathbf{1} \times X \mapsto X$                                                                                                  | $\mathbf{1} \times X \mapsto X$                                                                                                  | morphisms $X \times \mathbf{1} \mapsto X$ and $\mathbf{1} \times X \mapsto X$ in <b>Prof</b> <sub><math>\mathcal{V}</math></sub> , for any $\mathcal{V}$ -category $X$ . What are |
| fong-spivak-invitation_F01680 | 159  | ■ 0.465 | $\eta_X: \mathbf{1} \mapsto \mathcal{X}^{\operatorname{op}} \times X$                                                            | $\eta_X: \mathbf{1} \mapsto \mathcal{X}^{\operatorname{op}} \times X$                                                            | elaborate, the unit is a $\mathcal{V}$ -profunctor $\eta_X: \mathbf{1} \mapsto \mathcal{X}^{\operatorname{op}} \times X$ . By definition, this is a $\mathcal{V}$ -functor        |
| fong-spivak-invitation_F01681 | 159  | ■ 0.973 | $\eta_X(1, x, x') := \mathcal{X}(x, x')$                                                                                         | $\eta_X(1, x, x') := \mathcal{X}(x, x')$                                                                                         | we define it by $\eta_X(1, x, x') := \mathcal{X}(x, x')$ . Similarly, the counit is the profunctor $\epsilon_X: (X \times$                                                        |
| fong-spivak-invitation_F01682 | 159  | ■ 0.973 | $\epsilon_X: (X \times$                                                                                                          | $\epsilon_X: (X \times$                                                                                                          | we define it by $\eta_X(1, x, x') := \mathcal{X}(x, x')$ . Similarly, the counit is the profunctor $\epsilon_X: (X \times$                                                        |
| fong-spivak-invitation_F01683 | 159  | ■ 0.239 | $\mathcal{X}^{\operatorname{op}}) \rightarrow \mathbf{1}$                                                                        | $\mathcal{X}^{\operatorname{op}}) \rightarrow \mathbf{1}$                                                                        | $\mathcal{X}^{\operatorname{op}}) \mapsto \mathbf{1}$ , defined by $\epsilon_X(x, x', 1) := \mathcal{X}(x, x')$ .                                                                 |
| fong-spivak-invitation_F01684 | 159  | ■ 0.239 | $\epsilon_X(x, x', 1) := \mathcal{X}(x, x')$                                                                                     | $\epsilon_X(x, x', 1) := \mathcal{X}(x, x')$                                                                                     | $\mathcal{X}^{\operatorname{op}}) \mapsto \mathbf{1}$ , defined by $\epsilon_X(x, x', 1) := \mathcal{X}(x, x')$ .                                                                 |
| fong-spivak-invitation_F01685 | 162  | ■ 1.000 | $u = 15x$                                                                                                                        | $u = 15x$                                                                                                                        | – produces two output signals: $u = 15x$ (upper right) and $v = 3x + 21y$ (lower right).                                                                                          |
| fong-spivak-invitation_F01686 | 162  | ■ 1.000 | $v = 3x + 21y$                                                                                                                   | $v = 3x + 21y$                                                                                                                   | – produces two output signals: $u = 15x$ (upper right) and $v = 3x + 21y$ (lower right).                                                                                          |
| fong-spivak-invitation_F01687 | 162  | ■ 1.000 | $x + 7y$                                                                                                                         | $x + 7y$                                                                                                                         | the second copy of $x$ to the lower signal to get $x + 7y$ , and so on.                                                                                                           |
| fong-spivak-invitation_F01688 | 163  | ■ 0.608 | $(\mathcal{C}, 0, +)$                                                                                                            | $(\mathcal{C}, 0, +)$                                                                                                            | <b>Definition 5.1.</b> A <i>prop</i> is a symmetric strict monoidal category $(\mathcal{C}, 0, +)$ for which                                                                      |
| fong-spivak-invitation_F01689 | 163  | ■ 0.876 | $\operatorname{Ob}(\mathcal{C}) = \mathbb{N}$                                                                                    | $\operatorname{Ob}(\mathcal{C}) = \mathbb{N}$                                                                                    | $\operatorname{Ob}(\mathcal{C}) = \mathbb{N}$ , the monoidal unit is $0 \in \mathbb{N}$ , and the monoidal product on objects is given                                            |
| fong-spivak-invitation_F01690 | 163  | ■ 0.876 | $0 \in \mathbb{N}$                                                                                                               | $0 \in \mathbb{N}$                                                                                                               | $\operatorname{Ob}(\mathcal{C}) = \mathbb{N}$ , the monoidal unit is $0 \in \mathbb{N}$ , and the monoidal product on objects is given                                            |
| fong-spivak-invitation_F01691 | 163  | ■ 1.000 | $\mathcal{C}(m, n)$                                                                                                              | $\mathcal{C}(m, n)$                                                                                                              | (i) a set $\mathcal{C}(m, n)$ of morphisms $m \rightarrow n$ , for $m, n \in \mathbb{N}$ ;                                                                                        |

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| fong-spivak-invitation_F01692 | 163  | ■ 0.341 | $\operatorname{map}_{\mathrm{id}}: n \rightarrow n$    | $\operatorname{map}_{\mathrm{id}}: n \rightarrow n$    | (ii) for all $n \in \mathbb{N}$ , an identity map $\operatorname{id}_n: n \rightarrow n$ ;                                                                                             |
| fong-spivak-invitation_F01693 | 163  | ■ 1.000 | $\sigma_{m,n}: m+n \rightarrow n+m$                    | $\sigma_{m,n}: m+n \rightarrow n+m$                    | (iii) for all $m, n \in \mathbb{N}$ , a symmetry map $\sigma_{m,n}: m+n \rightarrow n+m$ ;                                                                                             |
| fong-spivak-invitation_F01694 | 163  | ■ 1.000 | $f: m \rightarrow n$                                   | $f: m \rightarrow n$                                   | (iv) a composition rule: given $f: m \rightarrow n$ and $g: n \rightarrow p$ ; a map $(f \circ g): m \rightarrow p$ ;                                                                  |
| fong-spivak-invitation_F01695 | 163  | ■ 1.000 | $g: n \rightarrow p$                                   | $g: n \rightarrow p$                                   | (iv) a composition rule: given $f: m \rightarrow n$ and $g: n \rightarrow p$ ; a map $(f \circ g): m \rightarrow p$ ;                                                                  |
| fong-spivak-invitation_F01696 | 163  | ■ 1.000 | $(f \circ g): m \rightarrow p$                         | $(f \circ g): m \rightarrow p$                         | (iv) a composition rule: given $f: m \rightarrow n$ and $g: n \rightarrow p$ ; a map $(f \circ g): m \rightarrow p$ ;                                                                  |
| fong-spivak-invitation_F01697 | 163  | ■ 1.000 | $f: m \rightarrow m^{\prime}$                          | $f: m \rightarrow m^{\prime}$                          | (v) a monoidal product on morphisms: given $f: m \rightarrow m^{\prime}$ and $g: n \rightarrow n^{\prime}$ , a map                                                                     |
| fong-spivak-invitation_F01698 | 163  | ■ 1.000 | $g: n \rightarrow n^{\prime}$                          | $g: n \rightarrow n^{\prime}$                          | (v) a monoidal product on morphisms: given $f: m \rightarrow m^{\prime}$ and $g: n \rightarrow n^{\prime}$ , a map                                                                     |
| fong-spivak-invitation_F01699 | 163  | ■ 0.987 | $(f+g): m+n \rightarrow m^{\prime}+n^{\prime}$         | $(f+g): m+n \rightarrow m^{\prime}+n^{\prime}$         | $(f+g): m+n \rightarrow m^{\prime}+n^{\prime}$ .                                                                                                                                       |
| fong-spivak-invitation_F01700 | 163  | ■ 1.000 | $\underline{m}=\{1, \ldots, m\}$                       | $\underline{m}=\{1, \ldots, m\}$                       | from $\underline{m}=\{1, \ldots, m\}$ to $\underline{n}=\{1, \ldots, n\}$ . (The identities, symmetries, and composition                                                               |
| fong-spivak-invitation_F01701 | 163  | ■ 1.000 | $\underline{n}=\{1, \ldots, n\}$                       | $\underline{n}=\{1, \ldots, n\}$                       | from $\underline{m}=\{1, \ldots, m\}$ to $\underline{n}=\{1, \ldots, n\}$ . (The identities, symmetries, and composition                                                               |
| fong-spivak-invitation_F01702 | 163  | ■ 1.000 | $f+g: m+n \rightarrow m^{\prime}+n^{\prime}$           | $f+g: m+n \rightarrow m^{\prime}+n^{\prime}$           | functions: that is, given $f: m \rightarrow m^{\prime}$ and $g: n \rightarrow n^{\prime}$ , we define $f+g: m+n \rightarrow m^{\prime}+n^{\prime}$                                     |
| fong-spivak-invitation_F01703 | 164  | ■ 1.000 | $f: 3 \rightarrow 2$                                   | $f: 3 \rightarrow 2$                                   | 1. Draw a morphism $f: 3 \rightarrow 2$ and a morphism $g: 2 \rightarrow 4$ in <b>FinSet</b> .                                                                                         |
| fong-spivak-invitation_F01704 | 164  | ■ 1.000 | $g: 2 \rightarrow 4$                                   | $g: 2 \rightarrow 4$                                   | 1. Draw a morphism $f: 3 \rightarrow 2$ and a morphism $g: 2 \rightarrow 4$ in <b>FinSet</b> .                                                                                         |
| fong-spivak-invitation_F01705 | 164  | ■ 1.000 | $f+g$                                                  | $f+g$                                                  | 2. Draw $f+g$ .                                                                                                                                                                        |
| fong-spivak-invitation_F01706 | 164  | ■ 0.997 | $\sigma_{m,n}$                                         | $\sigma_{m,n}$                                         | 5. Choose $m, n \in \mathbb{N}$ , and draw the symmetry map $\sigma_{m,n}$ in <b>FinSet</b> ?                                                                                          |
| fong-spivak-invitation_F01707 | 164  | ■ 0.995 | $\underline{m} \rightarrow \underline{n}$              | $\underline{m} \rightarrow \underline{n}$              | $\underline{m} \rightarrow \underline{n}$ . Note that in this case morphisms $m \rightarrow n$ only exist when $m = n$ ; when $m \neq n$                                               |
| fong-spivak-invitation_F01708 | 164  | ■ 0.995 | $m \neq n$                                             | $m \neq n$                                             | $\underline{m} \rightarrow \underline{n}$ . Note that in this case morphisms $m \rightarrow n$ only exist when $m = n$ ; when $m \neq n$                                               |
| fong-spivak-invitation_F01709 | 164  | ■ 0.762 | $\mathbf{Bij}(m, n)$                                   | $\mathbf{Bij}(m, n)$                                   | the homset $\mathbf{Bij}(m, n)$ is empty. Since <b>Bij</b> is a subcategory of <b>FinSet</b> , we can define the                                                                       |
| fong-spivak-invitation_F01710 | 164  | ■ 0.762 | $\mathbf{Bij}$                                         | <b>Bij</b>                                             | the homset $\mathbf{Bij}(m, n)$ is empty. Since <b>Bij</b> is a subcategory of <b>FinSet</b> , we can define the                                                                       |
| fong-spivak-invitation_F01711 | 164  | ■ 1.000 | $\underline{m} \sqcup \underline{n}$                   | $\underline{m} \sqcup \underline{n}$                   | are partitions on $\underline{m} \sqcup \underline{n}$ (see Example 4.46), is a prop.                                                                                                  |
| fong-spivak-invitation_F01712 | 164  | ■ 0.994 | $R \subseteq \underline{m} \times \underline{n}$       | $R \subseteq \underline{m} \times \underline{n}$       | <b>Example 5.6.</b> There is a prop <b>Rel</b> for which morphisms $m \rightarrow n$ are relations, $R \subseteq \underline{m} \times \underline{n}$ .                                 |
| fong-spivak-invitation_F01713 | 164  | ■ 1.000 | $S \subseteq \underline{n} \times \underline{p}$       | $S \subseteq \underline{n} \times \underline{p}$       | The composition of $R$ with $S \subseteq \underline{n} \times \underline{p}$ is                                                                                                        |
| fong-spivak-invitation_F01714 | 164  | ■ 0.859 | $(\mathbb{N}, \preceq)$                                | $(\mathbb{N}, \preceq)$                                | symmetric monoidal preorder of the form $(\mathbb{N}, \preceq)$ , for some poset relation $\preceq$ on $\mathbb{N}$ , where                                                            |
| fong-spivak-invitation_F01715 | 164  | ■ 0.859 | $\preceq$                                              | $\preceq$                                              | symmetric monoidal preorder of the form $(\mathbb{N}, \preceq)$ , for some poset relation $\preceq$ on $\mathbb{N}$ , where                                                            |
| fong-spivak-invitation_F01716 | 164  | ■ 0.999 | $R_1+R_2$                                              | $R_1+R_2$                                              | <sup>2</sup> The monoidal product $R_1+R_2$ of relations $R_1 \subseteq \underline{m}_1 \times \underline{n}_1$ and $R_2 \subseteq \underline{m}_2 \times \underline{n}_2$ is given by |
| fong-spivak-invitation_F01717 | 164  | ■ 0.999 | $R_1 \subseteq \underline{m}_1 \times \underline{n}_1$ | $R_1 \subseteq \underline{m}_1 \times \underline{n}_1$ | <sup>2</sup> The monoidal product $R_1+R_2$ of relations $R_1 \subseteq \underline{m}_1 \times \underline{n}_1$ and $R_2 \subseteq \underline{m}_2 \times \underline{n}_2$ is given by |
| fong-spivak-invitation_F01718 | 164  | ■ 0.999 | $R_2 \subseteq \underline{m}_2 \times \underline{n}_2$ | $R_2 \subseteq \underline{m}_2 \times \underline{n}_2$ | <sup>2</sup> The monoidal product $R_1+R_2$ of relations $R_1 \subseteq \underline{m}_1 \times \underline{n}_1$ and $R_2 \subseteq \underline{m}_2 \times \underline{n}_2$ is given by |

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| fong-spivak-invitation_F01719 | 164  | ■ 0.226 | $R_1 \sqcup R_2 \subseteq \left( \underline{m_1} \times \underline{n_1} \right) \sqcup \left( \underline{m_2} \times \underline{n_2} \right) \subseteq \left( \underline{m_1 \sqcup m_2} \right) \times \left( \underline{n_1 \sqcup n_2} \right)$ | $R_1 \sqcup R_2 \subseteq (\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1 \sqcup m_2}) \times (\underline{n_1 \sqcup n_2})$ | $R_1 \sqcup R_2 \subseteq (\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1 \sqcup m_2}) \times (\underline{n_1 \sqcup n_2})$ |
| fong-spivak-invitation_F01720 | 165  | ■ 0.727 | $F(n)=n$                                                                                                                                                                                                                                           | $F(n) = n$                                                                                                                                                                                     | (a) $F$ is identity-on-objects, i.e. $F(n) = n$ for all $n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$ , and                                                             |
| fong-spivak-invitation_F01721 | 165  | ■ 0.727 | $n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$                                                                                                                                                                               | $n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$                                                                                                                           | (a) $F$ is identity-on-objects, i.e. $F(n) = n$ for all $n \in \text{Ob}(\mathcal{C}) = \text{Ob}(\mathcal{D}) = \mathbb{N}$ , and                                                             |
| fong-spivak-invitation_F01722 | 165  | ■ 1.000 | $f: m_1 \rightarrow m_2$                                                                                                                                                                                                                           | $f: m_1 \rightarrow m_2$                                                                                                                                                                       | (b) for all $f: m_1 \rightarrow m_2$ and $g: n_1 \rightarrow n_2$ in $\mathcal{C}$ , we have $F(f) + F(g) = F(f + g)$                                                                          |
| fong-spivak-invitation_F01723 | 165  | ■ 1.000 | $g: n_1 \rightarrow n_2$                                                                                                                                                                                                                           | $g: n_1 \rightarrow n_2$                                                                                                                                                                       | (b) for all $f: m_1 \rightarrow m_2$ and $g: n_1 \rightarrow n_2$ in $\mathcal{C}$ , we have $F(f) + F(g) = F(f + g)$                                                                          |
| fong-spivak-invitation_F01724 | 165  | ■ 1.000 | $F(f)+F(g)=F(f+g)$                                                                                                                                                                                                                                 | $F(f) + F(g) = F(f + g)$                                                                                                                                                                       | (b) for all $f: m_1 \rightarrow m_2$ and $g: n_1 \rightarrow n_2$ in $\mathcal{C}$ , we have $F(f) + F(g) = F(f + g)$                                                                          |
| fong-spivak-invitation_F01725 | 165  | ■ 0.943 | $i: \text{Bij} \rightarrow \text{FinSet}$                                                                                                                                                                                                          | $i: \text{Bij} \rightarrow \text{FinSet}$                                                                                                                                                      | <b>Example 5.10.</b> The inclusion $i: \text{Bij} \rightarrow \text{FinSet}$ is a prop functor. Perhaps more interest-                                                                         |
| fong-spivak-invitation_F01726 | 165  | ■ 0.262 | $f: \underline{m} \rightarrow \underline{n}$                                                                                                                                                                                                       | $f: \underline{m} \rightarrow \underline{n}$                                                                                                                                                   | ingly, there is a prop functor $F: \text{FinSet} \rightarrow \text{Rel}_{\text{Fin}}$ . It sends a function $f: \underline{m} \rightarrow \underline{n}$ to the                                |
| fong-spivak-invitation_F01727 | 165  | ■ 1.000 | $F(f) := \{(i, j) \mid f(i) = j\} \subseteq \underline{m} \times \underline{n}$                                                                                                                                                                    | $F(f) := \{(i, j) \mid f(i) = j\} \subseteq \underline{m} \times \underline{n}$                                                                                                                | relation $F(f) := \{(i, j) \mid f(i) = j\} \subseteq \underline{m} \times \underline{n}$                                                                                                       |
| fong-spivak-invitation_F01728 | 165  | ■ 0.991 | $\iota$                                                                                                                                                                                                                                            | $\iota$                                                                                                                                                                                        | <b>Definition 5.11.</b> For $m, n \in \mathbb{N}$ , an $(m, n)$ -port graph $(V, in, out, \iota)$ is specified by                                                                              |
| fong-spivak-invitation_F01729 | 165  | ■ 0.896 | $V \rightarrow \mathbb{N}$                                                                                                                                                                                                                         | $V \rightarrow \mathbb{N}$                                                                                                                                                                     | (ii) functions $in, out: V \rightarrow \mathbb{N}$ , where $in(v)$ and $out(v)$ are called the <i>in degree</i> and <i>out</i>                                                                 |
| fong-spivak-invitation_F01730 | 165  | ■ 0.896 | $\text{in}(v)$                                                                                                                                                                                                                                     | $in(v)$                                                                                                                                                                                        | (ii) functions $in, out: V \rightarrow \mathbb{N}$ , where $in(v)$ and $out(v)$ are called the <i>in degree</i> and <i>out</i>                                                                 |
| fong-spivak-invitation_F01731 | 165  | ■ 0.896 | $\text{out}(v)$                                                                                                                                                                                                                                    | $out(v)$                                                                                                                                                                                       | (ii) functions $in, out: V \rightarrow \mathbb{N}$ , where $in(v)$ and $out(v)$ are called the <i>in degree</i> and <i>out</i>                                                                 |
| fong-spivak-invitation_F01732 | 165  | ■ 0.959 | $\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$                                                                                                                                                                         | $\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$                                                                                                                     | (iii) a bijection $\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$ , where $I = \{(v, i) \mid v \in V, 1 \leq i \leq in(v)\}$ is the                                 |
| fong-spivak-invitation_F01733 | 165  | ■ 0.959 | $I = \{(v, i) \mid v \in V, 1 \leq i \leq \text{in}(v)\}$                                                                                                                                                                                          | $I = \{(v, i) \mid v \in V, 1 \leq i \leq in(v)\}$                                                                                                                                             | (iii) a bijection $\iota: \underline{m} \sqcup O \xrightarrow{\cong} I \sqcup \underline{n}$ , where $I = \{(v, i) \mid v \in V, 1 \leq i \leq in(v)\}$ is the                                 |
| fong-spivak-invitation_F01734 | 165  | ■ 1.000 | $O = \{(v, i) \mid v \in V, 1 \leq i \leq \text{out}(v)\}$                                                                                                                                                                                         | $O = \{(v, i) \mid v \in V, 1 \leq i \leq out(v)\}$                                                                                                                                            | set of <i>vertex inputs</i> , and $O = \{(v, i) \mid v \in V, 1 \leq i \leq out(v)\}$ is the set of <i>vertex</i>                                                                              |
| fong-spivak-invitation_F01735 | 165  | ■ 1.000 | $e_{v,j}^{u,i}: u \rightarrow v$                                                                                                                                                                                                                   | $e_{v,j}^{u,i}: u \rightarrow v$                                                                                                                                                               | struct the graph with vertices $V$ and with an arrow $e_{v,j}^{u,i}: u \rightarrow v$ for every $i, j \in \mathbb{N}$ such                                                                     |
| fong-spivak-invitation_F01736 | 165  | ■ 1.000 | $i, j \in \mathbb{N}$                                                                                                                                                                                                                              | $i, j \in \mathbb{N}$                                                                                                                                                                          | struct the graph with vertices $V$ and with an arrow $e_{v,j}^{u,i}: u \rightarrow v$ for every $i, j \in \mathbb{N}$ such                                                                     |
| fong-spivak-invitation_F01737 | 165  | ■ 1.000 | $\iota(u, i) = (v, j)$                                                                                                                                                                                                                             | $\iota(u, i) = (v, j)$                                                                                                                                                                         | that $\iota(u, i) = (v, j)$ ; call it the <i>internal flow graph</i> . If the internal flow graph is acyclic                                                                                   |
| fong-spivak-invitation_F01738 | 165  | ■ 0.997 | $m=2$                                                                                                                                                                                                                                              | $m = 2$                                                                                                                                                                                        | <b>Example 5.12.</b> Here is an example of a $(2, 3)$ -port graph, i.e. with $m = 2$ and $n = 3$ :                                                                                             |
| fong-spivak-invitation_F01739 | 165  | ■ 0.997 | $n=3$                                                                                                                                                                                                                                              | $n = 3$                                                                                                                                                                                        | <b>Example 5.12.</b> Here is an example of a $(2, 3)$ -port graph, i.e. with $m = 2$ and $n = 3$ :                                                                                             |
| fong-spivak-invitation_F01740 | 166  | ■ 1.000 | $V = \{a, b, c\}$                                                                                                                                                                                                                                  | $V = \{a, b, c\}$                                                                                                                                                                              | outer box, and three on the right. The vertex set is $V = \{a, b, c\}$ and, for example                                                                                                        |
| fong-spivak-invitation_F01741 | 166  | ■ 1.000 | $\text{in}(a)=1$                                                                                                                                                                                                                                   | $in(a) = 1$                                                                                                                                                                                    | $in(a) = 1$ and $out(a) = 3$ , so we draw one port on the left-hand side and three ports on                                                                                                    |
| fong-spivak-invitation_F01742 | 166  | ■ 1.000 | $\text{out}(a)=3$                                                                                                                                                                                                                                  | $out(a) = 3$                                                                                                                                                                                   | $in(a) = 1$ and $out(a) = 3$ , so we draw one port on the left-hand side and three ports on                                                                                                    |
| fong-spivak-invitation_F01743 | 166  | ■ 0.359 | $(m, n)$                                                                                                                                                                                                                                           | $(m, n)$                                                                                                                                                                                       | Given an $(m, n)$ -port graph $(V, in, out, \iota)$ and an $(n, p)$ -port graph $(V', in', out', \iota')$ , we                                                                                 |



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| fong-spivak-invitation_F01771 | 168  | ■ 1.000 | $R(x, y)$                                                                    | $R(x, y)$                                                                    | 1. Suppose that for every $x, y \in P$ , if $R(x, y)$ then $f(x) \leq f(y)$ . Show that $f$ defines                                                         |
| fong-spivak-invitation_F01772 | 168  | ■ 1.000 | $f(x) \leq_Q f(y)$                                                           | $f(x) \leq_Q f(y)$                                                           | every $x, y \in P$ , if $R(x, y)$ then $f(x) \leq_Q f(y)$ .                                                                                                 |
| fong-spivak-invitation_F01773 | 168  | ■ 1.000 | $P, Q, R$                                                                    | $P, Q, R$                                                                    | <b>Exercise 5.17.</b> Let $P, Q, R$ , etc. be as in Exercise 5.16. We want to see that the universal                                                        |
| fong-spivak-invitation_F01774 | 168  | ■ 1.000 | $a, b \in Q$                                                                 | $a, b \in Q$                                                                 | 1. Suppose that for every $a, b \in Q$ , if $a \leq b$ then $(g(a), g(b)) \in R$ . Is it automatically                                                      |
| fong-spivak-invitation_F01775 | 168  | ■ 1.000 | $(g(a), g(b)) \in R$                                                         | $(g(a), g(b)) \in R$                                                         | 1. Suppose that for every $a, b \in Q$ , if $a \leq b$ then $(g(a), g(b)) \in R$ . Is it automatically                                                      |
| fong-spivak-invitation_F01776 | 168  | ■ 1.000 | $g: (Q, \leq_Q) \rightarrow (P, \leq_P)$                                     | $g: (Q, \leq_Q) \rightarrow (P, \leq_P)$                                     | true that $g$ defines a monotone map $g: (Q, \leq_Q) \rightarrow (P, \leq_P)$ ?                                                                             |
| fong-spivak-invitation_F01777 | 168  | ■ 1.000 | $\operatorname{Mor}(\mathcal{C})$                                            | $\operatorname{Mor}(\mathcal{C})$                                            | Let $\mathcal{C}$ be another category whose set of morphisms is denoted $\operatorname{Mor}(\mathcal{C})$ .                                                 |
| fong-spivak-invitation_F01778 | 168  | ■ 1.000 | $\operatorname{Mor}(\mathcal{C}) \rightarrow \operatorname{Ob}(\mathcal{C})$ | $\operatorname{Mor}(\mathcal{C}) \rightarrow \operatorname{Ob}(\mathcal{C})$ | $\operatorname{Mor}(\mathcal{C}) \rightarrow \operatorname{Ob}(\mathcal{C})$ ; interpret this statement.                                                    |
| fong-spivak-invitation_F01779 | 168  | ■ 1.000 | $\mathcal{G} \rightarrow \mathcal{C}$                                        | $\mathcal{G} \rightarrow \mathcal{C}$                                        | 2. Show that the set of functors $\mathcal{G} \rightarrow \mathcal{C}$ are in one-to-one correspondence with the set                                        |
| fong-spivak-invitation_F01780 | 168  | ■ 1.000 | $f: V \rightarrow \operatorname{Ob}(\mathcal{C})$                            | $f: V \rightarrow \operatorname{Ob}(\mathcal{C})$                            | of pairs of functions $(f, g)$ , where $f: V \rightarrow \operatorname{Ob}(\mathcal{C})$ and $g: A \rightarrow \operatorname{Mor}(\mathcal{C})$ , for which |
| fong-spivak-invitation_F01781 | 168  | ■ 1.000 | $g: A \rightarrow \operatorname{Mor}(\mathcal{C})$                           | $g: A \rightarrow \operatorname{Mor}(\mathcal{C})$                           | of pairs of functions $(f, g)$ , where $f: V \rightarrow \operatorname{Ob}(\mathcal{C})$ and $g: A \rightarrow \operatorname{Mor}(\mathcal{C})$ , for which |
| fong-spivak-invitation_F01782 | 168  | ■ 1.000 | $\operatorname{dom}(g(a)) = f(s(a))$                                         | $\operatorname{dom}(g(a)) = f(s(a))$                                         | $\operatorname{dom}(g(a)) = f(s(a))$ and $\operatorname{cod}(g(a)) = f(t(a))$ for all $a$ .                                                                 |
| fong-spivak-invitation_F01783 | 168  | ■ 1.000 | $\operatorname{cod}(g(a)) = f(t(a))$                                         | $\operatorname{cod}(g(a)) = f(t(a))$                                         | $\operatorname{dom}(g(a)) = f(s(a))$ and $\operatorname{cod}(g(a)) = f(t(a))$ for all $a$ .                                                                 |
| fong-spivak-invitation_F01784 | 169  | ■ 1.000 | $\operatorname{Loop}(A)$                                                     | $\operatorname{Loop}(A)$                                                     | one-object categories. For any set $A$ , there is a graph $\operatorname{Loop}(A)$ with one vertex and with                                                 |
| fong-spivak-invitation_F01785 | 169  | ■ 1.000 | $A = \{a\}$                                                                  | $A = \{a\}$                                                                  | 1. What are the elements of the free monoid on the set $A = \{a\}$ ?                                                                                        |
| fong-spivak-invitation_F01786 | 169  | ■ 1.000 | $\{a\}$                                                                      | $\{a\}$                                                                      | 2. Can you find a well-known monoid that is isomorphic to the free monoid on $\{a\}$ ?                                                                      |
| fong-spivak-invitation_F01787 | 169  | ■ 0.993 | $G, s, t$                                                                    | $G, s, t$                                                                    | <b>Definition 5.21.</b> A <i>prop signature</i> is a tuple $(G, s, t)$ , where $G$ is a set and $s, t: G \rightarrow \mathbb{N}$                            |
| fong-spivak-invitation_F01788 | 169  | ■ 0.993 | $s, t: G \rightarrow \mathbb{N}$                                             | $s, t: G \rightarrow \mathbb{N}$                                             | <b>Definition 5.21.</b> A <i>prop signature</i> is a tuple $(G, s, t)$ , where $G$ is a set and $s, t: G \rightarrow \mathbb{N}$                            |
| fong-spivak-invitation_F01789 | 169  | ■ 1.000 | $g \in G$                                                                    | $g \in G$                                                                    | are functions; each element $g \in G$ is called a <i>generator</i> and $s(g), t(g) \in \mathbb{N}$ are called                                               |
| fong-spivak-invitation_F01790 | 169  | ■ 1.000 | $s(g), t(g) \in \mathbb{N}$                                                  | $s(g), t(g) \in \mathbb{N}$                                                  | are functions; each element $g \in G$ is called a <i>generator</i> and $s(g), t(g) \in \mathbb{N}$ are called                                               |
| fong-spivak-invitation_F01791 | 169  | ■ 0.844 | $s, t$                                                                       | $s, t$                                                                       | its <i>in-arity</i> and <i>out-arity</i> . We often denote $(G, s, t)$ simply by $G$ , taking $s, t$ to be                                                  |
| fong-spivak-invitation_F01792 | 169  | ■ 0.990 | $\Gamma = (V$                                                                | $\Gamma = (V$                                                                | A <i>G-labeling</i> of a port graph $\Gamma = (V, in, out, \iota)$ is a function $\ell: V \rightarrow G$ such that                                          |
| fong-spivak-invitation_F01793 | 169  | ■ 0.990 | $\ell: V \rightarrow G$                                                      | $\ell: V \rightarrow G$                                                      | A <i>G-labeling</i> of a port graph $\Gamma = (V, in, out, \iota)$ is a function $\ell: V \rightarrow G$ such that                                          |
| fong-spivak-invitation_F01794 | 169  | ■ 0.982 | $s(\ell(v)) = \operatorname{in}(v)$                                          | $s(\ell(v)) = \operatorname{in}(v)$                                          | the arities agree: $s(\ell(v)) = \operatorname{in}(v)$ and $t(\ell(v)) = \operatorname{out}(v)$ for each $v \in V$ .                                        |
| fong-spivak-invitation_F01795 | 169  | ■ 0.982 | $t(\ell(v)) = \operatorname{out}(v)$                                         | $t(\ell(v)) = \operatorname{out}(v)$                                         | the arities agree: $s(\ell(v)) = \operatorname{in}(v)$ and $t(\ell(v)) = \operatorname{out}(v)$ for each $v \in V$ .                                        |
| fong-spivak-invitation_F01796 | 169  | ■ 1.000 | $\ell$                                                                       | $\ell$                                                                       | port graphs <b>PG</b> (see Eq. (5.4)); the labelings (the $\ell$ 's) are just carried along.                                                                |
| fong-spivak-invitation_F01797 | 169  | ■ 1.000 | $(m, n) \in \mathbb{N} \times \mathbb{N}$                                    | $(m, n) \in \mathbb{N} \times \mathbb{N}$                                    | <sup>3</sup> The arity of a prop morphism is a pair $(m, n) \in \mathbb{N} \times \mathbb{N}$ , where $m$ is the number of inputs and $n$ is the            |

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| fong-spivak-invitation_F01798 | 170  | ■ 1.000 | $G=\{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$   | $G = \{f : 1 \rightarrow 1, g : 2 \rightarrow 2, h : 2 \rightarrow 1\}$ | with the same generator. For example, if $G = \{f : 1 \rightarrow 1, g : 2 \rightarrow 2, h : 2 \rightarrow 1\}$ , then                   |
| fong-spivak-invitation_F01799 | 170  | ■ 1.000 | $3 \rightarrow 2$                                                    | $3 \rightarrow 2$                                                       | the following is a morphism $3 \rightarrow 2$ in $\mathbf{Free}(G)$ :                                                                     |
| fong-spivak-invitation_F01800 | 170  | ■ 1.000 | $V \rightarrow \varnothing$                                          | $V \rightarrow \varnothing$                                             | must have a labeling function of the form $V \rightarrow \varnothing$ , and hence we must have $V =$                                      |
| fong-spivak-invitation_F01801 | 170  | ■ 1.000 | $(n, m)$                                                             | $(n, m)$                                                                | $\varnothing$ ; see Exercise 1.20. Thus the only morphisms $(n, m)$ are those given by port graphs                                        |
| fong-spivak-invitation_F01802 | 170  | ■ 0.992 | $(\varnothing, !, !, \sigma)$                                        | $(\varnothing, !, !, \sigma)$                                           | $(\varnothing, !, !, \sigma)$ , where $\sigma : n \rightarrow m$ is a bijection.                                                          |
| fong-spivak-invitation_F01803 | 170  | ■ 0.992 | $\sigma: n \rightarrow m$                                            | $\sigma : n \rightarrow m$                                              | $(\varnothing, !, !, \sigma)$ , where $\sigma: n \rightarrow m$ is a bijection.                                                           |
| fong-spivak-invitation_F01804 | 170  | ■ 0.959 | $(m, n) \in \mathbb{N}^2$                                            | $(m, n) \in \mathbb{N}^2$                                               | i.e. having one generating morphism for each $(m, n) \in \mathbb{N}^2$ . Show that $\mathbf{Free}(G)$ is the                              |
| fong-spivak-invitation_F01805 | 170  | ■ 0.860 | $(G, s, t)$                                                          | $(G, s, t)$                                                             | <b>Proposition 5.24.</b> The free prop $\mathbf{Free}(G)$ on a signature $(G, s, t)$ has the property that,                               |
| fong-spivak-invitation_F01806 | 170  | ■ 0.974 | $\mathbf{Free}(G) \rightarrow \mathcal{C}$                           | $\mathbf{Free}(G) \rightarrow \mathcal{C}$                              | for any prop $\mathcal{C}$ , the prop functors $\mathbf{Free}(G) \rightarrow \mathcal{C}$ are in one-to-one correspondence with           |
| fong-spivak-invitation_F01807 | 170  | ■ 1.000 | $G \rightarrow \mathcal{C}$                                          | $G \rightarrow \mathcal{C}$                                             | functions $G \rightarrow \mathcal{C}$ that send each $g \in G$ to a morphism $s(g) \rightarrow t(g)$ in $\mathcal{C}$ .                   |
| fong-spivak-invitation_F01808 | 170  | ■ 1.000 | $s(g) \rightarrow t(g)$                                              | $s(g) \rightarrow t(g)$                                                 | functions $G \rightarrow \mathcal{C}$ that send each $g \in G$ to a morphism $s(g) \rightarrow t(g)$ in $\mathcal{C}$ .                   |
| fong-spivak-invitation_F01810 | 171  | ■ 1.000 | $e: m \rightarrow n$                                                 | $e : m \rightarrow n$                                                   | a <i>G-generated prop expression</i> , or simply <i>expression</i> $e: m \rightarrow n$ , where $m, n \in \mathbb{N}$ ,                   |
| fong-spivak-invitation_F01811 | 171  | ■ 1.000 | $\mathrm{id}_0: 0 \rightarrow 0$                                     | $\mathrm{id}_0 : 0 \rightarrow 0$                                       | • The empty morphism $\mathrm{id}_0: 0 \rightarrow 0$ , the identity morphism $\mathrm{id}_1: 1 \rightarrow 1$ , and the                  |
| fong-spivak-invitation_F01812 | 171  | ■ 1.000 | $\mathrm{id}_1: 1 \rightarrow 1$                                     | $\mathrm{id}_1 : 1 \rightarrow 1$                                       | • The empty morphism $\mathrm{id}_0: 0 \rightarrow 0$ , the identity morphism $\mathrm{id}_1: 1 \rightarrow 1$ , and the                  |
| fong-spivak-invitation_F01813 | 171  | ■ 1.000 | $\sigma: 2 \rightarrow 2$                                            | $\sigma : 2 \rightarrow 2$                                              | symmetry $\sigma : 2 \rightarrow 2$ are expressions. <sup>4</sup>                                                                         |
| fong-spivak-invitation_F01814 | 171  | ■ 1.000 | $g: s(g) \rightarrow t(g)$                                           | $g : s(g) \rightarrow t(g)$                                             | • The generators $g \in G$ are expressions $g: s(g) \rightarrow t(g)$ .                                                                   |
| fong-spivak-invitation_F01815 | 171  | ■ 1.000 | $\alpha: m \rightarrow n$                                            | $\alpha : m \rightarrow n$                                              | • If $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $\alpha + \beta: m + p \rightarrow n + q$ is an         |
| fong-spivak-invitation_F01816 | 171  | ■ 1.000 | $\beta: p \rightarrow q$                                             | $\beta : p \rightarrow q$                                               | • If $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $\alpha + \beta: m + p \rightarrow n + q$ is an         |
| fong-spivak-invitation_F01817 | 171  | ■ 1.000 | $\alpha + \beta: m + p \rightarrow n + q$                            | $\alpha + \beta : m + p \rightarrow n + q$                              | • If $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $\alpha + \beta: m + p \rightarrow n + q$ is an         |
| fong-spivak-invitation_F01818 | 171  | ■ 0.988 | $\beta: n \rightarrow p$                                             | $\beta : n \rightarrow p$                                               | • If $\alpha: m \rightarrow n$ and $\beta: n \rightarrow p$ are expressions, then $\alpha \circ \beta: m \rightarrow p$ is an expression. |
| fong-spivak-invitation_F01819 | 171  | ■ 0.988 | $\alpha; \beta: m \rightarrow p$                                     | $\alpha; \beta : m \rightarrow p$                                       | • If $\alpha: m \rightarrow n$ and $\beta: n \rightarrow p$ are expressions, then $\alpha \circ \beta: m \rightarrow p$ is an expression. |
| fong-spivak-invitation_F01820 | 171  | ■ 1.000 | $\mathbf{Expr}(G)$                                                   | $\mathbf{Expr}(G)$                                                      | We write $\mathbf{Expr}(G)$ for the set of expressions in $G$ . If $e: m \rightarrow n$ is an expression, we                              |
| fong-spivak-invitation_F01821 | 171  | ■ 1.000 | $f: 1 \rightarrow 1$                                                 | $f : 1 \rightarrow 1$                                                   | • $f: 1 \rightarrow 1$ ,                                                                                                                  |
| fong-spivak-invitation_F01822 | 171  | ■ 0.988 | $f: \mathrm{id}_1: 1 \rightarrow 1$                                  | $f : \mathrm{id}_1 : 1 \rightarrow 1$                                   | • $f \circ \mathrm{id}_1: 1 \rightarrow 1$ ,                                                                                              |
| fong-spivak-invitation_F01823 | 171  | ■ 0.999 | $h + \mathrm{id}_1: 3 \rightarrow 2$                                 | $h + \mathrm{id}_1 : 3 \rightarrow 2$                                   | • $h + \mathrm{id}_1: 3 \rightarrow 2$ , and                                                                                              |
| fong-spivak-invitation_F01824 | 171  | ■ 0.866 | $\left(h + \mathrm{id}_1\right); \sigma; g; \sigma: 3 \rightarrow 2$ | $(h + \mathrm{id}_1); \sigma; g; \sigma : 3 \rightarrow 2$              | • $(h + \mathrm{id}_1) \circ \sigma \circ g \circ \sigma: 3 \rightarrow 2$                                                                |
| fong-spivak-invitation_F01825 | 171  | ■ 0.678 | $f \circ \mathrm{id}_1$                                              | $f \circ \mathrm{id}_1$                                                 | phism. For example, we want to consider $f \circ \mathrm{id}_1$ and $f$ to be the same morphism, since                                    |

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| fong-spivak-invitation_F01826 | 171  | ■ 0.953 | $f ; \mathrm{id}_1 = f$                                                                                                                                                                                                      | $f; \mathrm{id}_1 = f$                                                                                                                                                                  | the unitality axiom for categories says $f \circ \mathrm{id}_1 = f$ . Nonetheless, we only consider                                                                                                                                           |
| fong-spivak-invitation_F01827 | 171  | ■ 0.997 | $\sigma$                                                                                                                                                                                                                     | $\sigma$                                                                                                                                                                                | <sup>4</sup> One can think of $\sigma$ as the “swap” icon $\bowtie: 2 \rightarrow 2$ .                                                                                                                                                        |
| fong-spivak-invitation_F01828 | 171  | ■ 0.997 | $\mathrm{X}: 2 \rightarrow 2$                                                                                                                                                                                                | $X: 2 \rightarrow 2$                                                                                                                                                                    | <sup>4</sup> One can think of $\sigma$ as the “swap” icon $\bowtie: 2 \rightarrow 2$ .                                                                                                                                                        |
| fong-spivak-invitation_F01829 | 172  | ■ 0.616 | $\left(g + \mathrm{id}_1\right) \stackrel{\circ}{\circ} \left(\mathrm{id}_1 + \sigma\right) \stackrel{\circ}{\circ} \left(\mathrm{id}_1 + g\right) \stackrel{\circ}{\circ} \left(h + \mathrm{id}_1\right)$                   | $(g + \mathrm{id}_1) \overset{\circ}{\circ} (\mathrm{id}_1 + \sigma) \overset{\circ}{\circ} (\mathrm{id}_1 + g) \overset{\circ}{\circ} (h + \mathrm{id}_1)$                             | Here the result is $(g + \mathrm{id}_1) \overset{\circ}{\circ} (\mathrm{id}_1 + \sigma) \overset{\circ}{\circ} (\mathrm{id}_1 + g) \overset{\circ}{\circ} (h + \mathrm{id}_1)$ .                                                              |
| fong-spivak-invitation_F01830 | 172  | ■ 1.000 | $G = \{f: 1 \rightarrow 1, g: 2 \rightarrow$                                                                                                                                                                                 | $G = \{f: 1 \rightarrow 1, g: 2 \rightarrow$                                                                                                                                            | <b>Exercise 5.27.</b> Consider again the free prop on generators $G = \{f: 1 \rightarrow 1, g: 2 \rightarrow$                                                                                                                                 |
| fong-spivak-invitation_F01831 | 172  | ■ 0.605 | $h: 2 \rightarrow 1\}$                                                                                                                                                                                                       | $h: 2 \rightarrow 1\}$                                                                                                                                                                  | $2, h: 2 \rightarrow 1\}$ . Draw a picture of $(f + \mathrm{id}_1 + \mathrm{id}_1) \overset{\circ}{\circ} (\sigma + \mathrm{id}_1) \overset{\circ}{\circ} (\mathrm{id}_1 + h) \overset{\circ}{\circ} \sigma \overset{\circ}{\circ} g$ , where |
| fong-spivak-invitation_F01832 | 172  | ■ 0.605 | $\left(f + \mathrm{id}_1 + \mathrm{id}_1\right) \stackrel{\circ}{\circ} \left(\sigma + \mathrm{id}_1\right) \stackrel{\circ}{\circ} \left(\mathrm{id}_1 + h\right) \stackrel{\circ}{\circ} \sigma \stackrel{\circ}{\circ} g$ | $(f + \mathrm{id}_1 + \mathrm{id}_1) \overset{\circ}{\circ} (\sigma + \mathrm{id}_1) \overset{\circ}{\circ} (\mathrm{id}_1 + h) \overset{\circ}{\circ} \sigma \overset{\circ}{\circ} g$ | $2, h: 2 \rightarrow 1\}$ . Draw a picture of $(f + \mathrm{id}_1 + \mathrm{id}_1) \overset{\circ}{\circ} (\sigma + \mathrm{id}_1) \overset{\circ}{\circ} (\mathrm{id}_1 + h) \overset{\circ}{\circ} \sigma \overset{\circ}{\circ} g$ , where |
| fong-spivak-invitation_F01833 | 172  | ■ 0.666 | $(G, s, t, E)$                                                                                                                                                                                                               | $(G, s, t, E)$                                                                                                                                                                          | <b>Rough Definition 5.28.</b> A <i>presentation</i> $(G, s, t, E)$ for a prop is a set $G$ , functions                                                                                                                                        |
| fong-spivak-invitation_F01834 | 172  | ■ 1.000 | $E \subseteq \mathrm{Expr}(G) \times \mathrm{Expr}(G)$                                                                                                                                                                       | $E \subseteq \mathrm{Expr}(G) \times \mathrm{Expr}(G)$                                                                                                                                  | $s, t: G \rightarrow \mathbb{N}$ , and a set $E \subseteq \mathrm{Expr}(G) \times \mathrm{Expr}(G)$ of pairs of $G$ -generated prop expres-                                                                                                   |
| fong-spivak-invitation_F01835 | 172  | ■ 1.000 | $e_1$                                                                                                                                                                                                                        | $e_1$                                                                                                                                                                                   | sions, such that $e_1$ and $e_2$ have the same arity for each $(e_1, e_2) \in E$ . We refer to $G$ as                                                                                                                                         |
| fong-spivak-invitation_F01836 | 172  | ■ 1.000 | $e_2$                                                                                                                                                                                                                        | $e_2$                                                                                                                                                                                   | sions, such that $e_1$ and $e_2$ have the same arity for each $(e_1, e_2) \in E$ . We refer to $G$ as                                                                                                                                         |
| fong-spivak-invitation_F01837 | 172  | ■ 1.000 | $\left(e_1, e_2\right) \in E$                                                                                                                                                                                                | $(e_1, e_2) \in E$                                                                                                                                                                      | sions, such that $e_1$ and $e_2$ have the same arity for each $(e_1, e_2) \in E$ . We refer to $G$ as                                                                                                                                         |
| fong-spivak-invitation_F01838 | 172  | ■ 0.998 | $e_1 = e_2$                                                                                                                                                                                                                  | $e_1 = e_2$                                                                                                                                                                             | are elements in $\mathrm{Expr}(G)$ , quotiented by both the equations $e_1 = e_2$ where $(e_1, e_2) \in E$ ,                                                                                                                                  |
| fong-spivak-invitation_F01839 | 173  | ■ 1.000 | $g \in G, f(g)$                                                                                                                                                                                                              | $g \in G, f(g)$                                                                                                                                                                         | • for all $g \in G$ , $f(g)$ is a morphism $s(g) \rightarrow t(g)$ , and                                                                                                                                                                      |
| fong-spivak-invitation_F01840 | 173  | ■ 1.000 | $f\left(e_1\right)=f\left(e_2\right)$                                                                                                                                                                                        | $f\left(e_1\right)=f\left(e_2\right)$                                                                                                                                                   | • for all $(e_1, e_2) \in E$ , we have that $f\left(e_1\right)=f\left(e_2\right)$ in $\mathcal{C}$ , where $f(e)$ denotes the                                                                                                                 |
| fong-spivak-invitation_F01841 | 173  | ■ 1.000 | $f(e)$                                                                                                                                                                                                                       | $f(e)$                                                                                                                                                                                  | • for all $(e_1, e_2) \in E$ , we have that $f\left(e_1\right)=f\left(e_2\right)$ in $\mathcal{C}$ , where $f(e)$ denotes the                                                                                                                 |
| fong-spivak-invitation_F01842 | 173  | ■ 0.658 | $(G, s, t, \varnothing)$                                                                                                                                                                                                     | $(G, s, t, \emptyset)$                                                                                                                                                                  | nition 5.21, is the same thing as the prop presented by $(G, s, t, \emptyset)$ , having no relations,                                                                                                                                         |
| fong-spivak-invitation_F01843 | 173  | ■ 0.976 | $R, 0,+, 1, *$                                                                                                                                                                                                               | $R, 0, +, 1, *$                                                                                                                                                                         | <b>Definition 5.31.</b> A <i>rig</i> is a tuple $(R, 0, +, 1, *)$ , where $R$ is a set, $0, 1 \in R$ are elements,                                                                                                                            |
| fong-spivak-invitation_F01844 | 173  | ■ 0.976 | $0,1 \in R$                                                                                                                                                                                                                  | $0, 1 \in R$                                                                                                                                                                            | <b>Definition 5.31.</b> A <i>rig</i> is a tuple $(R, 0, +, 1, *)$ , where $R$ is a set, $0, 1 \in R$ are elements,                                                                                                                            |
| fong-spivak-invitation_F01845 | 173  | ■ 0.986 | $+, *: R \times R \rightarrow R$                                                                                                                                                                                             | $+, *: R \times R \rightarrow R$                                                                                                                                                        | and $+, *: R \times R \rightarrow R$ are functions, such that                                                                                                                                                                                 |
| fong-spivak-invitation_F01846 | 173  | ■ 0.989 | $(R,+, 0)$                                                                                                                                                                                                                   | $(R, +, 0)$                                                                                                                                                                             | (a) $(R, +, 0)$ is a commutative monoid,                                                                                                                                                                                                      |
| fong-spivak-invitation_F01847 | 173  | ■ 0.615 | $R, *, 1$                                                                                                                                                                                                                    | $R, *, 1$                                                                                                                                                                               | (b) $(R, *, 1)$ is a monoid, <sup>7</sup>                                                                                                                                                                                                     |
| fong-spivak-invitation_F01848 | 173  | ■ 1.000 | $a *(b+c)=a * b+a * c$                                                                                                                                                                                                       | $a * (b + c) = a * b + a * c$                                                                                                                                                           | (c) $a * (b + c) = a * b + a * c$ and $(a + b) * c = a * c + b * c$ for all $a, b, c \in R$ ,                                                                                                                                                 |
| fong-spivak-invitation_F01849 | 173  | ■ 1.000 | $(a+b) * c=a * c+b * c$                                                                                                                                                                                                      | $(a + b) * c = a * c + b * c$                                                                                                                                                           | (c) $a * (b + c) = a * b + a * c$ and $(a + b) * c = a * c + b * c$ for all $a, b, c \in R$ ,                                                                                                                                                 |
| fong-spivak-invitation_F01850 | 173  | ■ 1.000 | $a, b, c \in R$                                                                                                                                                                                                              | $a, b, c \in R$                                                                                                                                                                         | (c) $a * (b + c) = a * b + a * c$ and $(a + b) * c = a * c + b * c$ for all $a, b, c \in R$ ,                                                                                                                                                 |

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| fong-spivak-invitation_F01851 | 173  | ■ 1.000 | $a * 0 = 0 = 0 * a$                                             | $a * 0 = 0 = 0 * a$                                             | (d) $a * 0 = 0 = 0 * a$ for all $a \in R$ .                                                                                                                  |
| fong-spivak-invitation_F01852 | 173  | ■ 1.000 | $a \in R$                                                       | $a \in R$                                                       | (d) $a * 0 = 0 = 0 * a$ for all $a \in R$ .                                                                                                                  |
| fong-spivak-invitation_F01853 | 173  | ■ 0.733 | $(\mathbb{N}, 0, +, 1, *)$                                      | $(\mathbb{N}, 0, +, 1, *)$                                      | <b>Example 5.32.</b> The natural numbers form a rig $(\mathbb{N}, 0, +, 1, *)$ .                                                                             |
| fong-spivak-invitation_F01854 | 173  | ■ 0.999 | $(R, *, 1)$                                                     | $(R, *, 1)$                                                     | <sup>7</sup> Note that we did not demand that $(R, *, 1)$ be commutative; we shall see a naturally arising example where                                     |
| fong-spivak-invitation_F01855 | 174  | ■ 0.898 | $(V, 0, \vee, I, \otimes)$                                      | $(V, 0, \vee, I, \otimes)$                                      | <b>Example 5.34.</b> Any quantale $\mathcal{V} = (V, \leq, I, \otimes)$ determines a rig $(V, 0, \vee, I, \otimes)$ , where                                  |
| fong-spivak-invitation_F01856 | 174  | ■ 1.000 | $0 = \bigvee \varnothing$                                       | $0 = \bigvee \varnothing$                                       | $0 = \bigvee \varnothing$ is the empty join. See Definition 2.57.                                                                                            |
| fong-spivak-invitation_F01857 | 174  | ■ 1.000 | $\operatorname{Mat}_n(R)$                                       | $\operatorname{Mat}_n(R)$                                       | <b>Example 5.35.</b> If $R$ is a rig and $n \in \mathbb{N}$ is any natural number, then the set $\operatorname{Mat}_n(R)$ of                                 |
| fong-spivak-invitation_F01858 | 174  | ■ 1.000 | $(n \times n)$                                                  | $(n \times n)$                                                  | $(n \times n)$ -matrices in $R$ forms a rig. A matrix $M \in \operatorname{Mat}_n(R)$ is a function $M : \underline{n} \times \underline{n} \rightarrow R$ . |
| fong-spivak-invitation_F01859 | 174  | ■ 1.000 | $M \in \operatorname{Mat}_n(R)$                                 | $M \in \operatorname{Mat}_n(R)$                                 | $(n \times n)$ -matrices in $R$ forms a rig. A matrix $M \in \operatorname{Mat}_n(R)$ is a function $M : \underline{n} \times \underline{n} \rightarrow R$ . |
| fong-spivak-invitation_F01860 | 174  | ■ 1.000 | $M : \underline{n} \times \underline{n} \rightarrow R$          | $M : \underline{n} \times \underline{n} \rightarrow R$          | $(n \times n)$ -matrices in $R$ forms a rig. A matrix $M \in \operatorname{Mat}_n(R)$ is a function $M : \underline{n} \times \underline{n} \rightarrow R$ . |
| fong-spivak-invitation_F01861 | 174  | ■ 1.000 | $M + N$                                                         | $M + N$                                                         | Addition $M + N$ of matrices is given by $(M + N)(i, j) := M(i, j) + N(i, j)$ and                                                                            |
| fong-spivak-invitation_F01862 | 174  | ■ 1.000 | $(M + N)(i, j) := M(i, j) + N(i, j)$                            | $(M + N)(i, j) := M(i, j) + N(i, j)$                            | Addition $M + N$ of matrices is given by $(M + N)(i, j) := M(i, j) + N(i, j)$ and                                                                            |
| fong-spivak-invitation_F01863 | 174  | ■ 0.999 | $(M * N)(i, j) := \sum_{k \in \underline{n}} M(i, k) * N(k, j)$ | $(M * N)(i, j) := \sum_{k \in \underline{n}} M(i, k) * N(k, j)$ | multiplication $M * N$ is given by $(M * N)(i, j) := \sum_{k \in \underline{n}} M(i, k) * N(k, j)$ . The 0-matrix                                            |
| fong-spivak-invitation_F01864 | 174  | ■ 1.000 | $0(i, j) := 0$                                                  | $0(i, j) := 0$                                                  | is $0(i, j) := 0$ for all $i, j \in \underline{n}$ . Note that $\operatorname{Mat}_n(R)$ is generally not commutative.                                       |
| fong-spivak-invitation_F01865 | 174  | ■ 1.000 | $i, j \in \underline{n}$                                        | $i, j \in \underline{n}$                                        | is $0(i, j) := 0$ for all $i, j \in \underline{n}$ . Note that $\operatorname{Mat}_n(R)$ is generally not commutative.                                       |
| fong-spivak-invitation_F01866 | 174  | ■ 1.000 | $1 \in \operatorname{Mat}_n(R)$                                 | $1 \in \operatorname{Mat}_n(R)$                                 | multiplicative identity $1 \in \operatorname{Mat}_n(R)$ ?                                                                                                    |
| fong-spivak-invitation_F01867 | 174  | ■ 1.000 | $\operatorname{Mat}_n(\mathbb{N})$                              | $\operatorname{Mat}_n(\mathbb{N})$                              | $\operatorname{Mat}_n(\mathbb{N})$ is not commutative, where $\mathbb{N}$ is as in Example 5.32.                                                             |
| fong-spivak-invitation_F01868 | 174  | ■ 0.998 | $(\mathbb{R}, 0, +, 1, *)$                                      | $\mathbb{R}, 0, +, 1, *$                                        | <b>Example 5.37.</b> Any ring forms a rig. In particular, the real numbers $(\mathbb{R}, 0, +, 1, *)$ are a                                                  |
| fong-spivak-invitation_F01869 | 174  | ■ 1.000 | $M = R$                                                         | $M = R$                                                         | $R$ -module $M$ . We will not discuss this here, keeping to the simpler requirement that $M = R$ .                                                           |
| fong-spivak-invitation_F01870 | 175  | ■ 1.000 | $R = \mathbb{N}$                                                | $R = \mathbb{N}$                                                | For example, consider the rig $R = \mathbb{N}$ from Example 5.32, where the scalars are the                                                                  |
| fong-spivak-invitation_F01871 | 175  | ■ 1.000 | $a = 15x$                                                       | $a = 15x$                                                       | $a = 15x$ and $b = 3x + 21y$ .                                                                                                                               |
| fong-spivak-invitation_F01872 | 175  | ■ 1.000 | $b = 3x + 21y$                                                  | $b = 3x + 21y$                                                  | $a = 15x$ and $b = 3x + 21y$ .                                                                                                                               |
| fong-spivak-invitation_F01873 | 176  | ■ 1.000 | $15x$                                                           | $15x$                                                           | amplified by 3. Thus, if $x$ is the first input signal, then this contributes $15x$ to the first                                                             |
| fong-spivak-invitation_F01874 | 176  | ■ 1.000 | $3x = x + 2x$                                                   | $3x = x + 2x$                                                   | two paths to it from each input. The top input thus contributes $3x = x + 2x$ , whereas                                                                      |
| fong-spivak-invitation_F01875 | 176  | ■ 0.954 | $*7$                                                            | $*7$                                                            | the bottom input, passing through an additional $*7$ amplification, contributes $21y$ .                                                                      |
| fong-spivak-invitation_F01876 | 176  | ■ 0.954 | $21y$                                                           | $21y$                                                           | the bottom input, passing through an additional $*7$ amplification, contributes $21y$ .                                                                      |
| fong-spivak-invitation_F01877 | 176  | ■ 0.999 | $(x, y, z)$                                                     | $(x, y, z)$                                                     | the movement of a particle. For example, consider a particle whose position $(x, y, z)$ in                                                                   |

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| fong-spivak-invitation_F01878 | 177  | ■ 1.000 | $s, t: G_{\{R\}} \rightarrow \mathbb{N}$                                    | $s, t: G_R \rightarrow \mathbb{N}$                      | and let $s, t: G_R \rightarrow \mathbb{N}$ be given by the number of dangling wires on the left and right                                    |
| fong-spivak-invitation_F01879 | 177  | ■ 0.403 | $\boldsymbol{\operatorname{Free}}(G_{\{R\}})$                               | $\mathbf{Free}(G_R)$                                    | free prop $\mathbf{Free}(G_R)$ on this set $G_R$ of generators. We define $\mathbf{SFG}_R := \mathbf{Free}(G_R)$ .                           |
| fong-spivak-invitation_F01880 | 177  | ■ 0.403 | $G_{\{R\}}$                                                                 | $G_R$                                                   | free prop $\mathbf{Free}(G_R)$ on this set $G_R$ of generators. We define $\mathbf{SFG}_R := \mathbf{Free}(G_R)$ .                           |
| fong-spivak-invitation_F01881 | 177  | ■ 0.403 | $\operatorname{SFG}_{\{R\}} := \boldsymbol{\operatorname{Free}}(G_{\{R\}})$ | $\mathbf{SFG}_R := \mathbf{Free}(G_R)$                  | free prop $\mathbf{Free}(G_R)$ on this set $G_R$ of generators. We define $\mathbf{SFG}_R := \mathbf{Free}(G_R)$ .                           |
| fong-spivak-invitation_F01882 | 177  | ■ 0.576 | $\operatorname{Free}(G_{\{R\}})$                                            | $\mathbf{Free}(G_R)$                                    | port graphs, morphisms in $\mathbf{Free}(G_R)$ should be drawn slightly differently. For example,                                            |
| fong-spivak-invitation_F01883 | 178  | ■ 0.996 | $(m \times n)$                                                              | $(m \times n)$                                          | An $(m \times n)$ -matrix $M$ with values in $R$ is a function $M: (\underline{m} \times \underline{n}) \rightarrow R$ . Given an            |
| fong-spivak-invitation_F01884 | 178  | ■ 0.996 | $M: (\underline{m} \times \underline{n}) \rightarrow R$                     | $M: (\underline{m} \times \underline{n}) \rightarrow R$ | An $(m \times n)$ -matrix $M$ with values in $R$ is a function $M: (\underline{m} \times \underline{n}) \rightarrow R$ . Given an            |
| fong-spivak-invitation_F01885 | 178  | ■ 0.982 | $(n \times p)$                                                              | $(n \times p)$                                          | $(m \times n)$ -matrix $M$ and an $(n \times p)$ -matrix $N$ , their <i>composite</i> is the $(m \times p)$ -matrix $M \circ N$              |
| fong-spivak-invitation_F01886 | 178  | ■ 0.982 | $(m \times p)$                                                              | $(m \times p)$                                          | $(m \times n)$ -matrix $M$ and an $(n \times p)$ -matrix $N$ , their <i>composite</i> is the $(m \times p)$ -matrix $M \circ N$              |
| fong-spivak-invitation_F01887 | 178  | ■ 1.000 | $a \in \underline{m}$                                                       | $a \in \underline{m}$                                   | defined as follows for any $a \in \underline{m}$ and $c \in \underline{p}$ :                                                                 |
| fong-spivak-invitation_F01888 | 178  | ■ 1.000 | $c \in \underline{p}$                                                       | $c \in \underline{p}$                                   | defined as follows for any $a \in \underline{m}$ and $c \in \underline{p}$ :                                                                 |
| fong-spivak-invitation_F01889 | 178  | ■ 0.748 | $\sum_{b \in \underline{n}}$                                                | $\sum_{b \in \underline{n}}$                            | Here the $\sum_{b \in \underline{n}}$ just means repeated addition (using the rig $R$ 's $+$ operation), as usual.                           |
| fong-spivak-invitation_F01890 | 178  | ■ 1.000 | $A v$                                                                       | $Av$                                                    | $v$ by multiplication in the order $Av$ , where $v$ is a column vector. In keeping with our                                                  |
| fong-spivak-invitation_F01891 | 178  | ■ 0.997 | $v ; A$                                                                     | $v; A$                                                  | composition convention, we use the opposite order, $v \circ A$ , where $v$ is a row vector. See                                              |
| fong-spivak-invitation_F01892 | 178  | ■ 0.918 | $\operatorname{Mat}(R)$                                                     | $\mathbf{Mat}(R)$                                       | <b>Definition 5.43.</b> Let $R$ be a rig. We define the <i>prop of <math>R</math>-matrices</i> , denoted $\mathbf{Mat}(R)$ ,                 |
| fong-spivak-invitation_F01893 | 178  | ■ 0.999 | $A: m \rightarrow n$                                                        | $A: m \rightarrow n$                                    | monoidal product is given by the direct sum of matrices: given matrices $A: m \rightarrow n$ and                                             |
| fong-spivak-invitation_F01894 | 178  | ■ 1.000 | $b: p \rightarrow q$                                                        | $b: p \rightarrow q$                                    | $b: p \rightarrow q$ , we define $A + B: m + p \rightarrow n + q$ to be the block matrix                                                     |
| fong-spivak-invitation_F01895 | 178  | ■ 1.000 | $A+B: m+p \rightarrow n+q$                                                  | $A + B: m + p \rightarrow n + q$                        | $b: p \rightarrow q$ , we define $A + B: m + p \rightarrow n + q$ to be the block matrix                                                     |
| fong-spivak-invitation_F01896 | 178  | ■ 0.979 | $m \times q$                                                                | $m \times q$                                            | where each 0 represents a matrix of zeros of the appropriate dimension $(m \times q$ and $n \times p)$ .                                     |
| fong-spivak-invitation_F01897 | 178  | ■ 0.979 | $n \times p$                                                                | $n \times p$                                            | where each 0 represents a matrix of zeros of the appropriate dimension $(m \times q$ and $n \times p)$ .                                     |
| fong-spivak-invitation_F01898 | 178  | ■ 1.000 | $A+B$                                                                       | $A + B$                                                 | What is the direct sum matrix $A + B$ ?                                                                                                      |
| fong-spivak-invitation_F01900 | 179  | ■ 1.000 | $(1 \times 1)$                                                              | $(1 \times 1)$                                          | $(1 \times 1)$ -matrix $(a): 1 \rightarrow 1$ : it is an operation that takes an input $x \in R$ and returns $a * x$ .                       |
| fong-spivak-invitation_F01901 | 179  | ■ 1.000 | $(a): 1 \rightarrow 1$                                                      | $(a): 1 \rightarrow 1$                                  | $(1 \times 1)$ -matrix $(a): 1 \rightarrow 1$ : it is an operation that takes an input $x \in R$ and returns $a * x$ .                       |
| fong-spivak-invitation_F01902 | 179  | ■ 1.000 | $x \in R$                                                                   | $x \in R$                                               | $(1 \times 1)$ -matrix $(a): 1 \rightarrow 1$ : it is an operation that takes an input $x \in R$ and returns $a * x$ .                       |
| fong-spivak-invitation_F01903 | 179  | ■ 1.000 | $a * x$                                                                     | $a * x$                                                 | $(1 \times 1)$ -matrix $(a): 1 \rightarrow 1$ : it is an operation that takes an input $x \in R$ and returns $a * x$ .                       |
| fong-spivak-invitation_F01904 | 179  | ■ 0.588 | $\supset$                                                                   | $\supset$                                               | Similarly, we can interpret $\supset$ as the $(2 \times 1)$ -matrix $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ : it is an operation that takes a |
| fong-spivak-invitation_F01905 | 179  | ■ 0.588 | $(2 \times 1)$                                                              | $(2 \times 1)$                                          | Similarly, we can interpret $\supset$ as the $(2 \times 1)$ -matrix $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ : it is an operation that takes a |

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| fong-spivak-invitation_F01906 | 179  | 0.588 | $\binom{1}{1}$                                                                            | $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$                                                    | Similarly, we can interpret $\triangleright$ as the $(2 \times 1)$ -matrix $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ : it is an operation that takes a |
| fong-spivak-invitation_F01907 | 179  | 1.000 | $x+y$                                                                                     | $x + y$                                                                                   | row vector consisting of two inputs, $x$ and $y$ , and returns $x + y$ . Here is a table showing                                                    |
| fong-spivak-invitation_F01908 | 222  | 0.793 | $1 \rightarrow 1$                                                                         | $1 \rightarrow 1$                                                                         | Since this and the circuit in (6.17) are both morphisms $\underline{1} \rightarrow \underline{1}$ , we may compose them                             |
| fong-spivak-invitation_F01913 | 179  | 1.000 | $0 \times n$                                                                              | $0 \times n$                                                                              | dimensions. In linear algebra it is unusual to consider matrices of the form $0 \times n$ or $n \times 0$                                           |
| fong-spivak-invitation_F01914 | 179  | 1.000 | $n \times 0$                                                                              | $n \times 0$                                                                              | dimensions. In linear algebra it is unusual to consider matrices of the form $0 \times n$ or $n \times 0$                                           |
| fong-spivak-invitation_F01915 | 179  | 0.996 | $(0 \times 3)$                                                                            | $(0 \times 3)$                                                                            | can multiply a $(0 \times 3)$ -matrix by a $(3 \times n)$ -matrix for any $n$ , but you can not multiply it                                         |
| fong-spivak-invitation_F01916 | 179  | 0.996 | $(3 \times n)$                                                                            | $(3 \times n)$                                                                            | can multiply a $(0 \times 3)$ -matrix by a $(3 \times n)$ -matrix for any $n$ , but you can not multiply it                                         |
| fong-spivak-invitation_F01917 | 179  | 1.000 | $(2 \times n)$                                                                            | $(2 \times n)$                                                                            | by a $(2 \times n)$ -matrix.                                                                                                                        |
| fong-spivak-invitation_F01918 | 180  | 0.999 | $S: \mathbf{SFG}_R \rightarrow \mathbf{Mat}(R)$                                           | $S: \mathbf{SFG}_R \rightarrow \mathbf{Mat}(R)$                                           | <b>Theorem 5.45.</b> There is a prop functor $S: \mathbf{SFG}_R \rightarrow \mathbf{Mat}(R)$ that sends the generators                              |
| fong-spivak-invitation_F01919 | 180  | 1.000 | $S(g)$                                                                                    | $S(g)$                                                                                    | We have now constructed a matrix $S(g)$ from any signal flow graph $g$ . But how can we                                                             |
| fong-spivak-invitation_F01920 | 180  | 1.000 | $m \times n$                                                                              | $m \times n$                                                                              | for the generators in Definition 5.40, the associated matrix has dimension $m \times n$ , where                                                     |
| fong-spivak-invitation_F01921 | 180  | 0.999 | $(i, j)$                                                                                  | $(i, j)$                                                                                  | $m$ is the number of inputs and $n$ the number of outputs, with $(i, j)$ th entry describing the                                                    |
| fong-spivak-invitation_F01922 | 180  | 1.000 | $\mathrm{id}_0: 0 \rightarrow 0, \mathrm{id}_1: 1 \rightarrow 1, \sigma: 2 \rightarrow 2$ | $\mathrm{id}_0: 0 \rightarrow 0, \mathrm{id}_1: 1 \rightarrow 1, \sigma: 2 \rightarrow 2$ | built from the morphisms $\mathrm{id}_0: 0 \rightarrow 0, \mathrm{id}_1: 1 \rightarrow 1, \sigma: 2 \rightarrow 2$ , and the generators in          |
| fong-spivak-invitation_F01923 | 180  | 1.000 | $(\alpha+\beta):(m+p) \rightarrow (n+q)$                                                  | $(\alpha + \beta): (m + p) \rightarrow (n + q)$                                           | • If $\alpha: m \rightarrow n$ and $\beta: p \rightarrow q$ are expressions, then $(\alpha + \beta): (m + p) \rightarrow (n + q)$ is                |
| fong-spivak-invitation_F01924 | 180  | 1.000 | $\mathrm{id}_0, \mathrm{id}_1$                                                            | $\mathrm{id}_0, \mathrm{id}_1$                                                            | true when $g$ is equal to $\mathrm{id}_0, \mathrm{id}_1$ , and $\sigma$ .                                                                           |
| fong-spivak-invitation_F01925 | 180  | 0.853 | $0 \rightarrow 0$                                                                         | $0 \rightarrow 0$                                                                         | $(): 0 \rightarrow 0$ . The morphisms $\mathrm{id}_1, \sigma$ , and $a \in R$ map to the identity matrix, the swap                                  |
| fong-spivak-invitation_F01926 | 180  | 0.853 | $\mathrm{id}_1, \sigma$                                                                   | $\mathrm{id}_1, \sigma$                                                                   | $(): 0 \rightarrow 0$ . The morphisms $\mathrm{id}_1, \sigma$ , and $a \in R$ map to the identity matrix, the swap                                  |
| fong-spivak-invitation_F01927 | 180  | 0.607 | $n=p$                                                                                     | $n = p$                                                                                   | holds for (i) $\alpha \circ \beta$ (when $n = p$ ) and for (ii) $\alpha + \beta$ (in general).                                                      |
| fong-spivak-invitation_F01928 | 180  | 0.607 | $\alpha+\beta$                                                                            | $\alpha + \beta$                                                                          | holds for (i) $\alpha \circ \beta$ (when $n = p$ ) and for (ii) $\alpha + \beta$ (in general).                                                      |
| fong-spivak-invitation_F01929 | 181  | 1.000 | $\beta: n \rightarrow q$                                                                  | $\beta: n \rightarrow q$                                                                  | Here $\alpha: m \rightarrow n$ and $\beta: n \rightarrow q$ are signal flow graphs, assumed to obey the proposition.                                |
| fong-spivak-invitation_F01930 | 181  | 0.798 | $j \in \underline{n}$                                                                     | $j \in \underline{n}$                                                                     | to $k$ – of the amplification along that path. So let’s also fix some $j \in \underline{n}$ , and consider                                          |
| fong-spivak-invitation_F01931 | 181  | 0.627 | $\alpha /$                                                                                | $\alpha /$                                                                                | some $j$ th output of $\alpha$ /input of $\beta$ , the amplification that $i$ contributes to $k$ is                                                 |
| fong-spivak-invitation_F01932 | 181  | 0.983 | $S(\alpha) ; S(\beta)$                                                                    | $S(\alpha); S(\beta)$                                                                     | This is exactly the formula for matrix multiplication, which is composition $S(\alpha) \circ S(\beta)$                                              |
| fong-spivak-invitation_F01933 | 181  | 0.959 | $\mathbf{Mat}(R)$                                                                         | $\mathbf{Mat}(R)$                                                                         | in the prop $\mathbf{Mat}(R)$ ; see Definition 5.43. So $\alpha \circ \beta$ obeys the proposition when $\alpha$ and                                |
| fong-spivak-invitation_F01934 | 181  | 0.666 | $(m+i)$                                                                                   | $(m + i)$                                                                                 | output of $\alpha + \beta$ , the $i$ th input of $\beta$ is the $(m + i)$ th output of $\alpha + \beta$ , and the $j$ th out-                       |
| fong-spivak-invitation_F01935 | 181  | 0.556 | $n+j$                                                                                     | $n + j$                                                                                   | put of $\beta$ is the $(n + j)$ th output of $\alpha + \beta$ . This means that the matrix with $(i, j)$ th                                         |
| fong-spivak-invitation_F01936 | 181  | 1.000 | $S(\alpha)+S(\beta)=S(\alpha+\beta)$                                                      | $S(\alpha) + S(\beta) = S(\alpha + \beta)$                                                | $S(\alpha) + S(\beta) = S(\alpha + \beta)$ , as in Definition 5.43. This proves the proposition.                                                    |

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| fong-spivak-invitation_F01937 | 183  | ■ 1.000 | $M \in \operatorname{Mat}(R)$                                                                     | $M \in \operatorname{Mat}(R)$                                                                       | <b>Proposition 5.48.</b> Given any matrix $M \in \operatorname{Mat}(R)$ , there exists a signal flow graph                                         |
| fong-spivak-invitation_F01938 | 183  | ■ 1.000 | $g \in \mathbf{SFG}_R$                                                                            | $g \in \mathbf{SFG}_R$                                                                              | $g \in \mathbf{SFG}_R$ such that $S(g) = M$ .                                                                                                      |
| fong-spivak-invitation_F01939 | 183  | ■ 1.000 | $S(g)=M$                                                                                          | $S(g) = M$                                                                                          | $g \in \mathbf{SFG}_R$ such that $S(g) = M$ .                                                                                                      |
| fong-spivak-invitation_F01940 | 183  | ■ 1.000 | $S(g)(i, j)$                                                                                      | $S(g)(i, j)$                                                                                        | such that $S(g) = M$ . In particular, to compute $S(g)(i, j)$ , we know that we can sim-                                                           |
| fong-spivak-invitation_F01941 | 183  | ■ 1.000 | $M(i, j)$                                                                                         | $M(i, j)$                                                                                           | $M(i, j)$ .                                                                                                                                        |
| fong-spivak-invitation_F01942 | 183  | ■ 0.488 | $\left(\begin{array}{cc} a & b \\ c & d \end{array}\right)$                                       | $\left(\begin{array}{cc} a & b \\ c & d \end{array}\right)$                                         | $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . Then we define $g$ to be the signal flow graph                                                    |
| fong-spivak-invitation_F01943 | 183  | ■ 1.000 | $\left(\begin{array}{c} 0 \\ 1 \\ 2 \end{array}\right)$                                           | $\left(\begin{array}{c} 0 \\ 1 \\ 2 \end{array}\right)$                                             | 1. $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$                                                                                                     |
| fong-spivak-invitation_F01944 | 183  | ■ 1.000 | $\left(\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right)$                                       | $\left(\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right)$                                         | 2. $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$                                                                                                  |
| fong-spivak-invitation_F01945 | 183  | ■ 1.000 | $\left(\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right)$                              | $\left(\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right)$                                | 3. $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$                                                                                          |
| fong-spivak-invitation_F01946 | 184  | ■ 1.000 | $\mathbf{SFG}_R$                                                                                  | $\mathbf{SFG}_R$                                                                                    | the same as the set of generators for $\mathbf{SFG}_R$ ; see Definition 5.40.                                                                      |
| fong-spivak-invitation_F01947 | 184  | ■ 1.000 | $a, b \in R$                                                                                      | $a, b \in R$                                                                                        | We have the following equations for any $a, b \in R$ :                                                                                             |
| fong-spivak-invitation_F01948 | 185  | ■ 0.974 | $\binom{0}{6}$                                                                                    | $\binom{0}{6}$                                                                                      | <b>Example 5.52.</b> Both of the signal flow graphs below represent the same matrix, $\binom{0}{6}$ :                                              |
| fong-spivak-invitation_F01949 | 185  | ■ 0.665 | $R=(\mathbb{N}, 0, +, 1, *)$                                                                      | $R = (\mathbb{N}, 0, +, 1, *)$                                                                      | 1. Let $R = (\mathbb{N}, 0, +, 1, *)$ . By examining the presentation of $\operatorname{Mat}(R)$ in Theorem 5.51,                                  |
| fong-spivak-invitation_F01950 | 185  | ■ 1.000 | $R=\mathbb{N} / 3 \mathbb{N}$                                                                     | $R = \mathbb{N} / 3 \mathbb{N}$                                                                     | 2. Now suppose the rig is $R = \mathbb{N} / 3 \mathbb{N}$ ; if you do not know what this means, just replace                                       |
| fong-spivak-invitation_F01951 | 186  | ■ 0.780 | $(M, \mu, \eta)$                                                                                  | $(M, \mu, \eta)$                                                                                    | <b>Definition 5.55.</b> A monoid object $(M, \mu, \eta)$ in a symmetric monoidal category                                                          |
| fong-spivak-invitation_F01952 | 186  | ■ 1.000 | $\mu: M \otimes M \rightarrow M$                                                                  | $\mu: M \otimes M \rightarrow M$                                                                    | $(\mathcal{C}, I, \otimes)$ is an object $M$ of $\mathcal{C}$ together with morphisms $\mu: M \otimes M \rightarrow M$ and $\eta: I \rightarrow M$ |
| fong-spivak-invitation_F01953 | 186  | ■ 1.000 | $\eta: I \rightarrow M$                                                                           | $\eta: I \rightarrow M$                                                                             | $(\mathcal{C}, I, \otimes)$ is an object $M$ of $\mathcal{C}$ together with morphisms $\mu: M \otimes M \rightarrow M$ and $\eta: I \rightarrow M$ |
| fong-spivak-invitation_F01954 | 186  | ■ 0.993 | $(\mu \otimes \operatorname{id}) ; \mu=(\operatorname{id} \otimes \mu) ; \mu$                     | $(\mu \otimes \operatorname{id}); \mu = (\operatorname{id} \otimes \mu); \mu$                       | (a) $\mu \otimes \operatorname{id} \circ \mu = (\operatorname{id} \otimes \mu) \circ \mu$ ,                                                        |
| fong-spivak-invitation_F01955 | 186  | ■ 1.000 | $(\eta \otimes \operatorname{id}) ; \mu=\operatorname{id}=(\operatorname{id} \otimes \eta) ; \mu$ | $(\eta \otimes \operatorname{id}); \mu = \operatorname{id} = (\operatorname{id} \otimes \eta); \mu$ | (b) $\eta \otimes \operatorname{id} \circ \mu = \operatorname{id} = (\operatorname{id} \otimes \eta) \circ \mu$ .                                  |
| fong-spivak-invitation_F01956 | 186  | ■ 0.528 | $\sigma_{M, M}: \mu=\mu$                                                                          | $\sigma_{M, M}: \mu = \mu$                                                                          | (c) $\sigma_{M, M} \circ \mu = \mu$ ,                                                                                                              |
| fong-spivak-invitation_F01957 | 186  | ■ 1.000 | $\sigma_{M, M}$                                                                                   | $\sigma_{M, M}$                                                                                     | where $\sigma_{M, M}$ is the swap map on $M$ in $\mathcal{C}$ . We often denote it simply by $\sigma$ .                                            |
| fong-spivak-invitation_F01958 | 186  | ■ 1.000 | $\mu: M \times M \rightarrow M$                                                                   | $\mu: M \times M \rightarrow M$                                                                     | Example 2.5; see also Example 3.7. That is, it is a set $M$ , a function $\mu: M \times M \rightarrow M$ ,                                         |
| fong-spivak-invitation_F01959 | 186  | ■ 1.000 | $\eta(1) \in M$                                                                                   | $\eta(1) \in M$                                                                                     | which we denote by infix notation $*$ , and an element $\eta(1) \in M$ , which we denote by $e$ ,                                                  |
| fong-spivak-invitation_F01960 | 186  | ■ 1.000 | $(a * b) * c=a *(b * c)$                                                                          | $(a * b) * c = a * (b * c)$                                                                         | satisfying $a * b * c = a * (b * c)$ and $a * e = a = e * a$ .                                                                                     |
| fong-spivak-invitation_F01961 | 186  | ■ 1.000 | $a * e=a=e * a$                                                                                   | $a * e = a = e * a$                                                                                 | satisfying $(a * b) * c = a * (b * c)$ and $a * e = a = e * a$ .                                                                                   |

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| fong-spivak-invitation_F01962 | 186  | ■ 1.000 | $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ | $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ | 1. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a * b$ and if $\eta \in \mathbb{R}$ is defined to |
| fong-spivak-invitation_F01963 | 186  | ■ 1.000 | $\mu(a, b) = a * b$                                        | $\mu(a, b) = a * b$                                        | 1. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a * b$ and if $\eta \in \mathbb{R}$ is defined to |
| fong-spivak-invitation_F01964 | 186  | ■ 1.000 | $\eta \in \mathbb{R}$                                      | $\eta \in \mathbb{R}$                                      | 1. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a * b$ and if $\eta \in \mathbb{R}$ is defined to |
| fong-spivak-invitation_F01965 | 186  | ■ 0.997 | $\eta = 1$                                                 | $\eta = 1$                                                 | be $\eta = 1$ , then $(\mathbb{R}, *, 1)$ satisfies all three conditions of Definition 5.55.                                                            |
| fong-spivak-invitation_F01966 | 186  | ■ 0.997 | $(\mathbb{R}, *, 1)$                                       | $(\mathbb{R}, *, 1)$                                       | be $\eta = 1$ , then $(\mathbb{R}, *, 1)$ satisfies all three conditions of Definition 5.55.                                                            |
| fong-spivak-invitation_F01967 | 186  | ■ 0.999 | $\mu(a, b) = a + b$                                        | $\mu(a, b) = a + b$                                        | 2. Show that if $\mu: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is defined by $\mu(a, b) = a + b$ and if $\eta \in \mathbb{R}$ is defined    |
| fong-spivak-invitation_F01968 | 186  | ■ 0.892 | $\eta = 0$                                                 | $\eta = 0$                                                 | to be $\eta = 0$ , then $(\mathbb{R}, +, 0)$ satisfies all three conditions of Definition 5.55.                                                         |
| fong-spivak-invitation_F01969 | 186  | ■ 0.892 | $(\mathbb{R}, +, 0)$                                       | $(\mathbb{R}, +, 0)$                                       | to be $\eta = 0$ , then $(\mathbb{R}, +, 0)$ satisfies all three conditions of Definition 5.55.                                                         |
| fong-spivak-invitation_F01970 | 186  | ■ 0.361 | $\mu = \supset -$                                          | $\mu = \supset -$                                          | <b>Example 5.58.</b> Graphically, we can depict $\mu = \supset -$ and $\eta = \circ$ . Then axioms (a), (b), and                                        |
| fong-spivak-invitation_F01971 | 186  | ■ 0.361 | $\eta = \circ$                                             | $\eta = \circ$                                             | <b>Example 5.58.</b> Graphically, we can depict $\mu = \supset -$ and $\eta = \circ$ . Then axioms (a), (b), and                                        |
| fong-spivak-invitation_F01972 | 187  | ■ 0.998 | $U: \operatorname{Mat}(R) \rightarrow$                     | $U: \operatorname{Mat}(R) \rightarrow$                     | <b>Exercise 5.59.</b> For any rig $R$ , there is a functor $U: \mathbf{Mat}(R) \rightarrow \mathbf{Set}$ , sending the object                           |
| fong-spivak-invitation_F01973 | 187  | ■ 1.000 | $R^n$                                                      | $R^n$                                                      | $n \in \mathbb{N}$ to the set $R^n$ , and sending a morphism (matrix) $M: m \rightarrow n$ to the function                                              |
| fong-spivak-invitation_F01974 | 187  | ■ 1.000 | $M: m \rightarrow n$                                       | $M: m \rightarrow n$                                       | $n \in \mathbb{N}$ to the set $R^n$ , and sending a morphism (matrix) $M: m \rightarrow n$ to the function                                              |
| fong-spivak-invitation_F01975 | 187  | ■ 1.000 | $R^m \rightarrow R^n$                                      | $R^m \rightarrow R^n$                                      | $R^m \rightarrow R^n$ given by vector-matrix multiplication.                                                                                            |
| fong-spivak-invitation_F01976 | 187  | ■ 0.932 | $\{1\}$                                                    | $\{1\}$                                                    | is $\{1\}$ , a set with one element, and the monoidal product is $\times$ (see Example 4.38).                                                           |
| fong-spivak-invitation_F01977 | 187  | ■ 0.773 | $(U(M), U(\mu), U(\eta))$                                  | $(U(M), U(\mu), U(\eta))$                                  | 2. Show that if $(M, \mu, \eta)$ is a monoid object in $\mathbf{Mat}(R)$ then $(U(M), U(\mu), U(\eta))$ is a                                            |
| fong-spivak-invitation_F01978 | 187  | ■ 0.537 | $\operatorname{prop} \mathbf{Mat}(R)$                      | $\operatorname{prop} \mathbf{Mat}(R)$                      | in the prop $\mathbf{Mat}(R)$ . If $R = \mathbb{R}$ is the rig of real numbers, this means that we have                                                 |
| fong-spivak-invitation_F01979 | 187  | ■ 0.537 | $R = \mathbb{R}$                                           | $R = \mathbb{R}$                                           | in the prop $\mathbf{Mat}(R)$ . If $R = \mathbb{R}$ is the rig of real numbers, this means that we have                                                 |
| fong-spivak-invitation_F01980 | 187  | ■ 0.739 | $(1, -\subset, \rightarrow)$                               | $(1, -\subset, \rightarrow)$                               | <b>Example 5.60.</b> The triple $(1, -\subset, \rightarrow)$ in $\mathbf{Mat}(R)$ forms a commutative monoid object in                                  |
| fong-spivak-invitation_F01981 | 187  | ■ 0.428 | $\operatorname{Mat}(R)^{\operatorname{op}}$                | $\operatorname{Mat}(R)^{\operatorname{op}}$                | $\mathbf{Mat}(R)^{\operatorname{op}}$ . We hence also say that $(1, -\subset, \rightarrow)$ forms a <i>co-commutative comonoid object</i>               |
| fong-spivak-invitation_F01982 | 187  | ■ 1.000 | $\mu: 2 \rightarrow 1$                                     | $\mu: 2 \rightarrow 1$                                     | <b>Remark 5.64.</b> If we present a prop $\mathcal{M}$ using two generators $\mu: 2 \rightarrow 1$ and $\eta: 0 \rightarrow 1$ ,                        |
| fong-spivak-invitation_F01983 | 187  | ■ 1.000 | $\eta: 0 \rightarrow 1$                                    | $\eta: 0 \rightarrow 1$                                    | <b>Remark 5.64.</b> If we present a prop $\mathcal{M}$ using two generators $\mu: 2 \rightarrow 1$ and $\eta: 0 \rightarrow 1$ ,                        |
| fong-spivak-invitation_F01984 | 188  | ■ 1.000 | $\mathcal{M} \rightarrow \mathcal{C}$                      | $\mathcal{M} \rightarrow \mathcal{C}$                      | in $\mathcal{C}$ correspond to strict monoidal functors $\mathcal{M} \rightarrow \mathcal{C}$ . This sort of idea leads to the                          |
| fong-spivak-invitation_F01985 | 188  | ■ 0.503 | $-\subset$                                                 | $-\subset$                                                 | normally draw the copy icon like so, $-\subset$ , we could just as easily reverse it and draw an                                                        |
| fong-spivak-invitation_F01986 | 188  | ■ 1.000 | $g: m \rightarrow n$                                       | $g: m \rightarrow n$                                       | A signal flow graph $g: m \rightarrow n$ takes an input $x \in R^m$ and gives an output $y \in R^n$ .                                                   |
| fong-spivak-invitation_F01987 | 188  | ■ 1.000 | $x \in R^m$                                                | $x \in R^m$                                                | A signal flow graph $g: m \rightarrow n$ takes an input $x \in R^m$ and gives an output $y \in R^n$ .                                                   |
| fong-spivak-invitation_F01988 | 188  | ■ 1.000 | $y \in R^n$                                                | $y \in R^n$                                                | A signal flow graph $g: m \rightarrow n$ takes an input $x \in R^m$ and gives an output $y \in R^n$ .                                                   |

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| fong-spivak-invitation_F01989 | 188  | ■ 0.998 | $\mathrm{B}(g) \subseteq R^m \times R^n$                                      | $B(g) \subseteq R^m \times R^n$                           | <i>behavior</i> of $g$ and denote it $B(g) \subseteq R^m \times R^n$ . For example, the “copy” flow graph                                            |
| fong-spivak-invitation_F01990 | 188  | ■ 1.000 | $(x, (x, x))$                                                                 | $(x, (x, x))$                                             | behavior. Similarly, $(x, (x, x))$ is copy behavior for every $x \in R$ , thus we have                                                               |
| fong-spivak-invitation_F01991 | 188  | ■ 0.986 | $g^{\mathrm{op}}$                                                             | $g^{\mathrm{op}}$                                         | the inputs with the outputs, so if we denote this reversed icon by $g^{\mathrm{op}}$ , we must have                                                  |
| fong-spivak-invitation_F01992 | 188  | ■ 0.869 | $g^{\mathrm{op}}: n \rightarrow m$                                            | $g^{\mathrm{op}}: n \rightarrow m$                        | $g^{\mathrm{op}}: n \rightarrow m$ . Thus if $B(g) \subseteq R^m \times R^n$ then we need $B(g^{\mathrm{op}}) \subseteq R^n \times R^m$ . One simple |
| fong-spivak-invitation_F01993 | 188  | ■ 0.869 | $\mathrm{B}(\left(g^{\mathrm{op}}\right) \subseteq R^n \times R^m \times R^m$ | $B(g^{\mathrm{op}}) \subseteq R^n \times R^m$             | $g^{\mathrm{op}}: n \rightarrow m$ . Thus if $B(g) \subseteq R^m \times R^n$ then we need $B(g^{\mathrm{op}}) \subseteq R^n \times R^m$ . One simple |
| fong-spivak-invitation_F01994 | 188  | ■ 0.997 | $(b, a)$                                                                      | $(b, a)$                                                  | way to do this is to replace each $(a, b)$ with $(b, a)$ , so we would have                                                                          |
| fong-spivak-invitation_F01995 | 188  | ■ 0.473 | $\mathrm{B}(-\subset)$                                                        | $B(-\subset)$                                             | 1. What is the behavior $B(-\subset)$ of the reversed addition icon $\subset: 1 \rightarrow 2$ ?                                                     |
| fong-spivak-invitation_F01996 | 188  | ■ 0.473 | $-\subset: 1 \rightarrow 2$                                                   | $-\subset: 1 \rightarrow 2$                               | 1. What is the behavior $B(-\subset)$ of the reversed addition icon $\subset: 1 \rightarrow 2$ ?                                                     |
| fong-spivak-invitation_F01997 | 188  | ■ 0.237 | $\mathrm{B}(-)$                                                               | $B(-)$                                                    | 2. What is the behavior $B(\supset)$ of the reversed copy icon $\supset: 2 \rightarrow 1$ ?                                                          |
| fong-spivak-invitation_F01998 | 188  | ■ 0.237 | $-: 2 \rightarrow 1$                                                          | $-: 2 \rightarrow 1$                                      | 2. What is the behavior $B(\supset)$ of the reversed copy icon $\supset: 2 \rightarrow 1$ ?                                                          |
| fong-spivak-invitation_F01999 | 189  | ■ 1.000 | $B \subseteq R^m \times R^n$                                                  | $B \subseteq R^m \times R^n$                              | of a signal flow graph $m \rightarrow n$ is a subset $B \subseteq R^m \times R^n$ , i.e. a relation. Why not try to                                  |
| fong-spivak-invitation_F02000 | 189  | ■ 1.000 | $N: n \rightarrow p$                                                          | $N: n \rightarrow p$                                      | flow graphs with matrices $M: m \rightarrow n$ and $N: n \rightarrow p$ , we see that their behaviors are                                            |
| fong-spivak-invitation_F02001 | 189  | ■ 1.000 | $B_1 := \left\{ (x, Mx) \mid x \in R^m \right\}$                              | $B_1 := \{(x, Mx) \mid x \in R^m\}$                       | the relations $B_1 := \{(x, Mx) \mid x \in R^m\}$ and $B_2 := \{(y, Ny) \mid y \in R^n\}$ , while the                                                |
| fong-spivak-invitation_F02002 | 189  | ■ 1.000 | $B_2 := \left\{ (y, Ny) \mid y \in R^n \right\}$                              | $B_2 := \{(y, Ny) \mid y \in R^n\}$                       | the relations $B_1 := \{(x, Mx) \mid x \in R^m\}$ and $B_2 := \{(y, Ny) \mid y \in R^n\}$ , while the                                                |
| fong-spivak-invitation_F02003 | 189  | ■ 0.412 | $M \circ_9 N$                                                                 | $M \circ_9 N$                                             | behavior of $M \circ_9 N$ is the relation $\{(x, x \circ_9 M \circ_9 N) \mid x \in R^m\}$ . This is a case of relation                               |
| fong-spivak-invitation_F02004 | 189  | ■ 0.412 | $\left( \left( x, x \circ_9 M \circ_9 N \right) \mid x \in R^m \right)$       | $\{(x, x \circ_9 M \circ_9 N) \mid x \in R^m\}$           | behavior of $M \circ_9 N$ is the relation $\{(x, x \circ_9 M \circ_9 N) \mid x \in R^m\}$ . This is a case of relation                               |
| fong-spivak-invitation_F02005 | 189  | ■ 1.000 | $B_1 \subseteq R^m \times R^n$                                                | $B_1 \subseteq R^m \times R^n$                            | composition. Given relations $B_1 \subseteq R^m \times R^n$ and $B_2 \subseteq R^n \times R^p$ , their composite                                     |
| fong-spivak-invitation_F02006 | 189  | ■ 1.000 | $B_2 \subseteq R^n \times R^p$                                                | $B_2 \subseteq R^n \times R^p$                            | composition. Given relations $B_1 \subseteq R^m \times R^n$ and $B_2 \subseteq R^n \times R^p$ , their composite                                     |
| fong-spivak-invitation_F02007 | 189  | ■ 0.493 | $B_1 \circ B_2 \subseteq R^m \times R^p$                                      | $B_1 \circ B_2 \subseteq R^m \times R^p$                  | $B_1 \circ B_2 \subseteq R^m \times R^p$ is given by                                                                                                 |
| fong-spivak-invitation_F02008 | 189  | ■ 0.999 | $\mathbf{Rel}_R$                                                              | $\mathbf{Rel}_R$                                          | <b>Definition 5.66.</b> Let $R$ be a rig. We define the prop $\mathbf{Rel}_R$ of $R$ -relations to have sub-                                         |
| fong-spivak-invitation_F02009 | 189  | ■ 1.000 | $C \subseteq R^p \times R^q$                                                  | $C \subseteq R^p \times R^q$                              | prop $\mathbf{Rel}_R$ . If $B \subseteq R^m \times R^n$ and $C \subseteq R^p \times R^q$ are morphisms in $\mathbf{Rel}_R$ , write down              |
| fong-spivak-invitation_F02010 | 189  | ■ 1.000 | $B + C$                                                                       | $B + C$                                                   | $B + C$ in set notation.                                                                                                                             |
| fong-spivak-invitation_F02011 | 189  | ■ 0.689 | $g \in G_R$                                                                   | $g \in G_R$                                               | flow graphs. In Section 5.4.3, we wrote $g^{\mathrm{op}}$ for the mirror image of $g$ , for each $g \in G_R$ .                                       |
| fong-spivak-invitation_F02012 | 189  | ■ 1.000 | $G_R^{\mathrm{op}} := \{g^{\mathrm{op}} \mid g \in G_R\}$                     | $G_R^{\mathrm{op}} := \{g^{\mathrm{op}} \mid g \in G_R\}$ | So let’s write $G_R^{\mathrm{op}} := \{g^{\mathrm{op}} \mid g \in G_R\}$ for the set of all the mirror images of generators.                         |
| fong-spivak-invitation_F02013 | 189  | ■ 1.000 | $\mathbf{SFG}_R^+$                                                            | $\mathbf{SFG}_R^+$                                        | We call a morphism in the prop $\mathbf{SFG}_R^+$ a ( <i>non-simplified</i> ) <i>signal flow graph</i> : these extend                                |
| fong-spivak-invitation_F02014 | 190  | ■ 1.000 | $g: m \rightarrow n, h: \ell \rightarrow n$                                   | $g: m \rightarrow n, h: \ell \rightarrow n$               | <b>Exercise 5.68.</b> Let $g: m \rightarrow n, h: \ell \rightarrow n$ be signal flow graphs. Note that $h^{\mathrm{op}}: n \rightarrow \ell$         |
| fong-spivak-invitation_F02015 | 190  | ■ 1.000 | $h^{\mathrm{op}}: n \rightarrow \ell$                                         | $h^{\mathrm{op}}: n \rightarrow \ell$                     | <b>Exercise 5.68.</b> Let $g: m \rightarrow n, h: \ell \rightarrow n$ be signal flow graphs. Note that $h^{\mathrm{op}}: n \rightarrow \ell$         |

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| fong-spivak-invitation_F02016 | 190  | ■ 0.882 | $\left(h^{\mathrm{op}}\right)$                             | $(h^{\mathrm{op}})$                             | is a signal flow graph, and we can form the composite $g \circ (h^{\mathrm{op}})$ :                                                                    |
| fong-spivak-invitation_F02017 | 190  | ■ 0.615 | $\left(h^{\mathrm{op}}\right) \subseteq R^m \times R^\ell$ | $(h^{\mathrm{op}}) \subseteq R^m \times R^\ell$ | Show that the behavior of $g \circ (h^{\mathrm{op}}) \subseteq R^m \times R^\ell$ is equal to                                                          |
| fong-spivak-invitation_F02018 | 190  | ■ 0.986 | $g: m \rightarrow n, h: m \rightarrow p$                   | $g: m \rightarrow n, h: m \rightarrow p$        | <b>Exercise 5.69.</b> Let $g: m \rightarrow n, h: m \rightarrow p$ be signal flow graphs. Note that $(g^{\mathrm{op}}): n \rightarrow$                 |
| fong-spivak-invitation_F02019 | 190  | ■ 0.986 | $\left(g^{\mathrm{op}}\right): n \rightarrow$              | $(g^{\mathrm{op}}): n \rightarrow$              | <b>Exercise 5.69.</b> Let $g: m \rightarrow n, h: m \rightarrow p$ be signal flow graphs. Note that $(g^{\mathrm{op}}): n \rightarrow$                 |
| fong-spivak-invitation_F02020 | 190  | ■ 0.971 | $g^{\mathrm{op}}; h$                                       | $g^{\mathrm{op}}; h$                            | $m$ is a signal flow graph, and we can form the composite $g^{\mathrm{op}} \circ h$                                                                    |
| fong-spivak-invitation_F02021 | 190  | ■ 0.864 | $S(g) \in \operatorname{Mat}(R)$                           | $S(g) \in \operatorname{Mat}(R)$                | $S(g) \in \operatorname{Mat}(R)$ be the associated $(m \times n)$ -matrix (see Theorem 5.45).                                                          |
| fong-spivak-invitation_F02022 | 191  | ■ 0.981 | $b_1, b_2 \in \operatorname{B}(g)$                         | $b_1, b_2 \in \operatorname{B}(g)$              | subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$ , for any $r \in R$ . $\diamond$                        |
| fong-spivak-invitation_F02023 | 191  | ■ 0.981 | $b_1 + b_2$                                                | $b_1 + b_2$                                     | subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$ , for any $r \in R$ . $\diamond$                        |
| fong-spivak-invitation_F02024 | 191  | ■ 0.981 | $r * b_1$                                                  | $r * b_1$                                       | subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$ , for any $r \in R$ . $\diamond$                        |
| fong-spivak-invitation_F02025 | 191  | ■ 0.981 | $r \in R$                                                  | $r \in R$                                       | subspace. That is, if $b_1, b_2 \in \operatorname{B}(g)$ then so are $b_1 + b_2$ and $r * b_1$ , for any $r \in R$ . $\diamond$                        |
| fong-spivak-invitation_F02026 | 191  | ■ 0.840 | $\mathbf{LinRel}_R$                                        | $\mathbf{LinRel}_R$                             | <b>Exercise 5.71.</b> One might want to show that linear relations on $R$ form a prop, $\mathbf{LinRel}_R$ .                                           |
| fong-spivak-invitation_F02027 | 191  | ■ 0.997 | $(x, y) \in B$                                             | $(x, y) \in B$                                  | that $B$ is linear. In other words, where for any $(x, y) \in B$ and $r \in R$ the element                                                             |
| fong-spivak-invitation_F02028 | 191  | ■ 0.987 | $(r * x, r * y) \in R^m \times R^n$                        | $(r * x, r * y) \in R^m \times R^n$             | $(r * x, r * y) \in R^m \times R^n$ is in $B$ , and for any $(x', y') \in B$ , the element $(x + x', y + y')$                                          |
| fong-spivak-invitation_F02029 | 191  | ■ 0.987 | $\left(x^{\prime}, y^{\prime}\right) \in B$                | $(x', y') \in B$                                | $(r * x, r * y) \in R^m \times R^n$ is in $B$ , and for any $(x', y') \in B$ , the element $(x + x', y + y')$                                          |
| fong-spivak-invitation_F02030 | 191  | ■ 0.987 | $\left(x+x^{\prime}, y+y^{\prime}\right)$                  | $(x + x', y + y')$                              | $(r * x, r * y) \in R^m \times R^n$ is in $B$ , and for any $(x', y') \in B$ , the element $(x + x', y + y')$                                          |
| fong-spivak-invitation_F02031 | 191  | ■ 0.644 | $G_R \sqcup G_R^{\mathrm{op}}$                             | $G_R \sqcup G_R^{\mathrm{op}}$                  | $R$ . Moreover, it is possible to give such a presentation whose generating set is $G_R \sqcup G_R^{\mathrm{op}}$                                      |
| fong-spivak-invitation_F02032 | 192  | ■ 1.000 | $\eta_1$                                                   | $\eta_1$                                        | from Eq. (5.15) serve as the $\eta_1$ and $\epsilon_1$ from Definition 4.44. Using monoidal products                                                   |
| fong-spivak-invitation_F02033 | 192  | ■ 1.000 | $\epsilon_1$                                               | $\epsilon_1$                                    | from Eq. (5.15) serve as the $\eta_1$ and $\epsilon_1$ from Definition 4.44. Using monoidal products                                                   |
| fong-spivak-invitation_F02034 | 192  | ■ 1.000 | $n = n^*$                                                  | $n = n^*$                                       | $n \in \mathbb{N}$ is dual to itself, $n = n^*$ .                                                                                                      |
| fong-spivak-invitation_F02035 | 192  | ■ 0.999 | $\mathbb{R}[s, s^{-1}]$                                    | $\mathbb{R}[s, s^{-1}]$                         | speed $u$ . We'll work in signal flow diagrams over the rig $\mathbb{R}[s, s^{-1}]$ of polynomials in $s$                                              |
| fong-spivak-invitation_F02036 | 192  | ■ 1.000 | $s^{-1}$                                                   | $s^{-1}$                                        | and $s^{-1}$ with coefficients in $\mathbb{R}$ and where $ss^{-1} = s^{-1}s = 1$ . This is standard in control                                         |
| fong-spivak-invitation_F02037 | 192  | ■ 1.000 | $ss^{-1} = s^{-1}s = 1$                                    | $ss^{-1} = s^{-1}s = 1$                         | and $s^{-1}$ with coefficients in $\mathbb{R}$ and where $ss^{-1} = s^{-1}s = 1$ . This is standard in control                                         |
| fong-spivak-invitation_F02038 | 192  | ■ 1.000 | $a * u + b * v$                                            | $a * u + b * v$                                 | will take some linear combination $a * u + b * v$ of the desired speed and actual speed,                                                               |
| fong-spivak-invitation_F02039 | 192  | ■ 1.000 | $\{(F, u, v) \mid u = v\}$                                 | $\{(F, u, v) \mid u = v\}$                      | this signal flow graph close to the relation $\{(F, u, v) \mid u = v\}$ ? By phrasing problems                                                         |
| fong-spivak-invitation_F02040 | 196  | ■ 0.824 | $\operatorname{Cospan}_{\mathcal{C}}$                      | $\operatorname{Cospan}_{\mathcal{C}}$           | that of <i>cospan</i> s in $\mathcal{C}$ , denoted $\mathbf{Cospan}_{\mathcal{C}}$ – that can conveniently package $\mathcal{C}$ 's colimits in        |
| fong-spivak-invitation_F02041 | 196  | ■ 1.000 | $s_{m, n}: m \rightarrow n$                                | $s_{m, n}: m \rightarrow n$                     | an icon $s_{m, n}: m \rightarrow n$ . These icons, known as <i>spiders</i> , <sup>3</sup> are drawn as follows:                                        |
| fong-spivak-invitation_F02042 | 197  | ■ 0.988 | $\varnothing \in \mathcal{C}$                              | $\varnothing \in \mathcal{C}$                   | <b>Definition 6.1.</b> Let $\mathcal{C}$ be a category. An <i>initial object</i> in $\mathcal{C}$ is an object $\varnothing \in \mathcal{C}$ such that |

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| fong-spivak-invitation_F02043 | 197  | ■ 1.000 | $!_{\{T\}}: \varnothing \rightarrow T$    | $!_T: \varnothing \rightarrow T$          | for each object $T$ in $\mathcal{C}$ there exists a unique morphism $!_T: \varnothing \rightarrow T$ .            |
| fong-spivak-invitation_F02044 | 198  | ■ 0.992 | $A, \leq$                                 | $A, \leq$                                 | 1. $(A, \leq)$ has no initial object.                                                                             |
| fong-spivak-invitation_F02045 | 198  | ■ 0.996 | $(A, \leq)$                               | $(A, \leq)$                               | 3. $(A, \leq)$ has two initial objects.                                                                           |
| fong-spivak-invitation_F02046 | 198  | ■ 1.000 | $\varnothing \rightarrow T$               | $\varnothing \rightarrow T$               | is a unique function $\varnothing \rightarrow T$ , since $\varnothing$ has no elements.                           |
| fong-spivak-invitation_F02047 | 198  | ■ 0.860 | $T \in \mathrm{Ob}(\mathcal{C})$          | $T \in \mathrm{Ob}(\mathcal{C})$          | need to show that for each object $T \in \mathrm{Ob}(\mathcal{C})$ (i.e. for both $T = A$ and $T = B$ )           |
| fong-spivak-invitation_F02048 | 198  | ■ 0.860 | $T=A$                                     | $T = A$                                   | need to show that for each object $T \in \mathrm{Ob}(\mathcal{C})$ (i.e. for both $T = A$ and $T = B$ )           |
| fong-spivak-invitation_F02049 | 198  | ■ 0.860 | $T=B$                                     | $T = B$                                   | need to show that for each object $T \in \mathrm{Ob}(\mathcal{C})$ (i.e. for both $T = A$ and $T = B$ )           |
| fong-spivak-invitation_F02050 | 198  | ■ 0.791 | $\varnothing =$                           | $\varnothing =$                           | there is a unique morphism $\varnothing \rightarrow T$ . Trying the case $\varnothing =^? A$ this condition fails |
| fong-spivak-invitation_F02051 | 198  | ■ 1.000 | $\varnothing =^? B$                       | $\varnothing =^? B$                       | $\varnothing =^? B$ this condition fails when $T = A$ : there are zero morphisms $B \rightarrow A$ , not          |
| fong-spivak-invitation_F02052 | 198  | ■ 1.000 | $B \rightarrow A$                         | $B \rightarrow A$                         | $\varnothing =^? B$ this condition fails when $T = A$ : there are zero morphisms $B \rightarrow A$ , not          |
| fong-spivak-invitation_F02053 | 199  | ■ 0.977 | $\left( R, 0_R, +_R, 1_R, *_R \right)$    | $(R, 0_R, +_R, 1_R, *_R)$                 | $(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) =$    |
| fong-spivak-invitation_F02054 | 199  | ■ 0.977 | $\left( S, 0_S, +_S, 1_S, *_S \right)$    | $(S, 0_S, +_S, 1_S, *_S)$                 | $(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) =$    |
| fong-spivak-invitation_F02055 | 199  | ■ 0.977 | $f: R \rightarrow S$                      | $f: R \rightarrow S$                      | $(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) =$    |
| fong-spivak-invitation_F02056 | 199  | ■ 0.977 | $f(0_R) =$                                | $f(0_R) =$                                | $(R, 0_R, +_R, 1_R, *_R)$ to $(S, 0_S, +_S, 1_S, *_S)$ is a function $f: R \rightarrow S$ such that $f(0_R) =$    |
| fong-spivak-invitation_F02057 | 199  | ■ 1.000 | $0_S, f(r_1 +_R r_2) = f(r_1) +_S f(r_2)$ | $0_S, f(r_1 +_R r_2) = f(r_1) +_S f(r_2)$ | $0_S, f(r_1 +_R r_2) = f(r_1) +_S f(r_2)$ , etc.                                                                  |
| fong-spivak-invitation_F02058 | 199  | ■ 1.000 | $c_{\{1\}}$                               | $c_1$                                     | <b>Exercise 6.10.</b> Let $\mathcal{C}$ be a category, and suppose that $c_1$ and $c_2$ are initial objects. Find |
| fong-spivak-invitation_F02059 | 199  | ■ 1.000 | $c_{\{2\}}$                               | $c_2$                                     | <b>Exercise 6.10.</b> Let $\mathcal{C}$ be a category, and suppose that $c_1$ and $c_2$ are initial objects. Find |
| fong-spivak-invitation_F02060 | 199  | ■ 0.998 | $\iota_A: A \rightarrow$                  | $\iota_A: A \rightarrow$                  | is an object, which we denote $A + B$ , together with a pair of morphisms $(\iota_A: A \rightarrow$               |
| fong-spivak-invitation_F02061 | 199  | ■ 0.900 | $A+B, \iota_B: B \rightarrow A+B$         | $A + B, \iota_B: B \rightarrow A + B$     | $(A + B, \iota_B: B \rightarrow A + B)$ such that, for all objects $T$ and pairs of morphisms $(f: A \rightarrow$ |
| fong-spivak-invitation_F02062 | 199  | ■ 0.900 | $f: A \rightarrow$                        | $f: A \rightarrow$                        | $A + B, \iota_B: B \rightarrow A + B)$ such that, for all objects $T$ and pairs of morphisms $(f: A \rightarrow$  |
| fong-spivak-invitation_F02063 | 199  | ■ 0.999 | $T, g: B \rightarrow T$                   | $T, g: B \rightarrow T$                   | $T, g: B \rightarrow T)$ , there exists a unique morphism $[f, g]: A + B \rightarrow T$ such that the             |
| fong-spivak-invitation_F02064 | 199  | ■ 0.999 | $[f, g]: A+B \rightarrow T$               | $[f, g]: A + B \rightarrow T$             | $T, g: B \rightarrow T)$ , there exists a unique morphism $[f, g]: A + B \rightarrow T$ such that the             |
| fong-spivak-invitation_F02065 | 199  | ■ 1.000 | $[f, g]$                                  | $[f, g]$                                  | We call $[f, g]$ the <i>copairing</i> of $f$ and $g$ .                                                            |
| fong-spivak-invitation_F02066 | 200  | ■ 0.997 | $\iota_A: A \rightarrow A \sqcup B$       | $\iota_A: A \rightarrow A \sqcup B$       | the disjoint union $A \sqcup B$ together with the inclusion functions $\iota_A: A \rightarrow A \sqcup B$ and     |
| fong-spivak-invitation_F02067 | 200  | ■ 1.000 | $\iota_B: B \rightarrow A \sqcup B$       | $\iota_B: B \rightarrow A \sqcup B$       | $\iota_B: B \rightarrow A \sqcup B$ .                                                                             |
| fong-spivak-invitation_F02068 | 200  | ■ 1.000 | $f: A \rightarrow T$                      | $f: A \rightarrow T$                      | Suppose we have functions $f: A \rightarrow T$ and $g: B \rightarrow T$ for some other set                        |
| fong-spivak-invitation_F02069 | 200  | ■ 1.000 | $g: B \rightarrow T$                      | $g: B \rightarrow T$                      | Suppose we have functions $f: A \rightarrow T$ and $g: B \rightarrow T$ for some other set                        |

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| fong-spivak-invitation_F02876 | 200  | ■ 0.804 | $[f, g]: A \sqcup B \rightarrow T$                       | $[f, g]: A \sqcup B \rightarrow T$                       | $[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$ . What is it? Any           |
| fong-spivak-invitation_F02871 | 200  | ■ 0.804 | $\iota_A$                                                | $\iota_A$                                                | $[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$ . What is it? Any           |
| fong-spivak-invitation_F02872 | 200  | ■ 0.804 | $[f, g]=f$                                               | $[f, g] = f$                                             | $[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$ . What is it? Any           |
| fong-spivak-invitation_F02873 | 200  | ■ 0.804 | $\iota_B$                                                | $\iota_B$                                                | $[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$ . What is it? Any           |
| fong-spivak-invitation_F02874 | 200  | ■ 0.804 | $[f, g]=g$                                               | $[f, g] = g$                                             | $[f, g]: A \sqcup B \rightarrow T$ such that $\iota_A \circ [f, g] = f$ and $\iota_B \circ [f, g] = g$ . What is it? Any           |
| fong-spivak-invitation_F02875 | 200  | ■ 0.997 | $x \in A \sqcup B$                                       | $x \in A \sqcup B$                                       | element $x \in A \sqcup B$ is either “from $A$ ” or “from $B$ ,” i.e. either there is some $a \in A$ with                          |
| fong-spivak-invitation_F02876 | 200  | ■ 0.998 | $x=\iota_A(a)$                                           | $x = \iota_A(a)$                                         | $x = \iota_A(a)$ or there is some $b \in B$ with $x = \iota_B(b)$ . By Eq. (6.1), we must have:                                    |
| fong-spivak-invitation_F02877 | 200  | ■ 0.998 | $b \in B$                                                | $b \in B$                                                | $x = \iota_A(a)$ or there is some $b \in B$ with $x = \iota_B(b)$ . By Eq. (6.1), we must have:                                    |
| fong-spivak-invitation_F02878 | 200  | ■ 0.998 | $x=\iota_B(b)$                                           | $x = \iota_B(b)$                                         | $x = \iota_A(a)$ or there is some $b \in B$ with $x = \iota_B(b)$ . By Eq. (6.1), we must have:                                    |
| fong-spivak-invitation_F02879 | 200  | ■ 1.000 | $T=\{a, b, c, \ldots, z\}$                               | $T = \{a, b, c, \ldots, z\}$                             | <b>Exercise 6.14.</b> Suppose $T = \{a, b, c, \ldots, z\}$ is the set of letters in the alphabet, and                              |
| fong-spivak-invitation_F02880 | 200  | ■ 0.999 | $f($                                                     | $f($                                                     | element of $A$ to the first letter of its label, e.g. $f(\text{apple}) = a$ . Let $g: B \rightarrow T$ be the                      |
| fong-spivak-invitation_F02881 | 200  | ■ 0.999 | $)=a$                                                    | $) = a$                                                  | element of $A$ to the first letter of its label, e.g. $f(\text{apple}) = a$ . Let $g: B \rightarrow T$ be the                      |
| fong-spivak-invitation_F02882 | 200  | ■ 0.993 | $g($                                                     | $g($                                                     | function sending each element of $B$ to the last letter of its label, e.g. $g(\text{apple}) = e$ .                                 |
| fong-spivak-invitation_F02883 | 200  | ■ 0.993 | $)=e$                                                    | $) = e$                                                  | function sending each element of $B$ to the last letter of its label, e.g. $g(\text{apple}) = e$ .                                 |
| fong-spivak-invitation_F02884 | 200  | ■ 1.000 | $[f, g](x)$                                              | $[f, g](x)$                                              | Write down the function $[f, g](x)$ for all eight elements of $A \sqcup B$ .                                                       |
| fong-spivak-invitation_F02885 | 200  | ■ 1.000 | $f: A \rightarrow C, g: B \rightarrow C$                 | $f: A \rightarrow C, g: B \rightarrow C$                 | <b>Exercise 6.15.</b> Let $f: A \rightarrow C, g: B \rightarrow C$ , and $h: C \rightarrow D$ be morphisms in a                    |
| fong-spivak-invitation_F02886 | 200  | ■ 0.359 | $\iota_A; [f, g]=f$                                      | $\iota_A; [f, g] = f$                                    | 1. $\boxed{\iota_A \circ [f, g] = f}$ ,                                                                                            |
| fong-spivak-invitation_F02887 | 200  | ■ 0.859 | $[f, g] ; h=[f ; h, g ; h]$                              | $[f, g]; h = [f; h, g; h]$                               | 3. $\boxed{[f, g] \circ h = [f \circ h, g \circ h]}$ ,                                                                             |
| fong-spivak-invitation_F02888 | 200  | ■ 1.000 | $\left[\iota_A, \iota_B\right]=\mathrm{id}_{A+B}$        | $[\iota_A, \iota_B] = \mathrm{id}_{A+B}$                 | 4. $\boxed{[\iota_A, \iota_B] = \mathrm{id}_{A+B}}$ .                                                                              |
| fong-spivak-invitation_F02889 | 200  | ■ 0.981 | $(\mathcal{C}, +, \varnothing)$                          | $(\mathcal{C}, +, \varnothing)$                          | denoted $\varnothing$ . Then $(\mathcal{C}, +, \varnothing)$ is a symmetric monoidal category (recall Definition 4.34).            |
| fong-spivak-invitation_F02890 | 200  | ■ 0.807 | $\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ | $\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ | 1. Show that $+$ extends to a functor $\mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ . In particular, how does it act on |
| fong-spivak-invitation_F02891 | 200  | ■ 0.963 | $\mathcal{C} \times \mathcal{C}$                         | $\mathcal{C} \times \mathcal{C}$                         | morphisms in $\mathcal{C} \times \mathcal{C}$ ?                                                                                    |
| fong-spivak-invitation_F02892 | 201  | ■ 1.000 | $A+\varnothing \rightarrow A$                            | $A + \varnothing \rightarrow A$                          | isomorphisms $A + \varnothing \rightarrow A$ and $\varnothing + A \rightarrow A$ .                                                 |
| fong-spivak-invitation_F02893 | 201  | ■ 1.000 | $\varnothing+A \rightarrow A$                            | $\varnothing + A \rightarrow A$                          | isomorphisms $A + \varnothing \rightarrow A$ and $\varnothing + A \rightarrow A$ .                                                 |
| fong-spivak-invitation_F02894 | 201  | ■ 1.000 | $(A+B)+C \rightarrow A+(B+C)$                            | $(A + B) + C \rightarrow A + (B + C)$                    | (a) $\boxed{(A + B) + C \rightarrow A + (B + C)}$ ,                                                                                |
| fong-spivak-invitation_F02895 | 201  | ■ 1.000 | $A+B \rightarrow B+A$                                    | $A + B \rightarrow B + A$                                | (b) $\boxed{A + B \rightarrow B + A}$ .                                                                                            |
| fong-spivak-invitation_F02896 | 201  | ■ 1.000 | $f: A \rightarrow X$                                     | $f: A \rightarrow X$                                     | <b>Definition 6.17.</b> Let $\mathcal{C}$ be a category and let $f: A \rightarrow X$ and $g: A \rightarrow Y$ be morphisms         |

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| fong-spivak-invitation_F02097 | 201  | ■ 1.000 | $g: A \rightarrow Y$                                                  | $g: A \rightarrow Y$                                  | <b>Definition 6.17.</b> Let $\mathcal{C}$ be a category and let $f: A \rightarrow X$ and $g: A \rightarrow Y$ be morphisms                                                |
| fong-spivak-invitation_F02098 | 201  | ■ 0.996 | $X + \{ \}_{A} Y$                                                     | $X + {}_A Y$                                          | in $\mathcal{C}$ that have a common domain. The <i>pushout</i> $X + {}_A Y$ is the colimit of the diagram                                                                 |
| fong-spivak-invitation_F02099 | 201  | ■ 0.712 | $X + \{A\} Y$                                                         | $X + {}_A Y$                                          | In more detail, a pushout consists of (i) an object $X + {}_A Y$ and (ii) morphisms $\iota_X: X \rightarrow$                                                              |
| fong-spivak-invitation_F02100 | 201  | ■ 0.712 | $\iota_X: X \rightarrow$                                              | $\iota_X: X \rightarrow$                              | In more detail, a pushout consists of (i) an object $X + {}_A Y$ and (ii) morphisms $\iota_X: X \rightarrow$                                                              |
| fong-spivak-invitation_F02101 | 201  | ■ 0.997 | $\iota_Y: Y \rightarrow X + \{ \}_{A} Y$                              | $\iota_Y: Y \rightarrow X + {}_A Y$                   | $X + {}_A Y$ and $\iota_Y: Y \rightarrow X + {}_A Y$ satisfying (a) and (b) below.                                                                                        |
| fong-spivak-invitation_F02102 | 201  | ■ 0.989 | $\ulcorner$                                                           | $\ulcorner$                                           | commutes. (We will explain the ‘ $\ulcorner$ ’ symbol below.)                                                                                                             |
| fong-spivak-invitation_F02103 | 201  | ■ 1.000 | $x: X \rightarrow T, y: Y \rightarrow T$                              | $x: X \rightarrow T, y: Y \rightarrow T$              | (b) For all objects $T$ and morphisms $x: X \rightarrow T, y: Y \rightarrow T$ , if the diagram                                                                           |
| fong-spivak-invitation_F02104 | 201  | ■ 0.955 | $t: X + \{ \}_{A} Y \rightarrow T$                                    | $t: X + {}_A Y \rightarrow T$                         | commutes, then there exists a unique morphism $t: X + {}_A Y \rightarrow T$ such that                                                                                     |
| fong-spivak-invitation_F02105 | 202  | ■ 1.000 | $\iota_X$                                                             | $\iota_X$                                             | We further call $\iota_X$ the <i>pushout of <math>g</math> along <math>f</math></i> , and similarly $\iota_Y$ the <i>pushout of <math>f</math> along <math>g</math></i> . |
| fong-spivak-invitation_F02106 | 202  | ■ 1.000 | $\iota_Y$                                                             | $\iota_Y$                                             | We further call $\iota_X$ the <i>pushout of <math>g</math> along <math>f</math></i> , and similarly $\iota_Y$ the <i>pushout of <math>f</math> along <math>g</math></i> . |
| fong-spivak-invitation_F02107 | 202  | ■ 1.000 | $B \leftarrow A \rightarrow C$                                        | $B \leftarrow A \rightarrow C$                        | pushout of a diagram $B \leftarrow A \rightarrow C$ is equal to the coproduct $B \sqcup C$ : namely, both                                                                 |
| fong-spivak-invitation_F02108 | 202  | ■ 1.000 | $B \sqcup C$                                                          | $B \sqcup C$                                          | pushout of a diagram $B \leftarrow A \rightarrow C$ is equal to the coproduct $B \sqcup C$ : namely, both                                                                 |
| fong-spivak-invitation_F02109 | 202  | ■ 1.000 | $B \vee C$                                                            | $B \vee C$                                            | are equal to the join $B \vee C$ .                                                                                                                                        |
| fong-spivak-invitation_F02110 | 202  | ■ 1.000 | $i: A \rightarrow A^{\prime}$                                         | $i: A \rightarrow A'$                                 | $i: A \rightarrow A'$ and $j: X \rightarrow X'$ , we can take $X'$ to be the pushout $X + {}_A A'$ , i.e. the following                                                   |
| fong-spivak-invitation_F02111 | 202  | ■ 1.000 | $j: X \rightarrow X^{\prime}$                                         | $j: X \rightarrow X'$                                 | $i: A \rightarrow A'$ and $j: X \rightarrow X'$ , we can take $X'$ to be the pushout $X + {}_A A'$ , i.e. the following                                                   |
| fong-spivak-invitation_F02112 | 202  | ■ 1.000 | $X^{\prime}$                                                          | $X'$                                                  | $i: A \rightarrow A'$ and $j: X \rightarrow X'$ , we can take $X'$ to be the pushout $X + {}_A A'$ , i.e. the following                                                   |
| fong-spivak-invitation_F02113 | 202  | ■ 1.000 | $X + \{ \}_{A} A^{\prime}$                                            | $X + {}_A A'$                                         | $i: A \rightarrow A'$ and $j: X \rightarrow X'$ , we can take $X'$ to be the pushout $X + {}_A A'$ , i.e. the following                                                   |
| fong-spivak-invitation_F02114 | 202  | ■ 0.659 | $f^{\prime} := i^{-1} \circ f \circ j$                                | $f' := i^{-1} \circ f \circ j$                        | where $f' := i^{-1} \circ f \circ j$ . To see this, observe that if there is any object $T$ such that the                                                                 |
| fong-spivak-invitation_F02115 | 202  | ■ 0.568 | $x=i$                                                                 | $x = i$                                               | then $f \circ x = i \circ a$ , and so we are forced to take $x': X \rightarrow T$ to be $x' := j^{-1} \circ x$ . This                                                     |
| fong-spivak-invitation_F02116 | 202  | ■ 0.568 | $x^{\prime}: X \rightarrow T$                                         | $x': X \rightarrow T$                                 | then $f \circ x = i \circ a$ , and so we are forced to take $x': X \rightarrow T$ to be $x' := j^{-1} \circ x$ . This                                                     |
| fong-spivak-invitation_F02117 | 202  | ■ 0.568 | $x^{\prime}:=j^{-1} ; x$                                              | $x' := j^{-1}; x$                                     | then $f \circ x = i \circ a$ , and so we are forced to take $x': X \rightarrow T$ to be $x' := j^{-1} \circ x$ . This                                                     |
| fong-spivak-invitation_F02118 | 202  | ■ 0.978 | $f^{\prime}; x^{\prime}=i^{-1} ; f ; j ; j^{-1} ; x=i^{-1} ; i ; a=a$ | $f'; x' = i^{-1}; f; j; j^{-1}; x = i^{-1}; i; a = a$ | because $f' \circ x' = i^{-1} \circ f \circ j \circ j^{-1} \circ x = i^{-1} \circ i \circ a = a$ .                                                                        |
| fong-spivak-invitation_F02119 | 203  | ■ 1.000 | $\{f(a) \sim g(a) \mid a \in A\}$                                     | $\{f(a) \sim g(a) \mid a \in A\}$                     | relation $\{f(a) \sim g(a) \mid a \in A\}$ .                                                                                                                              |
| fong-spivak-invitation_F02120 | 203  | ■ 1.000 | $f(a)$                                                                | $f(a)$                                                | a connection between $f(a)$ in $X$ and $g(a)$ in $Y$ . The pushout is the set of connected                                                                                |
| fong-spivak-invitation_F02121 | 203  | ■ 1.000 | $g(a)$                                                                | $g(a)$                                                | a connection between $f(a)$ in $X$ and $g(a)$ in $Y$ . The pushout is the set of connected                                                                                |
| fong-spivak-invitation_F02122 | 203  | ■ 1.000 | $f: \underline{4} \rightarrow \underline{5}$                          | $f: \underline{4} \rightarrow \underline{5}$          | <b>Exercise 6.22.</b> What is the pushout of the functions $f: \underline{4} \rightarrow \underline{5}$ and $g: \underline{4} \rightarrow \underline{3}$ pictured         |
| fong-spivak-invitation_F02123 | 203  | ■ 1.000 | $g: \underline{4} \rightarrow \underline{3}$                          | $g: \underline{4} \rightarrow \underline{3}$          | <b>Exercise 6.22.</b> What is the pushout of the functions $f: \underline{4} \rightarrow \underline{5}$ and $g: \underline{4} \rightarrow \underline{3}$ pictured         |

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| fong-spivak-invitation_F02124 | 203  | ■ 1.000 | $X, Y \in$                                                           | $X, Y \in$                                     | <b>Example 6.23.</b> Suppose a category $\mathcal{C}$ has an initial object $\emptyset$ . For any two objects $X, Y \in$                      |
| fong-spivak-invitation_F02125 | 203  | ■ 0.884 | $f: \varnothing \rightarrow X$                                       | $f: \emptyset \rightarrow X$                   | $\text{Ob } \mathcal{C}$ , there is a unique morphism $f: \emptyset \rightarrow X$ and a unique morphism $g: \emptyset \rightarrow Y$ ; this  |
| fong-spivak-invitation_F02126 | 203  | ■ 0.884 | $g: \varnothing \rightarrow Y$                                       | $g: \emptyset \rightarrow Y$                   | $\text{Ob } \mathcal{C}$ , there is a unique morphism $f: \emptyset \rightarrow X$ and a unique morphism $g: \emptyset \rightarrow Y$ ; this  |
| fong-spivak-invitation_F02127 | 203  | ■ 0.923 | $X \stackrel{f}{\leftarrow} \varnothing \stackrel{g}{\rightarrow} Y$ | $X \xleftarrow{f} \emptyset \xrightarrow{g} Y$ | The diagram $X \xleftarrow{f} \emptyset \xrightarrow{g} Y$ has a pushout in $\mathcal{C}$ iff $X$ and $Y$ have a coproduct in $\mathcal{C}$ , |
| fong-spivak-invitation_F02128 | 203  | ■ 1.000 | $X+Y$                                                                | $X + Y$                                        | coproduct $X + Y$ ; then the diagram to the left below                                                                                        |
| fong-spivak-invitation_F02129 | 203  | ■ 1.000 | $X+Y \rightarrow T$                                                  | $X + Y \rightarrow T$                          | right, there is a unique map $X + Y \rightarrow T$ making the diagram as in Eq. (6.4) commute                                                 |
| fong-spivak-invitation_F02130 | 203  | ■ 0.968 | $\{ \}_{2}$                                                          | $2$                                            | (why? <sub>2</sub> ). This shows that $X + Y$ is a pushout, $X +_{\emptyset} Y \cong X + Y$ .                                                 |
| fong-spivak-invitation_F02131 | 203  | ■ 0.968 | $X+\varnothing Y \cong X+Y$                                          | $X + \emptyset Y \cong X + Y$                  | (why? <sub>2</sub> ). This shows that $X + Y$ is a pushout, $X +_{\emptyset} Y \cong X + Y$ .                                                 |
| fong-spivak-invitation_F02132 | 203  | ■ 1.000 | $X+\varnothing Y$                                                    | $X + \emptyset Y$                              | Similarly, if a pushout $X +_{\emptyset} Y$ exists, then it satisfies the universal property of the                                           |
| fong-spivak-invitation_F02133 | 203  | ■ 0.997 | $\{ \}_{3}$                                                          | $3$                                            | coproduct (why? <sub>3</sub> ).                                                                                                               |
| fong-spivak-invitation_F02134 | 204  | ■ 1.000 | $A=X=Y=\mathbb{N}$                                                   | $A = X = Y = \mathbb{N}$                       | <b>Example 6.25.</b> Let $A = X = Y = \mathbb{N}$ . Consider the functions $f: A \rightarrow X$ and $g: A \rightarrow Y$                      |
| fong-spivak-invitation_F02135 | 204  | ■ 0.988 | $f(a):=\lfloor a / 2 \rfloor$                                        | $f(a) := \lfloor a/2 \rfloor$                  | given by the “floor” functions, $f(a) := \lfloor a/2 \rfloor$ and $g(a) := \lfloor (a+1)/2 \rfloor$ .                                         |
| fong-spivak-invitation_F02136 | 204  | ■ 0.988 | $g(a):=\lfloor (a+1) / 2 \rfloor$                                    | $g(a) := \lfloor (a+1)/2 \rfloor$              | given by the “floor” functions, $f(a) := \lfloor a/2 \rfloor$ and $g(a) := \lfloor (a+1)/2 \rfloor$ .                                         |
| fong-spivak-invitation_F02137 | 204  | ■ 1.000 | $x: X \rightarrow T$                                                 | $x: X \rightarrow T$                           | If $T$ is any other set and we have maps $x: X \rightarrow T$ and $y: Y \rightarrow T$ that commute with                                      |
| fong-spivak-invitation_F02138 | 204  | ■ 1.000 | $y: Y \rightarrow T$                                                 | $y: Y \rightarrow T$                           | If $T$ is any other set and we have maps $x: X \rightarrow T$ and $y: Y \rightarrow T$ that commute with                                      |
| fong-spivak-invitation_F02139 | 204  | ■ 1.000 | $f \cong x=g \cong y$                                                | $f \cong x = g \cong y$                        | $f$ and $g$ , i.e. $f \cong x = g \cong y$ , then this commutativity implies that                                                             |
| fong-spivak-invitation_F02140 | 204  | ■ 1.000 | $f(1)=0$                                                             | $f(1) = 0$                                     | In other words, $Y$ ’s 0 and $X$ ’s 0 go to the same place in $T$ , say $t$ . But since $f(1) = 0$                                            |
| fong-spivak-invitation_F02141 | 204  | ■ 1.000 | $g(1)=1$                                                             | $g(1) = 1$                                     | and $g(1) = 1$ , we also have that $t = x(0) = x(f(1)) = y(g(1)) = y(1)$ . This means                                                         |
| fong-spivak-invitation_F02142 | 204  | ■ 1.000 | $t=x(0)=x(f(1))=y(g(1))=y(1)$                                        | $t = x(0) = x(f(1)) = y(g(1)) = y(1)$          | and $g(1) = 1$ , we also have that $t = x(0) = x(f(1)) = y(g(1)) = y(1)$ . This means                                                         |
| fong-spivak-invitation_F02143 | 204  | ■ 1.000 | $g(2)=1$                                                             | $g(2) = 1$                                     | $Y$ ’s 1 goes to $t$ also. But since $g(2) = 1$ and $f(2) = 1$ , we also have that $t = g(1) =$                                               |
| fong-spivak-invitation_F02144 | 204  | ■ 1.000 | $f(2)=1$                                                             | $f(2) = 1$                                     | $Y$ ’s 1 goes to $t$ also. But since $g(2) = 1$ and $f(2) = 1$ , we also have that $t = g(1) =$                                               |
| fong-spivak-invitation_F02145 | 204  | ■ 1.000 | $t=g(1)=$                                                            | $t = g(1) =$                                   | $Y$ ’s 1 goes to $t$ also. But since $g(2) = 1$ and $f(2) = 1$ , we also have that $t = g(1) =$                                               |
| fong-spivak-invitation_F02146 | 204  | ■ 0.986 | $y(g(2))=x(f(2))=x(1)$                                               | $y(g(2)) = x(f(2)) = x(1)$                     | $y(g(2)) = x(f(2)) = x(1)$ , which means that $X$ ’s 1 also goes to $t$ . One can keep                                                        |
| fong-spivak-invitation_F02147 | 204  | ■ 1.000 | $X \sqcup_A Y \cong \{1\}$                                           | $X \sqcup_A Y \cong \{1\}$                     | $X \sqcup_A Y \cong \{1\}$ .                                                                                                                  |
| fong-spivak-invitation_F02148 | 204  | ■ 0.862 | $j \in \operatorname{Ob} \mathcal{J}$                                | $j \in \text{Ob } \mathcal{J}$                 | have finitely many objects too, because each object $j \in \text{Ob } \mathcal{J}$ has its own identity morphism $\text{id}_j$ .              |
| fong-spivak-invitation_F02149 | 204  | ■ 0.862 | $\operatorname{id}_j$                                                | $\text{id}_j$                                  | have finitely many objects too, because each object $j \in \text{Ob } \mathcal{J}$ has its own identity morphism $\text{id}_j$ .              |
| fong-spivak-invitation_F02150 | 205  | ■ 0.997 | $\operatorname{colim}_{\mathcal{J}} D$                               | $\text{colim}_{\mathcal{J}} D$                 | <b>Definition 6.26.</b> We say that a category $\mathcal{C}$ has finite colimits if a colimit, $\text{colim}_{\mathcal{J}} D$ ,               |

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| fong-spivak-invitation_F02151 | 205  | 0.820 | $\mathfrak{d}$                                                                                          | $\mathfrak{d}$                                                                                          | exists whenever $\mathcal{J}$ is a finite category and $D: \mathcal{J} \rightarrow \mathcal{C}$ is a diagram.                                                                 |
| fong-spivak-invitation_F02152 | 205  | 0.963 | $!: \mathbf{0} \rightarrow \mathcal{C}$                                                                 | $!: \mathbf{0} \rightarrow \mathcal{C}$                                                                 | $!: \mathbf{0} \rightarrow \mathcal{C}$ , where $\mathbf{0}$ is the category with no objects and no morphisms, and $!$ is the unique                                          |
| fong-spivak-invitation_F02153 | 206  | 0.997 | $(v, d) \sim (w, e)$                                                                                    | $(v, d) \sim (w, e)$                                                                                    | generated by putting $(v, d) \sim (w, e)$ if there is an arrow $a: v \rightarrow w$ in $J$ such that                                                                          |
| fong-spivak-invitation_F02154 | 206  | 0.997 | $a: v \rightarrow w$                                                                                    | $a: v \rightarrow w$                                                                                    | generated by putting $(v, d) \sim (w, e)$ if there is an arrow $a: v \rightarrow w$ in $J$ such that                                                                          |
| fong-spivak-invitation_F02155 | 206  | 0.991 | $D(a)(d)=e$                                                                                             | $D(a)(d) = e$                                                                                           | $D(a)(d) = e$ . Then this set, together with the functions $\iota_v: D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$ given                                             |
| fong-spivak-invitation_F02156 | 206  | 0.991 | $\iota_v: D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$                                        | $\iota_v: D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$                                        | $D(a)(d) = e$ . Then this set, together with the functions $\iota_v: D(v) \rightarrow \operatorname{colim}_{\mathcal{J}} D$ given                                             |
| fong-spivak-invitation_F02157 | 206  | 0.996 | $d \in D(v)$                                                                                            | $d \in D(v)$                                                                                            | by sending $d \in D(v)$ to its equivalence class, constitutes a colimit of $D$ .                                                                                              |
| fong-spivak-invitation_F02158 | 206  | 0.508 | $\mathfrak{J} = \begin{array}{ cc } \hline v_1 & v_2 \\ \hline \bullet & \bullet \\ \hline \end{array}$ | $\mathfrak{J} = \begin{array}{ cc } \hline v_1 & v_2 \\ \hline \bullet & \bullet \\ \hline \end{array}$ | <b>Example 6.34.</b> A coproduct is a colimit on the graph $\mathcal{J} = \begin{array}{ cc } \hline v_1 & v_2 \\ \hline \bullet & \bullet \\ \hline \end{array}$ . A functor |
| fong-spivak-invitation_F02159 | 206  | 0.990 | $X := D(\overline{v_1})$                                                                                | $X := D(v_1)$                                                                                           | $D: \mathcal{J} \rightarrow \mathbf{Set}$ can be identified with a choice of two sets, $X := D(\overline{v_1})$ and $Y := D(v_2)$ .                                           |
| fong-spivak-invitation_F02160 | 206  | 0.990 | $Y := D(\overline{v_2})$                                                                                | $Y := D(v_2)$                                                                                           | $D: \mathcal{J} \rightarrow \mathbf{Set}$ can be identified with a choice of two sets, $X := D(\overline{v_1})$ and $Y := D(v_2)$ .                                           |
| fong-spivak-invitation_F02161 | 207  | 0.479 | $\mathbf{1} = \boxed{\bullet}$                                                                          | $\mathbf{1} = \boxed{\bullet}$                                                                          | <b>Example 6.35.</b> If $\mathcal{J}$ is the category $\mathbf{1} = \boxed{\bullet}$ , the formula in Theorem 6.32 yields the set                                             |
| fong-spivak-invitation_F02162 | 207  | 1.000 | $D(v)$                                                                                                  | $D(v)$                                                                                                  | This is isomorphic to the set $D(v)$ . In other words, if $X$ is a set considered as a diagram                                                                                |
| fong-spivak-invitation_F02163 | 207  | 0.995 | $X: \mathbf{1} \rightarrow$                                                                             | $X: \mathbf{1} \rightarrow$                                                                             | $X: \mathbf{1} \rightarrow \mathbf{Set}$ , then its colimit (like its limit) is just $X$ again.                                                                               |
| fong-spivak-invitation_F02164 | 207  | 1.000 | $X \stackrel{f}{\leftarrow} N \stackrel{g}{\rightarrow} Y$                                              | $X \stackrel{f}{\leftarrow} N \stackrel{g}{\rightarrow} Y$                                              | diagram $X \stackrel{f}{\leftarrow} N \stackrel{g}{\rightarrow} Y$ – agree with the description we gave in Example 6.21.                                                      |
| fong-spivak-invitation_F02165 | 207  | 0.985 | $X \underset{g}{\stackrel{f}{\rightrightarrows}} Y$                                                     | $X \underset{g}{\stackrel{f}{\rightrightarrows}} Y$                                                     | Consider some diagram $X \underset{g}{\stackrel{f}{\rightrightarrows}} Y$ on this graph in $\mathbf{Set}$ . The coequalizer of                                                |
| fong-spivak-invitation_F02166 | 207  | 1.000 | $y \sim y'$                                                                                             | $y \sim y'$                                                                                             | erated by declaring $y \sim y'$ whenever there exists $x$ in $X$ such that $f(x) = y$ and                                                                                     |
| fong-spivak-invitation_F02167 | 207  | 1.000 | $f(x)=y$                                                                                                | $f(x) = y$                                                                                              | erated by declaring $y \sim y'$ whenever there exists $x$ in $X$ such that $f(x) = y$ and                                                                                     |
| fong-spivak-invitation_F02168 | 207  | 1.000 | $g(x)=y'$                                                                                               | $g(x) = y'$                                                                                             | $g(x) = y'$ .                                                                                                                                                                 |
| fong-spivak-invitation_F02169 | 207  | 0.883 | $f, g: X \rightarrow Y$                                                                                 | $f, g: X \rightarrow Y$                                                                                 | $\bullet$ , the blue and red arrows respectively define functions $f, g: X \rightarrow Y$ . Let's leave the                                                                   |
| fong-spivak-invitation_F02170 | 208  | 1.000 | $A \rightarrow N \leftarrow B$                                                                          | $A \rightarrow N \leftarrow B$                                                                          | common object $A \rightarrow N \leftarrow B$ . The common object $N$ is called the <i>apex</i> of the cospan                                                                  |
| fong-spivak-invitation_F02171 | 208  | 1.000 | $N \stackrel{g}{\leftarrow} B \stackrel{h}{\rightarrow} P$                                              | $N \stackrel{g}{\leftarrow} B \stackrel{h}{\rightarrow} P$                                              | Then, if a pushout of $N \stackrel{g}{\leftarrow} B \stackrel{h}{\rightarrow} P$ exists in $\mathcal{C}$ , as shown on the left, we can extract a                             |
| fong-spivak-invitation_F02172 | 208  | 0.822 | $C \in \operatorname{Ob} \mathcal{C}$                                                                   | $C \in \operatorname{Ob} \mathcal{C}$                                                                   | we need an identity on every object $C \in \operatorname{Ob} \mathcal{C}$ ; but that's not hard: use $C \rightarrow C \leftarrow C$                                           |
| fong-spivak-invitation_F02173 | 208  | 0.822 | $C \rightarrow C \leftarrow C$                                                                          | $C \rightarrow C \leftarrow C$                                                                          | we need an identity on every object $C \in \operatorname{Ob} \mathcal{C}$ ; but that's not hard: use $C \rightarrow C \leftarrow C$                                           |
| fong-spivak-invitation_F02174 | 208  | 1.000 | $A \rightarrow B \rightarrow C$                                                                         | $A \rightarrow B \rightarrow C$                                                                         | pushouts, so we don't know that every two sequential morphisms $A \rightarrow B \rightarrow C$ can be                                                                         |
| fong-spivak-invitation_F02175 | 208  | 1.000 | $A \rightarrow P \leftarrow B$                                                                          | $A \rightarrow P \leftarrow B$                                                                          | between their respective apexes. That is, the cospan $A \rightarrow P \leftarrow B$ and $A \rightarrow P' \leftarrow B$ in                                                    |

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| fong-spivak-invitation_F02176              | 208  | ■ 1.000 | $A \rightarrowtail P^{\prime} \leftarrowtail B$                                                                                                                           | $A \rightarrow P' \leftarrow B$                                     | between their respective apexes. That is, the cospan $A \rightarrowtail P \leftarrowtail B$ and $A \rightarrowtail P' \leftarrowtail B$ in                              |
| fong-spivak-invitation_F02177              | 208  | ■ 1.000 | $P \cong P^{\prime}$                                                                                                                                                      | $P \cong P'$                                                        | the diagram shown left below are equivalent iff there is an isomorphism $P \cong P'$ making                                                                             |
| fong-spivak-invitation_F02178              | 208  | ■ 0.824 | $\{ \}_{{\text{\texttt{C}}}}$                                                                                                                                             | $\mathcal{C}$                                                       | business: $\mathbf{Cospan}_{\mathcal{C}}$ will form a category. But in fact, we are very close to getting more.                                                         |
| fong-spivak-invitation_F02179              | 209  | ■ 0.500 | $_{\mathcal{C}}$                                                                                                                                                          | $\mathcal{C}$                                                       | $\mathbf{Cospan}_{\mathcal{C}}$ with the same objects as $\mathcal{C}$ , i.e. $\text{Ob}(\mathbf{Cospan}_{\mathcal{C}}) = \text{Ob}(\mathcal{C})$ , where the morphisms |
| fong-spivak-invitation_F02180              | 209  | ■ 0.500 | $\operatorname{Ob}\{\left(\operatorname{\mathbf{Cospan}}_{\mathcal{C}}\right)\}_{\mathcal{C}}=\operatorname{Ob}\{\left(\operatorname{\mathcal{C}}\right)\}_{\mathcal{C}}$ | $\text{Ob}(\mathbf{Cospan}_{\mathcal{C}}) = \text{Ob}(\mathcal{C})$ | $\mathbf{Cospan}_{\mathcal{C}}$ with the same objects as $\mathcal{C}$ , i.e. $\text{Ob}(\mathbf{Cospan}_{\mathcal{C}}) = \text{Ob}(\mathcal{C})$ , where the morphisms |
| fong-spivak-invitation_F02181              | 209  | ■ 0.907 | $_{\mathcal{C}}, \varnothing, +$                                                                                                                                          | $\mathcal{C}, \varnothing, +$                                       | $(\mathbf{Cospan}_{\mathcal{C}}, \varnothing, +)$ . The monoidal unit is the initial object $\varnothing \in \mathcal{C}$ and the monoidal                              |
| fong-spivak-invitation_F02182              | 209  | ■ 1.000 | $A+\varnothing \cong A$                                                                                                                                                   | $A + \varnothing \cong A$                                           | product is given by coproduct. The coherence isomorphisms, e.g. $A + \varnothing \cong A$ , can be                                                                      |
| fong-spivak-invitation_F02183 [conf 0.098] | 209  | ■ 0.098 | $\operatorname{\mathbf{Cospan}}_{\mathcal{C}}, \varnothing, +$                                                                                                            | $\mathbf{Cospan}_{\mathcal{C}}, \varnothing, +$                     | It is a straightforward but time-consuming exercise to verify that $(\mathbf{Cospan}_{\mathcal{C}}, \varnothing, +)$                                                    |
| fong-spivak-invitation_F02184              | 209  | ■ 0.749 | $_{\text{\texttt{FinSet}}}$                                                                                                                                               | $\text{FinSet}$                                                     | define a symmetric monoidal category $\mathbf{Cospan}_{\text{FinSet}}$ . What does it look like? It looks a                                                             |
| fong-spivak-invitation_F02185              | 209  | ■ 0.986 | $\{ \}_{{\text{\texttt{FinSet}}}}$                                                                                                                                        | $\text{FinSet}$                                                     | The objects of $\mathbf{Cospan}_{\text{FinSet}}$ are finite sets; here let's draw them as collections of •s.                                                            |
| fong-spivak-invitation_F02186              | 209  | ■ 1.000 | $A \xrightarrow{f} N \xleftarrow{g} B$                                                                                                                                    | $A \xrightarrow{f} N \xleftarrow{g} B$                              | six-element set. Below are two depictions of a cospan $A \xrightarrow{f} N \xleftarrow{g} B$ .                                                                          |
| fong-spivak-invitation_F02187              | 209  | ■ 1.000 | $g(b)$                                                                                                                                                                    | $g(b)$                                                              | arrows from each $a \in A$ to $f(a)$ and each $b \in B$ to $g(b)$ . In the depiction on the                                                                             |
| fong-spivak-invitation_F02188              | 210  | ■ 0.997 | $A+B \rightarrowtail B+C$                                                                                                                                                 | $A + B \rightarrowtail B + C$                                       | $\mathbf{Cospan}_{\text{FinSet}}$ . Draw their monoidal product as a morphism $A + B \rightarrowtail B + C$ in                                                          |
| fong-spivak-invitation_F02189              | 211  | ■ 0.995 | $(X, \mu, \eta)$                                                                                                                                                          | $(X, \mu, \eta)$                                                    | To make this precise, recall from Definition 5.55 that a commutative monoid $(X, \mu, \eta)$                                                                            |
| fong-spivak-invitation_F02190              | 211  | ■ 0.531 | $\mathcal{C}, I, \otimes$                                                                                                                                                 | $\mathcal{C}, I, \otimes$                                           | in symmetric monoidal category $(\mathcal{C}, I, \otimes)$ is an object $X$ of $\mathcal{C}$ together with (noteworthy)                                                 |
| fong-spivak-invitation_F02191              | 211  | —       | $\mu: X \otimes X \rightarrowtail X$                                                                                                                                      | $\mu: X \otimes X \rightarrow X$                                    |                                                                                    |
| fong-spivak-invitation_F02192              | 211  | —       | $\eta: I \rightarrowtail X$                                                                                                                                               | $\eta: I \rightarrow X$                                             | —                                                                                                                                                                       |
| fong-spivak-invitation_F02193              | 211  | ■ 0.786 | $X \otimes X$                                                                                                                                                             | $X \otimes X$                                                       | where $\bowtie$ is the symmetry on $X \otimes X$ . A cocommutative comonoid $(X, \delta, \epsilon)$ is an object                                                        |
| fong-spivak-invitation_F02194              | 211  | ■ 0.786 | $(X, \delta, \epsilon)$                                                                                                                                                   | $(X, \delta, \epsilon)$                                             | where $\bowtie$ is the symmetry on $X \otimes X$ . A cocommutative comonoid $(X, \delta, \epsilon)$ is an object                                                        |
| fong-spivak-invitation_F02195              | 211  | ■ 1.000 | $\delta: X \rightarrowtail X \otimes X, \epsilon: X \rightarrowtail I$                                                                                                    | $\delta: X \rightarrow X \otimes X, \epsilon: X \rightarrow I$      | $X$ with maps $\delta: X \rightarrow X \otimes X, \epsilon: X \rightarrow I$ , obeying the mirror images of the laws in                                                 |
| fong-spivak-invitation_F02196              | 211  | ■ 1.000 | $\mu, \eta, \delta$                                                                                                                                                       | $\mu, \eta, \delta$                                                 | comonoid, and consider a wiring diagram built only from the icons $\mu, \eta, \delta$ , and $\epsilon$ ,                                                                |
| fong-spivak-invitation_F02197              | 211  | ■ 1.000 | $\epsilon$                                                                                                                                                                | $\epsilon$                                                          | comonoid, and consider a wiring diagram built only from the icons $\mu, \eta, \delta$ , and $\epsilon$ ,                                                                |
| fong-spivak-invitation_F02198              | 211  | ■ 1.000 | $\mu$                                                                                                                                                                     | $\mu$                                                               | how the morphisms $\mu$ and $\delta$ , the merger and the splitter, interact.                                                                                           |
| fong-spivak-invitation_F02199              | 211  | ■ 1.000 | $\delta$                                                                                                                                                                  | $\delta$                                                            | how the morphisms $\mu$ and $\delta$ , the merger and the splitter, interact.                                                                                           |
| fong-spivak-invitation_F02200              | 211  | ■ 0.986 | $(\mathcal{C}, \otimes, I)$                                                                                                                                               | $(\mathcal{C}, \otimes, I)$                                         | <b>Definition 6.43.</b> Let $X$ be an object in a symmetric monoidal category $(\mathcal{C}, \otimes, I)$ . A                                                           |
| fong-spivak-invitation_F02201              | 211  | ■ 0.902 | $\mu, \eta, \delta, \epsilon$                                                                                                                                             | $\mu, \eta, \delta, \epsilon$                                       | <i>Frobenius structure</i> on $X$ consists of a 4-tuple $(\mu, \eta, \delta, \epsilon)$ such that $(X, \mu, \eta)$ is a                                                 |
| fong-spivak-invitation_F02202              | 211  | ■ 0.902 | $X, \mu, \eta$                                                                                                                                                            | $X, \mu, \eta$                                                      | <i>Frobenius structure</i> on $X$ consists of a 4-tuple $(\mu, \eta, \delta, \epsilon)$ such that $(X, \mu, \eta)$ is a                                                 |

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| fong-spivak-invitation_F02283 | 212  | ■ 1.000 | $X^{\otimes m} \rightarrow X^{\otimes n}$               | $X^{\otimes m} \rightarrow X^{\otimes n}$          | With these two equations, it turns out that two morphisms $X^{\otimes m} \rightarrow X^{\otimes n}$ – defined                         |
| fong-spivak-invitation_F02284 | 212  | ■ 1.000 | $\mu, \delta$                                           | $\mu, \delta$                                      | by composing and tensoring identities on $X$ and the noteworthy morphisms $\mu, \delta$ , etc. –                                      |
| fong-spivak-invitation_F02285 | 212  | ■ 0.889 | $X, \mu, \eta, \delta, \epsilon$                        | $X, \mu, \eta, \delta, \epsilon$                   | <b>Definition 6.44.</b> Let $(X, \mu, \eta, \delta, \epsilon)$ be a Frobenius monoid in a monoidal category $(\mathcal{C},$           |
| fong-spivak-invitation_F02286 | 212  | ■ 1.000 | $I, \otimes)$                                           | $I, \otimes)$                                      | $I, \otimes)$ . Let $m, n \in \mathbb{N}$ . Define $s_{m,n}: X^{\otimes m} \rightarrow X^{\otimes n}$ to be the following morphism    |
| fong-spivak-invitation_F02287 | 212  | ■ 1.000 | $s_{\{m, n\}}: X^{\otimes m} \rightarrow X^{\otimes n}$ | $s_{m,n}: X^{\otimes m} \rightarrow X^{\otimes n}$ | $I, \otimes)$ . Let $m, n \in \mathbb{N}$ . Define $s_{m,n}: X^{\otimes m} \rightarrow X^{\otimes n}$ to be the following morphism    |
| fong-spivak-invitation_F02288 | 212  | ■ 0.998 | $(m-1) \mu \mathrm{~}\mathit{\mathrm{s}}$               | $(m-1)\mu s$                                       | It can be written formally as $(m-1) \mu s$ followed by $(n-1) \delta s$ , with special cases when                                    |
| fong-spivak-invitation_F02289 | 212  | ■ 0.998 | $(n-1) \delta \mathrm{~}\mathit{\mathrm{s}}$            | $(n-1)\delta s$                                    | It can be written formally as $(m-1) \mu s$ followed by $(n-1) \delta s$ , with special cases when                                    |
| fong-spivak-invitation_F02210 | 212  | ■ 1.000 | $m=0$                                                   | $m = 0$                                            | $m = 0$ or $n = 0$ .                                                                                                                  |
| fong-spivak-invitation_F02211 | 212  | ■ 0.993 | $s_{\{m, n\}}$                                          | $s_{m,n}$                                          | We call $s_{m,n}$ the <i>spider of type <math>(m, n)</math></i> , and can draw it more simply as the icon                             |
| fong-spivak-invitation_F02212 | 212  | ■ 1.000 | $\mu, \eta, \delta, \epsilon, \sigma$                   | $\mu, \eta, \delta, \epsilon, \sigma$              | a wire is spiderable, it means that we have a bunch of morphisms $\mu, \eta, \delta, \epsilon, \sigma$ that we                        |
| fong-spivak-invitation_F02213 | 212  | ■ 0.973 | $f: X^{\otimes m} \rightarrow X^{\otimes n}$            | $f: X^{\otimes m} \rightarrow X^{\otimes n}$       | $I, \otimes)$ . Suppose that we have a map $f: X^{\otimes m} \rightarrow X^{\otimes n}$ constructed from spiders and the              |
| fong-spivak-invitation_F02214 | 212  | ■ 1.000 | $\sigma: X^{\otimes 2} \rightarrow X^{\otimes 2}$       | $\sigma: X^{\otimes 2} \rightarrow X^{\otimes 2}$  | symmetry map $\sigma: X^{\otimes 2} \rightarrow X^{\otimes 2}$ using composition and the monoidal product, and such                   |
| fong-spivak-invitation_F02215 | 212  | ■ 1.000 | $f=s_{\{m, n\}}$                                        | $f = s_{m,n}$                                      | $f = s_{m,n}$ .                                                                                                                       |
| fong-spivak-invitation_F02216 | 212  | ■ 1.000 | $X \otimes X \rightarrow X \otimes X \otimes X$         | $X \otimes X \rightarrow X \otimes X \otimes X$    | morphisms $X \otimes X \rightarrow X \otimes X \otimes X$ in the following list are necessarily equal?                                |
| fong-spivak-invitation_F02217 | 213  | ■ 0.999 | $G:=\{\mu, \eta, \delta, \epsilon\}$                    | $G := \{\mu, \eta, \delta, \epsilon\}$             | <b>Theorem 6.48.</b> Consider the four-element set $G := \{\mu, \eta, \delta, \epsilon\}$ and define                                  |
| fong-spivak-invitation_F02218 | 213  | ■ 1.000 | $G \rightarrow \mathbb{N}$                              | $G \rightarrow \mathbb{N}$                         | $in, out: G \rightarrow \mathbb{N}$ as follows:                                                                                       |
| fong-spivak-invitation_F02219 | 213  | ■ 0.970 | $(G, E)$                                                | $(G, E)$                                           | Then the free prop on $(G, E)$ is equivalent, as a symmetric monoidal category, <sup>5</sup> to                                       |
| fong-spivak-invitation_F02220 | 215  | ■ 0.970 | $X, \mu_X, \delta_X, \eta_X, \epsilon_X$                | $X, \mu_X, \delta_X, \eta_X, \epsilon_X$           | which each object $X$ is equipped with a Frobenius structure $(X, \mu_X, \delta_X, \eta_X, \epsilon_X)$ such                          |
| fong-spivak-invitation_F02221 | 215  | ■ 0.996 | $\eta_I=\mathrm{id}_I, \epsilon_I=\epsilon_I$           | $\eta_I = \mathrm{id}_I = \epsilon_I$              | for all objects $X, Y$ , and such that $\eta_I = \mathrm{id}_I = \epsilon_I$ .                                                        |
| fong-spivak-invitation_F02222 | 215  | ■ 1.000 | $\mu_X, \delta_X, \eta_X, \epsilon_X$                   | $\mu_X, \delta_X, \eta_X, \epsilon_X$              | The Frobenius morphisms $\mu_X, \delta_X, \eta_X, \epsilon_X$ for each object $X$ are constructed using the                           |
| fong-spivak-invitation_F02223 | 215  | ■ 0.532 | $\operatorname{Cospan}_{\text{FinSet}}$                 | $\operatorname{Cospan}_{\text{FinSet}}$            | <b>Exercise 6.52.</b> By Example 6.51, the category $\operatorname{Cospan}_{\text{FinSet}}$ is a hypergraph cate-                     |
| fong-spivak-invitation_F02224 | 215  | ■ 0.571 | $\varnothing, \sqcup)$                                  | $\varnothing, \sqcup)$                             | <b>Example 6.54.</b> Recall the monoidal category $(\mathbf{Corel}, \varnothing, \sqcup)$ from Example 4.46; its                      |
| fong-spivak-invitation_F02225 | 215  | ■ 1.000 | $\mu_X: X \sqcup X \rightarrow X$                       | $\mu_X: X \sqcup X \rightarrow X$                  | the corelation $\mu_X: X \sqcup X \rightarrow X$ such that two elements of $X \sqcup X \sqcup X$ are equivalent if                    |
| fong-spivak-invitation_F02226 | 215  | ■ 1.000 | $X \sqcup X \sqcup X$                                   | $X \sqcup X \sqcup X$                              | the corelation $\mu_X: X \sqcup X \rightarrow X$ such that two elements of $X \sqcup X \sqcup X$ are equivalent if                    |
| fong-spivak-invitation_F02227 | 215  | ■ 1.000 | $\delta_X: X \rightarrow X \sqcup X$                    | $\delta_X: X \rightarrow X \sqcup X$               | and only if they come from the same underlying element of $X$ . Define $\delta_X: X \rightarrow X \sqcup X$                           |
| fong-spivak-invitation_F02228 | 215  | ■ 1.000 | $\eta_X: \varnothing \rightarrow X$                     | $\eta_X: \varnothing \rightarrow X$                | in the same way, and define $\eta_X: \varnothing \rightarrow X$ and $\epsilon_X: X \rightarrow \varnothing$ such that no two elements |
| fong-spivak-invitation_F02229 | 215  | ■ 1.000 | $\epsilon_X: X \rightarrow \varnothing$                 | $\epsilon_X: X \rightarrow \varnothing$            | in the same way, and define $\eta_X: \varnothing \rightarrow X$ and $\epsilon_X: X \rightarrow \varnothing$ such that no two elements |

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| fong-spivak-invitation_F02230 | 215  | ■ 1.000 | $X=\varnothing \sqcup X=X \sqcup \varnothing$                                                | $X = \emptyset \sqcup X = X \sqcup \emptyset$                 | of $X = \emptyset \sqcup X = X \sqcup \emptyset$ are equivalent.                                                                                                            |
| fong-spivak-invitation_F02231 | 216  | ■ 1.000 | $\left(X, \mu_X, \eta_X, \delta_X, \epsilon_X \right)$                                       | $(X, \mu_X, \eta_X, \delta_X, \epsilon_X)$                    | These maps define a special commutative Frobenius monoid $(X, \mu_X, \eta_X, \delta_X, \epsilon_X)$ ,                                                                       |
| fong-spivak-invitation_F02232 | 217  | ■ 0.456 | $\left(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{P}} \right)$                          | $(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{P}})$       | <b>Rough Definition 6.58.</b> Let $(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{C}})$ and $(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}})$ be symmetric monoidal |
| fong-spivak-invitation_F02233 | 217  | ■ 0.456 | $\left(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}} \right)$                          | $(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}})$       | <b>Rough Definition 6.58.</b> Let $(\mathcal{C}, I_{\mathcal{C}}, \otimes_{\mathcal{C}})$ and $(\mathcal{D}, I_{\mathcal{D}}, \otimes_{\mathcal{D}})$ be symmetric monoidal |
| fong-spivak-invitation_F02234 | 217  | ■ 0.992 | $(F, \varphi)$                                                                               | $(F, \varphi)$                                                | categories. To specify a <i>symmetric monoidal functor</i> $(F, \varphi)$ between them,                                                                                     |
| fong-spivak-invitation_F02235 | 217  | ■ 0.986 | $\varphi_I\colon I_{\mathcal{D}} \rightarrow F(I_{\mathcal{C}})$                             | $\varphi_I : I_{\mathcal{D}} \rightarrow F(I_{\mathcal{C}})$  | (ii) one specifies a morphism $\varphi_I : I_{\mathcal{D}} \rightarrow F(I_{\mathcal{C}})$ ,                                                                                |
| fong-spivak-invitation_F02236 | 217  | ■ 0.865 | $c_1, c_2 \in \mathrm{Ob}(\mathcal{C})$                                                      | $c_1, c_2 \in \mathrm{Ob}(\mathcal{C})$                       | (iii) for each $c_1, c_2 \in \mathrm{Ob}(\mathcal{C})$ , one specifies a morphism                                                                                           |
| fong-spivak-invitation_F02237 | 217  | ■ 1.000 | $\varphi$                                                                                    | $\varphi$                                                     | We call the various maps $\varphi$ <i>coherence maps</i> . We require the coherence maps to obey                                                                            |
| fong-spivak-invitation_F02238 | 217  | ■ 0.990 | $\varphi_I$                                                                                  | $\varphi_I$                                                   | monoidal structures on $\mathcal{C}$ and $\mathcal{D}$ . If $\varphi_I$ and $\varphi_{c_1, c_2}$ are isomorphisms for all $c_1, c_2$ , we say                               |
| fong-spivak-invitation_F02239 | 217  | ■ 0.990 | $\varphi_{c_1, c_2}$                                                                         | $\varphi_{c_1, c_2}$                                          | monoidal structures on $\mathcal{C}$ and $\mathcal{D}$ . If $\varphi_I$ and $\varphi_{c_1, c_2}$ are isomorphisms for all $c_1, c_2$ , we say                               |
| fong-spivak-invitation_F02240 | 217  | ■ 0.990 | $c_1, c_2$                                                                                   | $c_1, c_2$                                                    | monoidal structures on $\mathcal{C}$ and $\mathcal{D}$ . If $\varphi_I$ and $\varphi_{c_1, c_2}$ are isomorphisms for all $c_1, c_2$ , we say                               |
| fong-spivak-invitation_F02241 | 217  | ■ 1.000 | $S \in$                                                                                      | $S \in$                                                       | sending a set $S \in \mathbf{Set}$ to its set of subsets $P(S) := \{R \subseteq S\}$ . It acts on morphisms                                                                 |
| fong-spivak-invitation_F02242 | 217  | ■ 1.000 | $\mathrm{P}(S) := \{R \subseteq S\}$                                                         | $P(S) := \{R \subseteq S\}$                                   | sending a set $S \in \mathbf{Set}$ to its set of subsets $P(S) := \{R \subseteq S\}$ . It acts on morphisms                                                                 |
| fong-spivak-invitation_F02243 | 217  | ■ 0.954 | $\mathrm{im}_f\colon \mathrm{P}(S) \rightarrow \mathrm{P}(T)$                                | $\mathrm{im}_f : P(S) \rightarrow P(T)$                       | by sending a function $f : S \rightarrow T$ to the image map $\mathrm{im}_f : P(S) \rightarrow P(T)$ , which maps                                                           |
| fong-spivak-invitation_F02244 | 217  | ■ 1.000 | $R \subseteq S$                                                                              | $R \subseteq S$                                               | $R \subseteq S$ to $\{f(r) \mid r \in R\} \subseteq T$ .                                                                                                                    |
| fong-spivak-invitation_F02245 | 217  | ■ 1.000 | $\{f(r) \mid r \in R\} \subseteq T$                                                          | $\{f(r) \mid r \in R\} \subseteq T$                           | $R \subseteq S$ to $\{f(r) \mid r \in R\} \subseteq T$ .                                                                                                                    |
| fong-spivak-invitation_F02246 | 217  | ■ 0.983 | $(\{1\}, \times)$                                                                            | $(\{1\}, \times)$                                             | Now consider the symmetric monoidal structure $(\{1\}, \times)$ on $\mathbf{Set}$ from Example 4.38.                                                                        |
| fong-spivak-invitation_F02247 | 217  | ■ 1.000 | $\varphi_I\colon \{1\} \rightarrow$                                                          | $\varphi_I : \{1\} \rightarrow$                               | To make $P$ a symmetric monoidal functor, we need to specify a function $\varphi_I : \{1\} \rightarrow$                                                                     |
| fong-spivak-invitation_F02248 | 217  | ■ 1.000 | $\mathrm{P}(\{1\})$                                                                          | $P(\{1\})$                                                    | $P(\{1\})$ and for all sets $S$ and $T$ , a function $\varphi_{S, T} : P(S) \times P(T) \rightarrow P(S \times T)$ . One possibility                                        |
| fong-spivak-invitation_F02249 | 217  | ■ 1.000 | $\varphi_{S, T}\colon \mathrm{P}(S) \times \mathrm{P}(T) \rightarrow \mathrm{P}(S \times T)$ | $\varphi_{S, T} : P(S) \times P(T) \rightarrow P(S \times T)$ | $P(\{1\})$ and for all sets $S$ and $T$ , a function $\varphi_{S, T} : P(S) \times P(T) \rightarrow P(S \times T)$ . One possibility                                        |
| fong-spivak-invitation_F02250 | 217  | ■ 1.000 | $\varphi_I(1)$                                                                               | $\varphi_I(1)$                                                | is to define $\varphi_I(1)$ to be the maximal subset $\{1\} \subseteq \{1\}$ , and given subsets $A \subseteq S$ and                                                        |
| fong-spivak-invitation_F02251 | 217  | ■ 1.000 | $\{1\} \subseteq \{1\}$                                                                      | $\{1\} \subseteq \{1\}$                                       | is to define $\varphi_I(1)$ to be the maximal subset $\{1\} \subseteq \{1\}$ , and given subsets $A \subseteq S$ and                                                        |
| fong-spivak-invitation_F02252 | 217  | ■ 1.000 | $A \subseteq S$                                                                              | $A \subseteq S$                                               | is to define $\varphi_I(1)$ to be the maximal subset $\{1\} \subseteq \{1\}$ , and given subsets $A \subseteq S$ and                                                        |
| fong-spivak-invitation_F02253 | 217  | ■ 1.000 | $B \subseteq T$                                                                              | $B \subseteq T$                                               | $B \subseteq T$ , to define $\varphi_{S, T}(A, B)$ to be the product subset $A \times B \subseteq S \times T$ . With these                                                  |
| fong-spivak-invitation_F02254 | 217  | ■ 1.000 | $\varphi_{S, T}(A, B)$                                                                       | $\varphi_{S, T}(A, B)$                                        | $B \subseteq T$ , to define $\varphi_{S, T}(A, B)$ to be the product subset $A \times B \subseteq S \times T$ . With these                                                  |
| fong-spivak-invitation_F02255 | 217  | ■ 1.000 | $A \times B \subseteq S \times T$                                                            | $A \times B \subseteq S \times T$                             | $B \subseteq T$ , to define $\varphi_{S, T}(A, B)$ to be the product subset $A \times B \subseteq S \times T$ . With these                                                  |
| fong-spivak-invitation_F02256 | 217  | ■ 0.969 | $\mathrm{P}, \varphi$                                                                        | $P, \varphi$                                                  | definitions, $(P, \varphi)$ is a symmetric monoidal functor.                                                                                                                |

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| fong-spivak-invitation_F02257 | 217  | ■ 1.000 | $\varphi_{S, T}$                                      | $\varphi_{S, T}$                                 | <b>Exercise 6.60.</b> Check that the maps $\varphi_{S, T}$ defined in Example 6.59 are natural in $S$ and                             |
| fong-spivak-invitation_F02258 | 218  | ■ 1.000 | $2 \ \Omega$                                          | $2\Omega$                                        | as a resistor of resistance $2\Omega$ ). For this circuit to be a morphism in some category, i.e.                                     |
| fong-spivak-invitation_F02259 | 218  | ■ 1.000 | $A \rightarrowtail N$                                 | $A \rightarrow N$                                | functions $A \rightarrow N$ and $B \rightarrow N$ are indicated by the gray arrows. This forms a cospan                               |
| fong-spivak-invitation_F02260 | 218  | ■ 1.000 | $B \rightarrowtail N$                                 | $B \rightarrow N$                                | functions $A \rightarrow N$ and $B \rightarrow N$ are indicated by the gray arrows. This forms a cospan                               |
| fong-spivak-invitation_F02261 | 219  | ■ 0.997 | $(F, \varphi): (\mathcal{C}, +) \rightarrowtail$      | $(F, \varphi): (\mathcal{C}, +) \longrightarrow$ | <b>Definition 6.61.</b> Let $\mathcal{C}$ be a category with finite colimits, and $(F, \varphi): (\mathcal{C}, +) \longrightarrow$    |
| fong-spivak-invitation_F02262 | 219  | ■ 0.621 | $A \rightarrowtail N \stackrel{o}{\leftarrow} B$      | $A \xrightarrow{i} N \xleftarrow{o} B$           | ting of a cospan $A \xrightarrow{i} N \xleftarrow{o} B$ in $\mathcal{C}$ together with an element $s \in F(N)$ . <sup>6</sup> We call |
| fong-spivak-invitation_F02263 | 219  | ■ 0.621 | $s \in F(N)$                                          | $s \in F(N)$ . <sup>6</sup>                      | ting of a cospan $A \xrightarrow{i} N \xleftarrow{o} B$ in $\mathcal{C}$ together with an element $s \in F(N)$ . <sup>6</sup> We call |
| fong-spivak-invitation_F02264 | 219  | ■ 0.637 | $\mathcal{C} =$                                       | $\mathcal{C} =$                                  | The intuition here is to use $\mathcal{C} = \mathbf{FinSet}$ , and, for each object $N \in \mathbf{FinSet}$ , the functor             |
| fong-spivak-invitation_F02265 | 219  | ■ 0.637 | $N \in$                                               | $N \in$                                          | The intuition here is to use $\mathcal{C} = \mathbf{FinSet}$ , and, for each object $N \in \mathbf{FinSet}$ , the functor             |
| fong-spivak-invitation_F02266 | 219  | ■ 1.000 | $F(N)$                                                | $F(N)$                                           | available decorations on $N$ nodes, taken from the set $F(N)$ .                                                                       |
| fong-spivak-invitation_F02267 | 219  | ■ 1.000 | $(A \rightarrowtail N \stackrel{g}{\leftarrow} B, s)$ | $(A \xrightarrow{f} N \xleftarrow{g} B, s)$      | Let $(A \xrightarrow{f} N \xleftarrow{g} B, s)$ and $(B \xrightarrow{h} P \xleftarrow{k} C, t)$ represent decorated cospans. Their    |
| fong-spivak-invitation_F02268 | 219  | ■ 1.000 | $(B \rightarrowtail P \stackrel{k}{\leftarrow} C, t)$ | $(B \xrightarrow{h} P \xleftarrow{k} C, t)$      | Let $(A \xrightarrow{f} N \xleftarrow{g} B, s)$ and $(B \xrightarrow{h} P \xleftarrow{k} C, t)$ represent decorated cospans. Their    |
| fong-spivak-invitation_F02269 | 219  | ■ 1.000 | $B \rightarrowtail$                                   | $B \xrightarrow{h}$                              | composite is represented by the composite of the cospans $A \xrightarrow{f} N \xleftarrow{g} B$ and $B \xrightarrow{h}$               |
| fong-spivak-invitation_F02270 | 219  | ■ 1.000 | $P \stackrel{k}{\leftarrow} C$                        | $P \xleftarrow{k} C$                             | $P \xleftarrow{k} C$ , paired with the following element of $F(N +_B P)$ :                                                            |
| fong-spivak-invitation_F02271 | 219  | ■ 1.000 | $F(N + \{ \}_B P)$                                    | $F(N +_B P)$                                     | $P \xleftarrow{k} C$ , paired with the following element of $F(N +_B P)$ :                                                            |
| fong-spivak-invitation_F02272 | 220  | ■ 0.559 | $(F, \varphi): (\mathcal{C}, +) \rightarrowtail$      | $(F, \varphi): (\mathcal{C}, +) \longrightarrow$ | functor $(F, \varphi): (\mathcal{C}, +) \longrightarrow (\mathbf{Set}, \times)$ , there is a hypergraph category $\mathbf{Cospan}_F$  |
| fong-spivak-invitation_F02273 | 220  | ■ 0.559 | $\times)$                                             | $\times)$                                        | functor $(F, \varphi): (\mathcal{C}, +) \longrightarrow (\mathbf{Set}, \times)$ , there is a hypergraph category $\mathbf{Cospan}_F$  |
| fong-spivak-invitation_F02274 | 220  | ■ 0.559 | $_F\}$                                                | $F$                                              | functor $(F, \varphi): (\mathcal{C}, +) \longrightarrow (\mathbf{Set}, \times)$ , there is a hypergraph category $\mathbf{Cospan}_F$  |
| fong-spivak-invitation_F02275 | 220  | ■ 0.660 | $\operatorname{Cospan}_F\}$                           | $\operatorname{Cospan}_F$                        | tion $\mathbf{Cospan}_F$ , even though they’re very different. An expert tells you “they’re not so                                    |
| fong-spivak-invitation_F02276 | 220  | ■ 1.000 | $F(c) := \{*\}$                                       | $F(c) := \{*\}$                                  | different; one is a special case of the other. Just use the constant functor $F(c) := \{*\}$ .”                                       |
| fong-spivak-invitation_F02277 | 220  | ■ 1.000 | $\Omega$                                              | $\Omega$                                         | To be clear, the $\Omega$ are just labels; the above set is isomorphic to $\{\text{light}, \text{switch},$                            |
| fong-spivak-invitation_F02278 | 220  | ■ 0.926 | $\sqcup \mathbb{R}^+$                                 | $\sqcup \mathbb{R}^+$                            | battery $\} \sqcup \mathbb{R}^+$ . But we write $C$ this way to remind us that it consists of circuit                                 |
| fong-spivak-invitation_F02279 | 220  | ■ 1.000 | $\ell: A \rightarrowtail C$                           | $\ell: A \rightarrow C$                          | source and target functions, together with a function $\ell: A \rightarrowtail C$ labeling each edge with                             |
| fong-spivak-invitation_F02280 | 220  | ■ 1.000 | $V = \{1, 2\}, A = \{e\}, s(e) = 1$                   | $V = \{1, 2\}, A = \{e\}, s(e) = 1$              | For example, we might have the simple case of $V = \{1, 2\}, A = \{e\}, s(e) = 1$ ,                                                   |
| fong-spivak-invitation_F02281 | 220  | ■ 0.808 | $t(e) = 2$                                            | $t(e) = 2$                                       | $t(e) = 2$ – so $e$ is an edge from 1 to 2 – and $\ell(e) = 3\Omega$ . This represents a resistor with                                |
| fong-spivak-invitation_F02282 | 220  | ■ 0.808 | $\ell(e) = 3 \ \Omega$                                | $\ell(e) = 3\Omega$                              | $t(e) = 2$ – so $e$ is an edge from 1 to 2 – and $\ell(e) = 3\Omega$ . This represents a resistor with                                |

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| fong-spivak-invitation_F02283 | 220  | ■ 1.000 | $3 \ \Omega$                                                                                                            | $3\Omega$                                                                                                               | resistance $3\Omega$ :                                                                                                                                               |
| fong-spivak-invitation_F02284 | 220  | ■ 1.000 | $V=\{1,2\}, \ A=\{e\}$                                                                                                  | $V = \{1, 2\}, A = \{e\}$                                                                                               | could also be represented by the ‘reversed graph’, with data $V = \{1, 2\}, A = \{e\}$ ,                                                                             |
| fong-spivak-invitation_F02285 | 220  | ■ 1.000 | $s(e)=2, \ t(e)=1$                                                                                                      | $s(e) = 2, t(e) = 1$                                                                                                    | $s(e) = 2, t(e) = 1$ , and $\ell(e) = 3\Omega$ .                                                                                                                     |
| fong-spivak-invitation_F02286 | 221  | ■ 0.978 | $V, \ A, \ s, \ t, \ \ell$                                                                                              | $V, A, s, t, \ell$                                                                                                      | <b>Exercise 6.64.</b> Write a tuple $(V, A, s, t, \ell)$ that represents the circuit in Eq. (6.11). $\diamond$                                                       |
| fong-spivak-invitation_F02287 | 221  | ■ 0.996 | $\psi$                                                                                                                  | $\psi$                                                                                                                  | as in Definition 6.61. We’ll call the functor $(\text{Circ}, \psi)$ . We start by defining the functor part                                                          |
| fong-spivak-invitation_F02288 | 221  | ■ 0.995 | $f: V \rightarrow V^{\prime}$                                                                                           | $f : V \rightarrow V'$                                                                                                  | On morphisms, $\text{Circ}$ sends a function $f: V \rightarrow V'$ to the function                                                                                   |
| fong-spivak-invitation_F02289 | 221  | ■ 0.989 | $c \in \text{Circ}(4)$                                                                                                  | $c \in \text{Circ}(4)$                                                                                                  | <b>Exercise 6.65.</b> To understand this functor better, let $c \in \text{Circ}(4)$ be the circuit                                                                   |
| fong-spivak-invitation_F02290 | 221  | ■ 1.000 | $f: \underline{4} \rightarrow \underline{3}$                                                                            | $f : \underline{4} \rightarrow \underline{3}$                                                                           | and let $f: \underline{4} \rightarrow \underline{3}$ be the function                                                                                                 |
| fong-spivak-invitation_F02291 | 221  | ■ 0.998 | $\text{Circ}(f)(c)$                                                                                                     | $\text{Circ}(f)(c)$                                                                                                     | Draw a picture of the circuit $\text{Circ}(f)(c)$ .                                                                                                                  |
| fong-spivak-invitation_F02292 | 221  | ■ 1.000 | $\psi_{V, V^{\prime}}$                                                                                                  | $\psi_{V, V'}$                                                                                                          | coherence maps $\psi_{V, V'}$ for finite sets $V, V'$ , we define                                                                                                    |
| fong-spivak-invitation_F02293 | 221  | ■ 1.000 | $V, \ V^{\prime}$                                                                                                       | $V, V'$                                                                                                                 | coherence maps $\psi_{V, V'}$ for finite sets $V, V'$ , we define                                                                                                    |
| fong-spivak-invitation_F02294 | 221  | ■ 1.000 | $V+V^{\prime}$                                                                                                          | $V + V'$                                                                                                                | considers them together as a circuit on the disjoint union of vertices $V + V'$ .                                                                                    |
| fong-spivak-invitation_F02295 | 221  | ■ 0.978 | $\psi_{\underline{2}, \underline{2}}(b, s) \in \text{Circ}(\underline{2} + \underline{2}) = \text{Circ}(\underline{4})$ | $\psi_{\underline{2}, \underline{2}}(b, s) \in \text{Circ}(\underline{2} + \underline{2}) = \text{Circ}(\underline{4})$ | $\psi_{\underline{2}, \underline{2}}(b, s) \in \text{Circ}(\underline{2} + \underline{2}) = \text{Circ}(\underline{4})$ should be, and draw a picture.               |
| fong-spivak-invitation_F02296 | 222  | ■ 0.885 | $\text{Circ}$                                                                                                           | $\text{Circ}$                                                                                                           | Our notation for this category is <b>Cospan</b> <sub>Circ</sub> . Following Theorem 6.62, the objects of                                                             |
| fong-spivak-invitation_F02297 | 222  | ■ 0.668 | $\underline{2}$                                                                                                         | $\underline{2}$                                                                                                         | tions of white circles $\circ$ . For example, we’ll represent the object $\underline{2}$ of <b>Cospan</b> <sub>Circ</sub> as two                                     |
| fong-spivak-invitation_F02298 | 222  | ■ 0.668 | $\mathbf{Cospan}_{\text{Circ}}$                                                                                         | <b>Cospan</b> <sub>Circ</sub>                                                                                           | tions of white circles $\circ$ . For example, we’ll represent the object $\underline{2}$ of <b>Cospan</b> <sub>Circ</sub> as two                                     |
| fong-spivak-invitation_F02299 | 222  | ■ 0.692 | $\text{Cospan}_{\text{Circ}}$                                                                                           | $\text{Cospan}_{\text{Circ}}$                                                                                           | More interesting than the objects, however, are the morphisms in <b>Cospan</b> <sub>Circ</sub> . These                                                               |
| fong-spivak-invitation_F02300 | 222  | ■ 1.000 | $\underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$                                                      | $\underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$                                                      | is, cospan $\underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$ together with an element $c$ of $\text{Circ}(\underline{p})$ . As an example, consider |
| fong-spivak-invitation_F02301 | 222  | ■ 1.000 | $\text{Circ}(\underline{p})$                                                                                            | $\text{Circ}(\underline{p})$                                                                                            | is, cospan $\underline{m} \rightarrow \underline{p} \leftarrow \underline{n}$ together with an element $c$ of $\text{Circ}(\underline{p})$ . As an example, consider |
| fong-spivak-invitation_F02302 | 222  | ■ 0.998 | $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$                                          | $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$                                          | the cospan $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$ where $i_1(1) = 1$ and $i_2(1) = 2$ , equipped with the battery            |
| fong-spivak-invitation_F02303 | 222  | ■ 0.998 | $i_1(1)=1$                                                                                                              | $i_1(1) = 1$                                                                                                            | the cospan $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$ where $i_1(1) = 1$ and $i_2(1) = 2$ , equipped with the battery            |
| fong-spivak-invitation_F02304 | 222  | ■ 0.998 | $i_2(1)=2$                                                                                                              | $i_2(1) = 2$                                                                                                            | the cospan $\underline{1} \xrightarrow{i_1} \underline{2} \xleftarrow{i_2} \underline{1}$ where $i_1(1) = 1$ and $i_2(1) = 2$ , equipped with the battery            |
| fong-spivak-invitation_F02305 | 222  | ■ 0.987 | $(V, \ A, \ s, \ t, \ \ell) \in \text{Circ}(2)$                                                                         | $(V, A, s, t, \ell) \in \text{Circ}(2)$                                                                                 | some $C$ -circuit $(V, A, s, t, \ell) \in \text{Circ}(2)$ . What is it?                                                                                              |
| fong-spivak-invitation_F02307 | 222  | ■ 0.795 | $\underline{1}$                                                                                                         | $\underline{1}$                                                                                                         | First we’ll consider composition. Consider the following decorated cospan from $\underline{1}$ to $\underline{1}$ :                                                  |
| fong-spivak-invitation_F02308 | 222  | ■ 1.000 | $\text{Circ}([\iota_N, \iota_P])(\psi_{N, P}(s, t))$                                                                    | $\text{Circ}([\iota_N, \iota_P])(\psi_{N, P}(s, t))$                                                                    | oration, we use the formula $\text{Circ}([\iota_N, \iota_P])(\psi_{N, P}(s, t))$ from (6.15). Again, this is rather                                                  |
| fong-spivak-invitation_F02309 | 223  | ■ 0.971 | $\underline{1}=\{\circ\}$                                                                                               | $\underline{1} = \{\circ\}$                                                                                             | (We simply ignore the decorations for now.) If we pushout over the common set $\underline{1} = \{\circ\}$ ,                                                          |
| fong-spivak-invitation_F02310 | 223  | ■ 1.000 | $\psi_{\underline{2}, \underline{2}}$                                                                                   | $\psi_{\underline{2}, \underline{2}}$                                                                                   | tion: you told us what the map $\psi_{\underline{2}, \underline{2}}$ does in Exercise 6.66. It takes the two decorations,                                            |

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| fong-spivak-invitation_F02311 | 223  | ■ 0.912 | $\psi_{\{N, P\}}(s, t)$                                                        | $\psi_{N,P}(s, t)$                                                     | in $\text{Circ}(\underline{4})$ . So this is what the $\psi_{N,P}(s, t)$ part means. What does $[\iota_N, \iota_P]$ mean? Recall                               |
| fong-spivak-invitation_F02312 | 223  | ■ 0.912 | $\text{left}[\iota_{\{N\}}, \iota_{\{P\}}\text{right}]$                        | $[\iota_N, \iota_P]$                                                   | in $\text{Circ}(\underline{4})$ . So this is what the $\psi_{N,P}(s, t)$ part means. What does $[\iota_N, \iota_P]$ mean? Recall                               |
| fong-spivak-invitation_F02313 | 223  | ■ 1.000 | $A \rightarrowtail M \leftarrowtail B$                                         | $A \rightarrow M \leftarrow B$                                         | given two cospans $A \rightarrow M \leftarrow B$ and $C \rightarrow N \leftarrow D$ , we take their coproduct cospan                                           |
| fong-spivak-invitation_F02314 | 223  | ■ 1.000 | $C \rightarrowtail N \leftarrowtail D$                                         | $C \rightarrow N \leftarrow D$                                         | given two cospans $A \rightarrow M \leftarrow B$ and $C \rightarrow N \leftarrow D$ , we take their coproduct cospan                                           |
| fong-spivak-invitation_F02315 | 223  | ■ 1.000 | $A+C \rightarrowtail M+N \leftarrowtail B+D$                                   | $A + C \rightarrow M + N \leftarrow B + D$                             | $A + C \rightarrow M + N \leftarrow B + D$ . And for decorations, we use the map $\psi_{M,N} : \text{Circ}(M) \times$                                          |
| fong-spivak-invitation_F02316 | 223  | ■ 1.000 | $\psi_{\{M, N\}} : \operatorname{Circ}(M) \times$                              | $\psi_{M,N} : \text{Circ}(M) \times$                                   | $A + C \rightarrow M + N \leftarrow B + D$ . And for decorations, we use the map $\psi_{M,N} : \text{Circ}(M) \times$                                          |
| fong-spivak-invitation_F02317 | 224  | ■ 1.000 | $\operatorname{Circ}(N) \rightarrowtail \operatorname{Circ}(M+N)$              | $\text{Circ}(N) \rightarrow \text{Circ}(M + N)$                        | $\text{Circ}(N) \rightarrow \text{Circ}(M + N)$ . So, for example, suppose we want to take the monoidal product                                                |
| fong-spivak-invitation_F02318 | 224  | ■ 1.000 | $\eta : \emptyset \rightarrowtail 2$                                           | $\eta : 0 \rightarrow 2$                                               | <b>Exercise 6.69.</b> Write $x$ for the open circuit in (6.19). Also define cospans $\eta : 0 \rightarrow 2$                                                   |
| fong-spivak-invitation_F02319 | 224  | ■ 0.848 | $\eta : 2 \rightarrowtail \emptyset$                                           | $\eta : 2 \rightarrow 0$                                               | and $\eta : 2 \rightarrow 0$ as follows:                                                                                                                       |
| fong-spivak-invitation_F02320 | 224  | ■ 0.878 | $(\underline{1}, \emptyset, !, !, !) \in \operatorname{Circ}(\underline{1})$ . | $(\underline{1}, \emptyset, !, !, !) \in \text{Circ}(\underline{1})$ . | where each of these are decorated by the empty circuit $(\underline{1}, \emptyset, !, !, !) \in \text{Circ}(\underline{1})$ .                                  |
| fong-spivak-invitation_F02321 | 224  | ■ 0.956 | $\eta ; x ; \epsilon$                                                          | $\eta; x; \epsilon$                                                    | Compute the composite $\eta \circ x \circ \epsilon \in \mathbf{Cospan}_{\text{Circ}}$ . This is a morphism $\underline{0} \rightarrow \underline{0}$ ; we call |
| fong-spivak-invitation_F02322 | 224  | ■ 0.956 | $\mathbf{Cospan}_{\text{Circ}}$                                                | $\mathbf{Cospan}_{\text{Circ}}$                                        | Compute the composite $\eta \circ x \circ \epsilon \in \mathbf{Cospan}_{\text{Circ}}$ . This is a morphism $\underline{0} \rightarrow \underline{0}$ ; we call |
| fong-spivak-invitation_F02323 | 224  | ■ 0.956 | $\underline{\emptyset} \rightarrowtail \underline{\emptyset}$                  | $\underline{0} \rightarrow \underline{0}$                              | Compute the composite $\eta \circ x \circ \epsilon \in \mathbf{Cospan}_{\text{Circ}}$ . This is a morphism $\underline{0} \rightarrow \underline{0}$ ; we call |
| fong-spivak-invitation_F02324 | 225  | ■ 0.890 | $\mathcal{O} \rightarrowtail$                                                  | $\mathcal{O} \rightarrow$                                              | structures from operad functors $\mathcal{O} \rightarrow \mathbf{Set}$ .                                                                                       |
| fong-spivak-invitation_F02325 | 226  | ■ 1.000 | $f, g, h, k$                                                                   | $f, g, h, k$                                                           | The cost is that the “boxes” $f, g, h, k$ in Eq. (6.21) no longer have a left/right                                                                            |
| fong-spivak-invitation_F02326 | 226  | ■ 0.999 | $\mathcal{M} \rightarrowtail$                                                  | $\mathcal{M} \rightarrow$                                              | theory of monoids as a prop $\mathcal{M}$ , and defined monoids using functors $\mathcal{M} \rightarrow \mathbf{Set}$ ; see                                    |
| fong-spivak-invitation_F02327 | 226  | ■ 0.578 | $\mathbf{Cospan}_{\text{FinSet}}$                                              | $\mathbf{Cospan}_{\text{FinSet}}$                                      | Remark 5.64. In the same way, we can view the category $\mathbf{Cospan}_{\text{FinSet}}$ as some sort                                                          |
| fong-spivak-invitation_F02328 | 226  | ■ 0.989 | $\text{FinSet} \rightarrowtail$                                                | $\text{FinSet} \rightarrow$                                            | $\mathbf{Cospan}_{\text{FinSet}} \rightarrow \mathbf{Set}$ .                                                                                                   |
| fong-spivak-invitation_F02329 | 226  | ■ 1.000 | $\mathcal{O}$                                                                  | $\mathcal{O}$                                                          | So that’s the idea. An operad $\mathcal{O}$ will define a theory or grammar of composition, and                                                                |
| fong-spivak-invitation_F02330 | 226  | ■ 0.864 | $t_1, \dots, t_n, t$                                                           | $t_1, \dots, t_n, t$                                                   | (ii) for each tuple $(t_1, \dots, t_n, t)$ of types, one specifies a set $\mathcal{O}(t_1, \dots, t_n, t)$ , whose                                             |
| fong-spivak-invitation_F02331 | 226  | ■ 0.864 | $\mathcal{O}(t_1, \dots, t_n, t)$                                              | $\mathcal{O}(t_1, \dots, t_n, t)$                                      | (ii) for each tuple $(t_1, \dots, t_n, t)$ of types, one specifies a set $\mathcal{O}(t_1, \dots, t_n, t)$ , whose                                             |
| fong-spivak-invitation_F02332 | 226  | ■ 0.990 | $t_1, \dots, t_n ; t$                                                          | $t_1, \dots, t_n ; t$                                                  | elements are called <i>operations of arity</i> $(t_1, \dots, t_n ; t)$ ;                                                                                       |
| fong-spivak-invitation_F02333 | 226  | ■ 0.882 | $s_1, \dots, s_m, t_i$                                                         | $s_1, \dots, s_m, t_i$                                                 | (iii) for each pair of tuples $(s_1, \dots, s_m, t_i)$ and $(t_1, \dots, t_n, t)$ , one specifies a function                                                   |
| fong-spivak-invitation_F02334 | 227  | ■ 0.991 | $\text{id}_t \in \mathcal{O}(t ; t)$                                           | $\text{id}_t \in \mathcal{O}(t; t)$                                    | (iv) for each type $t$ , one specifies an operation $\text{id}_t \in \mathcal{O}(t; t)$ called the <i>identity</i>                                             |
| fong-spivak-invitation_F02335 | 227  | ■ 1.000 | $m+n-1$                                                                        | $m + n - 1$                                                            | of arity $m + n - 1$ : we have $m$ arguments to fill in $m$ , and the remaining $n - 1$ arguments                                                              |
| fong-spivak-invitation_F02336 | 227  | ■ 1.000 | $n-1$                                                                          | $n - 1$                                                                | of arity $m + n - 1$ : we have $m$ arguments to fill in $m$ , and the remaining $n - 1$ arguments                                                              |
| fong-spivak-invitation_F02337 | 227  | ■ 0.648 | $\circ_i$                                                                      | $\circ_i$                                                              | substitution function $\circ_i$ , which says we obtain the operation $f \circ_i g \in \mathcal{O}(m + n - 1)$ . The                                            |
| fong-spivak-invitation_F02338 | 227  | ■ 0.648 | $f \circ_i g \in \mathcal{O}(m+n-1)$                                           | $f \circ g \in \mathcal{O}(m + n - 1)$                                 | substitution function $\circ_i$ , which says we obtain the operation $f \circ_i g \in \mathcal{O}(m + n - 1)$ . The                                            |

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| fong-spivak-invitation_F02339 | 228  | ■ 0.998 | $X_{\{1\}} \times \cdots \times X_{\{n\}} \rightarrow Y$                                                                            | $X_1 \times \cdots \times X_n \rightarrow Y$                                                                               | (ii) functions $X_1 \times \cdots \times X_n \rightarrow Y$ as operations of arity $(X_1, \dots, X_n; Y)$ ,                                                                                                |
| fong-spivak-invitation_F02340 | 228  | ■ 0.998 | $X_{\{1\}}, \ldots, X_{\{n\}} ; Y$                                                                                                  | $X_1, \dots, X_n; Y$                                                                                                       | (ii) functions $X_1 \times \cdots \times X_n \rightarrow Y$ as operations of arity $(X_1, \dots, X_n; Y)$ ,                                                                                                |
| fong-spivak-invitation_F02341 | 228  | ■ 0.929 | $f \in \operatorname{Set}(W_{\{1\}}, \ldots, W_{\{m\}} ; X_{\{i\}})$ , $g \in \operatorname{Set}(X_{\{1\}}, \ldots, X_{\{n\}} ; Y)$ | $f \in \operatorname{Set}(W_1, \dots, W_m; X_i), g \in \operatorname{Set}(X_1, \dots, X_n; Y)$                             | where $f \in \operatorname{Set}(W_1, \dots, W_m; X_i), g \in \operatorname{Set}(X_1, \dots, X_n; Y)$ , and hence $g \circ_i f$ is a                                                                        |
| fong-spivak-invitation_F02342 | 228  | ■ 0.929 | $g \circ_{\{i\}} f$                                                                                                                 | $g \circ_i f$                                                                                                              | where $f \in \operatorname{Set}(W_1, \dots, W_m; X_i), g \in \operatorname{Set}(X_1, \dots, X_n; Y)$ , and hence $g \circ_i f$ is a                                                                        |
| fong-spivak-invitation_F02343 | 228  | ■ 0.780 | $\operatorname{identities} \operatorname{id}_X \in \operatorname{Set}(X; X)$                                                        | identities $\operatorname{id}_X \in \operatorname{Set}(X; X)$                                                              | (iv) identities $\operatorname{id}_X \in \operatorname{Set}(X; X)$ given by the identity function $\operatorname{id}_X : X \rightarrow X$ .                                                                |
| fong-spivak-invitation_F02344 | 228  | ■ 0.780 | $\operatorname{id}_X : X \rightarrow X$                                                                                             | $\operatorname{id}_X : X \rightarrow X$                                                                                    | (iv) identities $\operatorname{id}_X \in \operatorname{Set}(X; X)$ given by the identity function $\operatorname{id}_X : X \rightarrow X$ .                                                                |
| fong-spivak-invitation_F02345 | 228  | ■ 1.000 | $a \in \mathbb{N}$                                                                                                                  | $a \in \mathbb{N}$                                                                                                         | (i) natural numbers $a \in \mathbb{N}$ as types,                                                                                                                                                           |
| fong-spivak-invitation_F02346 | 228  | ■ 0.810 | $\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$                                     | $\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$                            | (ii) cospans $\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$ of finite sets as operations of arity $(a_1, \dots, a_n; b)$ ,                                |
| fong-spivak-invitation_F02347 | 228  | ■ 0.810 | $\left( \underline{a_1}, \ldots, \underline{a_n} ; b \right)$                                                                       | $(a_1, \dots, a_n; b)$                                                                                                     | (ii) cospans $\underline{a_1} + \cdots + \underline{a_n} \rightarrow \underline{p} \leftarrow \underline{b}$ of finite sets as operations of arity $(a_1, \dots, a_n; b)$ ,                                |
| fong-spivak-invitation_F02348 | 228  | ■ 0.185 | $\operatorname{id}_a \in \operatorname{Set}(a ; a)$                                                                                 | $\operatorname{id}_a \in \operatorname{Set}(a; a)$                                                                         | (iv) identities $\operatorname{id}_a \in \operatorname{Set}(a; a)$ just given by the identity $\operatorname{id}_a : a \rightarrow a$ .                                                                    |
| fong-spivak-invitation_F02349 | 228  | ■ 0.185 | $\operatorname{cospans} \operatorname{id}_a \in \operatorname{Set}(a; a)$                                                           | (not rendered)                                                                                                             | (iv) identities $\operatorname{id}_a \in \operatorname{Set}(a; a)$ just given by the identity $\operatorname{id}_a : a \rightarrow a$ .                                                                    |
| fong-spivak-invitation_F02350 | 228  | ■ 0.922 | $\operatorname{FinSet}, 0, +$                                                                                                       | $\operatorname{FinSet}, 0, +$                                                                                              | This is the operadic analogue of the monoidal category $(\operatorname{Cospan}_{\operatorname{FinSet}}, 0, +)$ .                                                                                           |
| fong-spivak-invitation_F02351 | 229  | ■ 0.536 | $\underline{3}, \underline{3}, \underline{4}, \underline{2} ; \underline{3}$                                                        | $\underline{3}, \underline{3}, \underline{4}, \underline{2} ; \underline{3}$                                               | This is an operation of arity $(\underline{3}, \underline{3}, \underline{4}, \underline{2}; \underline{3})$ . Why? The circles marked $f$ and $g$ have 3                                                   |
| fong-spivak-invitation_F02352 | 229  | ■ 0.999 | $\underline{3} + \underline{3} + \underline{4} + \underline{2} \rightarrow \underline{p} \leftarrow \underline{3}$                  | $\underline{3} + \underline{3} + \underline{4} + \underline{2} \xrightarrow{a} \underline{p} \xleftarrow{b} \underline{3}$ | arity is, by (ii), a cospan $\underline{3} + \underline{3} + \underline{4} + \underline{2} \xrightarrow{a} \underline{p} \xleftarrow{b} \underline{3}$ . In the diagram above, the apex $\underline{p}$ is |
| fong-spivak-invitation_F02353 | 229  | ■ 0.999 | $\underline{p}$                                                                                                                     | $\underline{p}$                                                                                                            | arity is, by (ii), a cospan $\underline{3} + \underline{3} + \underline{4} + \underline{2} \xrightarrow{a} \underline{p} \xleftarrow{b} \underline{3}$ . In the diagram above, the apex $\underline{p}$ is |
| fong-spivak-invitation_F02354 | 229  | ■ 0.973 | $\underline{7}$                                                                                                                     | $\underline{7}$                                                                                                            | the set $\underline{7}$ , because there are 7 nodes $\bullet$ in the diagram. The function $a$ sends each port on                                                                                          |
| fong-spivak-invitation_F02355 | 229  | ■ 0.972 | $f \in \operatorname{Cospan}(2, 2 ; 2)$                                                                                             | $f \in \operatorname{Cospan}(2, 2; 2)$                                                                                     | 1. Consider the following cospan $f \in \operatorname{Cospan}(2, 2; 2)$ :                                                                                                                                  |
| fong-spivak-invitation_F02356 | 229  | ■ 1.000 | $g \in$                                                                                                                             | $g \in$                                                                                                                    | 2. Draw the wiring diagram corresponding to the following cospan $g \in$                                                                                                                                   |
| fong-spivak-invitation_F02357 | 229  | ■ 0.999 | $\operatorname{Cospan}(2, 2, 2 ; 0)$                                                                                                | $\operatorname{Cospan}(2, 2, 2; 0)$                                                                                        | $\operatorname{Cospan}(2, 2, 2; 0)$ :                                                                                                                                                                      |
| fong-spivak-invitation_F02358 | 229  | ■ 0.706 | $g \circ_{\{1\}} f$                                                                                                                 | $g \circ_1 f$                                                                                                              | 3. Compute the cospan $g \circ_1 f$ . What is its arity?                                                                                                                                                   |
| fong-spivak-invitation_F02359 | 229  | ■ 0.589 | $\mathcal{O}_{\mathcal{C}}$                                                                                                         | $\mathcal{O}_{\mathcal{C}}$                                                                                                | $\mathcal{O}_{\mathcal{C}}$ , called the <i>operad underlying</i> $\mathcal{C}$ , defined as having:                                                                                                       |
| fong-spivak-invitation_F02360 | 229  | ■ 0.957 | $\operatorname{Ob}(\mathcal{C})$                                                                                                    | $\operatorname{Ob}(\mathcal{C})$                                                                                           | (i) $\operatorname{Ob}(\mathcal{C})$ as types,                                                                                                                                                             |
| fong-spivak-invitation_F02361 | 229  | ■ 0.940 | $C_{\{1\}} \otimes \cdots \otimes C_{\{n\}} \rightarrow D$                                                                          | $C_1 \otimes \cdots \otimes C_n \rightarrow D$                                                                             | (ii) morphisms $C_1 \otimes \cdots \otimes C_n \rightarrow D$ in $\mathcal{C}$ as the operations of arity $(C_1, \dots, C_n; D)$ ,                                                                         |
| fong-spivak-invitation_F02362 | 229  | ■ 0.940 | $C_{\{1\}}, \ldots, C_{\{n\}} ; D$                                                                                                  | $C_1, \dots, C_n; D$                                                                                                       | (ii) morphisms $C_1 \otimes \cdots \otimes C_n \rightarrow D$ in $\mathcal{C}$ as the operations of arity $(C_1, \dots, C_n; D)$ ,                                                                         |
| fong-spivak-invitation_F02363 | 230  | ■ 0.800 | $\operatorname{id}_a \in \mathcal{O}_{\mathcal{C}}(a ; a)$                                                                          | $\operatorname{id}_a \in \mathcal{O}_{\mathcal{C}}(a; a)$                                                                  | (iv) identities $\operatorname{id}_a \in \mathcal{O}_{\mathcal{C}}(a; a)$ defined by $\operatorname{id}_a$ .                                                                                               |

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| fong-spivak-invitation_F02364 | 230  | 0.800 | $\mathrm{id}_{\{a\}}$                                              | $\mathrm{id}_a$                                             | (iv) identities $\mathrm{id}_a \in \mathcal{O}_{\mathcal{C}}(a; a)$ defined by $\mathrm{id}_a$ .                                            |
| fong-spivak-invitation_F02365 | 230  | 0.997 | $F: \mathcal{O} \rightarrow \mathcal{P}$                           | $F: \mathcal{O} \rightarrow \mathcal{P}$                    | collections $T$ and $U$ respectively. To specify an operad functor $F: \mathcal{O} \rightarrow \mathcal{P}$ ,                               |
| fong-spivak-invitation_F02366 | 230  | 1.000 | $f: T \rightarrow U$                                               | $f: T \rightarrow U$                                        | (i) one specifies a function $f: T \rightarrow U$ ,                                                                                         |
| fong-spivak-invitation_F02367 | 230  | 1.000 | $F: \mathcal{O} \rightarrow$                                       | $F: \mathcal{O} \rightarrow$                                | <b>Definition 6.77.</b> An algebra for an operad $\mathcal{O}$ is an operad functor $F: \mathcal{O} \rightarrow \mathbf{Set}$ .             |
| fong-spivak-invitation_F02368 | 230  | 1.000 | $F(t)$                                                             | $F(t)$                                                      | operad $\mathcal{O}$ and an algebra $F: \mathcal{O} \rightarrow \mathbf{Set}$ will take a type $t$ and return a set $F(t)$ of fillers       |
| fong-spivak-invitation_F02369 | 230  | 1.000 | $f \in \mathcal{O}(\mathrm{t}_1, \dots, \mathrm{t}_n; \mathrm{t})$ | $f \in \mathcal{O}(t_1, \dots, t_n; t)$                     | for box $t$ . Moreover, given an operation (i.e. a wiring diagram) $f \in \mathcal{O}(t_1, \dots, t_n; t)$ , we                             |
| fong-spivak-invitation_F02370 | 230  | 1.000 | $F(f)$                                                             | $F(f)$                                                      | get a function $F(f)$ that takes an element of each set $F(t_i)$ , and returns an element of                                                |
| fong-spivak-invitation_F02371 | 230  | 1.000 | $F(\mathrm{t}_i)$                                                  | $F(t_i)$                                                    | get a function $F(f)$ that takes an element of each set $F(t_i)$ , and returns an element of                                                |
| fong-spivak-invitation_F02372 | 230  | 1.000 | $\mathrm{t}_1, \dots, \mathrm{t}_n$                                | $t_1, \dots, t_n$                                           | $F(t)$ . For example, it takes $n$ circuits with interface $t_1, \dots, t_n$ , respectively, and returns                                    |
| fong-spivak-invitation_F02373 | 230  | 0.998 | $T = \mathrm{Ob}$                                                  | $T = \mathrm{Ob}$                                           | <b>Example 6.78.</b> For electric circuits, the types are again finite sets, $T = \mathrm{Ob}(\mathbf{FinSet})$ ,                           |
| fong-spivak-invitation_F02374 | 230  | 0.691 | $\mathrm{Circ}(t)$                                                 | $\mathrm{Circ}(t)$                                          | have a set $\mathrm{Circ}(t)$ of fillers, namely the set of electric circuits with that $t$ -marked ter-                                    |
| fong-spivak-invitation_F02375 | 231  | 0.623 | $\mathrm{Circ}(\varphi) \in \mathrm{Set}(\mathrm{Circ}(2))$        | $\mathrm{Circ}(\varphi) \in \mathrm{Set}(\mathrm{Circ}(2))$ | above-drawn case, we would get a morphism $\mathrm{Circ}(\varphi) \in \mathbf{Set}(\mathrm{Circ}(2), \mathrm{Circ}(2), \mathrm{Circ}(2))$ ; |
| fong-spivak-invitation_F02376 | 231  | 0.623 | $\mathrm{Circ}(2)$                                                 | $\mathrm{Circ}(2)$                                          | above-drawn case, we would get a morphism $\mathrm{Circ}(\varphi) \in \mathbf{Set}(\mathrm{Circ}(2), \mathrm{Circ}(2), \mathrm{Circ}(2))$ ; |
| fong-spivak-invitation_F02377 | 231  | 0.490 | $\mathrm{Circ}(0)$                                                 | $\mathrm{Circ}(0)$                                          | $\mathrm{Circ}(0)$ , i.e. a function                                                                                                        |
| fong-spivak-invitation_F02378 | 233  | 1.000 | $\in \mathbb{R}$                                                   | $\in \mathbb{R}$                                            | safe $\in \mathbb{R}$ from other objects. To do so, several components must interact: a sensor                                              |
| fong-spivak-invitation_F02379 | 233  | 1.000 | $S^{\prime}$                                                       | $S^{\prime}$                                                | that approximates the real distance by an internal variable $S^{\prime}$ , a controller that uses $S^{\prime}$ to                           |
| fong-spivak-invitation_F02380 | 234  | 0.953 | $S \geq$                                                           | $S \geq$                                                    | In order for the system to ensure $S \geq \text{safe}$ , we need each of the components                                                     |
| fong-spivak-invitation_F02381 | 234  | 0.922 | $S < \text{safe}$                                                  | $S < \text{safe}$                                           | signal $A$ , e.g. that if at any time $S < \text{safe}$ , then within one second it will emit the sig-                                      |
| fong-spivak-invitation_F02382 | 234  | 0.984 | $A =$                                                              | $A =$                                                       | nal $A = \text{go}$ . The motor maintains a relationship between $A$ and $S$ , e.g. that $A$ dictates                                       |
| fong-spivak-invitation_F02383 | 235  | 0.911 | $[a, b]$                                                           | $[a, b]$                                                    | • $S$ (wire): The behavior of $S$ over a time-interval $[a, b]$ is that of all continuous real-                                             |
| fong-spivak-invitation_F02384 | 235  | 1.000 | $[a, b] \rightarrow \mathbb{R}$                                    | $[a, b] \rightarrow \mathbb{R}$                             | valued functions $[a, b] \rightarrow \mathbb{R}$ .                                                                                          |
| fong-spivak-invitation_F02385 | 235  | 0.998 | $\left\{ \left( S^{\prime}, A \right) \mid \right.$                | $\{(S^{\prime}, A) \mid$                                    | • controller (box): The relation $\{(S^{\prime}, A) \mid \text{Eq. (7.2)}\}$ , i.e. all behavioral pairs $(S^{\prime}, A)$                  |
| fong-spivak-invitation_F02386 | 235  | 0.998 | $\left. \left( S^{\prime}, A \right) \right\}$                     | $\left. (S^{\prime}, A) \right\}$                           | • controller (box): The relation $\{(S^{\prime}, A) \mid \text{Eq. (7.2)}\}$ , i.e. all behavioral pairs $(S^{\prime}, A)$                  |
| fong-spivak-invitation_F02387 | 236  | 0.929 | $\in P$                                                            | $\in P$                                                     | people to the set $\mathbb{B} = \{\text{false}, \text{true}\}$ of truth values, where Brendan $\in P$ maps to                               |
| fong-spivak-invitation_F02388 | 236  | 1.000 | $\mathcal{E}$                                                      | $\mathcal{E}$                                               | in many ways. Here are some facts that are true of any topos $\mathcal{E}$ :                                                                |
| fong-spivak-invitation_F02389 | 236  | 1.000 | $1 \rightarrow \Omega$                                             | $1 \xrightarrow{\text{true}} \Omega$                        | 5. $\mathcal{E}$ has a subobject classifier $1 \xrightarrow{\text{true}} \Omega$ .                                                          |
| fong-spivak-invitation_F02390 | 236  | 0.996 | $\mathcal{E} =$                                                    | $\mathcal{E} =$                                             | In particular, since $\mathbf{Set}$ is a topos, all of the above facts are true for $\mathcal{E} = \mathbf{Set}$ . Our first                |

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| fong-spivak-invitation_F02391 | 237  | ■ 0.833 | $\lrcorner$                                  | $\lrcorner$                                  | In the left-hand square, the corner symbol $\lrcorner$ unambiguously means that the square                                           |
| fong-spivak-invitation_F02392 | 237  | ■ 0.999 | $(B, C, E, F)$                               | $(B, C, E, F)$                               | $(B, C, E, F)$ is a pullback. But in the right-hand square, does the corner symbol mean                                              |
| fong-spivak-invitation_F02393 | 237  | ■ 0.876 | $(A, B, D, E)$                               | $(A, B, D, E)$                               | that $(A, B, D, E)$ is a pullback or that $(A, C, D, F)$ is a pullback? It's ambiguous but,                                          |
| fong-spivak-invitation_F02394 | 237  | ■ 0.876 | $(A, C, D, F)$                               | $(A, C, D, F)$                               | that $(A, B, D, E)$ is a pullback or that $(A, C, D, F)$ is a pullback? It's ambiguous but,                                          |
| fong-spivak-invitation_F02395 | 237  | ■ 0.989 | $\left(B, C, B^{\prime}, C^{\prime}\right)$  | $(B, C, B', C')$                             | <b>Proposition 7.1.</b> In the commutative diagram below, suppose that the $(B, C, B', C')$                                          |
| fong-spivak-invitation_F02396 | 237  | ■ 0.811 | $A, B, A^{\prime}, B^{\prime}$               | $A, B, A', B'$                               | Then the $(A, B, A', B')$ square is a pullback iff the $(A, C, A', C')$ rectangle is a                                               |
| fong-spivak-invitation_F02397 | 237  | ■ 0.811 | $A, C, A^{\prime}, C^{\prime}$               | $A, C, A', C'$                               | Then the $(A, B, A', B')$ square is a pullback iff the $(A, C, A', C')$ rectangle is a                                               |
| fong-spivak-invitation_F02398 | 238  | ■ 1.000 | $i: B^{\prime} \rightarrow B$                | $i: B' \rightarrow B$                        | That is, suppose that $i: B' \rightarrow B$ is an isomorphism and $f: A \rightarrow B$ is any morphism.                              |
| fong-spivak-invitation_F02399 | 238  | ■ 1.000 | $i^{\prime}$                                 | $i'$                                         | Show that $i'$ is an isomorphism, in the following diagram:                                                                          |
| fong-spivak-invitation_F02400 | 238  | ■ 1.000 | $f^{\prime}$                                 | $f'$                                         | is $f'$ .                                                                                                                            |
| fong-spivak-invitation_F02401 | 238  | ■ 1.000 | $f: C \rightarrow D$                         | $f: C \rightarrow D$                         | mono factorizations are. We say that a morphism $f: C \rightarrow D$ in $\mathcal{E}$ has an epi-mono                                |
| fong-spivak-invitation_F02402 | 238  | ■ 1.000 | $\operatorname{im}(f)$                       | $\operatorname{im}(f)$                       | factorization if it has an “image”; that is, there is an object $\operatorname{im}(f)$ , an epimorphism                              |
| fong-spivak-invitation_F02403 | 238  | ■ 0.811 | $C \rightarrow \operatorname{im}(f)$         | $C \rightarrow \operatorname{im}(f)$         | $C \twoheadrightarrow \operatorname{im}(f)$ , and a monomorphism $\operatorname{im}(f) \hookrightarrow D$ , whose composite is $f$ . |
| fong-spivak-invitation_F02404 | 238  | ■ 0.811 | $\operatorname{im}(f) \rightarrowtail D$     | $\operatorname{im}(f) \hookrightarrow D$     | $C \twoheadrightarrow \operatorname{im}(f)$ , and a monomorphism $\operatorname{im}(f) \hookrightarrow D$ , whose composite is $f$ . |
| fong-spivak-invitation_F02405 | 238  | ■ 0.938 | $\operatorname{im}(f)=\{f(c) \mid$           | $\operatorname{im}(f)=\{f(c) \mid$           | tion $f: C \rightarrow D$ may be factored as a surjective function onto its image $\operatorname{im}(f)=\{f(c)$                      |
| fong-spivak-invitation_F02406 | 238  | ■ 1.000 | $c \in C\}$                                  | $c \in C\}$                                  | $c \in C\}$ , followed by the inclusion of this image into the codomain $D$ . Moreover, this                                         |
| fong-spivak-invitation_F02407 | 239  | ■ 1.000 | $f: \underline{3} \rightarrow \underline{3}$ | $f: \underline{3} \rightarrow \underline{3}$ | <b>Exercise 7.7.</b> Factor the following function $f: \underline{3} \rightarrow \underline{3}$ as an epimorphism followed by        |
| fong-spivak-invitation_F02408 | 239  | ■ 1.000 | $f=(e ; m)$                                  | $f=(e ; m)$                                  | epimorphism $e$ and a monomorphism $m$ such that $f=(e \circ m)$ is their composite.                                                 |
| fong-spivak-invitation_F02409 | 239  | ■ 1.000 | $C, D \in \mathcal{C}$                       | $C, D \in \mathcal{C}$                       | two objects $C, D \in \mathcal{C}$ , there is a “hom-object” $D^C \in \mathcal{C}$ and a natural isomorphism for                     |
| fong-spivak-invitation_F02410 | 239  | ■ 1.000 | $D^{\mathcal{C}} \in \mathcal{C}$            | $D^C \in \mathcal{C}$                        | two objects $C, D \in \mathcal{C}$ , there is a “hom-object” $D^C \in \mathcal{C}$ and a natural isomorphism for                     |
| fong-spivak-invitation_F02411 | 239  | ■ 1.000 | $A \in \mathcal{C}$                          | $A \in \mathcal{C}$                          | any $A \in \mathcal{C}$ :                                                                                                            |
| fong-spivak-invitation_F02412 | 239  | ■ 1.000 | $f(-,-): A \times C \rightarrow D$           | $f(-,-): A \times C \rightarrow D$           | game $f(-,-): A \times C \rightarrow D$ ; for whatever action $a \in A$ that you take and action $c \in C$                           |
| fong-spivak-invitation_F02413 | 239  | ■ 1.000 | $f(a, c)$                                    | $f(a, c)$                                    | that I take, $f(a, c)$ is some value in $D$ . Since you're self-centered but loving, you think                                       |
| fong-spivak-invitation_F02414 | 239  | ■ 1.000 | $f(a,-): C \rightarrow D$                    | $f(a,-): C \rightarrow D$                    | you make a game $f(a,-): C \rightarrow D$ for me alone. In the formalism, $D^C$ represents the                                       |
| fong-spivak-invitation_F02415 | 239  | ■ 1.000 | $D^{\mathcal{C}}$                            | $D^C$                                        | you make a game $f(a,-): C \rightarrow D$ for me alone. In the formalism, $D^C$ represents the                                       |
| fong-spivak-invitation_F02416 | 239  | ■ 1.000 | $v, w, x \in V$                              | $v, w, x \in V$                              | Definition 2.66 – and suppose it satisfies the following for all $v, w, x \in V$ :                                                   |
| fong-spivak-invitation_F02417 | 239  | ■ 1.000 | $v \leq I$                                   | $v \leq I$                                   | • $v \leq I$ ,                                                                                                                       |

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| fong-spivak-invitation_F02418 | 239  | ■ 1.000 | $v \otimes w \leq v$                                        | $v \otimes w \leq v$                                        | • $v \otimes w \leq v$ and $v \otimes w \leq w$ ,                                                                                                 |
| fong-spivak-invitation_F02419 | 239  | ■ 1.000 | $v \otimes w \leq w$                                        | $v \otimes w \leq w$                                        | • $v \otimes w \leq v$ and $v \otimes w \leq w$ ,                                                                                                 |
| fong-spivak-invitation_F02420 | 239  | ■ 1.000 | $x \leq v$                                                  | $x \leq v$                                                  | • if $x \leq v$ and $x \leq w$ then $x \leq v \otimes w$ .                                                                                        |
| fong-spivak-invitation_F02421 | 239  | ■ 1.000 | $x \leq w$                                                  | $x \leq w$                                                  | • if $x \leq v$ and $x \leq w$ then $x \leq v \otimes w$ .                                                                                        |
| fong-spivak-invitation_F02422 | 239  | ■ 1.000 | $x \leq v \otimes w$                                        | $x \leq v \otimes w$                                        | • if $x \leq v$ and $x \leq w$ then $x \leq v \otimes w$ .                                                                                        |
| fong-spivak-invitation_F02423 | 240  | ■ 0.781 | $X \hookrightarrow Y$                                       | $X \hookrightarrow Y$                                       | are injections, and any injection $X \hookrightarrow Y$ is isomorphic to a subset of $Y$ . This gives                                             |
| fong-spivak-invitation_F02424 | 240  | ■ 0.822 | $\Omega \in \mathcal{E}$                                    | $\Omega \in \mathcal{E}$                                    | nal object $1$ . A <i>subobject classifier</i> in $\mathcal{E}$ consists of an object $\Omega \in \mathcal{E}$ , together with a                  |
| fong-spivak-invitation_F02425 | 240  | ■ 1.000 | $1 \rightarrow \Omega$                                      | $1 \rightarrow \Omega$                                      | monomorphism $\text{true}: 1 \rightarrow \Omega$ , satisfying the following property: for any objects $X$                                         |
| fong-spivak-invitation_F02426 | 240  | ■ 0.496 | $m: X \rightrightarrows Y$                                  | $m: X \rightrightarrows Y$                                  | and $Y$ and monomorphism $m: X \rightarrow Y$ in $\mathcal{E}$ , there is a unique morphism $\ulcorner m \urcorner: Y \rightarrow \Omega$         |
| fong-spivak-invitation_F02427 | 240  | ■ 0.496 | $\ulcorner m \urcorner: Y \rightarrow \Omega$               | $\ulcorner m \urcorner: Y \rightarrow \Omega$               | and $Y$ and monomorphism $m: X \rightarrow Y$ in $\mathcal{E}$ , there is a unique morphism $\ulcorner m \urcorner: Y \rightarrow \Omega$         |
| fong-spivak-invitation_F02428 | 240  | ■ 0.999 | $\ulcorner m \urcorner$                                     | $\ulcorner m \urcorner$                                     | We refer to $\ulcorner m \urcorner$ as the <i>characteristic map</i> of $m$ , or we say that $\ulcorner m \urcorner$ <i>classifies</i> $m$ . Con- |
| fong-spivak-invitation_F02429 | 240  | ■ 0.999 | $p: Y \rightarrow \Omega$                                   | $p: Y \rightarrow \Omega$                                   | versely, given any map $p: Y \rightarrow \Omega$ , we denote the pullback of $\text{true}$ as on the right of                                     |
| fong-spivak-invitation_F02430 | 240  | ■ 1.000 | $Y \rightarrow \Omega$                                      | $Y \rightarrow \Omega$                                      | A <i>predicate</i> on $Y$ is a morphism $Y \rightarrow \Omega$ .                                                                                  |
| fong-spivak-invitation_F02431 | 241  | ■ 1.000 | $1 \rightarrow$                                             | $1 \rightarrow$                                             | generically called $\text{true}: 1 \rightarrow \Omega$ ; in the case of <b>Set</b> it is played by the function $1 \rightarrow$                   |
| fong-spivak-invitation_F02432 | 241  | ■ 0.976 | $m: X \hookrightarrow Y$                                    | $m: X \hookrightarrow Y$                                    | tion 7.9 says that, for any injective function $m: X \hookrightarrow Y$ between sets, we are supposed                                             |
| fong-spivak-invitation_F02433 | 241  | ■ 0.968 | $\ulcorner m \urcorner: Y \rightarrow \{$                   | $\ulcorner m \urcorner: Y \rightarrow \{$                   | to be able to find a characteristic function $\ulcorner m \urcorner: Y \rightarrow \{\text{true}, \text{false}\}$ with some sort                  |
| fong-spivak-invitation_F02434 | 241  | ■ 1.000 | $\ulcorner m \urcorner(y)$                                  | $\ulcorner m \urcorner(y)$                                  | In other words, if we think of $X$ as a subset of $Y$ , then we make $\ulcorner m \urcorner(y)$ equal to $\text{true}$                            |
| fong-spivak-invitation_F02435 | 241  | ■ 1.000 | $y \in X$                                                   | $y \in X$                                                   | iff $y \in X$ .                                                                                                                                   |
| fong-spivak-invitation_F02436 | 241  | ■ 1.000 | $p: Y \rightarrow \mathbb{B}$                               | $p: Y \rightarrow \mathbb{B}$                               | $p: Y \rightarrow \mathbb{B}$ , and vice versa. How it works is encoded in Definition 7.9, but the basic idea                                     |
| fong-spivak-invitation_F02437 | 241  | ■ 1.000 | $\{Y \mid p\}$                                              | $\{Y \mid p\}$                                              | This might help explain our abstract notation $\{Y \mid p\}$ in Eq. (7.4).                                                                        |
| fong-spivak-invitation_F02438 | 241  | ■ 1.000 | $X = \mathbb{N} = \{0, 1, 2, \dots\}$                       | $X = \mathbb{N} = \{0, 1, 2, \dots\}$                       | <b>Exercise 7.10.</b> Let $X = \mathbb{N} = \{0, 1, 2, \dots\}$ and $Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$ ; we                         |
| fong-spivak-invitation_F02439 | 241  | ■ 1.000 | $Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$            | $Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$            | <b>Exercise 7.10.</b> Let $X = \mathbb{N} = \{0, 1, 2, \dots\}$ and $Y = \mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$ ; we                         |
| fong-spivak-invitation_F02440 | 241  | ■ 0.995 | $m: X \succ Y$                                              | $m: X \succ Y$                                              | have $X \subseteq Y$ , so consider it as a monomorphism $m: X \hookrightarrow Y$ . It has a characteristic                                        |
| fong-spivak-invitation_F02441 | 241  | ■ 1.000 | $\ulcorner m \urcorner: Y \rightarrow \mathbb{B}$           | $\ulcorner m \urcorner: Y \rightarrow \mathbb{B}$           | function $\ulcorner m \urcorner: Y \rightarrow \mathbb{B}$ , as in Definition 7.9.                                                                |
| fong-spivak-invitation_F02442 | 241  | ■ 1.000 | $\ulcorner m \urcorner(-5) \in \mathbb{B}$                  | $\ulcorner m \urcorner(-5) \in \mathbb{B}$                  | 1. What is $\ulcorner m \urcorner(-5) \in \mathbb{B}$ ?                                                                                           |
| fong-spivak-invitation_F02443 | 241  | ■ 1.000 | $\ulcorner m \urcorner(0) \in \mathbb{B}$                   | $\ulcorner m \urcorner(0) \in \mathbb{B}$                   | 2. What is $\ulcorner m \urcorner(0) \in \mathbb{B}$ ?                                                                                            |
| fong-spivak-invitation_F02444 | 241  | ■ 0.998 | $\text{id}_{\mathbb{N}}: \mathbb{N} \rightarrow \mathbb{N}$ | $\text{id}_{\mathbb{N}}: \mathbb{N} \rightarrow \mathbb{N}$ | 1. Consider the identity function $\text{id}_{\mathbb{N}}: \mathbb{N} \rightarrow \mathbb{N}$ . It is an injection, so it has a characteristic    |

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| fong-spivak-invitation_F02445 | 241  | ■ 0.808 | $\left\langle \ulcorner \mathrm{id}_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B} \right\rangle$ | $\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$ | function $\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$ . Give a concrete description of $\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner$ , i.e. its exact value for |
| fong-spivak-invitation_F02446 | 241  | ■ 0.808 | $\left\langle \ulcorner \mathrm{id}_{\mathbb{N}} \urcorner \right\rangle$                                     | $\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner$                                     | function $\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$ . Give a concrete description of $\ulcorner \mathrm{id}_{\mathbb{N}} \urcorner$ , i.e. its exact value for |
| fong-spivak-invitation_F02447 | 241  | ■ 1.000 | $!_{\mathbb{N}} : \emptyset \rightarrow \mathbb{N}$                                                           | $!_{\mathbb{N}} : \emptyset \rightarrow \mathbb{N}$                                | 2. Consider the unique function $!_{\mathbb{N}} : \emptyset \rightarrow \mathbb{N}$ from the empty set. Give a concrete                                                                                |
| fong-spivak-invitation_F02448 | 241  | ■ 0.654 | $\left\langle \ulcorner !_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B} \right\rangle$           | $\ulcorner !_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$           | description of $\ulcorner !_{\mathbb{N}} \urcorner : \mathbb{N} \rightarrow \mathbb{B}$ .                                                                                                              |
| fong-spivak-invitation_F02449 | 242  | ■ 1.000 | $1 \rightarrow \mathbb{B} \times \mathbb{B}$                                                                  | $1 \rightarrow \mathbb{B} \times \mathbb{B}$                                       | Consider the function $1 \rightarrow \mathbb{B} \times \mathbb{B}$ picking out the element $(\mathrm{true}, \mathrm{true})$ . This is a                                                                |
| fong-spivak-invitation_F02450 | 242  | ■ 0.994 | $\urcorner : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$                                             | $\urcorner : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$                  | monomorphism, so it defines a characteristic function $\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ .                                     |
| fong-spivak-invitation_F02451 | 242  | ■ 1.000 | $\mathbb{B} \times \mathbb{B}$                                                                                | $\mathbb{B} \times \mathbb{B}$                                                     | What function is it? By Eq. (7.6) the only element of $\mathbb{B} \times \mathbb{B}$ that can be sent to $\mathrm{true}$                                                                               |
| fong-spivak-invitation_F02452 | 242  | ■ 0.944 | $\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner (P, Q) \in \mathbb{B}$                                    | $\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner (P, Q) \in \mathbb{B}$         | is $(\mathrm{true}, \mathrm{true})$ . Thus $\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner (P, Q) \in \mathbb{B}$ must be given by the following                                                   |
| fong-spivak-invitation_F02453 | 242  | ■ 0.944 | $\urcorner (P, Q) \in \mathbb{B}$                                                                             | $\urcorner (P, Q) \in \mathbb{B}$                                                  | is $(\mathrm{true}, \mathrm{true})$ . Thus $\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner (P, Q) \in \mathbb{B}$ must be given by the following                                                   |
| fong-spivak-invitation_F02455 | 242  | ■ 1.000 | $P \wedge Q$                                                                                                  | $P \wedge Q$                                                                       | This is exactly the truth table for the AND of $P$ and $Q$ , i.e. for $P \wedge Q$ . In other words,                                                                                                   |
| fong-spivak-invitation_F02456 | 242  | ■ 0.929 | $\urcorner = \wedge$                                                                                          | $\urcorner = \wedge$                                                               | $\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner = \wedge$ . Note that this defines $\wedge$ as a function $\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ , and we                  |
| fong-spivak-invitation_F02457 | 242  | ■ 0.929 | $\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$                                                | $\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$                     | $\ulcorner (\mathrm{true}, \mathrm{true}) \urcorner = \wedge$ . Note that this defines $\wedge$ as a function $\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ , and we                  |
| fong-spivak-invitation_F02458 | 242  | ■ 1.000 | $x \wedge y := \wedge(x, y)$                                                                                  | $x \wedge y := \wedge(x, y)$                                                       | use the usual infix notation $x \wedge y := \wedge(x, y)$ .                                                                                                                                            |
| fong-spivak-invitation_F02459 | 242  | ■ 1.000 | $\vee : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$                                                  | $\vee : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$                       | function $\vee : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ defining OR, is:                                                                                                                 |
| fong-spivak-invitation_F02461 | 242  | ■ 1.000 | $m : X \subseteq \mathbb{B} \times \mathbb{B}$                                                                | $m : X \subseteq \mathbb{B} \times \mathbb{B}$                                     | set $m : X \subseteq \mathbb{B} \times \mathbb{B}$ , what subset would $X$ be? By Eq. (7.6), $X$ should be the set of                                                                                  |
| fong-spivak-invitation_F02462 | 242  | ■ 0.912 | $\mathbb{B} \times$                                                                                           | $\mathbb{B} \times$                                                                | tion 7.2.1. In fact, one can think of $X$ as the union of $\{\mathrm{true}\} \times \mathbb{B}$ and $\mathbb{B} \times$                                                                                |
| fong-spivak-invitation_F02463 | 242  | ■ 1.000 | $\Omega \times \Omega$                                                                                        | $\Omega \times \Omega$                                                             | gous subobject of $\Omega \times \Omega$ , and hence classify a map $\Omega \times \Omega \rightarrow \Omega$ , in any                                                                                 |
| fong-spivak-invitation_F02464 | 242  | ■ 1.000 | $\Omega \times \Omega \rightarrow \Omega$                                                                     | $\Omega \times \Omega \rightarrow \Omega$                                          | gous subobject of $\Omega \times \Omega$ , and hence classify a map $\Omega \times \Omega \rightarrow \Omega$ , in any                                                                                 |
| fong-spivak-invitation_F02465 | 243  | ■ 1.000 | $\neg : \mathbb{B} \rightarrow \mathbb{B}$                                                                    | $\neg : \mathbb{B} \rightarrow \mathbb{B}$                                         | false. The function $\neg : \mathbb{B} \rightarrow \mathbb{B}$ is the characteristic function of some thing, (*?*)                                                                                     |
| fong-spivak-invitation_F02466 | 243  | ■ 1.000 | $P \Rightarrow Q$                                                                                             | $P \Rightarrow Q$                                                                  | <b>Exercise 7.13.</b> Given two booleans $P, Q$ , define $P \Rightarrow Q$ to mean $P = (P \wedge Q)$ .                                                                                                |
| fong-spivak-invitation_F02467 | 243  | ■ 1.000 | $P = (P \wedge Q)$                                                                                            | $P = (P \wedge Q)$                                                                 | <b>Exercise 7.13.</b> Given two booleans $P, Q$ , define $P \Rightarrow Q$ to mean $P = (P \wedge Q)$ .                                                                                                |
| fong-spivak-invitation_F02468 | 243  | ■ 1.000 | $m : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$                                                     | $m : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$                          | 3. What is the characteristic function $m : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ for $P \Rightarrow Q$ ?                                                                               |
| fong-spivak-invitation_F02469 | 243  | ■ 0.999 | $E := \{n \in \mathbb{N} \mid n \text{ is even}\}$                                                            | $E := \{n \in \mathbb{N} \mid n$                                                   | <b>Exercise 7.14.</b> Consider the sets $E := \{n \in \mathbb{N} \mid n \text{ is even}\}$ , $P := \{n \in \mathbb{N} \mid n \text{ is prime}\}$ ,                                                     |
| fong-spivak-invitation_F02470 | 243  | ■ 0.999 | $\}, P := \{n \in \mathbb{N} \mid n$                                                                          | $\}, P := \{n \in \mathbb{N} \mid n$                                               | <b>Exercise 7.14.</b> Consider the sets $E := \{n \in \mathbb{N} \mid n \text{ is even}\}$ , $P := \{n \in \mathbb{N} \mid n \text{ is prime}\}$ ,                                                     |
| fong-spivak-invitation_F02471 | 243  | ■ 1.000 | $T := \{n \in \mathbb{N} \mid n \geq 10\}$                                                                    | $T := \{n \in \mathbb{N} \mid n \geq 10\}$                                         | and $T := \{n \in \mathbb{N} \mid n \geq 10\}$ . Each is a subset of $\mathbb{N}$ , so defines a function $\mathbb{N} \rightarrow \mathbb{B}$ .                                                        |
| fong-spivak-invitation_F02472 | 243  | ■ 1.000 | $\mathbb{N} \rightarrow \mathbb{B}$                                                                           | $\mathbb{N} \rightarrow \mathbb{B}$                                                | and $T := \{n \in \mathbb{N} \mid n \geq 10\}$ . Each is a subset of $\mathbb{N}$ , so defines a function $\mathbb{N} \rightarrow \mathbb{B}$ .                                                        |
| fong-spivak-invitation_F02473 | 243  | ■ 0.992 | $\ulcorner E \urcorner(17)$                                                                                   | $\ulcorner E \urcorner(17)$                                                        | 1. What is $\ulcorner E \urcorner(17)$ ?                                                                                                                                                               |
| fong-spivak-invitation_F02474 | 243  | ■ 0.995 | $\ulcorner P \urcorner(17)$                                                                                   | $\ulcorner P \urcorner(17)$                                                        | 2. What is $\ulcorner P \urcorner(17)$ ?                                                                                                                                                               |

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| fong-spivak-invitation_F02475 | 243  | ■ 1.000 | $\ulcorner T \urcorner(17)$                                                       | $\lceil T \rceil(17)$                                                | 3. What is $\lceil T \rceil(17)$ ?                                                                                                                  |
| fong-spivak-invitation_F02476 | 243  | ■ 1.000 | $(\ulcorner E \urcorner \wedge \ulcorner P \urcorner) \vee \ulcorner T \urcorner$ | $(\lceil E \rceil \wedge \lceil P \rceil) \vee \lceil T \rceil$      | 4. Name the smallest three elements in the set classified by $\lceil E \rceil \wedge \lceil P \rceil \vee \lceil T \rceil$ .                        |
| fong-spivak-invitation_F02477 | 243  | ■ 1.000 | $\wedge, \vee, \Rightarrow: \Omega \times \Omega \rightarrow \Omega$              | $\wedge, \vee, \Rightarrow: \Omega \times \Omega \rightarrow \Omega$ | how to understand logic in terms of $\Omega$ : there are various maps $\wedge, \vee, \Rightarrow: \Omega \times \Omega \rightarrow \Omega$          |
| fong-spivak-invitation_F02478 | 243  | ■ 1.000 | $\neg: \Omega \rightarrow \Omega$                                                 | $\neg: \Omega \rightarrow \Omega$                                    | and $\neg: \Omega \rightarrow \Omega$ , etc., which serve as logical connectives. These are operations on truth                                     |
| fong-spivak-invitation_F02479 | 244  | ■ 1.000 | $\mathbb{R}^2$                                                                    | $\mathbb{R}^2$                                                       | of the plane $\mathbb{R}^2$ as a space, we might examine only points and regions in it. But if we                                                   |
| fong-spivak-invitation_F02480 | 244  | ■ 0.163 | $\mathcal{C}^{\mathrm{op}} \rightarrow$                                           | $\mathcal{C}^{\mathrm{op}} \rightarrow$                              | $\mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$ . <sup>5</sup> Such functors are called <i>presheaves on <math>\mathcal{C}</math></i> .        |
| fong-spivak-invitation_F02481 | 244  | ■ 0.605 | $P: \mathcal{C}^{\mathrm{op}} \rightarrow$                                        | $P: \mathcal{C}^{\mathrm{op}} \rightarrow$                           | $P: \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$ . To each object $c \in \mathcal{C}$ , we refer to the set $P(c)$ as <i>the set of</i>      |
| fong-spivak-invitation_F02482 | 244  | ■ 0.605 | $P(c)$                                                                            | $P(c)$                                                               | $P: \mathcal{C}^{\mathrm{op}} \rightarrow \mathbf{Set}$ . To each object $c \in \mathcal{C}$ , we refer to the set $P(c)$ as <i>the set of</i>      |
| fong-spivak-invitation_F02483 | 244  | ■ 1.000 | $P(f): P(c) \rightarrow P(c')$                                                    | $P(f): P(c) \rightarrow P(c')$                                       | $P(f): P(c) \rightarrow P(c')$ as the <i>restriction map along <math>f</math></i> . For any section $s \in P(c)$ ,                                  |
| fong-spivak-invitation_F02484 | 244  | ■ 1.000 | $s \in P(c)$                                                                      | $s \in P(c)$                                                         | $P(f): P(c) \rightarrow P(c')$ as the <i>restriction map along <math>f</math></i> . For any section $s \in P(c)$ ,                                  |
| fong-spivak-invitation_F02485 | 244  | ■ 0.458 | $\mathcal{C}^{\mathrm{op}}$                                                       | $\mathcal{C}^{\mathrm{op}}$                                          | <sup>5</sup> The category of functors $\mathcal{C} \rightarrow \mathbf{Set}$ is also a topos: use $\mathcal{C}^{\mathrm{op}}$ as the defining site. |
| fong-spivak-invitation_F02486 | 245  | ■ 1.000 | $P(f)(s) \in P(c')$                                                               | $P(f)(s) \in P(c')$                                                  | we may denote $P(f)(s) \in P(c')$ , i.e. its restriction along $f$ , by $s _f$ . If $P$ and $Q$                                                     |
| fong-spivak-invitation_F02487 | 245  | ■ 1.000 | $s _f$                                                                            | $s _f$                                                               | we may denote $P(f)(s) \in P(c')$ , i.e. its restriction along $f$ , by $s _f$ . If $P$ and $Q$                                                     |
| fong-spivak-invitation_F02488 | 245  | ■ 1.000 | $\alpha: P \rightarrow Q$                                                         | $\alpha: P \rightarrow Q$                                            | are presheaves, a <i>morphism</i> $\alpha: P \rightarrow Q$ between them is a natural transformation of                                             |
| fong-spivak-invitation_F02489 | 245  | ■ 0.402 | $\mathbf{ArShp}^{\mathrm{op}} \rightarrow$                                        | $I: \mathbf{ArShp}^{\mathrm{op}} \rightarrow$                        | A presheaf on $\mathbf{ArShp}$ is a functor $I: \mathbf{ArShp}^{\mathrm{op}} \rightarrow \mathbf{Set}$ , which is a database instance on            |
| fong-spivak-invitation_F02490 | 246  | ■ 1.000 | $P(X) = \{U \subseteq X\}$                                                        | $P(X) = \{U \subseteq X\}$                                           | <b>Definition 7.17.</b> Let $X$ be a set, and let $P(X) = \{U \subseteq X\}$ denote its set of subsets. A                                           |
| fong-spivak-invitation_F02491 | 246  | ■ 0.999 | $\mathbf{Op} \subseteq P(X)$                                                      | $\mathbf{Op} \subseteq P(X)$                                         | <i>topology</i> on $X$ is a subset $\mathbf{Op} \subseteq P(X)$ , elements of which we call <i>open sets</i> , <sup>6</sup> satisfying              |
| fong-spivak-invitation_F02492 | 246  | ■ 1.000 | $X \in \mathbf{Op}$                                                               | $X \subseteq X$                                                      | (a) Whole set: the subset $X \subseteq X$ is open, i.e. $X \in \mathbf{Op}$ .                                                                       |
| fong-spivak-invitation_F02493 | 246  | ■ 1.000 | $X \in \mathbf{Op}$                                                               | $X \in \mathbf{Op}$                                                  | (a) Whole set: the subset $X \subseteq X$ is open, i.e. $X \in \mathbf{Op}$ .                                                                       |
| fong-spivak-invitation_F02494 | 246  | ■ 0.984 | $U, V \in \mathbf{Op}$                                                            | $U, V \in \mathbf{Op}$                                               | (b) Binary intersections: if $U, V \in \mathbf{Op}$ then $(U \cap V) \in \mathbf{Op}$ .                                                             |
| fong-spivak-invitation_F02495 | 246  | ■ 0.984 | $(U \cap V) \in \mathbf{Op}$                                                      | $(U \cap V) \in \mathbf{Op}$                                         | (b) Binary intersections: if $U, V \in \mathbf{Op}$ then $(U \cap V) \in \mathbf{Op}$ .                                                             |
| fong-spivak-invitation_F02496 | 246  | ■ 1.000 | $U_i \in \mathbf{Op}$                                                             | $U_i \in \mathbf{Op}$                                                | (c) Arbitrary unions: if $I$ is a set and if we are given an open set $U_i \in \mathbf{Op}$ for each                                                |
| fong-spivak-invitation_F02497 | 246  | ■ 1.000 | $\left(\bigcup_{i \in I} U_i\right) \in \mathbf{Op}$                              | $\left(\bigcup_{i \in I} U_i\right) \in \mathbf{Op}$                 | $i$ , then their union is also open, $\left(\bigcup_{i \in I} U_i\right) \in \mathbf{Op}$ . We interpret the particular case                        |
| fong-spivak-invitation_F02498 | 246  | ■ 0.915 | $I = \emptyset$                                                                   | $I = \emptyset$                                                      | where $I = \emptyset$ to mean that the empty set is open: $\emptyset \in \mathbf{Op}$ .                                                             |
| fong-spivak-invitation_F02499 | 246  | ■ 0.915 | $\emptyset \in \mathbf{Op}$                                                       | $\emptyset \in \mathbf{Op}$                                          | where $I = \emptyset$ to mean that the empty set is open: $\emptyset \in \mathbf{Op}$ .                                                             |
| fong-spivak-invitation_F02500 | 246  | ■ 1.000 | $U = \bigcup_{i \in I} U_i$                                                       | $U = \bigcup_{i \in I} U_i$                                          | If $U = \bigcup_{i \in I} U_i$ , we say that $(U_i)_{i \in I}$ <i>covers</i> $U$ .                                                                  |

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| fong-spivak-invitation_F02501 | 246  | ■ 1.000 | $\left(U_{\{i\}}\right)_{\{i \in I\}}$                                                                                     | $(U_i)_{i \in I}$                                                                                                          | If $U = \bigcup_{i \in I} U_i$ , we say that $(U_i)_{i \in I}$ covers $U$ .                                                                 |
| fong-spivak-invitation_F02502 | 246  | ■ 0.932 | $X, \mathbf{0\ p}$                                                                                                         | $X, \mathbf{Op}$                                                                                                           | A pair $(X, \mathbf{Op})$ , where $X$ is a set and $\mathbf{Op}$ is a topology on $X$ , is called a <i>topological</i>                      |
| fong-spivak-invitation_F02503 | 246  | ■ 0.932 | $\mathbf{0\ p}$                                                                                                            | $\mathbf{Op}$                                                                                                              | A pair $(X, \mathbf{Op})$ , where $X$ is a set and $\mathbf{Op}$ is a topology on $X$ , is called a <i>topological</i>                      |
| fong-spivak-invitation_F02504 | 246  | ■ 1.000 | $X, \mathbf{0\ p}_{\{X\}}$                                                                                                 | $X, \mathbf{Op}_X$                                                                                                         | A <i>continuous function</i> between topological spaces $(X, \mathbf{Op}_X)$ and $(Y, \mathbf{Op}_Y)$ is a func-                            |
| fong-spivak-invitation_F02505 | 246  | ■ 1.000 | $Y, \mathbf{0\ p}_{\{Y\}}$                                                                                                 | $Y, \mathbf{Op}_Y$                                                                                                         | A <i>continuous function</i> between topological spaces $(X, \mathbf{Op}_X)$ and $(Y, \mathbf{Op}_Y)$ is a func-                            |
| fong-spivak-invitation_F02506 | 246  | ■ 0.999 | $U \in \mathbf{0\ p}_{\{Y\}}$                                                                                              | $U \in \mathbf{Op}_Y$                                                                                                      | tion $f: X \rightarrow Y$ such that, for every $U \in \mathbf{Op}_Y$ , the preimage $f^{-1}(U)$ is in                                       |
| fong-spivak-invitation_F02507 | 246  | ■ 0.999 | $f^{-1}(U)$                                                                                                                | $f^{-1}(U)$                                                                                                                | tion $f: X \rightarrow Y$ such that, for every $U \in \mathbf{Op}_Y$ , the preimage $f^{-1}(U)$ is in                                       |
| fong-spivak-invitation_F02508 | 246  | ■ 1.000 | $\mathbf{0\ p}_{\{X\}}$                                                                                                    | $\mathbf{Op}_X$                                                                                                            | $\mathbf{Op}_X$ .                                                                                                                           |
| fong-spivak-invitation_F02509 | 246  | ■ 0.999 | $U \in \mathbf{0\ p}$                                                                                                      | $U \in \mathbf{Op}$                                                                                                        | <sup>6</sup> In other words, we refer to a subset $U \subseteq X$ as <i>open</i> if $U \in \mathbf{Op}$ .                                   |
| fong-spivak-invitation_F02510 | 247  | ■ 0.936 | $\epsilon \in \mathbb{R}$                                                                                                  | $\epsilon \in \mathbb{R}$                                                                                                  | <b>Example 7.18.</b> The usual topology $\mathbf{Op}$ on $\mathbb{R}^2$ is based on “ $\epsilon$ -balls.” For any $\epsilon \in \mathbb{R}$ |
| fong-spivak-invitation_F02511 | 247  | ■ 0.985 | $\epsilon > 0$                                                                                                             | $\epsilon > 0$                                                                                                             | with $\epsilon > 0$ , and any point $p = (x, y) \in \mathbb{R}^2$ , define the $\epsilon$ -ball centered at $p$                             |
| fong-spivak-invitation_F02512 | 247  | ■ 0.985 | $p=(x, y) \in \mathbb{R}^{\{2\}}$                                                                                          | $p = (x, y) \in \mathbb{R}^2$                                                                                              | with $\epsilon > 0$ , and any point $p = (x, y) \in \mathbb{R}^2$ , define the $\epsilon$ -ball centered at $p$                             |
| fong-spivak-invitation_F02513 | 247  | ■ 1.000 | $B(x, y ; \epsilon)$                                                                                                       | $B(x, y; \epsilon)$                                                                                                        | In other words, $B(x, y; \epsilon)$ is the set of all points within $\epsilon$ of $(x, y)$ .                                                |
| fong-spivak-invitation_F02514 | 247  | ■ 0.796 | $U \subseteq \mathbb{R}^{\{2\}}$                                                                                           | $U \subseteq \mathbb{R}^2$                                                                                                 | For an arbitrary subset $U \subseteq \mathbb{R}^2$ , we call it open and put it in $\mathbf{Op}$ if, for every $(x, y) \in$                 |
| fong-spivak-invitation_F02515 | 247  | ■ 0.796 | $(x, y) \in$                                                                                                               | $(x, y) \in$                                                                                                               | For an arbitrary subset $U \subseteq \mathbb{R}^2$ , we call it open and put it in $\mathbf{Op}$ if, for every $(x, y) \in$                 |
| fong-spivak-invitation_F02516 | 247  | ■ 1.000 | $B(x, y ; \epsilon) \subseteq U$                                                                                           | $B(x, y; \epsilon) \subseteq U$                                                                                            | $U$ , there exists a (small enough) $\epsilon > 0$ such that $B(x, y; \epsilon) \subset U$ .                                                |
| fong-spivak-invitation_F02517 | 247  | ■ 1.000 | $d\left(x_{\{1\}}, x_{\{2\}}\right):=\left x_{\{1\}}-x_{\{2\}}\right $                                                     | $d\left(x_1, x_2\right):=\left x_1-x_2\right $                                                                             | <b>Exercise 7.19.</b> Consider the set $\mathbb{R}$ . It is a metric space with $d\left(x_1, x_2\right):=\left x_1-x_2\right $ .            |
| fong-spivak-invitation_F02518 | 247  | ■ 1.000 | $x \in \mathbb{R}$                                                                                                         | $x \in \mathbb{R}$                                                                                                         | for each $x \in \mathbb{R}$ , define $B(x, \epsilon)$ .                                                                                     |
| fong-spivak-invitation_F02519 | 247  | ■ 1.000 | $B(x, \epsilon)$                                                                                                           | $B(x, \epsilon)$                                                                                                           | for each $x \in \mathbb{R}$ , define $B(x, \epsilon)$ .                                                                                     |
| fong-spivak-invitation_F02520 | 247  | ■ 1.000 | $U \subseteq \mathbb{R}$                                                                                                   | $U \subseteq \mathbb{R}$                                                                                                   | 2. When is an arbitrary subset $U \subseteq \mathbb{R}$ called open, in analogy with Example 7.18?                                          |
| fong-spivak-invitation_F02521 | 247  | ■ 1.000 | $U_{\{1\}}, U_{\{2\}}$                                                                                                     | $U_1, U_2$                                                                                                                 | 3. Find three open sets $U_1, U_2$ , and $U$ in $\mathbb{R}$ , such that $(U_i)_{i \in \{1,2\}}$ covers $U$ .                               |
| fong-spivak-invitation_F02522 | 247  | ■ 1.000 | $\left(U_{\{i\}}\right)_{\{i \in \{1,2\}\}}$                                                                               | $(U_i)_{i \in \{1,2\}}$                                                                                                    | 3. Find three open sets $U_1, U_2$ , and $U$ in $\mathbb{R}$ , such that $(U_i)_{i \in \{1,2\}}$ covers $U$ .                               |
| fong-spivak-invitation_F02523 | 247  | ■ 1.000 | $\mathbf{0\ p}_{\{\text {crse }\}}=(\varnothing, X)$                                                                       | $\mathbf{Op}_{\text {crse }}=(\varnothing, X)$                                                                             | possible: $\mathbf{Op}_{\text {crse }}=(\varnothing, X)$ . There is also a “finest” topology, having as many open sets                      |
| fong-spivak-invitation_F02524 | 247  | ■ 0.992 | $\mathbf{0\ p}_{\{\text {fine }\}}=\mathrm{P}(X)$                                                                          | $\mathbf{Op}_{\text {fine }}=\mathrm{P}(X)$                                                                                | as possible: $\mathbf{Op}_{\text {fine }}=\mathrm{P}(X)$ . The latter, $(X, \mathrm{P}(X))$ , is called the <i>discrete space on the</i>    |
| fong-spivak-invitation_F02525 | 247  | ■ 0.992 | $(X, \mathrm{P}(X))$                                                                                                       | $(X, \mathrm{P}(X))$                                                                                                       | as possible: $\mathbf{Op}_{\text {fine }}=\mathrm{P}(X)$ . The latter, $(X, \mathrm{P}(X))$ , is called the <i>discrete space on the</i>    |
| fong-spivak-invitation_F02526 | 247  | ■ 1.000 | $d\left((x, y),\left(x^{\prime}, y^{\prime}\right)\right):=\sqrt{\left(x-x^{\prime}\right)^2+\left(y-y^{\prime}\right)^2}$ | $d\left((x, y),\left(x^{\prime}, y^{\prime}\right)\right):=\sqrt{\left(x-x^{\prime}\right)^2+\left(y-y^{\prime}\right)^2}$ | <sup>7</sup> Here, $d((x, y), (x', y')) := \sqrt{(x - x')^2 + (y - y')^2}$ is the usual “Euclidean distance” between two points.            |

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| fong-spivak-invitation_F02527 | 248  | ■ 1.000 | $\mathbf{0\ p}_{\{\text{crse}\}}$            | $\mathbf{Op}_{\text{crse}}$         | 1. Verify that, for any set $X$ , what we called $\mathbf{Op}_{\text{crse}}$ in Example 7.20 really is a topology,                                                        |
| fong-spivak-invitation_F02528 | 248  | ■ 1.000 | $\mathbf{0\ p}_{\{\text{fine}\}}$            | $\mathbf{Op}_{\text{fine}}$         | 2. Verify also that $\mathbf{Op}_{\text{fine}}$ really is a topology.                                                                                                     |
| fong-spivak-invitation_F02529 | 248  | ■ 0.899 | $\left(Y, \mathbf{0}_{\{Y\}}\right)$         | $(Y, \mathbf{O}_Y)$                 | 3. Show that if $(X, \mathbf{P}(X))$ is discrete and $(Y, \mathbf{Op}_Y)$ is any topological space, then every                                                            |
| fong-spivak-invitation_F02530 | 248  | ■ 1.000 | $X \rightarrowtail Y$                        | $X \rightarrow Y$                   | function $X \rightarrow Y$ is continuous.                                                                                                                                 |
| fong-spivak-invitation_F02531 | 248  | ■ 1.000 | $X=\{1,2\}$                                  | $X = \{1, 2\}$                      | <b>Example 7.22.</b> There are four topologies possible on $X = \{1, 2\}$ . Two are $\mathbf{Op}_{\text{crse}}$ and                                                       |
| fong-spivak-invitation_F02532 | 248  | ■ 0.333 | $\{1,2\}, \mathbf{0} \ \mathbf{0}_{\{p\}_1}$ | $\{1, 2\}, \mathbf{Op}_1$           | The two topological spaces $(\{1, 2\}, \mathbf{Op}_1)$ and $(\{1, 2\}, \mathbf{Op}_2)$ are isomorphic; either one                                                         |
| fong-spivak-invitation_F02533 | 248  | ■ 0.333 | $\{1,2\}, \mathbf{0} \ \mathbf{0}_{\{p\}_2}$ | $\{1, 2\}, \mathbf{Op}_2$           | The two topological spaces $(\{1, 2\}, \mathbf{Op}_1)$ and $(\{1, 2\}, \mathbf{Op}_2)$ are isomorphic; either one                                                         |
| fong-spivak-invitation_F02534 | 248  | ■ 0.459 | $(X, \mathbf{0\ p})$                         | $(X, \mathbf{Op})$                  | Given a topological space $(X, \mathbf{Op})$ , the set $\mathbf{Op}$ has the structure of a preorder using                                                                |
| fong-spivak-invitation_F02535 | 248  | ■ 0.794 | $\mathbf{0\ p}, \subseteq$                   | $\mathbf{Op}, \subseteq$            | the subset relation, $(\mathbf{Op}, \subseteq)$ . It is reflexive because $U \subseteq U$ for any $U \in \mathbf{Op}$ , and it is                                         |
| fong-spivak-invitation_F02536 | 248  | ■ 0.794 | $U \subseteq U$                              | $U \subseteq U$                     | the subset relation, $(\mathbf{Op}, \subseteq)$ . It is reflexive because $U \subseteq U$ for any $U \in \mathbf{Op}$ , and it is                                         |
| fong-spivak-invitation_F02537 | 248  | ■ 1.000 | $U \subseteq V$                              | $U \subseteq V$                     | transitive because if $U \subseteq V$ and $V \subseteq W$ then $U \subseteq W$ .                                                                                          |
| fong-spivak-invitation_F02538 | 248  | ■ 1.000 | $V \subseteq W$                              | $V \subseteq W$                     | transitive because if $U \subseteq V$ and $V \subseteq W$ then $U \subseteq W$ .                                                                                          |
| fong-spivak-invitation_F02539 | 248  | ■ 1.000 | $U \subseteq W$                              | $U \subseteq W$                     | transitive because if $U \subseteq V$ and $V \subseteq W$ then $U \subseteq W$ .                                                                                          |
| fong-spivak-invitation_F02540 | 248  | ■ 0.936 | $U, \ V$                                     | $U, V$                              | gory: its objects are the open sets $U$ and for any $U, V$ the set of morphisms $\mathbf{Op}(U, V)$                                                                       |
| fong-spivak-invitation_F02541 | 248  | ■ 0.936 | $\mathbf{0\ p}(U, \ V)$                      | $\mathbf{Op}(U, V)$                 | gory: its objects are the open sets $U$ and for any $U, V$ the set of morphisms $\mathbf{Op}(U, V)$                                                                       |
| fong-spivak-invitation_F02542 | 248  | ■ 1.000 | $U \nsubseteq V$                             | $U \not\subseteq V$                 | is empty if $U \not\subseteq V$ and it has one element if $U \subseteq V$ .                                                                                               |
| fong-spivak-invitation_F02543 | 248  | ■ 0.623 | $X, \mathbf{0\ p}_{\{1\}}$                   | $X, \mathbf{Op}_1$                  | <b>Exercise 7.23.</b> Recall the Sierpinski space, say $(X, \mathbf{Op}_1)$ from Example 7.22.                                                                            |
| fong-spivak-invitation_F02544 | 248  | ■ 0.558 | $Y \subseteq X$                              | $Y \subseteq X$                     | <b>Exercise 7.24.</b> Given any topological space $(X, \mathbf{Op})$ , any subset $Y \subseteq X$ can be given                                                            |
| fong-spivak-invitation_F02545 | 248  | ■ 0.971 | $\mathbf{0\ p}_{\{?\cap Y\}}$                | $\mathbf{Op}_{? \cap Y}$            | the <i>subspace topology</i> , call it $\mathbf{Op}_{? \cap Y}$ . This topology defines any $A \subseteq Y$ to be open,                                                   |
| fong-spivak-invitation_F02546 | 248  | ■ 0.971 | $A \subseteq Y$                              | $A \subseteq Y$                     | the <i>subspace topology</i> , call it $\mathbf{Op}_{? \cap Y}$ . This topology defines any $A \subseteq Y$ to be open,                                                   |
| fong-spivak-invitation_F02547 | 248  | ■ 0.999 | $A \in \mathbf{0\ p}_{\{?\cap Y\}}$          | $A \in \mathbf{Op}_{? \cap Y}$      | $A \in \mathbf{Op}_{? \cap Y}$ , if there is an open set $B \in \mathbf{Op}$ such that $A = B \cap Y$ .                                                                   |
| fong-spivak-invitation_F02548 | 248  | ■ 0.999 | $B \in \mathbf{0\ p}$                        | $B \in \mathbf{Op}$                 | $A \in \mathbf{Op}_{? \cap Y}$ , if there is an open set $B \in \mathbf{Op}$ such that $A = B \cap Y$ .                                                                   |
| fong-spivak-invitation_F02549 | 248  | ■ 0.999 | $A=B \cap Y$                                 | $A = B \cap Y$                      | $A \in \mathbf{Op}_{? \cap Y}$ , if there is an open set $B \in \mathbf{Op}$ such that $A = B \cap Y$ .                                                                   |
| fong-spivak-invitation_F02550 | 248  | ■ 1.000 | $Y \in \mathbf{0\ p}_{\{?\cap Y\}}$          | $Y \in \mathbf{Op}_{? \cap Y}$      | 1. Find a $B \in \mathbf{Op}$ that shows that the whole set $Y$ is open, i.e. $Y \in \mathbf{Op}_{? \cap Y}$ .                                                            |
| fong-spivak-invitation_F02551 | 248  | ■ 1.000 | $Y \hookrightarrowtail X$                    | $Y \hookrightarrow X$               | 3. Show that the inclusion function $Y \hookrightarrow X$ is a continuous function.                                                                                       |
| fong-spivak-invitation_F02552 | 248  | ■ 0.958 | $(\mathbf{0\ p}, \subseteq)$                 | $(\mathbf{Op}, \subseteq)$          | $(\mathbf{Op}, \subseteq)$ of open sets. It turns out that $(\mathbf{Op}, \subseteq, X, \cap)$ is always a quantale in the sense                                          |
| fong-spivak-invitation_F02553 | 248  | ■ 0.958 | $(\mathbf{0\ p}, \subseteq, X, \cap)$        | $(\mathbf{Op}, \subseteq, X, \cap)$ | $(\mathbf{Op}, \subseteq)$ of open sets. It turns out that $(\mathbf{Op}, \subseteq, X, \cap)$ is always a quantale in the sense                                          |
| fong-spivak-invitation_F02554 | 248  | ■ 0.585 | $U_{\{i\}} \subseteq X$                      | $U_i \subseteq X$                   | <sup>8</sup> Hint 1: for any set $I$ , collection of sets $(U_i)_{i \in I}$ with $U_i \subseteq X$ , and set $V \subseteq X$ , one has $(\bigcup_{i \in I} U_i) \cap V =$ |

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| fong-spivak-invitation_F02555 | 248  | ■ 0.585 | $\left(\bigcup_{i \in I} U_i\right) \cap V =$                           | $\left(\bigcup_{i \in I} U_i\right) \cap V =$              | <sup>8</sup> Hint 1: for any set $I$ , collection of sets $(U_i)_{i \in I}$ with $U_i \subseteq X$ , and set $V \subseteq X$ , one has $\left(\bigcup_{i \in I} U_i\right) \cap V =$ |
| fong-spivak-invitation_F02556 | 248  | ■ 1.000 | $\bigcup_{i \in I} (U_i \cap V)$                                        | $\bigcup_{i \in I} (U_i \cap V)$                           | $\bigcup_{i \in I} (U_i \cap V)$ . Hint 2: for any $U, V, W \subseteq X$ , one has $(U \cap W) \cap (V \cap W) = (U \cap V) \cap \bar{W}$ .                                          |
| fong-spivak-invitation_F02557 | 248  | ■ 1.000 | $U, V, W \subseteq X$                                                   | $U, V, W \subseteq X$                                      | $\bigcup_{i \in I} (U_i \cap V)$ . Hint 2: for any $U, V, W \subseteq X$ , one has $(U \cap W) \cap (V \cap W) = (U \cap V) \cap \bar{W}$ .                                          |
| fong-spivak-invitation_F02558 | 248  | ■ 1.000 | $(U \cap W) \cap (V \cap W) = (U \cap V) \cap W$                        | $(U \cap W) \cap (V \cap W) = (U \cap V) \cap W$           | $\bigcup_{i \in I} (U_i \cap V)$ . Hint 2: for any $U, V, W \subseteq X$ , one has $(U \cap W) \cap (V \cap W) = (U \cap V) \cap \bar{W}$ .                                          |
| fong-spivak-invitation_F02559 | 249  | ■ 0.945 | $P: \mathbf{0}^{\mathrm{op}} \rightarrow$                               | $P: \mathbf{Op}^{\mathrm{op}} \rightarrow$                 | $P: \mathbf{Op}^{\mathrm{op}} \rightarrow \mathbf{Set}$ . Thus to every open set $U \in \mathbf{Op}$ we have a set $P(U)$ , called the set                                           |
| fong-spivak-invitation_F02560 | 249  | ■ 0.945 | $P(U)$                                                                  | $P(U)$                                                     | $P: \mathbf{Op}^{\mathrm{op}} \rightarrow \mathbf{Set}$ . Thus to every open set $U \in \mathbf{Op}$ we have a set $P(U)$ , called the set                                           |
| fong-spivak-invitation_F02561 | 249  | ■ 1.000 | $V \subseteq U$                                                         | $V \subseteq U$                                            | of sections over $U$ , and to every inclusion of open sets $V \subseteq U$ we have a function                                                                                        |
| fong-spivak-invitation_F02562 | 249  | ■ 1.000 | $P(U) \rightarrow P(V)$                                                 | $P(U) \rightarrow P(V)$                                    | $P(U) \rightarrow P(V)$ called the <i>restriction</i> . If $s \in P(U)$ is a section over $U$ , we may denote                                                                        |
| fong-spivak-invitation_F02563 | 249  | ■ 1.000 | $s \in P(U)$                                                            | $s \in P(U)$                                               | $P(U) \rightarrow P(V)$ called the <i>restriction</i> . If $s \in P(U)$ is a section over $U$ , we may denote                                                                        |
| fong-spivak-invitation_F02564 | 249  | ■ 1.000 | $s _V$                                                                  | $s _V$                                                     | its restriction to $V$ by $s _V$ . Recall that we say a collection of open sets $(U_i)_{i \in I}$ covers an                                                                          |
| fong-spivak-invitation_F02565 | 249  | ■ 1.000 | $(s_i)_{i \in I}$                                                       | $(s_i)_{i \in I}$                                          | Let $(U_i)_{i \in I}$ be a collection of open sets $U_i \in \mathbf{Op}$ covering $U$ . A <i>matching family</i> $(s_i)_{i \in I}$                                                   |
| fong-spivak-invitation_F02566 | 249  | ■ 1.000 | $s_i \in P(U_i)$                                                        | $s_i \in P(U_i)$                                           | of $P$ -sections over $(U_i)_{i \in I}$ consists of a section $s_i \in P(U_i)$ for each $i \in I$ , such that for                                                                    |
| fong-spivak-invitation_F02567 | 249  | ■ 1.000 | $i \in I$                                                               | $i \in I$                                                  | of $P$ -sections over $(U_i)_{i \in I}$ consists of a section $s_i \in P(U_i)$ for each $i \in I$ , such that for                                                                    |
| fong-spivak-invitation_F02568 | 249  | ■ 1.000 | $i, j \in I$                                                            | $i, j \in I$                                               | every $i, j \in I$ , we have                                                                                                                                                         |
| fong-spivak-invitation_F02569 | 249  | ■ 1.000 | $s _{U_i} = s_i$                                                        | $s _{U_i} = s_i$                                           | is a <i>gluing</i> , or <i>glued section</i> , of the matching family if $s _{U_i} = s_i$ holds for all $i \in I$ .                                                                  |
| fong-spivak-invitation_F02570 | 249  | ■ 0.997 | $\mathbf{Shv}(X, \mathbf{0}^{\mathrm{op}})$                             | $\mathbf{Shv}(X, \mathbf{Op})$                             | We denote by $\mathbf{Shv}(X, \mathbf{Op})$ , or more briefly $\mathbf{Shv}(X)$ , the category of sheaves on                                                                         |
| fong-spivak-invitation_F02571 | 249  | ■ 0.997 | $\mathbf{Shv}(X)$                                                       | $\mathbf{Shv}(X)$                                          | We denote by $\mathbf{Shv}(X, \mathbf{Op})$ , or more briefly $\mathbf{Shv}(X)$ , the category of sheaves on                                                                         |
| fong-spivak-invitation_F02572 | 250  | ■ 1.000 | $U = \emptyset \subseteq X$                                             | $U = \emptyset \subseteq X$                                | take $U = \emptyset \subseteq X$ and $I = \emptyset$ , then the empty collection of subsets (one for each $i \in I$ ,                                                                |
| fong-spivak-invitation_F02573 | 250  | ■ 0.948 | $P(\emptyset) \cong \{()\}$                                             | $P(\emptyset) \cong \{()\}$                                | (but rarely sufficient) condition is that $P(\emptyset) \cong \{()\}$ , i.e. $P(\emptyset)$ must be a set with one                                                                   |
| fong-spivak-invitation_F02574 | 250  | ■ 0.948 | $P(\emptyset)$                                                          | $P(\emptyset)$                                             | (but rarely sufficient) condition is that $P(\emptyset) \cong \{()\}$ , i.e. $P(\emptyset)$ must be a set with one                                                                   |
| fong-spivak-invitation_F02575 | 250  | ■ 0.984 | $f(a_1) = f(a_2) = a, f(b_1) = b$                                       | $f(a_1) = f(a_2) = a, f(b_1) = b$                          | the element of $Y$ immediately below it. For example, $f(a_1) = f(a_2) = a, f(b_1) = b$ ,                                                                                            |
| fong-spivak-invitation_F02576 | 250  | ■ 1.000 | $f^{-1}(y) \subseteq X$                                                 | $f^{-1}(y) \subseteq X$                                    | For each point $y \in Y$ , the preimage set $f^{-1}(y) \subseteq X$ above it is often called the <i>fiber</i>                                                                        |
| fong-spivak-invitation_F02577 | 250  | ■ 1.000 | $f': X \rightarrow Y$                                                   | $f': X \rightarrow Y$                                      | 4. Give an example of a function $f': X \rightarrow Y$ for which every fiber has either one or                                                                                       |
| fong-spivak-invitation_F02578 | 251  | ■ 0.952 | $\mathbf{Sec}_f: \mathbf{0}^{\mathrm{op}}(Y)^{\mathrm{op}} \rightarrow$ | $\mathbf{Sec}_f: \mathbf{Op}(Y)^{\mathrm{op}} \rightarrow$ | we begin by building a presheaf – i.e. a functor <del><math>\mathbf{Sec}_f: \mathbf{Op}(Y)^{\mathrm{op}} \rightarrow</math></del> $\mathbf{Set}$ – and then we'll                    |
| fong-spivak-invitation_F02579 | 251  | ■ 1.000 | $\mathbf{Sec}_f$                                                        | $\mathbf{Sec}_f$                                           | Define the presheaf $\mathbf{Sec}_f$ on an arbitrary subset $U \subseteq Y$ by:                                                                                                      |
| fong-spivak-invitation_F02580 | 251  | ■ 1.000 | $U \subseteq Y$                                                         | $U \subseteq Y$                                            | Define the presheaf $\mathbf{Sec}_f$ on an arbitrary subset $U \subseteq Y$ by:                                                                                                      |
| fong-spivak-invitation_F02581 | 251  | ■ 1.000 | $\mathbf{Sec}_f(U)$                                                     | $\mathbf{Sec}_f(U)$                                        | One might describe $\mathbf{Sec}_f(U)$ as the set of all ways to pick a “cross-section” of the $f$                                                                                   |

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| fong-spivak-invitation_F02582 | 251  | ■ 1.000 | $s \in \operatorname{Sec}_f(U)$            | $s \in \operatorname{Sec}_f(U)$            | arrangement over $U$ . That is, an element $s \in \operatorname{Sec}_f(U)$ is a choice of one element per              |
| fong-spivak-invitation_F02583 | 251  | ■ 1.000 | $U=\{a, b\}$                               | $U = \{a, b\}$                             | As an example, let's say $U = \{a, b\}$ . How many such $s$ 's are there in $\operatorname{Sec}_f(U)$ ? To             |
| fong-spivak-invitation_F02584 | 251  | ■ 0.570 | $V_1=\{a, b, c\}$                          | $V_1 = \{a, b, c\}$                        | 1. Let $V_1 = \{a, b, c\}$ . Draw all the sections over it, i.e. all elements of $\operatorname{Sec}_f(V_1)$ , as we   |
| fong-spivak-invitation_F02585 | 251  | ■ 0.570 | $\operatorname{Sec}_f(V_1)$                | $\operatorname{Sec}_f(V_1)$                | 1. Let $V_1 = \{a, b, c\}$ . Draw all the sections over it, i.e. all elements of $\operatorname{Sec}_f(V_1)$ , as we   |
| fong-spivak-invitation_F02586 | 251  | ■ 0.993 | $V_2=\{a, b, c, d\}$                       | $V_2 = \{a, b, c, d\}$                     | 2. Let $V_2 = \{a, b, c, d\}$ . Again draw all the sections, $\operatorname{Sec}_f(V_2)$ .                             |
| fong-spivak-invitation_F02587 | 251  | ■ 0.993 | $\operatorname{Sec}_f(V_2)$                | $\operatorname{Sec}_f(V_2)$                | 2. Let $V_2 = \{a, b, c, d\}$ . Again draw all the sections, $\operatorname{Sec}_f(V_2)$ .                             |
| fong-spivak-invitation_F02588 | 251  | ■ 0.990 | $V_3=\{a, b, d, e\}$                       | $V_3 = \{a, b, d, e\}$                     | 3. Let $V_3 = \{a, b, d, e\}$ . How many sections (elements of $\operatorname{Sec}_f(V_3)$ ) are there?                |
| fong-spivak-invitation_F02589 | 251  | ■ 0.990 | $\operatorname{Sec}_f(V_3)$                | $\operatorname{Sec}_f(V_3)$                | 3. Let $V_3 = \{a, b, d, e\}$ . How many sections (elements of $\operatorname{Sec}_f(V_3)$ ) are there?                |
| fong-spivak-invitation_F02590 | 251  | ■ 1.000 | $V=\{a\}$                                  | $V = \{a\}$                                | restriction maps are easy: if $V \subseteq U$ , say $V = \{a\}$ and $U = \{a, b\}$ , then given a section              |
| fong-spivak-invitation_F02591 | 251  | ■ 1.000 | $\operatorname{Sec}_f(\{a, b, c\})$        | $\operatorname{Sec}_f(\{a, b, c\})$        | 1. Write out the sets of sections $\operatorname{Sec}_f(\{a, b, c\})$ and $\operatorname{Sec}_f(\{a, c\})$ .           |
| fong-spivak-invitation_F02592 | 251  | ■ 1.000 | $\operatorname{Sec}_f(\{a, c\})$           | $\operatorname{Sec}_f(\{a, c\})$           | 1. Write out the sets of sections $\operatorname{Sec}_f(\{a, b, c\})$ and $\operatorname{Sec}_f(\{a, c\})$ .           |
| fong-spivak-invitation_F02593 | 252  | ■ 1.000 | $U_1=\{a, b\}$                             | $U_1 = \{a, b\}$                           | condition, consider the set $U_1 = \{a, b\}$ and $U_2 = \{b, e\}$ . These cover the set $U =$                          |
| fong-spivak-invitation_F02594 | 252  | ■ 1.000 | $U_2=\{b, e\}$                             | $U_2 = \{b, e\}$                           | condition, consider the set $U_1 = \{a, b\}$ and $U_2 = \{b, e\}$ . These cover the set $U =$                          |
| fong-spivak-invitation_F02595 | 252  | ■ 1.000 | $U=$                                       | $U =$                                      | condition, consider the set $U_1 = \{a, b\}$ and $U_2 = \{b, e\}$ . These cover the set $U =$                          |
| fong-spivak-invitation_F02596 | 252  | ■ 1.000 | $\{a, b, e\}=U_1 \cup U_2$                 | $\{a, b, e\} = U_1 \cup U_2$               | $\{a, b, e\} = U_1 \cup U_2$ . By Definition 7.27, a matching family for this cover consists of a                      |
| fong-spivak-invitation_F02597 | 252  | ■ 1.000 | $U_1$                                      | $U_1$                                      | section over $U_1$ and a section over $U_2$ that agree on the overlap set, $U_1 \cap U_2 = \{b\}$ .                    |
| fong-spivak-invitation_F02598 | 252  | ■ 1.000 | $U_2$                                      | $U_2$                                      | section over $U_1$ and a section over $U_2$ that agree on the overlap set, $U_1 \cap U_2 = \{b\}$ .                    |
| fong-spivak-invitation_F02599 | 252  | ■ 1.000 | $U_1 \cap U_2 = \{b\}$                     | $U_1 \cap U_2 = \{b\}$                     | section over $U_1$ and a section over $U_2$ that agree on the overlap set, $U_1 \cap U_2 = \{b\}$ .                    |
| fong-spivak-invitation_F02600 | 252  | ■ 1.000 | $s_1 \in \operatorname{Sec}_f(U_1)$        | $s_1 \in \operatorname{Sec}_f(U_1)$        | So consider $s_1 \in \operatorname{Sec}_f(U_1)$ and $s_2 \in \operatorname{Sec}_f(U_2)$ shown below.                   |
| fong-spivak-invitation_F02601 | 252  | ■ 1.000 | $s_2 \in \operatorname{Sec}_f(U_2)$        | $s_2 \in \operatorname{Sec}_f(U_2)$        | So consider $s_1 \in \operatorname{Sec}_f(U_1)$ and $s_2 \in \operatorname{Sec}_f(U_2)$ shown below.                   |
| fong-spivak-invitation_F02602 | 252  | ■ 1.000 | $g_1$                                      | $g_1$                                      | Since sections $g_1$ and $g_2$ agree on the overlap – they both send $b$ to $b_2$ – the two sections                   |
| fong-spivak-invitation_F02603 | 252  | ■ 1.000 | $b_2$                                      | $b_2$                                      | Since sections $g_1$ and $g_2$ agree on the overlap – they both send $b$ to $b_2$ – the two sections                   |
| fong-spivak-invitation_F02604 | 252  | ■ 1.000 | $U=\{a, b, e\}$                            | $U = \{a, b, e\}$                          | shown in Eq. (7.12) can be glued to form a single section over $U = \{a, b, e\}$ :                                     |
| fong-spivak-invitation_F02605 | 252  | ■ 0.524 | $\left(s_1, s_2\right)$                    | $(s_1, s_2)$                               | 2. For your answer $(s_1, s_2)$ in part 1, can you find a section $s \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that |
| fong-spivak-invitation_F02606 | 252  | ■ 0.524 | $s \in \operatorname{Sec}_f(U_1 \cup U_2)$ | $s \in \operatorname{Sec}_f(U_1 \cup U_2)$ | 2. For your answer $(s_1, s_2)$ in part 1, can you find a section $s \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that |
| fong-spivak-invitation_F02607 | 252  | ■ 1.000 | $s _{U_1}=s_1$                             | $s _{U_1} = s_1$                           | $s _{U_1} = s_1$ and $s _{U_2} = s_2$ ?                                                                                |
| fong-spivak-invitation_F02608 | 252  | ■ 1.000 | $s _{U_2}=s_2$                             | $s _{U_2} = s_2$                           | $s _{U_1} = s_1$ and $s _{U_2} = s_2$ ?                                                                                |

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| fong-spivak-invitation_F02609 | 252  | ■ 0.982 | $h_{\{1\}} \in \operatorname{Sec}_{\{f\}}(\left(U_{\{1\}}\right))$                   | $h_1 \in \operatorname{Sec}_f(U_1)$                             | 3. Find a section $h_1 \in \operatorname{Sec}_f(U_1)$ and a section $h_2 \in \operatorname{Sec}_f(U_2)$ that <i>do</i> agree on the |
| fong-spivak-invitation_F02610 | 252  | ■ 0.982 | $h_{\{2\}} \in \operatorname{Sec}_{\{f\}}(\left(U_{\{2\}}\right))$                   | $h_2 \in \operatorname{Sec}_f(U_2)$                             | 3. Find a section $h_1 \in \operatorname{Sec}_f(U_1)$ and a section $h_2 \in \operatorname{Sec}_f(U_2)$ that <i>do</i> agree on the |
| fong-spivak-invitation_F02611 | 252  | ■ 0.982 | d o                                                                                  | <i>do</i>                                                       | 3. Find a section $h_1 \in \operatorname{Sec}_f(U_1)$ and a section $h_2 \in \operatorname{Sec}_f(U_2)$ that <i>do</i> agree on the |
| fong-spivak-invitation_F02612 | 252  | ■ 1.000 | $h \in \operatorname{Sec}_{\{f\}}(\left(U_{\{1\}} \cup U_{\{2\}}\right))$            | $h \in \operatorname{Sec}_f(U_1 \cup U_2)$                      | 4. Can you find a section $h \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that $h _{U_1} = h_1$ and $h _{U_2} = h_2$ ? $\diamond$   |
| fong-spivak-invitation_F02613 | 252  | ■ 1.000 | $\left.h\right _{U_{\{1\}}}=h_{\{1\}}$                                               | $h _{U_1} = h_1$                                                | 4. Can you find a section $h \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that $h _{U_1} = h_1$ and $h _{U_2} = h_2$ ? $\diamond$   |
| fong-spivak-invitation_F02614 | 252  | ■ 1.000 | $\left.h\right _{U_{\{2\}}}=h_{\{2\}}$                                               | $h _{U_2} = h_2$                                                | 4. Can you find a section $h \in \operatorname{Sec}_f(U_1 \cup U_2)$ such that $h _{U_1} = h_1$ and $h _{U_2} = h_2$ ? $\diamond$   |
| fong-spivak-invitation_F02615 | 253  | ■ 1.000 | $f:\left(X, \mathbf{p}_{\{X\}}\right) \rightarrow\left(Y, \mathbf{p}_{\{Y\}}\right)$ | $f:(X, \mathbf{Op}_X) \rightarrow (Y, \mathbf{Op}_Y)$           | <b>Example 7.33.</b> Let $f:(X, \mathbf{Op}_X) \rightarrow (Y, \mathbf{Op}_Y)$ be a continuous function. Consider the               |
| fong-spivak-invitation_F02616 | 253  | ■ 0.965 | $\operatorname{Sec}_{\{f\}}: \mathbf{p}_{\{Y\}}^{\mathrm{op}} \rightarrow$           | $\operatorname{Sec}_f: \mathbf{Op}_Y^{\mathrm{op}} \rightarrow$ | functor $\operatorname{Sec}_f: \mathbf{Op}_Y^{\mathrm{op}} \rightarrow \mathbf{Set}$ given by                                       |
| fong-spivak-invitation_F02617 | 253  | ■ 1.000 | $\mathbf{p}_{\{Y\}}$                                                                 | $\mathbf{Op}_Y$                                                 | The morphisms of $\mathbf{Op}_Y$ are inclusions $V \subseteq U$ . Given $g: U \rightarrow X$ and $V \subseteq U$ , what             |
| fong-spivak-invitation_F02618 | 253  | ■ 1.000 | $g: U \rightarrow X$                                                                 | $g: U \rightarrow X$                                            | The morphisms of $\mathbf{Op}_Y$ are inclusions $V \subseteq U$ . Given $g: U \rightarrow X$ and $V \subseteq U$ , what             |
| fong-spivak-invitation_F02619 | 253  | ■ 1.000 | $T M$                                                                                | $TM$                                                            | $TM$ whose points are pairs $(m, v)$ , where $m \in M$ is a point in the manifold and $v$ is a                                      |
| fong-spivak-invitation_F02620 | 253  | ■ 1.000 | $(m, v)$                                                                             | $(m, v)$                                                        | $TM$ whose points are pairs $(m, v)$ , where $m \in M$ is a point in the manifold and $v$ is a                                      |
| fong-spivak-invitation_F02621 | 253  | ■ 1.000 | $m \in M$                                                                            | $m \in M$                                                       | $TM$ whose points are pairs $(m, v)$ , where $m \in M$ is a point in the manifold and $v$ is a                                      |
| fong-spivak-invitation_F02622 | 253  | ■ 1.000 | $\pi: T M \rightarrow$                                                               | $\pi: TM \rightarrow$                                           | The tangent bundle $TM$ on a manifold $M$ comes with a continuous map $\pi: TM \rightarrow$                                         |
| fong-spivak-invitation_F02623 | 253  | ■ 1.000 | $(m, v) \mapsto m$                                                                   | $(m, v) \mapsto m$                                              | $M$ back down to the manifold, sending $(m, v) \mapsto m$ . One might say that $\pi$                                                |
| fong-spivak-invitation_F02624 | 253  | ■ 1.000 | $\pi$                                                                                | $\pi$                                                           | $M$ back down to the manifold, sending $(m, v) \mapsto m$ . One might say that $\pi$                                                |
| fong-spivak-invitation_F02625 | 253  | ■ 0.846 | $\operatorname{Sec}_{\{\pi\}}$                                                       | $\operatorname{Sec}_{\pi}$                                      | Example 7.33, $\pi$ defines a sheaf $\operatorname{Sec}_{\pi}$ . It could be called the sheaf of “tangent                           |
| fong-spivak-invitation_F02626 | 254  | ■ 1.000 | $U \subseteq M$                                                                      | $U \subseteq M$                                                 | Earth, above. Given an open subset $U \subseteq M$ , an element $v \in \operatorname{Sec}_{\pi}(U)$ is called                       |
| fong-spivak-invitation_F02627 | 254  | ■ 1.000 | $v \in \operatorname{Sec}_{\{\pi\}}(U)$                                              | $v \in \operatorname{Sec}_{\pi}(U)$                             | Earth, above. Given an open subset $U \subseteq M$ , an element $v \in \operatorname{Sec}_{\pi}(U)$ is called                       |
| fong-spivak-invitation_F02628 | 254  | ■ 1.000 | $v(u)$                                                                               | $v(u)$                                                          | a vector field over $U$ because it continuously assigns a tangent vector $v(u)$ to each                                             |
| fong-spivak-invitation_F02629 | 254  | ■ 1.000 | $U \neq M$                                                                           | $U \neq M$                                                      | possible to prove that for every open set $U$ , as long as $U \neq M$ , there is a vector field                                     |
| fong-spivak-invitation_F02630 | 254  | ■ 1.000 | $U \neq X$                                                                           | $U \neq X$                                                      | that hair can be well combed whenever $U \neq X$ but cannot be well combed when $U = X$                                             |
| fong-spivak-invitation_F02631 | 254  | ■ 1.000 | $U=X$                                                                                | $U = X$                                                         | that hair can be well combed whenever $U \neq X$ but cannot be well combed when $U = X$                                             |
| fong-spivak-invitation_F02632 | 254  | ■ 0.997 | $\mathbf{Shv}\left(\mathbf{S h v}\right)(M)$                                         | $\mathbf{Shv}(M)$                                               | that we can consider the category $\mathbf{Shv}(M)$ of sheaves on $M$ ; in fact, such categories are                                |
| fong-spivak-invitation_F02633 | 254  | ■ 0.884 | $\{1,2\}, \mathbf{p}_{\{1\}}$                                                        | $\{1,2\}, \mathbf{Op}_1$                                        | <b>Exercise 7.37.</b> Consider the Sierpinski space $(\{1,2\}, \mathbf{Op}_1)$ from Example 7.22.                                   |
| fong-spivak-invitation_F02634 | 255  | ■ 1.000 | $X=\mathbf{1}$                                                                       | $X = \mathbf{1}$                                                | one-point space $X = \mathbf{1}$ with its unique topology, we find that the category $\mathbf{Set}$ is a topos,                     |
| fong-spivak-invitation_F02635 | 255  | ■ 1.000 | $\wedge, \vee, \neg, \rightarrow, \exists, \forall$                                  | $\wedge, \vee, \neg, \Rightarrow, \exists, \forall$             | a topological space. There, the same logical symbols $\wedge, \vee, \neg, \Rightarrow, \exists, \forall$ become opera-              |

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| fong-spivak-invitation_F02636 | 255  | ■ 0.934 | $\Omega=\mathbb{B}$                              | $\Omega = \mathbb{B}$                            | classifier is the set of booleans $\Omega = \mathbb{B}$ . In a sheaf topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$ , the object            |
| fong-spivak-invitation_F02637 | 255  | ■ 0.934 | $\mathcal{E}=\operatorname{Shv}(X, \mathbf{Op})$ | $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$     | classifier is the set of booleans $\Omega = \mathbb{B}$ . In a sheaf topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$ , the object            |
| fong-spivak-invitation_F02638 | 255  | ■ 0.809 | $\Omega: \mathbf{Op}^{\mathrm{op}} \rightarrow$  | $\Omega : \mathbf{Op}^{\mathrm{op}} \rightarrow$ | presheaf that satisfies the sheaf condition. As a presheaf it is just a functor $\Omega: \mathbf{Op}^{\mathrm{op}} \rightarrow$                 |
| fong-spivak-invitation_F02639 | 255  | ■ 1.000 | $\Omega(U)$                                      | $\Omega(U)$                                      | <b>Set</b> ; it assigns a set $\Omega(U)$ to each open $U \subseteq X$ and comes with a restriction map                                         |
| fong-spivak-invitation_F02640 | 255  | ■ 1.000 | $\Omega(U) \rightarrow \Omega(V)$                | $\Omega(U) \rightarrow \Omega(V)$                | $\Omega(U) \rightarrow \Omega(V)$ whenever $V \subseteq U$ . So in our quest to understand $\Omega$ , we first ask the                          |
| fong-spivak-invitation_F02641 | 256  | ■ 1.000 | $X=\{1\}$                                        | $X = \{1\}$                                      | <b>Exercise 7.38.</b> Let $X = \{1\}$ be the one-point space. We said above that the subobject                                                  |
| fong-spivak-invitation_F02642 | 256  | ■ 0.966 | $\mathbf{Shv}(1)$                                | $\mathbf{Shv}(1)$                                | classifier of $\mathbf{Shv}(1)$ is the set $\mathbb{B}$ of booleans, but how does that align with the definition                                |
| fong-spivak-invitation_F02643 | 256  | ■ 1.000 | $W \subseteq V \subseteq U$                      | $W \subseteq V \subseteq U$                      | i.e. that whenever $W \subseteq V \subseteq U$ , the restriction map $\Omega(U) \rightarrow \Omega(V)$ fol-                                     |
| fong-spivak-invitation_F02644 | 256  | ■ 1.000 | $\Omega(V) \rightarrow \Omega(W)$                | $\Omega(V) \rightarrow \Omega(W)$                | lowed by the restriction map $\Omega(V) \rightarrow \Omega(W)$ is the same as the restriction map                                               |
| fong-spivak-invitation_F02645 | 256  | ■ 1.000 | $\Omega(U) \rightarrow \Omega(W)$                | $\Omega(U) \rightarrow \Omega(W)$                | $\Omega(U) \rightarrow \Omega(W)$ .                                                                                                             |
| fong-spivak-invitation_F02646 | 256  | ■ 1.000 | $V_i \in \Omega(U_i)$                            | $V_i \in \Omega(U_i)$                            | an element $V_i \in \Omega(U_i)$ , i.e. an open set $V_i \subseteq U_i$ , for each $i \in I$ . Suppose further                                  |
| fong-spivak-invitation_F02647 | 256  | ■ 1.000 | $V_i \subseteq U_i$                              | $V_i \subseteq U_i$                              | an element $V_i \in \Omega(U_i)$ , i.e. an open set $V_i \subseteq U_i$ , for each $i \in I$ . Suppose further                                  |
| fong-spivak-invitation_F02648 | 256  | ■ 1.000 | $V_i \cap U_j = V_j \cap U_i$                    | $V_i \cap U_j = V_j \cap U_i$                    | that, for all $i, j \in I$ , it is the case that $V_i \cap U_j = V_j \cap U_i$ , i.e. that the elements form                                    |
| fong-spivak-invitation_F02649 | 256  | ■ 1.000 | $V := \bigcup_{i \in I} V_i$                     | $V := \bigcup_{i \in I} V_i$                     | a matching family. Define $V := \bigcup_{i \in I} V_i$ ; it is an open subset of $U$ , so we can con-                                           |
| fong-spivak-invitation_F02650 | 256  | ■ 1.000 | $(V_i)_{i \in I}$                                | $(V_i)_{i \in I}$                                | the $(V_i)_{i \in I}$ :                                                                                                                         |
| fong-spivak-invitation_F02651 | 256  | ■ 1.000 | $V \cap U_j = V_j$                               | $V \cap U_j = V_j$                               | In other words $V \cap U_j = V_j$ for any $j \in I$ . So our $\Omega$ has been upgraded from presheaf                                           |
| fong-spivak-invitation_F02652 | 256  | ■ 1.000 | $j \in I$                                        | $j \in I$                                        | In other words $V \cap U_j = V_j$ for any $j \in I$ . So our $\Omega$ has been upgraded from presheaf                                           |
| fong-spivak-invitation_F02653 | 256  | ■ 0.908 | $\{1\} \rightarrow \Omega$                       | $\{1\} \rightarrow \Omega$                       | standing, we also need to give a sheaf morphism $\mathrm{true}: \{1\} \rightarrow \Omega$ . Here $\{1\}: \mathbf{Op}^{\mathrm{op}} \rightarrow$ |
| fong-spivak-invitation_F02654 | 256  | ■ 0.908 | $\{1\}: \mathbf{Op}^{\mathrm{op}} \rightarrow$   | $\{1\}: \mathbf{Op}^{\mathrm{op}} \rightarrow$   | standing, we also need to give a sheaf morphism $\mathrm{true}: \{1\} \rightarrow \Omega$ . Here $\{1\}: \mathbf{Op}^{\mathrm{op}} \rightarrow$ |
| fong-spivak-invitation_F02655 | 256  | ■ 1.000 | $\{1\} = \{1\}(U) \rightarrow \Omega(U)$         | $\{1\} = \{1\}(U) \rightarrow \Omega(U)$         | that assigns, for every $U \in \mathbf{Op}$ , the function $\{1\} = \{1\}(U) \rightarrow \Omega(U)$ sending $1 \mapsto U$ ,                     |
| fong-spivak-invitation_F02656 | 256  | ■ 1.000 | $1 \mapsto U$                                    | $1 \mapsto U$                                    | that assigns, for every $U \in \mathbf{Op}$ , the function $\{1\} = \{1\}(U) \rightarrow \Omega(U)$ sending $1 \mapsto U$ ,                     |
| fong-spivak-invitation_F02657 | 256  | ■ 1.000 | $U=\varnothing$                                  | $U = \varnothing$                                | $\mathrm{true}$ ; if $U$ is nothing, $U = \varnothing$ , then $P$ is really false. But, in general, it's just true in some                      |
| fong-spivak-invitation_F02658 | 257  | ■ 0.944 | $(V, V; A)$                                      | $(V, V; A)$                                      | $\mathrm{true}: 1 \rightarrow \Omega$ sends that loop to $(V, V; A)$ .                                                                          |
| fong-spivak-invitation_F02659 | 257  | ■ 1.000 | $i: H \subseteq G$                               | $i: H \subseteq G$                               | Given any graph $G$ and subgraph $i: H \subseteq G$ , we need to construct a graph homo-                                                        |
| fong-spivak-invitation_F02660 | 257  | ■ 0.546 | $\lceil H \rceil: G \rightarrow \Omega$          | $\lceil H \rceil: G \rightarrow \Omega$          | morphism $\lceil H \rceil: G \rightarrow \Omega$ classifying $H$ . The idea is that, for each part of $G$ , we decide                           |
| fong-spivak-invitation_F02661 | 257  | ■ 0.577 | $(V, 0; 0), (0, V; 0), (V, V; 0)$                | $(V, 0; 0), (0, V; 0), (V, V; 0)$                | we send $a$ to $(V, 0; 0), (0, V; 0), (V, V; 0)$ , or $(0, 0; 0)$ , respectively.                                                               |
| fong-spivak-invitation_F02662 | 257  | ■ 0.577 | $0; 0$                                           | $0; 0$                                           | we send $a$ to $(V, 0; 0), (0, V; 0), (V, V; 0)$ , or $(0, 0; 0)$ , respectively.                                                               |

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| fong-spivak-invitation_F02663 | 257  | ■ 1.000 | $H \subseteq G$                           | $H \subseteq G$                           | <b>Exercise 7.41.</b> Consider the subgraph $H \subset G$ shown here:                                                      |
| fong-spivak-invitation_F02664 | 257  | ■ 1.000 | $U, V \in \Omega(X)$                      | $U, V \in \Omega(X)$                      | ues $U, V \in \Omega(X)$ , then their conjunction “ $U$ AND $V$ ” is their intersection, and their                         |
| fong-spivak-invitation_F02665 | 257  | ■ 1.000 | $\cap$                                    | $\cap$                                    | These formulas are easy to remember, because $\wedge$ looks like $\cap$ and $\vee$ looks like $\cup$ . The                 |
| fong-spivak-invitation_F02666 | 257  | ■ 1.000 | $\cup$                                    | $\cup$                                    | These formulas are easy to remember, because $\wedge$ looks like $\cap$ and $\vee$ looks like $\cup$ . The                 |
| fong-spivak-invitation_F02667 | 257  | ■ 1.000 | $U \Rightarrow V$                         | $U \Rightarrow V$                         | implication $U \Rightarrow V$ is the largest open set $R$ such that $R \cap U \subseteq V$ , i.e.                          |
| fong-spivak-invitation_F02668 | 257  | ■ 1.000 | $R \cap U \subseteq V$                    | $R \cap U \subseteq V$                    | implication $U \Rightarrow V$ is the largest open set $R$ such that $R \cap U \subseteq V$ , i.e.                          |
| fong-spivak-invitation_F02669 | 258  | ■ 0.933 | $\neg U :=$                               | $\neg U :=$                               | Finally, the negation of $U$ is given by $\neg U := (U \Rightarrow \text{false})$ , and this turns out to                  |
| fong-spivak-invitation_F02670 | 258  | ■ 0.933 | $U \Rightarrow$                           | $U \Rightarrow$                           | Finally, the negation of $U$ is given by $\neg U := (U \Rightarrow \text{false})$ , and this turns out to                  |
| fong-spivak-invitation_F02671 | 258  | ■ 1.000 | $R \cap U = \varnothing$                  | $R \cap U = \varnothing$                  | be relatively simple. By Eq. (7.16), it is the union of all $R$ such that $R \cap U = \varnothing$ , i.e.                  |
| fong-spivak-invitation_F02672 | 258  | ■ 1.000 | $\neg U$                                  | $\neg U$                                  | recognize that $\neg U$ is the “interior of the complement of $U$ .”                                                       |
| fong-spivak-invitation_F02673 | 258  | ■ 1.000 | $X = \mathbb{R}$                          | $X = \mathbb{R}$                          | <b>Example 7.42.</b> Consider the real line $X = \mathbb{R}$ as a topological space (see Exercise 7.19).                   |
| fong-spivak-invitation_F02674 | 258  | ■ 1.000 | $U = \{x \in \mathbb{R} \mid x < 3\}$     | $U = \{x \in \mathbb{R} \mid x < 3\}$     | Let $U, V \in \Omega(X)$ be the open sets $U = \{x \in \mathbb{R} \mid x < 3\}$ and $V = \{x \in \mathbb{R} \mid -4 < x <$ |
| fong-spivak-invitation_F02675 | 258  | ■ 1.000 | $V = \{x \in \mathbb{R} \mid -4 < x <$    | $V = \{x \in \mathbb{R} \mid -4 < x <$    | Let $U, V \in \Omega(X)$ be the open sets $U = \{x \in \mathbb{R} \mid x < 3\}$ and $V = \{x \in \mathbb{R} \mid -4 < x <$ |
| fong-spivak-invitation_F02676 | 258  | ■ 1.000 | $U = (-\infty, 3)$                        | $U = (-\infty, 3)$                        | 4). Using interval notation, $U = (-\infty, 3)$ and $V = (-4, 4)$ . Then                                                   |
| fong-spivak-invitation_F02677 | 258  | ■ 1.000 | $V = (-4, 4)$                             | $V = (-4, 4)$                             | 4). Using interval notation, $U = (-\infty, 3)$ and $V = (-4, 4)$ . Then                                                   |
| fong-spivak-invitation_F02678 | 258  | ■ 1.000 | $U \wedge V = (-4, 3)$                    | $U \wedge V = (-4, 3)$                    | • $U \wedge V = (-4, 3)$ ,                                                                                                 |
| fong-spivak-invitation_F02679 | 258  | ■ 1.000 | $U \vee V = (-\infty, 4)$                 | $U \vee V = (-\infty, 4)$                 | • $U \vee V = (-\infty, 4)$ ,                                                                                              |
| fong-spivak-invitation_F02680 | 258  | ■ 1.000 | $\neg U = (3, \infty)$                    | $\neg U = (3, \infty)$                    | • $\neg U = (3, \infty)$ ,                                                                                                 |
| fong-spivak-invitation_F02681 | 258  | ■ 1.000 | $\neg V = (-\infty, -4) \cup (4, \infty)$ | $\neg V = (-\infty, -4) \cup (4, \infty)$ | • $\neg V = (-\infty, -4) \cup (4, \infty)$ ,                                                                              |
| fong-spivak-invitation_F02682 | 258  | ■ 1.000 | $(U \Rightarrow V) = (-4, \infty)$        | $(U \Rightarrow V) = (-4, \infty)$        | • $U \Rightarrow V = (-4, \infty)$ ,                                                                                       |
| fong-spivak-invitation_F02683 | 258  | ■ 1.000 | $(V \Rightarrow U) = U$                   | $(V \Rightarrow U) = U$                   | • $V \Rightarrow U = U$ .                                                                                                  |
| fong-spivak-invitation_F02684 | 258  | ■ 1.000 | $U = \mathbb{R} - \{0\}$                  | $U = \mathbb{R} - \{0\}$                  | subset $U = \mathbb{R} - \{0\}$ .                                                                                          |
| fong-spivak-invitation_F02685 | 258  | ■ 1.000 | $\neg \neg U$                             | $\neg \neg U$                             | 2. What open subset is $\neg \neg U$ ?                                                                                     |
| fong-spivak-invitation_F02686 | 258  | ■ 1.000 | $U \subseteq \neg \neg U$                 | $U \subseteq \neg \neg U$                 | 3. Is it true that $U \subset \neg \neg U$ ?                                                                               |
| fong-spivak-invitation_F02687 | 258  | ■ 1.000 | $\neg \neg U \subseteq U$                 | $\neg \neg U \subseteq U$                 | 4. Is it true that $\neg \neg U \subseteq U$ ?                                                                             |
| fong-spivak-invitation_F02688 | 258  | ■ 0.983 | $\wedge U = U$                            | $\wedge U = U$                            | $U \in \mathbf{Op}$ , we have $(\text{true} \wedge U) = U$ . Which open set is it?                                         |
| fong-spivak-invitation_F02689 | 258  | ■ 0.748 | $\vee U = \text{true}$                    | $\vee U = \text{true}$                    | 2. Other things we should expect from $\text{true}$ include $(\text{true} \vee U) = \text{true}$ and $(U \Rightarrow$      |
| fong-spivak-invitation_F02690 | 258  | ■ 0.748 | $(U \Rightarrow$                          | $(U \Rightarrow$                          | 2. Other things we should expect from $\text{true}$ include $(\text{true} \vee U) = \text{true}$ and $(U \Rightarrow$      |

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| fong-spivak-invitation_F02691 | 258  | ■ 0.840 | )=                                           | ) =                                          | $\text{true}) = \text{true}$ and $(\text{true} \Rightarrow U) = U$ . Do these hold for your answer to 1?                                    |
| fong-spivak-invitation_F02692 | 258  | ■ 0.840 | (                                            | (                                            | $\text{true}) = \text{true}$ and $(\text{true} \Rightarrow U) = U$ . Do these hold for your answer to 1?                                    |
| fong-spivak-invitation_F02693 | 258  | ■ 0.840 | $\Rightarrow U)=U$                           | $\Rightarrow U) = U$                         | $\text{true}) = \text{true}$ and $(\text{true} \Rightarrow U) = U$ . Do these hold for your answer to 1?                                    |
| fong-spivak-invitation_F02694 | 258  | ■ 0.999 | $\vee U)=U$                                  | $\vee U) = U$                                | $U \in \mathbf{Op}$ , we have $(\text{false} \vee U) = U$ . Which open set is it?                                                           |
| fong-spivak-invitation_F02695 | 258  | ■ 0.900 | $\wedge U$                                   | $\wedge U$                                   | 4. Other things we should expect from $\text{false}$ include $(\text{false} \wedge U) =$                                                    |
| fong-spivak-invitation_F02696 | 259  | ■ 1.000 | $\pi: E \rightarrow X$                       | $\pi: E \rightarrow X$                       | <b>Example 7.45.</b> For a vector bundle $\pi: E \rightarrow X$ over a space $X$ , the corresponding sheaf                                  |
| fong-spivak-invitation_F02697 | 259  | ■ 0.988 | $i_U: U \subseteq X$                         | $i_U: U \subseteq X$                         | is $\text{Sec}_\pi$ corresponding to its sections: to each open set $i_U: U \subseteq X$ , we associate the set                             |
| fong-spivak-invitation_F02698 | 259  | ■ 0.691 | $s: U \rightarrow E$                         | $s: U \rightarrow E$                         | of functions $s: U \rightarrow E$ for which $s \circ \pi = i_U$ . For example, in the case of the tangent                                   |
| fong-spivak-invitation_F02699 | 259  | ■ 0.691 | $s; \pi = i_U$                               | $s; \pi = i_U$                               | of functions $s: U \rightarrow E$ for which $s \circ \pi = i_U$ . For example, in the case of the tangent                                   |
| fong-spivak-invitation_F02700 | 259  | ■ 1.000 | $\pi: TM \rightarrow M$                      | $\pi: TM \rightarrow M$                      | bundle $\pi: TM \rightarrow M$ (see Example 7.34), the corresponding sheaf, call it $\mathbf{VF}$ , associates                              |
| fong-spivak-invitation_F02701 | 259  | ■ 0.998 | $\mathbf{VF}(U)$                             | $\mathbf{VF}(U)$                             | to each $U$ the set $\mathbf{VF}(U)$ of vector fields on $U$ .                                                                              |
| fong-spivak-invitation_F02702 | 259  | ■ 0.813 | $\rightarrow \Omega$                         | $\rightarrow \Omega$                         | For example, one could have a predicate $\text{Grad}: \mathbf{VF} \rightarrow \Omega$ that asks for the largest sub-                        |
| fong-spivak-invitation_F02703 | 259  | ■ 0.998 | $\text{Grad}(v)$                             | $\text{Grad}(v)$                             | space $\text{Grad}(v)$ on which a given vector field $v$ comes from the gradient of some scalar                                             |
| fong-spivak-invitation_F02704 | 259  | ■ 0.887 | $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$ | $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$ | In a topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$ , a predicate is a sheaf morphism $p: S \rightarrow \Omega$ where                   |
| fong-spivak-invitation_F02705 | 259  | ■ 0.887 | $p: S \rightarrow \Omega$                    | $p: S \rightarrow \Omega$                    | In a topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$ , a predicate is a sheaf morphism $p: S \rightarrow \Omega$ where                   |
| fong-spivak-invitation_F02706 | 259  | ■ 1.000 | $S \in \mathcal{E}$                          | $S \in \mathcal{E}$                          | $S \in \mathcal{E}$ is a sheaf and $\Omega \in \mathcal{E}$ is the subobject classifier, the sheaf of truth values. By                      |
| fong-spivak-invitation_F02707 | 259  | ■ 1.000 | $p(U): S(U) \rightarrow \Omega(U)$           | $p(U): S(U) \rightarrow \Omega(U)$           | Definition 7.27 we get a function $p(U): S(U) \rightarrow \Omega(U)$ for any open set $U \subseteq X$ . In                                  |
| fong-spivak-invitation_F02708 | 259  | ■ 0.972 | $\text{Bob} \in S(U)$                        | $\text{Bob} \in S(U)$                        | sheaf of people (people come and go over time), $\text{Bob} \in S(U)$ is a person existing over a                                           |
| fong-spivak-invitation_F02709 | 259  | ■ 0.998 | $s \in S(U)$                                 | $s \in S(U)$                                 | the weather,” acts on sections $s \in S(U)$ , say $s = \text{Bob}$ . But, by Definition 7.9, any                                            |
| fong-spivak-invitation_F02710 | 259  | ■ 0.998 | $s =$                                        | $s =$                                        | the weather,” acts on sections $s \in S(U)$ , say $s = \text{Bob}$ . But, by Definition 7.9, any                                            |
| fong-spivak-invitation_F02711 | 260  | ■ 1.000 | $\{S \mid p\} \subseteq S$                   | $\{S \mid p\} \subseteq S$                   | predicate $p: S \rightarrow \Omega$ also defines a subobject of $\{S \mid p\} \subseteq S$ . Describe the sections of                       |
| fong-spivak-invitation_F02712 | 260  | ■ 0.763 | $ \Omega^E  =$                               | $ \Omega^E  =$                               | For a topos $\mathcal{E} = \mathbf{Shv}(X, \mathbf{Op})$ and object (sheaf) $S \in \mathcal{E}$ , the set of $S$ -predicates $ \Omega^E  =$ |
| fong-spivak-invitation_F02713 | 260  | ■ 1.000 | $\mathcal{E}(S, \Omega)$                     | $\mathcal{E}(S, \Omega)$                     | $\mathcal{E}(S, \Omega)$ is naturally given the structure of a poset, which we denote                                                       |
| fong-spivak-invitation_F02714 | 260  | ■ 1.000 | $p, q: S \rightarrow \Omega$                 | $p, q: S \rightarrow \Omega$                 | Given two predicates $p, q: S \rightarrow \Omega$ , we say that $p \leq^S q$ if the first implies the second.                               |
| fong-spivak-invitation_F02715 | 260  | ■ 1.000 | $p \leq^S q$                                 | $p \leq^S q$                                 | Given two predicates $p, q: S \rightarrow \Omega$ , we say that $p \leq^S q$ if the first implies the second.                               |
| fong-spivak-invitation_F02716 | 260  | ■ 1.000 | $p(s) \subseteq U$                           | $p(s) \subseteq U$                           | $p(s) \subseteq U$ and $q(s) \subseteq U$ . We say that $p \leq^S q$ if $p(s) \subseteq q(s)$ for all $U \in \mathbf{Op}$ and               |
| fong-spivak-invitation_F02717 | 260  | ■ 1.000 | $q(s) \subseteq U$                           | $q(s) \subseteq U$                           | $p(s) \subseteq U$ and $q(s) \subseteq U$ . We say that $p \leq^S q$ if $p(s) \subseteq q(s)$ for all $U \in \mathbf{Op}$ and               |

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| fong-spivak-invitation_F02718 | 260  | ■ 1.000 | $p(s) \subseteq q(s)$                                    | $p(s) \subseteq q(s)$                                    | $p(s) \subseteq U$ and $q(s) \subseteq U$ . We say that $p \leq^S q$ if $p(s) \subseteq q(s)$ for all $U \in \mathbf{Op}$ and                                 |
| fong-spivak-invitation_F02719 | 260  | ■ 1.000 | $\leq^S$                                                 | $\leq^S$                                                 | $s \in S(U)$ . We often drop the superscript from $\leq^S$ and simply write $\leq$ . In formal logic                                                          |
| fong-spivak-invitation_F02720 | 260  | ■ 1.000 | $S=1$                                                    | $S = 1$                                                  | In particular, if $S = 1$ is the terminal object, we denote $ \Omega^S $ by $ \Omega $ , and refer to elements                                                |
| fong-spivak-invitation_F02721 | 260  | ■ 1.000 | $ \Omega^S $                                             | $ \Omega^S $                                             | In particular, if $S = 1$ is the terminal object, we denote $ \Omega^S $ by $ \Omega $ , and refer to elements                                                |
| fong-spivak-invitation_F02722 | 260  | ■ 1.000 | $ \Omega $                                               | $ \Omega $                                               | In particular, if $S = 1$ is the terminal object, we denote $ \Omega^S $ by $ \Omega $ , and refer to elements                                                |
| fong-spivak-invitation_F02723 | 260  | ■ 1.000 | $p \in  \Omega $                                         | $p \in  \Omega $                                         | $p \in  \Omega $ as <i>propositions</i> . They are just morphisms $p: 1 \rightarrow \Omega$ .                                                                 |
| fong-spivak-invitation_F02724 | 260  | ■ 1.000 | $p: 1 \rightarrow \Omega$                                | $p: 1 \rightarrow \Omega$                                | $p \in  \Omega $ as <i>propositions</i> . They are just morphisms $p: 1 \rightarrow \Omega$ .                                                                 |
| fong-spivak-invitation_F02725 | 260  | ■ 1.000 | $p=q$                                                    | $p = q$                                                  | $p = q$ . The reason is that for any subsets $U, V \subseteq X$ , if $U \subseteq V$ and $V \subseteq U$ then $U = V$ .                                       |
| fong-spivak-invitation_F02726 | 260  | ■ 1.000 | $U, V \subseteq X$                                       | $U, V \subseteq X$                                       | $p = q$ . The reason is that for any subsets $U, V \subseteq X$ , if $U \subseteq V$ and $V \subseteq U$ then $U = V$ .                                       |
| fong-spivak-invitation_F02727 | 260  | ■ 1.000 | $U=V$                                                    | $U = V$                                                  | $p = q$ . The reason is that for any subsets $U, V \subseteq X$ , if $U \subseteq V$ and $V \subseteq U$ then $U = V$ .                                       |
| fong-spivak-invitation_F02728 | 260  | ■ 0.813 | $S \in \mathbf{Shv}(X)$                                  | $S \in \mathbf{Shv}(X)$                                  | <b>Exercise 7.47.</b> Give an example of a space $X$ , a sheaf $S \in \mathbf{Shv}(X)$ , and two predicates                                                   |
| fong-spivak-invitation_F02729 | 260  | ■ 0.977 | $p(s) \vdash_{s:S} q(s)$                                 | $p(s) \vdash_{s:S} q(s)$                                 | $p, q: S \rightarrow \Omega$ for which $p(s) \vdash_{s:S} q(s)$ holds. You do not have to be formal.                                                          |
| fong-spivak-invitation_F02730 | 260  | ■ 0.999 | $\wedge, \vee, \rightarrow, \neg$                        | $\wedge, \vee, \Rightarrow, \neg$                        | All of the logical symbols ( <code>true</code> , <code>false</code> , $\wedge, \vee, \Rightarrow, \neg$ ) from Section 7.4.2 make                             |
| fong-spivak-invitation_F02731 | 260  | ■ 1.000 | $(p \wedge q): S \rightarrow \Omega$                     | $(p \wedge q): S \rightarrow \Omega$                     | $(p \wedge q): S \rightarrow \Omega$ by $(p \wedge q)(s) := p(s) \wedge q(s)$ , and similarly for $\vee$ . Thus one says                                      |
| fong-spivak-invitation_F02732 | 260  | ■ 1.000 | $(p \wedge q)(s) := p(s) \wedge q(s)$                    | $(p \wedge q)(s) := p(s) \wedge q(s)$                    | $(p \wedge q): S \rightarrow \Omega$ by $(p \wedge q)(s) := p(s) \wedge q(s)$ , and similarly for $\vee$ . Thus one says                                      |
| fong-spivak-invitation_F02733 | 260  | ■ 1.000 | $n+1$                                                    | $n + 1$                                                  | exists.” Each takes in a predicate of $n + 1$ variables and returns a predicate of $n$ variables.                                                             |
| fong-spivak-invitation_F02734 | 260  | ■ 0.979 | $S, T \in \mathbf{Shv}(X, \mathbf{Op})$                  | $S, T \in \mathbf{Shv}(X, \mathbf{Op})$                  | <b>Example 7.48.</b> Suppose we have two sheaves $S, T \in \mathbf{Shv}(X, \mathbf{Op})$ and a predicate                                                      |
| fong-spivak-invitation_F02735 | 260  | ■ 1.000 | $p: S \times T \rightarrow \Omega$                       | $p: S \times T \rightarrow \Omega$                       | $p: S \times T \rightarrow \Omega$ . Let’s say $T$ represents what’s considered newsworthy and $S$ is again                                                   |
| fong-spivak-invitation_F02736 | 260  | ■ 1.000 | $t \in T(U)$                                             | $t \in T(U)$                                             | the set of people. So for a subset of time $U$ , a section $t \in T(U)$ is something that’s                                                                   |
| fong-spivak-invitation_F02737 | 261  | ■ 1.000 | $p(s, t)$                                                | $p(s, t)$                                                | subset of times on which $p(s, t)$ is true.                                                                                                                   |
| fong-spivak-invitation_F02738 | 261  | ■ 0.992 | $T, \dots$                                               | $T, \dots$                                               | “For all $t$ in $T$ , ... is worried about $t$ ” is itself a predicate on just one variable, $S$ ,                                                            |
| fong-spivak-invitation_F02739 | 261  | ■ 0.998 | $\exists(t: T). p(s, t)$                                 | $\exists(t: T). p(s, t)$                                 | also a predicate on $S$ , which we denote $\exists(t: T). p(s, t)$ . If we apply this predicate to                                                            |
| fong-spivak-invitation_F02740 | 261  | ■ 0.999 | $p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$ | $p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$ | <b>Exercise 7.49.</b> In the topos <b>Set</b> , where $\Omega = \mathbb{B}$ , consider the predicate $p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$ |
| fong-spivak-invitation_F02741 | 261  | ■ 1.000 | $\forall(z: \mathbb{Z}). p(n, z)$                        | $\forall(z: \mathbb{Z}). p(n, z)$                        | 1. What is the set of $n \in \mathbb{N}$ for which the predicate $\forall(z: \mathbb{Z}). p(n, z)$ holds?                                                     |
| fong-spivak-invitation_F02742 | 261  | ■ 1.000 | $\exists(z: \mathbb{Z}). p(n, z)$                        | $\exists(z: \mathbb{Z}). p(n, z)$                        | 2. What is the set of $n \in \mathbb{N}$ for which the predicate $\exists(z: \mathbb{Z}). p(n, z)$ holds?                                                     |
| fong-spivak-invitation_F02743 | 261  | ■ 0.999 | $z \in \mathbb{Z}$                                       | $z \in \mathbb{Z}$                                       | 3. What is the set of $z \in \mathbb{Z}$ for which the predicate $\forall(n: \mathbb{N}). p(n, z)$ holds?                                                     |
| fong-spivak-invitation_F02744 | 261  | ■ 0.999 | $\forall(n: \mathbb{N}). p(n, z)$                        | $\forall(n: \mathbb{N}). p(n, z)$                        | 3. What is the set of $z \in \mathbb{Z}$ for which the predicate $\forall(n: \mathbb{N}). p(n, z)$ holds?                                                     |

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| fong-spivak-invitation_F02745 | 261  | ■ 0.979 | $\exists(n : \mathbb{N}) . p(n, z)$                                                 | $\exists(n : \mathbb{N}).p(n, z)$                                                   | 4. What is the set of $z \in \mathbb{Z}$ for which the predicate $\exists(n : \mathbb{N}). p(n, z)$ holds?                        |
| fong-spivak-invitation_F02746 | 261  | ■ 0.817 | $\forall(t :$                                                                       | $\forall(t :$                                                                       | So, given $p$ , we have a universally and an existentially quantified predicate $\forall(t :$                                     |
| fong-spivak-invitation_F02747 | 261  | ■ 1.000 | $S \rightarrow \Omega$                                                              | $S \rightarrow \Omega$                                                              | morphisms $S \rightarrow \Omega$ or, equivalently, as subsheaves of $S$ ?                                                         |
| fong-spivak-invitation_F02748 | 261  | ■ 1.000 | $\forall(t : T) . p(s, t)$                                                          | $\forall(t : T).p(s, t)$                                                            | Given a predicate $p : S \times T \rightarrow \Omega$ , the universally quantified predicate $\forall(t : T). p(s, t)$            |
| fong-spivak-invitation_F02749 | 261  | ■ 1.000 | $V \in \Omega(U)$                                                                   | $V \in \Omega(U)$                                                                   | takes a section $s \in S(U)$ , for any open set $U$ , and returns a certain open set $V \in \Omega(U)$ .                          |
| fong-spivak-invitation_F02750 | 261  | ■ 1.000 | $p(\text{left}(\text{left.s} \mid_{\{V\}}, t \mid_{\text{right}}))=V$               | $p(s _V, t) = V$                                                                    | Namely, it returns the largest open set $V \subseteq U$ for which $p(s _V, t) = V$ holds for all                                  |
| fong-spivak-invitation_F02751 | 261  | ■ 1.000 | $t \in T(V)$                                                                        | $t \in T(V)$                                                                        | $t \in T(V)$ .                                                                                                                    |
| fong-spivak-invitation_F02752 | 261  | ■ 0.998 | $p(s, t)=$                                                                          | $p(s, t) =$                                                                         | above definition to the example $p(s, t) =$ “person $s$ is worried about news $t$ ” from                                          |
| fong-spivak-invitation_F02753 | 261  | ■ 1.000 | $T(V)$                                                                              | $T(V)$                                                                              | Example 7.48. Here, $T(V)$ is the set of items that are in the news throughout the                                                |
| fong-spivak-invitation_F02754 | 262  | ■ 0.516 | $p^{\{\text{prime}\}} : S \rightarrow \Omega^T$                                     | $p' : S \rightarrow \Omega^T$                                                       | where $p' : S \rightarrow \Omega^T$ is the currying of $S \times T \rightarrow \Omega$ and $\text{true}^T$ is the currying of the |
| fong-spivak-invitation_F02755 | 262  | ■ 0.516 | $S \times T \rightarrow \Omega$                                                     | $S \times T \rightarrow \Omega$                                                     | where $p' : S \rightarrow \Omega^T$ is the currying of $S \times T \rightarrow \Omega$ and $\text{true}^T$ is the currying of the |
| fong-spivak-invitation_F02756 | 262  | ■ 0.516 | $\{ \}^T$                                                                           | $T$                                                                                 | where $p' : S \rightarrow \Omega^T$ is the currying of $S \times T \rightarrow \Omega$ and $\text{true}^T$ is the currying of the |
| fong-spivak-invitation_F02757 | 262  | ■ 0.817 | $1 \times T \stackrel{!}{\rightarrow} 1 \stackrel{\text{true}}{\rightarrow} \Omega$ | $1 \times T \stackrel{!}{\rightarrow} 1 \stackrel{\text{true}}{\rightarrow} \Omega$ | composite $1 \times T \stackrel{!}{\rightarrow} 1 \stackrel{\text{true}}{\rightarrow} \Omega$ . See Eq. (7.3).                    |
| fong-spivak-invitation_F02758 | 262  | ■ 1.000 | $V = \bigcup_i V_i$                                                                 | $V = \bigcup_i V_i$                                                                 | namely the union $V = \bigcup_i V_i$ of all the open sets $V_i$ for which there exists some $t_i \in$                             |
| fong-spivak-invitation_F02759 | 262  | ■ 1.000 | $V_i$                                                                               | $V_i$                                                                               | namely the union $V = \bigcup_i V_i$ of all the open sets $V_i$ for which there exists some $t_i \in$                             |
| fong-spivak-invitation_F02760 | 262  | ■ 1.000 | $t_i \in$                                                                           | $t_i \in$                                                                           | namely the union $V = \bigcup_i V_i$ of all the open sets $V_i$ for which there exists some $t_i \in$                             |
| fong-spivak-invitation_F02761 | 262  | ■ 0.999 | $T(\text{left}(V_i \mid_{\text{right}}))$                                           | $T(V_i)$                                                                            | $T(V_i)$ satisfying $p(s _{V_i}, t_i) = V_i$ . If the result is $U$ itself, you might be tempted to                               |
| fong-spivak-invitation_F02762 | 262  | ■ 0.999 | $p(\text{left}(\text{left.s} \mid_{\{V_i\}}, t_i \mid_{\text{right}}))=V_i$         | $p(s _{V_i}, t_i) = V_i$                                                            | $T(V_i)$ satisfying $p(s _{V_i}, t_i) = V_i$ . If the result is $U$ itself, you might be tempted to                               |
| fong-spivak-invitation_F02763 | 262  | ■ 1.000 | $p(t)$                                                                              | $p(t)$                                                                              | think “ah, so there exists some $t \in T(U)$ satisfying $p(t)$ ,” but that is not necessarily so.                                 |
| fong-spivak-invitation_F02764 | 262  | ■ 1.000 | $U = \bigcup U_i$                                                                   | $U = \bigcup U_i$                                                                   | There is just a cover of $U = \bigcup U_i$ and local sections $t_i \in T(U_i)$ , each satisfying $p$ , as                         |
| fong-spivak-invitation_F02765 | 262  | ■ 1.000 | $t_i \in T(\text{left}(U_i \mid_{\text{right}}))$                                   | $t_i \in T(U_i)$                                                                    | There is just a cover of $U = \bigcup U_i$ and local sections $t_i \in T(U_i)$ , each satisfying $p$ , as                         |
| fong-spivak-invitation_F02766 | 262  | ■ 1.000 | $\{(s, t) \in S \times T \mid p(s, t)\} \subseteq S \times T$                       | $\{(s, t) \in S \times T \mid p(s, t)\} \subseteq S \times T$                       | with the subobject classified by $p$ , namely $\boxed{\{(s, t) \in S \times T \mid p(s, t)\} \subseteq S \times T}$ compose       |
| fong-spivak-invitation_F02767 | 262  | ■ 1.000 | $\pi_S : S \times T \rightarrow S$                                                  | $\pi_S : S \times T \rightarrow S$                                                  | with the projection $\pi_S : S \times T \rightarrow S$ as on the upper right; then take the epi-mono                              |
| fong-spivak-invitation_F02768 | 262  | ■ 0.983 | $( \Omega^S , \leq^S)$                                                              | $( \Omega^S , \leq^S)$                                                              | poset $( \Omega^S , \leq^S)$ of predicates on $S$ , where $ \Omega^S  = \mathcal{E}(S, \Omega)$ is just the set of morphisms      |
| fong-spivak-invitation_F02769 | 262  | ■ 0.983 | $ \Omega^S  = \mathcal{E}(S, \Omega)$                                               | $ \Omega^S  = \mathcal{E}(S, \Omega)$                                               | poset $( \Omega^S , \leq^S)$ of predicates on $S$ , where $ \Omega^S  = \mathcal{E}(S, \Omega)$ is just the set of morphisms      |
| fong-spivak-invitation_F02770 | 263  | ■ 0.996 | $j : \Omega \rightarrow \Omega$                                                     | $j : \Omega \rightarrow \Omega$                                                     | <b>Definition 7.52.</b> A modality in $\mathbf{Shv}(X)$ is a sheaf morphism $j : \Omega \rightarrow \Omega$ satisfying three      |
| fong-spivak-invitation_F02771 | 263  | ■ 1.000 | $p, q \in \Omega(U)$                                                                | $p, q \in \Omega(U)$                                                                | properties for all $U \subseteq X$ and $p, q \in \Omega(U)$ :                                                                     |

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| fong-spivak-invitation_F02772 | 263  | ■ 0.965 | $(j ; j)(p) \leq j(p)$                               | $(j;j)(p) \leq j(p)$                 | (b) $(\overline{j} \circ j)(p) < j(p)$ ,                                                                                                     |
| fong-spivak-invitation_F02773 | 263  | ■ 1.000 | $j(p \wedge q)=j(p) \wedge j(q)$                     | $j(p \wedge q) = j(p) \wedge j(q)$   | (c) $\overline{j}(p \wedge q) = j(p) \wedge j(q)$ .                                                                                          |
| fong-spivak-invitation_F02774 | 263  | ■ 1.000 | $p \in \Omega(U)$                                    | $p \in \Omega(U)$                    | holds for all $U \subseteq X$ and $p \in \Omega(U)$ . Show that for all $q \in \Omega(U)$ we have $j(j(q)) \leq$                             |
| fong-spivak-invitation_F02775 | 263  | ■ 1.000 | $q \in \Omega(U)$                                    | $q \in \Omega(U)$                    | holds for all $U \subseteq X$ and $p \in \Omega(U)$ . Show that for all $q \in \Omega(U)$ we have $j(j(q)) \leq$                             |
| fong-spivak-invitation_F02776 | 263  | ■ 1.000 | $j(j(q)) \leq$                                       | $j(j(q)) \leq$                       | holds for all $U \subseteq X$ and $p \in \Omega(U)$ . Show that for all $q \in \Omega(U)$ we have $j(j(q)) \leq$                             |
| fong-spivak-invitation_F02777 | 263  | ■ 1.000 | $j(q)$                                               | $j(q)$                               | $j(q)$ iff $j(j(q)) = j(q)$ .                                                                                                                |
| fong-spivak-invitation_F02778 | 263  | ■ 1.000 | $j(j(q))=j(q)$                                       | $j(j(q)) = j(q)$                     | $j(q)$ iff $\overline{j}(j(q)) = j(q)$ .                                                                                                     |
| fong-spivak-invitation_F02779 | 263  | ■ 0.999 | $p, \ldots$                                          | $p, \dots$                           | Diego,” there is a modal operator “assuming $p, \dots$ ” Now we are in a position to make                                                    |
| fong-spivak-invitation_F02780 | 263  | ■ 1.000 | $\Omega \rightarrow \Omega$                          | $\Omega \rightarrow \Omega$          | (a) the sheaf morphism $\Omega \rightarrow \Omega$ given by sending $q$ to $p \Rightarrow q$ is a modality,                                  |
| fong-spivak-invitation_F02781 | 263  | ■ 1.000 | $p \vee q$                                           | $p \vee q$                           | (b) the sheaf morphism $\Omega \rightarrow \Omega$ given by sending $q$ to $p \vee q$ is a modality,                                         |
| fong-spivak-invitation_F02782 | 263  | ■ 0.431 | $q \Rightarrow p$                                    | $q \Rightarrow p$                    | (c) the sheaf morphism $\Omega \rightarrow \Omega$ given by sending $q$ to $(q \Rightarrow p) \Rightarrow p$ is a modality.                  |
| fong-spivak-invitation_F02783 | 263  | ■ 1.000 | $p(s)$                                               | $p(s)$                               | $p(s)$ , and what does it mean?                                                                                                              |
| fong-spivak-invitation_F02784 | 263  | ■ 0.999 | $j(p(s))$                                            | $j(p(s))$                            | 3. What sort of thing is $j(p(s))$ and what does it mean?                                                                                    |
| fong-spivak-invitation_F02785 | 263  | ■ 1.000 | $p(s) \leq j(p(s))$                                  | $p(s) \leq j(p(s))$                  | 4. Is it true that $\overline{p}(s) < j(p(s))$ ? Explain briefly.                                                                            |
| fong-spivak-invitation_F02786 | 263  | ■ 1.000 | $j(j(p(s)))=j(p(s))$                                 | $j(j(p(s))) = j(p(s))$               | 5. Is it true that $\overline{j}(j(p(s))) = j(p(s))$ ? Explain briefly.                                                                      |
| fong-spivak-invitation_F02787 | 263  | ■ 1.000 | $q: S \rightarrow \Omega$                            | $q: S \rightarrow \Omega$            | 6. Choose another predicate $q: S \rightarrow \Omega$ . Is it true that $j(p \wedge q) = j(p) \wedge j(q)$ ? Explain                         |
| fong-spivak-invitation_F02788 | 264  | ■ 0.515 | $t \in T(c)$                                         | $t \in T(c)$                         | Is it true that for every table $c \in \text{Ob}(\mathcal{C})$ and every row $t \in T(c)$ there exists a row $s \in S(c)$ such               |
| fong-spivak-invitation_F02789 | 264  | ■ 0.515 | $s \in S(c)$                                         | $s \in S(c)$                         | Is it true that for every table $c \in \text{Ob}(\mathcal{C})$ and every row $t \in T(c)$ there exists a row $s \in S(c)$ such               |
| fong-spivak-invitation_F02790 | 264  | ■ 1.000 | $f(s)=t$                                             | $f(s) = t$                           | that $f(s) = t$ ?                                                                                                                            |
| fong-spivak-invitation_F02791 | 264  | ■ 1.000 | $t \in T(u)$                                         | $t \in T(u)$                         | Is it true that for every open set $U \subseteq X$ and every section $t \in T(u)$ of the bundle $T$ , there exists                           |
| fong-spivak-invitation_F02792 | 264  | ■ 0.999 | $\left(U_{\{i\}} \subseteq U\right)_{i \in I}$       | $(U_i \subseteq U)_{i \in I}$        | an open covering of $\overline{U_i \subseteq U}_{i \in I}$ of $U$ and a section $s_i \in S(U_i)$ of the bundle $S$ for each $i \in I$ ,      |
| fong-spivak-invitation_F02793 | 264  | ■ 0.999 | $s_{\{i\}} \in S\left(U_{\{i\}}\right)$              | $s_i \in S\left(U_i\right)$          | an open covering of $(U_i \subseteq U)_{i \in I}$ of $U$ and a section $s_i \in S(U_i)$ of the bundle $S$ for each $i \in I$ ,               |
| fong-spivak-invitation_F02794 | 264  | ■ 0.999 | $f\left(s_{\{i\}}\right)=\left.t\right _{u_{\{i\}}}$ | $f\left(s_i\right)=t _{u_i}$         | such that $f(s_i) = t _{u_i}$ is the restriction of $s$ to $U_i$ ?                                                                           |
| fong-spivak-invitation_F02795 | 264  | ■ 0.999 | $U_{\{i\}}$                                          | $U_i$                                | such that $f(s_i) = t _{u_i}$ is the restriction of $s$ to $U_i$ ?                                                                           |
| fong-spivak-invitation_F02796 | 265  | ■ 1.000 | $\mathbb{I} \mathbb{R}$                              | $\mathbb{I}\mathbb{R}$               | The interval domain $\mathbb{I}\mathbb{R}$ is a specific topological space, which we will use to model inter-                                |
| fong-spivak-invitation_F02797 | 265  | ■ 0.680 | $\mathbf{Shv}(\mathbb{I} \mathbb{R})$                | $\mathbf{Shv}(\mathbb{I}\mathbb{R})$ | vals of time. In other words, we will be interested in the category $\mathbf{Shv}(\mathbb{I}\mathbb{R})$ of sheaves                          |
| fong-spivak-invitation_F02798 | 265  | ■ 0.898 | $a < b$                                              | $a < b$                              | For $a < b$ in $\mathbb{R}$ , let $o_{[a,b]}$ denote the set $o_{[a,b]} := \{[d,u] \in \mathbb{I}\mathbb{R} \mid a < d \leq u < b\}$ ; these |

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| fong-spivak-invitation_F02799 | 265  | 0.898 | $\mathsf{o}_{\{[a, b]\}}$                                                       | $\mathcal{O}_{[a,b]}$                                                    | For $a < b$ in $\mathbb{R}$ , let $\mathcal{O}_{[a,b]}$ denote the set $\mathcal{O}_{[a,b]} := \{[d, u] \in \mathbb{IR} \mid a < d \leq u < b\}$ ; these           |
| fong-spivak-invitation_F02800 | 265  | 0.898 | $\mathsf{o}_{\{[a, b]\}} := \{[d, u] \in \mathbb{IR} \mid a < d \leq u < b\}$   | $\mathcal{O}_{[a,b]} := \{[d, u] \in \mathbb{R} \mid a < d \leq u < b\}$ | For $a < b$ in $\mathbb{R}$ , let $\mathcal{O}_{[a,b]}$ denote the set $\mathcal{O}_{[a,b]} := \{[d, u] \in \mathbb{IR} \mid a < d < u < b\}$ ; these              |
| fong-spivak-invitation_F02801 | 265  | 1.000 | $\mathsf{o}_{\{[0,5]\}}$                                                        | $\mathcal{O}_{[0,5]}$                                                    | Thus for example, $\mathcal{O}_{[0,5]}$ is an open set: it contains every $[d, u]$ contained in the open                                                           |
| fong-spivak-invitation_F02802 | 265  | 1.000 | $[d, u]$                                                                        | $[d, u]$                                                                 | Thus for example, $\mathcal{O}_{[0,5]}$ is an open set: it contains every $[d, u]$ contained in the open                                                           |
| fong-spivak-invitation_F02803 | 265  | 0.979 | $\{x \in \mathbb{IR} \mid 0 < x < 5\}$                                          | $\{x \in \mathbb{R} \mid 0 < x < 5\}$                                    | interval $\{x \in \mathbb{R} \mid 0 < x < 5\}$ . Similarly $\mathcal{O}_{[4,8]}$ is an open set, but note that $\mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]} \neq$ |
| fong-spivak-invitation_F02804 | 265  | 0.979 | $\mathsf{o}_{\{[4,8]\}}$                                                        | $\mathcal{O}_{[4,8]}$                                                    | interval $\{x \in \mathbb{R} \mid 0 < x < 5\}$ . Similarly $\mathcal{O}_{[4,8]}$ is an open set, but note that $\mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]} \neq$ |
| fong-spivak-invitation_F02805 | 265  | 0.979 | $\mathsf{o}_{\{[0,5]\}} \cup \mathsf{o}_{\{[4,8]\}} \neq$                       | $\mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]} \neq$                      | interval $\{x \in \mathbb{R} \mid 0 < x < 5\}$ . Similarly $\mathcal{O}_{[4,8]}$ is an open set, but note that $\mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]} \neq$ |
| fong-spivak-invitation_F02806 | 265  | 0.993 | $\mathsf{o}_{\{[0,8]\}}$                                                        | $\mathcal{O}_{[0,8]}$                                                    | $\mathcal{O}_{[0,8]}$ . Indeed, the interval $[2, 6]$ is in the right-hand side but not the left.                                                                  |
| fong-spivak-invitation_F02807 | 265  | 1.000 | $[2,6] \in \mathsf{o}_{\{[0,8]\}}$                                              | $[2, 6] \in \mathcal{O}_{[0,8]}$                                         | 1. Explain why $[2, 6] \in \mathcal{O}_{[0,8]}$ .                                                                                                                  |
| fong-spivak-invitation_F02808 | 265  | 0.977 | $\mathsf{notin} \mathsf{o}_{\{[0,5]\}} \cup \mathsf{o}_{\{[4,8]\}}$             | $\notin \mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]}$                    | 2. Explain why $[2, 6] \notin \mathcal{O}_{[0,5]} \cup \mathcal{O}_{[4,8]}$ .                                                                                      |
| fong-spivak-invitation_F02809 | 265  | 0.505 | $\mathbf{B\ T} := \operatorname{Shv}(\mathbb{IR} \mid \mathbb{R}, \mathbf{Op})$ | $\mathbf{BT} := \operatorname{Shv}(\mathbb{IR}, \mathbf{Op})$            | Let $\mathbf{Op}$ denote the open sets of $\mathbb{IR}$ , as described above, and let $\mathbf{BT} := \operatorname{Shv}(\mathbb{IR}, \mathbf{Op})$                |
| fong-spivak-invitation_F02810 | 265  | 0.963 | $\mathbb{R} \subseteq \mathbb{R}$                                               | $\mathbb{R} \subseteq \mathbb{R}$                                        | <b>Exercise 7.59.</b> Show that a subset $U \subseteq \mathbb{R}$ is open in the subspace topology of $\mathbb{R} \subseteq \mathbb{IR}$                           |
| fong-spivak-invitation_F02811 | 265  | 1.000 | $U \cap \mathbb{R}$                                                             | $U \cap \mathbb{R}$                                                      | iff $U \cap \mathbb{R}$ is open in the usual topology on $\mathbb{R}$ defined in Example 7.18.                                                                     |
| fong-spivak-invitation_F02812 | 265  | 0.912 | $\mathbf{B\ T} = \operatorname{Shv}(\mathbb{IR} \mid \mathbb{R}, \mathbf{Op})$  | $\mathbf{BT} = \operatorname{Shv}(\mathbb{IR}, \mathbf{Op})$             | We cannot go into much depth about the sheaf topos $\mathbf{BT} = \operatorname{Shv}(\mathbb{IR}, \mathbf{Op})$ , for reasons                                      |
| fong-spivak-invitation_F02813 | 266  | 0.569 | $\mathbb{IR}$                                                                   | $\mathbb{IR}$                                                            | discuss what it means to be a sheaf on $\mathbb{IR}$ , giving a few examples including that of the                                                                 |
| fong-spivak-invitation_F02814 | 266  | 0.576 | $\mathbb{IR}, \mathbf{Op}$                                                      | $\mathbb{IR}, \mathbf{Op}$                                               | A sheaf $S$ on the interval domain $(\mathbb{IR}, \mathbf{Op})$ is a functor $S: \mathbf{Op}^{\operatorname{op}} \rightarrow \mathbf{Set}$ : it assigns to         |
| fong-spivak-invitation_F02815 | 266  | 0.576 | $S: \mathbf{Op}^{\operatorname{op}} \rightarrow$                                | $S: \mathbf{Op}^{\operatorname{op}} \rightarrow$                         | A sheaf $S$ on the interval domain $(\mathbb{IR}, \mathbf{Op})$ is a functor $S: \mathbf{Op}^{\operatorname{op}} \rightarrow \mathbf{Set}$ : it assigns to         |
| fong-spivak-invitation_F02816 | 266  | 1.000 | $S(U)$                                                                          | $S(U)$                                                                   | each open set $U$ a set $S(U)$ ; how should we interpret this? An element $s \in S(U)$ is                                                                          |
| fong-spivak-invitation_F02817 | 266  | 1.000 | $\left(s_i\right)$                                                              | $(s_i)$                                                                  | we can find matching $U_i$ -events $(s_i)$ for each $i \in I$ , then the sheaf condition (Defini-                                                                  |
| fong-spivak-invitation_F02818 | 266  | 0.871 | $U \subseteq \mathbb{R}$                                                        | $U \subseteq \mathbb{R}$                                                 | We said in Section 7.5.1 that every open set $U \subseteq \mathbb{IR}$ can be written as the union                                                                 |
| fong-spivak-invitation_F02819 | 266  | 1.000 | $S(\mathcal{O}_{[a,b]})$                                                        | $S(\mathcal{O}_{[a,b]})$                                                 | $S(\mathcal{O}_{[a,b]})$ on these basic open sets. The sheaf condition furthermore implies that these                                                              |
| fong-spivak-invitation_F02820 | 266  | 0.991 | $A \in \operatorname{Shv}(\mathbb{IR} \mid \mathbb{R})$                         | $A \in \operatorname{Shv}(\mathbb{IR})$                                  | <b>Example 7.60.</b> For any set $A$ there is a sheaf $A \in \operatorname{Shv}(\mathbb{IR})$ that assigns to each open set                                        |
| fong-spivak-invitation_F02821 | 266  | 1.000 | $A(U) := A$                                                                     | $A(U) := A$                                                              | $U$ the set $A(U) := A$ . This allows us to refer to integers, or real numbers, or letters of an                                                                   |
| fong-spivak-invitation_F02822 | 266  | 1.000 | $7 \in \mathbb{N}$                                                              | $7 \in \mathbb{N}$                                                       | alphabet, as though they were behaviors. What sort of behavior is $7 \in \mathbb{N}$ ? It is the sort of                                                           |
| fong-spivak-invitation_F02823 | 266  | 0.758 | $F_X$                                                                           | $F_X$                                                                    | <b>Example 7.61.</b> Fix any topological space $(X, \mathbf{Op}_X)$ . Then there is a sheaf $F_X$ of <i>local</i>                                                  |
| fong-spivak-invitation_F02824 | 266  | 0.787 | $\mathbb{R}$                                                                    | $\mathbb{R}$                                                             | <i>functions from <math>\mathbb{IR}</math> to <math>X</math></i> . That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$ , we assign the set $F_X(U) :=$     |
| fong-spivak-invitation_F02825 | 266  | 0.787 | $U \in \mathbf{Op}_{\mathbb{R}}$                                                | $U \in \mathbf{Op}_{\mathbb{R}}$                                         | <i>functions from <math>\mathbb{IR}</math> to <math>X</math></i> . That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$ , we assign the set $F_X(U) :=$     |

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| fong-spivak-invitation_F02826 | 266  | 0.787 | $F_{\{X\}}(U) :=$                                 | $F_X(U) :=$                                   | <i>functions from <math>\mathbb{IR}</math> to <math>X</math>. That is, for any open set <math>U \in \mathbf{Op}_{\mathbb{IR}}</math>, we assign the set <math>F_X(U) :=</math></i> |
| fong-spivak-invitation_F02827 | 266  | 0.759 | $\{f: U \rightarrowtail X \mid f$                 | $\{f: U \rightarrow X \mid f$                 | $\{f: U \rightarrow X \mid f$ is continuous}. There is also the sheaf $G_X$ of local functions on the                                                                              |
| fong-spivak-invitation_F02828 | 266  | 0.759 | $G_{\{X\}}$                                       | $G_X$                                         | $\{f: U \rightarrow X \mid f$ is continuous}. There is also the sheaf $G_X$ of local functions on the                                                                              |
| fong-spivak-invitation_F02829 | 266  | 0.519 | $\mathbb{R} \subseteq \mathbb{R}$                 | $\mathbb{R} \subseteq \mathbb{R}$             | subspace $\mathbb{R} \subseteq \mathbb{IR}$ . That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$ , we assign the set $G_X(U) :=$                                          |
| fong-spivak-invitation_F02830 | 266  | 0.519 | $U \in \mathbf{Op}_{\mathbb{R}}$                  | $U \in \mathbf{Op}_{\mathbb{R}}$              | subspace $\mathbb{R} \subseteq \mathbb{IR}$ . That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$ , we assign the set $G_X(U) :=$                                          |
| fong-spivak-invitation_F02831 | 266  | 0.519 | $G_{\{X\}}(U) :=$                                 | $G_X(U) :=$                                   | subspace $\mathbb{R} \subseteq \mathbb{IR}$ . That is, for any open set $U \in \mathbf{Op}_{\mathbb{IR}}$ , we assign the set $G_X(U) :=$                                          |
| fong-spivak-invitation_F02832 | 266  | 0.985 | $\{f: U \cap \mathbb{R} \rightarrowtail X \mid f$ | $\{f: U \cap \mathbb{R} \rightarrow X \mid f$ | $\{f: U \cap \mathbb{R} \rightarrow X \mid f$ is continuous}.                                                                                                                      |
| fong-spivak-invitation_F02833 | 266  | 0.846 | $(X, \mathbf{Op}_X)$                              | $(X, \mathbf{Op}_X)$                          | $(X, \mathbf{Op}_X)$ and any subset $R \subseteq \mathbb{IR}$ of the interval domain. Define $H_X(U) := \{f: U \cap$                                                               |
| fong-spivak-invitation_F02834 | 266  | 0.846 | $R \subseteq \mathbb{R}$                          | $R \subseteq \mathbb{R}$                      | $(X, \mathbf{Op}_X)$ and any subset $R \subseteq \mathbb{IR}$ of the interval domain. Define $H_X(U) := \{f: U \cap$                                                               |
| fong-spivak-invitation_F02835 | 266  | 0.846 | $H_{\{X\}}(U) := \{f: U \cap$                     | $H_X(U) := \{f: U \cap$                       | $(X, \mathbf{Op}_X)$ and any subset $R \subseteq \mathbb{IR}$ of the interval domain. Define $H_X(U) := \{f: U \cap$                                                               |
| fong-spivak-invitation_F02836 | 266  | 1.000 | $R \rightarrowtail X \mid f$                      | $R \rightarrow X \mid f$                      | $R \rightarrow X \mid f$ is continuous}.                                                                                                                                           |
| fong-spivak-invitation_F02837 | 266  | 1.000 | $H_{\{X\}}$                                       | $H_X$                                         | 1. Is $H_X$ a presheaf? If not, why not; if so, what are the restriction maps?                                                                                                     |
| fong-spivak-invitation_F02838 | 267  | 0.740 | $(d, d+3)$                                        | $(d, d+3)$                                    | idea is straightforward: for any interval of time, say a three-day interval $[d, d+3]$ , let                                                                                       |
| fong-spivak-invitation_F02839 | 267  | 1.000 | $L(d, d+3)$                                       | $L(d, d+3)$                                   | $L(d, d+3)$ denote the set of all possible behaviors a person who is in love could possibly                                                                                        |
| fong-spivak-invitation_F02840 | 267  | 0.458 | $\mathbb{I}\mathbb{R}, \mathbf{Op}$               | $\mathbb{I}\mathbb{R}, \mathbf{Op}$           | <b>BT</b> is the topos of sheaves on the space $(\mathbb{IR}, \mathbf{Op})$ , as defined in Section 7.5.1. As                                                                      |
| fong-spivak-invitation_F02841 | 267  | 1.000 | $(0, 5)$                                          | $(0, 5)$                                      | Suppose Bob likes the weather throughout the interval $(0, 5)$ and throughout the                                                                                                  |
| fong-spivak-invitation_F02842 | 267  | 0.806 | $(4, 8)$                                          | $(4, 8)$                                      | remained watching position $p$ .” Then just because it’s true on $(0, 5)$ and on $(4, 8)$ , does                                                                                   |
| fong-spivak-invitation_F02843 | 267  | 1.000 | $(0, 8)$                                          | $(0, 8)$                                      | be true on $(0, 5)$ and true on $(4, 8)$ but not on $(0, 8)$ . In order to capture the semantics of                                                                                |
| fong-spivak-invitation_F02844 | 268  | 1.000 | $\theta \in \mathbb{R}_{\geq 0}$                  | $\theta \in \mathbb{R}_{\geq 0}$              | behavior types: the altimeter is a variable whose value $\theta \in \mathbb{R}_{\geq 0}$ is changing continuously                                                                  |
| fong-spivak-invitation_F02845 | 268  | 1.000 | $p(D, T)$                                         | $p(D, T)$                                     | $p$ , one could imagine the predicate $p(D, T)$ for $D: \text{dials}$ and $T: \text{thrusters}$ as                                                                                 |
| fong-spivak-invitation_F02846 | 268  | 0.968 | $@_t$                                             | $@_t$                                         | Here $@_t$ is a modality, as we discussed in Definition 7.52; in fact it turns out to be one                                                                                       |
| fong-spivak-invitation_F02847 | 268  | 1.000 | $@_t(q)$                                          | $@_t(q)$                                      | statement $@_t(q)$ says that $q$ is true in some small enough neighborhood around $t$ . So                                                                                         |
| fong-spivak-invitation_F02848 | 268  | 0.999 | $D \in$                                           | $D \in$                                       | Given an actual playing-out-of-events over a time period $U$ , i.e. actual section $D \in$                                                                                         |
| fong-spivak-invitation_F02849 | 268  | 1.000 | $\text{dials}(U)$                                 | $\text{dials}(U)$                             | $\text{dials}(U)$ and section $T \in \text{thrusters}(U)$ , the predicate Eq. (7.19) will hold on                                                                                  |
| fong-spivak-invitation_F02850 | 268  | 1.000 | $T \in \text{thrusters}(U)$                       | $T \in \text{thrusters}(U)$                   | $\text{dials}(U)$ and section $T \in \text{thrusters}(U)$ , the predicate Eq. (7.19) will hold on                                                                                  |
| fong-spivak-invitation_F02851 | 270  | 1.000 | $f(x)=x+5$                                        | $f(x) = x + 5$                                | <b>Order-preserving:</b> Take $f(x) = x + 5$ ; if $x \leq y$ then $x + 5 \leq y + 5$ , so $f$ is order-preserving.                                                                 |
| fong-spivak-invitation_F02852 | 270  | 1.000 | $x+5 \leq y+5$                                    | $x + 5 \leq y + 5$                            | <b>Order-preserving:</b> Take $f(x) = x + 5$ ; if $x \leq y$ then $x + 5 \leq y + 5$ , so $f$ is order-preserving.                                                                 |

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| fong-spivak-invitation_F02853 | 270  | ■ 0.498 | $g(x) := -x$                                  | $g(x) := -x$                                  | Take $g(x) := -x$ ; even though $1 \leq 2$ , the required inequality $-1 \leq^? -2$ does not hold, so $g$                                                   |
| fong-spivak-invitation_F02854 | 270  | ■ 0.498 | $1 \leq 2$                                    | $1 \leq 2$                                    | Take $g(x) := -x$ ; even though $1 \leq 2$ , the required inequality $-1 \leq^? -2$ does not hold, so $g$                                                   |
| fong-spivak-invitation_F02855 | 270  | ■ 0.498 | $-1 \leq ?$                                   | $-1 \leq ?$                                   | Take $g(x) := -x$ ; even though $1 \leq 2$ , the required inequality $-1 \leq^? -2$ does not hold, so $g$                                                   |
| fong-spivak-invitation_F02856 | 270  | ■ 0.992 | $f(x) := x+5$                                 | $f(x) := x+5$                                 | <b>Metric-preserving:</b> Take $f(x) := x+5$ ; for any $x, y$ we have $ x-y  =  (x+5)-(y+5) $                                                               |
| fong-spivak-invitation_F02857 | 270  | ■ 0.992 | $ x-y  =  (x+5)-(y+5) $                       | $ x-y  =  (x+5)-(y+5) $                       | <b>Metric-preserving:</b> Take $f(x) := x+5$ ; for any $x, y$ we have $ x-y  =  (x+5)-(y+5) $                                                               |
| fong-spivak-invitation_F02858 | 270  | ■ 1.000 | $g(x) := 2 * x$                               | $g(x) := 2 * x$                               | $g(x) := 2 * x$ ; then with $x = 1$ and $y = 2$ we have $ x-y  = 1$ but $ 2x-2y  = 2$ , so $g$ does                                                         |
| fong-spivak-invitation_F02859 | 270  | ■ 1.000 | $x=1$                                         | $x = 1$                                       | $g(x) := 2 * x$ ; then with $x = 1$ and $y = 2$ we have $ x-y  = 1$ but $ 2x-2y  = 2$ , so $g$ does                                                         |
| fong-spivak-invitation_F02860 | 270  | ■ 1.000 | $y=2$                                         | $y = 2$                                       | $g(x) := 2 * x$ ; then with $x = 1$ and $y = 2$ we have $ x-y  = 1$ but $ 2x-2y  = 2$ , so $g$ does                                                         |
| fong-spivak-invitation_F02861 | 270  | ■ 1.000 | $ x-y =1$                                     | $ x-y  = 1$                                   | $g(x) := 2 * x$ ; then with $x = 1$ and $y = 2$ we have $ x-y  = 1$ but $ 2x-2y  = 2$ , so $g$ does                                                         |
| fong-spivak-invitation_F02862 | 270  | ■ 1.000 | $ 2x-2y =2$                                   | $ 2x-2y  = 2$                                 | $g(x) := 2 * x$ ; then with $x = 1$ and $y = 2$ we have $ x-y  = 1$ but $ 2x-2y  = 2$ , so $g$ does                                                         |
| fong-spivak-invitation_F02863 | 270  | ■ 1.000 | $f(x) := 3 * x$                               | $f(x) := 3 * x$                               | <b>Addition-preserving:</b> Take $f(x) := 3 * x$ ; for any $x, y$ we have $3 * (x+y) = (3 * x) + (3 * y)$ , so                                              |
| fong-spivak-invitation_F02864 | 270  | ■ 1.000 | $3 * (x+y) = (3 * x) + (3 * y)$               | $3 * (x+y) = (3 * x) + (3 * y)$               | <b>Addition-preserving:</b> Take $f(x) := 3 * x$ ; for any $x, y$ we have $3 * (x+y) = (3 * x) + (3 * y)$ , so                                              |
| fong-spivak-invitation_F02865 | 270  | ■ 1.000 | $g(x) := x+1$                                 | $g(x) := x+1$                                 | $f$ preserves addition. Take $g(x) := x+1$ ; then with $x = 0$ and $y = 0$ , we have $g(x+y) = 1$ ,                                                         |
| fong-spivak-invitation_F02866 | 270  | ■ 1.000 | $x=0$                                         | $x = 0$                                       | $f$ preserves addition. Take $g(x) := x+1$ ; then with $x = 0$ and $y = 0$ , we have $g(x+y) = 1$ ,                                                         |
| fong-spivak-invitation_F02867 | 270  | ■ 1.000 | $y=0$                                         | $y = 0$                                       | $f$ preserves addition. Take $g(x) := x+1$ ; then with $x = 0$ and $y = 0$ , we have $g(x+y) = 1$ ,                                                         |
| fong-spivak-invitation_F02868 | 270  | ■ 1.000 | $g(x+y)=1$                                    | $g(x+y) = 1$                                  | $f$ preserves addition. Take $g(x) := x+1$ ; then with $x = 0$ and $y = 0$ , we have $g(x+y) = 1$ ,                                                         |
| fong-spivak-invitation_F02869 | 270  | ■ 1.000 | $g(x)+g(y)=2$                                 | $g(x) + g(y) = 2$                             | but $g(x) + g(y) = 2$ , so $g$ does not preserve addition.                                                                                                  |
| fong-spivak-invitation_F02870 | 271  | ■ 0.995 | $\{1,2,3,4\}$                                 | $\{1,2,3,4\}$                                 | 2. Here is a picture (using text, rather than circles) for partitions of the set $\{1, 2, 3, 4\}$ :                                                         |
| fong-spivak-invitation_F02871 | 271  | ■ 1.000 | $A=(12)(3)(4)$                                | $A = (12)(3)(4)$                              | For the remaining parts, we choose $A = (12)(3)(4)$ and $B = (13)(2)(4)$ .                                                                                  |
| fong-spivak-invitation_F02872 | 271  | ■ 1.000 | $B=(13)(2)(4)$                                | $B = (13)(2)(4)$                              | For the remaining parts, we choose $A = (12)(3)(4)$ and $B = (13)(2)(4)$ .                                                                                  |
| fong-spivak-invitation_F02873 | 271  | ■ 1.000 | $A \vee B = (123)(4)$                         | $A \vee B = (123)(4)$                         | 3. $A \vee B = (123)(4)$ .                                                                                                                                  |
| fong-spivak-invitation_F02874 | 271  | ■ 1.000 | $0 \notin \{n \in \mathbb{Z} \mid n \geq 1\}$ | $0 \notin \{n \in \mathbb{Z} \mid n \geq 1\}$ | 2. This is false: $0 \in \mathbb{N}$ but $0 \notin \{n \in \mathbb{Z} \mid n \geq 1\}$ .                                                                    |
| fong-spivak-invitation_F02875 | 271  | ■ 0.936 | $\{1,2,3\}$                                   | $\{1,2,3\}$                                   | 2. The union of $\{1, 2, 3\}$ and $\{1\}$ is $\{1, 2, 3\} \cup \{1\} = \{1, 2, 3\}$ .                                                                       |
| fong-spivak-invitation_F02876 | 271  | ■ 0.936 | $\{1,2,3\} \cup \{1\} = \{1,2,3\}$            | $\{1,2,3\} \cup \{1\} = \{1,2,3\}$            | 2. The union of $\{1, 2, 3\}$ and $\{1\}$ is $\{1, 2, 3\} \cup \{1\} = \{1, 2, 3\}$ .                                                                       |
| fong-spivak-invitation_F02877 | 271  | ■ 0.994 | $\{h, 1\} \times \{1,2,3\}$                   | $\{h, 1\} \times \{1,2,3\}$                   | 3. The six elements of $\{h, 1\} \times \{1, 2, 3\}$ are                                                                                                    |
| fong-spivak-invitation_F02878 | 271  | ■ 0.996 | $\{h, 1\} \sqcup \{1,2,3\}$                   | $\{h, 1\} \sqcup \{1,2,3\}$                   | 4. The five elements of $\{h, 1\} \sqcup \{1, 2, 3\}$ are                                                                                                   |
| fong-spivak-invitation_F02879 | 271  | ■ 1.000 | $\{h, 1\} \cup \{1,2,3\}$                     | $\{h, 1\} \cup \{1,2,3\}$                     | 5. The four elements of $\{h, 1\} \cup \{1, 2, 3\}$ are                                                                                                     |
| fong-spivak-invitation_F02880 | 272  | ■ 1.000 | $p_1^{\prime}, p_2^{\prime} \in P^{\prime}$   | $p_1^{\prime}, p_2^{\prime} \in P^{\prime}$   | 1. Given $p \in P$ , suppose we had $p_1^{\prime}, p_2^{\prime} \in P^{\prime}$ such that $A_p = A_{p_1^{\prime}}$ and $A_p = A_{p_2^{\prime}}$ . Well then |

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| fong-spivak-invitation_F02881 | 272  | ■ 1.000 | $A_{\{p\}}=A_{\{p_1\}^{\wedge\prime}}^{\wedge\prime}$                                        | $A_p = A'_{p_1}$                     | 1. Given $p \in P$ , suppose we had $p'_1, p'_2 \in P'$ such that $A_p = A'_{p'_1}$ and $A_p = A'_{p'_2}$ . Well then                       |
| fong-spivak-invitation_F02882 | 272  | ■ 1.000 | $A_{\{p\}}=A_{\{p_2\}^{\wedge\prime}}^{\wedge\prime}$                                        | $A_p = A'_{p'_2}$                    | 1. Given $p \in P$ , suppose we had $p'_1, p'_2 \in P'$ such that $A_p = A'_{p'_1}$ and $A_p = A'_{p'_2}$ . Well then                       |
| fong-spivak-invitation_F02883 | 272  | ■ 1.000 | $A_{\{p_1\}^{\wedge\prime}}=A_{\{p_2\}^{\wedge\prime}}$                                      | $A_{p'_1} = A_{p'_2}$                | $A_{p'_1} = A_{p'_2}$ , so in particular $A_{p'_1} \cap A_{p'_2} = A_{p'_1}$ . By Definition 1.10 of partition, $A_{p'_1} \neq \emptyset$ , |
| fong-spivak-invitation_F02884 | 272  | ■ 1.000 | $A_{\{p_1\}^{\wedge\prime}} \cap A_{\{p_2\}^{\wedge\prime}}=A_{\{p_1\}^{\wedge\prime}}$      | $A_{p'_1} \cap A_{p'_2} = A_{p'_1}$  | $A_{p'_1} = A_{p'_2}$ , so in particular $A_{p'_1} \cap A_{p'_2} = A_{p'_1}$ . By Definition 1.10 of partition, $A_{p'_1} \neq \emptyset$ , |
| fong-spivak-invitation_F02885 | 272  | ■ 1.000 | $A_{\{p_1\}^{\wedge\prime}} \neq \varnothing$                                                | $A_{p'_1} \neq \emptyset$            | $A_{p'_1} = A_{p'_2}$ , so in particular $A_{p'_1} \cap A_{p'_2} = A_{p'_1}$ . By Definition 1.10 of partition, $A_{p'_1} \neq \emptyset$ , |
| fong-spivak-invitation_F02886 | 272  | ■ 1.000 | $p_1 \neq p_2$                                                                               | $p_1 \neq p_2$                       | and yet if $p_1 \neq p_2$ then $A_{p'_1} \cap A_{p'_2} = \emptyset$ . This can't be, so we must have $p'_1 = p'_2$ , as                     |
| fong-spivak-invitation_F02887 | 272  | ■ 1.000 | $A_{\{p_1\}^{\wedge\prime}} \cap A_{\{p_2\}^{\wedge\prime}}=\varnothing$                     | $A_{p'_1} \cap A_{p'_2} = \emptyset$ | and yet if $p_1 \neq p_2$ then $A_{p'_1} \cap A_{p'_2} = \emptyset$ . This can't be, so we must have $p'_1 = p'_2$ , as                     |
| fong-spivak-invitation_F02888 | 272  | ■ 1.000 | $p_1^{\wedge\prime}=p_2^{\wedge\prime}$                                                      | $p'_1 = p'_2$                        | and yet if $p_1 \neq p_2$ then $A_{p'_1} \cap A_{p'_2} = \emptyset$ . This can't be, so we must have $p'_1 = p'_2$ , as                     |
| fong-spivak-invitation_F02889 | 272  | ■ 1.000 | $A_{\{p^{\wedge\prime}\}^{\wedge\prime}} \neq \varnothing$                                   | $A'_{p'} \neq \emptyset$             | Since $A'_{p'} \neq \emptyset$ is nonempty by definition, we can pick some $a \in A'_{p'}$ ; since $A'_{p'} \subseteq A$ , we               |
| fong-spivak-invitation_F02890 | 272  | ■ 1.000 | $a \in A_{\{p^{\wedge\prime}\}^{\wedge\prime}}$                                              | $a \in A'_{p'}$                      | Since $A'_{p'} \neq \emptyset$ is nonempty by definition, we can pick some $a \in A'_{p'}$ ; since $A'_{p'} \subseteq A$ , we               |
| fong-spivak-invitation_F02891 | 272  | ■ 1.000 | $A_{\{p^{\wedge\prime}\}^{\wedge\prime}} \subseteq A$                                        | $A'_{p'} \subseteq A$                | Since $A'_{p'} \neq \emptyset$ is nonempty by definition, we can pick some $a \in A'_{p'}$ ; since $A'_{p'} \subseteq A$ , we               |
| fong-spivak-invitation_F02892 | 272  | ■ 1.000 | $p^{\wedge\prime} \in P^{\wedge\prime}$                                                      | $p'' \in P'$                         | $p$ ; now we show that $A_p = A'_{p'}$ . By assumption there is some $p'' \in P'$ with $A_p = A'_{p''}$ ,                                   |
| fong-spivak-invitation_F02893 | 272  | ■ 1.000 | $A_{\{p\}}=A_{\{p^{\wedge\prime}\}^{\wedge\prime}}$                                          | $A_p = A'_{p''}$                     | $p$ ; now we show that $A_p = A'_{p'}$ . By assumption there is some $p'' \in P'$ with $A_p = A'_{p''}$ ,                                   |
| fong-spivak-invitation_F02894 | 272  | ■ 1.000 | $a \in A_{\{p^{\wedge\prime}\}^{\wedge\prime}}$                                              | $a \in A'_{p''}$                     | so now $a \in A'_{p''}$ and $a \in A'_{p'}$ , so $a \in A'_{p'} \cap A'_{p''}$ . Again by definition, having a nonempty                     |
| fong-spivak-invitation_F02895 | 272  | ■ 1.000 | $a \in A_{\{p^{\wedge\prime}\}^{\wedge\prime}} \cap A_{\{p^{\wedge\prime}\}^{\wedge\prime}}$ | $a \in A'_{p'} \cap A'_{p''}$        | so now $a \in A'_{p''}$ and $a \in A'_{p'}$ , so $a \in A'_{p'} \cap A'_{p''}$ . Again by definition, having a nonempty                     |
| fong-spivak-invitation_F02896 | 272  | ■ 1.000 | $p^{\wedge\prime}=p^{\wedge\prime}$                                                          | $p' = p''$                           | intersection means $p' = p''$ . So we conclude that $A_p = A_{p'}$ .                                                                        |
| fong-spivak-invitation_F02897 | 272  | ■ 1.000 | $A_{\{p\}}=A_{\{p^{\wedge\prime}\}}$                                                         | $A_p = A_{p'}$                       | intersection means $p' = p''$ . So we conclude that $A_p = A_{p'}$ .                                                                        |
| fong-spivak-invitation_F02898 | 272  | ■ 1.000 | $a \in A_{\{p\}} \cap A_{\{q\}}$                                                             | $a \in A_p \cap A_q$                 | suppose otherwise and derive a contradiction. So suppose there exists $a \in A_p \cap A_q$ ; we will                                        |
| fong-spivak-invitation_F02899 | 272  | ■ 1.000 | $A_{\{p\}}=A_{\{q\}}$                                                                        | $A_p = A_q$                          | show that $A_p = A_q$ , which contradicts an earlier hypothesis. To show that these two subsets                                             |
| fong-spivak-invitation_F02900 | 272  | ■ 1.000 | $a^{\wedge\prime} \in A_{\{p\}}$                                                             | $a' \in A_p$                         | are equal, it suffices to show that $a' \in A_p$ iff $a' \in A_q$ for all $a' \in A$ . Suppose $a' \in A_p$ ;                               |
| fong-spivak-invitation_F02901 | 272  | ■ 1.000 | $a^{\wedge\prime} \in A_{\{q\}}$                                                             | $a' \in A_q$                         | are equal, it suffices to show that $a' \in A_p$ iff $a' \in A_q$ for all $a' \in A$ . Suppose $a' \in A_p$ ;                               |
| fong-spivak-invitation_F02902 | 272  | ■ 1.000 | $a^{\wedge\prime} \in A$                                                                     | $a' \in A$                           | are equal, it suffices to show that $a' \in A_p$ iff $a' \in A_q$ for all $a' \in A$ . Suppose $a' \in A_p$ ;                               |
| fong-spivak-invitation_F02903 | 272  | ■ 1.000 | $a \sim a^{\wedge\prime}$                                                                    | $a \sim a'$                          | then because $A_p$ is connected, we have $a \sim a'$ . And because $A_q$ is closed, $a' \in A_q$ . In a                                     |
| fong-spivak-invitation_F02904 | 272  | ■ 1.000 | $X:=\left\{a^{\wedge\prime} \in A \mid a^{\wedge\prime} \sim a\right\}$                      | $X := \{a' \in A \mid a' \sim a\}$   | show that there is some closed and connected subset containing $a$ . Let $X := \{a' \in A \mid a' \sim a\}$ ;                               |
| fong-spivak-invitation_F02905 | 272  | ■ 0.962 | $a^{\wedge\prime} \in X$                                                                     | $a' \in X$                           | we claim it is closed and connected and contains $a$ . To see $X$ is closed, suppose $a' \in X$ and                                         |

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| fong-spivak-invitation_F02906 | 272  | ■ 0.999 | $b \sim a^{\prime}$                            | $b \sim a'$                                    | $b \sim a'$ ; then $b \sim a$ by transitivity and symmetry of $\sim$ , so $b \in X$ . To see that $X$ is connected,                           |
| fong-spivak-invitation_F02907 | 272  | ■ 0.999 | $b \sim a$                                     | $b \sim a$                                     | $b \sim a'$ ; then $b \sim a$ by transitivity and symmetry of $\sim$ , so $b \in X$ . To see that $X$ is connected,                           |
| fong-spivak-invitation_F02908 | 272  | ■ 0.999 | $b \in X$                                      | $b \in X$                                      | $b \sim a'$ ; then $b \sim a$ by transitivity and symmetry of $\sim$ , so $b \in X$ . To see that $X$ is connected,                           |
| fong-spivak-invitation_F02909 | 272  | ■ 0.999 | $b, c \in X$                                   | $b, c \in X$                                   | suppose $b, c \in X$ ; then $b \sim a$ and $c \sim a$ so $b \sim c$ by the transitivity and symmetry of $\sim$ .                              |
| fong-spivak-invitation_F02910 | 272  | ■ 0.999 | $c \sim a$                                     | $c \sim a$                                     | suppose $b, c \in X$ ; then $b \sim a$ and $c \sim a$ so $b \sim c$ by the transitivity and symmetry of $\sim$ .                              |
| fong-spivak-invitation_F02911 | 272  | ■ 0.896 | $a \in X$                                      | $a \in X$                                      | Finally, $a \in X$ by the reflexivity of $\sim$ .                                                                                             |
| fong-spivak-invitation_F02912 | 272  | ■ 1.000 | $\varnothing \rightarrow \{1\}$                | $\emptyset \rightarrow \{1\}$                  | 1. The unique function $\emptyset \rightarrow \{1\}$ is injective but not surjective.                                                         |
| fong-spivak-invitation_F02913 | 272  | ■ 1.000 | $\{a, b\} \rightarrow \{1\}$                   | $\{a, b\} \rightarrow \{1\}$                   | 2. The unique function $\{a, b\} \rightarrow \{1\}$ is surjective but not injective.                                                          |
| fong-spivak-invitation_F02914 | 273  | ■ 1.000 | $F \subseteq A \times \varnothing$             | $F \subseteq A \times \emptyset$               | By Definition 1.17, a function $f: A \rightarrow \emptyset$ is a subset $F \subseteq A \times \emptyset$ such that, for all $a \in A$ , there |
| fong-spivak-invitation_F02915 | 273  | ■ 0.862 | $b \in \varnothing$                            | $b \in \emptyset$                              | exists a unique $b \in \emptyset$ with $(a, b) \in F$ . But there are no elements $b \in \emptyset$ , so if $F$ is to have the                |
| fong-spivak-invitation_F02916 | 273  | ■ 0.862 | $(a, b) \in F$                                 | $(a, b) \in F$                                 | exists a unique $b \in \emptyset$ with $(a, b) \in F$ . But there are no elements $b \in \emptyset$ , so if $F$ is to have the                |
| fong-spivak-invitation_F02917 | 273  | ■ 0.862 | $b \in^? \varnothing$                          | $b \in^? \emptyset$                            | exists a unique $b \in \emptyset$ with $(a, b) \in F$ . But there are no elements $b \in \emptyset$ , so if $F$ is to have the                |
| fong-spivak-invitation_F02918 | 273  | ■ 0.999 | $\{\bullet, *, \circ\}$                        | $\{\bullet, *, \circ\}$                        | Below each partition, we draw a corresponding surjection out of $\{\bullet, *, \circ\}$ :                                                     |
| fong-spivak-invitation_F02919 | 273  | ■ 0.962 | $1 \rightarrow 3$                              | $1 \rightarrow 3$                              | The graph $G$ from Exercise 1.33 is a strange Hasse diagram because it has two arrows $1 \rightarrow 3$ and                                   |
| fong-spivak-invitation_F02920 | 273  | ■ 0.582 | $P := V = \{1, 2, 3, 4\}$                      | $P := V = \{1, 2, 3, 4\}$                      | formula from working. The preorder $(P, \leq)$ is given by taking $P := V = \{1, 2, 3, 4\}$ and writing                                       |
| fong-spivak-invitation_F02921 | 274  | ■ 1.000 | $(\bullet)(\circ)(*) \leq (\bullet)(\circ)(*)$ | $(\bullet)(\circ)(*) \leq (\bullet)(\circ)(*)$ | $(\bullet)(\circ)(*) < (\bullet)(\circ)(*)$ ,                                                                                                 |
| fong-spivak-invitation_F02922 | 274  | ■ 1.000 | $(\bullet)(\circ)(*) \leq (\bullet\circ)(*)$   | $(\bullet)(\circ)(*) \leq (\bullet\circ)(*)$   | $(\bullet)(\circ)(*) < (\bullet\circ)(*)$ ,                                                                                                   |
| fong-spivak-invitation_F02923 | 274  | ■ 0.982 | $(\bullet)(\circ)(*) \leq (\bullet*)(\circ)$   | $(\bullet)(\circ)(*) \leq (\bullet*)(\circ)$   | $(\bullet)(\circ)(*) < (\bullet*)(\circ)$ ,                                                                                                   |
| fong-spivak-invitation_F02924 | 274  | ■ 0.950 | $(\bullet)(\circ)(*) \leq (\bullet)(\circ*)$   | $(\bullet)(\circ)(*) \leq (\bullet)(\circ*)$   | $(\bullet)(\circ)(*) < (\bullet)(\circ*)$ ,                                                                                                   |
| fong-spivak-invitation_F02925 | 274  | ■ 0.651 | $(\bullet)(\circ)(*) \leq (\bullet\circ*)$     | $(\bullet)(\circ)(*) \leq (\bullet\circ*)$     | $(\bullet)(\circ)(*) < (\bullet\circ*)$ ,                                                                                                     |
| fong-spivak-invitation_F02926 | 274  | ■ 1.000 | $(\bullet\circ)(*) \leq (\bullet\circ)(*)$     | $(\bullet\circ)(*) \leq (\bullet\circ)(*)$     | $(\bullet\circ)(*) < (\bullet\circ)(*)$ ,                                                                                                     |
| fong-spivak-invitation_F02927 | 274  | ■ 1.000 | $(\bullet\circ)(*) \leq (\bullet\circ*)$       | $(\bullet\circ)(*) \leq (\bullet\circ*)$       | $(\bullet\circ)(*) < (\bullet\circ*)$ ,                                                                                                       |
| fong-spivak-invitation_F02928 | 274  | ■ 1.000 | $(\bullet*)(\circ) \leq (\bullet*)(\circ)$     | $(\bullet*)(\circ) \leq (\bullet*)(\circ)$     | $(\bullet*)(\circ) < (\bullet*)(\circ)$ ,                                                                                                     |
| fong-spivak-invitation_F02929 | 274  | ■ 1.000 | $(\bullet*)(\circ) \leq (\bullet\circ*)$       | $(\bullet*)(\circ) \leq (\bullet\circ*)$       | $(\bullet*)(\circ) < (\bullet\circ*)$ ,                                                                                                       |
| fong-spivak-invitation_F02930 | 274  | ■ 0.999 | $(\bullet)(\circ*) \leq (\bullet)(\circ*)$     | $(\bullet)(\circ*) \leq (\bullet)(\circ*)$     | $(\bullet)(\circ*) < (\bullet)(\circ*)$ ,                                                                                                     |
| fong-spivak-invitation_F02931 | 274  | ■ 0.896 | $(\bullet)(\circ*) \leq (\bullet\circ*)$       | $(\bullet)(\circ*) \leq (\bullet\circ*)$       | $(\bullet)(\circ*) < (\bullet\circ*)$ ,                                                                                                       |
| fong-spivak-invitation_F02932 | 274  | ■ 0.860 | $(\bullet\circ*) \leq (\bullet\circ*)$         | $(\bullet\circ*) \leq (\bullet\circ*)$         | $(\bullet\circ*) < (\bullet\circ*)$ .                                                                                                         |
| fong-spivak-invitation_F02933 | 274  | ■ 1.000 | $4 \not\leq 6$                                 | $4 \not\leq 6$                                 | No, it is not a total order; for example $4 \not\leq 6$ and $6 \not\leq 4$ .                                                                  |
| fong-spivak-invitation_F02934 | 274  | ■ 1.000 | $6 \not\leq 4$                                 | $6 \not\leq 4$                                 | No, it is not a total order; for example $4 \not\leq 6$ and $6 \not\leq 4$ .                                                                  |

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| fong-spivak-invitation_F02935 | 274  | ■ 1.000 | $a, b \in \mathbb{R}$                               | $a, b \in \mathbb{R}$                               | Yes, the usual $\leq$ ordering is a total order on $\mathbb{R}$ : for every $a, b \in \mathbb{R}$ either $a \leq b$ or $b \leq a$ .                       |
| fong-spivak-invitation_F02936 | 274  | ■ 1.000 | $!: S \rightarrow \{1\}$                            | $!: S \rightarrow \{1\}$                            | The coarsest partition on $S$ corresponds to the unique function $!: S \rightarrow \{1\}$ . The finest partition on                                       |
| fong-spivak-invitation_F02937 | 274  | ■ 0.953 | $\mathrm{U}(X)$                                     | $\mathrm{U}(X)$                                     | $q$ such that $p \leq q$ is $p$ itself, so $q$ is definitely in $U$ ! This means that $\mathrm{U}(X)$ contains all subsets of                             |
| fong-spivak-invitation_F02938 | 274  | ■ 1.000 | $\mathrm{U}(X) = \mathrm{P}(X)$                     | $\mathrm{U}(X) = \mathrm{P}(X)$                     | $X$ , so it's <b>exactly the power</b> set, $\mathrm{U}(X) = \mathrm{P}(X)$ .                                                                             |
| fong-spivak-invitation_F02939 | 275  | ■ 1.000 | $0 \leq \cdots \leq 3$                              | $0 \leq \cdots \leq 3$                              | With $X = \{0, 1, 2\}$ , the Hasse diagram for $\mathrm{P}(X)$ , the preorder $0 \leq \cdots \leq 3$ , and the cardinality                                |
| fong-spivak-invitation_F02940 | 275  | ■ 1.000 | $q \in \uparrow p$                                  | $q \in \uparrow p$                                  | 1. Let $q \in \uparrow p$ , and suppose $q \leq q'$ . Since $q \in \uparrow p$ , we have $p \leq q$ . Thus by transitivity $p \leq q'$ ,                  |
| fong-spivak-invitation_F02941 | 275  | ■ 1.000 | $p \leq q^{\prime}$                                 | $p \leq q'$                                         | 1. Let $q \in \uparrow p$ , and suppose $q \leq q'$ . Since $q \in \uparrow p$ , we have $p \leq q$ . Thus by transitivity $p \leq q'$ ,                  |
| fong-spivak-invitation_F02942 | 275  | ■ 0.835 | $q^{\prime} \in \uparrow p$                         | $q' \in \uparrow p$                                 | so $q' \in \uparrow p$ . Thus $\uparrow p$ is an upper set.                                                                                               |
| fong-spivak-invitation_F02943 | 275  | ■ 0.835 | $\uparrow p$                                        | $\uparrow p$                                        | so $q' \in \uparrow p$ . Thus $\uparrow p$ is an upper set.                                                                                               |
| fong-spivak-invitation_F02944 | 275  | ■ 0.966 | $q \leq^{\mathrm{op}} p$                            | $q \leq^{\mathrm{op}} p$                            | 2. Suppose $p \leq q$ in $P$ ; this means that $q \leq^{\mathrm{op}} p$ in $P^{\mathrm{op}}$ . We must show that $\uparrow q \subseteq \uparrow p$ . Take |
| fong-spivak-invitation_F02945 | 275  | ■ 0.966 | $P^{\mathrm{op}}$                                   | $P^{\mathrm{op}}$                                   | 2. Suppose $p \leq q$ in $P$ ; this means that $q \leq^{\mathrm{op}} p$ in $P^{\mathrm{op}}$ . We must show that $\uparrow q \subseteq \uparrow p$ . Take |
| fong-spivak-invitation_F02946 | 275  | ■ 0.966 | $\uparrow q \subseteq \uparrow p$                   | $\uparrow q \subseteq \uparrow p$                   | 2. Suppose $p \leq q$ in $P$ ; this means that $q \leq^{\mathrm{op}} p$ in $P^{\mathrm{op}}$ . We must show that $\uparrow q \subseteq \uparrow p$ . Take |
| fong-spivak-invitation_F02947 | 275  | ■ 1.000 | $q^{\prime} \in \uparrow q$                         | $q' \in \uparrow q$                                 | any $q' \in \uparrow q$ . Then $q \leq q'$ , so by transitivity $p \leq q'$ , and hence $q' \in \uparrow p$ . Thus $\uparrow q \subseteq \uparrow p$ .    |
| fong-spivak-invitation_F02948 | 275  | ■ 1.000 | $p \not\leq p^{\prime}$                             | $p \not\leq p'$                                     | that if $p \not\leq p'$ then $\uparrow(p') \not\subseteq \uparrow(p)$ . This is straightforward, since by reflexivity we always have                      |
| fong-spivak-invitation_F02949 | 275  | ■ 1.000 | $\uparrow(p') \not\subseteq \uparrow(p)$            | $\uparrow(p') \not\subseteq \uparrow(p)$            | that if $p \not\leq p'$ then $\uparrow(p') \not\subseteq \uparrow(p)$ . This is straightforward, since by reflexivity we always have                      |
| fong-spivak-invitation_F02950 | 275  | ■ 1.000 | $p' \in \uparrow(p')$                               | $p' \in \uparrow(p')$                               | $p' \in \uparrow(p')$ , but if $p \not\leq p'$ , then $p' \notin \uparrow(p)$ , so $\uparrow(p') \not\subseteq \uparrow(p)$ .                             |
| fong-spivak-invitation_F02951 | 275  | ■ 1.000 | $p' \notin \uparrow(p)$                             | $p' \notin \uparrow(p)$                             | $p' \in \uparrow(p')$ , but if $p \not\leq p'$ , then $p' \notin \uparrow(p)$ , so $\uparrow(p') \not\subseteq \uparrow(p)$ .                             |
| fong-spivak-invitation_F02952 | 276  | ■ 0.506 | $P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$         | $P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$         | 4. The map $\uparrow: P^{\mathrm{op}} \rightarrow \mathrm{U}(P)$ can be depicted:                                                                         |
| fong-spivak-invitation_F02953 | 276  | ■ 1.000 | $p_1 \leq_P p_2$                                    | $p_1 \leq_P p_2$                                    | every function $f: P \rightarrow Q$ is monotone, i.e. that if $p_1 \leq_P p_2$ then $f(p_1) \leq_Q f(p_2)$ . But in                                       |
| fong-spivak-invitation_F02954 | 276  | ■ 1.000 | $f(p_1) \leq_Q f(p_2)$                              | $f(p_1) \leq_Q f(p_2)$                              | every function $f: P \rightarrow Q$ is monotone, i.e. that if $p_1 \leq_P p_2$ then <b><math>f(p_1) \leq_Q f(p_2)</math></b> . But in                     |
| fong-spivak-invitation_F02955 | 276  | ■ 1.000 | $f(p_1) = f(p_2)$                                   | $f(p_1) = f(p_2)$                                   | $f(p_1) = f(p_2)$ , so $f(p_1) \leq f(p_2)$ .                                                                                                             |
| fong-spivak-invitation_F02956 | 276  | ■ 1.000 | $f(p_1) \leq f(p_2)$                                | $f(p_1) \leq f(p_2)$                                | $f(p_1) = f(p_2)$ , so $f(p_1) \leq f(p_2)$ .                                                                                                             |
| fong-spivak-invitation_F02957 | 276  | ■ 0.999 | $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$ | $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$ | Let $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$ be the set of all integers, and let $Y = \{n, z, p\}$ ; let $f: X \rightarrow$                    |
| fong-spivak-invitation_F02958 | 276  | ■ 0.999 | $Y = \{n, z, p\}$                                   | $Y = \{n, z, p\}$                                   | Let $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$ be the set of all integers, and let $Y = \{n, z, p\}$ ; let $f: X \rightarrow$                    |
| fong-spivak-invitation_F02959 | 276  | ■ 0.999 | $f: X \rightarrow$                                  | $f: X \rightarrow$                                  | Let $X = \mathbb{Z} = \{\ldots, -2, -1, 0, 1, \ldots\}$ be the set of all integers, and let $Y = \{n, z, p\}$ ; let $f: X \rightarrow$                    |
| fong-spivak-invitation_F02960 | 276  | ■ 1.000 | $P := (nz)(p)$                                      | $P := (nz)(p)$                                      | We consider two partitions of $Y$ , namely $P := (nz)(p)$ and $Q := (np)(z)$ . Technically, these are                                                     |
| fong-spivak-invitation_F02961 | 276  | ■ 1.000 | $Q := (np)(z)$                                      | $Q := (np)(z)$                                      | We consider two partitions of $Y$ , namely $P := (nz)(p)$ and $Q := (np)(z)$ . Technically, these are                                                     |
| fong-spivak-invitation_F02962 | 276  | ■ 1.000 | $\{\{n, z\}, \{p\}\}$                               | $\{\{n, z\}, \{p\}\}$                               | notation for $\{\{n, z\}, \{p\}\}$ and $\{\{n, p\}, \{z\}\}$ as sets of disjoint subsets whose union is $Y$ . Their pulled                                |

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| fong-spivak-invitation_F02963 | 276  | ■ 1.000 | $\{\{n, p\}, \{z\}\}$                                                    | $\{\{n, p\}, \{z\}\}$                                                    | notation for $\{\{n, z\}, \{p\}\}$ and $\{\{n, p\}, \{z\}\}$ as sets of disjoint subsets whose union is $Y$ . Their pulled                              |
| fong-spivak-invitation_F02964 | 276  | ■ 1.000 | $f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$                               | $f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$                               | back partitions are $f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$ and $f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$ , or                                      |
| fong-spivak-invitation_F02965 | 276  | ■ 1.000 | $f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$                               | $f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$                               | back partitions are $f^*P = (\ldots, -2, -1, 0)(1, 2, \ldots)$ and $f^*Q = (0)(\ldots, -2, -1, 1, 2, \ldots)$ , or                                      |
| fong-spivak-invitation_F02966 | 276  | ■ 1.000 | $\left(P, \leq_P\right), \left(Q, \leq_Q\right)$                         | $(P, \leq_P), (Q, \leq_Q)$                                               | We have preorders $(P, \leq_P)$ , $(Q, \leq_Q)$ , and $(R, \leq_R)$ , and we have monotone maps $f: P \rightarrow Q$                                    |
| fong-spivak-invitation_F02967 | 276  | ■ 0.812 | $\mathrm{id}_P$                                                          | $\mathrm{id}_P$                                                          | 1. To see that $\mathrm{id}_P$ is monotone, we need to show that if $p_1 \leq_P p_2$ then $\mathrm{id}_P(p_1) \leq \mathrm{id}_P(p_2)$ . But            |
| fong-spivak-invitation_F02968 | 276  | ■ 0.812 | $\operatorname{id}_P(p_1) \leq \operatorname{id}_P(p_2)$                 | $\mathrm{id}_P(p_1) \leq \mathrm{id}_P(p_2)$                             | 1. To see that $\mathrm{id}_P$ is monotone, we need to show that if $p_1 \leq_P p_2$ then $\mathrm{id}_P(p_1) \leq \mathrm{id}_P(p_2)$ . But            |
| fong-spivak-invitation_F02969 | 276  | ■ 1.000 | $\operatorname{id}_P(p) = p$                                             | $\mathrm{id}_P(p) = p$                                                   | $\mathrm{id}_P(p) = p$ for all $p \in P$ , so this is clear.                                                                                            |
| fong-spivak-invitation_F02970 | 276  | ■ 0.988 | $q_1 \leq_Q q_2$                                                         | $q_1 \leq_Q q_2$                                                         | 2. We have that $p_1 \leq_P p_2$ implies $f(p_1) \leq_Q f(p_2)$ and that $q_1 \leq_Q q_2$ implies $g(q_1) \leq_R$                                       |
| fong-spivak-invitation_F02971 | 276  | ■ 0.988 | $g(q_1) \leq_R$                                                          | $g(q_1) \leq_R$                                                          | 2. We have that $p_1 \leq_P p_2$ implies $f(p_1) \leq_Q f(p_2)$ and that $q_1 \leq_Q q_2$ implies $g(q_1) \leq_R$                                       |
| fong-spivak-invitation_F02972 | 276  | ■ 1.000 | $g(q_2)$                                                                 | $g(q_2)$                                                                 | $g(q_2)$ . By substitution, $p_1 \leq_P p_2$ implies $g(f(p_1)) \leq_R g(f(p_2))$ which is exactly what is                                              |
| fong-spivak-invitation_F02973 | 276  | ■ 1.000 | $g(f(p_1)) \leq_R g(f(p_2))$                                             | $g(f(p_1)) \leq_R g(f(p_2))$                                             | $g(q_2)$ . By substitution, $p_1 \leq_P p_2$ implies $g(f(p_1)) \leq_R g(f(p_2))$ which is exactly what is                                              |
| fong-spivak-invitation_F02974 | 276  | ■ 0.679 | $(f; g)$                                                                 | $(f; g)$                                                                 | required for $(f \circ g)$ to be monotone.                                                                                                              |
| fong-spivak-invitation_F02975 | 276  | ■ 0.998 | $p_2 \leq p_1$                                                           | $p_2 \leq p_1$                                                           | skeletal, i.e. $p_1 \leq p_2$ and $p_2 \leq p_1$ implies $p_1 = p_2$ . And suppose it is dagger, i.e. $p_1 \leq p_2$                                    |
| fong-spivak-invitation_F02976 | 277  | ■ 1.000 | $\Phi: \operatorname{Prt}(\{\bullet, *, \circ\}) \rightarrow \mathbb{B}$ | $\Phi: \operatorname{Prt}(\{\bullet, *, \circ\}) \rightarrow \mathbb{B}$ | $\Phi: \operatorname{Prt}(\{\bullet, *, \circ\}) \rightarrow \mathbb{B}$ . So suppose $P, Q$ are partitions with $P \leq Q$ ; we need to show that if   |
| fong-spivak-invitation_F02977 | 277  | ■ 0.994 | $\Phi(P) =$                                                              | $\Phi(P) =$                                                              | $\Phi(P) = \text{true}$ then $\Phi(Q) = \text{true}$ .                                                                                                  |
| fong-spivak-invitation_F02978 | 277  | ■ 0.994 | $\Phi(Q) =$                                                              | $\Phi(Q) =$                                                              | $\Phi(P) = \text{true}$ then $\Phi(Q) = \text{true}$ .                                                                                                  |
| fong-spivak-invitation_F02979 | 277  | ■ 1.000 | $x \sim_P y$                                                             | $x \sim_P y$                                                             | lumps more stuff together. Technically, $x \sim_P y$ implies $x \sim_Q y$ for all $x, y \in \{\bullet, *, \circ\}$ . Applying                           |
| fong-spivak-invitation_F02980 | 277  | ■ 1.000 | $x \sim_Q y$                                                             | $x \sim_Q y$                                                             | lumps more stuff together. Technically, $x \sim_P y$ implies $x \sim_Q y$ for all $x, y \in \{\bullet, *, \circ\}$ . Applying                           |
| fong-spivak-invitation_F02981 | 277  | ■ 1.000 | $x, y \in \{\bullet, *, \circ\}$                                         | $x, y \in \{\bullet, *, \circ\}$                                         | lumps more stuff together. Technically, $x \sim_P y$ implies $x \sim_Q y$ for all $x, y \in \{\bullet, *, \circ\}$ . Applying                           |
| fong-spivak-invitation_F02982 | 277  | ■ 0.980 | $U \mapsto f^{-1}(U)$                                                    | $U \mapsto f^{-1}(U)$                                                    | Given a function $f: P \rightarrow Q$ , we have $f^*: \mathcal{U}(Q) \rightarrow \mathcal{U}(P)$ given by $U \mapsto f^{-1}(U)$ . But upper             |
| fong-spivak-invitation_F02983 | 277  | ■ 1.000 | $u: Q \rightarrow \mathbb{B}$                                            | $u: Q \rightarrow \mathbb{B}$                                            | sets in $Q$ are classified by monotone maps $u: Q \rightarrow \mathbb{B}$ , and similarly for $P$ ; our job is to show that                             |
| fong-spivak-invitation_F02984 | 277  | ■ 1.000 | $f^*(U)$                                                                 | $f^*(U)$                                                                 | $f^*(U)$ is given by composing the classifier $u$ with $f$ .                                                                                            |
| fong-spivak-invitation_F02985 | 277  | ■ 0.904 | $q \mapsto$                                                              | $q \mapsto$                                                              | $q \mapsto \text{true}$ iff $q \in U$ . Then $(f \circ u): P \rightarrow \mathbb{B}$ sends $p \mapsto \text{true}$ iff $f(p) \in U$ ; it corresponds to |
| fong-spivak-invitation_F02986 | 277  | ■ 0.904 | $p \mapsto$                                                              | $p \mapsto$                                                              | $q \mapsto \text{true}$ iff $q \in U$ . Then $(f \circ u): P \rightarrow \mathbb{B}$ sends $p \mapsto \text{true}$ iff $f(p) \in U$ ; it corresponds to |
| fong-spivak-invitation_F02987 | 277  | ■ 0.904 | $f(p) \in U$                                                             | $f(p) \in U$                                                             | $q \mapsto \text{true}$ iff $q \in U$ . Then $(f \circ u): P \rightarrow \mathbb{B}$ sends $p \mapsto \text{true}$ iff $f(p) \in U$ ; it corresponds to |
| fong-spivak-invitation_F02988 | 277  | ■ 1.000 | $\{p \in P \mid f(p) \in U\}$                                            | $\{p \in P \mid f(p) \in U\}$                                            | the upper set $\{p \in P \mid f(p) \in U\}$ which is exactly $f^{-1}(U)$ .                                                                              |
| fong-spivak-invitation_F02989 | 277  | ■ 1.000 | $S = \left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\}$               | $S = \left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\}$               | 1. 0 is a lower bound for $S = \{\frac{1}{n+1} \mid n \in \mathbb{N}\}$ because $0 \leq \frac{1}{n+1}$ for any $n \in \mathbb{N}$ .                     |
| fong-spivak-invitation_F02990 | 277  | ■ 1.000 | $0 \leq \frac{1}{n+1}$                                                   | $0 \leq \frac{1}{n+1}$                                                   | 1. 0 is a lower bound for $S = \{\frac{1}{n+1} \mid n \in \mathbb{N}\}$ because $0 \leq \frac{1}{n+1}$ for any $n \in \mathbb{N}$ .                     |



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| fong-spivak-invitation_F03019 | 278  | ■ 1.000 | $f(p)=1$                                                                                            | $f(p) = 1$                            | When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$ , so both $f(p) = 1 \leq 1 = q$ and                                                                                                               |
| fong-spivak-invitation_F03020 | 278  | ■ 1.000 | $g(q)=2$                                                                                            | $g(q) = 2$                            | When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$ , so both $f(p) = 1 \leq 1 = q$ and                                                                                                               |
| fong-spivak-invitation_F03021 | 278  | ■ 1.000 | $f(p)=1 \leq 1=q$                                                                                   | $f(p) = 1 \leq 1 = q$                 | When $p = q = 1$ we have $f(p) = 1$ and $g(q) = 2$ , so both $f(p) = 1 \leq 1 = q$ and                                                                                                               |
| fong-spivak-invitation_F03022 | 278  | ■ 1.000 | $p=1 \leq g(q)$                                                                                     | $p = 1 \leq g(q)$                     | $p = 1 \leq g(q)$ ; it works! The same sort of story happens when $(p, q)$ is (1, 2), (1, 3), (2, 1),                                                                                                |
| fong-spivak-invitation_F03023 | 278  | ■ 1.000 | $(p, q)$                                                                                            | $(p, q)$                              | $p = 1 \leq g(q)$ ; it works! The same sort of story happens when $(p, q)$ is (1, 2), (1, 3), (2, 1),                                                                                                |
| fong-spivak-invitation_F03024 | 278  | ■ 1.000 | $(1,2),(1,3),(2,1)$                                                                                 | $(1,2),(1,3),(2,1)$                   | $p = 1 \leq g(q)$ ; it works! The same sort of story happens when $(p, q)$ is <span style="border: 1px solid red;">(1,2),(1,3),(2,1)</span> ,                                                        |
| fong-spivak-invitation_F03025 | 278  | ■ 0.991 | $p=3, q=1$                                                                                          | $p = 3, q = 1$                        | (2, 2), (2, 3), and (3, 3). A different story happens for $p = 3, q = 1$ and $p = 3, q = 2$ . In                                                                                                     |
| fong-spivak-invitation_F03026 | 278  | ■ 0.991 | $p=3, q=2$                                                                                          | $p = 3, q = 2$                        | (2, 2), (2, 3), and (3, 3). A different story happens for $p = 3, q = 1$ and $p = 3, q = 2$ . In                                                                                                     |
| fong-spivak-invitation_F03027 | 278  | ■ 1.000 | $f(p)=3$                                                                                            | $f(p) = 3$                            | those cases $f(p) = 3$ and $g(q) = 2$ , and neither inequality holds: $f(p) \not\leq q$ and $p \not\leq g(q)$ .                                                                                      |
| fong-spivak-invitation_F03028 | 278  | ■ 1.000 | $f(p) \not\leq q$                                                                                   | $f(p) \not\leq q$                     | those cases $f(p) = 3$ and $g(q) = 2$ , and neither inequality holds: $f(p) \not\leq q$ and $p \not\leq g(q)$ .                                                                                      |
| fong-spivak-invitation_F03029 | 278  | ■ 1.000 | $p \not\leq g(q)$                                                                                   | $p \not\leq g(q)$                     | those cases $f(p) = 3$ and $g(q) = 2$ , and neither inequality holds: $f(p) \not\leq q$ and $p \not\leq g(q)$ .                                                                                      |
| fong-spivak-invitation_F03030 | 278  | ■ 1.000 | $f(2) \not\leq 1$                                                                                   | $f(2) \not\leq 1$                     | Here $f$ is <i>not</i> left adjoint to $g$ because $f(2) \not\leq 1$ but $2 \leq g(1)$ .                                                                                                             |
| fong-spivak-invitation_F03031 | 278  | ■ 1.000 | $2 \leq g(1)$                                                                                       | $2 \leq g(1)$                         | Here $f$ is <i>not</i> left adjoint to $g$ because $f(2) \not\leq 1$ but $2 \leq g(1)$ .                                                                                                             |
| fong-spivak-invitation_F03032 | 278  | ■ 1.000 | $C(r):=\lceil r / 3\rceil$                                                                          | $C(r) := \lceil r/3 \rceil$           | happens. Writing <span style="border: 1px solid red;"><math>C(r) := \lceil r/3 \rceil</math></span> , then for all $z \in \mathbb{Z}$ and $r \in \mathbb{R}$ we have $L(z) \leq r$ iff $z \leq C(r)$ |
| fong-spivak-invitation_F03033 | 278  | ■ 1.000 | $L(z) \leq r$                                                                                       | $L(z) \leq r$                         | happens. Writing $C(r) := \lceil r/3 \rceil$ , then for all $z \in \mathbb{Z}$ and $r \in \mathbb{R}$ we have <span style="border: 1px solid red;"><math>L(z) \leq r</math></span> iff $z \leq C(r)$ |
| fong-spivak-invitation_F03034 | 278  | ■ 1.000 | $z \leq C(r)$                                                                                       | $z \leq C(r)$                         | happens. Writing $C(r) := \lceil r/3 \rceil$ , then for all $z \in \mathbb{Z}$ and $r \in \mathbb{R}$ we have $L(z) \leq r$ iff $z \leq C(r)$                                                        |
| fong-spivak-invitation_F03035 | 278  | ■ 1.000 | $z=1$                                                                                               | $z = 1$                               | by definition of adjunction. So take $z = 1$ and $r = .01$ ; then $\lceil r/3 \rceil = 1$ so $z \leq C(r)$ , and                                                                                     |
| fong-spivak-invitation_F03036 | 278  | ■ 1.000 | $r=.01$                                                                                             | $r = .01$                             | by definition of adjunction. So take $z = 1$ and $r = .01$ ; then $\lceil r/3 \rceil = 1$ so $z \leq C(r)$ , and                                                                                     |
| fong-spivak-invitation_F03037 | 278  | ■ 1.000 | $\lceil r / 3\rceil=1$                                                                              | $\lceil r/3 \rceil = 1$               | by definition of adjunction. So take $z = 1$ and $r = .01$ ; then $\lceil r/3 \rceil = 1$ so $z \leq C(r)$ , and                                                                                     |
| fong-spivak-invitation_F03038 | 278  | ■ 1.000 | $L(1) \leq 0.01$                                                                                    | $L(1) \leq 0.01$                      | hence $L(z) \leq r$ , i.e. $L(1) \leq 0.01$ . In the same way $L(1) \leq r$ for all $r > 0$ , so $L(1) \leq 0$ . By                                                                                  |
| fong-spivak-invitation_F03039 | 278  | ■ 1.000 | $L(1) \leq r$                                                                                       | $L(1) \leq r$                         | hence $L(z) \leq r$ , i.e. $L(1) \leq 0.01$ . In the same way $L(1) \leq r$ for all $r > 0$ , so $L(1) \leq 0$ . By                                                                                  |
| fong-spivak-invitation_F03040 | 278  | ■ 1.000 | $r>0$                                                                                               | $r > 0$                               | hence $L(z) \leq r$ , i.e. $L(1) \leq 0.01$ . In the same way $L(1) \leq r$ for all $r > 0$ , so $L(1) \leq 0$ . By                                                                                  |
| fong-spivak-invitation_F03041 | 278  | ■ 1.000 | $L(1) \leq 0$                                                                                       | $L(1) \leq 0$                         | hence $L(z) \leq r$ , i.e. $L(1) \leq 0.01$ . In the same way $L(1) \leq r$ for all $r > 0$ , so $L(1) \leq 0$ . By                                                                                  |
| fong-spivak-invitation_F03042 | 278  | ■ 1.000 | $1 \leq C(0)=\lceil 0 / 3\rceil=0$                                                                  | $1 \leq C(0) = \lceil 0/3 \rceil = 0$ | definition of adjunction <span style="border: 1px solid red;"><math>1 \leq C(0) = \lceil 0/3 \rceil = 0</math></span> , a contradiction.                                                             |
| fong-spivak-invitation_F03043 | 278  | ■ 1.000 | $c_{\{1\}}, c_{\{2\}}, c_{\{3\}}, c_{\{4\}}$                                                        | $c_1, c_2, c_3, c_4$                  | see Example 1.96. Take <span style="border: 1px solid red;"><math>c_1, c_2, c_3, c_4</math></span> to be the following partitions:                                                                   |
| fong-spivak-invitation_F03044 | 278  | ■ 0.997 | $g_{\{!\}\left(c_{\{1\}}\right)}, g_{\{!\}\left(c_{\{2\}}\right)}, g_{\{!\}\left(c_{\{3\}}\right)}$ | $g_{!}(c_1), g_{!}(c_2), g_{!}(c_3)$  | Then the induced partitions $g_{!}(c_1), g_{!}(c_2), g_{!}(c_3)$ , and $g_{!}(c_4)$ on $T$ are:                                                                                                      |
| fong-spivak-invitation_F03045 | 278  | ■ 0.997 | $g_{\{!\}\left(c_{\{4\}}\right)}$                                                                   | $g_{!}(c_4)$                          | Then the induced partitions $g_{!}(c_1), g_{!}(c_2), g_{!}(c_3)$ , and $g_{!}(c_4)$ on $T$ are:                                                                                                      |
| fong-spivak-invitation_F03046 | 279  | ■ 1.000 | $S=\{1,2,3,4\}$                                                                                     | $S = \{1, 2, 3, 4\}$                  | Here are the partitions on $S = \{1, 2, 3, 4\}$ induced via $g^*$ by the five partitions on $T =$                                                                                                    |

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| fong-spivak-invitation_F03047 | 279  | ■ 0.980 | $c \leq$                             | $c \leq$                             | 5. Comparing $c$ , the left-hand partition in part 1., with $g^*(d)$ and $g^*(e)$ , we indeed have $c \leq$                                  |
| fong-spivak-invitation_F03048 | 279  | ■ 1.000 | $f: P \rightarrowtail Q: g$          | $f: P \rightarrowtail Q: g$          | Suppose $P$ and $Q$ are preorders, and that $f: P \rightarrowtail Q: g$ are monotone maps.                                                   |
| fong-spivak-invitation_F03049 | 279  | ■ 1.000 | $g(q) \leq g(q)$                     | $g(q) \leq g(q)$                     | and $q \in Q$ . Then starting with the reflexivity fact $g(q) \leq g(q)$ , the definition with $p := g(q)$                                   |
| fong-spivak-invitation_F03050 | 279  | ■ 1.000 | $p := g(q)$                          | $p := g(q)$                          | and $q \in Q$ . Then starting with the reflexivity fact $g(q) \leq g(q)$ , the definition with $p := g(q)$                                   |
| fong-spivak-invitation_F03051 | 279  | ■ 1.000 | $f(p) \leq f(g(q))$                  | $f(p) \leq f(g(q))$                  | both sides, we have $f(p) \leq f(g(q))$ , and then by transitivity $f(g(q)) \leq q$ implies $f(p) \leq q$ ,                                  |
| fong-spivak-invitation_F03052 | 280  | ■ 1.000 | $g, g^{\prime}: Q \rightarrowtail P$ | $g, g^{\prime}: Q \rightarrowtail P$ | 1. Suppose that $f: P \rightarrowtail Q$ has two right adjoints, $g, g^{\prime}: Q \rightarrowtail P$ . We want to show that                 |
| fong-spivak-invitation_F03053 | 280  | ■ 1.000 | $g(q) \leq g^{\prime}(q)$            | $g(q) \leq g^{\prime}(q)$            | $g(q) \cong g^{\prime}(q)$ for all $q \in Q$ . We will prove $g(q) \leq g^{\prime}(q)$ ; the inequality $g^{\prime}(q) \leq g(q)$            |
| fong-spivak-invitation_F03054 | 280  | ■ 1.000 | $g^{\prime}(q) \leq g(q)$            | $g^{\prime}(q) \leq g(q)$            | $g(q) \cong g^{\prime}(q)$ for all $q \in Q$ . We will prove $g(q) \leq g^{\prime}(q)$ ; the inequality $g^{\prime}(q) \leq g(q)$            |
| fong-spivak-invitation_F03055 | 280  | ■ 1.000 | $p \leq g^{\prime}(f(p))$            | $p \leq g^{\prime}(f(p))$            | is similar. To do this, we use the fact that $p \leq g^{\prime}(f(p))$ and $f(g(q)) \leq q$ for all $p, q$ by                                |
| fong-spivak-invitation_F03056 | 280  | ■ 1.000 | $j := \bigvee A$                     | $j := \bigvee A$                     | Suppose $f: P \rightarrowtail Q$ is left adjoint to $g: Q \rightarrowtail P$ . Let $A \subseteq P$ be any subset and let $j := \bigvee A$ be |
| fong-spivak-invitation_F03057 | 280  | ■ 1.000 | $f(a) \leq f(j)$                     | $f(a) \leq f(j)$                     | its join. Then since $f$ is monotone $f(a) \leq f(j)$ for all $a \in A$ , so $f(j)$ is an upper bound for the                                |
| fong-spivak-invitation_F03058 | 280  | ■ 1.000 | $f(j)$                               | $f(j)$                               | its join. Then since $f$ is monotone $f(a) \leq f(j)$ for all $a \in A$ , so $f(j)$ is an upper bound for the                                |
| fong-spivak-invitation_F03059 | 280  | ■ 1.000 | $f(a) \leq b$                        | $f(a) \leq b$                        | $f(A)$ , meaning we have $f(a) \leq b$ for all $a \in A$ . Then by definition of adjunction, we also have                                    |
| fong-spivak-invitation_F03060 | 280  | ■ 1.000 | $a \leq g(b)$                        | $a \leq g(b)$                        | $a \leq g(b)$ for all $a \in A$ . By definition of join, we have $j \leq g(b)$ . Again by definition of adjunction                           |
| fong-spivak-invitation_F03061 | 280  | ■ 1.000 | $j \leq g(b)$                        | $j \leq g(b)$                        | $a \leq g(b)$ for all $a \in A$ . By definition of join, we have $j \leq g(b)$ . Again by definition of adjunction                           |
| fong-spivak-invitation_F03062 | 280  | ■ 1.000 | $f(j) \leq b$                        | $f(j) \leq b$                        | $f(j) \leq b$ , as desired.                                                                                                                  |
| fong-spivak-invitation_F03063 | 281  | ■ 1.000 | $f(p)$                               | $f(p)$                               | The former uses that $f(g(q)) \leq q$ , with $f(p)$ substituted for $q$ : this gives $f(g(f(p))) \leq f(p)$ ,                                |
| fong-spivak-invitation_F03066 | 281  | ■ 0.977 | $B_{\{1\}} := \{a, b\}$              | $B_1 := \{a, b\}$                    | 1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$ . Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$ .                                      |
| fong-spivak-invitation_F03067 | 281  | ■ 0.977 | $B_{\{2\}} := \{c\}$                 | $B_2 := \{c\}$                       | 1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$ . Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$ .                                      |
| fong-spivak-invitation_F03068 | 281  | ■ 0.977 | $f^*(B_{\{1\}}) = \{a_1\}$           | $f^*(B_1) = \{a_1\}$                 | 1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$ . Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$ .                                      |
| fong-spivak-invitation_F03069 | 281  | ■ 0.977 | $f^*(B_{\{2\}}) = \{c_1, c_2\}$      | $f^*(B_2) = \{c_1, c_2\}$            | 1. Let $B_1 := \{a, b\}$ and $B_2 := \{c\}$ . Then $f^*(B_1) = \{a_1\}$ and $f^*(B_2) = \{c_1, c_2\}$ .                                      |
| fong-spivak-invitation_F03070 | 281  | ■ 1.000 | $A_{\{1\}} := \varnothing$           | $A_1 := \varnothing$                 | 2. Let $A_1 := \varnothing$ and $A_2 := \{a_1, c_1\}$ . Then $f_!(A_1) = \varnothing$ and $f_!(A_2) = \{a, c\}$ .                            |
| fong-spivak-invitation_F03071 | 281  | ■ 1.000 | $A_{\{2\}} := \{a_1, c_1\}$          | $A_2 := \{a_1, c_1\}$                | 2. Let $A_1 := \varnothing$ and $A_2 := \{a_1, c_1\}$ . Then $f_!(A_1) = \varnothing$ and $f_!(A_2) = \{a, c\}$ .                            |
| fong-spivak-invitation_F03072 | 281  | ■ 1.000 | $f_!(A_{\{1\}}) = \varnothing$       | $f_!(A_1) = \varnothing$             | 2. Let $A_1 := \varnothing$ and $A_2 := \{a_1, c_1\}$ . Then $f_!(A_1) = \varnothing$ and $f_!(A_2) = \{a, c\}$ .                            |
| fong-spivak-invitation_F03073 | 281  | ■ 1.000 | $f_!(A_{\{2\}}) = \{a, c\}$          | $f_!(A_2) = \{a, c\}$                | 2. Let $A_1 := \varnothing$ and $A_2 := \{a_1, c_1\}$ . Then $f_!(A_1) = \varnothing$ and $f_!(A_2) = \{a, c\}$ .                            |
| fong-spivak-invitation_F03074 | 281  | ■ 0.999 | $A_{\{1\}}$                          | $A_1$                                | 3. With the same $A_1$ and $A_2$ , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$ .                                                 |
| fong-spivak-invitation_F03075 | 281  | ■ 0.999 | $A_{\{2\}}$                          | $A_2$                                | 3. With the same $A_1$ and $A_2$ , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$ .                                                 |
| fong-spivak-invitation_F03076 | 281  | ■ 0.999 | $f_*(A_{\{1\}}) = \{b\}$             | $f_*(A_1) = \{b\}$                   | 3. With the same $A_1$ and $A_2$ , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$ .                                                 |

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| fong-spivak-invitation_F03077 | 281  | ■ 0.999 | $f_{\{*\}}\left(A_{\{2\}}\right)=\{a, b\}$                         | $f_*(A_2) = \{a, b\}$                           | 3. With the same $A_1$ and $A_2$ , we compute $f_*(A_1) = \{b\}$ and $f_*(A_2) = \{a, b\}$ .                                                   |
| fong-spivak-invitation_F03078 | 281  | ■ 1.000 | $g(f(p))$                                                          | $g(f(p))$                                       | 1. By Proposition 1.101 we have that $p \leq g(f(p))$ , and of course $g(f(p))$ and $(f \circ g)(p)$ mean                                      |
| fong-spivak-invitation_F03079 | 281  | ■ 1.000 | $(f \ ; \ g)(p)$                                                   | $(f;g)(p)$                                      | 1. By Proposition 1.101 we have that $p \leq g(f(p))$ , and of course $g(f(p))$ and $(f \circ g)(p)$ mean                                      |
| fong-spivak-invitation_F03080 | 281  | ■ 1.000 | $g(f(g(f(p)))) \ \leq g(f(p))$                                     | $g(f(g(f(p)))) \leq g(f(p))$                    | 2. We want to show that $g(f(g(f(p)))) < g(f(p))$ and $g(f(p)) \leq g(f(g(f(p))))$ for all $p$ .                                               |
| fong-spivak-invitation_F03081 | 281  | ■ 1.000 | $g(f(p)) \ \leq g(f(g(f(p))))$                                     | $g(f(p)) \leq g(f(g(f(p))))$                    | 2. We want to show that $g(f(g(f(p)))) \leq g(f(p))$ and $g(f(p)) \leq g(f(g(f(p))))$ for all $p$ .                                            |
| fong-spivak-invitation_F03082 | 281  | ■ 1.000 | $p^{\{\prime\}} \ \leq g\left(f\left(p^{\{\prime\}}\right)\right)$ | $p' \leq g(f(p'))$                              | The latter is just the fact that $p' \leq g(f(p'))$ for any $p'$ , applied with $g(f(p))$ in place of $p'$ .                                   |
| fong-spivak-invitation_F03083 | 281  | ■ 1.000 | $p^{\{\prime\}}$                                                   | $p'$                                            | The latter is just the fact that $p' \leq g(f(p'))$ for any $p'$ , applied with $g(f(p))$ in place of $p'$ .                                   |
| fong-spivak-invitation_F03084 | 281  | ■ 1.000 | $f(g(f(p))) \ \leq f(p)$                                           | $f(g(f(p))) \leq f(p)$                          | The former uses that $f(g(q)) \leq q$ , with $f(p)$ substituted for $q$ : this gives $f(g(f(p))) < f(p)$ ,                                     |
| fong-spivak-invitation_F03085 | 281  | ■ 0.657 | $a \ b$                                                            | $ab$                                            | We denote tuples $(a, b)$ by $ab$ for space reasons. So the relation $\{(1, 1), (1, 2), (2, 1)\}$ will be                                      |
| fong-spivak-invitation_F03086 | 281  | ■ 0.657 | $\{(1,1),(1,2),(2,1)\}$                                            | $\{(1,1),(1,2),(2,1)\}$                         | We denote tuples $(a, b)$ by $ab$ for space reasons. So the relation $\{(1,1),(1,2),(2,1)\}$ will be                                           |
| fong-spivak-invitation_F03087 | 281  | ■ 1.000 | $S:=\{1,2,3\}$                                                     | $S := \{1, 2, 3\}$                              | Let $S := \{1, 2, 3\}$ .                                                                                                                       |
| fong-spivak-invitation_F03088 | 281  | ■ 0.997 | $1 \ \leq 1, 2 \ \leq 2$                                           | $1 \leq 1, 2 \leq 2$                            | 1. Let $\leq$ be the preorder with $1 \leq 2$ , and of course $1 \leq 1, 2 \leq 2$ , and $3 \leq 3$ . Then $U(\leq) =$                         |
| fong-spivak-invitation_F03089 | 281  | ■ 0.997 | $3 \ \leq 3$                                                       | $3 \leq 3$                                      | 1. Let $\leq$ be the preorder with $1 \leq 2$ , and of course $1 \leq 1, 2 \leq 2$ , and $3 \leq 3$ . Then $U(\leq) =$                         |
| fong-spivak-invitation_F03090 | 281  | ■ 0.997 | $U(\leq)=$                                                         | $U(\leq) =$                                     | 1. Let $\leq$ be the preorder with $1 \leq 2$ , and of course $1 \leq 1, 2 \leq 2$ , and $3 \leq 3$ . Then $U(\leq) =$                         |
| fong-spivak-invitation_F03091 | 281  | ■ 0.876 | $\{(1,1),(1,2),(2,2),(3,3)\}$                                      | $\{(1,1),(1,2),(2,2),(3,3)\}$                   | $\{(1,1),(1,2),(2,2),(3,3)\}$                                                                                                                  |
| fong-spivak-invitation_F03092 | 281  | ■ 1.000 | $Q:=\{(1,1)\}$                                                     | $Q := \{(1,1)\}$                                | 2. Let $Q := \{(1,1)\}$ and $Q' := \{(2,1)\}$ .                                                                                                |
| fong-spivak-invitation_F03093 | 281  | ■ 1.000 | $Q^{\{\prime\}}:=\{(2,1)\}$                                        | $Q' := \{(2,1)\}$                               | 2. Let $Q := \{(1,1)\}$ and $Q' := \{(2,1)\}$ .                                                                                                |
| fong-spivak-invitation_F03094 | 282  | ■ 0.812 | $\mathrm{Cl}(Q)$                                                   | $\mathrm{Cl}(Q)$                                | 3. The closure $\mathrm{Cl}(Q)$ of $Q$ is the smallest preorder containing $(1,1)$ , which is $\mathrm{Cl}(Q) =$                               |
| fong-spivak-invitation_F03095 | 282  | ■ 0.812 | $\mathrm{Cl}(Q)=$                                                  | $\mathrm{Cl}(Q) =$                              | 3. The closure $\mathrm{Cl}(Q)$ of $Q$ is the smallest preorder containing $(1,1)$ , which is $\mathrm{Cl}(Q) =$                               |
| fong-spivak-invitation_F03096 | 282  | ■ 1.000 | $\{(1,1),(2,2),(3,3)\}$                                            | $\{(1,1),(2,2),(3,3)\}$                         | $\{(1,1),(2,2),(3,3)\}$ . Similarly, $\mathrm{Cl}(Q') = \{(1,1),(2,1),(2,2),(3,3)\}$ . It is easy to see that                                  |
| fong-spivak-invitation_F03097 | 282  | ■ 1.000 | $\mathrm{Cl}(Q^{\{\prime\}})=\{(1,1),(2,1),(2,2),(3,3)\}$          | $\mathrm{Cl}(Q') = \{(1,1),(2,1),(2,2),(3,3)\}$ | $\{(1,1),(2,2),(3,3)\}$ . Similarly, $\mathrm{Cl}(Q') = \{(1,1),(2,1),(2,2),(3,3)\}$ . It is easy to see that                                  |
| fong-spivak-invitation_F03098 | 282  | ■ 0.995 | $(2,1)$                                                            | $(2,1)$                                         | 4. It is easy to see that $\mathrm{Cl}(Q') \not\subseteq \leq$ because the ordered pair $(2,1)$ is in $\mathrm{Cl}(Q')$ but is not in $\leq$ . |
| fong-spivak-invitation_F03099 | 282  | ■ 0.995 | $\mathrm{Cl}(Q^{\{\prime\}})$                                      | $\mathrm{Cl}(Q')$                               | 4. It is easy to see that $\mathrm{Cl}(Q') \not\subseteq \leq$ because the ordered pair $(2,1)$ is in $\mathrm{Cl}(Q')$ but is not in $\leq$ . |
| fong-spivak-invitation_F03100 | 282  | ■ 1.000 | $x_{\{1\}}=-1, \ x_{\{2\}}=0, \ y_{\{1\}}=-1$                      | $x_1 = -1, x_2 = 0, y_1 = -1$                   | The expert is right! The proposal violates property (a) when $x_1 = -1, x_2 = 0, y_1 = -1$ , and                                               |
| fong-spivak-invitation_F03101 | 282  | ■ 1.000 | $y_{\{2\}}=1$                                                      | $y_2 = 1$                                       | $y_2 = 1$ . Indeed $-1 \leq -1$ and $0 \leq 1$ , but $-1 * 0 = 0 \not\leq -1 = -1 * 1$ .                                                       |
| fong-spivak-invitation_F03102 | 282  | ■ 1.000 | $-1 \ \leq -1$                                                     | $-1 \leq -1$                                    | $y_2 = 1$ . Indeed $-1 \leq -1$ and $0 \leq 1$ , but $-1 * 0 = 0 \not\leq -1 = -1 * 1$ .                                                       |
| fong-spivak-invitation_F03103 | 282  | ■ 1.000 | $-1 \ * \ 0=0 \ \not \leq -1=-1 \ * \ 1$                           | $-1 * 0 = 0 \not\leq -1 = -1 * 1$               | $y_2 = 1$ . Indeed $-1 \leq -1$ and $0 \leq 1$ , but $-1 * 0 = 0 \not\leq -1 = -1 * 1$ .                                                       |
| fong-spivak-invitation_F03104 | 282  | ■ 0.988 | $(\operatorname{Disc}(M), =, *, e)$                                | $(\mathrm{Disc}(M), =, *, e)$                   | To check that $(\mathbf{Disc}(M), =, *, e)$ is a symmetric monoidal preorder, we need to check our pro-                                        |
| fong-spivak-invitation_F03105 | 282  | ■ 1.000 | $x_{\{1\}} \ \otimes x_{\{2\}}=x_{\{1\}} \ \otimes x_{\{2\}}$      | $x_1 \otimes x_2 = x_1 \otimes x_2$             | $x_1 \otimes x_2 = x_1 \otimes x_2$ , conditions (b) and (c) are precisely the equations Eq. (2.2), and (d) is the                             |

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| fong-spivak-invitation_F03106 | 282  | ■ 1.000 | $u \leq u, v \leq v$                              | $u \leq u, v \leq v$                        | 2. We use reflexivity when we assert that $u \leq u, v \leq v$ and $z \leq z$ , and use transitivity to assert                                                    |
| fong-spivak-invitation_F03107 | 282  | ■ 1.000 | $z \leq z$                                        | $z \leq z$                                  | 2. We use reflexivity when we assert that $u \leq u, v \leq v$ and $z \leq z$ , and use transitivity to assert                                                    |
| fong-spivak-invitation_F03108 | 282  | ■ 1.000 | $t+u \leq y+z$                                    | $t + u \leq y + z$                          | that the above sequence of inequalities implies the single inequality $t+u \leq y+z$ .                                                                            |
| fong-spivak-invitation_F03109 | 282  | ■ 1.000 | $z \rightarrow w$                                 | $z \rightarrow w$                           | Condition (a), monotonicity, says that if $x \rightarrow y$ and $z \rightarrow w$ are reactions, then $x + z \rightarrow y + w$ is a                              |
| fong-spivak-invitation_F03110 | 282  | ■ 1.000 | $x+z \rightarrow y+w$                             | $x + z \rightarrow y + w$                   | Condition (a), monotonicity, says that if $x \rightarrow y$ and $z \rightarrow w$ are reactions, then $x + z \rightarrow y + w$ is a                              |
| fong-spivak-invitation_F03111 | 283  | ■ 0.954 | $1 \mid 1$                                        | $1 \mid 1$                                  | This proposed monoidal product is not monotonic: we have $1 1$ and $1 2$ , but $(1 + 1) \nmid (1 + 2)$ .                                                          |
| fong-spivak-invitation_F03112 | 283  | ■ 0.954 | $1 \mid 2$                                        | $1 \mid 2$                                  | This proposed monoidal product is not monotonic: we have $1 1$ and $1 2$ , but $(1 + 1) \nmid (1 + 2)$ .                                                          |
| fong-spivak-invitation_F03113 | 283  | ■ 0.954 | $(1+1) \nmid (1+2)$                               | $(1 + 1) \nmid (1 + 2)$                     | This proposed monoidal product is not monotonic: we have $1 1$ and $1 2$ , but $(1 + 1) \nmid (1 + 2)$ .                                                          |
| fong-spivak-invitation_F03114 | 283  | ■ 0.964 | $z \leq w$                                        | $z \leq w$                                  | 2. We need to show that (a) if $x \leq y$ and $z \leq w$ , then $\min(x, z) \leq \min(y, w)$ , (b) $\min(x, \text{yes}) =$                                        |
| fong-spivak-invitation_F03115 | 283  | ■ 0.964 | $\min(x, z) \leq \min(y, w)$                      | $\min(x, z) \leq \min(y, w)$                | 2. We need to show that (a) if $x \leq y$ and $z \leq w$ , then $\min(x, z) \leq \min(y, w)$ , (b) $\min(x, \text{yes}) =$                                        |
| fong-spivak-invitation_F03116 | 283  | ■ 0.964 | $\min(x$                                          | $\min(x$                                    | 2. We need to show that (a) if $x \leq y$ and $z \leq w$ , then $\min(x, z) \leq \min(y, w)$ , (b) $\min(x, \text{yes}) =$                                        |
| fong-spivak-invitation_F03117 | 283  | ■ 0.766 | $x=\min(y \text{ e } s, x)$                       | $x = \min(\text{yes}, x)$                   | $x = \min(\text{yes}, x)$ , (c) $\min(\min(x, y), z) = \min(x, \min(y, z))$ , and (d) $\min(x, y) = \min(y, x)$ .                                                 |
| fong-spivak-invitation_F03118 | 283  | ■ 0.766 | $\min(\min(x, y), z)=\min(x, \min(y, z))$         | $\min(\min(x, y), z) = \min(x, \min(y, z))$ | $x = \min(\text{yes}, x)$ , (c) $\min(\min(x, y), z) = \min(x, \min(y, z))$ , and (d) $\min(x, y) = \min(y, x)$ .                                                 |
| fong-spivak-invitation_F03119 | 283  | ■ 0.766 | $\min(x, y)=\min(y, x)$                           | $\min(x, y) = \min(y, x)$                   | $x = \min(\text{yes}, x)$ , (c) $\min(\min(x, y), z) = \min(x, \min(y, z))$ , and (d) $\min(x, y) = \min(y, x)$ .                                                 |
| fong-spivak-invitation_F03120 | 283  | ■ 0.984 | $(\mathrm{P}(S), \leq, S, \cap)$                  | $(P(S), \leq, S, \cap)$                     | Yes, $(P(S), \leq, S, \cap)$ is a symmetric monoidal preorder.                                                                                                    |
| fong-spivak-invitation_F03121 | 283  | ■ 1.000 | $(P \wedge Q)(n)$                                 | $(P \wedge Q)(n)$                           | statements $P$ and $Q$ and makes the statement $P \wedge Q$ , where $(P \wedge Q)(n)$ is true if $P(n)$ and                                                       |
| fong-spivak-invitation_F03122 | 283  | ■ 0.601 | $\left(\mathrm{Prop}^{\mathbb{N}}, \leq, \right.$ | $(\mathrm{Prop}^{\mathbb{N}}, \leq$         | $Q(n)$ are true, and false otherwise. Then $(\mathrm{Prop}^{\mathbb{N}}, \leq, \text{true}, \wedge)$ forms a symmetric monoidal                                   |
| fong-spivak-invitation_F03123 | 283  | ■ 0.601 | $\left.\wedge\right)$                             | $\wedge)$                                   | $Q(n)$ are true, and false otherwise. Then $(\mathrm{Prop}^{\mathbb{N}}, \leq, \text{true}, \wedge)$ forms a symmetric monoidal                                   |
| fong-spivak-invitation_F03124 | 283  | ■ 0.997 | $n+10 \leq 1$                                     | $n + 10 \leq 1$                             | bers, such as “ $n + 10 \leq 1$ ” or “ $n$ is made of cheese.” We can also define $\vee$ such that $(P \vee Q)(n)$                                                |
| fong-spivak-invitation_F03125 | 283  | ■ 0.997 | $(P \vee Q)(n)$                                   | $(P \vee Q)(n)$                             | bers, such as “ $n + 10 \leq 1$ ” or “ $n$ is made of cheese.” We can also define $\vee$ such that $(P \vee Q)(n)$                                                |
| fong-spivak-invitation_F03126 | 283  | ■ 0.769 | $\left.\vee\right)$                               | $\vee)$                                     | is true if and only if at least one of $P(n)$ and $Q(n)$ is true. Then $(\mathrm{Prop}^{\mathbb{N}}, \leq, \text{false}, \vee)$ forms a                           |
| fong-spivak-invitation_F03127 | 283  | ■ 0.827 | $I \otimes x = x =$                               | $I \otimes x = x =$                         | These conditions have nothing to do with the order in $\mathcal{X}^{\mathrm{op}}$ : they simply state that $I \otimes x = x =$                                    |
| fong-spivak-invitation_F03128 | 283  | ■ 1.000 | $x \otimes I$                                     | $x \otimes I$                               | $x \otimes I$ , and $(x \otimes y) \otimes z = x \otimes (y \otimes z)$ , and $x \otimes y = y \otimes x$ . Since these are true in $\mathcal{X}$ , they are true |
| fong-spivak-invitation_F03129 | 283  | ■ 0.727 | $X^{\mathrm{op}}$                                 | $X^{\mathrm{op}}$                           | in $\mathcal{X}^{\mathrm{op}}$ .                                                                                                                                  |
| fong-spivak-invitation_F03130 | 283  | ■ 0.675 | $\mathrm{Cost}^{\mathrm{op}}$                     | $\mathrm{Cost}^{\mathrm{op}}$               | 1. The preorder $\mathbf{Cost}^{\mathrm{op}}$ has underlying set $[0, \infty]$ , and the usual increasing order on real numbers                                   |
| fong-spivak-invitation_F03131 | 284  | ■ 1.000 | $=\infty \geq 0=g$                                | $= \infty \geq 0 = g$                       | 1. The map $g$ is monotonic as $g(\text{false}) = \infty \geq 0 = g(\text{true})$ , satisfies condition (a) since                                                 |
| fong-spivak-invitation_F03132 | 284  | ■ 1.000 | $0 \geq 0=g$                                      | $0 \geq 0 = g$                              | $0 \geq 0 = g(\text{true})$ , and satisfies condition (b) since                                                                                                   |
| fong-spivak-invitation_F03133 | 284  | ■ 0.990 | $f(n)=1$                                          | $f(n) = 1$                                  | 2. The map $f(n) = 1$ for all $n \in \mathbb{N}$ defines a monoidal monotone $f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$ .                 |

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| fong-spivak-invitation_F03134 | 284  | ■ 0.990 | $f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$     | $f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$     | 2. The map $f(n) = 1$ for all $n \in \mathbb{N}$ defines a monoidal monotone $f: (\mathbb{N}, \leq, 0, +) \rightarrow (\mathbb{N}, \leq, 1, *)$ . |
| fong-spivak-invitation_F03135 | 284  | ■ 0.276 | $\mathbb{Z}, \leq, *, 1$                                               | $\mathbb{Z}, \leq, *, 1$                                               | 3. $(\mathbb{Z}, \leq, *, 1)$ is not a monoidal preorder because $*$ is not monotone. Indeed $-1 \leq 0$ but $(-1 * 1) \not\leq 1$ .              |
| fong-spivak-invitation_F03136 | 284  | ■ 0.276 | $\leq 0$                                                               | $\leq 0$                                                               | 3. $(\mathbb{Z}, \leq, *, 1)$ is not a monoidal preorder because $*$ is not monotone. Indeed $-1 \leq 0$ but $(-1 * 1) \not\leq 1$ .              |
| fong-spivak-invitation_F03137 | 284  | ■ 1.000 | $-1 \not\leq (0 * 0)$                                                  | $-1 \not\leq (0 * 0)$                                                  | $-1 \not\leq (0 * 0)$ .                                                                                                                           |
| fong-spivak-invitation_F03138 | 284  | ■ 0.382 | $\mathcal{X}_P$                                                        | $\mathcal{X}_P$                                                        | struct a <b>Bool</b> -category $\mathcal{X}_P$ with $P$ as its set of objects, and with $\mathcal{X}_P(p, q) := \text{true}$ if $p \leq q$ ,      |
| fong-spivak-invitation_F03139 | 284  | ■ 0.382 | $\mathcal{X}_P(p, q) := \text{true}$                                   | $\mathcal{X}_P(p, q) := \text{true}$                                   | struct a <b>Bool</b> -category $\mathcal{X}_P$ with $P$ as its set of objects, and with $\mathcal{X}_P(p, q) := \text{true}$ if $p \leq q$ ,      |
| fong-spivak-invitation_F03140 | 284  | ■ 0.579 | $\mathcal{X}_P(p, q) :=$                                               | $\mathcal{X}_P(p, q) :=$                                               | and $\mathcal{X}_P(p, q) := \text{false}$ otherwise. How do we turn this back into a preorder? Following the                                      |
| fong-spivak-invitation_F03141 | 284  | ■ 0.626 | $\text{Ob}(\mathcal{X}_P) = P$                                         | $\text{Ob}(\mathcal{X}_P) = P$                                         | proof of Theorem 2.32, we construct a preorder with underlying set $\text{Ob}(\mathcal{X}_P) = P$ , and with                                      |
| fong-spivak-invitation_F03142 | 284  | ■ 0.219 | $\mathcal{X}_P(p, q) = \text{true}$                                    | $\mathcal{X}_P(p, q) = \text{true}$                                    | $p \leq q$ if and only if $\mathcal{X}_P(p, q) = \text{true}$ . This is precisely the preorder $(P, \leq)$ !                                      |
| fong-spivak-invitation_F03143 | 284  | ■ 0.694 | $\text{Ob}(\mathcal{X})$                                               | $(\text{Ob}(\mathcal{X}), \leq)$                                       | 2. Let $\mathcal{X}$ be a <b>Bool</b> -category. By the proof of Theorem 2.32, we construct a preorder $(\text{Ob}(\mathcal{X}), \leq)$ ,         |
| fong-spivak-invitation_F03144 | 284  | ■ 0.897 | $\text{Ob}(X)$                                                         | $\text{Ob}(X)$                                                         | ple 2.31 in part 1, we construct a <b>Bool</b> -category $\mathcal{X}'$ whose set of objects is $\text{Ob}(\mathcal{X})$ , and such               |
| fong-spivak-invitation_F03145 | 284  | ■ 0.727 | $\mathcal{X}'(x, y) =$                                                 | $\mathcal{X}'(x, y) =$                                                 | that $\mathcal{X}'(x, y) = \text{true}$ if and only if $x \leq y$ in $(\text{Ob}(\mathcal{X}), \leq)$ . But, by construction, this means          |
| fong-spivak-invitation_F03146 | 284  | ■ 0.642 | $\mathcal{X}'(x, y) = \mathcal{X}(x, y)$                               | $\mathcal{X}'(x, y) = \mathcal{X}(x, y)$                               | $\mathcal{X}'(x, y) = \mathcal{X}(x, y)$ . So we get back the <b>Bool</b> -category we started with.                                              |
| fong-spivak-invitation_F03147 | 284  | ■ 0.492 | $d(\text{US})$                                                         | $d(\text{US})$                                                         | The distance $d(\text{US}, \text{Spain})$ is bigger: the distance from, for example, San Diego to anywhere in                                     |
| fong-spivak-invitation_F03148 | 284  | ■ 0.971 | $([0, \infty], \geq)$                                                  | $([0, \infty], \geq)$                                                  | The difference between a Lawvere metric space – that is, a category enriched over $[0, \infty]$ , $\geq$ ,                                        |
| fong-spivak-invitation_F03149 | 284  | ■ 0.558 | $0, +)$                                                                | $0, +)$                                                                | $0, +)$ – and a category enriched over $(\mathbb{R}_{\geq 0}, \geq, 0, +)$ is that, in the latter, infinite distances                             |
| fong-spivak-invitation_F03150 | 285  | ■ 0.575 | $\min(\mathcal{X}(x, y), \mathcal{X}(y, z)) \leq \mathcal{X}(x, z)$    | $\min(\mathcal{X}(x, y), \mathcal{X}(y, z)) \leq \mathcal{X}(x, z)$    | $\min(\mathcal{X}(x, y), \mathcal{X}(y, z)) \leq \mathcal{X}(x, z)$ for all $x, y, z$ . So a <b>NMY</b> -category can be thought of as set        |
| fong-spivak-invitation_F03151 | 285  | ■ 0.575 | $x, y, z$                                                              | $x, y, z$                                                              | $\min(\mathcal{X}(x, y), \mathcal{X}(y, z)) \leq \mathcal{X}(x, z)$ for all $x, y, z$ . So a <b>NMY</b> -category can be thought of as set        |
| fong-spivak-invitation_F03152 | 286  | ■ 0.684 | $\mathcal{X}(C, D)$                                                    | $\mathcal{X}(C, D)$                                                    | For example, to compute the hom-object $\mathcal{X}(C, D)$ , we notice that there are two paths: $C \rightarrow$                                  |
| fong-spivak-invitation_F03153 | 286  | ■ 0.684 | $C \rightarrow$                                                        | $C \rightarrow$                                                        | For example, to compute the hom-object $\mathcal{X}(C, D)$ , we notice that there are two paths: $C \rightarrow$                                  |
| fong-spivak-invitation_F03154 | 286  | ■ 1.000 | $A \rightarrow B \rightarrow D$                                        | $A \rightarrow B \rightarrow D$                                        | $A \rightarrow B \rightarrow D$ and $C \rightarrow D$ . For the first path, the intersection is the set $\{\text{boat}\}$ . For the second        |
| fong-spivak-invitation_F03155 | 286  | ■ 1.000 | $C \rightarrow D$                                                      | $C \rightarrow D$                                                      | $A \rightarrow B \rightarrow D$ and $C \rightarrow D$ . For the first path, the intersection is the set $\{\text{boat}\}$ . For the second        |
| fong-spivak-invitation_F03156 | 286  | ■ 0.515 | $\mathcal{X}(x, y) \cap \mathcal{X}(y, z) \subseteq \mathcal{X}(x, z)$ | $\mathcal{X}(x, y) \cap \mathcal{X}(y, z) \subseteq \mathcal{X}(x, z)$ | paths, we ensure that $\mathcal{X}(x, y) \cap \mathcal{X}(y, z) \subseteq \mathcal{X}(x, z)$ for all $x, y, z$ .                                  |
| fong-spivak-invitation_F03158 | 286  | ■ 1.000 | $\min(M(x, y), M(y, z)) \leq M(x, y)$                                  | $\min(M(x, y), M(y, z)) \leq M(x, y)$                                  | monoidal unit $\infty$ , and because $\min(M(x, y), M(y, z)) \leq M(x, y)$ for all $x, y, z \in \{A, B, C\}$ .                                    |
| fong-spivak-invitation_F03159 | 286  | ■ 1.000 | $x, y, z \in \{A, B, C\}$                                              | $x, y, z \in \{A, B, C\}$                                              | monoidal unit $\infty$ , and because $\min(M(x, y), M(y, z)) \leq M(x, y)$ for all $x, y, z \in \{A, B, C\}$ .                                    |
| fong-spivak-invitation_F03160 | 286  | ■ 0.507 | $d(A, B) = d(B, A) =$                                                  | $d(A, B) = d(B, A) =$                                                  | 2. Let $\mathcal{X}$ be the Lawvere metric space with two objects $A$ and $B$ , such that $d(A, B) = d(B, A) =$                                   |
| fong-spivak-invitation_F03161 | 286  | ■ 0.501 | $X_f = \mathcal{A} \bullet \mathcal{B}$                                | $X_f = \mathcal{A} \bullet \mathcal{B}$                                | 5. Then we have $X_f = \mathcal{A} \bullet \mathcal{B}$ while $X_g = \mathcal{A} \longrightarrow \mathcal{B}$ .                                   |
| fong-spivak-invitation_F03162 | 286  | ■ 0.501 | $X_g = \mathcal{A} \longrightarrow \mathcal{B}$                        | $X_g = \mathcal{A} \longrightarrow \mathcal{B}$                        | 5. Then we have $X_f = \mathcal{A} \bullet \mathcal{B}$ while $X_g = \mathcal{A} \longrightarrow \mathcal{B}$ .                                   |

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| fong-spivak-invitation_F03163 | 287  | ■ 1.000 | $d(x, y) \leq d(y, x)$                                                                             | $d(x, y) \leq d(y, x)$                                                                             | so for all $x, y$ we must have $d(x, y) \leq d(y, x)$ . But this means also that $d(y, x) \leq d(x, y)$ ,                                                                                     |
| fong-spivak-invitation_F03164 | 287  | ■ 1.000 | $d(y, x) \leq d(x, y)$                                                                             | $d(y, x) \leq d(x, y)$                                                                             | so for all $x, y$ we must have $d(x, y) \leq d(y, x)$ . But this means also that $d(y, x) \leq d(x, y)$ ,                                                                                     |
| fong-spivak-invitation_F03165 | 287  | ■ 1.000 | $d(y, x) =$                                                                                        | $d(y, x) =$                                                                                        | Now let's consider the skeletality condition. This says that if $d(x, y) = 0$ and $d(y, x) =$                                                                                                 |
| fong-spivak-invitation_F03166 | 287  | ■ 0.791 | $(x, y) \in \mathcal{X} \times \mathcal{Y}$                                                        | $(x, y) \in \mathcal{X} \times \mathcal{Y}$                                                        | 1. Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$ . Since $\mathcal{X}$ and $\mathcal{Y}$ are $\mathcal{V}$ -categories, we have $I \leq \mathcal{X}(x, x)$ and $I \leq \mathcal{Y}(y, y)$ . |
| fong-spivak-invitation_F03167 | 287  | ■ 0.791 | $I \leq y(y, y)$                                                                                   | $I \leq y(y, y)$                                                                                   | 1. Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$ . Since $\mathcal{X}$ and $\mathcal{Y}$ are $\mathcal{V}$ -categories, we have $I \leq \mathcal{X}(x, x)$ and $I \leq \mathcal{Y}(y, y)$ . |
| fong-spivak-invitation_F03168 | 287  | ■ 0.850 | $I = I \otimes I \leq X(x, x) \otimes y(y, y) = (\mathcal{X} \times \mathcal{Y})(x, y)$            | $I = I \otimes I \leq X(x, x) \otimes y(y, y) = (\mathcal{X} \times \mathcal{Y})(x, y)$            | Thus $I = I \otimes I \leq \mathcal{X}(x, x) \otimes \mathcal{Y}(y, y) = (\mathcal{X} \times \mathcal{Y})(x, y)$ .                                                                            |
| fong-spivak-invitation_F03169 | 287  | ■ 0.665 | $y(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) = \mathcal{X}(x_2, x_3) \otimes$                        | $y(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) = \mathcal{X}(x_2, x_3) \otimes$                        | 3. In particular, we use the symmetry to conclude that $\mathcal{Y}(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) = \mathcal{X}(x_2, x_3) \otimes$                                                  |
| fong-spivak-invitation_F03170 | 287  | ■ 0.998 | $y(y_1, y_2)$                                                                                      | $y(y_1, y_2)$                                                                                      | $\mathcal{Y}(y_1, y_2)$ .                                                                                                                                                                     |
| fong-spivak-invitation_F03171 | 287  | ■ 0.993 | $(\mathbb{R} \times \mathbb{R})(5, 6), (-1, 4) = \mathbb{R}(5, -1) + \mathbb{R}(6, 4) = 6 + 2 = 8$ | $(\mathbb{R} \times \mathbb{R})(5, 6), (-1, 4) = \mathbb{R}(5, -1) + \mathbb{R}(6, 4) = 6 + 2 = 8$ | We just apply Definition 2.53(ii): $(\mathbb{R} \times \mathbb{R})(5, 6), (-1, 4) = \mathbb{R}(5, -1) + \mathbb{R}(6, 4) = 6 + 2 = 8$ .                                                       |
| fong-spivak-invitation_F03172 | 287  | ■ 0.999 | $u \leq u'$                                                                                        | $u \leq u'$                                                                                        | 1. The function $- \otimes v: V \rightarrow V$ is monotone, because if $u \leq u'$ then $u \otimes v \leq u' \otimes v$ by the                                                                |
| fong-spivak-invitation_F03173 | 287  | ■ 0.999 | $u \otimes v \leq u' \otimes v$                                                                    | $u \otimes v \leq u' \otimes v$                                                                    | 1. The function $- \otimes v: V \rightarrow V$ is monotone, because if $u \leq u'$ then $u \otimes v \leq u' \otimes v$ by the                                                                |
| fong-spivak-invitation_F03174 | 287  | ■ 0.997 | $a := (v \multimap w)$                                                                             | $a := (v \multimap w)$                                                                             | 2. Let $a := (v \multimap w)$ in Eq. (2.23). The right-hand side is thus $(v \multimap w) \leq (v \multimap w)$ , which is                                                                    |
| fong-spivak-invitation_F03175 | 287  | ■ 0.975 | $(v \multimap u) \otimes v \leq u \leq u'$                                                         | $(v \multimap u) \otimes v \leq u \leq u'$                                                         | 3. Let $u \leq u'$ . Then, using part 2, $(v \multimap u) \otimes v \leq u \leq u'$ . Applying Eq. (2.23), we thus have                                                                       |
| fong-spivak-invitation_F03176 | 287  | ■ 1.000 | $(v \multimap u) \leq (v \multimap u')$                                                            | $(v \multimap u) \leq (v \multimap u')$                                                            | $(v \multimap u) \leq (v \multimap u')$ . This shows that the map $(v \multimap -): V \rightarrow V$ is monotone.                                                                             |
| fong-spivak-invitation_F03177 | 287  | ■ 1.000 | $(v \multimap -): V \rightarrow V$                                                                 | $(v \multimap -): V \rightarrow V$                                                                 | $(v \multimap u) \leq (v \multimap u')$ . This shows that the map $(v \multimap -): V \rightarrow V$ is monotone.                                                                             |
| fong-spivak-invitation_F03178 | 287  | ■ 0.998 | $(v \multimap -)$                                                                                  | $(v \multimap -)$                                                                                  | do not know $(- \otimes v)$ and $(v \multimap -)$ are monotone maps. We proved this, however, in parts 1                                                                                      |
| fong-spivak-invitation_F03179 | 288  | ■ 1.000 | $x \Rightarrow y$                                                                                  | $x \Rightarrow y$                                                                                  | We need to find the hom-element. This is given by implication. That is, define the function $x \Rightarrow y$                                                                                 |
| fong-spivak-invitation_F03180 | 288  | ■ 0.987 | $(a \wedge v) \leq w$                                                                              | $(a \wedge v) \leq w$                                                                              | Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$ . Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$ ,                                            |
| fong-spivak-invitation_F03181 | 288  | ■ 0.987 | $a \leq (v \Rightarrow w)$                                                                         | $a \leq (v \Rightarrow w)$                                                                         | Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$ . Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$ ,                                            |
| fong-spivak-invitation_F03182 | 288  | ■ 0.987 | $v =$                                                                                              | $v =$                                                                                              | Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$ . Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$ ,                                            |
| fong-spivak-invitation_F03183 | 288  | ■ 0.987 | $a \wedge$                                                                                         | $a \wedge$                                                                                         | Then $(a \wedge v) \leq w$ if and only if $a \leq (v \Rightarrow w)$ . Indeed, if $v = \text{false}$ then $a \wedge \text{false} = \text{false}$ ,                                            |
| fong-spivak-invitation_F03184 | 288  | ■ 0.570 | $\Rightarrow w$                                                                                    | $\Rightarrow w$                                                                                    | and so the left-hand side is always true. But $(\text{false} \Rightarrow w) = \text{true}$ , so the right-hand side                                                                           |
| fong-spivak-invitation_F03185 | 288  | ■ 0.672 | $v = \text{true}$                                                                                  | $v = \text{true}$                                                                                  | is always true too. If $v = \text{true}$ , then $a \wedge \text{true} = a$ so the left-hand side says $a \leq w$ . But                                                                        |
| fong-spivak-invitation_F03186 | 288  | ■ 0.672 | $= a$                                                                                              | $= a$                                                                                              | is always true too. If $v = \text{true}$ , then $a \wedge \text{true} = a$ so the left-hand side says $a \leq w$ . But                                                                        |
| fong-spivak-invitation_F03187 | 288  | ■ 0.672 | $a \leq w$                                                                                         | $a \leq w$                                                                                         | is always true too. If $v = \text{true}$ , then $a \wedge \text{true} = a$ so the left-hand side says $a \leq w$ . But                                                                        |
| fong-spivak-invitation_F03188 | 288  | ■ 0.969 | $\Rightarrow w) = w$                                                                               | $\Rightarrow w) = w$                                                                               | $(\text{true} \Rightarrow w) = w$ , so the right-hand side is the same. Thus $\Rightarrow$ defines a hom-element as per                                                                       |
| fong-spivak-invitation_F03189 | 288  | ■ 0.672 | $(\bigvee \varnothing) = \text{false}$                                                             | $(\bigvee \varnothing) = \text{false}$                                                             | 1a. In <b>Bool</b> , $(\bigvee \varnothing) = \text{false}$ , the least element.                                                                                                              |
| fong-spivak-invitation_F03190 | 288  | ■ 0.999 | $(\bigvee \varnothing) = \infty$                                                                   | $(\bigvee \varnothing) = \infty$                                                                   | 1b. In <b>Cost</b> , $(\bigvee \varnothing) = \infty$ . This is because we use the opposite order $\geq$ on $[0, \infty]$ , so $\bigvee \varnothing$ is the                                   |

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|-------------------------------|------|---------|-------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| fong-spivak-invitation_F03191 | 288  | ■ 1.000 | $(\bigvee \varnothing)=0$                 | $(\bigvee \varnothing) = 0$                 | we denote $(\bigvee \varnothing) = 0$ , is a bit confusing! Beware!                                                            |
| fong-spivak-invitation_F03192 | 288  | ■ 1.000 | $\min (x, y)$                             | $\min(x, y)$                                | 2b. In <b>Cost</b> , $x \vee y$ is the minimum $\min(x, y)$ . Again because we use the opposite order on $[0, \infty]$ ,       |
| fong-spivak-invitation_F03193 | 288  | ■ 0.764 | $(\mathrm{P}(S), \subseteq, S, \cap)$     | $(P(S), \subseteq, S, \cap)$                | Yes, the power-set monoidal preorder $(P(S), \subseteq, S, \cap)$ is a quantale. The hom-object $B \multimap C$ is             |
| fong-spivak-invitation_F03194 | 288  | ■ 0.764 | $B \multimap C$                           | $B \multimap C$                             | Yes, the power-set monoidal preorder $(P(S), \subseteq, S, \cap)$ is a quantale. The hom-object $B \multimap C$ is             |
| fong-spivak-invitation_F03195 | 288  | ■ 1.000 | $\bar{B} \cup C$                          | $\bar{B} \cup C$                            | given by $\bar{B} \cup C$ , where $\bar{B}$ is the <i>complement</i> of $B$ : it contains all elements of $S$ not contained in |
| fong-spivak-invitation_F03196 | 288  | ■ 1.000 | $\bar{B}$                                 | $\bar{B}$                                   | given by $\bar{B} \cup C$ , where $\bar{B}$ is the <i>complement</i> of $B$ : it contains all elements of $S$ not contained in |
| fong-spivak-invitation_F03197 | 288  | ■ 1.000 | $(A \cap B) \subseteq C$                  | $(A \cap B) \subseteq C$                    | $B$ . To see that this satisfies Eq. (2.23), note that if $(A \cap B) \subseteq C$ , then                                      |
| fong-spivak-invitation_F03198 | 288  | ■ 1.000 | $A \subseteq (\bar{B} \cup C)$            | $A \subseteq (\bar{B} \cup C)$              | On the other hand, if $A \subseteq (\bar{B} \cup C)$ , then                                                                    |
| fong-spivak-invitation_F03199 | 288  | ■ 1.000 | $v \in V, 0 \otimes v = 0$                | $v \in V, 0 \otimes v = 0$                  | 1. We first use Proposition 2.64 (2) and symmetry to show that, for all $v \in V, 0 \otimes v = 0$ .                           |
| fong-spivak-invitation_F03200 | 289  | ■ 0.998 | $\left(v, a_1, \ldots, a_n\right)$        | $(v, a_1, \ldots, a_n)$                     | define a path in $G$ to be a tuple of the form $(v, a_1, \ldots, a_n)$ where $v \in V$ is a vertex, $s(a_1) = v$ ,             |
| fong-spivak-invitation_F03201 | 289  | ■ 0.998 | $s\left(a_1\right)=v$                     | $s\left(a_1\right) = v$                     | define a path in $G$ to be a tuple of the form $(v, a_1, \ldots, a_n)$ where $v \in V$ is a vertex, $s(a_1) = v$ ,             |
| fong-spivak-invitation_F03202 | 290  | ■ 1.000 | $t\left(a_i\right)=s\left(a_{i+1}\right)$ | $t\left(a_i\right) = s\left(a_{i+1}\right)$ | and $t(a_i) = s(a_{i+1})$ for all $i \in \{1, \ldots, n-1\}$ ; the length of this path is $n$ , and this definition            |
| fong-spivak-invitation_F03203 | 290  | ■ 1.000 | $i \in\{1, \ldots, n-1\}$                 | $i \in \{1, \ldots, n-1\}$                  | and $t(a_i) = s(a_{i+1})$ for all $i \in \{1, \ldots, n-1\}$ ; the length of this path is $n$ , and this definition            |
| fong-spivak-invitation_F03204 | 290  | ■ 1.000 | $s(p):=v$                                 | $s(p) := v$                                 | makes sense for any $n \in \mathbb{N}$ . We say that the source of $p$ is $s(p) := v$ and the target of $p$ is defined         |
| fong-spivak-invitation_F03205 | 290  | ■ 0.998 | $p=\left(v, a_1, \ldots, a_m\right)$      | $p = (v, a_1, \ldots, a_m)$                 | Two paths $p = (v, a_1, \ldots, a_m)$ and $q = (w, b_1, \ldots, b_n)$ can be concatenated if $t(p) = s(q)$ , in                |
| fong-spivak-invitation_F03206 | 290  | ■ 0.998 | $q=\left(w, b_1, \ldots, b_n\right)$      | $q = (w, b_1, \ldots, b_n)$                 | Two paths $p = (v, a_1, \ldots, a_m)$ and $q = (w, b_1, \ldots, b_n)$ can be concatenated if $t(p) = s(q)$ , in                |
| fong-spivak-invitation_F03207 | 290  | ■ 0.998 | $t(p)=s(q)$                               | $t(p) = s(q)$                               | Two paths $p = (v, a_1, \ldots, a_m)$ and $q = (w, b_1, \ldots, b_n)$ can be concatenated if $t(p) = s(q)$ , in                |
| fong-spivak-invitation_F03208 | 290  | ■ 0.991 | $p=(v)$                                   | $p = (v)$                                   | $p = (v)$ for some $v \in V$ . It is easy to see from the above that $(v)$ and $(w, b_1, \ldots, b_n)$ can be                  |
| fong-spivak-invitation_F03209 | 290  | ■ 0.991 | $(v)$                                     | $(v)$                                       | $p = (v)$ for some $v \in V$ . It is easy to see from the above that $(v)$ and $(w, b_1, \ldots, b_n)$ can be                  |
| fong-spivak-invitation_F03210 | 290  | ■ 0.991 | $\left(w, b_1, \ldots, b_n\right)$        | $(w, b_1, \ldots, b_n)$                     | $p = (v)$ for some $v \in V$ . It is easy to see from the above that $(v)$ and $(w, b_1, \ldots, b_n)$ can be                  |
| fong-spivak-invitation_F03211 | 290  | ■ 1.000 | $v=w$                                     | $v = w$                                     | concatenated iff $v = w$ , in which case the result is $(w, b_1, \ldots, b_n)$ . Similarly $(v, a_1, \ldots, a_m)$             |
| fong-spivak-invitation_F03212 | 290  | ■ 1.000 | $\left(v, a_1, \ldots, a_m\right)$        | $(v, a_1, \ldots, a_m)$                     | concatenated iff $v = w$ , in which case the result is $(w, b_1, \ldots, b_n)$ . Similarly $(v, a_1, \ldots, a_m)$             |
| fong-spivak-invitation_F03213 | 290  | ■ 0.896 | $(w)$                                     | $(w)$                                       | and $(w)$ can be concatenated iff $w = t(a_m)$ , in which case the result is $(v, a_1, \ldots, a_m)$ . Finally,                |
| fong-spivak-invitation_F03214 | 290  | ■ 0.896 | $w=t\left(a_m\right)$                     | $w = t\left(a_m\right)$                     | and $(w)$ can be concatenated iff $w = t(a_m)$ , in which case the result is $(v, a_1, \ldots, a_m)$ . Finally,                |
| fong-spivak-invitation_F03215 | 290  | ■ 0.940 | $r=\left(x, c_1, \ldots, c_o\right)$      | $r = (x, c_1, \ldots, c_o)$                 | for associativity with $p$ and $q$ as above and $r = (x, c_1, \ldots, c_o)$ , the formula readily reads that                   |
| fong-spivak-invitation_F03216 | 290  | ■ 0.379 | $(p ; q)\{ \}_9^{\circ} r$                | $(p; q)_9^{\circ} r$                        | whichever way they are concatenated, $(p \circ q) \circ r$ or $p \circ (q \circ r)$ , the result is                            |
| fong-spivak-invitation_F03217 | 290  | ■ 0.379 | $p \circ (q ; r)$                         | $p \circ (q; r)$                            | whichever way they are concatenated, $(p \circ q) \circ r$ or $p \circ (q \circ r)$ , the result is                            |
| fong-spivak-invitation_F03219 | 293  | ■ 1.000 | $f_{\{2\}}$                               | $f_2$                                       | The other way reverses $f_1$ and $f_2$ .                                                                                       |

| Identifier                    | Page | Conf.   | LaTeX source                                                        | Rendered                            | Image                                                                                                                                                               |
|-------------------------------|------|---------|---------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fong-spivak-invitation_F03223 | 290  | ■ 1.000 | $1+2+\cdots+n$                                                      | $1 + 2 + \cdots + n$                | is $\boxed{1+2+\cdots+n}$ . This makes sense because (and the proof strategy would be to verify that)                                                               |
| fong-spivak-invitation_F03224 | 290  | ■ 0.759 | $n-i$                                                               | $n - i$                             | the above graph has $n$ paths of length 0, it has $n - 1$ paths of length 1, and so on: it has $n - i$                                                              |
| fong-spivak-invitation_F03225 | 290  | ■ 1.000 | $0 \leq i \leq n$                                                   | $0 \leq i \leq n$                   | paths of length $i$ for every $0 \leq i \leq n$ .                                                                                                                   |
| fong-spivak-invitation_F03226 | 291  | ■ 1.000 | $m+n$                                                               | $m + n$                             | with a path of length $n$ results in a path of length $m + n$ .                                                                                                     |
| fong-spivak-invitation_F03227 | 291  | ■ 0.325 | $h=j=A$                                                             | $h = j = A$                         | Note that $A \circ f \circ \boxed{h = j = A} \circ g \circ i$ .                                                                                                     |
| fong-spivak-invitation_F03228 | 291  | ■ 0.312 | $z, s, s \cdot s$                                                   | $z, s, s.s$                         | There are four morphisms in $\mathcal{D}$ , shown below, namely $z, s, s \circ s$ , and $s \circ s \circ s$ :                                                       |
| fong-spivak-invitation_F03229 | 291  | ■ 0.312 | $s \cdot s \cdot s$                                                 | $s.s.s$                             | There are four morphisms in $\mathcal{D}$ , shown below, namely $z, s, s \circ s$ , and $s \circ s \circ s$ :                                                       |
| fong-spivak-invitation_F03230 | 292  | ■ 0.968 | $(f(1), f(2))$                                                      | $(f(1), f(2))$                      | A function $f: \underline{2} \rightarrow \underline{3}$ can be described as an ordered pair $(f(1), f(2))$ . The nine such functions                                |
| fong-spivak-invitation_F03231 | 292  | ■ 0.970 | $n!$                                                                | $n!$                                | the number of isomorphisms between them is $n$ -factorial, often denoted $n!$ . So for example                                                                      |
| fong-spivak-invitation_F03232 | 292  | ■ 0.998 | $5 * 4 * 3 * 2 * 1=120$                                             | $5 * 4 * 3 * 2 * 1 = 120$           | there are $\boxed{5 * 4 * 3 * 2 * 1 = 120}$ isomorphisms between $\{1, 2, 3, 4, 5\}$ and $\{a, b, c, d, e\}$ .                                                      |
| fong-spivak-invitation_F03233 | 292  | ■ 0.998 | $\{1,2,3,4,5\}$                                                     | $\{1, 2, 3, 4, 5\}$                 | there are $5 * 4 * 3 * 2 * 1 = 120$ isomorphisms between $\{1, 2, 3, 4, 5\}$ and $\{a, b, c, d, e\}$ .                                                              |
| fong-spivak-invitation_F03234 | 292  | ■ 0.998 | $\{a, b, c, d, e\}$                                                 | $\{a, b, c, d, e\}$                 | there are $5 * 4 * 3 * 2 * 1 = 120$ isomorphisms between $\{1, 2, 3, 4, 5\}$ and $\{a, b, c, d, e\}$ .                                                              |
| fong-spivak-invitation_F03235 | 292  | ■ 0.755 | $f: c \rightarrowtail c$                                            | $f: c \rightarrow c$                | $f: c \rightarrow c$ such that $f \circ \mathrm{id}_c = \mathrm{id}_c$ and $\mathrm{id}_c \circ f = \mathrm{id}_c$ . Take $f = \mathrm{id}_c$ ; this works.         |
| fong-spivak-invitation_F03236 | 292  | ■ 0.755 | $f; \mathrm{id}_c = \mathrm{id}_c$                                  | $f; \mathrm{id}_c = \mathrm{id}_c$  | $f: c \rightarrow c$ such that $f \circ \mathrm{id}_c = \mathrm{id}_c$ and $\mathrm{id}_c \circ f = \mathrm{id}_c$ . Take $f = \mathrm{id}_c$ ; this works.         |
| fong-spivak-invitation_F03237 | 292  | ■ 0.755 | $\mathrm{id}_c; f = \mathrm{id}_c$                                  | $\mathrm{id}_c; f = \mathrm{id}_c$  | $f: c \rightarrow c$ such that $f \circ \mathrm{id}_c = \mathrm{id}_c$ and $\mathrm{id}_c \circ f = \mathrm{id}_c$ . Take $f = \mathrm{id}_c$ ; this works.         |
| fong-spivak-invitation_F03238 | 292  | ■ 0.755 | $f = \mathrm{id}_c$                                                 | $f = \mathrm{id}_c$                 | $f: c \rightarrow c$ such that $f \circ \mathrm{id}_c = \mathrm{id}_c$ and $\mathrm{id}_c \circ f = \mathrm{id}_c$ . Take $f = \mathrm{id}_c$ ; this works.         |
| fong-spivak-invitation_F03239 | 292  | ■ 1.000 | $s^{\{n\}}$                                                         | $s^n$                               | each morphism is of the form $s^n$ for some $n \in \mathbb{N}$ and composing it with $s$ gives $s^{n+1}$ , which                                                    |
| fong-spivak-invitation_F03240 | 292  | ■ 1.000 | $s^{\{n+1\}}$                                                       | $s^{n+1}$                           | each morphism is of the form $s^n$ for some $n \in \mathbb{N}$ and composing it with $s$ gives $s^{n+1}$ , which                                                    |
| fong-spivak-invitation_F03241 | 292  | ■ 1.000 | $s^{\{\emptyset\}}$                                                 | $s^0$                               | is never $s^0$ .                                                                                                                                                    |
| fong-spivak-invitation_F03242 | 292  | ■ 0.513 | $p \circ \mathrm{id}_9 = \mathrm{id}$                               | $p_9 q = \mathrm{id}$               | whose length is 0. In order for $p$ to be an isomorphism, there must be some $q$ such that $p \circ q = \mathrm{id}$                                                |
| fong-spivak-invitation_F03243 | 292  | ■ 0.384 | $q \circ \mathrm{id}_9^{\circ} = \mathrm{id}$                       | $q_9^{\circ} p = \mathrm{id}$       | and $q \circ p = \mathrm{id}$ , in which case the length of $p$ (or $q$ ) must be 0.                                                                                |
| fong-spivak-invitation_F03244 | 292  | ■ 0.917 | $\mathbf{2} \rightarrowtail \mathbf{3}$                             | $\mathbf{2} \rightarrow \mathbf{3}$ | The other three functors $\mathbf{2} \rightarrow \mathbf{3}$ are shown here:                                                                                        |
| fong-spivak-invitation_F03245 | 293  | ■ 0.979 | $g^{\prime}; i^{\prime} \mapsto f; h$                               | $g'; i' \mapsto f; h$               | If one of these seems different from the rest, it's probably $g' \circ i' \mapsto f \circ h$ . But note that in fact                                                |
| fong-spivak-invitation_F03246 | 293  | ■ 0.317 | $g^{\prime} \circ i^{\prime} \mapsto g \circ i$                     | $g'_9 i' \mapsto g_9 i$             | also $g' \circ i' \mapsto g \circ i$ because $g \circ i = f \circ h$ , so it's not an outlier after all.                                                            |
| fong-spivak-invitation_F03247 | 293  | ■ 0.317 | $g \circ i^{\circ} = f \circ h$                                     | $g_9 i = f_9^{\circ} h$             | also $g' \circ i' \mapsto g \circ i$ because $g \circ i = f \circ h$ , so it's not an outlier after all.                                                            |
| fong-spivak-invitation_F03248 | 293  | ■ 0.421 | $F, G$                                                              | $F, G$                              | We need to give two functors $F, G$ from $\mathbf{a} \xrightarrow{f} \mathbf{b}$ to $\mathbf{a}' \xrightarrow[f_2]{f_1} \mathbf{b}'$ whose on-objects parts are the |
| fong-spivak-invitation_F03249 | 293  | ■ 0.421 | $\stackrel{\mathrm{a}}{\rightarrowtail} \rightarrowtail \mathrm{b}$ | $\xrightarrow{a} \xrightarrow{b}$   | We need to give two functors $F, G$ from $\mathbf{a} \xrightarrow{f} \mathbf{b}$ to $\mathbf{a}' \xrightarrow[f_2]{f_1} \mathbf{b}'$ whose on-objects parts are the |

| Identifier                    | Page | Conf.   | LaTeX source                                                                                                                                                            | Rendered                                                                                                                                                                | Image                                                                                                                                                                                                                                                 |
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| fong-spivak-invitation_F03250 | 293  | ■ 0.421 | $\stackrel{a'}{\text{f}}_{\text{1}} \rightarrow \text{f}_{\text{2}}$<br>$\stackrel{b'}{\text{f}}_{\text{2}} \rightarrow \text{f}_{\text{1}}$                            | $\frac{a'}{f_1} \xrightarrow{b'} \frac{b'}{f_2}$                                                                                                                        | We need to give two functors $F, G$ from $\bullet \xrightarrow{f} \bullet$ to $\bullet \xrightarrow{\frac{f_1}{f_2}} \bullet$ whose on-objects parts are the                                                                                          |
| fong-spivak-invitation_F03251 | 293  | ■ 0.929 | $\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$                                                                                                  | $\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$                                                                                                  | 1. Let $\mathcal{C}$ be a category. Then defining $\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$ by $\operatorname{id}_{\mathcal{C}}(x) = x$ for every object and morphism                                                    |
| fong-spivak-invitation_F03252 | 293  | ■ 0.929 | $\operatorname{id}_{\mathcal{C}}(x) = x$                                                                                                                                | $\operatorname{id}_{\mathcal{C}}(x) = x$                                                                                                                                | 1. Let $\mathcal{C}$ be a category. Then defining $\operatorname{id}_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$ by $\operatorname{id}_{\mathcal{C}}(x) = x$ for every object and morphism                                                    |
| fong-spivak-invitation_F03253 | 293  | ■ 0.108 | $\operatorname{id}_{\mathcal{C}}(\operatorname{id}_{\mathcal{C}}) = \operatorname{id}_{\mathcal{C}} = \operatorname{id}_{\operatorname{id}_{\mathcal{C}}(\mathcal{C})}$ | $\operatorname{id}_{\mathcal{C}}(\operatorname{id}_{\mathcal{C}}) = \operatorname{id}_{\mathcal{C}} = \operatorname{id}_{\operatorname{id}_{\mathcal{C}}(\mathcal{C})}$ | in $\mathcal{C}$ is a functor because it preserves identities $\operatorname{id}_{\mathcal{C}}(\operatorname{id}_{\mathcal{C}}) = \operatorname{id}_{\mathcal{C}} = \operatorname{id}_{\operatorname{id}_{\mathcal{C}}(\mathcal{C})}$ for each object |
| fong-spivak-invitation_F03254 | 293  | ■ 0.166 | $\operatorname{id}_{\mathcal{C}}(f; g) = f; g = \operatorname{id}_{\mathcal{C}}(f)g \operatorname{id}_{\mathcal{U}(g)}$                                                 | $\operatorname{id}_{\mathcal{C}}(f; g) = f; g = \operatorname{id}_{\mathcal{C}}(f)g \operatorname{id}_{\mathcal{U}(g)}$                                                 | $c \in \operatorname{Ob}(\mathcal{C})$ , and it preserves composition $\operatorname{id}_{\mathcal{C}}(f \circ g) = \operatorname{id}_{\mathcal{C}}(f) \circ \operatorname{id}_{\mathcal{C}}(g) = f \circ g$ for each pair                            |
| fong-spivak-invitation_F03255 | 293  | ■ 0.468 | $F; G$                                                                                                                                                                  | $F; G$                                                                                                                                                                  | 2. Given functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{E}$ , we need to show that $F \circ G$ is a functor, i.e.                                                                                         |
| fong-spivak-invitation_F03256 | 293  | ■ 0.970 | $(F; G)(\operatorname{id}_{\mathcal{C}}) =$                                                                                                                             | $(F; G)(\operatorname{id}_{\mathcal{C}}) =$                                                                                                                             | that it preserves identities and compositions. If $c \in \mathcal{C}$ is an object then $(F \circ G)(\operatorname{id}_c) =$                                                                                                                          |
| fong-spivak-invitation_F03257 | 293  | ■ 0.452 | $G(F(\operatorname{id}_{\mathcal{C}})) = G(\operatorname{id}_{F(\mathcal{C})}) = \operatorname{id}_{G(F(\mathcal{C}))}$                                                 | $G(F(\operatorname{id}_{\mathcal{C}})) = G(\operatorname{id}_{F(\mathcal{C})}) = \operatorname{id}_{G(F(\mathcal{C}))}$                                                 | $G(F(\operatorname{id}_c)) = G(\operatorname{id}_{F(c)}) = \operatorname{id}_{G(F(c))}$ because $F$ and $G$ preserve identities. If $f, g$ are                                                                                                        |
| fong-spivak-invitation_F03258 | 293  | ■ 0.904 | $\operatorname{id}_{\mathcal{C}}$                                                                                                                                       | $\operatorname{id}_{\mathcal{C}}$                                                                                                                                       | functor and we pre-compose it as above with $\operatorname{id}_{\mathcal{C}}$ ; it is easy to see that the result will again be                                                                                                                       |
| fong-spivak-invitation_F03259 | 293  | ■ 0.987 | $\operatorname{id}_{\mathcal{D}}$                                                                                                                                       | $\operatorname{id}_{\mathcal{D}}$                                                                                                                                       | $F$ , and similarly if we post-compose $F$ with $\operatorname{id}_{\mathcal{D}}$ . This gives unitality, and associativity is just                                                                                                                   |
| fong-spivak-invitation_F03260 | 293  | ■ 1.000 | $H: \mathcal{E} \rightarrow \mathcal{F}$                                                                                                                                | $H: \mathcal{E} \rightarrow \mathcal{F}$                                                                                                                                | as easy, though more wordy. Given $F$ as above and $G: \mathcal{D} \rightarrow \mathcal{E}$ and $H: \mathcal{E} \rightarrow \mathcal{F}$ , we need to                                                                                                 |
| fong-spivak-invitation_F03261 | 293  | ■ 0.638 | $(F; G) \circ H = F; (G; H)$                                                                                                                                            | $(F; G) \circ H = F; (G; H)$                                                                                                                                            | show that $(F \circ G) \circ H = F \circ (G \circ H)$ . It's a simple application of the definition: for any $x \in \mathcal{C}$ ,                                                                                                                    |
| fong-spivak-invitation_F03262 | 293  | ■ 0.638 | $x \in \mathcal{C}$                                                                                                                                                     | $x \in \mathcal{C}$                                                                                                                                                     | show that $(F \circ G) \circ H = F \circ (G \circ H)$ . It's a simple application of the definition: for any $x \in \mathcal{C}$ ,                                                                                                                    |
| fong-spivak-invitation_F03263 | 293  | ■ 0.999 | $F_S(\operatorname{id}_1) = \operatorname{id}_S$                                                                                                                        | $F_S(\operatorname{id}_1) = \operatorname{id}_S$                                                                                                                        | Let $S \in \mathbf{Set}$ be a set. Define $F_S: \mathbf{1} \rightarrow \mathbf{Set}$ by $F_S(1) = S$ and $F_S(\operatorname{id}_1) = \operatorname{id}_S$ . With this definition,                                                                     |
| fong-spivak-invitation_F03264 | 293  | ■ 0.883 | $F_S$                                                                                                                                                                   | $F_S$                                                                                                                                                                   | $F_S$ preserves identities and compositions (the only compositions in $\mathbf{1}$ is the composite of the                                                                                                                                            |
| fong-spivak-invitation_F03265 | 293  | ■ 0.883 | $\mathbf{1}$                                                                                                                                                            | $\mathbf{1}$                                                                                                                                                            | $F_S$ preserves identities and compositions (the only compositions in $\mathbf{1}$ is the composite of the                                                                                                                                            |
| fong-spivak-invitation_F03266 | 294  | ■ 1.000 | $D(z)$                                                                                                                                                                  | $D(z)$                                                                                                                                                                  | 1. Data on this schema, i.e. a set-valued functor, assigns a set $D(z)$ and a function $D(s): D(z) \rightarrow$                                                                                                                                       |
| fong-spivak-invitation_F03267 | 294  | ■ 1.000 | $D(s): D(z) \rightarrow$                                                                                                                                                | $D(s): D(z) \rightarrow$                                                                                                                                                | 1. Data on this schema, i.e. a set-valued functor, assigns a set $D(z)$ and a function $D(s): D(z) \rightarrow$                                                                                                                                       |
| fong-spivak-invitation_F03268 | 294  | ■ 0.877 | $D_Z$                                                                                                                                                                   | $D_Z$                                                                                                                                                                   | <i>involution</i> of the set $D_z$ :                                                                                                                                                                                                                  |
| fong-spivak-invitation_F03269 | 294  | ■ 1.000 | $D(c)$                                                                                                                                                                  | $D(c)$                                                                                                                                                                  | 2. We could make $D(c)$ the set of people at a “secret Santa” Christmas party, where everyone                                                                                                                                                         |
| fong-spivak-invitation_F03270 | 294  | ■ 0.884 | $D(b)$                                                                                                                                                                  | $D(b)$                                                                                                                                                                  | gives a gift to someone, possibly themselves. Take $D(b)$ to be the set of gifts, $g$ the giver                                                                                                                                                       |
| fong-spivak-invitation_F03271 | 294  | ■ 1.000 | $D(a)$                                                                                                                                                                  | $D(a)$                                                                                                                                                                  | person), $D(a)$ is the set of people who give a gift to themselves, and $d(f): D(a) \rightarrow D(b)$ is                                                                                                                                              |
| fong-spivak-invitation_F03272 | 294  | ■ 1.000 | $d(f): D(a) \rightarrow D(b)$                                                                                                                                           | $d(f): D(a) \rightarrow D(b)$                                                                                                                                           | person), $D(a)$ is the set of people who give a gift to themselves, and $d(f): D(a) \rightarrow D(b)$ is                                                                                                                                              |
| fong-spivak-invitation_F03273 | 294  | ■ 1.000 | $F, G, H \in \mathcal{D}^{\mathcal{C}}$                                                                                                                                 | $F, G, H \in \mathcal{D}^{\mathcal{C}}$                                                                                                                                 | $F, G, H \in \mathcal{D}^{\mathcal{C}}$ ; these are functors $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$ . Morphisms $\alpha: F \rightarrow G$ and $\beta: G \rightarrow H$                                                                        |
| fong-spivak-invitation_F03274 | 294  | ■ 1.000 | $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$                                                                                                                          | $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$                                                                                                                          | $F, G, H \in \mathcal{D}^{\mathcal{C}}$ ; these are functors $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$ . Morphisms $\alpha: F \rightarrow G$ and $\beta: G \rightarrow H$                                                                        |
| fong-spivak-invitation_F03275 | 294  | ■ 1.000 | $\beta: G \rightarrow H$                                                                                                                                                | $\beta: G \rightarrow H$                                                                                                                                                | $F, G, H \in \mathcal{D}^{\mathcal{C}}$ ; these are functors $F, G, H: \mathcal{C} \rightarrow \mathcal{D}$ . Morphisms $\alpha: F \rightarrow G$ and $\beta: G \rightarrow H$                                                                        |
| fong-spivak-invitation_F03276 | 294  | ■ 0.983 | $c \in \mathcal{C}, \alpha$                                                                                                                                             | $c \in \mathcal{C}, \alpha$                                                                                                                                             | So for each $c \in \mathcal{C}$ , $\alpha_c: F(c) \rightarrow G(c)$ and $\beta$ has a component $\beta_c: G(c) \rightarrow$                                                                                                                           |

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|--------------------------------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fong-spivak-invitation_F03277              | 294  | ■ 0.983 | $\backslash\mathrm{beta}_{\{c\}}: G(c) \rightarrowtail$                                                                                                                                                                                      | $\beta_c : G(c) \rightarrow$                                                                                              | So for each $c \in \mathcal{C}$ , $\alpha$ has a component $\alpha_c: F(c) \rightarrow G(c)$ and $\beta$ has a component $\beta_c: G(c) \rightarrow$                                                                                                               |
| fong-spivak-invitation_F03278              | 294  | ■ 0.606 | $H(c)$                                                                                                                                                                                                                                       | $H(c)$                                                                                                                    | $H(c)$ in $\mathcal{D}$ . The expert has told us to define $(\alpha \circ \beta)_c := (\alpha_c \circ \beta_c)$ , and indeed that is a morphism                                                                                                                    |
| fong-spivak-invitation_F03279              | 294  | ■ 0.606 | $(\alpha ; \beta)_{\{C\}}:=\left(\alpha_{\{C\}} ; \beta_{\{C\}}\right)$                                                                                                                                                                      | $(\alpha; \beta)_C := (\alpha_C; \beta_C)$                                                                                | $H(c)$ in $\mathcal{D}$ . The expert has told us to define $(\alpha \circ \beta)_c := (\alpha_c \circ \beta_c)$ , and indeed that is a morphism                                                                                                                    |
| fong-spivak-invitation_F03280              | 294  | ■ 1.000 | $F(c) \rightarrowtail H(c)$                                                                                                                                                                                                                  | $F(c) \rightarrow H(c)$                                                                                                   | $F(c) \rightarrow H(c)$ .                                                                                                                                                                                                                                          |
| fong-spivak-invitation_F03281              | 295  | ■ 0.997 | $\backslash\mathrm{operatorname{id}}_{\{F\}}$                                                                                                                                                                                                | $\mathrm{id}_F$                                                                                                           | 2. We propose that the identity natural transformation $\mathrm{id}_F$ on a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ has as its                                                                                                                            |
| fong-spivak-invitation_F03282              | 295  | ■ 0.599 | $\left(\backslash\mathrm{operatorname{id}}_{\{F\}}\right)_{\{C\}}:=\backslash\mathrm{operatorname{id}}_{\{F(c)\}}$                                                                                                                           | $(\mathrm{id}_F)_C := \mathrm{id}_{F(c)}$                                                                                 | $c$ -component the morphism $(\mathrm{id}_F)_c := \mathrm{id}_{F(c)}$ in $\mathcal{D}$ , for any $c$ . The naturality square                                                                                                                                       |
| fong-spivak-invitation_F03283              | 295  | ■ 0.992 | $\mathrm{Id}_{\{F\}}$                                                                                                                                                                                                                        | $\mathrm{id}_F$                                                                                                           | obviously commutes for any $f: c \rightarrow c'$ . And it is unital: post-composing $\mathrm{id}_F$ with any                                                                                                                                                       |
| fong-spivak-invitation_F03284              | 295  | ■ 1.000 | $\backslash\mathrm{beta}: F \rightarrowtail G$                                                                                                                                                                                               | $\beta: F \rightarrow G$                                                                                                  | $\beta: F \rightarrow G$ (and similarly for precomposing with any $\alpha: E \rightarrow F$ ) results in a natural trans-                                                                                                                                          |
| fong-spivak-invitation_F03285              | 295  | ■ 1.000 | $\backslash\mathrm{alpha}: E \rightarrowtail F$                                                                                                                                                                                              | $\alpha: E \rightarrow F$                                                                                                 | $\beta: F \rightarrow G$ (and similarly for precomposing with any $\alpha: E \rightarrow F$ ) results in a natural trans-                                                                                                                                          |
| fong-spivak-invitation_F03286 [conf 0.093] | 295  | ■ 0.093 | $\left(\backslash\mathrm{operatorname{id}}_{\{F\}}\right)_{\{C\}}$                                                                                                                                                                           | $(\mathrm{id}_F)_C$                                                                                                       | formation $\mathrm{id}_F \circ \beta$ with components $(\mathrm{id}_F)_c \circ \beta_c = (\mathrm{id}_{F(c)} \circ \beta_c) = \beta_c$ , and this is just $\beta$ as                                                                                               |
| fong-spivak-invitation_F03287 [conf 0.093] | 295  | ■ 0.093 | $\backslash\mathrm{beta}_{\{C\}}:=\left(\backslash\mathrm{operatorname{id}}_{\{F(c)\}}\right)^{\circ}\backslash\mathrm{beta}_{\{C\}}\rightarrowtail\backslash\mathrm{beta}_{\{C\}}$                                                          | $\beta_C = (\mathrm{id}_{F(c)} \circ \beta_C) = \beta_C$                                                                  | formation $\mathrm{id}_F \circ \beta$ with components $(\mathrm{id}_F)_c \circ \beta_c = (\mathrm{id}_{F(c)} \circ \beta_c) = \beta_c$ , and this is just $\beta$ as                                                                                               |
| fong-spivak-invitation_F03288              | 295  | ■ 1.000 | $\backslash\mathrm{alpha}, \backslash\mathrm{beta}: F \rightarrowtail G$                                                                                                                                                                     | $\alpha, \beta: F \rightarrow G$                                                                                          | 1. Suppose that $F, G: \mathcal{C} \rightarrow \mathcal{P}$ are functors and $\alpha, \beta: F \rightarrow G$ are natural transformations; we                                                                                                                      |
| fong-spivak-invitation_F03289              | 295  | ■ 0.981 | $\backslash\mathrm{alpha}=\backslash\mathrm{beta}$                                                                                                                                                                                           | $\alpha = \beta$                                                                                                          | need to show that $\alpha = \beta$ . It suffices to check that $\alpha_c = \beta_c$ for each object $c \in \mathrm{Ob}(\mathcal{C})$ . But $\alpha_c$                                                                                                              |
| fong-spivak-invitation_F03290              | 295  | ■ 0.981 | $\backslash\mathrm{alpha}_{\{c\}}=\backslash\mathrm{beta}_{\{c\}}$                                                                                                                                                                           | $\alpha_c = \beta_c$                                                                                                      | need to show that $\alpha = \beta$ . It suffices to check that $\alpha_c = \beta_c$ for each object $c \in \mathrm{Ob}(\mathcal{C})$ . But $\alpha_c$                                                                                                              |
| fong-spivak-invitation_F03291              | 295  | ■ 0.728 | $\backslash\mathrm{beta}_{\{C\}}$                                                                                                                                                                                                            | $\beta_C$                                                                                                                 | and $\beta_c$ are morphisms $F(c) \rightarrow G(c)$ in $\mathcal{P}$ , which is a preorder, and the definition of a preorder                                                                                                                                       |
| fong-spivak-invitation_F03292              | 295  | ■ 0.728 | $F(c) \rightarrowtail G(c)$                                                                                                                                                                                                                  | $F(c) \rightarrow G(c)$                                                                                                   | and $\beta_c$ are morphisms $F(c) \rightarrow G(c)$ in $\mathcal{P}$ , which is a preorder, and the definition of a preorder                                                                                                                                       |
| fong-spivak-invitation_F03293              | 295  | ■ 0.999 | $\backslash\mathrm{alpha}_{\{C\}}=\backslash\mathrm{beta}_{\{C\}}$                                                                                                                                                                           | $\alpha_C = \beta_C$                                                                                                      | $\alpha_c = \beta_c$ , as desired.                                                                                                                                                                                                                                 |
| fong-spivak-invitation_F03294              | 295  | ■ 0.717 | $\backslash\mathrm{mathcal{P}}:=\backslash\mathrm{mathbf{1}}$                                                                                                                                                                                | $\mathcal{P} := \mathbf{1}$                                                                                               | 2. This is false. Let $\mathcal{P} := \mathbf{1}$ , let $\mathcal{C} := \left  \begin{smallmatrix} a & \xrightarrow{f_1} & b \\ \bullet & \xrightarrow{f_2} & \bullet \end{smallmatrix} \right $ , let $F(1) := a$ , let $G(1) := b$ , let $\alpha_1 := f_1$ , and |
| fong-spivak-invitation_F03295              | 295  | ■ 0.717 | $\backslash\mathrm{mathcal{C}}:=\backslash\mathrm{stackrel{a}{\bullet}}\backslash\mathrm{bullet}\rightarrowtail\backslash\mathrm{f}_{\{2\}}\{\backslash\mathrm{f}_{\{1\}}\}\backslash\mathrm{stackrel{b}{\bullet}}\backslash\mathrm{bullet}$ | $\mathcal{C} := \begin{smallmatrix} a & \xrightarrow{f_1} & b \\ \bullet & \xrightarrow{f_2} & \bullet \end{smallmatrix}$ | 2. This is false. Let $\mathcal{P} := \mathbf{1}$ , let $\mathcal{C} := \left  \begin{smallmatrix} a & \xrightarrow{f_1} & b \\ \bullet & \xrightarrow{f_2} & \bullet \end{smallmatrix} \right $ , let $F(1) := a$ , let $G(1) := b$ , let $\alpha_1 := f_1$ , and |
| fong-spivak-invitation_F03296              | 295  | ■ 0.717 | $F(1):=a$                                                                                                                                                                                                                                    | $F(1) := a$                                                                                                               | 2. This is false. Let $\mathcal{P} := \mathbf{1}$ , let $\mathcal{C} := \left  \begin{smallmatrix} a & \xrightarrow{f_1} & b \\ \bullet & \xrightarrow{f_2} & \bullet \end{smallmatrix} \right $ , let $F(1) := a$ , let $G(1) := b$ , let $\alpha_1 := f_1$ , and |
| fong-spivak-invitation_F03297              | 295  | ■ 0.717 | $G(1):=b$                                                                                                                                                                                                                                    | $G(1) := b$                                                                                                               | 2. This is false. Let $\mathcal{P} := \mathbf{1}$ , let $\mathcal{C} := \left  \begin{smallmatrix} a & \xrightarrow{f_1} & b \\ \bullet & \xrightarrow{f_2} & \bullet \end{smallmatrix} \right $ , let $F(1) := a$ , let $G(1) := b$ , let $\alpha_1 := f_1$ , and |
| fong-spivak-invitation_F03298              | 295  | ■ 0.717 | $\backslash\mathrm{alpha}_{\{1\}}:=\backslash\mathrm{f}_{\{1\}}$                                                                                                                                                                             | $\alpha_1 := f_1$                                                                                                         | 2. This is false. Let $\mathcal{P} := \mathbf{1}$ , let $\mathcal{C} := \left  \begin{smallmatrix} a & \xrightarrow{f_1} & b \\ \bullet & \xrightarrow{f_2} & \bullet \end{smallmatrix} \right $ , let $F(1) := a$ , let $G(1) := b$ , let $\alpha_1 := f_1$ , and |
| fong-spivak-invitation_F03299              | 295  | ■ 1.000 | $\backslash\mathrm{beta}_{\{1\}}:=\backslash\mathrm{g}_{\{2\}}$                                                                                                                                                                              | $\beta_1 := g_2$                                                                                                          | let $\beta_1 := g_2$ .                                                                                                                                                                                                                                             |
| fong-spivak-invitation_F03300              | 296  | ■ 1.000 | $\backslash\mathrm{alpha}_{\{\backslash\mathrm{text}{Arrow}\}}(b)=e$                                                                                                                                                                         | $\alpha_{\mathrm{Arrow}}(b) = e$                                                                                          | 1. We have $\alpha_{\mathrm{Arrow}}(b) = e$ and $\alpha_{\mathrm{Vertex}}(1) = 4$ , $\alpha_{\mathrm{Vertex}}(2) = 5$ , and $\alpha_{\mathrm{Vertex}}(3) = 5$ .                                                                                                    |
| fong-spivak-invitation_F03301              | 296  | ■ 1.000 | $\backslash\mathrm{alpha}_{\{\backslash\mathrm{text}{Vertex}\}}(1)=4, \backslash\mathrm{alpha}_{\{\backslash\mathrm{text}{Vertex}\}}(2)=5$                                                                                                   | $\alpha_{\mathrm{Vertex}}(1) = 4, \alpha_{\mathrm{Vertex}}(2) = 5$                                                        | 1. We have $\alpha_{\mathrm{Arrow}}(b) = e$ and $\alpha_{\mathrm{Vertex}}(1) = 4, \alpha_{\mathrm{Vertex}}(2) = 5$ , and $\alpha_{\mathrm{Vertex}}(3) = 5$ .                                                                                                       |

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| fong-spivak-invitation_F03302 | 296  | ■ 1.000 | $\alpha_{\text{Vertex}}(3)=5$                            | $\alpha_{\text{Vertex}}(3) = 5$                  | 1. We have $\alpha_{\text{Arrow}}(b) = e$ and $\alpha_{\text{Vertex}}(1) = 4$ , $\alpha_{\text{Vertex}}(2) = 5$ , and $\alpha_{\text{Vertex}}(3) = 5$ .                                                         |
| fong-spivak-invitation_F03303 | 297  | ■ 1.000 | $X \times B \rightarrowtail Y \times B$                  | $X \times B \rightarrow Y \times B$              | 1. Given an arbitrary function $f: X \rightarrow Y$ , we need a function $X \times B \rightarrowtail Y \times B$ . We suggest                                                                                   |
| fong-spivak-invitation_F03304 | 297  | ■ 0.965 | $f \times B$                                             | $f \times B$                                     | the function which might be denoted $f \times B$ ; it sends $(x, b)$ to $(f(x), b)$ . This assignment is                                                                                                        |
| fong-spivak-invitation_F03305 | 297  | ■ 0.965 | $(x, b)$                                                 | $(x, b)$                                         | the function which might be denoted $f \times B$ ; it sends $(x, b)$ to $(f(x), b)$ . This assignment is                                                                                                        |
| fong-spivak-invitation_F03306 | 297  | ■ 0.965 | $(f(x), b)$                                              | $(f(x), b)$                                      | the function which might be denoted $f \times B$ ; it sends $(x, b)$ to $(f(x), b)$ . This assignment is                                                                                                        |
| fong-spivak-invitation_F03307 | 297  | ■ 0.860 | $\mathrm{id}_{X \times B}$                               | $\mathrm{id}_{X \times B}$                       | functorial: applied to $\mathrm{id}_X$ it returns $\mathrm{id}_{X \times B}$ and it preserves composition.                                                                                                      |
| fong-spivak-invitation_F03308 | 297  | ■ 1.000 | $X^B \rightarrowtail Y^B$                                | $X^B \rightarrow Y^B$                            | 2. Given a function $f: X \rightarrow Y$ , we need a function $X^B \rightarrow Y^B$ . The canonical function would                                                                                              |
| fong-spivak-invitation_F03309 | 297  | ■ 0.885 | $f^B$                                                    | $f^B$                                            | be denoted $f^B$ ; it sends a function $g: B \rightarrow X$ to the composite $(g \circ f): B \rightarrow X \rightarrow Y$ . This is                                                                             |
| fong-spivak-invitation_F03310 | 297  | ■ 0.885 | $g: B \rightarrowtail X$                                 | $g: B \rightarrow X$                             | be denoted $f^B$ ; it sends a function $g: B \rightarrow X$ to the composite $(g \circ f): B \rightarrow X \rightarrow Y$ . This is                                                                             |
| fong-spivak-invitation_F03311 | 297  | ■ 0.885 | $(g \circ f): B \rightarrowtail X \rightarrowtail Y$     | $(g \circ f): B \rightarrow X \rightarrow Y$     | be denoted $f^B$ ; it sends a function $g: B \rightarrow X$ to the composite $(g \circ f): B \rightarrow X \rightarrow Y$ . This is                                                                             |
| fong-spivak-invitation_F03312 | 297  | ■ 0.984 | $f^B(\mathrm{id}_X) = \mathrm{id}_{X^B}$                 | $f^B(\mathrm{id}_X) = \mathrm{id}_{X^B}$         | functorial: applied to $\mathrm{id}_X$ it sends $g$ to $g$ , i.e. $f^B(\mathrm{id}_X) = \mathrm{id}_{X^B}$ , and applied to the composite                                                                       |
| fong-spivak-invitation_F03313 | 297  | ■ 0.773 | $(f_1 \circ f_2): X \rightarrowtail Y \rightarrowtail Z$ | $(f_1 \circ f_2): X \rightarrow Y \rightarrow Z$ | $(f_1 \circ f_2): X \rightarrow Y \rightarrow Z$ , we have                                                                                                                                                      |
| fong-spivak-invitation_F03314 | 297  | ■ 1.000 | $g \in X^B$                                              | $g \in X^B$                                      | for any $g \in X^B$ .                                                                                                                                                                                           |
| fong-spivak-invitation_F03315 | 297  | ■ 0.993 | $\mathbb{N}^{\mathbb{N}}$                                | $\mathbb{N}^{\mathbb{N}}$                        | 3. If $p: \mathbb{N} \rightarrow \mathbb{N}^{\mathbb{N}}$ is the result of currying $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ , then $p(3)$ is an element of $\mathbb{N}^{\mathbb{N}}$ , i.e. we |
| fong-spivak-invitation_F03316 | 297  | ■ 0.974 | $p(3): \mathbb{N} \rightarrowtail \mathbb{N}$            | $p(3): \mathbb{N} \rightarrow \mathbb{N}$        | have $p(3): \mathbb{N} \rightarrow \mathbb{N}$ ; what function is it? It is the function that adds 3. That is $p(3)(n) := n + 3$ .                                                                              |
| fong-spivak-invitation_F03317 | 297  | ■ 0.974 | $p(3)(n) := n + 3$                                       | $p(3)(n) := n + 3$                               | have $p(3): \mathbb{N} \rightarrow \mathbb{N}$ ; what function is it? It is the function that adds 3. That is $p(3)(n) := n + 3$ .                                                                              |
| fong-spivak-invitation_F03318 | 297  | ■ 0.989 | $!: \mathcal{C} \rightarrowtail \mathbf{1}$              | $!: \mathcal{C} \rightarrow \mathbf{1}$          | The functor $!: \mathcal{C} \rightarrowtail \mathbf{1}$ from Eq. (3.14) sends each object $c \in \mathcal{C}$ to the unique object $1 \in \mathbf{1}$ and                                                       |
| fong-spivak-invitation_F03319 | 297  | ■ 0.989 | $1 \in \mathbf{1}$                                       | $1 \in \mathbf{1}$                               | The functor $!: \mathcal{C} \rightarrow \mathbf{1}$ from Eq. (3.14) sends each object $c \in \mathcal{C}$ to the unique object $1 \in \mathbf{1}$ and                                                           |
| fong-spivak-invitation_F03320 | 297  | ■ 0.987 | $\mathrm{B} = \mathrm{Bob}$ , $3 = \mathrm{Em}_3$        | $B = \text{Bob}, 3 = \text{Em}_3$                | Here it is, with names and emails shortened (e.g. B=Bob, 3=Em_3):                                                                                                                                               |
| fong-spivak-invitation_F03321 | 297  | ■ 1.000 | $c \rightarrowtail z$                                    | $c \rightarrow z$                                | $\Rightarrow$ . When $\mathcal{C}$ is the category underlying a preorder, there is at most one morphism between any                                                                                             |
| fong-spivak-invitation_F03322 | 298  | ■ 0.969 | $\mathcal{C} \in \mathbf{Cat}$                           | $\mathcal{C} \in \mathbf{Cat}$                   | $\mathcal{C} \rightarrow \mathbf{1}$ for any object (category) $\mathcal{C} \in \mathbf{Cat}$ .                                                                                                                 |
| fong-spivak-invitation_F03323 | 298  | ■ 0.944 | $2V := \bullet \bullet$                                  | $2V := \bullet \bullet$                          | Consider the graph $2V := \boxed{\bullet \bullet}$ with two vertices and no arrows, and let $\mathcal{C} = \mathbf{Free}(2V)$ ; it has                                                                          |
| fong-spivak-invitation_F03324 | 298  | ■ 0.944 | $\mathcal{C} = \mathbf{Free}(2V)$                        | $\mathcal{C} = \mathbf{Free}(2V)$                | Consider the graph $2V := \boxed{\bullet \bullet}$ with two vertices and no arrows, and let $\mathcal{C} = \mathbf{Free}(2V)$ ; it has                                                                          |
| fong-spivak-invitation_F03325 | 298  | ■ 1.000 | $z \in \mathcal{P}$                                      | $z \in \mathcal{P}$                              | A product of $x$ and $y$ in $\mathcal{P}$ is an object $z \in \mathcal{P}$ equipped with maps $z \rightarrow x$ and $z \rightarrow y$ such that                                                                 |
| fong-spivak-invitation_F03326 | 298  | ■ 1.000 | $z \rightarrowtail x$                                    | $z \rightarrow x$                                | A product of $x$ and $y$ in $\mathcal{P}$ is an object $z \in \mathcal{P}$ equipped with maps $z \rightarrow x$ and $z \rightarrow y$ such that                                                                 |
| fong-spivak-invitation_F03327 | 298  | ■ 1.000 | $z \rightarrowtail y$                                    | $z \rightarrow y$                                | A product of $x$ and $y$ in $\mathcal{P}$ is an object $z \in \mathcal{P}$ equipped with maps $z \rightarrow x$ and $z \rightarrow y$ such that                                                                 |
| fong-spivak-invitation_F03328 | 298  | ■ 1.000 | $z^{\prime}$                                             | $z'$                                             | for any other object $z'$ and maps $z' \rightarrow x$ and $z' \rightarrow y$ , there is a unique morphism $z' \rightarrow z$ making                                                                             |
| fong-spivak-invitation_F03329 | 298  | ■ 1.000 | $z^{\prime} \rightarrowtail x$                           | $z' \rightarrow x$                               | for any other object $z'$ and maps $z' \rightarrow x$ and $z' \rightarrow y$ , there is a unique morphism $z' \rightarrow z$ making                                                                             |



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| fong-spivak-invitation-F03356 | 298  | 0.832 | $f_{\{ \}_9 \ p_Y} = p_Y^{\prime}$                                                                                                              | $f_9 p_Y = p_Y'$                                                                                                                | $f \colon Z' \rightarrow Z$ making the triangles commute, $f \circ p_X = p_X'$ and $f \circ p_Y = p_Y'$ . But “an object                                                                                                                                         |
| fong-spivak-invitation-F03357 | 298  | 0.939 | $Z \in \operatorname{Cone}(X, Y)$                                                                                                               | $Z \in \operatorname{Cone}(X, Y)$                                                                                               | So the definition above becomes: a product of $X$ and $Y$ is an object $Z \in \mathbf{Cone}(X, Y)$ such that                                                                                                                                                     |
| fong-spivak-invitation-F03358 | 298  | 0.822 | $Z^{\prime} \rightarrowtail Z$                                                                                                                  | $Z' \rightarrowtail Z$                                                                                                          | for any other object $Z'$ there is a unique morphism $Z' \rightarrowtail Z$ in $\mathbf{Cone}(X, Y)$ . This is exactly the                                                                                                                                       |
| fong-spivak-invitation-F03359 | 298  | 0.108 | $\stackrel{v_1}{\bullet}$                                                                                                                       | $\bullet^{v_1}$                                                                                                                 | Suppose $\mathcal{J}$ is the graph $\begin{array}{ c } \hline \begin{array}{cc} v_1 & v_2 \\ \bullet & \bullet \end{array} \\ \hline \end{array}$ and $D \colon \mathcal{J} \rightarrow \mathbf{Set}$ is given by two sets, $D(v_1) = X$ and $D(v_2) = Y$        |
| fong-spivak-invitation-F03360 | 298  | 0.108 | $\{ \ }^{\{v\}}$                                                                                                                                | $v$                                                                                                                             | Suppose $\mathcal{J}$ is the graph $\begin{array}{ c } \hline \begin{array}{cc} v_1 & v_2 \\ \bullet & \bullet \end{array} \\ \hline \end{array}$ and $D \colon \mathcal{J} \rightarrow \mathbf{Set}$ is given by two sets, $D(v_1) = X$ and $D(v_2) = Y$        |
| fong-spivak-invitation-F03361 | 298  | 0.108 | $D \left( v_{\{1\}} \right) = X$                                                                                                                | $D \left( v_1 \right) = X$                                                                                                      | Suppose $\mathcal{J}$ is the graph $\begin{array}{ c } \hline \begin{array}{cc} v_1 & v_2 \\ \bullet & \bullet \end{array} \\ \hline \end{array}$ and $D \colon \mathcal{J} \rightarrow \mathbf{Set}$ is given by two sets, $D(v_1) = X$ and $D(v_2) = Y$        |
| fong-spivak-invitation-F03362 | 298  | 0.108 | $D \left( v_{\{2\}} \right) = Y$                                                                                                                | $D \left( v_2 \right) = Y$                                                                                                      | Suppose $\mathcal{J}$ is the graph $\begin{array}{ c } \hline \begin{array}{cc} v_1 & v_2 \\ \bullet & \bullet \end{array} \\ \hline \end{array}$ and $D \colon \mathcal{J} \rightarrow \mathbf{Set}$ is given by two sets, $D(v_1) = X$ and $D(v_2) = Y$        |
| fong-spivak-invitation-F03363 | 299  | 0.998 | $\lim_{\mathcal{J}} D := \left\{ \left( d_1, \dots, d_n \right) \mid d_i \in D \left( v_{\{i\}} \right) \right\}$                               | $\lim_{\mathcal{J}} D := \{ (d_1, \dots, d_n) \mid d_i \in D(v_i) \}$                                                           | $\lim_{\mathcal{J}} D := \{ (d_1, \dots, d_n) \mid d_i \in D(v_i) \}$ for all $1 \leq i \leq n$ and                                                                                                                                                              |
| fong-spivak-invitation-F03364 | 299  | 0.998 | $1 \leq i \leq n$                                                                                                                               | $1 \leq i \leq n$                                                                                                               | $\lim_{\mathcal{J}} D := \{ (d_1, \dots, d_n) \mid d_i \in D(v_i) \}$ for all $1 \leq i \leq n$ and                                                                                                                                                              |
| fong-spivak-invitation-F03365 | 299  | 0.902 | $a \colon v_{\{i\}} \rightarrowtail v_{\{j\}} \in A$                                                                                            | $a \colon v_i \rightarrowtail v_j \in A$                                                                                        | for all $a \colon v_i \rightarrowtail v_j \in A$ , we have $D(a)(d_i) = d_j$ .                                                                                                                                                                                   |
| fong-spivak-invitation-F03366 | 299  | 0.902 | $D(a) \left( d_{\{i\}} \right) = d_{\{j\}}$                                                                                                     | $D(a) \left( d_i \right) = d_j$                                                                                                 | for all $a \colon v_i \rightarrowtail v_j \in A$ , we have $D(a)(d_i) = d_j$ .                                                                                                                                                                                   |
| fong-spivak-invitation-F03367 | 299  | 0.720 | $F^{\operatorname{op}} \colon \mathcal{C}^{\operatorname{op}} \rightarrow \mathcal{D}^{\operatorname{op}}$                                      | $F^{\operatorname{op}} \colon \mathcal{C}^{\operatorname{op}} \rightarrow \mathcal{D}^{\operatorname{op}}$                      | Given a functor $F \colon \mathcal{C} \rightarrow \mathcal{D}$ , we define its opposite $F^{\operatorname{op}} \colon \mathcal{C}^{\operatorname{op}} \rightarrow \mathcal{D}^{\operatorname{op}}$ as follows. For each object                                   |
| fong-spivak-invitation-F03368 | 299  | 0.379 | $c \in \operatorname{Ob} \left( \mathcal{C}^{\operatorname{op}} \right) = \operatorname{Ob}(\mathcal{C})$                                       | $c \in \operatorname{Ob} \left( \mathcal{C}^{\operatorname{op}} \right) = \operatorname{Ob}(\mathcal{C})$                       | $c \in \operatorname{Ob}(\mathcal{C}^{\operatorname{op}}) = \operatorname{Ob}(\mathcal{C})$ , put $F^{\operatorname{op}}(c) := F(c)$ . For each morphism $f \colon c_1 \rightarrowtail c_2$ in $\mathcal{C}^{\operatorname{op}}$ , we have                       |
| fong-spivak-invitation-F03369 | 299  | 0.379 | $F^{\operatorname{op}}(c) := F(c)$                                                                                                              | $F^{\operatorname{op}}(c) := F(c)$                                                                                              | $c \in \operatorname{Ob}(\mathcal{C}^{\operatorname{op}}) = \operatorname{Ob}(\mathcal{C})$ , put $F^{\operatorname{op}}(c) := F(c)$ . For each morphism $f \colon c_1 \rightarrowtail c_2$ in $\mathcal{C}^{\operatorname{op}}$ , we have                       |
| fong-spivak-invitation-F03370 | 299  | 1.000 | $f^{\prime} \colon c_{\{2\}} \rightarrowtail c_{\{1\}}$                                                                                         | $f' \colon c_2 \rightarrowtail c_1$                                                                                             | a corresponding morphism $f' \colon c_2 \rightarrowtail c_1$ in $\mathcal{C}$ and thus a morphism $F(f') \colon F(c_2) \rightarrowtail F(c_1)$ in                                                                                                                |
| fong-spivak-invitation-F03371 | 299  | 1.000 | $F \left( f^{\prime} \right) \colon F \left( c_{\{2\}} \right) \rightarrowtail F \left( c_{\{1\}} \right)$                                      | $F \left( f' \right) \colon F \left( c_2 \right) \rightarrowtail F \left( c_1 \right)$                                          | a corresponding morphism $f' \colon c_2 \rightarrowtail c_1$ in $\mathcal{C}$ and thus a morphism $F(f') \colon F(c_2) \rightarrowtail F(c_1)$ in                                                                                                                |
| fong-spivak-invitation-F03372 | 299  | 1.000 | $F \left( f^{\prime} \right)^{\prime} \colon F^{\operatorname{op}} \left( c_1 \right) \rightarrowtail F^{\operatorname{op}} \left( c_2 \right)$ | $F \left( f' \right)' \colon F^{\operatorname{op}} \left( c_1 \right) \rightarrowtail F^{\operatorname{op}} \left( c_2 \right)$ | $\mathcal{D}$ , and thus a morphism $F \left( f' \right)' \colon F^{\operatorname{op}}(c_1) \rightarrowtail F^{\operatorname{op}}(c_2)$ . Hence we can define $F^{\operatorname{op}}(f) := F \left( f' \right)'$ .                                               |
| fong-spivak-invitation-F03373 | 299  | 1.000 | $F^{\operatorname{op}}(f) := F \left( f^{\prime} \right)^{\prime}$                                                                              | $F^{\operatorname{op}}(f) := F \left( f' \right)'$                                                                              | $\mathcal{D}$ , and thus a morphism $F \left( f' \right)' \colon F^{\operatorname{op}}(c_1) \rightarrowtail F^{\operatorname{op}}(c_2)$ . Hence we can define $F^{\operatorname{op}}(f) := F \left( f' \right)'$ .                                               |
| fong-spivak-invitation-F03374 | 299  | 0.702 | $\left( -^{\prime} \right)$                                                                                                                     | $(-')$                                                                                                                          | Note that the primes $(-')$ are pretty meaningless, we only put them there to differentiate between                                                                                                                                                              |
| fong-spivak-invitation-F03375 | 299  | 0.999 | $F^{\operatorname{op}}$                                                                                                                         | $F^{\operatorname{op}}$                                                                                                         | It is easy to check that our definition of $F^{\operatorname{op}}$ is functorial: it sends identities to identities and                                                                                                                                          |
| fong-spivak-invitation-F03376 | 299  | 0.588 | $\mathcal{X}^{\operatorname{op}} \times \mathcal{Y}$                                                                                            | $\mathcal{X}^{\operatorname{op}} \times \mathcal{Y}$                                                                            | 1. The Hasse diagram for $\mathcal{X}^{\operatorname{op}} \times \mathcal{Y}$ is shown here (ignore the colors):                                                                                                                                                 |
| fong-spivak-invitation-F03377 | 299  | 0.705 | $\mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \rightarrowtail \mathbb{B}$                                                                 | $\mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \rightarrowtail \mathbb{B}$                                                 | 2. There is a profunctor $\Lambda \colon \mathcal{X} \nrightarrowtail \mathcal{Y}$ , i.e. a functor $\mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \rightarrow \mathbb{B}$ , such that, in the picture above,                                               |
| fong-spivak-invitation-F03378 | 299  | 0.698 | $\mathcal{X}^{\operatorname{op}} \times \mathcal{Y}$                                                                                            | $\mathcal{X}^{\operatorname{op}} \times \mathcal{Y}$                                                                            | The preorder $\mathcal{X}^{\operatorname{op}} \times \mathcal{Y}$ describes tasks in decreasing difficulty. For example, (we hope) it is easier                                                                                                                  |
| fong-spivak-invitation-F03379 | 299  | 1.000 | $\Lambda$                                                                                                                                       | $\Lambda$                                                                                                                       | book. The profunctor $\Lambda$ describes possible states of knowledge for my aunt: she can describe                                                                                                                                                              |
| fong-spivak-invitation-F03380 | 300  | 0.534 | $\Phi \colon \mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \rightarrow \mathcal{V}$                                                        | $\Phi \colon \mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \rightarrow \mathcal{V}$                                        | Recall from Definition 2.25 that a $\mathcal{V}$ -functor $\Phi \colon \mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \rightarrow \mathcal{V}$ is a function $\Phi \colon \operatorname{Ob}(\mathcal{X}^{\operatorname{op}} \times \mathcal{Y}) \rightarrow$ |
| fong-spivak-invitation-F03381 | 300  | 0.534 | $\Phi \colon \operatorname{Ob} \left( \mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \right) \rightarrow$                                   | $\Phi \colon \operatorname{Ob} \left( \mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \right) \rightarrow$                   | Recall from Definition 2.25 that a $\mathcal{V}$ -functor $\Phi \colon \mathcal{X}^{\operatorname{op}} \times \mathcal{Y} \rightarrow \mathcal{V}$ is a function $\Phi \colon \operatorname{Ob}(\mathcal{X}^{\operatorname{op}} \times \mathcal{Y}) \rightarrow$ |

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|-------------------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fong-spivak-invitation_F03382 | 301  | ■ 0.995 | $\Phi:(T \times E) \longrightarrow \$$                                                                                              | $\Phi:(T \times E) \longrightarrow \$$                                                                                              | Yes, this is valid: it just means that the profunctor $\Phi:(T \times E) \rightarrow \$$ does not relate (good-natured,                                                         |
| fong-spivak-invitation_F03383 | 301  | ■ 0.996 | $\Phi(($                                                                                                                            | $\Phi(($                                                                                                                            | funny) to any element of $\$$ . More formally, it means that $\Phi((\text{good-natured, funny}), p) = \text{false}$                                                             |
| fong-spivak-invitation_F03384 | 301  | ■ 0.996 | $), p) =$                                                                                                                           | $), p) =$                                                                                                                           | funny) to any element of $\$$ . More formally, it means that $\Phi((\text{good-natured, funny}), p) = \text{false}$                                                             |
| fong-spivak-invitation_F03385 | 301  | ■ 1.000 | $p \in \$ = \{\$100 \mathrm{~K}, \$500 \mathrm{~K}, \$1 \mathrm{~M}\}$                                                              | $p \in \$ = \{\$100 \mathrm{~K}, \$500 \mathrm{~K}, \$1 \mathrm{~M}\}$                                                              | for all $p \in \$ = \{\$100\text{K}, \$500\text{K}, \$1\text{M}\}$ . We can interpret this to mean that it is not feasible to                                                   |
| fong-spivak-invitation_F03386 | 301  | ■ 0.932 | $(\Phi: \Psi)(-, -) =$                                                                                                              | $(\Phi: \Psi)(-, -) =$                                                                                                              | shortest paths go through the bridges from $D$ to $y$ and $y$ to $r$ , so in this case $\Phi(\Psi)(-, -) =$                                                                     |
| fong-spivak-invitation_F03387 | 301  | ■ 0.753 | $X(-, D) + 9 + Z(r, -)$                                                                                                             | $X(-, D) + 9 + Z(r, -)$                                                                                                             | $X(-, D) + 9 + Z(r, -)$ . This gives:                                                                                                                                           |
| fong-spivak-invitation_F03389 | 302  | ■ 0.951 | $U_{\mathcal{X}}$                                                                                                                   | $U_{\mathcal{X}}$                                                                                                                   | We choose the <b>Cost</b> -category $\mathcal{X}$ from Eq. (2.18). The unit profunctor $U_{\mathcal{X}}$ on $\mathcal{X}$ is described by the                                   |
| fong-spivak-invitation_F03390 | 302  | ■ 0.971 | $I \leq \mathcal{P}(p, p)$                                                                                                          | $I \leq \mathcal{P}(p, p)$                                                                                                          | city of $\otimes$ (Definition 2.1(a)) applied to the inequalities $I \leq \mathcal{P}(p, p)$ (the identity law for $\mathcal{P}$ at                                             |
| fong-spivak-invitation_F03391 | 302  | ■ 1.000 | $\Phi(p, q) \leq \Phi(p, q)$                                                                                                        | $\Phi(p, q) \leq \Phi(p, q)$                                                                                                        | $p$ , Definition 2.30(a)) and $\Phi(p, q) \leq \Phi(p, q)$ (reflexivity of preorder $\mathcal{V}$ , Definition 1.25(a)).                                                        |
| fong-spivak-invitation_F03392 | 302  | ■ 1.000 | $i_{\{0\}} \in I$                                                                                                                   | $i_0 \in I$                                                                                                                         | The third step uses the definition of join: for any set $I$ element $i_0 \in I$ , a subset $s_{i \in I}$ at a poset,                                                            |
| fong-spivak-invitation_F03393 | 302  | ■ 1.000 | $s_{\{i \in I\}}$                                                                                                                   | $s_{i \in I}$                                                                                                                       | The third step uses the definition of join: for any set $I$ element $i_0 \in I$ , a subset $s_{i \in I}$ at a poset,                                                            |
| fong-spivak-invitation_F03394 | 302  | ■ 0.998 | $s_{\{i_{\{0\}}\}} \leq \bigvee_{i \in I} s$                                                                                        | $s_{i_0} \leq \bigvee_{i \in I} s$                                                                                                  | we have $s_{i_0} \leq \bigvee_{i \in I} s$ . In our case, $Ii_0 = p$ , and $s_i = \mathcal{P}(p, i) \otimes \Phi$ . The final equality is just                                  |
| fong-spivak-invitation_F03395 | 302  | ■ 0.998 | $I i_{\{0\}} = p$                                                                                                                   | $Ii_0 = p$                                                                                                                          | we have $s_{i_0} \leq \bigvee_{i \in I} s$ . In our case, $Ii_0 = p$ , and $s_i = \mathcal{P}(p, i) \otimes \Phi$ . The final equality is just                                  |
| fong-spivak-invitation_F03396 | 302  | ■ 0.998 | $s_{\{i\}} = \mathcal{P}(p, i) \otimes \Phi$                                                                                        | $s_i = \mathcal{P}(p, i) \otimes \Phi$                                                                                              | we have $s_{i_0} \leq \bigvee_{i \in I} s$ . In our case, $Ii_0 = p$ , and $s_i = \mathcal{P}(p, i) \otimes \Phi$ . The final equality is just                                  |
| fong-spivak-invitation_F03397 | 302  | ■ 0.558 | $I = \text{true}$                                                                                                                   | $I = \text{true}$                                                                                                                   | 2. Note that, in <b>Bool</b> , $I = \text{true}$ . Since the identity law at $p$ says $\text{true} \leq \mathcal{P}(p, p)$ , and $\text{true}$ is                               |
| fong-spivak-invitation_F03398 | 302  | ■ 0.558 | $\text{true} \leq \mathcal{P}(p, p)$                                                                                                | $\text{true} \leq \mathcal{P}(p, p)$                                                                                                | 2. Note that, in <b>Bool</b> , $I = \text{true}$ . Since the identity law at $p$ says $\text{true} \leq \mathcal{P}(p, p)$ , and $\text{true}$ is                               |
| fong-spivak-invitation_F03399 | 302  | ■ 0.912 | $\mathcal{P}(p, p) = \text{tr}$                                                                                                     | $\mathcal{P}(p, p) = \text{tr}$                                                                                                     | the largest element of the preorder <b>Bool</b> , we thus have $\mathcal{P}(p, p) = \text{true}$ for all $p$ . This shows                                                       |
| fong-spivak-invitation_F03400 | 302  | ■ 0.888 | $\Phi(p, q)$                                                                                                                        | $\Phi(p, q)$                                                                                                                        | side of the inequality is equal to $\Phi(p, q)$ . We split into two cases. Suppose $\Phi(p, q) = \text{true}$ .                                                                 |
| fong-spivak-invitation_F03401 | 302  | ■ 0.888 | $\Phi(p, q) = \text{true}$                                                                                                          | $\Phi(p, q) = \text{true}$                                                                                                          | side of the inequality is equal to $\Phi(p, q)$ . We split into two cases. Suppose $\Phi(p, q) = \text{true}$ .                                                                 |
| fong-spivak-invitation_F03402 | 302  | ■ 0.449 | $\Phi(p, q) =$                                                                                                                      | $\Phi(p, q) =$                                                                                                                      | Next, suppose $\Phi(p, q) = \text{false}$ . Note that since $\Phi$ is a monotone map $\mathcal{P}^{\text{op}} \times \mathcal{Q} \rightarrow \mathbf{Bool}$ , if                |
| fong-spivak-invitation_F03403 | 302  | ■ 0.449 | $\mathcal{P}^{\text{op}} \times \mathcal{Q} \rightarrow$                                                                            | $\mathcal{P}^{\text{op}} \times \mathcal{Q} \rightarrow$                                                                            | Next, suppose $\Phi(p, q) = \text{false}$ . Note that since $\Phi$ is a monotone map $\mathcal{P}^{\text{op}} \times \mathcal{Q} \rightarrow \mathbf{Bool}$ , if                |
| fong-spivak-invitation_F03404 | 302  | ■ 0.987 | $p \leq p_1$                                                                                                                        | $p \leq p_1$                                                                                                                        | $p \leq p_1$ in $\mathcal{P}$ , then $\Phi(p_1, q) \leq \Phi(p, q)$ in <b>Bool</b> . Thus if $\mathcal{P}(p, p_1) = \text{true}$ then $\Phi(p_1, q) =$                          |
| fong-spivak-invitation_F03405 | 302  | ■ 0.987 | $\Phi(p_1, q) \leq \Phi(p, q)$                                                                                                      | $\Phi(p_1, q) \leq \Phi(p, q)$                                                                                                      | $p \leq p_1$ in $\mathcal{P}$ , then $\Phi(p_1, q) \leq \Phi(p, q)$ in <b>Bool</b> . Thus if $\mathcal{P}(p, p_1) = \text{true}$ then $\Phi(p_1, q) =$                          |
| fong-spivak-invitation_F03406 | 302  | ■ 0.987 | $\mathcal{P}(p, p_1) = \text{true}$                                                                                                 | $\mathcal{P}(p, p_1) = \text{true}$                                                                                                 | $p \leq p_1$ in $\mathcal{P}$ , then $\Phi(p_1, q) \leq \Phi(p, q)$ in <b>Bool</b> . Thus if $\mathcal{P}(p, p_1) = \text{true}$ then $\Phi(p_1, q) =$                          |
| fong-spivak-invitation_F03407 | 302  | ■ 0.987 | $\Phi(p_1, q) =$                                                                                                                    | $\Phi(p_1, q) =$                                                                                                                    | $p \leq p_1$ in $\mathcal{P}$ , then $\Phi(p_1, q) \leq \Phi(p, q)$ in <b>Bool</b> . Thus if $\mathcal{P}(p, p_1) = \text{true}$ then $\Phi(p_1, q) =$                          |
| fong-spivak-invitation_F03408 | 302  | ■ 0.672 | $\mathcal{P}(p, p_1) =$                                                                                                             | $\mathcal{P}(p, p_1) =$                                                                                                             | $\Phi(p, q) = \text{false}$ . This implies that for all $p_1 \in \mathcal{P}$ , we have either $\mathcal{P}(p, p_1) = \text{false}$ or                                          |
| fong-spivak-invitation_F03409 | 302  | ■ 0.775 | $\bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q) = \bigvee_{p_1 \in \mathcal{P}} \text{false} = \text{false}$ | $\bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q) = \bigvee_{p_1 \in \mathcal{P}} \text{false} = \text{false}$ | $\Phi(p_1, q) = \text{false}$ , and hence $\bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q) = \bigvee_{p_1 \in \mathcal{P}} \text{false} = \text{false}$ . |
| fong-spivak-invitation_F03410 | 302  | ■ 1.000 | $\Phi(p, q) = \bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q)$                                                | $\Phi(p, q) = \bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q)$                                                | Thus, in either case, we see that $\Phi(p, q) = \bigvee_{p_1 \in \mathcal{P}} \mathcal{P}(p, p_1) \wedge \Phi(p_1, q)$ , as required.                                           |

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| fong-spivak-invitation_F03411 | 302  | ■ 1.000 | $v \otimes I = v$                                                                   | $v \otimes I = v$                                                                   | 3. The first equation is unitality in monoidal categories, $v \otimes I = v$ for any $v \in V$ . The second                                                                     |
| fong-spivak-invitation_F03412 | 302  | ■ 1.000 | $I \leq Q(q, q)$                                                                    | $I \leq Q(q, q)$                                                                    | is that $I \leq Q(q, q)$ by unitality of enriched categories, see Definition 2.30, together with                                                                                |
| fong-spivak-invitation_F03413 | 302  | ■ 1.000 | $v_{\{1\}} \leq v_{\{2\}}$                                                          | $v_1 \leq v_2$                                                                      | monotonicity of monoidal product: $v_1 \leq v_2$ implies $v \otimes v_1 \leq v \otimes v_2$ . The third was shown in                                                            |
| fong-spivak-invitation_F03414 | 302  | ■ 1.000 | $v \otimes v_{\{1\}} \leq v \otimes v_{\{2\}}$                                      | $v \otimes v_1 \leq v \otimes v_2$                                                  | monotonicity of monoidal product: $v_1 \leq v_2$ implies $v \otimes v_1 \leq v \otimes v_2$ . The third was shown in                                                            |
| fong-spivak-invitation_F03415 | 303  | ■ 0.888 | $\widehat{\mathrm{id}}: \mathcal{P} \rightarrow \mathcal{P}$                        | $\widehat{\mathrm{id}}: \mathcal{P} \rightarrow \mathcal{P}$                        | This is very straightforward. We wish to check $\widehat{\mathrm{id}}: \mathcal{P} \rightarrow \mathcal{P}$ has the formula $\widehat{\mathrm{id}}(p, q) = \mathcal{P}(p, q)$ . |
| fong-spivak-invitation_F03416 | 303  | ■ 0.888 | $\widehat{\mathrm{id}}(p, q) = \mathcal{P}(p, q)$                                   | $\widehat{\mathrm{id}}(p, q) = \mathcal{P}(p, q)$                                   | This is very straightforward. We wish to check $\widehat{\mathrm{id}}: \mathcal{P} \rightarrow \mathcal{P}$ has the formula $\widehat{\mathrm{id}}(p, q) = \mathcal{P}(p, q)$ . |
| fong-spivak-invitation_F03417 | 303  | ■ 0.928 | $\widehat{\mathrm{id}}(p, q) := \mathcal{P}(\mathrm{id}(p), q) = \mathcal{P}(p, q)$ | $\widehat{\mathrm{id}}(p, q) := \mathcal{P}(\mathrm{id}(p), q) = \mathcal{P}(p, q)$ | By Definition 4.24, $\widehat{\mathrm{id}}(p, q) := \mathcal{P}(\mathrm{id}(p), q) = \mathcal{P}(p, q)$ . So they're the same.                                                  |
| fong-spivak-invitation_F03418 | 303  | ■ 0.743 | $\check{+}: \mathbb{R} \multimap \mathbb{R} \times \mathbb{R} \times \mathbb{R}$    | $\check{+}: \mathbb{R} \multimap \mathbb{R} \times \mathbb{R} \times \mathbb{R}$    | The conjoint $\check{+}: \mathbb{R} \multimap \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ sends $(a, b, c, d)$ to $\mathbb{R}(a, b + c + d)$ , which is true if $a \leq$    |
| fong-spivak-invitation_F03419 | 303  | ■ 0.743 | $\mathbb{R}(a, b + c + d)$                                                          | $\mathbb{R}(a, b + c + d)$                                                          | The conjoint $\check{+}: \mathbb{R} \multimap \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ sends $(a, b, c, d)$ to $\mathbb{R}(a, b + c + d)$ , which is true if $a \leq$    |
| fong-spivak-invitation_F03420 | 303  | ■ 0.743 | $a \leq$                                                                            | $a \leq$                                                                            | The conjoint $\check{+}: \mathbb{R} \multimap \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ sends $(a, b, c, d)$ to $\mathbb{R}(a, b + c + d)$ , which is true if $a \leq$    |
| fong-spivak-invitation_F03421 | 303  | ■ 1.000 | $b + c + d$                                                                         | $b + c + d$                                                                         | $b + c + d$ , and false otherwise.                                                                                                                                              |
| fong-spivak-invitation_F03422 | 303  | ■ 0.863 | $\widehat{F}(p, q) = Q(F(p), q)$                                                    | $\widehat{F}(p, q) = Q(F(p), q)$                                                    | 1. By Definition 4.24, $\widehat{F}(p, q) = Q(F(p), q)$ and $\check{G}(p, q) = Q(p, G(q))$ . Since $\mathcal{V}$ is skeletal, $F$                                               |
| fong-spivak-invitation_F03423 | 303  | ■ 0.863 | $\check{G}(p, q) = Q(p, G(q))$                                                      | $\check{G}(p, q) = Q(p, G(q))$                                                      | 1. By Definition 4.24, $\widehat{F}(p, q) = Q(F(p), q)$ and $\check{G}(p, q) = Q(p, G(q))$ . Since $\mathcal{V}$ is skeletal, $F$                                               |
| fong-spivak-invitation_F03424 | 303  | ■ 0.766 | $Q(F(p), q) = Q(p, G(q))$                                                           | $Q(F(p), q) = Q(p, G(q))$                                                           | and $G$ are $\mathcal{V}$ adjoints if and only if $Q(F(p), q) = Q(p, G(q))$ . Thus $F$ and $G$ are adjoints if                                                                  |
| fong-spivak-invitation_F03425 | 303  | ■ 0.948 | $\widehat{F} = \check{G}$                                                           | $\widehat{F} = \check{G}$                                                           | and only if $\widehat{F} = \check{G}$ .                                                                                                                                         |
| fong-spivak-invitation_F03426 | 303  | ■ 0.999 | $\otimes: \mathcal{P} \times \mathcal{P} \rightarrow \mathcal{P}$                   | $\otimes: \mathcal{P} \times \mathcal{P} \rightarrow \mathcal{P}$                   | Definition 2.1 calls for merely a function $\otimes: \mathcal{P} \times \mathcal{P} \rightarrow \mathcal{P}$ , while Definition 4.34 calls for a <i>functor</i> .               |
| fong-spivak-invitation_F03427 | 303  | ■ 0.999 | $(x_1, x_2) \leq (y_1, y_2)$                                                        | $(x_1, x_2) \leq (y_1, y_2)$                                                        | Definition 2.1: it says that if $(x_1, x_2) \leq (y_1, y_2)$ in $\mathcal{P} \otimes \mathcal{P}$ , then we must have $x_1 \otimes x_2 \leq y_1 \otimes y_2$                    |
| fong-spivak-invitation_F03428 | 303  | ■ 0.999 | $\mathcal{P} \otimes \mathcal{P}$                                                   | $\mathcal{P} \otimes \mathcal{P}$                                                   | Definition 2.1: it says that if $(x_1, x_2) \leq (y_1, y_2)$ in $\mathcal{P} \otimes \mathcal{P}$ , then we must have $x_1 \otimes x_2 \leq y_1 \otimes y_2$                    |
| fong-spivak-invitation_F03429 | 303  | ■ 1.000 | $2.1 \otimes$                                                                       | $2.1 \otimes$                                                                       | in $\mathcal{P}$ . So it is also the case that in Definition 2.1 $\otimes$ is a functor.                                                                                        |
| fong-spivak-invitation_F03430 | 303  | ■ 0.998 | $g_E(5, 3) =$                                                                       | $g_E(5, 3) =$                                                                       | 1. $g_E(5, 3) = \text{false}$ , $g_F(5, 3) = 2$ .                                                                                                                               |
| fong-spivak-invitation_F03431 | 303  | ■ 0.998 | $g_F(5, 3) = 2$                                                                     | $g_F(5, 3) = 2$                                                                     | 1. $g_E(5, 3) = \text{false}$ , $g_F(5, 3) = 2$ .                                                                                                                               |
| fong-spivak-invitation_F03432 | 303  | ■ 0.976 | $g_E(3, 5) =$                                                                       | $g_E(3, 5) =$                                                                       | 2. $g_E(3, 5) = \text{true}$ , $g_F(3, 5) = -2$ .                                                                                                                               |
| fong-spivak-invitation_F03433 | 303  | ■ 0.976 | $g_F(3, 5) = -2$                                                                    | $g_F(3, 5) = -2$                                                                    | 2. $g_E(3, 5) = \text{true}$ , $g_F(3, 5) = -2$ .                                                                                                                               |
| fong-spivak-invitation_F03434 | 304  | ■ 0.733 | $) = 5$                                                                             | $) = 5$                                                                             | 3. $h(5, \text{true}) = 5$ .                                                                                                                                                    |
| fong-spivak-invitation_F03435 | 304  | ■ 0.885 | $) = -5$                                                                            | $) = -5$                                                                            | 4. $h(-5, \text{true}) = -5$ .                                                                                                                                                  |
| fong-spivak-invitation_F03436 | 304  | ■ 0.693 | $) = 6$                                                                             | $) = 6$                                                                             | 5. $h(-5, \text{false}) = 6$ .                                                                                                                                                  |
| fong-spivak-invitation_F03437 | 304  | ■ 1.000 | $q_G(-2, 3) = 2, q_F(-2, 3) = -13$                                                  | $q_G(-2, 3) = 2, q_F(-2, 3) = -13$                                                  | 6. $q_G(-2, 3) = 2, q_F(-2, 3) = -13$ .                                                                                                                                         |
| fong-spivak-invitation_F03438 | 304  | ■ 1.000 | $q_G(2, 3) = -1, q_F(2, 3) = 7$                                                     | $q_G(2, 3) = -1, q_F(2, 3) = 7$                                                     | 7. $q_G(2, 3) = -1, q_F(2, 3) = 7$ .                                                                                                                                            |

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| fong-spivak-invitation_F03439 | 304  | 0.970 | $\mathcal{C}(x, y)$                                                                                | $\mathcal{C}(x, y)$                                                                                | $\mathcal{C}(x, y)$ of morphisms. Moreover, recall that the monoidal unit $I$ is the one-element set $\{1\}$ . This                                                                                                             |
| fong-spivak-invitation_F03440 | 304  | 0.483 | $\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$                                                   | $\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$                                                   | means a morphism $\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$ is a function $\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$ . This is the same data as                                                                 |
| fong-spivak-invitation_F03441 | 304  | 0.483 | $\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$                                               | $\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$                                               | means a morphism $\mathrm{id}_x: I \rightarrow \mathcal{C}(x, x)$ is a function $\mathrm{id}_x: \{1\} \rightarrow \mathcal{C}(x, x)$ . This is the same data as                                                                 |
| fong-spivak-invitation_F03442 | 304  | 0.919 | $\mathrm{id}_x = \mathrm{id}_x(1) \in \mathcal{C}(x, x)$                                           | $\mathrm{id}_x = \mathrm{id}_x(1) \in \mathcal{C}(x, x)$                                           | simply an element $\mathrm{id}_x = \mathrm{id}_x(1) \in \mathcal{C}(x, x)$ ; we call this data the identity morphism on $x$ . Finally,                                                                                          |
| fong-spivak-invitation_F03443 | 304  | 0.960 | $\{ \}_9: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$               | $g: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$                     | a morphism $\S: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$ is a function $\S: \mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$ ; this is                               |
| fong-spivak-invitation_F03444 | 304  | 0.960 | $\{ \}_9: \mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$                | $g: \mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$                      | a morphism $\S: \mathcal{C}(x, y) \otimes \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$ is a function $\S: \mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$ ; this is                               |
| fong-spivak-invitation_F03445 | 304  | 0.662 | $I \rightarrow \mathcal{X}(x, x)$                                                                  | $I \rightarrow \mathcal{X}(x, x)$                                                                  | An identity element in a <b>Cost</b> -category $\mathcal{X}$ is a morphism $I \rightarrow \mathcal{X}(x, x)$ in <b>Cost</b> = $([0, \infty], \geq,$                                                                             |
| fong-spivak-invitation_F03446 | 304  | 0.662 | $=([0, \infty], \geq$                                                                              | $=([0, \infty], \geq$                                                                              | An identity element in a <b>Cost</b> -category $\mathcal{X}$ is a morphism $I \rightarrow \mathcal{X}(x, x)$ in <b>Cost</b> $\equiv ([0, \infty], \geq,$                                                                        |
| fong-spivak-invitation_F03447 | 304  | 0.802 | $0 \geq \mathfrak{X}(x, x)$                                                                        | $0 \geq \mathfrak{X}(x, x)$                                                                        | 0, +), and hence the condition that $0 \geq \mathfrak{X}(x, x)$ . This implies that $\mathfrak{X}(x, x) = 0$ . In terms of                                                                                                      |
| fong-spivak-invitation_F03448 | 304  | 0.802 | $\mathcal{X}(x, x) = 0$                                                                            | $\mathcal{X}(x, x) = 0$                                                                            | 0, +), and hence the condition that $0 \geq \mathfrak{X}(x, x)$ . This implies that $\mathfrak{X}(x, x) = 0$ . In terms of                                                                                                      |
| fong-spivak-invitation_F03449 | 305  | 0.801 | $x \in \mathcal{X}$                                                                                | $x \in \mathcal{X}$                                                                                | resources, $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ , with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$ . That is, if $x$ is available                                                                             |
| fong-spivak-invitation_F03450 | 305  | 0.801 | $y \in \mathcal{Y}$                                                                                | $y \in \mathcal{Y}$                                                                                | resources, $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ , with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$ . That is, if $x$ is available                                                                             |
| fong-spivak-invitation_F03451 | 305  | 0.801 | $(x, y) \leq (x', y')$                                                                             | $(x, y) \leq (x', y')$                                                                             | resources, $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ , with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$ . That is, if $x$ is available                                                                             |
| fong-spivak-invitation_F03452 | 305  | 0.801 | $y \leq y'$                                                                                        | $y \leq y'$                                                                                        | resources, $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ , with $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$ . That is, if $x$ is available                                                                             |
| fong-spivak-invitation_F03453 | 305  | 0.920 | $\Phi: X_1 \mapsto X_2$                                                                            | $\Phi: X_1 \mapsto X_2$                                                                            | Given two profunctors $\Phi: \mathcal{X}_1 \multimap \mathcal{X}_2$ and $\Psi: \mathcal{Y}_1 \multimap \mathcal{Y}_2$ , the profunctor $\Phi \times \Psi$ represents their                                                      |
| fong-spivak-invitation_F03454 | 305  | 1.000 | $y_1$                                                                                              | $y_1$                                                                                              | conjunction, i.e. AND. In other words, if $y_1$ can be obtained given $x_1$ AND $y_2$ can be obtained                                                                                                                           |
| fong-spivak-invitation_F03455 | 305  | 1.000 | $x_1$                                                                                              | $x_1$                                                                                              | conjunction, i.e. AND. In other words, if $y_1$ can be obtained given $x_1$ AND $y_2$ can be obtained                                                                                                                           |
| fong-spivak-invitation_F03456 | 305  | 1.000 | $y_2$                                                                                              | $y_2$                                                                                              | conjunction, i.e. AND. In other words, if $y_1$ can be obtained given $x_1$ AND $y_2$ can be obtained                                                                                                                           |
| fong-spivak-invitation_F03457 | 305  | 0.627 | $x_2$                                                                                              | $x_2$                                                                                              | given $x_2$ , then $(y_1, y_2)$ can be obtained given $(x_1, x_2)$ .                                                                                                                                                            |
| fong-spivak-invitation_F03458 | 305  | 0.627 | $(y_1, y_2)$                                                                                       | $(y_1, y_2)$                                                                                       | given $x_2$ , then $(y_1, y_2)$ can be obtained given $(x_1, x_2)$ .                                                                                                                                                            |
| fong-spivak-invitation_F03459 | 305  | 0.627 | $(x_1, x_2)$                                                                                       | $(x_1, x_2)$                                                                                       | given $x_2$ , then $(y_1, y_2)$ can be obtained given $(x_1, x_2)$ .                                                                                                                                                            |
| fong-spivak-invitation_F03460 | 305  | 0.398 | $\mathcal{X} \times \mathbf{1} - \mathfrak{X}$                                                     | $\mathcal{X} \times \mathbf{1} - \mathfrak{X}$                                                     | The profunctor $\mathcal{X} \times \mathbf{1} \multimap \mathcal{X}$ defined by the functor $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ that maps $\alpha((x, 1), y) :=$ |
| fong-spivak-invitation_F03461 | 305  | 0.398 | $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ | $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ | The profunctor $\mathcal{X} \times \mathbf{1} \multimap \mathcal{X}$ defined by the functor $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ that maps $\alpha((x, 1), y) :=$ |
| fong-spivak-invitation_F03462 | 305  | 0.398 | $\alpha((x, 1), y) :=$                                                                             | $\alpha((x, 1), y) :=$                                                                             | The profunctor $\mathcal{X} \times \mathbf{1} \multimap \mathcal{X}$ defined by the functor $\alpha: (\mathcal{X} \times \mathbf{1})^{\mathrm{op}} \times \mathcal{X} \rightarrow \mathcal{V}$ that maps $\alpha((x, 1), y) :=$ |
| fong-spivak-invitation_F03463 | 305  | 0.732 | $X(x, y)$                                                                                          | $X(x, y)$                                                                                          | $\mathcal{X}(x, y)$ is an isomorphism. It has inverse $\alpha^{-1}: \mathcal{X} \multimap \mathcal{X} \times \mathbf{1}$ defined by $\alpha^{-1}(x, (y, 1)) := \mathcal{X}(x, y)$ .                                             |
| fong-spivak-invitation_F03464 | 305  | 0.732 | $\alpha^{-1}: X \mapsto X \times \mathbf{1}$                                                       | $\alpha^{-1}: X \mapsto X \times \mathbf{1}$                                                       | $\mathcal{X}(x, y)$ is an isomorphism. It has inverse $\alpha^{-1}: \mathcal{X} \multimap \mathcal{X} \times \mathbf{1}$ defined by $\alpha^{-1}(x, (y, 1)) := \mathcal{X}(x, y)$ .                                             |
| fong-spivak-invitation_F03465 | 305  | 0.732 | $\alpha^{-1}(x, (y, 1)) := X(x, y)$                                                                | $\alpha^{-1}(x, (y, 1)) := X(x, y)$                                                                | $\mathcal{X}(x, y)$ is an isomorphism. It has inverse $\alpha^{-1}: \mathcal{X} \multimap \mathcal{X} \times \mathbf{1}$ defined by $\alpha^{-1}(x, (y, 1)) := \mathcal{X}(x, y)$ .                                             |
| fong-spivak-invitation_F03466 | 305  | 0.844 | $\alpha^{-1}; \alpha = U_{\mathcal{X}}$                                                            | $\alpha^{-1}; \alpha = U_{\mathcal{X}}$                                                            | To see that $\alpha^{-1} \circ \alpha = U_{\mathcal{X}}$ , note first that the unit law for $\mathcal{X}$ at $z$ and the definition of join imply                                                                               |

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| fong-spivak-invitation_F03467 | 305  | 0.753 | $\alpha ; \alpha^{-1} = \mathrm{U}_{X \times \mathbf{1}}$  | $\alpha; \alpha^{-1} = U_{X \times \mathbf{1}}$ | Similarly we can show $\alpha \circ \alpha^{-1} = U_{X \times \mathbf{1}}$ , and hence that $\alpha$ is an isomorphism $X \times \mathbf{1} \rightarrowtail X$ . |
| fong-spivak-invitation_F03468 | 305  | 0.984 | $\beta((1, x), y) := \mathrm{cal}_X(x, y)$                 | $\beta((1, x), y) := \mathcal{X}(x, y)$         | Moreover, we can similarly show that $\beta((1, x), y) := X(x, y)$ defines an isomorphism $\beta: \mathbf{1} \times$                                             |
| fong-spivak-invitation_F03469 | 305  | 0.984 | $\beta: \mathbf{1} \times$                                 | $\beta: \mathbf{1} \times$                      | Moreover, we can similarly show that $\beta((1, x), y) := X(x, y)$ defines an isomorphism $\beta: \mathbf{1} \times$                                             |
| fong-spivak-invitation_F03470 | 305  | 0.641 | $x \rightarrowtail x$                                      | $x \rightarrow x$                               | $X \rightarrowtail X$ .                                                                                                                                          |
| fong-spivak-invitation_F03471 | 306  | 0.823 | $\alpha^{-1}$                                              | $\alpha^{-1}$                                   | is itself the identity (i.e. the unit profunctor on $X$ ), where $\alpha$ and $\alpha^{-1}$ are the isomorphisms defined                                         |
| fong-spivak-invitation_F03472 | 306  | 0.997 | $\Phi(x, y)$                                               | $\Phi(x, y)$                                    | Freely using the distributivity of $\otimes$ over $\vee$ , the value $\Phi(x, y)$ of this composite at $(x, y) \in$                                              |
| fong-spivak-invitation_F03473 | 306  | 0.443 | $\mathrm{scr}_X^{\mathrm{OP}} \times \mathrm{cal}_X$       | $\mathcal{X}^{\mathrm{OP}} \times \mathcal{X}$  | $X^{\mathrm{op}} \times X$ is given by                                                                                                                           |
| fong-spivak-invitation_F03474 | 306  | 0.996 | $\mathrm{U}_{\mathrm{cal}_X}(a, b) = \mathrm{cal}_X(a, b)$ | $U_{\mathcal{X}}(a, b) = \mathcal{X}(a, b)$     | with the unit profunctor $U_X(a, b) = X(a, b)$ is the identity.                                                                                                  |
| fong-spivak-invitation_F03475 | 306  | 1.000 | $m \rightarrowtail p$                                      | $m \rightarrow p$                               | $m \rightarrow p$ given by $(f \circ g)(i) = g(f(i))$ for all $1 \leq i \leq m$ .                                                                                |
| fong-spivak-invitation_F03476 | 306  | 1.000 | $(f ; g)(i) = g(f(i))$                                     | $(f;g)(i) = g(f(i))$                            | $m \rightarrow p$ given by $(f \circ g)(i) = g(f(i))$ for all $1 \leq i \leq m$ .                                                                                |
| fong-spivak-invitation_F03477 | 306  | 1.000 | $1 \leq i \leq m$                                          | $1 \leq i \leq m$                               | $m \rightarrow p$ given by $(f \circ g)(i) = g(f(i))$ for all $1 \leq i \leq m$ .                                                                                |
| fong-spivak-invitation_F03478 | 306  | 0.426 | $\mathrm{id}_m: m \rightarrowtail m$                       | $\mathrm{id}_m: m \rightarrow m$                | 4. The identity $\mathrm{id}_m: m \rightarrow m$ is given by $\mathrm{id}_m(i) = i$ for all $1 \leq i \leq m$ . Here is a picture of $\mathrm{id}_2$             |
| fong-spivak-invitation_F03479 | 306  | 0.426 | $\mathrm{id}_m(i) = i$                                     | $\mathrm{id}_m(i) = i$                          | 4. The identity $\mathrm{id}_m: m \rightarrow m$ is given by $\mathrm{id}_m(i) = i$ for all $1 \leq i \leq m$ . Here is a picture of $\mathrm{id}_2$             |
| fong-spivak-invitation_F03480 | 306  | 0.426 | $\mathrm{id}_2$                                            | $\mathrm{id}_2$                                 | 4. The identity $\mathrm{id}_m: m \rightarrow m$ is given by $\mathrm{id}_m(i) = i$ for all $1 \leq i \leq m$ . Here is a picture of $\mathrm{id}_2$             |
| fong-spivak-invitation_F03481 | 306  | 1.000 | $\mathrm{id}_8$                                            | $\mathrm{id}_8$                                 | and $\mathrm{id}_8$ :                                                                                                                                            |
| fong-spivak-invitation_F03482 | 307  | 1.000 | $\sigma_{3,5}: 8 \rightarrowtail 8$                        | $\sigma_{3,5}: 8 \rightarrow 8$                 | 5. Here is a picture of the symmetry $\sigma_{3,5}: 8 \rightarrow 8$ :                                                                                           |
| fong-spivak-invitation_F03483 | 307  | 0.996 | $m \preceq n$                                              | $m \preceq n$                                   | whose order is denoted $m \preceq n$ , and with the property that whenever $m_1 \preceq n_1$ and $m_2 \preceq n_2$ hold                                          |
| fong-spivak-invitation_F03484 | 307  | 0.996 | $m_1 \preceq n_1$                                          | $m_1 \preceq n_1$                               | whose order is denoted $m \preceq n$ , and with the property that whenever $m_1 \preceq n_1$ and $m_2 \preceq n_2$ hold                                          |
| fong-spivak-invitation_F03485 | 307  | 0.996 | $m_2 \preceq n_2$                                          | $m_2 \preceq n_2$                               | whose order is denoted $m \preceq n$ , and with the property that whenever $m_1 \preceq n_1$ and $m_2 \preceq n_2$ hold                                          |
| fong-spivak-invitation_F03486 | 307  | 1.000 | $m_1 + m_2 \preceq n_1 + n_2$                              | $m_1 + m_2 \preceq n_1 + n_2$                   | then $m_1 + m_2 \preceq n_1 + n_2$ does too.                                                                                                                     |
| fong-spivak-invitation_F03487 | 307  | 1.000 | $d \in \mathbb{N}$                                         | $d \in \mathbb{N}$                              | 2. Take $\preceq$ to be the usual order, $m \preceq n$ iff there exists $d \in \mathbb{N}$ with $d + m = n$ .                                                    |
| fong-spivak-invitation_F03488 | 307  | 1.000 | $d + m = n$                                                | $d + m = n$                                     | 2. Take $\preceq$ to be the usual order, $m \preceq n$ iff there exists $d \in \mathbb{N}$ with $d + m = n$ .                                                    |
| fong-spivak-invitation_F03489 | 307  | 0.999 | $m = n + d$                                                | $m = n + d$                                     | 3. Take $\preceq$ to be the reverse of the usual order, $m \preceq n$ iff there exists $d \in \mathbb{N}$ with $m = n + d$ .                                     |
| fong-spivak-invitation_F03490 | 307  | 1.000 | $q \in \mathbb{N}$                                         | $q \in \mathbb{N}$                              | 5. (Non-example.) Take $\preceq$ to be the division order, $m \preceq n$ iff there exists $q \in \mathbb{N}$ with $m * q = d$ .                                  |
| fong-spivak-invitation_F03491 | 307  | 1.000 | $m * q = d$                                                | $m * q = d$                                     | 5. (Non-example.) Take $\preceq$ to be the division order, $m \preceq n$ iff there exists $q \in \mathbb{N}$ with $m * q = d$ .                                  |
| fong-spivak-invitation_F03492 | 307  | 1.000 | $2 \preceq 4$                                              | $2 \preceq 4$                                   | This is a perfectly good poset, but it does not satisfy the monotonicity property: we have $2 \preceq 4$                                                         |
| fong-spivak-invitation_F03493 | 307  | 0.958 | $3 \preceq 3$                                              | $3 \preceq 3$                                   | and $5 \preceq 7$ but not $5 \preceq 7$ .                                                                                                                        |
| fong-spivak-invitation_F03494 | 307  | 0.958 | $5 \preceq^? 7$                                            | $5 \preceq^? 7$                                 | and $3 \preceq 3$ but not $5 \preceq^? 7$ .                                                                                                                      |

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| fong-spivak-invitation_F03495 | 307  | ■ 0.492 | $\mathbf{B i j}(m, n):=\{f: \underline{m} \rightarrow \underline{n} \mid f$ | $\mathbf{Bij}(m, n) := \{f : \underline{m} \rightarrow \underline{n} \mid f$ | 1. $\mathbf{Bij}(m, n) := \{f : \underline{m} \rightarrow \underline{n} \mid f$ is a bijection}. Note that $\mathbf{Bij}(m, n) = \emptyset$ if $m \neq n$ and it             |
| fong-spivak-invitation_F03496 | 307  | ■ 0.492 | $\mathbf{B i j}(m, n)=\varnothing$                                          | $\mathbf{Bij}(m, n) = \emptyset$                                             | 1. $\mathbf{Bij}(m, n) := \{f : \underline{m} \rightarrow \underline{n} \mid f$ is a bijection}. Note that $\mathbf{Bij}(m, n) = \emptyset$ if $m \neq n$ and it             |
| fong-spivak-invitation_F03497 | 307  | ■ 1.000 | $n \rightarrow n$                                                           | $n \rightarrow n$                                                            | 2. The identity map $n \rightarrow n$ is the bijection $\underline{n} \rightarrow \underline{n}$ sending $i \mapsto i$ .                                                     |
| fong-spivak-invitation_F03498 | 307  | ■ 1.000 | $\underline{n} \rightarrow \underline{n}$                                   | $\underline{n} \rightarrow \underline{n}$                                    | 2. The identity map $n \rightarrow n$ is the bijection $\underline{n} \rightarrow \underline{n}$ sending $i \mapsto i$ .                                                     |
| fong-spivak-invitation_F03499 | 307  | ■ 1.000 | $i \mapsto i$                                                               | $i \mapsto i$                                                                | 2. The identity map $n \rightarrow n$ is the bijection $\underline{n} \rightarrow \underline{n}$ sending $i \mapsto i$ .                                                     |
| fong-spivak-invitation_F03500 | 307  | ■ 1.000 | $m+n \rightarrow n+m$                                                       | $m+n \rightarrow n+m$                                                        | 3. The symmetry map $m+n \rightarrow n+m$ is the bijection $\sigma_{m,n} : \underline{m+n} \rightarrow \underline{n+m}$ given by                                             |
| fong-spivak-invitation_F03501 | 307  | ■ 1.000 | $\sigma_{m, n}: \underline{m+n} \rightarrow \underline{n+m}$                | $\sigma_{m,n} : \underline{m+n} \rightarrow \underline{n+m}$                 | 3. The symmetry map $m+n \rightarrow n+m$ is the bijection $\sigma_{m,n} : \underline{m+n} \rightarrow \underline{n+m}$ given by                                             |
| fong-spivak-invitation_F03502 | 307  | ■ 1.000 | $n \rightarrow p$                                                           | $n \rightarrow p$                                                            | 4. Composition of bijections $m \rightarrow n$ and $n \rightarrow p$ is just their composition as functions,                                                                 |
| fong-spivak-invitation_F03503 | 307  | ■ 1.000 | $(f+g):(m+$                                                                 | $(f+g):(m+$                                                                  | 5. Given bijections $f: m \rightarrow m'$ and $g: n \rightarrow n'$ , their monoidal product $(f+g):(m+$                                                                     |
| fong-spivak-invitation_F03504 | 307  | ■ 1.000 | $n) \rightarrow (m^{\prime}+n^{\prime})$                                    | $n) \rightarrow (m' + n')$                                                   | $\underline{n}) \rightarrow (m' + n')$ is given by                                                                                                                           |
| fong-spivak-invitation_F03505 | 307  | ■ 1.000 | $i \sim j$                                                                  | $i \sim j$                                                                   | reflexive relation, i.e. where $i \sim j$ iff $i = j$ .                                                                                                                      |
| fong-spivak-invitation_F03506 | 307  | ■ 1.000 | $i=j$                                                                       | $i = j$                                                                      | reflexive relation, i.e. where $i \sim j$ iff $i = j$ .                                                                                                                      |
| fong-spivak-invitation_F03507 | 308  | ■ 0.868 | $\underline{m+n+n+m}$                                                       | $\underline{m+n+n+m}$                                                        | 3. The symmetry map $\sigma_{m,n}$ , as an equivalence relation on $\underline{m+n+n+m}$ is “the obvious                                                                     |
| fong-spivak-invitation_F03508 | 308  | ■ 1.000 | $ i-j =m+n+n$                                                               | $ i-j  = m+n+n$                                                              | • $\underline{i-j} \equiv m+n+n$ , or                                                                                                                                        |
| fong-spivak-invitation_F03509 | 308  | ■ 0.996 | $m+1 \leq i \leq m+n+n$                                                     | $m+1 \leq i \leq m+n+n$                                                      | • $\underline{m+1 \leq i \leq m+n+n}$ and $m+1 \leq j \leq m+n+n$ and $ i-j  = n$ .                                                                                          |
| fong-spivak-invitation_F03510 | 308  | ■ 0.996 | $m+1 \leq j \leq m+n+n$                                                     | $m+1 \leq j \leq m+n+n$                                                      | • $m+1 \leq i \leq m+n+n$ and $\underline{m+1 \leq j \leq m+n+n}$ and $ i-j  = n$ .                                                                                          |
| fong-spivak-invitation_F03511 | 308  | ■ 0.996 | $ i-j =n$                                                                   | $ i-j  = n$                                                                  | • $m+1 \leq i \leq m+n+n$ and $m+1 \leq j \leq m+n+n$ and $\underline{i-j} \equiv n$ .                                                                                       |
| fong-spivak-invitation_F03512 | 308  | ■ 0.808 | $\underline{m+n}$                                                           | $\underline{m+n}$                                                            | 5. Given equivalence relations $\sim$ on $\underline{m+n}$ and $\sim'$ on $\underline{m'+n'}$ , we need an equivalence rela-                                                 |
| fong-spivak-invitation_F03513 | 308  | ■ 0.808 | $\sim^{\prime}$                                                             | $\sim'$                                                                      | 5. Given equivalence relations $\sim$ on $\underline{m+n}$ and $\sim'$ on $\underline{m'+n'}$ , we need an equivalence rela-                                                 |
| fong-spivak-invitation_F03514 | 308  | ■ 0.808 | $\underline{m^{\prime}+n^{\prime}}$                                         | $\underline{m' + n'}$                                                        | 5. Given equivalence relations $\sim$ on $\underline{m+n}$ and $\sim'$ on $\underline{m'+n'}$ , we need an equivalence rela-                                                 |
| fong-spivak-invitation_F03515 | 308  | ■ 0.777 | $\left(\sim+\sim^{\prime}\right)$                                           | $(\sim + \sim')$                                                             | tion $(\sim + \sim')$ on $\underline{m+n+m'+n'}$ . We take it to be “the obvious thing,” namely “using                                                                       |
| fong-spivak-invitation_F03516 | 308  | ■ 0.777 | $m+n+m^{\prime}+n^{\prime}$                                                 | $m+n+m'+n'$                                                                  | tion $(\sim + \sim')$ on $\underline{m+n+m'+n'}$ . We take it to be “the obvious thing,” namely “using                                                                       |
| fong-spivak-invitation_F03517 | 308  | ■ 0.990 | $\quad i \leq m+n$                                                          | $i \leq m+n$                                                                 | • $i \leq m+n$ and $j \leq m+n$ and $i \sim j$ , or                                                                                                                          |
| fong-spivak-invitation_F03518 | 308  | ■ 0.990 | $j \leq m+n$                                                                | $j \leq m+n$                                                                 | • $i \leq m+n$ and $j \leq m+n$ and $i \sim j$ , or                                                                                                                          |
| fong-spivak-invitation_F03519 | 308  | ■ 0.793 | $m+n+1 \leq i$                                                              | $m+n+1 \leq i$                                                               | • $m+n+1 \leq i$ and $m+n+1 \leq j$ and $i \sim' j$ .                                                                                                                        |
| fong-spivak-invitation_F03520 | 308  | ■ 0.793 | $m+n+1 \leq j$                                                              | $m+n+1 \leq j$                                                               | • $m+n+1 \leq i$ and $m+n+1 \leq j$ and $i \sim' j$ .                                                                                                                        |
| fong-spivak-invitation_F03521 | 308  | ■ 0.793 | $i \sim^{\prime} j$                                                         | $i \sim' j$                                                                  | • $m+n+1 \leq i$ and $m+n+1 \leq j$ and $i \sim' j$ .                                                                                                                        |
| fong-spivak-invitation_F03522 | 308  | ■ 0.998 | $\boldsymbol{\operatorname{Rel}}(m, n)$                                     | $\mathbf{Rel}(m, n)$                                                         | 1. $\mathbf{Rel}(m, n)$ is the set of relations on the set $\underline{m} \times \underline{n}$ , i.e. the set of subsets of $\underline{m} \times \underline{n}$ , i.e. its |

| Identifier                    | Page | Conf.   | LaTeX source                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                               | Image                                                                                                                                                                                                                                                     |
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| fong-spivak-invitation_F03523 | 308  | ■ 0.998 | $\underline{m} \times \underline{n}$                                                                                                                                                                                   | $\underline{m} \times \underline{n}$                                                                                                                                                                                   | 1. $\mathbf{Rel}(m, n)$ is the set of relations on the set $\underline{m} \times \underline{n}$ , i.e. the set of subsets of $\underline{m} \times \underline{n}$ , i.e. its                                                                              |
| fong-spivak-invitation_F03524 | 308  | ■ 1.000 | $\{(i, j) \in \underline{n} \times \underline{n} \mid i = j\}$                                                                                                                                                         | $\{(i, j) \in \underline{n} \times \underline{n} \mid i = j\}$                                                                                                                                                         | 2. The identity map $n \rightarrow n$ is the subset $\{(i, j) \in \underline{n} \times \underline{n} \mid i = j\}$ .                                                                                                                                      |
| fong-spivak-invitation_F03525 | 308  | ■ 1.000 | $(i, j) \in (\underline{m+n}) \times (\underline{n+m})$                                                                                                                                                                | $(i, j) \in (\underline{m+n}) \times (\underline{n+m})$                                                                                                                                                                | 3. The symmetry map $m+n \rightarrow n+m$ is the subset of pairs $(i, j) \in (\underline{m+n}) \times (\underline{n+m})$ such                                                                                                                             |
| fong-spivak-invitation_F03526 | 308  | ■ 0.998 | $i \leq m$                                                                                                                                                                                                             | $i \leq m$                                                                                                                                                                                                             | • $i \leq m$ and $m+1 \leq j$ and $i+m = j$ , or                                                                                                                                                                                                          |
| fong-spivak-invitation_F03527 | 308  | ■ 0.998 | $m+1 \leq j$                                                                                                                                                                                                           | $m+1 \leq j$                                                                                                                                                                                                           | • $i \leq m$ and $m+1 \leq j$ and $i+m = j$ , or                                                                                                                                                                                                          |
| fong-spivak-invitation_F03528 | 308  | ■ 0.998 | $i+m=j$                                                                                                                                                                                                                | $i+m=j$                                                                                                                                                                                                                | • $i \leq m$ and $m+1 \leq j$ and $i+m = j$ , or                                                                                                                                                                                                          |
| fong-spivak-invitation_F03529 | 308  | ■ 0.974 | $m+1 \leq i$                                                                                                                                                                                                           | $m+1 \leq i$                                                                                                                                                                                                           | • $\underline{m+1} \leq$ and $j \leq m$ and $j+m = i$ .                                                                                                                                                                                                   |
| fong-spivak-invitation_F03530 | 308  | ■ 0.974 | $j \leq m$                                                                                                                                                                                                             | $j \leq m$                                                                                                                                                                                                             | • $m+1 \leq i$ and $j \leq m$ and $j+m = i$ .                                                                                                                                                                                                             |
| fong-spivak-invitation_F03531 | 308  | ■ 0.974 | $j+m=i$                                                                                                                                                                                                                | $j+m=i$                                                                                                                                                                                                                | • $m+1 \leq i$ and $j \leq m$ and $j+m = i$ .                                                                                                                                                                                                             |
| fong-spivak-invitation_F03532 | 308  | ■ 0.667 | $R^{\prime} \subseteq \underline{m}^{\prime} \times \underline{n}^{\prime}$                                                                                                                                            | $R' \subseteq \underline{m}' \times \underline{n}'$                                                                                                                                                                    | 5. Given a relation $R \subseteq \underline{m} \times \underline{n}$ and a relation $R' \subseteq \underline{m}' \times \underline{n}'$ , we need a relation $(R + R') \subseteq$                                                                         |
| fong-spivak-invitation_F03533 | 308  | ■ 0.667 | $(R + R^{\prime}) \subseteq$                                                                                                                                                                                           | $(R + R') \subseteq$                                                                                                                                                                                                   | 5. Given a relation $R \subseteq \underline{m} \times \underline{n}$ and a relation $R' \subseteq \underline{m}' \times \underline{n}'$ , we need a relation $(R + R') \subseteq$                                                                         |
| fong-spivak-invitation_F03534 | 308  | ■ 1.000 | $\underline{m+m^{\prime}} \times \underline{n+n^{\prime}}$                                                                                                                                                             | $\underline{m+m'} \times \underline{n+n'}$                                                                                                                                                                             | $\underline{m+m'} \times \underline{n+n'}$ . As stated in the example (footnote), this can be given by a universal                                                                                                                                        |
| fong-spivak-invitation_F03535 | 308  | ■ 0.580 | $R_1 \sqcup R_2 \subseteq (\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1} \sqcup \underline{m_2}) \times (\underline{n_1} \sqcup \underline{n_2})$ | $R_1 \sqcup R_2 \subseteq (\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1} \sqcup \underline{m_2}) \times (\underline{n_1} \sqcup \underline{n_2})$ | given by $\underline{R_1} \sqcup \underline{R_2} \subseteq (\underline{m_1} \times \underline{n_1}) \sqcup (\underline{m_2} \times \underline{n_2}) \subseteq (\underline{m_1} \sqcup \underline{m_2}) \times (\underline{n_1} \sqcup \underline{n_2})$ . |
| fong-spivak-invitation_F03536 | 308  | ■ 0.818 | $(n, p)$                                                                                                                                                                                                               | $(n, p)$                                                                                                                                                                                                               | Composition of an $(m, n)$ -port graph $G$ and an $(n, p)$ -port graph $H$ looks visually like sticking                                                                                                                                                   |
| fong-spivak-invitation_F03537 | 309  | ■ 1.000 | $\leq_P$                                                                                                                                                                                                               | $\leq_P$                                                                                                                                                                                                               | We have a relation $R \subseteq P \times P$ which generates a preorder $\leq_P$ on $P$ , we have an arbitrary preorder                                                                                                                                    |
| fong-spivak-invitation_F03538 | 309  | ■ 0.999 | $x \leq_P y$                                                                                                                                                                                                           | $x \leq_P y$                                                                                                                                                                                                           | monotone, i.e. that for every $x \leq_P y$ we have $f(x) \leq_Q f(y)$ . By definition of $P$ being the                                                                                                                                                    |
| fong-spivak-invitation_F03539 | 309  | ■ 0.993 | $x_{\{0\}}, \ldots, x_{\{n\}}$                                                                                                                                                                                         | $x_0, \dots, x_n$                                                                                                                                                                                                      | reflexive, transitive closure of $R$ , we have $x \leq_P y$ iff there exists $n \in \mathbb{N}$ and $\underline{x_0, \dots, x_n}$ in $P$                                                                                                                  |
| fong-spivak-invitation_F03540 | 309  | ■ 1.000 | $x_{\{0\}}=x$                                                                                                                                                                                                          | $x_0 = x$                                                                                                                                                                                                              | with $x_0 = x$ and $x_n = y$ and $R(x_i, x_{i+1})$ for each $0 \leq i \leq n-1$ . (The case $n = 0$ handles                                                                                                                                               |
| fong-spivak-invitation_F03541 | 309  | ■ 1.000 | $x_{\{n\}}=y$                                                                                                                                                                                                          | $x_n = y$                                                                                                                                                                                                              | with $x_0 = x$ and $x_n = y$ and $R(x_i, x_{i+1})$ for each $0 \leq i \leq n-1$ . (The case $n = 0$ handles                                                                                                                                               |
| fong-spivak-invitation_F03542 | 309  | ■ 1.000 | $R(x_{\{i\}}, x_{\{i+1\}})$                                                                                                                                                                                            | $R(x_i, x_{i+1})$                                                                                                                                                                                                      | with $x_0 = x$ and $x_n = y$ and $R(x_i, x_{i+1})$ for each $0 \leq i \leq n-1$ . (The case $n = 0$ handles                                                                                                                                               |
| fong-spivak-invitation_F03543 | 309  | ■ 1.000 | $0 \leq i \leq n-1$                                                                                                                                                                                                    | $0 \leq i \leq n-1$                                                                                                                                                                                                    | with $x_0 = x$ and $x_n = y$ and $R(x_i, x_{i+1})$ for each $\underline{0 \leq i \leq n-1}$ . (The case $n = 0$ handles                                                                                                                                   |
| fong-spivak-invitation_F03544 | 309  | ■ 1.000 | $f(x_{\{i\}}) \leq_Q f(x_{\{i+1\}})$                                                                                                                                                                                   | $f(x_i) \leq_Q f(x_{i+1})$                                                                                                                                                                                             | reflexivity.) But then, by assumption, $R(x_i, x_{i+1})$ implies $f(x_i) \leq_Q f(x_{i+1})$ for each $i$ . By                                                                                                                                             |
| fong-spivak-invitation_F03545 | 309  | ■ 1.000 | $f(x_{\{0\}}) \leq_Q f(x_{\{i\}})$                                                                                                                                                                                     | $f(x_0) \leq_Q f(x_i)$                                                                                                                                                                                                 | induction on $i$ we show that $f(x_0) \leq_Q f(x_i)$ for all $0 \leq i \leq n$ , at which point we are done.                                                                                                                                              |
| fong-spivak-invitation_F03546 | 309  | ■ 0.725 | $x \leq \leq_P y$                                                                                                                                                                                                      | $x \leq \leq_P y$                                                                                                                                                                                                      | 2. Suppose now that $f$ is monotone, and take $x, y \in P$ for which $R(x, y)$ holds. Then $x \leq_P y$                                                                                                                                                   |
| fong-spivak-invitation_F03547 | 309  | ■ 0.945 | $n=1, x_{\{0\}}=x$                                                                                                                                                                                                     | $n = 1, x_0 = x$                                                                                                                                                                                                       | the above description is with $\underline{n=1, x_0=x}$ , and $x_n = y$ , which we said implies $x \leq_P y$ .)                                                                                                                                            |
| fong-spivak-invitation_F03548 | 309  | ■ 0.983 | $R(g(a), g(b))$                                                                                                                                                                                                        | $R(g(a), g(b))$                                                                                                                                                                                                        | 1. If $R(g(a), g(b))$ holds for all $a \leq_Q b$ then $g$ is monotone, because $R(x, y)$ implies $x \leq_P y$ .                                                                                                                                           |
| fong-spivak-invitation_F03549 | 309  | ■ 0.983 | $a \leq_Q b$                                                                                                                                                                                                           | $a \leq_Q b$                                                                                                                                                                                                           | 1. If $R(g(a), g(b))$ holds for all $a \leq_Q b$ then $g$ is monotone, because $R(x, y)$ implies $x \leq_P y$ .                                                                                                                                           |

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| fong-spivak-invitation_F03550 | 309  | ■ 1.000 | $(g(a), g(b)) \notin R$                                                       | $(g(a), g(b)) \notin R$                                                       | $a \leq_Q b$ and $\boxed{g(a), g(b)} \notin R$ . Indeed, take $Q := \{1\}$ to be the free preorder on one element,                                                                               |
| fong-spivak-invitation_F03551 | 309  | ■ 1.000 | $Q := \{1\}$                                                                  | $Q := \{1\}$                                                                  | $a \leq_Q b$ and $(g(a), g(b)) \notin R$ . Indeed, take $Q := \{1\}$ to be the free preorder on one element,                                                                                     |
| fong-spivak-invitation_F03552 | 309  | ■ 1.000 | $P := \{1\}$                                                                  | $P := \{1\}$                                                                  | and take $P := \{1\}$ with $R = \emptyset$ . Then the unique function $g: Q \rightarrow P$ is monotone (because                                                                                  |
| fong-spivak-invitation_F03553 | 309  | ■ 1.000 | $R = \text{\varnothing}$                                                      | $R = \emptyset$                                                               | and take $P := \{1\}$ with $R = \emptyset$ . Then the unique function $g: Q \rightarrow P$ is monotone (because                                                                                  |
| fong-spivak-invitation_F03554 | 309  | ■ 1.000 | $(g(1), g(1)) \notin R$                                                       | $(g(1), g(1)) \notin R$                                                       | $\leq_P$ is reflexive even though $R$ is empty), and yet $\boxed{g(1), g(1)} \notin R$ .                                                                                                         |
| fong-spivak-invitation_F03555 | 309  | ■ 0.991 | $q \in \operatorname{Mor}(\mathcal{C})$                                       | $q \in \operatorname{Mor}(\mathcal{C})$                                       | exactly one element of $\operatorname{Ob}(\mathcal{C})$ . So for $\operatorname{dom}$ we take any $q \in \operatorname{Mor}(\mathcal{C})$ , view it as a morphism                                |
| fong-spivak-invitation_F03556 | 309  | ■ 0.997 | $q: y \rightarrow z$                                                          | $q: y \rightarrow z$                                                          | $\boxed{q: y \Rightarrow z}$ , and send it to its domain $y$ . Similarly for $\operatorname{cod}$ : we put $\operatorname{cod}(q) := z$ .                                                        |
| fong-spivak-invitation_F03557 | 309  | ■ 0.997 | $\operatorname{cod}(q) := z$                                                  | $\operatorname{cod}(q) := z$                                                  | $q: y \rightarrow z$ , and send it to its domain $y$ . Similarly for $\operatorname{cod}$ : we put $\operatorname{cod}(q) := z$ .                                                                |
| fong-spivak-invitation_F03558 | 309  | ■ 1.000 | $F: \mathcal{G} \rightarrow \mathcal{C}$                                      | $F: \mathcal{G} \rightarrow \mathcal{C}$                                      | 2. Suppose first that we are given a functor $F: \mathcal{G} \rightarrow \mathcal{C}$ . On objects we have a function                                                                            |
| fong-spivak-invitation_F03559 | 309  | ■ 0.990 | $\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$   | $\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$   | $\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$ , and this defines $f$ since $\operatorname{Ob}(\mathcal{G}) = V$ . On morphisms, first note that                    |
| fong-spivak-invitation_F03560 | 309  | ■ 0.990 | $\operatorname{Ob}(\mathcal{G}) = V$                                          | $\operatorname{Ob}(\mathcal{G}) = V$                                          | $\operatorname{Ob}(\mathcal{G}) \rightarrow \operatorname{Ob}(\mathcal{C})$ , and this defines $f$ since $\operatorname{Ob}(\mathcal{G}) = V$ . On morphisms, first note that                    |
| fong-spivak-invitation_F03561 | 309  | ■ 1.000 | $\operatorname{Mor}(\mathcal{G})$                                             | $\operatorname{Mor}(\mathcal{G})$                                             | the arrows of graph $G$ are exactly the length=1 paths in $G$ , whereas $\operatorname{Mor}(\mathcal{G})$ is the set of                                                                          |
| fong-spivak-invitation_F03562 | 309  | ■ 1.000 | $A \subseteq \operatorname{Mor}(\mathcal{G})$                                 | $A \subseteq \operatorname{Mor}(\mathcal{G})$                                 | all paths in $G$ , so we have an inclusion $A \subseteq \operatorname{Mor}(\mathcal{G})$ . The functor $F$ provides a function                                                                   |
| fong-spivak-invitation_F03563 | 310  | ■ 1.000 | $\operatorname{Mor}(\mathcal{G}) \rightarrow \operatorname{Mor}(\mathcal{C})$ | $\operatorname{Mor}(\mathcal{G}) \rightarrow \operatorname{Mor}(\mathcal{C})$ | $\operatorname{Mor}(\mathcal{G}) \rightarrow \operatorname{Mor}(\mathcal{C})$ , which we can restrict to $A$ to obtain $g: A \rightarrow \operatorname{Mor}(\mathcal{C})$ . All functors satisfy |
| fong-spivak-invitation_F03564 | 310  | ■ 1.000 | $\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$                         | $\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$                         | $\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$ and $\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$ for any $r: w \rightarrow x$ . In particular when                                |
| fong-spivak-invitation_F03565 | 310  | ■ 1.000 | $\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$                         | $\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$                         | $\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$ and $\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$ for any $r: w \rightarrow x$ . In particular when                                |
| fong-spivak-invitation_F03566 | 310  | ■ 1.000 | $r: w \rightarrow x$                                                          | $r: w \rightarrow x$                                                          | $\operatorname{dom}(F(r)) = F(\operatorname{dom}(r))$ and $\operatorname{cod}(F(r)) = F(\operatorname{cod}(r))$ for any $r: w \rightarrow x$ . In particular when                                |
| fong-spivak-invitation_F03567 | 310  | ■ 0.976 | $r \in A$                                                                     | $r \in A$                                                                     | $r \in A$ is an arrow we have $\operatorname{dom}(r) = s(r)$ and $\operatorname{cod}(r) = t(r)$ . Thus we have found $(f, g)$ with                                                               |
| fong-spivak-invitation_F03568 | 310  | ■ 0.976 | $\operatorname{dom}(r) = s(r)$                                                | $\operatorname{dom}(r) = s(r)$                                                | $r \in A$ is an arrow we have $\operatorname{dom}(r) = s(r)$ and $\operatorname{cod}(r) = t(r)$ . Thus we have found $(f, g)$ with                                                               |
| fong-spivak-invitation_F03569 | 310  | ■ 0.976 | $\operatorname{cod}(r) = t(r)$                                                | $\operatorname{cod}(r) = t(r)$                                                | $r \in A$ is an arrow we have $\operatorname{dom}(r) = s(r)$ and $\operatorname{cod}(r) = t(r)$ . Thus we have found $(f, g)$ with                                                               |
| fong-spivak-invitation_F03570 | 310  | ■ 1.000 | $p := (v_0, a_1, a_2, \dots, a_n)$                                            | $p := (v_0, a_1, a_2, \dots, a_n)$                                            | $\boxed{p := (v_0, a_1, a_2, \dots, a_n)}$ in $G$ , where $v_0 \in V$ , $a_i \in A$ , $v_0 = s(a_1)$ , and $t(a_i) = s(a_{i+1})$                                                                 |
| fong-spivak-invitation_F03571 | 310  | ■ 1.000 | $v_0 \in V, a_i \in A, v_0 = s(a_1)$                                          | $v_0 \in V, a_i \in A, v_0 = s(a_1)$                                          | $p := (v_0, a_1, a_2, \dots, a_n)$ in $G$ , where $\boxed{v_0 \in V, a_i \in A, v_0 = s(a_1)}$ and $t(a_i) = s(a_{i+1})$                                                                         |
| fong-spivak-invitation_F03572 | 310  | ■ 1.000 | $1 \leq i \leq n-1$                                                           | $1 \leq i \leq n-1$                                                           | for all $1 \leq i \leq n-1$ . Then $g(a_i)$ is a morphism in $\mathcal{C}$ whose domain is $f(v_0)$ and the                                                                                      |
| fong-spivak-invitation_F03573 | 310  | ■ 1.000 | $g(a_i)$                                                                      | $g(a_i)$                                                                      | for all $1 \leq i \leq n-1$ . Then $g(a_i)$ is a morphism in $\mathcal{C}$ whose domain is $f(v_0)$ and the                                                                                      |
| fong-spivak-invitation_F03574 | 310  | ■ 1.000 | $f(v_0)$                                                                      | $f(v_0)$                                                                      | for all $1 \leq i \leq n-1$ . Then $g(a_i)$ is a morphism in $\mathcal{C}$ whose domain is $f(v_0)$ and the                                                                                      |
| fong-spivak-invitation_F03575 | 310  | ■ 1.000 | $g(a_{i+1})$                                                                  | $g(a_{i+1})$                                                                  | morphisms $g(a_i)$ and $g(a_{i+1})$ are composable for every $1 \leq i \leq n-1$ . We then take                                                                                                  |
| fong-spivak-invitation_F03576 | 310  | ■ 0.442 | $F(p) := \operatorname{id}_{f(v_0)} ; g(a_1) ; \dots ; g(a_n)$                | $F(p) := \operatorname{id}_{f(v_0)} ; g(a_1) ; \dots ; g(a_n)$                | $F(p) := \operatorname{id}_{f(v_0)} \circ g(a_1) \circ \dots \circ g(a_n)$ to be the composite. It is easy to check that this is                                                                 |
| fong-spivak-invitation_F03577 | 310  | ■ 1.000 | $(f', g')$                                                                    | $(f', g')$                                                                    | turn them into a functor $F: \mathcal{G} \rightarrow \mathcal{C}$ and then extract the new pair of functions $\boxed{f', g'}$ , then                                                             |

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| fong-spivak-invitation_F03578 | 310  | ■ 0.985 | $f=f^{\prime}$                                                                                                                                                                      | $f = f'$                                                                                                                                                                            | $f = f'$ and $g = g'$ . Finally, given a functor $F: \mathcal{G} \rightarrow \mathcal{C}$ , we extract the pair of functions $(f, g)$                                                                                                     |
| fong-spivak-invitation_F03579 | 310  | ■ 0.985 | $g=g^{\prime}$                                                                                                                                                                      | $g = g'$                                                                                                                                                                            | $f = f'$ and $g = g'$ . Finally, given a functor $F: \mathcal{G} \rightarrow \mathcal{C}$ , we extract the pair of functions $(f, g)$                                                                                                     |
| fong-spivak-invitation_F03580 | 310  | ■ 1.000 | $F^{\prime}: \mathcal{G} \rightarrow \mathcal{C}$                                                                                                                                   | $F' : \mathcal{G} \rightarrow \mathcal{C}$                                                                                                                                          | as above and then turn them into a new functor $F': \mathcal{G} \rightarrow \mathcal{C}$ . It is clear that $F$ and $F'$ act the                                                                                                          |
| fong-spivak-invitation_F03581 | 310  | ■ 1.000 | $F^{\prime}$                                                                                                                                                                        | $F'$                                                                                                                                                                                | as above and then turn them into a new functor $F': \mathcal{G} \rightarrow \mathcal{C}$ . It is clear that $F$ and $F'$ act the                                                                                                          |
| fong-spivak-invitation_F03582 | 310  | ■ 0.366 | $\operatorname{Mor}(\mathcal{C}), \operatorname{Ob}(\mathcal{C})$                                                                                                                   | $\operatorname{Mor}(\mathcal{C}), \operatorname{Ob}(\mathcal{C})$                                                                                                                   | 3. $(\operatorname{Mor}(\mathcal{C}), \operatorname{Ob}(\mathcal{C}), \operatorname{dom}, \operatorname{cod})$ is a graph; let's denote it $U(\mathcal{C}) \in \mathbf{Grph}$ . We have functors                                          |
| fong-spivak-invitation_F03583 | 310  | ■ 0.366 | $U(\mathcal{C}) \in$                                                                                                                                                                | $U(\mathcal{C}) \in$                                                                                                                                                                | 3. $(\operatorname{Mor}(\mathcal{C}), \operatorname{Ob}(\mathcal{C}), \operatorname{dom}, \operatorname{cod})$ is a graph; let's denote it $U(\mathcal{C}) \in \mathbf{Grph}$ . We have functors                                          |
| fong-spivak-invitation_F03584 | 310  | ■ 1.000 | $a^i * a^j = a^{i+j}$                                                                                                                                                               | $a^i * a^j = a^{i+j}$                                                                                                                                                               | $a^i * a^j = a^{i+j}$ .                                                                                                                                                                                                                   |
| fong-spivak-invitation_F03585 | 310  | ■ 1.000 | $a^i \mapsto i$                                                                                                                                                                     | $a^i \mapsto i$                                                                                                                                                                     | 2. This is isomorphic to $\mathbb{N}$ , by sending $a^i \mapsto i$ .                                                                                                                                                                      |
| fong-spivak-invitation_F03586 | 310  | ■ 0.600 | $\{a, b\}$                                                                                                                                                                          | $\{a, b\}$                                                                                                                                                                          | 3. The elements of the free monoid on the set $\{a, b\}$ are 'words in $a$ and $b$ ,' each of which we will                                                                                                                               |
| fong-spivak-invitation_F03587 | 310  | ■ 0.714 | $\operatorname{Free}(G, s, t)$                                                                                                                                                      | $\mathbf{Free}(G, s, t)$                                                                                                                                                            | We have two props: the prop of port graphs and the free prop $\mathbf{Free}(G, s, t)$ where                                                                                                                                               |
| fong-spivak-invitation_F03588 | 310  | ■ 0.698 | $(\Gamma, \ell)$                                                                                                                                                                    | $(\Gamma, \ell)$                                                                                                                                                                    | $(\Gamma, \ell)$ , where $\Gamma = (V, in, out, \iota)$ is an $(m, n)$ -port graph and $\ell: V \rightarrow G$ is a function, such                                                                                                        |
| fong-spivak-invitation_F03589 | 311  | ■ 1.000 | $\ell(v) \in G$                                                                                                                                                                     | $\ell(v) \in G$                                                                                                                                                                     | box with some left-hand ports and some right-hand ports – an arity – and $\ell(v) \in G$ is sup-                                                                                                                                          |
| fong-spivak-invitation_F03590 | 311  | ■ 0.512 | $\Gamma$                                                                                                                                                                            | $\Gamma$                                                                                                                                                                            | words, a morphism in our particular $\mathbf{Free}(G)$ can be identified with an $(m, n)$ -port graph $\Gamma$ ,                                                                                                                          |
| fong-spivak-invitation_F03591 | 311  | ■ 0.186 | $(f + \operatorname{id}_1 + \operatorname{id}_1) \stackrel{\circ}{9} (\sigma + \operatorname{id}_1) \stackrel{\circ}{9} (\operatorname{id}_1 + h) \stackrel{\circ}{\sigma} \circ g$ | $(f + \operatorname{id}_1 + \operatorname{id}_1) \stackrel{\circ}{9} (\sigma + \operatorname{id}_1) \stackrel{\circ}{9} (\operatorname{id}_1 + h) \stackrel{\circ}{\sigma} \circ g$ | Here is a picture of $(f + \operatorname{id}_1 + \operatorname{id}_1) \stackrel{\circ}{9} (\sigma + \operatorname{id}_1) \stackrel{\circ}{9} (\operatorname{id}_1 + h) \stackrel{\circ}{\sigma} \circ g$ , in the free prop on generators |
| fong-spivak-invitation_F03592 | 311  | ■ 0.990 | $G = \{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$                                                                                                                | $G = \{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$                                                                                                                | $G = \{f: 1 \rightarrow 1, g: 2 \rightarrow 2, h: 2 \rightarrow 1\}$                                                                                                                                                                      |
| fong-spivak-invitation_F03593 | 311  | ■ 0.999 | $\{1\}, \{2\}, \{3\}$                                                                                                                                                               | $\{1\}, \{2\}, \{3\}$                                                                                                                                                               | is the set $\{1\}, \{2\}, \{3\}$ . It is subtly different from $S$ , but the two are naturally isomorphic, and                                                                                                                            |
| fong-spivak-invitation_F03594 | 311  | ■ 0.554 | $(R, 0, +, 1, *)$                                                                                                                                                                   | $(R, 0, +, 1, *)$                                                                                                                                                                   | 1. If $(R, 0, +, 1, *)$ is a rig, then the multiplicative identity $1 \in \operatorname{Mat}_n(R)$ is the usual $n$ -by- $n$                                                                                                              |
| fong-spivak-invitation_F03595 | 311  | ■ 1.000 | $n=4$                                                                                                                                                                               | $n = 4$                                                                                                                                                                             | those elements of $R$ ). So for $n = 4$ it is:                                                                                                                                                                                            |
| fong-spivak-invitation_F03596 | 311  | ■ 1.000 | $A, B \in \operatorname{Mat}_2(\mathbb{N})$                                                                                                                                         | $A, B \in \operatorname{Mat}_2(\mathbb{N})$                                                                                                                                         | 2. We choose $n = 2$ and hence need to find two elements $A, B \in \operatorname{Mat}_2(\mathbb{N})$ such that $A * B \neq$                                                                                                               |
| fong-spivak-invitation_F03597 | 311  | ■ 1.000 | $A * B \neq$                                                                                                                                                                        | $A * B \neq$                                                                                                                                                                        | 2. We choose $n = 2$ and hence need to find two elements $A, B \in \operatorname{Mat}_2(\mathbb{N})$ such that $A * B \neq$                                                                                                               |
| fong-spivak-invitation_F03598 | 311  | ■ 1.000 | $B * A$                                                                                                                                                                             | $B * A$                                                                                                                                                                             | $B * A$ .                                                                                                                                                                                                                                 |
| fong-spivak-invitation_F03599 | 311  | ■ 0.937 | $(A * B)(1, 1) = 0 * 0 + 1 * 1 = 1$                                                                                                                                                 | $(A * B)(1, 1) = 0 * 0 + 1 * 1 = 1$                                                                                                                                                 | One can calculate from the multiplication formula (recalled in Example 5.35) that $(A * B)(1, 1) = 0 * 0 + 1 * 1 = 1$ and $(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$ , which are not                                                             |
| fong-spivak-invitation_F03600 | 311  | ■ 1.000 | $(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$                                                                                                                                                 | $(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$                                                                                                                                                 | $(B * A)(1, 1) = 0 * 0 + 1 * 1 = 1$ and $(B * A)(1, 1) = 0 * 0 + 0 * 0 = 0$ , which are not                                                                                                                                               |
| fong-spivak-invitation_F03601 | 311  | ■ 1.000 | $(16x + 4y, x + 4y)$                                                                                                                                                                | $(16x + 4y, x + 4y)$                                                                                                                                                                | the resulting output signal is $(16x + 4y, x + 4y)$ .                                                                                                                                                                                     |
| fong-spivak-invitation_F03602 | 312  | ■ 0.997 | $(16x + 4y, x + 4y)$                                                                                                                                                                | $(16x + 4y, x + 4y)$                                                                                                                                                                |                                                                                                                                                                                                                                           |
| fong-spivak-invitation_F03603 | 312  | ■ 0.673 | $A = \begin{pmatrix} 3 & 3 & 1 \\ 2 & 0 & 4 \end{pmatrix}$                                                                                                                          | $A = \begin{pmatrix} 3 & 3 & 1 \\ 2 & 0 & 4 \end{pmatrix}$                                                                                                                          | The monoidal product of $A = \begin{pmatrix} 3 & 3 & 1 \\ 2 & 0 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 5 & 6 & 1 \end{pmatrix}$ is                                                                                               |

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| fong-spivak-invitation_F03604 | 312  | 0.673 | $B=\left(\begin{array}{llll}2&5&6&1\end{array}\right)$                                                   | $B = \left( \begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right)$     | The monoidal product of $A = \left( \begin{array}{ccc} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right)$ and $B = \left( \begin{array}{cccc} 2 & 5 & 6 & 1 \end{array} \right)$ is |
| fong-spivak-invitation_F03605 | 313  | 1.000 | $c_{\{n\}}$                                                                                              | $c_n$                                                                  | • For the first layer $g_1$ , take the monoidal product of $m$ copies of $c_n$ ,                                                                                             |
| fong-spivak-invitation_F03606 | 313  | 1.000 | $s_{\{a\}}: 1 \rightarrow 1$                                                                             | $s_a: 1 \rightarrow 1$                                                 | where $s_a: 1 \rightarrow 1$ is the signal flow graph generator “scalar multiplication by $a$ .” This layer                                                                  |
| fong-spivak-invitation_F03607 | 313  | 1.000 | $g_{\{3\}}: m \times n \rightarrow m \times n$                                                           | $g_3: m \times n \rightarrow m \times n$                               | signal flow graph $g_3: m \times n \rightarrow m \times n$ , i.e. the composite and monoidal product of swap and                                                             |
| fong-spivak-invitation_F03608 | 313  | 1.000 | $(i-1) m+j$                                                                                              | $(i-1)m+j$                                                             | identity maps, such that the $(i-1)m+j$ th input is sent to the $(j-1)n+i$ th output, for all                                                                                |
| fong-spivak-invitation_F03609 | 313  | 1.000 | $(j-1) n+i$                                                                                              | $(j-1)n+i$                                                             | identity maps, such that the $(i-1)m+j$ th input is sent to the $(j-1)n+i$ th output, for all                                                                                |
| fong-spivak-invitation_F03610 | 313  | 1.000 | $1 \leq j \leq m$                                                                                        | $1 \leq j \leq m$                                                      | $1 \leq i \leq n$ and $1 \leq j \leq m$ .                                                                                                                                    |
| fong-spivak-invitation_F03611 | 313  | 0.999 | $a_{\{m\}}$                                                                                              | $a_m$                                                                  | where $a_m$ is the signal flow graph that adds $m$ inputs to produce a single output:                                                                                        |
| fong-spivak-invitation_F03612 | 313  | 0.528 | $g=g_{\{1\}} \, g_{\{2\}}$                                                                               | $g = g_1 g_2$                                                          | Using Proposition 5.46, it is a straightforward but tedious calculation to show that $g = g_1 \circ g_2 \circ$                                                               |
| fong-spivak-invitation_F03613 | 313  | 0.735 | $g_{\{3\}}$                                                                                              | $g_3$                                                                  | $g_3 \circ g_4: m \rightarrow n$ has the property that $S(g) = M$ .                                                                                                          |
| fong-spivak-invitation_F03614 | 313  | 0.735 | $g_{\{4\}}: m \rightarrow n$                                                                             | $g_4: m \rightarrow n$                                                 | $g_3 \circ g_4: m \rightarrow n$ has the property that $S(g) = M$ .                                                                                                          |
| fong-spivak-invitation_F03615 | 315  | 0.723 | $(\mu \otimes \mathrm{id}) ; \mu = (\mathrm{id} \otimes \mu) \circ \mu$<br>$\stackrel{\circ}{\circ} \mu$ | $(\mu \otimes \mathrm{id}); \mu = (\mathrm{id} \otimes \mu) \circ \mu$ | (a) $(\mu \otimes \mathrm{id}) \circ \mu = (\mathrm{id} \otimes \mu) \circ \mu$                                                                                              |
| fong-spivak-invitation_F03616 | 315  | 1.000 | $R^{\{0\}}=\{()\}$                                                                                       | $R^0 = \{()\}$                                                         | $R$ . In particular, $R^0 = \{()\}$ consists of a single vector of dimension 0.                                                                                              |
| fong-spivak-invitation_F03617 | 315  | 0.549 | $\operatorname{prop}(\operatorname{Mat}(R))$                                                             | $\operatorname{prop}(\operatorname{Mat}(R))$                           | 1. $U$ preserves the monoidal unit because 0 is the monoidal unit of any $\operatorname{prop}(\mathbf{Mat}(R))$ is a $\operatorname{prop}$ ,                                 |
| fong-spivak-invitation_F03618 | 315  | 0.927 | $R^{\{0\}}$                                                                                              | $R^0$                                                                  | $\{1\}$ is the monoidal unit of $\mathbf{Set}$ , and $R^0$ is canonically isomorphic to $\{1\}$ . $U$ also preserves the                                                     |
| fong-spivak-invitation_F03619 | 315  | 0.927 | $\{1\} \cdot U$                                                                                          | $\{1\}.U$                                                              | $\{1\}$ is the monoidal unit of $\mathbf{Set}$ , and $R^0$ is canonically isomorphic to $\{1\}$ . $U$ also preserves the                                                     |
| fong-spivak-invitation_F03620 | 315  | 1.000 | $R^{\{m\}} \times R^{\{n\}} \cong R^{\{m+n\}}$                                                           | $R^m \times R^n \cong R^{m+n}$                                         | monoidal product because there is a canonical isomorphism $R^m \times R^n \cong R^{m+n}$ .                                                                                   |
| fong-spivak-invitation_F03621 | 315  | 0.876 | $(m, \mu, \eta)$                                                                                         | $(m, \mu, \eta)$                                                       | 2. A monoid object in $\mathbf{Mat}(R)$ is a tuple $(m, \mu, \eta)$ where $m \in \mathbb{N}$ , $\mu: m+m \rightarrow m$ , and $\eta: 0 \rightarrow m$                        |
| fong-spivak-invitation_F03622 | 315  | 0.876 | $m \in \mathbb{N}, \mu: m+m \rightarrow m$                                                               | $m \in \mathbb{N}, \mu: m+m \rightarrow m$                             | 2. A monoid object in $\mathbf{Mat}(R)$ is a tuple $(m, \mu, \eta)$ where $m \in \mathbb{N}$ , $\mu: m+m \rightarrow m$ , and $\eta: 0 \rightarrow m$                        |
| fong-spivak-invitation_F03623 | 315  | 0.876 | $\eta: 0 \rightarrow m$                                                                                  | $\eta: 0 \rightarrow m$                                                | 2. A monoid object in $\mathbf{Mat}(R)$ is a tuple $(m, \mu, \eta)$ where $m \in \mathbb{N}$ , $\mu: m+m \rightarrow m$ , and $\eta: 0 \rightarrow m$                        |
| fong-spivak-invitation_F03624 | 315  | 0.896 | $\mu(\eta, x)=x=\mu(x, \eta)$                                                                            | $\mu(\eta, x) = x = \mu(x, \eta)$                                      | satisfy the properties $\mu(\eta, x) = x = \mu(x, \eta)$ and $\mu(x, \mu(y, z)) = \mu(\mu(x, y), z)$ . Note that                                                             |
| fong-spivak-invitation_F03625 | 315  | 0.896 | $\mu(x, \mu(y, z))=\mu(\mu(x, y), z)$                                                                    | $\mu(x, \mu(y, z)) = \mu(\mu(x, y), z)$                                | satisfy the properties $\mu(\eta, x) = x = \mu(x, \eta)$ and $\mu(x, \mu(y, z)) = \mu(\mu(x, y), z)$ . Note that                                                             |
| fong-spivak-invitation_F03626 | 315  | 1.000 | $0 \rightarrow m$                                                                                        | $0 \rightarrow m$                                                      | there is only one morphism $0 \rightarrow m$ in $\mathbf{Mat}(R)$ for any $m$ . It is not hard to show that for any                                                          |
| fong-spivak-invitation_F03627 | 315  | 1.000 | $m \in \mathbb{N}$                                                                                       | $m \in \mathbb{N}$                                                     | $m \in \mathbb{N}$ there is only one monoid structure. For example, when $m = 2$ , $\mu$ must be the following                                                               |
| fong-spivak-invitation_F03628 | 315  | 1.000 | $m=2, \mu$                                                                                               | $m = 2, \mu$                                                           | $m \in \mathbb{N}$ there is only one monoid structure. For example, when $m = 2$ , $\mu$ must be the following                                                               |
| fong-spivak-invitation_F03629 | 315  | 1.000 | $U(\eta): R^{\{0\}} \rightarrow R^{\{m\}}$                                                               | $U(\eta): R^0 \rightarrow R^m$                                         | Anyway, for any monoid $(m, \mu, \eta)$ , the morphism $U(\eta): R^0 \rightarrow R^m$ is given by $U(\eta)(1) :=$                                                            |
| fong-spivak-invitation_F03630 | 315  | 1.000 | $U(\eta)(1) :=$                                                                                          | $U(\eta)(1) :=$                                                        | Anyway, for any monoid $(m, \mu, \eta)$ , the morphism $U(\eta): R^0 \rightarrow R^m$ is given by $U(\eta)(1) :=$                                                            |
| fong-spivak-invitation_F03631 | 315  | 0.508 | $(0, \ldots, 0)$                                                                                         | $(0, \dots, 0)$                                                        | $(0, \dots, 0)$ , and the morphism $U(\mu): R^m \times R^m \rightarrow R^m$ is given by                                                                                      |

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| fong-spivak-invitation-F03632 | 315  | 0.508 | $U(\mu): R^{\{m\}} \times R^{\{m\}} \rightarrow R^{\{m\}}$                                                                                                        | $U(\mu): R^m \times R^m \rightarrow R^m$             | $(0, \dots, 0)$ , and the morphism $U(\mu): R^m \times R^m \rightarrow R^m$ is given by                                                                 |
| fong-spivak-invitation-F03633 | 315  | 1.000 | $R^{\{m\}}$                                                                                                                                                       | $R^m$                                                | These give $R^m$ the structure of a monoid.                                                                                                             |
| fong-spivak-invitation-F03634 | 315  | 0.510 | $(1, \circ, -c)$                                                                                                                                                  | $(1, \circ, -c)$                                     | 3. The triple $(1, \circ, -c)$ corresponds to the additive monoid structure on $\mathbb{R}$ , e.g. with $(5, 3) \mapsto 8$ .                            |
| fong-spivak-invitation-F03635 | 315  | 0.510 | $(5, 3) \mapsto 8$                                                                                                                                                | $(5, 3) \mapsto 8$                                   | 3. The triple $(1, \circ, -c)$ corresponds to the additive monoid structure on $\mathbb{R}$ , e.g. with $(5, 3) \mapsto 8$ .                            |
| fong-spivak-invitation-F03636 | 316  | 0.732 | $\mathrm{B}(-<)$                                                                                                                                                  | $B(-<)$                                              | 1. The behavior $B(-<)$ of the reversed addition icon $-<: 1 \rightarrow 2$ is the relation $\{(x, y, z) \in R^3 \mid$                                  |
| fong-spivak-invitation-F03637 | 316  | 0.732 | $-<: 1 \rightarrow 2$                                                                                                                                             | $-<: 1 \rightarrow 2$                                | 1. The behavior $B(-<)$ of the reversed addition icon $-<: 1 \rightarrow 2$ is the relation $\{(x, y, z) \in R^3 \mid$                                  |
| fong-spivak-invitation-F03638 | 316  | 0.732 | $\left\{(x, y, z) \in R^{\{3\}} \mid \text{mid} \right\}$                                                                                                         | $\{(x, y, z) \in R^3 \mid$                           | 1. The behavior $B(-<)$ of the reversed addition icon $-<: 1 \rightarrow 2$ is the relation $\{(x, y, z) \in R^3 \mid$                                  |
| fong-spivak-invitation-F03639 | 316  | 0.982 | $x=y+z\}$                                                                                                                                                         | $x = y + z\}$                                        | $x = y + z\}$ .                                                                                                                                         |
| fong-spivak-invitation-F03640 | 316  | 0.141 | $\mathrm{B}(\supset^{\{-\}})$                                                                                                                                     | $B(\supset^-)$                                       | 2. The behavior $B(\supset^-)$ of the reversed copy icon, $\supset^-: 2 \rightarrow 1$ is the relation $\{(x, y, z) \in R^3 \mid$                       |
| fong-spivak-invitation-F03641 | 316  | 0.141 | $\exists^{\{-\}}: 2 \rightarrow 1$                                                                                                                                | $\exists^-: 2 \rightarrow 1$                         | 2. The behavior $B(\supset^-)$ of the reversed copy icon, $\supset^-: 2 \rightarrow 1$ is the relation $\{(x, y, z) \in R^3 \mid$                       |
| fong-spivak-invitation-F03642 | 316  | 0.968 | $x=y=z\}$                                                                                                                                                         | $x = y = z\}$                                        | $x = y = z\}$ .                                                                                                                                         |
| fong-spivak-invitation-F03643 | 316  | 1.000 | $B+C \subseteq R^{\{m+p\}} \times R^{\{n+q\}}$                                                                                                                    | $B + C \subseteq R^{m+p} \times R^{n+q}$             | If $B \subseteq R^m \times R^n$ and $C \subseteq R^p \times R^q$ are morphisms in $\mathbf{Rel}_R$ , then take $B + C \subseteq R^{m+p} \times R^{n+q}$ |
| fong-spivak-invitation-F03644 | 316  | 0.410 | $\mathrm{B}(\leftarrow g \rightarrow)$                                                                                                                            | $B(g$                                                | respectively, and, by Eq. (5.13), the composite $B(g \circ (h^{\mathrm{op}})) = B(g) \circ B(h^{\mathrm{op}})$ is:                                      |
| fong-spivak-invitation-F03645 | 316  | 0.410 | $\leftarrow \leftarrow (h^{\mathrm{op}} \rightarrow) \rightarrow = \mathrm{B}(g) ; \rightarrow ; \leftarrow \leftarrow (h^{\mathrm{op}} \rightarrow) \rightarrow$ | $(h^{\mathrm{op}})) = B(g); B(h^{\mathrm{op}})$      | respectively, and, by Eq. (5.13), the composite $B(g \circ (h^{\mathrm{op}})) = B(g) \circ B(h^{\mathrm{op}})$ is:                                      |
| fong-spivak-invitation-F03646 | 316  | 1.000 | $S(h)$                                                                                                                                                            | $S(h)$                                               | Since $S(g)$ and $S(h)$ are functions, the above immediately reduces to the desired formula:                                                            |
| fong-spivak-invitation-F03647 | 316  | 0.981 | $h: m \rightarrow p$                                                                                                                                              | $h: m \rightarrow p$                                 | The behavior of $g^{\mathrm{op}}: n \rightarrow m$ and $h: m \rightarrow p$ are                                                                         |
| fong-spivak-invitation-F03648 | 316  | 0.998 | $\mathrm{B}(\leftarrow (g^{\mathrm{op}} \rightarrow) ; h \rightarrow) = \mathrm{B}(\leftarrow (g^{\mathrm{op}} \rightarrow) ; \rightarrow) ; \mathrm{B}(h)$       | $B((g^{\mathrm{op}}); h) = B(g^{\mathrm{op}}); B(h)$ | respectively, and, by Eq. (5.13), the composite $B((g^{\mathrm{op}}) \circ h) = B(g^{\mathrm{op}}) \circ B(h)$ is:                                      |
| fong-spivak-invitation-F03649 | 316  | 0.555 | $\{y \in R \mid y=0\}$                                                                                                                                            | $\{y \in R \mid y = 0\}$                             | 1. The behavior of the 0-reverse $\dashv$ is the subset $\{y \in R \mid y = 0\}$ , and its $n$ -fold tensor is similarly                                |
| fong-spivak-invitation-F03650 | 316  | 1.000 | $\left\{y \in R^{\{n\}} \mid \text{mid } y=0 \right\}$                                                                                                            | $\{y \in R^n \mid y = 0\}$                           | $\{y \in R^n \mid y = 0\}$ . Composing this relation with $S(g) \subseteq R^m \times R^n$ gives $\{x \in R^m \mid S(g) = 0\}$ ,                         |
| fong-spivak-invitation-F03651 | 316  | 1.000 | $S(g) \subseteq R^{\{m\}} \times R^{\{n\}}$                                                                                                                       | $S(g) \subseteq R^m \times R^n$                      | $\{y \in R^n \mid y = 0\}$ . Composing this relation with $S(g) \subseteq R^m \times R^n$ gives $\{x \in R^m \mid S(g) = 0\}$ ,                         |
| fong-spivak-invitation-F03652 | 316  | 1.000 | $\left\{x \in R^{\{m\}} \mid \text{mid } S(g)=0 \right\}$                                                                                                         | $\{x \in R^m \mid S(g) = 0\}$                        | $\{y \in R^n \mid y = 0\}$ . Composing this relation with $S(g) \subseteq R^m \times R^n$ gives $\{x \in R^m \mid S(g) = 0\}$ ,                         |
| fong-spivak-invitation-F03653 | 316  | 0.868 | $\{x \in R\}$                                                                                                                                                     | $\{x \in R\}$                                        | 2. The behavior of the discard-inverse $\leftarrow$ is the subset $\{x \in R\}$ , i.e. the largest subset of $R$ , and                                  |
| fong-spivak-invitation-F03654 | 316  | 1.000 | $R^{\{n\}} \subseteq R^{\{n\}}$                                                                                                                                   | $R^n \subseteq R^n$                                  | similarly its $m$ -fold tensor is $R^n \subseteq R^n$ . Composing this relation with $S(g) \subseteq R^m \times R^n$ gives                              |
| fong-spivak-invitation-F03655 | 316  | 0.999 | $\left\{y \in R^{\{n\}} \mid \text{mid} \right\}$                                                                                                                 | $\{y \in R^n \mid$                                   | $\{y \in R^n \mid$ there exists $x \in R^m$ such that $S(g)(x) = y\}$ , which is exactly the image of $S(g)$ .                                          |
| fong-spivak-invitation-F03656 | 316  | 0.999 | $\leftarrow S(g)(x)=y \rightarrow$                                                                                                                                | $S(g)(x) = y\}$                                      | $\{y \in R^n \mid$ there exists $x \in R^m$ such that $S(g)(x) = y\}$ , which is exactly the image of $S(g)$ .                                          |
| fong-spivak-invitation-F03657 | 317  | 0.552 | $\mathrm{B}(g)=\{(x, y) \mid S(g)(x)=y\}$                                                                                                                         | $B(g) = \{(x, y) \mid S(g)(x) = y\}$                 | 3. For any $g: m \rightarrow n$ , we first claim that the behavior $B(g) = \{(x, y) \mid S(g)(x) = y\}$ is linear, i.e.                                 |
| fong-spivak-invitation-F03658 | 317  | 1.000 | $S(g)(x)=y$                                                                                                                                                       | $S(g)(x) = y$                                        | so if $S(g)(x) = y$ then $S(g)(rx) = ry$ and $S(g)(x_1 + x_2) = S(g)(x_1) + S(g)(x_2)$ . Thus we                                                        |
| fong-spivak-invitation-F03659 | 317  | 1.000 | $S(g)(r\ x)=r\ y$                                                                                                                                                 | $S(g)(rx) = ry$                                      | so if $S(g)(x) = y$ then $S(g)(rx) = ry$ and $S(g)(x_1 + x_2) = S(g)(x_1) + S(g)(x_2)$ . Thus we                                                        |

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| fong-spivak-invitation_F03660 | 317  | ■ 1.000 | $S(g)\left(x_{\{1\}}+x_{\{2\}}\right)=S(g)\left(x_{\{1\}}\right)+S(g)\left(x_{\{2\}}\right)$ | $S(g)\left(x_1+x_2\right)=S(g)\left(x_1\right)+S(g)\left(x_2\right)$ | so if $S(g)(x) = y$ then $S(g)(rx) = ry$ and $S(g)(x_1+x_2) = S(g)(x_1) + S(g)(x_2)$ . Thus we                                                |
| fong-spivak-invitation_F03661 | 317  | ■ 1.000 | $(x, y) \in \mathrm{B}(g)$                                                                   | $(x, y) \in \mathrm{B}(g)$                                           | conclude that $(x, y) \in \mathrm{B}(g)$ implies $(rx, ry) \in \mathrm{B}(g)$ , so it's closed under scalar multiplication,                   |
| fong-spivak-invitation_F03662 | 317  | ■ 1.000 | $(r x, r y) \in \mathrm{B}(g)$                                                               | $(rx, ry) \in \mathrm{B}(g)$                                         | conclude that $(x, y) \in \mathrm{B}(g)$ implies $(rx, ry) \in \mathrm{B}(g)$ , so it's closed under scalar multiplication,                   |
| fong-spivak-invitation_F03663 | 317  | ■ 1.000 | $\left(x_{\{1\}}, y_{\{1\}}\right), \left(x_{\{2\}}, y_{\{2\}}\right) \in \mathrm{B}(g)$     | $\left(x_1, y_1\right), \left(x_2, y_2\right) \in \mathrm{B}(g)$     | and $(x_1, y_1), (x_2, y_2) \in \mathrm{B}(g)$ implies $(x_1+x_2, y_1+y_2) \in \mathrm{B}(g)$ , so it's closed under addition.                |
| fong-spivak-invitation_F03664 | 317  | ■ 1.000 | $\left(x_{\{1\}}+x_{\{2\}}, y_{\{1\}}+y_{\{2\}}\right) \in \mathrm{B}(g)$                    | $\left(x_1+x_2, y_1+y_2\right) \in \mathrm{B}(g)$                    | and $(x_1, y_1), (x_2, y_2) \in \mathrm{B}(g)$ implies $(x_1+x_2, y_1+y_2) \in \mathrm{B}(g)$ , so it's closed under addition.                |
| fong-spivak-invitation_F03665 | 317  | ■ 0.999 | $\mathrm{B}\left(g^{\mathrm{op}}\right)$                                                     | $\mathrm{B}\left(g^{\mathrm{op}}\right)$                             | Similarly, the behavior $\mathrm{B}(g^{\mathrm{op}})$ is also linear; the proof is similar.                                                   |
| fong-spivak-invitation_F03666 | 317  | ■ 0.386 | $C \subseteq R^{\{n\}} \times R^{\{p\}}$                                                     | $C \subseteq R^m \times R^p$                                         | $B \subseteq R^m \times R^n$ and $C \subseteq R^n \times R^p$ are linear. Take $(x_1, z_1), (x_2, z_2) \in B \circ C$ and take $r \in R$ . By |
| fong-spivak-invitation_F03667 | 317  | ■ 0.386 | $\left(x_{\{1\}}, z_{\{1\}}\right), \left(x_{\{2\}}, z_{\{2\}}\right) \in B \circ C$         | $\left(x_1, z_1\right), \left(x_2, z_2\right) \in B \circ C$         | $B \subseteq R^m \times R^n$ and $C \subseteq R^n \times R^p$ are linear. Take $(x_1, z_1), (x_2, z_2) \in B \circ C$ and take $r \in R$ . By |
| fong-spivak-invitation_F03668 | 317  | ■ 0.999 | $y_{\{1\}}, y_{\{2\}} \in R^{\{n\}}$                                                         | $y_1, y_2 \in R^n$                                                   | definition, there exist $y_1, y_2 \in R^n$ such that $(x_1, y_1), (x_2, y_2) \in B$ and $(y_1, z_1), (y_2, z_2) \in C$ .                      |
| fong-spivak-invitation_F03669 | 317  | ■ 0.999 | $\left(x_{\{1\}}, y_{\{1\}}\right), \left(x_{\{2\}}, y_{\{2\}}\right) \in B$                 | $\left(x_1, y_1\right), \left(x_2, y_2\right) \in B$                 | definition, there exist $y_1, y_2 \in R^n$ such that $(x_1, y_1), (x_2, y_2) \in B$ and $(y_1, z_1), (y_2, z_2) \in C$ .                      |
| fong-spivak-invitation_F03670 | 317  | ■ 0.999 | $\left(y_{\{1\}}, z_{\{1\}}\right), \left(y_{\{2\}}, z_{\{2\}}\right) \in C$                 | $\left(y_1, z_1\right), \left(y_2, z_2\right) \in C$                 | definition, there exist $y_1, y_2 \in R^n$ such that $(x_1, y_1), (x_2, y_2) \in B$ and $(y_1, z_1), (y_2, z_2) \in C$ .                      |
| fong-spivak-invitation_F03671 | 317  | ■ 1.000 | $\left(r x_{\{1\}}, r y_{\{1\}}\right) \in B$                                                | $\left(r x_1, r y_1\right) \in B$                                    | Since $B$ and $C$ are linear, $(rx_1, ry_1) \in B$ and $(ry_1, rz_1) \in C$ , and also $(x_1+x_2, y_1+y_2) \in B$                             |
| fong-spivak-invitation_F03672 | 317  | ■ 1.000 | $\left(r y_{\{1\}}, r z_{\{1\}}\right) \in C$                                                | $\left(r y_1, r z_1\right) \in C$                                    | Since $B$ and $C$ are linear, $(rx_1, ry_1) \in B$ and $(ry_1, rz_1) \in C$ , and also $(x_1+x_2, y_1+y_2) \in B$                             |
| fong-spivak-invitation_F03673 | 317  | ■ 1.000 | $\left(x_{\{1\}}+x_{\{2\}}, y_{\{1\}}+y_{\{2\}}\right) \in B$                                | $\left(x_1+x_2, y_1+y_2\right) \in B$                                | Since $B$ and $C$ are linear, $(rx_1, ry_1) \in B$ and $(ry_1, rz_1) \in C$ , and also $\boxed{x_1+x_2, y_1+y_2} \in B$                       |
| fong-spivak-invitation_F03674 | 317  | ■ 0.328 | $\left(y_{\{1\}}+y_{\{2\}}, z_{\{1\}}+z_{\{2\}}\right) \in C$                                | $\left(y_1+y_2, z_1+z_2\right) \in C$                                | and $(y_1+y_2, z_1+z_2) \in C$ . Hence $(rx_1, rz_1) \in (B \circ C)$ and $(x_1+x_2, z_1+z_2) \in (B \circ C)$ , as                           |
| fong-spivak-invitation_F03675 | 317  | ■ 0.328 | $\left(r x_{\{1\}}, r z_{\{1\}}\right) \in (B ; C)$                                          | $\left(r x_1, r z_1\right) \in (B ; C)$                              | and $(y_1+y_2, z_1+z_2) \in C$ . Hence $\boxed{rx_1, rz_1} \in (B \circ C)$ and $(x_1+x_2, z_1+z_2) \in (B \circ C)$ , as                     |
| fong-spivak-invitation_F03676 | 317  | ■ 0.328 | $\left(x_{\{1\}}+x_{\{2\}}, z_{\{1\}}+z_{\{2\}}\right) \in (B \circ C)$                      | $\left(x_1+x_2, z_1+z_2\right) \in (B \circ C)$                      | and $(y_1+y_2, z_1+z_2) \in C$ . Hence $(rx_1, rz_1) \in (B \circ C)$ and $(x_1+x_2, z_1+z_2) \in (B \circ C)$ , as                           |
| fong-spivak-invitation_F03677 | 317  | ■ 0.902 | $(B ; C) \subseteq R^{\{m\}} \times R^{\{p\}}$                                               | $(B ; C) \subseteq R^m \times R^p$                                   | $(B \circ C) \subseteq R^m \times R^p$ consisting of all $(x, z)$ for which there exists $y \in R^n$ with $(x, y) \in B$ and                  |
| fong-spivak-invitation_F03678 | 317  | ■ 0.902 | $(x, z)$                                                                                     | $(x, z)$                                                             | $(B \circ C) \subseteq R^m \times R^p$ consisting of all $(x, z)$ for which there exists $y \in R^n$ with $(x, y) \in B$ and                  |
| fong-spivak-invitation_F03679 | 317  | ■ 0.929 | $(y, z) \in C$                                                                               | $(y, z) \in C$                                                       | $(y, z) \in C$ . We want to show that the set $(B \circ C)$ is linear, i.e. closed under scalar multiplication and                            |
| fong-spivak-invitation_F03680 | 317  | ■ 0.929 | $(B \circ C)$                                                                                | $(B \circ C)$                                                        | $(y, z) \in C$ . We want to show that the set $(B \circ C)$ is linear, i.e. closed under scalar multiplication and                            |
| fong-spivak-invitation_F03681 | 317  | ■ 0.956 | $(x, z) \in (B ; C)$                                                                         | $(x, z) \in (B ; C)$                                                 | For scalar multiplication, take an $(x, z) \in (B \circ C)$ and any $r \in R$ . Since $B$ is linear, we have                                  |
| fong-spivak-invitation_F03682 | 317  | ■ 0.905 | $(r * x, r * y) \in B$                                                                       | $(r * x, r * y) \in B$                                               | $(r * x, r * y) \in B$ and since $C$ is linear we have $(r * y, r * z) \in C$ , so then $(r * x, r * z) \in (B \circ C)$ .                    |
| fong-spivak-invitation_F03683 | 317  | ■ 0.905 | $(r * y, r * z) \in C$                                                                       | $(r * y, r * z) \in C$                                               | $(r * x, r * y) \in B$ and since $C$ is linear we have $(r * y, r * z) \in C$ , so then $(r * x, r * z) \in (B \circ C)$ .                    |
| fong-spivak-invitation_F03684 | 317  | ■ 0.905 | $(r * x, r * z) \in (B \circ C)$                                                             | $(r * x, r * z) \in (B \circ C)$                                     | $(r * x, r * y) \in B$ and since $C$ is linear we have $(r * y, r * z) \in C$ , so then $(r * x, r * z) \in (B \circ C)$ .                    |
| fong-spivak-invitation_F03685 | 317  | ■ 1.000 | $\left(x^{\prime}, z^{\prime}\right) \in (B ; C)$                                            | $\left(x', z'\right) \in (B ; C)$                                    | For addition, if we also have $(x', z') \in (B \circ C)$ then there is some $y' \in R^n$ with $(x', y') \in B$ and                            |
| fong-spivak-invitation_F03686 | 317  | ■ 1.000 | $y^{\prime} \in R^{\{n\}}$                                                                   | $y' \in R^n$                                                         | For addition, if we also have $(x', z') \in (B \circ C)$ then there is some $y' \in R^n$ with $(x', y') \in B$ and                            |
| fong-spivak-invitation_F03687 | 317  | ■ 0.999 | $\left(y^{\prime}, z^{\prime}\right) \in C$                                                  | $\left(y', z'\right) \in C$                                          | $\boxed{y', z'} \in C$ , so since $B$ and $C$ are linear we have $(x+x', y+y') \in B$ and $(y+y', z+z') \in C$ ,                              |

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| fong-spivak-invitation_F03688              | 317  | ■ 0.999 | $\text{\texttt{\left(x+x^{\prime}, y+y^{\prime}\right) \in B}}$                             | $(x + x', y + y') \in B$             | $(y', z') \in C$ , so since $B$ and $C$ are linear we have $(x + x', y + y') \in B$ and $\boxed{y + y', z + z' \in C}$ .                        |
| fong-spivak-invitation_F03689              | 317  | ■ 0.999 | $\text{\texttt{\left(y+y^{\prime}, z+z^{\prime}\right) \in C}}$                             | $(y + y', z + z') \in C$             | $(y', z') \in C$ , so since $B$ and $C$ are linear we have $(x + x', y + y') \in B$ and $\boxed{y + y', z + z' \in C}$ .                        |
| fong-spivak-invitation_F03690              | 317  | ■ 0.918 | $\text{\texttt{\left(x+x^{\prime}, z+z^{\prime}\right) \in(B ; C)}}$                        | $(x + x', z + z') \in (B; C)$        | hence $\boxed{x + x', z + z' \in (B \circ C)}$ .                                                                                                |
| fong-spivak-invitation_F03691              | 317  | ■ 1.000 | $\text{\texttt{a \not \leq b}}$                                                             | $a \not\leq b$                       | 1. The left-most (the discrete preorder on $A$ ) has no initial object, because $a \not\leq b$ and $b \not\leq a$ .                             |
| fong-spivak-invitation_F03692              | 317  | ■ 1.000 | $\text{\texttt{b \not \leq a}}$                                                             | $b \not\leq a$                       | 1. The left-most (the discrete preorder on $A$ ) has no initial object, because $a \not\leq b$ and $b \not\leq a$ .                             |
| fong-spivak-invitation_F03693              | 317  | ■ 0.999 | $\text{\texttt{f\left(1_R\right)=1_S}}$                                                     | $f(1_R) = 1_S$                       | 1. The remaining conditions are that $\boxed{f(1_R) = 1_S}$ , and that $f(r_1 * r_2) = f(r_1) * f(r_2)$ .                                       |
| fong-spivak-invitation_F03694              | 317  | ■ 0.999 | $\text{\texttt{f\left(r_{1} *_R r_{2}\right)=f\left(r_{1}\right) *_S f\left(r_{2}\right)}}$ | $f(r_1 *_R r_2) = f(r_1) *_S f(r_2)$ | 1. The remaining conditions are that $f(1_R) = 1_S$ , and that $\boxed{f(r_1 *_R r_2) = f(r_1) *_S f(r_2)}$ .                                   |
| fong-spivak-invitation_F03695              | 317  | ■ 0.998 | $\text{\texttt{R=\left(R, 0_R,+_R, 1_R, *_R\right)}}$                                       | $R = (R, 0_R, +_R, 1_R, *_R)$        | it is initial means that, for any other rig $R = (R, 0_R, +_R, 1_R, *_R)$ , there is a unique rig                                               |
| fong-spivak-invitation_F03696              | 317  | ■ 1.000 | $\text{\texttt{f: \mathbb{N} \rightarrow R}}$                                               | $f : \mathbb{N} \rightarrow R$       | homomorphism $\boxed{f : \mathbb{N} \rightarrow R}$ .                                                                                           |
| fong-spivak-invitation_F03697              | 318  | ■ 0.773 | $\text{\texttt{0_R, 1}}$                                                                    | $0_R, 1$                             | What is this homomorphism? Well, to be a rig homomorphism, $f$ must send 0 to $0_R$ , 1 to $1_R$ .                                              |
| fong-spivak-invitation_F03698              | 318  | ■ 0.773 | $\text{\texttt{1_R}}$                                                                       | $1_R$                                | What is this homomorphism? Well, to be a rig homomorphism, $f$ must send 0 to $0_R$ , 1 to $1_R$ .                                              |
| fong-spivak-invitation_F03699              | 318  | ■ 0.731 | $\text{\texttt{f(m+n)=f(m)+_R f(n)}}$                                                       | $f(m + n) = f(m) +_R f(n)$           | Furthermore, we must also have $f(m + n) = f(m) +_R f(n)$ , and hence                                                                           |
| fong-spivak-invitation_F03700              | 318  | ■ 1.000 | $\text{\texttt{f(m *_ n)=f(m) *_R f(n)}}$                                                   | $f(m *_ n) = f(m) *_R f(n)$          | It remains to check $f(m *_ n) = f(m) *_R f(n)$ , and this will follow from distributivity. Noting                                              |
| fong-spivak-invitation_F03701              | 318  | ■ 1.000 | $\text{\texttt{f(m *_ n)}}$                                                                 | $f(m *_ n)$                          | that $f(m *_ n)$ is equal to the sum of $mn$ copies of $1_R$ , we have                                                                          |
| fong-spivak-invitation_F03702              | 318  | ■ 1.000 | $\text{\texttt{m n}}$                                                                       | $mn$                                 | that $f(m *_ n)$ is equal to the sum of $mn$ copies of $1_R$ , we have                                                                          |
| fong-spivak-invitation_F03703              | 318  | ■ 1.000 | $\text{\texttt{\varnothing \rightarrow c}}$                                                 | $\emptyset \rightarrow c$            | there is a unique map $\emptyset \rightarrow c$ coming from the initial object.                                                                 |
| fong-spivak-invitation_F03704              | 318  | ■ 1.000 | $\text{\texttt{c_{1} \rightarrow c}}$                                                       | $c_1 \rightarrow c$                  | If $c_1$ is initial then by the universal property, for any $c$ there is a unique morphism $c_1 \rightarrow c$ ; in                             |
| fong-spivak-invitation_F03705              | 318  | ■ 1.000 | $\text{\texttt{c_{2} \rightarrow c_{1}}}}$                                                  | $c_2 \rightarrow c_1$                | a unique morphism $c_2 \rightarrow c_1$ , call it $g$ . But how do we know that $f$ and $g$ are mutually inverse?                               |
| fong-spivak-invitation_F03706              | 318  | ■ 0.999 | $\text{\texttt{c_{1} \rightarrow c_{1}}}}$                                                  | $c_1 \rightarrow c_1$                | Well since $c_1$ is initial there is a unique morphism $c_1 \rightarrow c_1$ . But we can think of two: $\text{id}_{c_1}$ and                   |
| fong-spivak-invitation_F03707              | 318  | ■ 0.999 | $\text{\texttt{\mathrm{id}_{c_{1}}}}$                                                       | $\text{id}_{c_1}$                    | Well since $c_1$ is initial there is a unique morphism $c_1 \rightarrow c_1$ . But we can think of two: $\text{id}_{c_1}$ and                   |
| fong-spivak-invitation_F03708 [conf 0.078] | 318  | ■ 0.078 | $\text{\texttt{=\mathrm{id}_{c_{1}}}}$                                                      | $= \text{id}_{c_1}$                  | $f \circ g$ . Thus they must be equal. Similarly for $c_2$ , so we have $f \circ g = \text{id}_{c_1}$ and $g \circ f = \text{id}_{c_2}$ , which |
| fong-spivak-invitation_F03709 [conf 0.078] | 318  | ■ 0.078 | $\text{\texttt{f=\mathrm{id}_{c_{2}}}}$                                                     | $f = \text{id}_{c_2}$                | $f \circ g$ . Thus they must be equal. Similarly for $c_2$ , so we have $f \circ g = \text{id}_{c_1}$ and $g \circ f = \text{id}_{c_2}$ , which |
| fong-spivak-invitation_F03710              | 318  | ■ 1.000 | $\text{\texttt{p+q}}$                                                                       | $p + q$                              | Translating Definition 6.11 to this case, a coproduct $p + q$ in $P$ is an element of $P$ such that                                             |
| fong-spivak-invitation_F03711              | 318  | ■ 1.000 | $\text{\texttt{p \leq p+q}}$                                                                | $p \leq p + q$                       | $p \leq p + q$ and $q \leq p + q$ , and such that for all elements $x \in P$ with maps $p \leq x$ and $q \leq x$ , we                           |
| fong-spivak-invitation_F03712              | 318  | ■ 1.000 | $\text{\texttt{q \leq p+q}}$                                                                | $q \leq p + q$                       | $p \leq p + q$ and $q \leq p + q$ , and such that for all elements $x \in P$ with maps $p \leq x$ and $q \leq x$ , we                           |
| fong-spivak-invitation_F03713              | 318  | ■ 1.000 | $\text{\texttt{p+q \leq x}}$                                                                | $p + q \leq x$                       | have $p + q \leq x$ . But this says exactly that $p + q$ is a join: it is a least element above both $p$ and                                    |
| fong-spivak-invitation_F03714              | 319  | ■ 0.996 | $\text{\texttt{o}}$                                                                         | $o$                                  | $\text{orange1} \mapsto o$                                                                                                                      |
| fong-spivak-invitation_F03715              | 319  | ■ 0.366 | $\text{\texttt{\iota_{A9}\{ }_9^{\circ}\{f, g\}=f}}$                                        | $\iota_{A9}^\circ[f, g] = f$         | 1. The equation $\iota_A \circ [f, g] = f$ is the commutativity of the left-hand triangle in the commutative                                    |

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| fong-spivak-invitation-F03716 | 319  | ■ 0.451 | $\iota_{B\{\circ\}}[f, g] = g$                                                                                                                                                 | $\iota_B^\circ[f, g] = g$                                                                                                                     | 2. The equation $\iota_B \circ [f, g] = g$ is the commutativity of the right-hand triangle in the commutative                                             |
| fong-spivak-invitation-F03717 | 319  | ■ 0.428 | $[f, g]: h = [f; h, g \circ h]$                                                                                                                                                | $[f, g]: h = [f; h, g \circ h]$                                                                                                               | 3. The equation $[f, g] \circ h = [f \circ h, g \circ h]$ follows from the universal property of the coproduct.                                           |
| fong-spivak-invitation-F03718 | 319  | ■ 0.998 | $[f; h, g; h]: A+B \rightarrow D$                                                                                                                                              | $[f; h, g; h]: A + B \rightarrow D$                                                                                                           | commutes, and the universal property says there is a unique map $[f \circ h, g \circ h]: A + B \rightarrow D$                                             |
| fong-spivak-invitation-F03719 | 319  | ■ 0.966 | $[f, g]; h = [f; h, g \circ h]$                                                                                                                                                | $[f, g]; h = [f; h, g \circ h]$                                                                                                               | for which this occurs. Hence we must have $[f, g] \circ h = [f \circ h, g \circ h]$ .                                                                     |
| fong-spivak-invitation-F03720 | 319  | ■ 0.997 | $+: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$                                                                                                                    | $+: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$                                                                                   | 1. To define a functor $+: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ we must define its action on objects and morphisms. In                 |
| fong-spivak-invitation-F03721 | 319  | ■ 0.943 | $(A, B)$                                                                                                                                                                       | $(A, B)$                                                                                                                                      | both cases, we just take the coproduct. If $(A, B)$ is an object of $\mathcal{C} \times \mathcal{C}$ , its image $A + B$ is, as                           |
| fong-spivak-invitation-F03722 | 319  | ■ 1.000 | $(f, g): (A, B) \rightarrow (C, D)$                                                                                                                                            | $(f, g): (A, B) \rightarrow (C, D)$                                                                                                           | usual, the coproduct of the two objects of $\mathcal{C}$ . If $(f, g): (A, B) \rightarrow (C, D)$ is a morphism, then                                     |
| fong-spivak-invitation-F03723 | 319  | ■ 0.969 | $f+g=\left[f\{\circ\}\iota_{\{C\}}, g\{\circ\}\iota_{\{D\}}\right]: A+B \rightarrow C+D$                                                                                       | $f + g = [f^\circ \iota_C, g^\circ \iota_D]: A + B \rightarrow C + D$                                                                         | we can form a morphism $f + g = [f \circ \iota_C, g \circ \iota_D]: A + B \rightarrow C + D$ , where $\iota_C: C \rightarrow C + D$                       |
| fong-spivak-invitation-F03724 | 319  | ■ 0.969 | $\iota_{\{C\}}: C \rightarrow C+D$                                                                                                                                             | $\iota_C: C \rightarrow C + D$                                                                                                                | we can form a morphism $f + g = [f \circ \iota_C, g \circ \iota_D]: A + B \rightarrow C + D$ , where $\iota_C: C \rightarrow C + D$                       |
| fong-spivak-invitation-F03725 | 319  | ■ 0.997 | $\iota_{\{D\}}: D \rightarrow C+D$                                                                                                                                             | $\iota_D: D \rightarrow C + D$                                                                                                                | and $\iota_D: D \rightarrow C + D$ are the canonical morphisms given by the definition of the coproduct                                                   |
| fong-spivak-invitation-F03726 | 320  | ■ 0.491 | $(h, k): (C, D) \rightarrow (E, F)$                                                                                                                                            | $(h, k): (C, D) \rightarrow (E, F)$                                                                                                           | have a morphism $(h, k): (C, D) \rightarrow (E, F)$ in $\mathcal{C} \times \mathcal{C}$ . We need to show that $(f + g) \circ (h + k) =$                  |
| fong-spivak-invitation-F03727 | 320  | ■ 0.491 | $(f+g)\{\circ\}(h+k)=$                                                                                                                                                         | $(f + g) \circ (h + k) =$                                                                                                                     | have a morphism $(h, k): (C, D) \rightarrow (E, F)$ in $\mathcal{C} \times \mathcal{C}$ . We need to show that $(f + g) \circ (h + k) =$                  |
| fong-spivak-invitation-F03728 | 320  | ■ 0.755 | $(f; h)+(g; k)$                                                                                                                                                                | $(f; h) + (g; k)$                                                                                                                             | $(f \circ h) + (g \circ k)$ . This is a slightly more complicated version of the argument in Exercise 6.15                                                |
| fong-spivak-invitation-F03729 | 320  | ■ 1.000 | $(f; h)+$                                                                                                                                                                      | $(f; h) +$                                                                                                                                    | Indeed, we again use the uniqueness of the copairing in (6.1), this time to show that $(f \circ h) +$                                                     |
| fong-spivak-invitation-F03730 | 320  | ■ 0.158 | $(g \circ k)=\left[f\{\circ\}h\{\circ\}\iota_{\{E\}}, g\{\circ\}k\{\circ\}\iota_{\{F\}}\right]=(f+g)\{\circ\}(h+k)$                                                            | $(g \circ k) = [f \circ h \circ \iota_E, g \circ k \circ \iota_F] = (f + g) \circ (h + k)$                                                    | $(f \circ h) + (g \circ k) = [f \circ h \circ \iota_E, g \circ k \circ \iota_F] = (f + g) \circ (h + k)$ , as required.                                   |
| fong-spivak-invitation-F03731 | 320  | ■ 0.998 | $!_A: \varnothing \rightarrow A$                                                                                                                                               | $!_A: \emptyset \rightarrow A$                                                                                                                | 2. Recall that the universal property of the initial object gives a unique map $!_A: \emptyset \rightarrow A$ . Then                                      |
| fong-spivak-invitation-F03732 | 320  | ■ 0.738 | $\left[\mathrm{id}_A, !_A\right]$                                                                                                                                              | $[\mathrm{id}_A, !_A]$                                                                                                                        | the copairing $[\mathrm{id}_A, !_A]$ is a map $A + \emptyset \rightarrow A$ . Moreover, it is an isomorphism with inverse                                 |
| fong-spivak-invitation-F03733 | 320  | ■ 0.592 | $\{\circ\} A: A \rightarrow A+\varnothing$                                                                                                                                     | $\circ A: A \rightarrow A + \emptyset$                                                                                                        | $\iota_A: A \rightarrow A + \emptyset$ . Indeed, using the properties in Exercise 6.15 and the universal property of                                      |
| fong-spivak-invitation-F03734 | 320  | ■ 0.679 | $\left[\mathrm{id}_A, !_A\right]=\mathrm{id}_A$                                                                                                                                | $[\mathrm{id}_A, !_A] = \mathrm{id}_A$                                                                                                        | the initial object, we have $\iota_A \circ [\mathrm{id}_A, !_A] = \mathrm{id}_A$ , and                                                                    |
| fong-spivak-invitation-F03735 | 320  | ■ 0.986 | $\left[!_A, \mathrm{id}_A\right]: \varnothing+A \rightarrow A$                                                                                                                 | $[_A, \mathrm{id}_A]: \emptyset + A \rightarrow A$                                                                                            | An analogous argument shows $[_A, \mathrm{id}_A]: \emptyset + A \rightarrow A$ is an isomorphism.                                                         |
| fong-spivak-invitation-F03736 | 320  | ■ 0.644 | $\left[\mathrm{id}_A+\iota_B, \iota_C\right]=\left[\left[\iota_A, \iota_B\{\circ\}\iota_{\{B+C\}}\right], \iota_C\{\circ\}\iota_{\{B+C\}}\right]: (A+B)+C \rightarrow A+(B+C)$ | $[\mathrm{id}_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \circ \iota_{B+C}], \iota_C \circ \iota_{B+C}]: (A + B) + C \rightarrow A + (B + C)$ | (a) The map $[\mathrm{id}_A + \iota_B, \iota_C] = [[\iota_A, \iota_B \circ \iota_{B+C}], \iota_C \circ \iota_{B+C}]: (A + B) + C \rightarrow A + (B + C)$ |
| fong-spivak-invitation-F03737 | 320  | ■ 0.999 | $\left[\iota_A, \iota_B+\mathrm{id}_C\right]: A+(B+C) \rightarrow (A+B)+C$                                                                                                     | $[\iota_A, \iota_B + \mathrm{id}_C]: A + (B + C) \rightarrow (A + B) + C$                                                                     | is an isomorphism, with inverse $[_A, \iota_B + \mathrm{id}_C]: A + (B + C) \rightarrow (A + B) + C$ .                                                    |
| fong-spivak-invitation-F03738 | 320  | ■ 1.000 | $\left[\iota_A, \iota_B\right]: A+B \rightarrow B+A$                                                                                                                           | $[\iota_A, \iota_B]: A + B \rightarrow B + A$                                                                                                 | (b) The map $[_A, \iota_B]: A + B \rightarrow B + A$ is an isomorphism. Note our notation here is slightly                                                |
| fong-spivak-invitation-F03739 | 320  | ■ 0.991 | $\iota_A: A \rightarrow A+B$                                                                                                                                                   | $\iota_A: A \rightarrow A + B$                                                                                                                | confusing: there are two maps named $\iota_A$ , (i) $\iota_A: A \rightarrow A + B$ , and (ii) $\iota_A: A \rightarrow B + A$ ,                            |
| fong-spivak-invitation-F03740 | 320  | ■ 0.991 | $\iota_A: A \rightarrow B+A$                                                                                                                                                   | $\iota_A: A \rightarrow B + A$                                                                                                                | confusing: there are two maps named $\iota_A$ , (i) $\iota_A: A \rightarrow A + B$ , and (ii) $\iota_A: A \rightarrow B + A$ ,                            |
| fong-spivak-invitation-F03741 | 320  | ■ 0.980 | $\left[\iota_B, \iota_A\right]: B+A \rightarrow$                                                                                                                               | $[\iota_B, \iota_A]: B + A \rightarrow$                                                                                                       | and similarly for $\iota_B$ . In the above we mean the map (ii). It has inverse $[_B, \iota_A]: B + A \rightarrow$                                        |
| fong-spivak-invitation-F03742 | 320  | ■ 1.000 | $A=B=C$                                                                                                                                                                        | $A = B = C$                                                                                                                                   | $A = B = C$ , and the square consisting of all identities is its pushout.                                                                                 |
| fong-spivak-invitation-F03743 | 320  | ■ 1.000 | $s^{\prime}$                                                                                                                                                                   | $s'$                                                                                                                                          | than one object, because if $s'$ is another object then there is a morphism $s \rightarrow s'$ , but the                                                  |

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| fong-spivak-invitation_F03744 | 320  | ■ 1.000 | $s \rightarrow s^{\prime}$                          | $s \rightarrow s'$                            | than one object, because if $s'$ is another object then there is a morphism $s \rightarrow s'$ , but the                                                  |
| fong-spivak-invitation_F03745 | 320  | ■ 1.000 | $s=s^{\prime}$                                      | $s = s'$                                      | only morphisms in $S$ are identities so $s = s'$ . Hence the set $S$ must consist of exactly one                                                          |
| fong-spivak-invitation_F03746 | 321  | ■ 0.990 | $\underline{5} \sqcup \underline{3}$                | $\underline{5} \sqcup \underline{3}$          | set of equivalence classes in $\underline{5} \sqcup \underline{3}$ generated by the relation $\{f(a) \sim g(a) \mid a \in \underline{4}\}$ . If we denote |
| fong-spivak-invitation_F03747 | 321  | ■ 0.990 | $\{f(a) \sim g(a) \mid a \in \underline{4}\}$       | $\{f(a) \sim g(a) \mid a \in \underline{4}\}$ | set of equivalence classes in $\underline{5} \sqcup \underline{3}$ generated by the relation $\{f(a) \sim g(a) \mid a \in \underline{4}\}$ . If we denote |
| fong-spivak-invitation_F03748 | 321  | ■ 1.000 | $\underline{5}$                                     | $\underline{5}$                               | elements of $\underline{5}$ as $\{1, \dots, 5\}$ and those of $\underline{3}$ as $\{1', 2', 3'\}$ , we can redraw the functions $f, g$ :                  |
| fong-spivak-invitation_F03749 | 321  | ■ 1.000 | $\{1, \ldots, 5\}$                                  | $\{1, \dots, 5\}$                             | elements of $\underline{5}$ as $\{1, \dots, 5\}$ and those of $\underline{3}$ as $\{1', 2', 3'\}$ , we can redraw the functions $f, g$ :                  |
| fong-spivak-invitation_F03750 | 321  | ■ 1.000 | $\underline{3}$                                     | $\underline{3}$                               | elements of $\underline{5}$ as $\{1, \dots, 5\}$ and those of $\underline{3}$ as $\{1', 2', 3'\}$ , we can redraw the functions $f, g$ :                  |
| fong-spivak-invitation_F03751 | 321  | ■ 1.000 | $\left\{1^{\prime}, 2^{\prime}, 3^{\prime}\right\}$ | $\{1', 2', 3'\}$                              | elements of $\underline{5}$ as $\{1, \dots, 5\}$ and those of $\underline{3}$ as $\{1', 2', 3'\}$ , we can redraw the functions $f, g$ :                  |
| fong-spivak-invitation_F03752 | 321  | ■ 1.000 | $5 \sim 3^{\prime}$                                 | $5 \sim 3'$                                   | $5 \sim 3'$ . The equivalence classes are $\{1, 1', 3\}$ , $\{2\}$ , $\{4\}$ , and $\{5, 2', 3'\}$ . These four are exactly the                           |
| fong-spivak-invitation_F03753 | 321  | ■ 1.000 | $\left\{1, 1^{\prime}, 3\right\},\{2\},\{4\}$       | $\{1, 1', 3\}, \{2\}, \{4\}$                  | $5 \sim 3'$ . The equivalence classes are $\{1, 1', 3\}$ , $\{2\}$ , $\{4\}$ , and $\{5, 2', 3'\}$ . These four are exactly the                           |
| fong-spivak-invitation_F03754 | 321  | ■ 1.000 | $\left\{5, 2^{\prime}, 3^{\prime}\right\}$          | $\{5, 2', 3'\}$                               | $5 \sim 3'$ . The equivalence classes are $\{1, 1', 3\}$ , $\{2\}$ , $\{4\}$ , and $\{5, 2', 3'\}$ . These four are exactly the                           |
| fong-spivak-invitation_F03755 | 321  | ■ 1.000 | $\varnothing \rightarrow X+Y$                       | $\emptyset \rightarrow X + Y$                 | 1. The diagram to the left commutes because $\emptyset$ is initial, and so has a unique map $\emptyset \rightarrow X + Y$ .                               |
| fong-spivak-invitation_F03756 | 321  | ■ 0.635 | $f\{ \}_{9} \iota_X=g\{ \}_{9} \iota_Y$             | $f_{9\iota_X} = g_{9\iota_Y}$                 | This implies we must have $f \circ \iota_X = g \circ \iota_Y$ .                                                                                           |
| fong-spivak-invitation_F03757 | 321  | ■ 0.479 | $t: X+\varnothing Y \rightarrow T$                  | $t: X + \emptyset Y \rightarrow T$            | map $t: X + \emptyset Y \rightarrow T$ such that $\iota_X \circ t = x$ and $\iota_Y \circ t = y$ . Thus $X + \emptyset Y$ is the coproduct                |
| fong-spivak-invitation_F03758 | 321  | ■ 0.479 | $t=x$                                               | $t = x$                                       | map $t: X + \emptyset Y \rightarrow T$ such that $\iota_X \circ t = x$ and $\iota_Y \circ t = y$ . Thus $X + \emptyset Y$ is the coproduct                |
| fong-spivak-invitation_F03759 | 321  | ■ 0.479 | $t=y$                                               | $t = y$                                       | map $t: X + \emptyset Y \rightarrow T$ such that $\iota_X \circ t = x$ and $\iota_Y \circ t = y$ . Thus $X + \emptyset Y$ is the coproduct                |
| fong-spivak-invitation_F03760 | 322  | ■ 1.000 | $A, B, X, Y$                                        | $A, B, X, Y$                                  | That is, we need to show that $S$ , together with the maps from $A, B, X, Y$ , and $Z$ , has the                                                          |
| fong-spivak-invitation_F03761 | 322  | ■ 1.000 | $S \rightarrow T$                                   | $S \rightarrow T$                             | We need to show there is a unique map $S \rightarrow T$ making everything commute. Since $Q$ is a                                                         |
| fong-spivak-invitation_F03762 | 322  | ■ 1.000 | $X \leftarrow A \rightarrow Y$                      | $X \leftarrow A \rightarrow Y$                | pushout of $X \leftarrow A \rightarrow Y$ , there is a unique map $Q \rightarrow T$ making a commutative triangle with                                    |
| fong-spivak-invitation_F03763 | 322  | ■ 1.000 | $Q \rightarrow T$                                   | $Q \rightarrow T$                             | pushout of $X \leftarrow A \rightarrow Y$ , there is a unique map $Q \rightarrow T$ making a commutative triangle with                                    |
| fong-spivak-invitation_F03764 | 322  | ■ 1.000 | $Y \leftarrow B \rightarrow Z$                      | $Y \leftarrow B \rightarrow Z$                | $Y$ , and since $R$ is the pushout of $Y \leftarrow B \rightarrow Z$ , there is a unique map $R \rightarrow T$ making a com-                              |
| fong-spivak-invitation_F03765 | 322  | ■ 1.000 | $R \rightarrow T$                                   | $R \rightarrow T$                             | $Y$ , and since $R$ is the pushout of $Y \leftarrow B \rightarrow Z$ , there is a unique map $R \rightarrow T$ making a com-                              |
| fong-spivak-invitation_F03766 | 322  | ■ 1.000 | $(Y, Q, R, T)$                                      | $(Y, Q, R, T)$                                | mutative triangle with $Y$ . This implies that there is a commuting $(Y, Q, R, T)$ square, and hence                                                      |
| fong-spivak-invitation_F03767 | 322  | ■ 0.996 | $X+\{ \}_{N} Y$                                     | $X +_N Y$                                     | The formula in Theorem 6.32 says that the pushout $X +_N Y$ is given by the set of equivalence                                                            |
| fong-spivak-invitation_F03768 | 322  | ■ 1.000 | $X \sqcup N \sqcup Y$                               | $X \sqcup N \sqcup Y$                         | classes of $X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$ , and $y \sim n$                                    |
| fong-spivak-invitation_F03769 | 322  | ■ 1.000 | $x \sim n$                                          | $x \sim n$                                    | classes of $X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$ , and $y \sim n$                                    |
| fong-spivak-invitation_F03770 | 322  | ■ 1.000 | $x=f(n)$                                            | $x = f(n)$                                    | classes of $X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$ , and $y \sim n$                                    |
| fong-spivak-invitation_F03771 | 322  | ■ 1.000 | $y \sim n$                                          | $y \sim n$                                    | classes of $X \sqcup N \sqcup Y$ under the equivalence relation generated by $x \sim n$ if $x = f(n)$ , and $y \sim n$                                    |

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| fong-spivak-invitation_F03772              | 322  | ■ 1.000 | $y=g(n)$                                                                      | $y = g(n)$                                                                | if $y = g(n)$ , where $x \in X, y \in Y, n \in N$ . Since for every $n \in N$ there exists an $x \in X$ such that                                                                 |
| fong-spivak-invitation_F03773              | 322  | ■ 1.000 | $x \in X, y \in Y, n \in N$                                                   | $x \in X, y \in Y, n \in N$                                               | if $y = g(n)$ , where $x \in X, y \in Y, n \in N$ . Since for every $n \in N$ there exists an $x \in X$ such that                                                                 |
| fong-spivak-invitation_F03774              | 322  | ■ 1.000 | $n \in N$                                                                     | $n \in N$                                                                 | if $y = g(n)$ , where $x \in X, y \in Y, n \in N$ . Since for every $n \in N$ there exists an $x \in X$ such that                                                                 |
| fong-spivak-invitation_F03775              | 323  | ■ 1.000 | $B \rightarrowtail D \otimes D$                                               | $B \rightarrow D \otimes D$                                               | morphism $B \rightarrowtail D \otimes D$ .                                                                                                                                        |
| fong-spivak-invitation_F03776              | 323  | ■ 1.000 | $g: X \rightarrowtail Y$                                                      | $g: X \rightarrow Y$                                                      | $f: X \rightarrow Y$ and $g: X \rightarrow Y$ such that                                                                                                                           |
| fong-spivak-invitation_F03777              | 324  | ■ 0.998 | $\iota_1: X \rightarrowtail X+X$                                              | $\iota_1: X \rightarrow X + X$                                            | Write $\iota_1: X \rightarrow X + X$ for the map into the first copy of $X$ in $X + X$ , given by the definition of                                                               |
| fong-spivak-invitation_F03778              | 324  | ■ 0.998 | $X+X$                                                                         | $X + X$                                                                   | Write $\iota_1: X \rightarrow X + X$ for the map into the first copy of $X$ in $X + X$ , given by the definition of                                                               |
| fong-spivak-invitation_F03779              | 324  | ■ 0.446 | $\iota_1 \circ \{\mathrm{id}, \mathrm{id}\} = \mathrm{id}$                    | $\iota_1 \circ [\mathrm{id}, \mathrm{id}] = \mathrm{id}$                  | coproduct. Then, using the fact that $\iota_1 \circ [\mathrm{id}, \mathrm{id}] = \mathrm{id}$ from Exercise 6.15 1, and the commutativity                                         |
| fong-spivak-invitation_F03780 [conf 0.080] | 324  | ■ 0.080 | $f = \iota_1$                                                                 | $f = \iota_1$                                                             | of the above square, we have $f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] \circ f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] \circ g = g$ . This means that $f: X \rightarrow T$ |
| fong-spivak-invitation_F03781 [conf 0.080] | 324  | ■ 0.080 | $\{\circlearrowleft\} f = \iota_1$                                            | $\circlearrowleft f = \iota_1$                                            | of the above square, we have $f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] \circ f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] \circ g = g$ . This means that $f: X \rightarrow T$ |
| fong-spivak-invitation_F03782 [conf 0.080] | 324  | ■ 0.080 | $\left[\right]$                                                               | $[$                                                                       | of the above square, we have $f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] \circ f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] \circ g = g$ . This means that $f: X \rightarrow T$ |
| fong-spivak-invitation_F03783 [conf 0.080] | 324  | ■ 0.080 | $\{\circlearrowleft\} g = g$                                                  | $\circlearrowleft g = g$                                                  | of the above square, we have $f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] \circ f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] \circ g = g$ . This means that $f: X \rightarrow T$ |
| fong-spivak-invitation_F03784 [conf 0.080] | 324  | ■ 0.080 | $f: X \rightarrowtail T$                                                      | $f: X \rightarrow T$                                                      | of the above square, we have $f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] \circ f = \iota_1 \circ [\mathrm{id}, \mathrm{id}] \circ g = g$ . This means that $f: X \rightarrow T$ |
| fong-spivak-invitation_F03785              | 324  | ■ 0.971 | $X \leftarrowtail N \rightarrowtail Y$                                        | $X \leftarrow N \rightarrow Y$                                            | the objects of $\mathcal{C}$ . Next, observe that a morphism in $\mathbf{Cospan}_F$ is a cospan $X \leftarrow N \rightarrow Y$ in $\mathcal{C}$                                   |
| fong-spivak-invitation_F03786              | 324  | ■ 0.980 | $FN = \{*\}$                                                                  | $FN = \{*\}$                                                              | together with an element of $FN = \{*\}$ . But $FN$ also has a unique element, $*$ ! So there's no                                                                                |
| fong-spivak-invitation_F03787              | 324  | ■ 0.980 | $F N$                                                                         | $FN$                                                                      | together with an element of $FN = \{*\}$ . But $FN$ also has a unique element, $*$ ! So there's no                                                                                |
| fong-spivak-invitation_F03788              | 324  | ■ 0.966 | $\operatorname{Cospan}_F = \operatorname{Cospan}_{\mathcal{C}}$               | $\operatorname{Cospan}_F = \operatorname{Cospan}_{\mathcal{C}}$           | so $\mathbf{Cospan}_F = \mathbf{Cospan}_{\mathcal{C}}$ .                                                                                                                          |
| fong-spivak-invitation_F03789              | 324  | ■ 0.974 | $\operatorname{Cospan}_{\mathcal{C}} \rightarrowtail \operatorname{Cospan}_F$ | $\operatorname{Cospan}_{\mathcal{C}} \rightarrow \operatorname{Cospan}_F$ | morphism is the identity-on-objects functor $\mathbf{Cospan}_{\mathcal{C}} \rightarrow \mathbf{Cospan}_F$ that simply decorates each                                              |
| fong-spivak-invitation_F03790              | 325  | ■ 0.848 | $V = \{\mathrm{ul}, \mathrm{ur}, \mathrm{dl}, \mathrm{dr}\}$                  | $V = \{\mathrm{ul}, \mathrm{ur}, \mathrm{dl}, \mathrm{dr}\}$              | We can represent the circuit in Eq. (6.11) by the tuple $(V, A, s, t, \ell)$ where $V = \{\mathrm{ul}, \mathrm{ur}, \mathrm{dl}, \mathrm{dr}\}$ .                                 |
| fong-spivak-invitation_F03791              | 325  | ■ 0.928 | $A = \{\mathrm{r}1, \mathrm{r}2, \mathrm{r}3, \mathrm{c}1, \mathrm{i}1\}$     | $A = \{\mathrm{r}1, \mathrm{r}2, \mathrm{r}3, \mathrm{c}1, \mathrm{i}1\}$ | $A = \{\mathrm{r}1, \mathrm{r}2, \mathrm{r}3, \mathrm{c}1, \mathrm{i}1\}$ , and $s, t$ , and $\ell$ are defined by the table                                                      |
| fong-spivak-invitation_F03798              | 325  | ■ 1.000 | $\psi_{\underline{2}, \underline{2}}(b, s)$                                   | $\psi_{\underline{2}, \underline{2}}(b, s)$                               | The circuit $\psi_{\underline{2}, \underline{2}}(b, s)$ is the disjoint union of the two labeled graphs $b$ and $s$ :                                                             |
| fong-spivak-invitation_F03799              | 325  | ■ 0.555 | $1 \rightarrowtail \underline{2} \stackrel{g}{\leftarrowtail} \underline{1}$  | $1 \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$            | The cospan is the cospan $\underline{1} \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$ , where $f(1) = 1$ and $g(1) = 2$ . The decoration is the                     |
| fong-spivak-invitation_F03800              | 325  | ■ 0.555 | $f(1)=1$                                                                      | $f(1) = 1$                                                                | The cospan is the cospan $\underline{1} \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$ , where $f(1) = 1$ and $g(1) = 2$ . The decoration is the                     |
| fong-spivak-invitation_F03801              | 325  | ■ 0.555 | $g(1)=2$                                                                      | $g(1) = 2$                                                                | The cospan is the cospan $\underline{1} \xrightarrow{f} \underline{2} \xleftarrow{g} \underline{1}$ , where $f(1) = 1$ and $g(1) = 2$ . The decoration is the                     |
| fong-spivak-invitation_F03802              | 325  | ■ 0.923 | $\underline{2}, \{a\}, s, t, \ell$                                            | $\underline{2}, \{a\}, s, t, \ell$                                        | $C$ -circuit $(\underline{2}, \{a\}, s, t, \ell)$ , where $s(a) = 1, t(a) = 2$ and $\ell(a) = \text{battery}$ .                                                                   |
| fong-spivak-invitation_F03803              | 325  | ■ 0.923 | $s(a)=1, t(a)=2$                                                              | $s(a) = 1, t(a) = 2$                                                      | $C$ -circuit $(\underline{2}, \{a\}, s, t, \ell)$ , where $s(a) = 1, t(a) = 2$ and $\ell(a) = \text{battery}$ .                                                                   |
| fong-spivak-invitation_F03804              | 325  | ■ 0.923 | $\ell(a)=$                                                                    | $\ell(a) =$                                                               | $C$ -circuit $(\underline{2}, \{a\}, s, t, \ell)$ , where $s(a) = 1, t(a) = 2$ and $\ell(a) = \text{battery}$ .                                                                   |
| fong-spivak-invitation_F03805              | 325  | ■ 1.000 | $C := (V, A, s, t, \ell)$                                                     | $C := (V, A, s, t, \ell)$                                                 | Recall the circuit $C := (V, A, s, t, \ell)$ from the solution to Exercise 6.64. Then the first decorated                                                                         |

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| fong-spivak-invitation_F03806 | 325  | 0.940 | $\underline{1} \xrightarrow{\quad} V \stackrel{g}{\leftarrow} \underline{2}, f(1) = \text{ul}, g(1) =$<br>$\underline{2}, f(1) = \text{ul}, g(1) =$ | $\underline{1} \xrightarrow{f} V \xleftarrow{g} \underline{2}, f(1) = \text{ul}, g(1) =$ | cospan is given by the cospan $\underline{1} \xrightarrow{f} V \xleftarrow{g} \underline{2}, f(1) = \text{ul}, g(1) =$ ur, and $g(2) = \text{ur}$ , decorated                               |
| fong-spivak-invitation_F03807 | 325  | 0.940 | $g(2) =$                                                                                                                                            | $g(2) =$                                                                                 | cospan is given by the cospan $\underline{1} \xrightarrow{f} V \xleftarrow{g} \underline{2}, f(1) = \text{ul}, g(1) = \text{ur}$ , and $g(2) = \text{ur}$ , decorated                       |
| fong-spivak-invitation_F03808 | 325  | 0.975 | $\underline{2} \xrightarrow{f'} V' \xleftarrow{g'} \underline{2}$                                                                                   | $\underline{2} \xrightarrow{f'} V' \xleftarrow{g'} \underline{2}$                        | by circuit C. The second decorated cospan is given by the cospan $\underline{2} \xrightarrow{f'} V' \xleftarrow{g'} \underline{2}$ and the circuit                                          |
| fong-spivak-invitation_F03809 | 325  | 1.000 | $C^{\prime} := (V^{\prime}, A^{\prime}, s^{\prime}, t^{\prime}, \ell^{\prime})$                                                                     | $C' := (V', A', s', t', \ell')$                                                          | $C' := (V', A', s', t', \ell')$ , where $V' = \{l, r, d\}$ , $A' = \{r1', r2'\}$ , and the functions are given by the                                                                       |
| fong-spivak-invitation_F03810 | 325  | 1.000 | $V^{\prime} = \{l, r, d\}, A^{\prime} = \{\text{r1}', \text{r2}'\}$                                                                                 | $V' = \{l, r, d\}, A' = \{r1', r2'\}$                                                    | $C' := (V', A', s', t', \ell')$ , where $V' = \{l, r, d\}, A' = \{r1', r2'\}$ , and the functions are given by the                                                                          |
| fong-spivak-invitation_F03815 | 325  | 0.989 | $V \stackrel{g}{\leftarrow} \underline{2} \xrightarrow{f'} V'$                                                                                      | $V \xleftarrow{g} \underline{2} \xrightarrow{f'} V'$                                     | To compose these, we first take the pushout of $V \xleftarrow{g} \underline{2} \xrightarrow{f'} V'$ . This gives the a new apex                                                             |
| fong-spivak-invitation_F03816 | 325  | 0.998 | $V^{\prime} = \{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$                                                                              | $V'' = \{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$                          | $V'' = \{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$ with five elements, and composite cospan $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$ given by $h(1) =$ |
| fong-spivak-invitation_F03817 | 325  | 0.998 | $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$                                                                                    | $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$                         | $V'' = \{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$ with five elements, and composite cospan $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$ given by $h(1) =$ |
| fong-spivak-invitation_F03818 | 325  | 0.998 | $h(1) =$                                                                                                                                            | $h(1) =$                                                                                 | $V'' = \{\text{ul}, \text{dl}, \text{dr}, \text{m}, \text{r}\}$ with five elements, and composite cospan $\underline{1} \xrightarrow{h} V'' \xleftarrow{k} \underline{2}$ given by $h(1) =$ |
| fong-spivak-invitation_F03819 | 325  | 0.388 | $k(1) = \text{r}$                                                                                                                                   | $k(1) = \text{r}$                                                                        | ul, $k(1) = \text{r}$ and $k(2) = \text{m}$ . The new circuit is given by $(V, ''A + A', s, ''t, ''\ell'')$ where the                                                                       |
| fong-spivak-invitation_F03820 | 325  | 0.388 | $k(2) = \text{m}$                                                                                                                                   | $k(2) = \text{m}$                                                                        | ul, $k(1) = \text{r}$ and $k(2) = \text{m}$ . The new circuit is given by $(V, ''A + A', s, ''t, ''\ell'')$ where the                                                                       |
| fong-spivak-invitation_F03821 | 325  | 0.388 | $(V, ''A + A', s, ''t, ''\ell'')$                                                                                                                   | $(V, ''A + A', s, ''t, ''\ell'')$                                                        | ul, $k(1) = \text{r}$ and $k(2) = \text{m}$ . The new circuit is given by $(V, ''A + A', s, ''t, ''\ell'')$ where the                                                                       |
| fong-spivak-invitation_F03825 | 326  | 1.000 | $\eta$                                                                                                                                              | $\eta$                                                                                   | Composing $\eta$ and $x$ we have                                                                                                                                                            |
| fong-spivak-invitation_F03826 | 327  | 0.637 | $2, 2, 2, 2 ; \emptyset$                                                                                                                            | $2, 2, 2, 2; \emptyset$                                                                  | 3. The composite $g \circ_1 f$ has arity $(2, 2, 2, 2; \emptyset)$ ; there is a depiction on the left:                                                                                      |
| fong-spivak-invitation_F03827 | 327  | 0.898 | $(A, B, A', B')$                                                                                                                                    | $(A, B, A', B')$                                                                         | We need to show that the $(A, B, A', B')$ square is a pullback iff the $(A, C, A', C')$ rectangle is a                                                                                      |
| fong-spivak-invitation_F03828 | 327  | 0.898 | $(A, C, A', C')$                                                                                                                                    | $(A, C, A', C')$                                                                         | We need to show that the $(A, B, A', B')$ square is a pullback iff the $(A, C, A', C')$ rectangle is a                                                                                      |
| fong-spivak-invitation_F03829 | 327  | 0.608 | $X, p, q$                                                                                                                                           | $X, p, q$                                                                                | Suppose first that $(A, B, A', B')$ is a pullback, and take any $(X, p, q)$ as in the following                                                                                             |
| fong-spivak-invitation_F03830 | 327  | 0.274 | $g^{\prime} = p$                                                                                                                                    | $g' = p$                                                                                 | where $q \circ f' \circ g' = p \circ h_3$ . Then by the universal property of the $(B, C, B', C')$ pullback, we get                                                                         |
| fong-spivak-invitation_F03831 | 327  | 0.274 | $h_3$                                                                                                                                               | $h_3$                                                                                    | where $q \circ f' \circ g' = p \circ h_3$ . Then by the universal property of the $(B, C, B', C')$ pullback, we get                                                                         |
| fong-spivak-invitation_F03832 | 327  | 0.229 | $r ; h_2 = g$                                                                                                                                       | $r; h_2 = g$                                                                             | In other words $r \circ h_2 = g \circ f'$ and $r \circ g = p$ . Then by the universal property of the $(A, B, A', B')$                                                                      |
| fong-spivak-invitation_F03833 | 327  | 0.229 | $r ; g = p$                                                                                                                                         | $r; g = p$                                                                               | In other words $r \circ h_2 = g \circ f'$ and $r \circ g = p$ . Then by the universal property of the $(A, B, A', B')$                                                                      |
| fong-spivak-invitation_F03834 | 327  | 1.000 | $r^{\prime} : X \rightarrow A$                                                                                                                      | $r' : X \rightarrow A$                                                                   | pullback, we get a unique dotted arrow $r' : X \rightarrow A$ making the right-hand diagram commute, i.e.                                                                                   |
| fong-spivak-invitation_F03835 | 327  | 0.999 | $r^{\prime} ; f = r$                                                                                                                                | $r'; f = r$                                                                              | $r' \circ f = r$ and $r' \circ h_1 = q$ . This gives the existence of an $r$ with the required property, $r' \circ f = r$                                                                   |
| fong-spivak-invitation_F03836 | 327  | 0.999 | $r^{\prime} ; h_1 = q$                                                                                                                              | $r'; h_1 = q$                                                                            | $r' \circ f = r$ and $r' \circ h_1 = q$ . This gives the existence of an $r$ with the required property, $r' \circ f = r$                                                                   |
| fong-spivak-invitation_F03837 | 328  | 1.000 | $r^{\prime} ; f ; g = r ; g = p$                                                                                                                    | $r'; f; g = r; g = p$                                                                    | and $r' \circ f \circ g = r \circ g = p$ . To see uniqueness, suppose given another morphisms $r_0$ such that                                                                               |
| fong-spivak-invitation_F03838 | 328  | 1.000 | $r_0$                                                                                                                                               | $r_0$                                                                                    | and $r' \circ f \circ g = r \circ g = p$ . To see uniqueness, suppose given another morphisms $r_0$ such that                                                                               |

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| fong-spivak-invitation_F03839 | 328  | ■ 0.279 | $r_{\{0\}} \stackrel{\circ}{\circ} \text{\circ} f \text{\circ} g = p$                       | $r_0 \stackrel{\circ}{\circ} f \text{\circ} g = p$ | $r_0 \circledcirc f \circledcirc g = p$ and $r_0 \circledcirc h_1 = q$ :                                                                                   |
| fong-spivak-invitation_F03840 | 328  | ■ 0.279 | $r_{\{0\}} \stackrel{\circ}{\circ} \text{\circ} h_1 = q$                                    | $r_0 \stackrel{\circ}{\circ} h_1 = q$              | $r_0 \circledcirc f \circledcirc g = p$ and $r_0 \circledcirc h_1 = q$ :                                                                                   |
| fong-spivak-invitation_F03841 | 328  | ■ 0.413 | $r_{\{0\}} \stackrel{\circ}{\circ} \text{\circ} f = r$                                      | $r_0 \stackrel{\circ}{\circ} f = r$                | Then by the uniqueness of $r$ , we must have $r_0 \circledcirc f = r$ , and then by the uniqueness of $r'$ , we must                                       |
| fong-spivak-invitation_F03842 | 328  | ■ 0.413 | $r^{\wedge\{\prime\}}$                                                                      | $r'$                                               | Then by the uniqueness of $r$ , we must have $r_0 \circledcirc f = r$ , and then by the uniqueness of $r'$ , we must                                       |
| fong-spivak-invitation_F03843 | 328  | ■ 1.000 | $r_{\{0\}} = r^{\wedge\{\prime\}}$                                                          | $r_0 = r'$                                         | have $r_0 = r'$ . This proves the first result.                                                                                                            |
| fong-spivak-invitation_F03844 | 328  | ■ 0.823 | $B, C, B^{\wedge\{\prime\}}, C^{\wedge\{\prime\}}$                                          | $B, C, B', C'$                                     | The second is similar. Suppose that $(A, C, A', C')$ and $(B, C, B', C')$ are pullbacks and                                                                |
| fong-spivak-invitation_F03845 | 328  | ■ 0.824 | $r ; h_{\{2\}} = q ; f^{\wedge\{\prime\}}$                                                  | $r; h_2 = q; f'$                                   | i.e. where $r \circledcirc h_2 = q \circledcirc f'$ . Then letting $p := r \circledcirc g$ , we have                                                       |
| fong-spivak-invitation_F03846 | 328  | ■ 0.824 | $p := r ; g$                                                                                | $p := r; g$                                        | i.e. where $r \circledcirc h_2 = q \circledcirc f'$ . Then letting $p := r \circledcirc g$ , we have                                                       |
| fong-spivak-invitation_F03847 | 328  | ■ 0.666 | $r^{\wedge\{\prime\}}: X \rightarrow$                                                       | $r': X \rightarrow$                                | so by the universal property of the $(A, C, A', C')$ pullback, there is a unique morphism $r': X \rightarrow$                                              |
| fong-spivak-invitation_F03848 | 328  | ■ 0.427 | $r^{\wedge\{\prime\}} ; f ; g = p$                                                          | $r'; f; g = p$                                     | $A$ such that $r' \circledcirc f \circledcirc g = p$ and $r_0 \circledcirc h_1 = q$ , as shown:                                                            |
| fong-spivak-invitation_F03849 | 328  | ■ 0.427 | $r_{\{0\}}: h_{\{1\}} = q$                                                                  | $r_0 : h_1 = q$                                    | $A$ such that $r' \circledcirc f \circledcirc g = p$ and $r_0 \circledcirc h_1 = q$ , as shown:                                                            |
| fong-spivak-invitation_F03850 | 328  | ■ 0.104 | $r_{\{0\}} := r^{\wedge\{\prime\}} \{ \}_{9}^{\wedge\{\circ\}} f$                           | $r_0 := r' \circledcirc f$                         | But now let $r_0 := r' \circledcirc f$ . It satisfies $r_0 \circledcirc g = p$ and $r_0 \circledcirc h_2 = q \circledcirc f'$ , and $r$ satisfies the same |
| fong-spivak-invitation_F03851 | 328  | ■ 0.104 | $r_{\{0\}} \{ \}_{9}^{\wedge\{\circ\}} g = p$                                               | $r_0 \circledcirc g = p$                           | But now let $r_0 := r' \circledcirc f$ . It satisfies $r_0 \circledcirc g = p$ and $r_0 \circledcirc h_2 = q \circledcirc f'$ , and $r$ satisfies the same |
| fong-spivak-invitation_F03852 | 328  | ■ 0.104 | $r_{\{0\}} \{ \}_{9}^{\wedge\{\circ\}} h_2 = q$                                             | $r_0 \circledcirc h_2 = q$                         | But now let $r_0 := r' \circledcirc f$ . It satisfies $r_0 \circledcirc g = p$ and $r_0 \circledcirc h_2 = q \circledcirc f'$ , and $r$ satisfies the same |
| fong-spivak-invitation_F03853 | 328  | ■ 1.000 | $a_{\{1\}}, a_{\{2\}} \in A$                                                                | $a_1, a_2 \in A$                                   | A function $f: A \rightarrow B$ is injective iff for all $\boxed{a_1, a_2 \in A}$ , if $f(a_1) = f(a_2)$ then $a_1 = a_2$ . It is a                        |
| fong-spivak-invitation_F03854 | 328  | ■ 1.000 | $f \text{\left(} a_{\{1\}} \text{\right)} = f \text{\left(} a_{\{2\}} \text{\right)}$       | $f(a_1) = f(a_2)$                                  | A function $f: A \rightarrow B$ is injective iff for all $a_1, a_2 \in A$ , if $\boxed{f(a_1) = f(a_2)}$ then $a_1 = a_2$ . It is a                        |
| fong-spivak-invitation_F03855 | 328  | ■ 1.000 | $a_{\{1\}} = a_{\{2\}}$                                                                     | $a_1 = a_2$                                        | A function $f: A \rightarrow B$ is injective iff for all $a_1, a_2 \in A$ , if $f(a_1) = f(a_2)$ then $a_1 = a_2$ . It is a                                |
| fong-spivak-invitation_F03856 | 328  | ■ 0.908 | $g_{\{1\}}, g_{\{2\}}: X \rightarrow A$                                                     | $g_1, g_2: X \rightarrow A$                        | monomorphism iff for all sets $X$ and functions $g_1, g_2: X \rightarrow A$ , if $g_1 \circledcirc f = g_2 \circledcirc f$ then $g_1 = g_2$ .              |
| fong-spivak-invitation_F03857 | 328  | ■ 0.908 | $f = g_{\{2\}}$                                                                             | $f = g_2$                                          | monomorphism iff for all sets $X$ and functions $g_1, g_2: X \rightarrow A$ , if $g_1 \circledcirc f = g_2 \circledcirc f$ then $g_1 = g_2$ .              |
| fong-spivak-invitation_F03858 | 328  | ■ 0.908 | $g_{\{1\}} = g_{\{2\}}$                                                                     | $g_1 = g_2$                                        | monomorphism iff for all sets $X$ and functions $g_1, g_2: X \rightarrow A$ , if $g_1 \circledcirc f = g_2 \circledcirc f$ then $g_1 = g_2$ .              |
| fong-spivak-invitation_F03859 | 329  | ■ 1.000 | $X = \{*\}$                                                                                 | $X = \{*\}$                                        | Let $X = \{*\}$ be a one-element set, and let $g_1, g_2: X \rightarrow A$ be given by $g_1(*) := a_1$ and                                                  |
| fong-spivak-invitation_F03860 | 329  | ■ 1.000 | $g_{\{1\}}(*) := a_{\{1\}}$                                                                 | $g_1(*) := a_1$                                    | Let $X = \{*\}$ be a one-element set, and let $g_1, g_2: X \rightarrow A$ be given by $g_1(*) := a_1$ and                                                  |
| fong-spivak-invitation_F03861 | 329  | ■ 0.509 | $g_{\{2\}}(*) := a_{\{2\}}$                                                                 | $g_2(*) := a_2$                                    | $\boxed{g_2(*) := a_2}$ . Then $g_1 \circledcirc f = g_2 \circledcirc f$ , so $g_1 = g_2$ , so $a_1 = a_2$ .                                               |
| fong-spivak-invitation_F03862 | 329  | ■ 0.378 | $f =$                                                                                       | $f =$                                              | 2. Suppose that $f$ is an injection, let $X$ be any set, and let $g_1, g_2: X \rightarrow A$ be such that $g_1 \circledcirc f =$                           |
| fong-spivak-invitation_F03863 | 329  | ■ 0.653 | $g_{\{1\}}(x) = g_{\{2\}}(x)$                                                               | $g_1(x) = g_2(x)$                                  | $g_2 \circledcirc f$ . We will have $g_1 = g_2$ if we can show that $\boxed{g_1(x) = g_2(x)}$ for every $x \in X$ . So take any                            |
| fong-spivak-invitation_F03864 | 329  | ■ 1.000 | $f \text{\left(} g_{\{1\}}(x) \text{\right)} = f \text{\left(} g_{\{2\}}(x) \text{\right)}$ | $f(g_1(x)) = f(g_2(x))$                            | $x \in X$ ; since $\boxed{f(g_1(x)) = f(g_2(x))}$ and $f$ is injective, we have $g_1(x) = g_2(x)$ as desired.                                              |
| fong-spivak-invitation_F03865 | 329  | ■ 0.605 | $j := i^{\wedge\{-1\}}$                                                                     | $j := i^{-1}$                                      | Let $j := i^{-1}$ be the inverse of $i$ , and consider $g := (f \circledcirc j): A \rightarrow B'$ . Then $g \circledcirc i = f$ , so by the               |
| fong-spivak-invitation_F03866 | 329  | ■ 0.605 | $g := (f: j): A \rightarrow B^{\wedge\{\prime\}}$                                           | $g := (f: j): A \rightarrow B'$                    | Let $j := i^{-1}$ be the inverse of $i$ , and consider $g := (f \circledcirc j): A \rightarrow B'$ . Then $g \circledcirc i = f$ , so by the               |

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| fong-spivak-invitation_F03867 | 329  | 0.605 | $g: i = f$                                                                                                                      | $g : i = f$                                                             | Let $j := i^{-1}$ be the inverse of $i$ , and consider $g := (f \circ j) : A \rightarrow B'$ . Then $g \circ i = f$ , so by the                                          |
| fong-spivak-invitation_F03868 | 329  | 0.990 | $j^{\prime} : A \rightarrow A^{\prime}$                                                                                         | $j' : A \rightarrow A'$                                                 | existence part of the universal property, there is a map $j' : A \rightarrow A'$ such that $j' \circ i' = \text{id}_A$ and                                               |
| fong-spivak-invitation_F03869 | 329  | 0.990 | $j^{\prime} ; i^{\prime} = \mathrm{id}_A$                                                                                       | $j'; i' = \text{id}_A$                                                  | existence part of the universal property, there is a map $j' : A \rightarrow A'$ such that $j' \circ i' = \text{id}_A$ and                                               |
| fong-spivak-invitation_F03870 | 329  | 0.357 | $j^{\prime} \circ f^{\prime} = f^{\circ} j$                                                                                     | $j' \circ f' = f \circ j$                                               | $j' \circ f' = f \circ j$ . We will be done if we can show $i' \circ j' = \text{id}_{A'}$ . One checks that $(i' \circ j') \circ i' = i'$ and                            |
| fong-spivak-invitation_F03871 | 329  | 0.357 | $i^{\prime} \circ j^{\prime} = \mathrm{id}_{A^{\prime}}$                                                                        | $i' \circ j' = \text{id}_{A'}$                                          | $j' \circ f' = f \circ j$ . We will be done if we can show $i' \circ j' = \text{id}_{A'}$ . One checks that $(i' \circ j') \circ i' = i'$ and                            |
| fong-spivak-invitation_F03872 | 329  | 0.357 | $\left( i^{\prime} \circ j^{\prime} \right) \circ i^{\prime} = i^{\prime}$                                                      | $(i' \circ j') \circ i' = i'$                                           | $j' \circ f' = f \circ j$ . We will be done if we can show $i' \circ j' = \text{id}_{A'}$ . One checks that $(i' \circ j') \circ i' = i'$ and                            |
| fong-spivak-invitation_F03873 | 329  | 0.184 | $\left( i^{\prime} \circ j^{\prime} \right) \circ f^{\prime} = i^{\prime} \circ f^{\circ} j = f^{\circ} i \circ j = f^{\prime}$ | $(i' \circ j') \circ f' = i' \circ f \circ j = f' \circ i \circ j = f'$ | that $(i' \circ j') \circ f' = i' \circ f \circ j = f' \circ i \circ j = f'$ . But $\text{id}_{A'}$ also satisfies those properties: $\text{id}_{A'} \circ i' = i'$      |
| fong-spivak-invitation_F03874 | 329  | 0.184 | $\mathrm{id}_{A^{\prime}}$                                                                                                      | $\text{id}_{A'}$                                                        | that $(i' \circ j') \circ f' = i' \circ f \circ j = f' \circ i \circ j = f'$ . But $\text{id}_{A'}$ also satisfies those properties: $\text{id}_{A'} \circ i' = i'$      |
| fong-spivak-invitation_F03875 | 329  | 0.184 | $\mathrm{id}_{A^{\prime}} \circ i^{\prime} = i^{\prime}$                                                                        | $\text{id}_{A'} \circ i' = i'$                                          | that $(i' \circ j') \circ f' = i' \circ f \circ j = f' \circ i \circ j = f'$ . But $\text{id}_{A'}$ also satisfies those properties: $\text{id}_{A'} \circ i' = i'$      |
| fong-spivak-invitation_F03876 | 329  | 0.774 | $\mathrm{id}_{A^{\prime}} \circ f^{\prime} = f^{\prime}$                                                                        | $\text{id}_{A'} \circ f' = f'$                                          | and $\text{id}_{A'} \circ f' = f'$ , so by the uniqueness part of the universal property, $(i' \circ j') = \text{id}_{A'}$ .                                             |
| fong-spivak-invitation_F03877 | 329  | 0.774 | $\left( i^{\prime} ; j^{\prime} \right) = \mathrm{id}_{A^{\prime}}$                                                             | $(i'; j') = \text{id}_{A'}$                                             | and $\text{id}_{A'} \circ f' = f'$ , so by the uniqueness part of the universal property, $(i' \circ j') = \text{id}_{A'}$ .                                             |
| fong-spivak-invitation_F03878 | 329  | 0.669 | $g: X \rightarrow A$                                                                                                            | $g : X \rightarrow A$                                                   | So take any object $X$ and morphisms $g : X \rightarrow A$ and $h : X \rightarrow B$ such that $g \circ f = h \circ \text{id}_B$ .                                       |
| fong-spivak-invitation_F03879 | 329  | 0.669 | $h: X \rightarrow B$                                                                                                            | $h : X \rightarrow B$                                                   | So take any object $X$ and morphisms $g : X \rightarrow A$ and $h : X \rightarrow B$ such that $g \circ f = h \circ \text{id}_B$ .                                       |
| fong-spivak-invitation_F03880 | 329  | 0.669 | $f=h$                                                                                                                           | $f = h$                                                                 | So take any object $X$ and morphisms $g : X \rightarrow A$ and $h : X \rightarrow B$ such that $g \circ f = h \circ \text{id}_B$ .                                       |
| fong-spivak-invitation_F03881 | 329  | 0.669 | $\mathrm{id}_B$                                                                                                                 | $\text{id}_B$                                                           | So take any object $X$ and morphisms $g : X \rightarrow A$ and $h : X \rightarrow B$ such that $g \circ f = h \circ \text{id}_B$ .                                       |
| fong-spivak-invitation_F03882 | 329  | 0.650 | $r: X \rightarrow A$                                                                                                            | $r : X \rightarrow A$                                                   | We need to show there is a unique morphism $r : X \rightarrow A$ such that $r \circ \text{id}_A = g$ and $r \circ f = h$ .                                               |
| fong-spivak-invitation_F03883 | 329  | 0.650 | $r \circ \text{id}_A = g$                                                                                                       | $r \circ \text{id}_A = g$                                               | We need to show there is a unique morphism $r : X \rightarrow A$ such that $r \circ \text{id}_A = g$ and $r \circ f = h$ .                                               |
| fong-spivak-invitation_F03884 | 329  | 0.650 | $r: f=h$                                                                                                                        | $r : f = h$                                                             | We need to show there is a unique morphism $r : X \rightarrow A$ such that $r \circ \text{id}_A = g$ and $r \circ f = h$ .                                               |
| fong-spivak-invitation_F03885 | 329  | 1.000 | $r=g$                                                                                                                           | $r = g$                                                                 | That's easy: the first requirement forces $r = g$ and the second requirement is then fulfilled.                                                                          |
| fong-spivak-invitation_F03886 | 330  | 1.000 | $v \otimes w$                                                                                                                   | $v \otimes w$                                                           | • $v \otimes w$ serves as a meet operation, i.e. it satisfies the same universal property as $\wedge$ , namely                                                           |
| fong-spivak-invitation_F03887 | 330  | 0.958 | $v \leq (w \multimap x)$                                                                                                        | $v \leq (w \multimap x)$                                                | does, namely $v \leq (w \multimap x)$ iff $v \otimes w \leq x$ . So $\mathcal{V}$ is a cartesian closed category, and of course                                          |
| fong-spivak-invitation_F03888 | 330  | 0.958 | $v \otimes W \leq x$                                                                                                            | $v \otimes W \leq x$                                                    | does, namely $v \leq (w \multimap x)$ iff $v \otimes w \leq x$ . So $\mathcal{V}$ is a cartesian closed category, and of course                                          |
| fong-spivak-invitation_F03889 | 330  | 0.991 | $\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$                                                                          | $\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$                  | This is the product preorder $\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$ : its objects are pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ with $(a, b) \leq$ |
| fong-spivak-invitation_F03890 | 330  | 0.991 | $(a, b) \in \mathbb{N} \times \mathbb{N}$                                                                                       | $(a, b) \in \mathbb{N} \times \mathbb{N}$                               | This is the product preorder $\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$ : its objects are pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ with $(a, b) \leq$ |
| fong-spivak-invitation_F03891 | 330  | 0.991 | $(a, b) \leq$                                                                                                                   | $(a, b) \leq$                                                           | This is the product preorder $\mathbb{N}^{\text{op}} \times \mathbb{N}^{\text{op}}$ : its objects are pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ with $(a, b) \leq$ |
| fong-spivak-invitation_F03892 | 330  | 0.600 | $(a^{\prime}, b^{\prime})$                                                                                                      | $(a', b')$                                                              | $(a', b')$ iff, in the usual ordering on $\mathbb{N}$ , we have $a' \leq a$ and $b' \leq b$ . But you can just look at the                                               |
| fong-spivak-invitation_F03893 | 330  | 0.600 | $a^{\prime} \leq a$                                                                                                             | $a' \leq a$                                                             | $(a', b')$ iff, in the usual ordering on $\mathbb{N}$ , we have $a' \leq a$ and $b' \leq b$ . But you can just look at the                                               |

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| fong-spivak-invitation_F03894 | 330  | 0.600 | $b^{\{\prime\}}$ $\leq b$                                                                                | $b' \leq b$                                                                                              | $(a', b')$ iff, in the usual ordering on $\mathbb{N}$ , we have $a' \leq a$ and $b' \leq b$ . But you can just look at the                                                                               |
| fong-spivak-invitation_F03895 | 330  | 0.859 | $(a, b) \wedge (a^{\{\prime\}}, b^{\{\prime\}}) = (\max(a, a^{\{\prime\}}), \max(b, b^{\{\prime\}}))$    | $(a, b) \wedge (a', b') = (\max(a, a'), \max(b, b'))$                                                    | It has a top element, $(0, 0)$ , and it has binary meets, $(a, b) \wedge (a', b') = (\max(a, a'), \max(b, b'))$ .                                                                                        |
| fong-spivak-invitation_F03896 | 330  | 0.998 | $x \multimap y$                                                                                          | $x \multimap y$                                                                                          | for each $x, y$ , the hom-object $x \multimap y$ exists. The formula for it is $x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$ ,                                                                  |
| fong-spivak-invitation_F03897 | 330  | 0.998 | $x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$                                                   | $x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$                                                   | for each $x, y$ , the hom-object $x \multimap y$ exists. The formula for it is $x \multimap y = \bigvee \{w \mid w \wedge x \leq y\}$ .                                                                  |
| fong-spivak-invitation_F03898 | 330  | 1.000 | $y \wedge x \leq y$                                                                                      | $y \wedge x \leq y$                                                                                      | i.e. we need these particular joins to exist. Since $y \wedge x \leq y$ , we have $y \leq x \multimap y$ . So we can                                                                                     |
| fong-spivak-invitation_F03899 | 330  | 1.000 | $y \leq x \multimap y$                                                                                   | $y \leq x \multimap y$                                                                                   | i.e. we need these particular joins to exist. Since $y \wedge x \leq y$ , we have $y \leq x \multimap y$ . So we can                                                                                     |
| fong-spivak-invitation_F03900 | 330  | 1.000 | $x \multimap y = \bigvee \{w \mid y \leq w \text{ and } w \wedge x \leq y\}$                             | $x \multimap y = \bigvee \{w \mid y \leq w$                                                              | replace the formula with $x \multimap y = \bigvee \{w \mid y \leq w \text{ and } w \wedge x \leq y\}$ . But the set of elements                                                                          |
| fong-spivak-invitation_F03901 | 330  | 1.000 | $w \wedge x \leq y\}$                                                                                    | $w \wedge x \leq y\}$                                                                                    | replace the formula with $x \multimap y = \bigvee \{w \mid y \leq w \text{ and } w \wedge x \leq y\}$ . But the set of elements                                                                          |
| fong-spivak-invitation_F03902 | 330  | 0.543 | $\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$                                               | $\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$                                               | in $\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$ that are bigger than $y$ is finite and nonempty. <sup>2</sup> So this is a finite nonempty join,                                           |
| fong-spivak-invitation_F03903 | 330  | 0.651 | $\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$                                               | $\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$                                               | and $\mathbb{N}^{\mathrm{OP}} \times \mathbb{N}^{\mathrm{OP}}$ has all finite nonempty joins: they are given by inf.                                                                                     |
| fong-spivak-invitation_F03904 | 330  | 1.000 | $y = (a, b)$                                                                                             | $y = (a, b)$                                                                                             | <sup>2</sup> If $y = (a, b)$ then there are exactly $(a + 1) * (b + 1)$ elements $y'$ for which $y \leq y'$ .                                                                                            |
| fong-spivak-invitation_F03905 | 330  | 1.000 | $(a+1) * (b+1)$                                                                                          | $(a + 1) * (b + 1)$                                                                                      | <sup>2</sup> If $y = (a, b)$ then there are exactly $(a + 1) * (b + 1)$ elements $y'$ for which $y \leq y'$ .                                                                                            |
| fong-spivak-invitation_F03906 | 331  | 0.997 | $m: \mathbb{Z} \rightarrow \mathbb{B}$                                                                   | $m: \mathbb{Z} \rightarrow \mathbb{B}$                                                                   | Let $m: \mathbb{Z} \rightarrow \mathbb{B}$ be the characteristic function of the inclusion $\mathbb{N} \subseteq \mathbb{Z}$ .                                                                           |
| fong-spivak-invitation_F03907 | 331  | 0.997 | $\mathbb{N} \subseteq \mathbb{Z}$                                                                        | $\mathbb{N} \subseteq \mathbb{Z}$                                                                        | Let $m: \mathbb{Z} \rightarrow \mathbb{B}$ be the characteristic function of the inclusion $\mathbb{N} \subseteq \mathbb{Z}$ .                                                                           |
| fong-spivak-invitation_F03908 | 331  | 0.992 | $\ulcorner m \urcorner(-5) =$                                                                            | $\ulcorner m \urcorner(-5) =$                                                                            | 1. $\ulcorner m \urcorner(-5) = \text{false}$ .                                                                                                                                                          |
| fong-spivak-invitation_F03909 | 331  | 1.000 | $\ulcorner m \urcorner(0) =$                                                                             | $\ulcorner m \urcorner(0) =$                                                                             | 2. $\ulcorner m \urcorner(0) = \text{true}$ .                                                                                                                                                            |
| fong-spivak-invitation_F03910 | 331  | 0.615 | $\ulcorner !\mathbb{N} \urcorner: \mathbb{N} \rightarrow \mathbb{B}$                                     | $\ulcorner !\mathbb{N} \urcorner: \mathbb{N} \rightarrow \mathbb{B}$                                     | 2. Let $!\mathbb{N}: \emptyset \rightarrow \mathbb{N}$ be the inclusion of the empty set. The characteristic function $\ulcorner !\mathbb{N} \urcorner: \mathbb{N} \rightarrow \mathbb{B}$               |
| fong-spivak-invitation_F03911 | 331  | 1.000 | $A \subseteq \mathbb{B}$                                                                                 | $A \subseteq \mathbb{B}$                                                                                 | 1. The sort of thing $(*)$ we're looking for is a subobject of $\mathbb{B}$ , say $A \subseteq \mathbb{B}$ . This would have a                                                                           |
| fong-spivak-invitation_F03912 | 331  | 0.999 | $\ulcorner \Rightarrow \urcorner: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$                   | $\ulcorner \Rightarrow \urcorner: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$                   | 3. The characteristic function for $P \Rightarrow Q$ is the function $\ulcorner \Rightarrow \urcorner: \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$ given by the first,                          |
| fong-spivak-invitation_F03913 | 331  | 0.859 | $\subseteq \mathbb{B} \times \mathbb{B}$                                                                 | $\subseteq \mathbb{B} \times \mathbb{B}$                                                                 | 4. It classifies the subset $\{(\text{true}, \text{true}), (\text{false}, \text{true}), (\text{false}, \text{false})\} \subseteq \mathbb{B} \times \mathbb{B}$ .                                         |
| fong-spivak-invitation_F03914 | 331  | 0.980 | $\ulcorner E \urcorner, \ulcorner P \urcorner, \ulcorner T \urcorner: \mathbb{N} \rightarrow \mathbb{B}$ | $\ulcorner E \urcorner, \ulcorner P \urcorner, \ulcorner T \urcorner: \mathbb{N} \rightarrow \mathbb{B}$ | Say that $\ulcorner E \urcorner, \ulcorner P \urcorner, \ulcorner T \urcorner: \mathbb{N} \rightarrow \mathbb{B}$ classify respectively the subsets $E := \{n \in \mathbb{N} \mid n \text{ is even}\}$ , |
| fong-spivak-invitation_F03915 | 331  | 1.000 | $P := \{n \in \mathbb{N} \mid n \text{ is prime}\}$                                                      | $P := \{n \in \mathbb{N} \mid n$                                                                         | $P := \{n \in \mathbb{N} \mid n \text{ is prime}\}$ , and $T := \{n \in \mathbb{N} \mid n \geq 10\}$ of $\mathbb{N}$ .                                                                                   |
| fong-spivak-invitation_F03916 | 331  | 0.999 | $\ulcorner E \urcorner(17) =$                                                                            | $\ulcorner E \urcorner(17) =$                                                                            | 1. $\ulcorner E \urcorner(17) = \text{false}$ because 17 is not even.                                                                                                                                    |
| fong-spivak-invitation_F03917 | 331  | 0.865 | $\ulcorner P \urcorner(17) =$                                                                            | $\ulcorner P \urcorner(17) =$                                                                            | 2. $\ulcorner P \urcorner(17) = \text{true}$ because 17 is prime.                                                                                                                                        |
| fong-spivak-invitation_F03918 | 331  | 0.690 | $\ulcorner T \urcorner(17) = \text{true}$                                                                | $\ulcorner T \urcorner(17) = \text{true}$                                                                | 3. $\ulcorner T \urcorner(17) = \text{true}$ because $17 \geq 10$ .                                                                                                                                      |
| fong-spivak-invitation_F03919 | 331  | 0.690 | $17 \geq 10$                                                                                             | $17 \geq 10$                                                                                             | 3. $\ulcorner T \urcorner(17) = \text{true}$ because $17 \geq 10$ .                                                                                                                                      |
| fong-spivak-invitation_F03920 | 331  | 0.995 | $B(x, \epsilon) := \{x' \in \mathbb{R} \mid$                                                             | $B(x, \epsilon) := \{x' \in \mathbb{R} \mid$                                                             | 1. The one-dimensional analogue of an $\epsilon$ -ball around a point $x \in \mathbb{R}$ is $B(x, \epsilon) := \{x' \in \mathbb{R} \mid$                                                                 |
| fong-spivak-invitation_F03921 | 331  | 1.000 | $ x - x'  < \epsilon\}$                                                                                  | $ x - x'  < \epsilon\}$                                                                                  | $ x - x'  < \epsilon$ , i.e. the set of all points within $\epsilon$ of $x$ .                                                                                                                            |

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| fong-spivak-invitation_F03922 | 331  | ■ 1.000 | $x \in U$                                                                                                                                    | $x \in U$                                                               | 2. A subset $U \subseteq \mathbb{R}$ is open if, for every $x \in U$ , there is some $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U$ . |
| fong-spivak-invitation_F03923 | 331  | ■ 1.000 | $B(x, \epsilon) \subseteq U$                                                                                                                 | $B(x, \epsilon) \subseteq U$                                            | 2. A subset $U \subseteq \mathbb{R}$ is open if, for every $x \in U$ , there is some $\epsilon > 0$ such that $B(x, \epsilon) \subseteq U$ . |
| fong-spivak-invitation_F03924 | 331  | ■ 1.000 | $U_{\{1\}} := \{x \in \mathbb{R} \mid 0 < x < 2\}$                                                                                           | $U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$                            | 3. Let $U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$ and $U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$ . Then $U := U_1 \cup U_2 = \{x \in$    |
| fong-spivak-invitation_F03925 | 331  | ■ 1.000 | $U_{\{2\}} := \{x \in \mathbb{R} \mid 1 < x < 3\}$                                                                                           | $U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$                            | 3. Let $U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$ and $U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$ . Then $U := U_1 \cup U_2 = \{x \in$    |
| fong-spivak-invitation_F03926 | 331  | ■ 1.000 | $U := U_{\{1\}} \cup U_{\{2\}} = \{x \in$                                                                                                    | $U := U_1 \cup U_2 = \{x \in$                                           | 3. Let $U_1 := \{x \in \mathbb{R} \mid 0 < x < 2\}$ and $U_2 := \{x \in \mathbb{R} \mid 1 < x < 3\}$ . Then $U := U_1 \cup U_2 = \{x \in$    |
| fong-spivak-invitation_F03927 | 331  | ■ 1.000 | $\mathbb{R} \mid 0 < x < 3\}$                                                                                                                | $\mathbb{R} \mid 0 < x < 3\}$                                           | $\mathbb{R} \mid 0 < x < 3\}$ .                                                                                                              |
| fong-spivak-invitation_F03928 | 332  | ■ 1.000 | $I = \{1, 2, 3, 4, \dots\}$                                                                                                                  | $I = \{1, 2, 3, 4, \dots\}$                                             | 4. Let $I = \{1, 2, 3, 4, \dots\}$ and for each $i \in I$ let $U_i := \{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\}$ , so we have            |
| fong-spivak-invitation_F03929 | 332  | ■ 1.000 | $U_{\{i\}} := \left\{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\right\}$                                                                      | $U_i := \left\{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\right\}$       | 4. Let $I = \{1, 2, 3, 4, \dots\}$ and for each $i \in I$ let $U_i := \{x \in \mathbb{R} \mid \frac{1}{i} < x < 1\}$ , so we have            |
| fong-spivak-invitation_F03930 | 332  | ■ 1.000 | $U_{\{1\}} \subseteq U_{\{2\}} \subseteq U_{\{3\}} \subseteq \dots$                                                                          | $U_1 \subseteq U_2 \subseteq U_3 \subseteq \dots$                       | $U_1 \subseteq U_2 \subseteq U_3 \subseteq \dots$ . Their union is $U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$ .      |
| fong-spivak-invitation_F03931 | 332  | ■ 1.000 | $U := \bigcup_{i \in I} U_{\{i\}} = \{x \in \mathbb{R} \mid 0 < x < 1\}$                                                                     | $U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$      | $U_1 \subseteq U_2 \subseteq U_3 \subseteq \dots$ . Their union is $U := \bigcup_{i \in I} U_i = \{x \in \mathbb{R} \mid 0 < x < 1\}$ .      |
| fong-spivak-invitation_F03932 | 332  | ■ 1.000 | $\varnothing \subseteq X$                                                                                                                    | $\emptyset \subseteq X$                                                 | 1. The coarse topology on $X$ is the one whose only open sets are $X \subseteq X$ and $\emptyset \subseteq X$ . This is a                    |
| fong-spivak-invitation_F03933 | 332  | ■ 1.000 | $\bigcup_{i \in I} A_{\{i\}}$                                                                                                                | $\bigcup_{i \in I} A_i$                                                 | (the union $\bigcup_{i \in I} A_i$ is $X$ iff $A_i = X$ for some $i \in I$ ).                                                                |
| fong-spivak-invitation_F03934 | 332  | ■ 1.000 | $A_{\{i\}} = X$                                                                                                                              | $A_i = X$                                                               | (the union $\bigcup_{i \in I} A_i$ is $X$ iff $A_i = X$ for some $i \in I$ ).                                                                |
| fong-spivak-invitation_F03935 | 332  | ■ 1.000 | $A \subseteq X$                                                                                                                              | $A \subseteq X$                                                         | 2. The fine topology on $X$ is the one where every subset $A \subseteq X$ is considered open. All the                                        |
| fong-spivak-invitation_F03936 | 332  | ■ 1.000 | $f^{-1}(U) \subseteq X$                                                                                                                      | $f^{-1}(U) \subseteq X$                                                 | $f^{-1}(U) \subseteq X$ is open, and everything in $X$ is open.                                                                              |
| fong-spivak-invitation_F03937 | 332  | ■ 1.000 | $\varnothing \rightarrow \{1\} \rightarrow \{1, 2\}$                                                                                         | $\emptyset \rightarrow \{1\} \rightarrow \{1, 2\}$                      | 1. The Hasse diagram for the Sierpinski topology is $\emptyset \rightarrow \{1\} \rightarrow \{1, 2\}$ .                                     |
| fong-spivak-invitation_F03938 | 332  | ■ 1.000 | $U_{\{i\}} = U$                                                                                                                              | $U_i = U$                                                               | • or $U_i = U$ for some $i \in I$ .                                                                                                          |
| fong-spivak-invitation_F03939 | 332  | ■ 0.996 | $Y = B \cap Y$                                                                                                                               | $Y = B \cap Y$                                                          | 1. We want to show that $Y \in \mathbf{Op}_{? \cap Y}$ . We need to find $B \in \mathbf{Op}$ such that $Y = B \cap Y$ ; this is easy,        |
| fong-spivak-invitation_F03940 | 332  | ■ 1.000 | $B = X$                                                                                                                                      | $B = X$                                                                 | take $B = X$ .                                                                                                                               |
| fong-spivak-invitation_F03941 | 332  | ■ 1.000 | $\varnothing = \varnothing \cap Y$                                                                                                           | $\emptyset = \emptyset \cap Y$                                          | trary union. Note $\emptyset = \emptyset \cap Y$ , so according to the formula, $\emptyset \in \mathbf{Op}_{? \cap Y}$ . Suppose that        |
| fong-spivak-invitation_F03942 | 332  | ■ 1.000 | $\varnothing \in \mathbf{Op}_{? \cap Y}$                                                                                                     | $\emptyset \in \mathbf{Op}_{? \cap Y}$                                  | trary union. Note $\emptyset = \emptyset \cap Y$ , so according to the formula, $\emptyset \in \mathbf{Op}_{? \cap Y}$ . Suppose that        |
| fong-spivak-invitation_F03943 | 332  | ■ 0.929 | $A_{\{1\}}, A_{\{2\}} \in \mathbf{Op}_{? \cap Y}$                                                                                            | $A_1, A_2 \in \mathbf{Op}_{? \cap Y}$                                   | $A_1, A_2 \in \mathbf{Op}_{? \cap Y}$ . Then there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$ .         |
| fong-spivak-invitation_F03944 | 332  | ■ 0.929 | $B_{\{1\}}, B_{\{2\}} \in \mathbf{Op}$                                                                                                       | $B_1, B_2 \in \mathbf{Op}$                                              | $A_1, A_2 \in \mathbf{Op}_{? \cap Y}$ . Then there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$ .         |
| fong-spivak-invitation_F03945 | 332  | ■ 0.929 | $A_{\{1\}} = B_{\{1\}} \cap Y$                                                                                                               | $A_1 = B_1 \cap Y$                                                      | $A_1, A_2 \in \mathbf{Op}_{? \cap Y}$ . Then there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$ .         |
| fong-spivak-invitation_F03946 | 332  | ■ 0.929 | $A_{\{2\}} = B_{\{2\}} \cap Y$                                                                                                               | $A_2 = B_2 \cap Y$                                                      | $A_1, A_2 \in \mathbf{Op}_{? \cap Y}$ . Then there exist $B_1, B_2 \in \mathbf{Op}$ with $A_1 = B_1 \cap Y$ and $A_2 = B_2 \cap Y$ .         |
| fong-spivak-invitation_F03947 | 332  | ■ 0.892 | $A_{\{1\}} \cap A_{\{2\}} = \left(B_{\{1\}} \cap Y\right) \cap \left(B_{\{2\}} \cap Y\right) = \left(B_{\{1\}} \cap B_{\{2\}}\right) \cap Y$ | $A_1 \cap A_2 = (B_1 \cap Y) \cap (B_2 \cap Y) = (B_1 \cap B_2) \cap Y$ | But then $A_1 \cap A_2 = (B_1 \cap Y) \cap (B_2 \cap Y) = (B_1 \cap B_2) \cap Y$ , so it is in $\mathbf{Op}_{? \cap Y}$ since                |
| fong-spivak-invitation_F03948 | 332  | ■ 0.999 | $B_{\{1\}} \cap B_{\{2\}} \in \mathbf{Op}$                                                                                                   | $B_1 \cap B_2 \in \mathbf{Op}$                                          | $B_1 \cap B_2 \in \mathbf{Op}$ . The same idea works for arbitrary unions: given a set $I$ and $A_i$ for each $i \in I$ ,                    |

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| fong-spivak-invitation-F03949 | 332  | ■ 0.999 | $A_{\{i\}}$                                                                 | $A_i$                                                        | $B_1 \cap B_2 \in \mathbf{Op}$ . The same idea works for arbitrary unions: given a set $I$ and $A_i$ for each $i \in I$ ,                             |
| fong-spivak-invitation-F03950 | 332  | ■ 1.000 | $A_{\{i\}} = B_{\{i\}} \cap Y$                                              | $A_i = B_i \cap Y$                                           | we have $A_i = B_i \cap Y$ for some $B_i \in \mathbf{Op}$ , and                                                                                       |
| fong-spivak-invitation-F03951 | 332  | ■ 1.000 | $B_{\{i\}} \in \mathbf{Op}$                                                 | $B_i \in \mathbf{Op}$                                        | we have $A_i = B_i \cap Y$ for some $B_i \in \mathbf{Op}$ , and                                                                                       |
| fong-spivak-invitation-F03952 | 332  | ■ 1.000 | $U \subseteq \mathbf{Op}$                                                   | $U \subseteq \mathbf{Op}$                                    | each labeled by an open set $U \subseteq \mathbf{Op}$ . It might look something like this:                                                            |
| fong-spivak-invitation-F03953 | 333  | ■ 1.000 | $\mathcal{C}(B, C) = \left( (U_3 \wedge U_1) \vee (U_4 \wedge U_2) \right)$ | $\mathcal{C}(B, C) = (U_3 \wedge U_1) \vee (U_4 \wedge U_2)$ | $\mathcal{C}(B, C) = (U_3 \wedge U_1) \vee (U_4 \wedge U_2)$ , because $\wedge$ is the monoidal product in $\mathcal{V}$ .                            |
| fong-spivak-invitation-F03954 | 333  | ■ 1.000 | $\mathcal{C}(a, b)$                                                         | $\mathcal{C}(a, b)$                                          | In general, we can thus imagine the open set $\mathcal{C}(a, b)$ as a kind of “size restriction” for getting                                          |
| fong-spivak-invitation-F03955 | 333  | ■ 1.000 | $\left\{ a_{\{1\}}, a_{\{2\}} \right\}$                                     | $\{a_1, a_2\}$                                               | 1. The fiber of $f$ over $a$ is $\{a_1, a_2\}$ .                                                                                                      |
| fong-spivak-invitation-F03956 | 333  | ■ 1.000 | $\left\{ c_{\{1\}} \right\}$                                                | $\{c_1\}$                                                    | 2. The fiber of $f$ over $c$ is $\{c_1\}$ .                                                                                                           |
| fong-spivak-invitation-F03957 | 334  | ■ 0.999 | $\operatorname{Sec}_f(V_2) = \varnothing$                                   | $\operatorname{Sec}_f(V_2) = \varnothing$                    | 2. When $V_2 = \{a, b, c, d\}$ , there are no sections: $\operatorname{Sec}_f(V_2) = \varnothing$ .                                                   |
| fong-spivak-invitation-F03958 | 334  | ■ 0.827 | $2 * 3 * 1 * 2 = 12$                                                        | $2 * 3 * 1 * 2 = 12$                                         | 3. When $V_3 = \{a, b, d, e\}$ , the set $\operatorname{Sec}_f(V_3)$ has $2 * 3 * 1 * 2 = 12$ elements.                                               |
| fong-spivak-invitation-F03959 | 334  | ■ 0.998 | $g_{\{1\}} := \left( a_{\{1\}}, b_{\{1\}} \right)$                          | $g_1 := (a_1, b_1)$                                          | 1. Let $g_1 := (a_1, b_1)$ and $g_2 := (b_2, e_1)$ ; these do not agree on the overlap.                                                               |
| fong-spivak-invitation-F03960 | 334  | ■ 0.998 | $g_{\{2\}} := \left( b_{\{2\}}, e_{\{1\}} \right)$                          | $g_2 := (b_2, e_1)$                                          | 1. Let $g_1 := (a_1, b_1)$ and $g_2 := (b_2, e_1)$ ; these do not agree on the overlap.                                                               |
| fong-spivak-invitation-F03961 | 334  | ■ 1.000 | $g \in \operatorname{Sec}_f(U_1 \cup U_2)$                                  | $g \in \operatorname{Sec}_f(U_1 \cup U_2)$                   | 2. No, there’s no section $g \in \operatorname{Sec}_f(U_1 \cup U_2)$ for which $g _{U_1} = g_1$ and $g _{U_2} = g_2$ .                                |
| fong-spivak-invitation-F03962 | 334  | ■ 1.000 | $g _{U_1} = g_1$                                                            | $g _{U_1} = g_1$                                             | 2. No, there’s no section $g \in \operatorname{Sec}_f(U_1 \cup U_2)$ for which $g _{U_1} = g_1$ and $g _{U_2} = g_2$ .                                |
| fong-spivak-invitation-F03963 | 334  | ■ 1.000 | $g _{U_2} = g_2$                                                            | $g _{U_2} = g_2$                                             | 2. No, there’s no section $g \in \operatorname{Sec}_f(U_1 \cup U_2)$ for which $g _{U_1} = g_1$ and $g _{U_2} = g_2$ .                                |
| fong-spivak-invitation-F03964 | 335  | ■ 0.997 | $F(\{1, 2\}) \rightarrow F(\{1\}) \rightarrow$                              | $F(\{1, 2\}) \rightarrow F(\{1\}) \rightarrow$               | 2. A presheaf $F$ on $\mathbf{Op}$ consists of any three sets and any two functions $F(\{1, 2\}) \rightarrow F(\{1\}) \rightarrow$                    |
| fong-spivak-invitation-F03965 | 335  | ■ 1.000 | $F(\varnothing)$                                                            | $F(\varnothing)$                                             | $F(\varnothing)$ between them.                                                                                                                        |
| fong-spivak-invitation-F03966 | 335  | ■ 1.000 | $F(\varnothing) \cong \{1\}$                                                | $F(\varnothing) \cong \{1\}$                                 | 4. As explained in Example 7.28, $F$ will be a sheaf iff $F(\varnothing) \cong \{1\}$ . Thus the category of sheaves                                  |
| fong-spivak-invitation-F03967 | 335  | ■ 1.000 | $F(\{1, 2\}) \rightarrow F(\{1\})$                                          | $F(\{1, 2\}) \rightarrow F(\{1\})$                           | is equivalent to that of just two sets and one function $F(\{1, 2\}) \rightarrow F(\{1\})$ .                                                          |
| fong-spivak-invitation-F03968 | 335  | ■ 0.748 | $S \in \mathbf{Shv}(X)$                                                     | $S \in \mathbf{Shv}(X)$                                      | The one-point space $X = \{1\}$ has two open sets, $\varnothing$ and $\{1\}$ , and every sheaf $S \in \mathbf{Shv}(X)$ assigns                        |
| fong-spivak-invitation-F03969 | 335  | ■ 0.999 | $S(\varnothing) = \{()\}$                                                   | $S(\varnothing) = \{()\}$                                    | $S(\varnothing) = \{()\}$ by the sheaf condition (see Example 7.28). So the only data in a sheaf $S \in \mathbf{Shv}(X)$ is                           |
| fong-spivak-invitation-F03970 | 335  | ■ 1.000 | $S(\{1\})$                                                                  | $S(\{1\})$                                                   | the set $S(\{1\})$ . This is how we get the correspondence between sets and sheaves on the one-point                                                  |
| fong-spivak-invitation-F03971 | 335  | ■ 0.862 | $\Omega : \mathbf{Op}(X)^{\mathrm{op}} \rightarrow$                         | $\Omega : \mathbf{Op}(X)^{\mathrm{op}} \rightarrow$          | According to Eq. (7.13), the subobject classifier $\Omega : \mathbf{Op}(X)^{\mathrm{op}} \rightarrow \mathbf{Set}$ in $\mathbf{Shv}(X)$ should be the |
| fong-spivak-invitation-F03972 | 335  | ■ 0.862 | $\mathbf{Shv}(X)$                                                           | $\mathbf{Shv}(X)$                                            | According to Eq. (7.13), the subobject classifier $\Omega : \mathbf{Op}(X)^{\mathrm{op}} \rightarrow \mathbf{Set}$ in $\mathbf{Shv}(X)$ should be the |
| fong-spivak-invitation-F03973 | 335  | ■ 0.945 | $\Omega(\{1\})$                                                             | $\Omega(\{1\})$                                              | functor where $\Omega(\{1\})$ is the set of open sets of $\{1\}$ . So we’re hoping to see that there is a one-                                        |
| fong-spivak-invitation-F03974 | 335  | ■ 0.445 | $\mathbf{Op}(\{1\})$                                                        | $\mathbf{Op}(\{1\})$                                         | to-one correspondence between the set $\mathbf{Op}(\{1\})$ and the set $\mathbb{B} = \{\text{true}, \text{false}\}$ of booleans.                      |
| fong-spivak-invitation-F03975 | 335  | ■ 1.000 | $\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}$                   | $\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}$    | By Eqs. (7.13) and (7.14) the definition of $\Omega(U)$ is $\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}$ , and the                        |
| fong-spivak-invitation-F03976 | 335  | ■ 1.000 | $U' \mapsto U' \cap V$                                                      | $U' \mapsto U' \cap V$                                       | definition of the restriction map for $V \subseteq U$ is $U' \mapsto U' \cap V$ .                                                                     |

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| fong-spivak-invitation_F03977 | 335  | ■ 0.996 | $U^{\{\prime\}}$ $\subseteq U$                                                            | $U' \subseteq U$                                                                          | 1. It is functorial: given $W \subseteq V \subseteq U$ and $U' \subseteq U$ , we indeed have $(U' \cap V) \cap W = U' \cap W$ ,                               |
| fong-spivak-invitation_F03978 | 335  | ■ 0.996 | $\left(U^{\{\prime\}} \cap V\right) \cap W = U^{\{\prime\}} \cap W$                       | $(U' \cap V) \cap W = U' \cap W$                                                          | 1. It is functorial: given $W \subseteq V \subseteq U$ and $U' \subseteq U$ , we indeed have $\boxed{U' \cap V \cap W = U' \cap W}$ ,                         |
| fong-spivak-invitation_F03979 | 335  | ■ 1.000 | $W \subseteq V$                                                                           | $W \subseteq V$                                                                           | since $W \subseteq V$ . For functoriality, we also need preservation of identities, and this amounts to                                                       |
| fong-spivak-invitation_F03980 | 335  | ■ 1.000 | $U^{\{\prime\}} \cap U = U^{\{\prime\}}$                                                  | $U' \cap U = U'$                                                                          | $U' \cap U = U'$ for all $U' \subseteq U$ .                                                                                                                   |
| fong-spivak-invitation_F03981 | 335  | ■ 1.000 | $G^{\{\prime\}}$                                                                          | $G'$                                                                                      | There is only one that classifies $G'$ , and here it is. Let's write $\gamma := \ulcorner G' \urcorner$ .                                                     |
| fong-spivak-invitation_F03982 | 335  | ■ 1.000 | $\gamma := \left\langle G^{\{\prime\}} \right\rangle$                                     | $\gamma := \ulcorner G' \urcorner$                                                        | There is only one that classifies $G'$ , and here it is. Let's write $\gamma := \ulcorner G' \urcorner$ .                                                     |
| fong-spivak-invitation_F03983 | 335  | ■ 1.000 | $\gamma(D) = 0$                                                                           | $\gamma(D) = 0$                                                                           | • Since $D$ is missing from $G'$ , we have $\gamma(D) = 0$ (vertex: missing).                                                                                 |
| fong-spivak-invitation_F03984 | 335  | ■ 1.000 | $A, B, C$                                                                                 | $A, B, C$                                                                                 | • Since vertices $A, B, C$ are present in $G'$ we have $\gamma(A) = \gamma(B) = \gamma(C) = V$ (vertex: present).                                             |
| fong-spivak-invitation_F03985 | 335  | ■ 1.000 | $\gamma(A) = \gamma(B) = \gamma(C) = V$                                                   | $\gamma(A) = \gamma(B) = \gamma(C) = V$                                                   | • Since vertices $A, B, C$ are present in $G'$ we have $\boxed{\gamma(A) = \gamma(B) = \gamma(C) = V}$ (vertex: present).                                     |
| fong-spivak-invitation_F03986 | 335  | ■ 1.000 | $\gamma(i) = (V, 0; 0)$                                                                   | $\gamma(i) = (V, 0; 0)$                                                                   | • The above forces $\boxed{\gamma(i) = (V, 0; 0)}$ (arrow from present vertex to missing vertex: missing).                                                    |
| fong-spivak-invitation_F03987 | 335  | ■ 1.000 | $\gamma(f) = (V, V; A)$                                                                   | $\gamma(f) = (V, V; A)$                                                                   | • Since the arrow $f$ is in $G'$ , we have $\boxed{\gamma(f) = (V, V; A)}$ (arrow from present vertex to present                                              |
| fong-spivak-invitation_F03988 | 336  | ■ 1.000 | $\gamma(g) = \gamma(h) = (V, V; 0)$                                                       | $\gamma(g) = \gamma(h) = (V, V; 0)$                                                       | • Since the arrows $g$ and $h$ are missing in $G'$ , we have $\gamma(g) = \gamma(h) = (V, V; 0)$ (arrow from                                                  |
| fong-spivak-invitation_F03989 | 336  | ■ 1.000 | $U = \mathbb{R} - \{0\} \subseteq \mathbb{R}$                                             | $U = \mathbb{R} - \{0\} \subseteq \mathbb{R}$                                             | With $\boxed{U = \mathbb{R} - \{0\} \subseteq \mathbb{R}}$ , we have the following.                                                                           |
| fong-spivak-invitation_F03990 | 336  | ■ 1.000 | $\mathbb{R} - U = \{0\}$                                                                  | $\mathbb{R} - U = \{0\}$                                                                  | 1. The complement of $U$ is $\mathbb{R} - U = \{0\}$ and $\neg U$ is its interior, which is $\neg U = \emptyset$ .                                            |
| fong-spivak-invitation_F03991 | 336  | ■ 1.000 | $\neg U = \varnothing$                                                                    | $\neg U = \emptyset$                                                                      | 1. The complement of $U$ is $\mathbb{R} - U = \{0\}$ and $\neg U$ is its interior, which is $\neg U = \emptyset$ .                                            |
| fong-spivak-invitation_F03992 | 336  | ■ 1.000 | $\mathbb{R} - \neg U = \mathbb{R}$                                                        | $\mathbb{R} - \emptyset = \mathbb{R}$                                                     | 2. The complement of $\neg U$ is $\mathbb{R} - \emptyset = \mathbb{R}$ , and this is open, so $\neg \neg U = \mathbb{R}$ .                                    |
| fong-spivak-invitation_F03993 | 336  | ■ 1.000 | $\neg \neg U = \mathbb{R}$                                                                | $\neg \neg U = \mathbb{R}$                                                                | 2. The complement of $\neg U$ is $\mathbb{R} - \emptyset = \mathbb{R}$ , and this is open, so $\neg \neg U = \mathbb{R}$ .                                    |
| fong-spivak-invitation_F03994 | 336  | ■ 0.869 | $\neg \neg U \subseteq U$                                                                 | $\neg \neg U \subseteq U$                                                                 | 4. It is false that $\neg \neg U \subseteq U$ .                                                                                                               |
| fong-spivak-invitation_F03995 | 336  | ■ 0.976 | $V \in \mathbf{Op}$                                                                       | $V \in \mathbf{Op}$                                                                       | 1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$ ,                                  |
| fong-spivak-invitation_F03996 | 336  | ■ 0.976 | $\top \wedge V = V$                                                                       | $\top \wedge V = V$                                                                       | 1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$ ,                                  |
| fong-spivak-invitation_F03997 | 336  | ■ 0.976 | $V = X$                                                                                   | $V = X$                                                                                   | 1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$ ,                                  |
| fong-spivak-invitation_F03998 | 336  | ■ 0.976 | $\top \wedge X := \top \cap X = X$                                                        | $\top \wedge X := \top \cap X = X$                                                        | 1. If for any $V \in \mathbf{Op}$ we have $\top \wedge V = V$ then when $V = X$ we have $\top \wedge X := \top \cap X = X$ ,                                  |
| fong-spivak-invitation_F03999 | 336  | ■ 1.000 | $\top = \top \cap X = X$                                                                  | $\top = \top \cap X = X$                                                                  | but anything intersected with $X$ is itself, so $\top = \top \cap X = X$ .                                                                                    |
| fong-spivak-invitation_F04000 | 336  | ■ 0.897 | $(\top \vee V) := (X \cup V) = X$                                                         | $(\top \vee V) := (X \cup V) = X$                                                         | 2. $(\top \vee V) := (X \cup V) = X$ holds, $(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$ holds because               |
| fong-spivak-invitation_F04001 | 336  | ■ 0.897 | $(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$     | $(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$     | 2. $(\top \vee V) := (X \cup V) = X$ holds, $(V \Rightarrow X) = \bigcup_{\{R \in \mathbf{Op} \mid R \cap V \subseteq X\}} R = X$ holds because               |
| fong-spivak-invitation_F04002 | 336  | ■ 1.000 | $(X \cap V) \subseteq X$                                                                  | $(X \cap V) \subseteq X$                                                                  | $(X \cap V) \subseteq X$ , and $(X \Rightarrow V) = V = \bigcup_{\{R \in \mathbf{Op} \mid R \cap X \subseteq V\}} R = Y$ holds because $R \cap X = R$ .       |
| fong-spivak-invitation_F04003 | 336  | ■ 1.000 | $(X \Rightarrow V) = V = \bigcup_{\{R \in \mathbf{Op} \mid R \cap X \subseteq V\}} R = Y$ | $(X \Rightarrow V) = V = \bigcup_{\{R \in \mathbf{Op} \mid R \cap X \subseteq V\}} R = Y$ | $(X \cap V) \subseteq X$ , and $\boxed{X \Rightarrow V = V = \bigcup_{\{R \in \mathbf{Op} \mid R \cap X \subseteq V\}} R = Y}$ holds because $R \cap X = R$ . |
| fong-spivak-invitation_F04004 | 336  | ■ 1.000 | $R \cap X = R$                                                                            | $R \cap X = R$                                                                            | $(X \cap V) \subseteq X$ , and $(X \Rightarrow V) = V = \bigcup_{\{R \in \mathbf{Op} \mid R \cap X \subseteq V\}} R = Y$ holds because $R \cap X = R$ .       |

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| fong-spivak-invitation_F04005 | 336  | ■ 0.992 | $(\perp \vee V) = V$                                                                       | $(\perp \vee V) = V$                                                                       | 3. If for any set $V \in \mathbf{Op}$ we have $(\perp \vee V) = V$ , then when $V = \emptyset$ we have $(\perp \vee \emptyset) := (\perp \cup \emptyset) =$                |
| fong-spivak-invitation_F04006 | 336  | ■ 0.992 | $V = \emptyset$                                                                            | $V = \emptyset$                                                                            | 3. If for any set $V \in \mathbf{Op}$ we have $(\perp \vee V) = V$ , then when $V = \emptyset$ we have $(\perp \vee \emptyset) := (\perp \cup \emptyset) =$                |
| fong-spivak-invitation_F04007 | 336  | ■ 0.992 | $(\perp \vee \varnothing) := (\perp \cup \varnothing) =$                                   | $(\perp \vee \emptyset) := (\perp \cup \emptyset) =$                                       | 3. If for any set $V \in \mathbf{Op}$ we have $(\perp \vee V) = V$ , then when $V = \emptyset$ we have $(\perp \vee \emptyset) := (\perp \cup \emptyset) =$                |
| fong-spivak-invitation_F04008 | 336  | ■ 0.679 | $\perp = \perp \cup \varnothing = \varnothing$                                             | $\perp = \perp \cup \emptyset = \emptyset$                                                 | $\emptyset$ , but anything in a union with $\emptyset$ is itself, so $\perp = \perp \cup \emptyset = \emptyset$ .                                                          |
| fong-spivak-invitation_F04009 | 336  | ■ 0.995 | $(\perp \wedge V) = (\varnothing \cap V) = \varnothing$                                    | $(\perp \wedge V) = (\emptyset \cap V) = \emptyset$                                        | 4. $\perp \wedge V = (\emptyset \cap V) = \emptyset$ holds, and $(\perp \Rightarrow V) = \bigcup_{R \in \mathbf{Op}   R \cap \emptyset \subseteq V} R = X$ holds because   |
| fong-spivak-invitation_F04010 | 336  | ■ 0.995 | $(\perp \Rightarrow V) = \bigcup_{R \in \mathbf{Op}   R \cap \emptyset \subseteq V} R = X$ | $(\perp \Rightarrow V) = \bigcup_{R \in \mathbf{Op}   R \cap \emptyset \subseteq V} R = X$ | 4. $(\perp \wedge V) = (\emptyset \cap V) = \emptyset$ holds, and $(\perp \Rightarrow V) = \bigcup_{R \in \mathbf{Op}   R \cap \emptyset \subseteq V} R = X$ holds because |
| fong-spivak-invitation_F04011 | 336  | ■ 0.835 | $(X \cap \varnothing) \subseteq V$                                                         | $(X \cap \emptyset) \subseteq V$                                                           | $(X \cap \emptyset) \subseteq V$ .                                                                                                                                         |
| fong-spivak-invitation_F04012 | 336  | ■ 1.000 | $\{S \mid p\}$                                                                             | $\{S \mid p\}$                                                                             | time is a person who is alive throughout that interval. A section in the subobject $\{S \mid p\}$ over any                                                                 |
| fong-spivak-invitation_F04013 | 336  | ■ 0.508 | $S = \mathbb{N}$                                                                           | $S = \mathbb{N}$                                                                           | ing to the set $S = \mathbb{N}$ , let $p(s)$ be the predicate “ $24 \leq s \leq 28$ ,” and let $q(s)$ be the predicate “ $s$ is                                            |
| fong-spivak-invitation_F04014 | 336  | ■ 0.508 | $\leq s \leq 28$                                                                           | $\leq s \leq 28$                                                                           | ing to the set $S = \mathbb{N}$ , let $p(s)$ be the predicate “ $24 \leq s \leq 28$ ,” and let $q(s)$ be the predicate “ $s$ is                                            |
| fong-spivak-invitation_F04015 | 336  | ■ 0.508 | $q(s)$                                                                                     | $q(s)$                                                                                     | ing to the set $S = \mathbb{N}$ , let $p(s)$ be the predicate “ $24 \leq s \leq 28$ ,” and let $q(s)$ be the predicate “ $s$ is                                            |
| fong-spivak-invitation_F04016 | 336  | ■ 1.000 | $p(n, z)$                                                                                  | $p(n, z)$                                                                                  | We have the predicate $p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$ given by $p(n, z)$ iff $n \leq  z $ .                                                       |
| fong-spivak-invitation_F04017 | 336  | ■ 1.000 | $n \leq  z $                                                                               | $n \leq  z $                                                                               | We have the predicate $p: \mathbb{N} \times \mathbb{Z} \rightarrow \mathbb{B}$ given by $p(n, z)$ iff $n \leq  z $ .                                                       |
| fong-spivak-invitation_F04018 | 336  | ■ 1.000 | $\{0\} \subseteq \mathbb{N}$                                                               | $\{0\} \subseteq \mathbb{N}$                                                               | 1. The predicate $\forall (z: \mathbb{Z}). p(n, z)$ holds for $\{0\} \subseteq \mathbb{N}$ .                                                                               |
| fong-spivak-invitation_F04019 | 336  | ■ 1.000 | $\mathbb{N} \subseteq \mathbb{N}$                                                          | $\mathbb{N} \subseteq \mathbb{N}$                                                          | 2. The predicate $\exists (z: \mathbb{Z}). p(n, z)$ holds for $\mathbb{N} \subseteq \mathbb{N}$ .                                                                          |
| fong-spivak-invitation_F04020 | 336  | ■ 1.000 | $\varnothing \subseteq \mathbb{Z}$                                                         | $\emptyset \subseteq \mathbb{Z}$                                                           | 3. The predicate $\forall (n: \mathbb{N}). p(n, z)$ holds for $\emptyset \subseteq \mathbb{Z}$ .                                                                           |
| fong-spivak-invitation_F04021 | 336  | ■ 1.000 | $\mathbb{Z} \subseteq \mathbb{Z}$                                                          | $\mathbb{Z} \subseteq \mathbb{Z}$                                                          | 4. The predicate $\exists (n: \mathbb{N}). p(n, z)$ holds for $\mathbb{Z} \subseteq \mathbb{Z}$ .                                                                          |
| fong-spivak-invitation_F04022 | 337  | ■ 0.532 | $\forall (t: T)$                                                                           | $\forall (t: T)$                                                                           | 1. The formula says that $\forall (t: T). p(s, t)$ “returns the largest open set $V \subseteq U$ for which                                                                 |
| fong-spivak-invitation_F04023 | 337  | ■ 1.000 | $t_i \in T(V_i)$                                                                           | $t_i \in T(V_i)$                                                                           | for which there exists some $t_i \in T(V_i)$ satisfying $p(s _{V_i}, t_i) = V_i$ .” Substituting, this becomes                                                             |
| fong-spivak-invitation_F04024 | 337  | ■ 1.000 | $t_i$                                                                                      | $t_i$                                                                                      | “the union of all time intervals $V_i$ for which there is some item $t_i$ in the news about which $s$                                                                      |
| fong-spivak-invitation_F04025 | 337  | ■ 1.000 | $j(j(q)) \leq j(q)$                                                                        | $j(j(q)) \leq j(q)$                                                                        | It is clear that if $j(j(q)) = j(q)$ then $j(j(q)) \leq j(q)$ by reflexivity. On the other hand, assume                                                                    |
| fong-spivak-invitation_F04026 | 337  | ■ 0.999 | $p := j(q)$                                                                                | $p := j(q)$                                                                                | $p := j(q)$ we have both $j(p) \leq p$ and $p \leq j(p)$ . This means $p \cong j(p)$ , but $\Omega$ is a poset (not                                                        |
| fong-spivak-invitation_F04027 | 337  | ■ 0.999 | $j(p) \leq p$                                                                              | $j(p) \leq p$                                                                              | $p := j(q)$ we have both $j(p) \leq p$ and $p \leq j(p)$ . This means $p \cong j(p)$ , but $\Omega$ is a poset (not                                                        |
| fong-spivak-invitation_F04028 | 337  | ■ 0.999 | $p \cong j(p)$                                                                             | $p \cong j(p)$                                                                             | $p := j(q)$ we have both $j(p) \leq p$ and $p \leq j(p)$ . This means $p \cong j(p)$ , but $\Omega$ is a poset (not                                                        |
| fong-spivak-invitation_F04029 | 337  | ■ 1.000 | $p = j(p)$                                                                                 | $p = j(p)$                                                                                 | just a preorder) so $p = j(p)$ , i.e. $j(j(q)) = j(q)$ as desired.                                                                                                         |
| fong-spivak-invitation_F04030 | 337  | ■ 0.912 | $s \in S(U), p(s)$                                                                         | $s \in S(U), p(s)$                                                                         | 2. Let $U$ be the interval 2019/01/01 – 2019/02/01. For an arbitrary person $s \in S(U)$ , $p(s)$ is a                                                                     |
| fong-spivak-invitation_F04031 | 338  | ■ 0.916 | $j(j(p(s))) = j(p(s))$                                                                     | $j(j(p(s))) = j(p(s))$                                                                     | 5. It is true that $j(j(p(s))) = j(p(s))$ , because suppose we are given a time during which “if Bob                                                                       |
| fong-spivak-invitation_F04032 | 338  | ■ 0.890 | $0 \leq 2 \leq 6 \leq 8$                                                                   | $0 \leq 2 \leq 6 \leq 8$                                                                   | 1. Since $0 \leq 2 \leq 6 \leq 8$ , we have $[2, 6] \in o_{[0,8]}$ by the above formula.                                                                                   |

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| fong-spivak-invitation-F04033 | 338  | 0.890 | $[2,6] \in o_{\{[0,8]\}}$                                                             | $[2,6] \in O_{[0,8]}$                                         | 1. Since $0 \leq 2 \leq 6 \leq 8$ , we have $[2,6] \in o_{[0,8]}$ by the above formula.                                                                                           |
| fong-spivak-invitation-F04034 | 338  | 0.603 | $\text{in}^{\{\text{?}\}} o_{\{[0,5]\}} \cup o_{\{[4,8]\}}$                           | $\in^? o_{[0,5]} \cup o_{[4,8]}$                              | 2. In order to have $[2,6] \in^? o_{[0,5]} \cup o_{[4,8]}$ , we would need to have either $[2,6] \in^? o_{[0,5]}$ or                                                              |
| fong-spivak-invitation-F04035 | 338  | 0.603 | $\text{in}^{\{\text{?}\}} o_{\{[0,5]\}}$                                              | $\in^? o_{[0,5]}$                                             | 2. In order to have $[2,6] \in^? o_{[0,5]} \cup o_{[4,8]}$ , we would need to have either $[2,6] \in^? o_{[0,5]}$ or                                                              |
| fong-spivak-invitation-F04036 | 338  | 1.000 | $[2,6] \in^{\{?\}} o_{\{[4,8]\}}$                                                     | $[2,6] \in^? o_{[4,8]}$                                       | <span style="border: 1px solid red; padding: 2px;"><math>[2,6] \in^? o_{[4,8]}</math></span> . But the formula does not hold in either case.                                      |
| fong-spivak-invitation-F04037 | 338  | 0.984 | $\mathbb{R} \subseteq \mathbb{R}$                                                     | $\mathbb{R} \subseteq \mathbb{R}$                             | A subset $U \subseteq \mathbb{R}$ is open in the subspace topology of $\mathbb{R} \subseteq \mathbb{R}$ iff there is an open set $U' \subseteq \mathbb{R}$                        |
| fong-spivak-invitation-F04038 | 338  | 0.984 | $U^{\prime} \subseteq \mathbb{R}$                                                     | $U' \subseteq \mathbb{R}$                                     | A subset $U \subseteq \mathbb{R}$ is open in the subspace topology of $\mathbb{R} \subseteq \mathbb{R}$ iff there is an open set $U' \subseteq \mathbb{R}$                        |
| fong-spivak-invitation-F04039 | 338  | 1.000 | $U = U^{\prime} \cap \mathbb{R}$                                                      | $U = U' \cap \mathbb{R}$                                      | with <span style="border: 1px solid red; padding: 2px;"><math>U = U' \cap \mathbb{R}</math></span> . We want to show that this is the case iff $U$ is open in the usual topology. |
| fong-spivak-invitation-F04040 | 338  | 1.000 | $U^{\prime} = \bigcup_{i \in I} o_{\left[a_i, b_i\right]}$                            | $U' = \bigcup_{i \in I} o_{[a_i, b_i]}$                       | union of some basic opens, $U' = \bigcup_{i \in I} o_{[a_i, b_i]}$ , where $o_{[a_i, b_i]} = \{[d, u] \in \mathbb{R} \mid a_i < d < u <$                                          |
| fong-spivak-invitation-F04041 | 338  | 1.000 | $o_{\left[a_i, b_i\right]} = \left\{[d, u] \in \mathbb{R} \mid a_i < d < u < \right.$ | $o_{[a_i, b_i]} = \{[d, u] \in \mathbb{R} \mid a_i < d < u <$ | union of some basic opens, $U' = \bigcup_{i \in I} o_{[a_i, b_i]}$ , where $o_{[a_i, b_i]} = \{[d, u] \in \mathbb{R} \mid a_i < d < u <$                                          |
| fong-spivak-invitation-F04042 | 338  | 0.956 | $\left. b_i\right\}$                                                                  | $b_i\}$                                                       | $b_i\}$ . Since $\mathbb{R} = \{[x, x] \in \mathbb{R}\}$ , the intersection $U' \cap \mathbb{R}$ will then be                                                                     |
| fong-spivak-invitation-F04043 | 338  | 0.956 | $\mathbb{R} = \{[x, x] \in \mathbb{R}\}$                                              | $\mathbb{R} = \{[x, x] \in \mathbb{R}\}$                      | $b_i\}$ . Since $\mathbb{R} = \{[x, x] \in \mathbb{R}\}$ , the intersection $U' \cap \mathbb{R}$ will then be                                                                     |
| fong-spivak-invitation-F04044 | 338  | 0.956 | $U^{\prime} \cap \mathbb{R}$                                                          | $U' \cap \mathbb{R}$                                          | $b_i\}$ . Since $\mathbb{R} = \{[x, x] \in \mathbb{R}\}$ , the intersection $U' \cap \mathbb{R}$ will then be                                                                     |
| fong-spivak-invitation-F04045 | 338  | 1.000 | $B\left(m_i, r_i\right)$                                                              | $B\left(m_i, r_i\right)$                                      | and this is just the union of open balls $B\left(m_i, r_i\right)$ where $m_i := \frac{a_i+b_i}{2}$ is the midpoint and $r_i :=$                                                   |
| fong-spivak-invitation-F04046 | 338  | 1.000 | $m_i := \frac{a_i+b_i}{2}$                                                            | $m_i := \frac{a_i+b_i}{2}$                                    | and this is just the union of open balls $B\left(m_i, r_i\right)$ where $m_i := \frac{a_i+b_i}{2}$ is the midpoint and $r_i :=$                                                   |
| fong-spivak-invitation-F04047 | 338  | 1.000 | $r_i :=$                                                                              | $r_i :=$                                                      | and this is just the union of open balls $B\left(m_i, r_i\right)$ where $m_i := \frac{a_i+b_i}{2}$ is the midpoint and $r_i :=$                                                   |
| fong-spivak-invitation-F04048 | 338  | 0.500 | $\frac{b_i-a_i}{2}$                                                                   | $\frac{b_i-a_i}{2}$                                           | $\frac{b_i-a_i}{2}$ is the radius of the interval $(a_i, b_i)$ . The open balls $B\left(m_i, r_i\right)$ are open in the usual topology                                           |
| fong-spivak-invitation-F04049 | 338  | 0.500 | $\left(a_i, b_i\right)$                                                               | $\left(a_i, b_i\right)$                                       | $\frac{b_i-a_i}{2}$ is the radius of the interval $(a_i, b_i)$ . The open balls $B\left(m_i, r_i\right)$ are open in the usual topology                                           |
| fong-spivak-invitation-F04050 | 338  | 1.000 | $U = \bigcup_{j \in J} B\left(m_j, \epsilon_j\right)$                                 | $U = \bigcup_{j \in J} B\left(m_j, \epsilon_j\right)$         | Suppose that $U$ is open in the usual topology. Then $U = \bigcup_{j \in J} B\left(m_j, \epsilon_j\right)$ for some set $J$ . Let                                                 |
| fong-spivak-invitation-F04051 | 338  | 1.000 | $a_j := m_j - \epsilon_j$                                                             | $a_j := m_j - \epsilon_j$                                     | $a_j := m_j - \epsilon_j$ and $b_j := m_j + \epsilon_j$ . Then                                                                                                                    |
| fong-spivak-invitation-F04052 | 338  | 1.000 | $b_j := m_j + \epsilon_j$                                                             | $b_j := m_j + \epsilon_j$                                     | $a_j := m_j - \epsilon_j$ and $b_j := m_j + \epsilon_j$ . Then                                                                                                                    |
| fong-spivak-invitation-F04053 | 338  | 0.991 | $H_X(U) := \{f: U \cap R \rightarrow X \mid f$                                        | $H_X(U) := \{f: U \cap R \rightarrow X \mid f$                | <span style="border: 1px solid red; padding: 2px;"><math>H_X(U) := \{f: U \cap R \rightarrow X \mid f</math></span> is continuous}.                                               |
| fong-spivak-invitation-F04054 | 338  | 1.000 | $f: U \cap R \rightarrow$                                                             | $f: U \cap R \rightarrow$                                     | 1. $H_X$ is a presheaf: given $V \subseteq U$ the restriction map sends the continuous function $f: U \cap R \rightarrow$                                                         |
| fong-spivak-invitation-F04055 | 338  | 1.000 | $V \cap R \subseteq U \cap R$                                                         | $V \cap R \subseteq U \cap R$                                 | $X$ to its restriction along the subset <span style="border: 1px solid red; padding: 2px;"><math>V \cap R \subseteq U \cap R</math></span> .                                      |
| fong-spivak-invitation-F04056 | 338  | 1.000 | $U = \bigcup_i U_i$                                                                   | $U = \bigcup_i U_i$                                           | 2. It is a sheaf: given any family $U_i$ of open sets with $U = \bigcup_i U_i$ and a continuous function                                                                          |
| fong-spivak-invitation-F04057 | 338  | 1.000 | $f_i: U_i \cap R \rightarrow X$                                                       | $f_i: U_i \cap R \rightarrow X$                               | $f_i: U_i \cap R \rightarrow X$ for each $i$ , agreeing on overlaps, they can be glued together to give a                                                                         |
| fong-spivak-invitation-F04058 | 338  | 1.000 | $U \cap R$                                                                            | $U \cap R$                                                    | continuous function on all of $U \cap R$ , since $U \cap R = (\bigcup_i U_i) \cap R = \bigcup_i (U_i \cap R)$ .                                                                   |

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| fong-spivak-invitation_F04659 | 338  | ■ 1.000 | $U \cap R = \left( \bigcup_i U_i \right) \cap R = \bigcup_i (U_i \cap R)$ | $U \cap R = \left( \bigcup_i U_i \right) \cap R = \bigcup_i (U_i \cap R)$ | continuous function on all of $U \cap R$ , since $U \cap R = (\bigcup_i U_i) \cap R = \bigcup_i (U_i \cap R)$ . |